A CONCEPT-BASED MATHEMATICS COURSE

YEARS 3 · 4 · 5
MIDDLE YEARS PROGRAMME

STUDENT COURSEBOOK · VOLUME FIVE

MYP *Mathematics* 5

STUDENT COURSEBOOK
YEAR 5 · MODELLING · PROOF · TRANSITION



STRUCTURE

TRANSFORMATION

AUTHOR

Sadikou Moukoddim

Head of Mathematics Department
Mathematics Teacher · Category 1

ABOUT THIS VOLUME

The final volume of an original, inquiry-led course. It develops exponential and logarithmic functions, sequences and series, circle geometry, vectors and matrices, correlation and conditional probability, and closes with a sustained modelling capstone.

y = log x

MYP MATHEMATICS 5

A FIVE-VOLUME MATHEMATICS COURSE
FOR THE MIDDLE YEARS PROGRAMME

VOLUME FIVE

MYP *Mathematics* 5

STUDENT COURSEBOOK

Algebra and functions · Geometry and trigonometry · Statistics and probability · Modelling

First edition · 2026

MYP MATHEMATICS 5

A CONCEPT-BASED COURSE FOR YEARS 3–5

MYP Mathematics 5

STUDENT COURSEBOOK

Inquiry-led teaching, rigorous practice and authentic assessment for the final year of a coherent five-year mathematics series, with a deliberate bridge towards the Diploma Programme.

AUTHOR

Sadikou Moukoddim

Head of Mathematics Department
Mathematics Teacher · Category 1

VOLUME 5 OF 5

FIRST EDITION · 2026

PUBLICATION INFORMATION

MYP Mathematics 5

Student Coursebook

Author Sadikou Moukoddim, Head of Mathematics Department, Mathematics Teacher, Category 1.

Edition First edition, 2026. Volume 5 of a five-volume series.

Copyright © 2026 Sadikou Moukoddim. All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording or otherwise—without the prior written permission of the author, except for brief quotations in reviews or for classroom use within the purchasing institution.

Original work All chapter sequences, explanations, worked examples, investigations, exercises, assessment tasks, diagrams, cover art and design in this volume are original. No chapter structure, exercise text, graphic or page design has been copied from another textbook.

Curriculum note This is an independent publication. It has been designed with reference to the publicly available IB Middle Years Programme Mathematics framework (updated 2020) and the MYP assessment criteria A–D. The International Baccalaureate and MYP are trademarks of the International Baccalaureate Organization. This publication is not endorsed by, affiliated with, or sponsored by the IB. Year 5 objectives in this book are school-developed bridge objectives between MYP year 5 and the Diploma Programme, as the framework permits; they are not presented as an official IB syllabus, and this book does not guarantee admission to, or success in, any Diploma Programme course.

Design and production Cover and interior design created for this series. Diagrams are original vector graphics. Body text is set in Source Serif 4, display and labels in Inter and Bodoni Moda, and mathematics in KaTeX. Printed in A4 format.

Contact Mathematics Department. For permissions, corrections or adoption enquiries, contact the author through the issuing school.

“The essence of mathematics is not to make simple things complicated, but to make complicated things simple.” — S. Gudder

CONTENTS

The course

Page numbers refer to the main body of the book. Each chapter opens with an inquiry, develops concepts through worked examples and graduated practice, and closes with mixed practice, an assessment task, a summary and reflection.

01 Polynomials, Factor Theorem and Algebraic Fractions 1

- 1.1 Polynomials and their behaviour 3
- 1.2 Dividing polynomials 6
- 1.3 The remainder and factor theorems 9
- 1.4 Algebraic fractions 11
- 1.5 Rational equations and applications 14
- 1.6 Investigation: factors, roots and sign 17

02 Functions: Transformations, Composites and Inverses 23

- 2.1 Domain, range and function notation 25
- 2.2 Transformations of graphs 27
- 2.3 Composite functions 30
- 2.4 Inverse functions 32
- 2.5 Modelling with transformations 34
- 2.6 Investigation: when is a function invertible? 36

03 Exponential and Logarithmic Functions 43

- 3.1 Exponential growth and decay 45
- 3.2 The number e and continuous change 48
- 3.3 Logarithms and their laws 50
- 3.4 Solving exponential equations 53
- 3.5 Modelling with logarithms 55
- 3.6 Investigation: how many digits? 57

04 Sequences and Series 63

- 4.1 Arithmetic sequences 65
- 4.2 Arithmetic series and sigma notation 68
- 4.3 Geometric sequences 71
- 4.4 Geometric series and convergence 75
- 4.5 Annuities, loans and saving 78
- 4.6 Investigation: infinite sums 81

HOW THE BOOK WORKS

Every chapter follows the same rhythm: an authentic problem, a conceptual inquiry, precise definitions, worked reasoning, graduated practice, an investigation, a modelling task and a criterion-mapped assessment.

How to use this book vii

The MYP framework in Year 5 viii

Year 5 course map and progression x

Assumed Year 4 foundations xi

From MYP to the Diploma Programme xii

The chapters are sequenced so that each idea is met in a concrete or visual form before it is formalised. Later chapters deliberately reuse earlier tools: polynomial division from Chapter 1 underpins the factor theorem and rational functions; logarithms from Chapter 3 solve the financial equations of Chapter 4; vectors from Chapter 8 explain the transformations of Chapter 2.

CONTENTS

The course

05 Quadratic Inequalities and Linear Programming 89

5.1 Quadratic inequalities in one variable 91
5.2 Inequalities in two variables 94
5.3 Feasible regions and vertices 96
5.4 Linear programming 98
5.5 Modelling and optimisation 100
5.6 Investigation: integer solutions and the vertex 103

06 Circle Geometry and Geometric Proof 109

6.1 Angles in circles 111
6.2 Cyclic quadrilaterals 114
6.3 Tangents and radii 116
6.4 Chords, secants and the alternate segment 119
6.5 Constructing and writing proofs 122
6.6 Investigation: circle theorems under transformation 124

07 Sine and Cosine Rules; Trigonometry in Three Dimensions 129

7.1 The sine rule 131
7.2 The cosine rule 134
7.3 Area and the included angle 136
7.4 Bearings and navigation 138
7.5 Trigonometry in three dimensions 140
7.6 Investigation: measuring the unmeasurable 143

08 Vectors and Matrices 151

8.1 Vectors: notation and operations 153
8.2 Vectors and geometry 156
8.3 Vectors in three dimensions 158
8.4 Matrices and transformations 160
8.5 Determinants and inverse matrices 163
8.6 Investigation: area scale factors 166

REFERENCE AND REVISION

The final two chapters prepare you for cumulative assessment and for the transition to the Diploma Programme. Neither is a substitute for the official subject guide of any DP course.

Chapter 11 · Modelling capstone 217
Chapter 12 · Cumulative revision 233
Answers and assessment guidance 245
Formulae and notation 319
Glossary 323
Index 326

Chapters 1–4 build the algebraic and functional language of the year. Chapters 5–8 apply it to optimisation, geometry and linear algebra. Chapters 9–10 develop statistical and probabilistic reasoning. Chapter 11 integrates everything in a sustained modelling project; Chapter 12 revisits the whole course through unfamiliar problems.

CONTENTS

The course

09 **Statistics: Dispersion, Correlation and Regression** 173

9.1 Measuring spread 175
9.2 Standard deviation and grouped data 177
9.3 Comparing distributions 180
9.4 Scatter graphs and correlation 183
9.5 Regression and prediction 185
9.6 Investigation: correlation, causation and reliability 188

10 **Probability: Conditional Probability and Counting** 195

10.1 Sets, tables and probability 197
10.2 Conditional probability 200
10.3 Independence 203
10.4 Tree diagrams and total probability 205
10.5 Counting, permutations and combinations 208
10.6 Investigation: the birthday problem 210

11 **Capstone: A Mathematical Modelling Project** 217

11.1 The modelling cycle 219
11.2 Defining the problem 221
11.3 Building and testing a model 224
11.4 Optimising and evaluating 226
11.5 Presenting and defending 228

12 **Cumulative Revision and the Transition to DP Mathematics** 233

12.1 Cumulative practice: algebra and functions 235
12.2 Cumulative practice: geometry and trigonometry 236
12.3 Cumulative practice: statistics and probability 237
12.4 Unfamiliar problems and extended reasoning 238
12.5 Cumulative examination 239
12.6 From MYP to DP: expectations and readiness 241

REFERENCE

Answer entries give the decisive steps, not only results. Open tasks include sample responses and marking guidance.

Answers and assessment guidance 245
Formulae and notation 319
Glossary 323
Index 326

Chapter 12 is deliberately cumulative: it does not introduce new content, but asks you to select methods across the whole book. Its final section explains, without overclaiming, how MYP expectations differ from those of DP Mathematics courses.

ORIENTATION

How to use *this book*

The rhythm of a chapter

1. **Opening problem.** A real or mathematical situation that you cannot yet solve completely. It creates the need for the mathematics.
2. **Inquiry.** The statement of inquiry, key and related concepts, global context, and factual, conceptual and debatable questions frame the chapter.
3. **Prerequisite check.** A short diagnostic. If you cannot do these, use the refresher box before continuing.
4. **Concept development.** Definitions, representations, worked examples and your own explanations.
5. **Graduated practice.** Fluency, understanding, reasoning, multi-step application and extension. Do not skip the reasoning tier.
6. **Investigation.** You discover a pattern, generalise it and justify it.
7. **Modelling.** You use mathematics in a context where assumptions, units and accuracy matter.
8. **Mixed practice.** Cumulative retrieval across the whole chapter.
9. **Assessment.** A Criterion A–D task with a rubric. This is summative: complete it independently.
10. **Summary and reflection.** Check what you can do and what you still need to repair.

Components you will meet

INQUIRY
A question or problem that drives the section. It is not rhetorical: you should be able to attempt an answer by the end.

KEY IDEA
The smallest statement worth remembering. If you can explain it and use it, you have understood the concept.

WORKED EXAMPLE
A complete line of reasoning with the decisions made visible. The notes explain *why*, not just *what*.

MISCONCEPTION
A common error, stated plainly, with the reasoning that exposes it. Read these before practising.

GUIDED PRACTICE
Partially completed work. You supply the missing reasoning, then compare with a partner.

EXERCISE
Numbered practice. Tier labels indicate fluency, understanding, reasoning, application or extension.

ASSESSMENT
A summative task mapped to MYP criteria. The rubric tells you what evidence is required.

"An answer is not a solution until you can say why it must be true, what it means, and when it would fail."

USING THE ANSWER KEY
Attempt every question before checking. For a wrong answer, read the first line of the solution only, then retry. The key often shows more than one valid method; yours may differ and still be correct.

BEFORE A LESSON
Skim the inquiry and the first worked example. Write one question you want answered.

DURING A LESSON
Copy the reasoning, not just the answer. Annotate each step with *why* it is allowed.

AFTER A LESSON
Do the odd-numbered questions. Mark them. Repair one error before moving on.

THE PROGRAMME

The MYP framework *in Year 5*

The MYP is concept-based and inquiry-led. It asks students not only to calculate, but to notice structure, make generalisations, justify them and apply them responsibly. Mathematics in the MYP is organised through three key concepts—**form**, **logic** and **relationships**—and through related concepts that name the mathematical work more precisely. In the final year, the emphasis shifts towards transfer: choosing methods without being told which to use, and judging when a model is good enough.

KEY CONCEPT

Form

What something is like: its shape, structure, notation and pattern. In this course: polynomial form, exponential and logarithmic form, matrix form and distributional form.

KEY CONCEPT

Logic

What follows from what: deduction, proof, counterexample and the rules that make an argument valid.

KEY CONCEPT

Relationships

How quantities, representations and ideas connect: functions and inverses, trigonometric relationships, correlation and modelling.

Related concepts used in this volume

approximation · change · equivalence · generalisation · justification · measurement · models · patterns · quantity · representation · simplification · space · systems · validity

Global contexts

identities and relationships · orientation in space and time · personal and cultural expression · scientific and technical innovation · fairness and development · globalization and sustainability

Approaches to learning

Communication · Research · Thinking · Social · Self-management. Each chapter identifies the ATL clusters it develops and gives a concrete skill indicator.

The four assessment criteria

A **Knowing and understanding.** Select appropriate mathematics, apply it successfully, and solve problems correctly in familiar and unfamiliar contexts.

B **Investigating patterns.** Select problem-solving techniques, describe patterns as general rules, and prove or verify and justify those rules.

C **Communicating.** Use mathematical language and notation, represent information, move between representations, reason coherently and organise information logically.

D **Applying mathematics in real-life contexts.** Identify relevant elements, select and apply strategies, justify accuracy, and judge whether a solution makes sense.

The criterion summaries above are adapted, in plain language, from the MYP Mathematics framework (IB, updated 2020). The full official descriptors and achievement-level rubrics are held by the school. This book uses the criteria to design tasks and to explain what evidence a response should contain; it does not replace the official guide.

ASSESSMENT

Evidence of *understanding*

Formative and summative are not the same

Formative work is practice for learning. Exercises, guided examples, investigations and mixed practice are formative: you are expected to make errors, use the answer key and revise. **Summative** work is evidence of learning at a point in time. The assessment task at the end of each chapter is summative: it should be completed independently, without the answer key.

What strong evidence looks like

- **Criterion A:** correct selection and execution, with units and sensible accuracy.
- **Criterion B:** a table or diagram of cases, a general rule, and a verification or proof—not just three examples.
- **Criterion C:** notation used correctly; words, tables, graphs and algebra connected; a reader can follow the argument without guessing.
- **Criterion D:** assumptions stated, model applied, accuracy justified, and the answer interpreted in context.

Command terms

State / write down	Give a short answer without explanation.
Describe	Give a detailed account of the features.
Explain	Give reasons or causes; show how the result follows.
Calculate / determine	Obtain a numerical answer, showing sufficient working.
Verify / justify	Confirm with evidence; give a logical reason for a statement.
Prove	Establish truth by a general, logical argument.
Interpret / evaluate	Explain the meaning; make a judgement using evidence.
Model	Represent a situation mathematically, stating assumptions.

ACHIEVEMENT LEVELS 0–8

Each criterion is marked out of 8. The bands below are a study aid, not a substitute for the school's official rubric.

1–2	Simple, familiar situations; limited selection; support needed.
3–4	More complex familiar situations; appropriate selection; some justification.
5–6	Challenging familiar situations; successful application; general rules verified.
7–8	Challenging familiar and unfamiliar situations; precise, coherent, justified and evaluated.

Academic honesty

Collaboration is encouraged when learning, but summative work must be your own. Acknowledge data sources, tools and any help received. Copying a solution without understanding it is both dishonest and ineffective: the assessment asks you to transfer ideas to a new context.

After the task

Use the reflection page. Identify one conceptual error, one communication error and one habit to change. The answer key gives sample responses and common-error notes; compare your reasoning, not just your final answer.

SCOPE AND SEQUENCE

Year 5 *course map*

This is a school-designed Year 5 course aligned to the MYP framework. It is not a mandatory IB syllabus. The sequence assumes the Year 4 foundations listed on the next page and deliberately reserves Diploma Programme content for later study.

CH.	FOCUS	STRAND	KEY CONCEPT	CRITERIA
1	Polynomials and algebraic fractions	Algebra	Form	A, C
2	Functions and inverses	Algebra / functions	Relationships	A, B, C
3	Exponential and logarithmic functions	Number / algebra	Change	A, C, D
4	Sequences and series	Number / algebra	Form	A, B, C
5	Inequalities and linear programming	Algebra	Relationships	A, C, D
6	Circle geometry and proof	Geometry	Logic	B, C
7	Sine and cosine rules; 3D trigonometry	Geometry / trigonometry	Relationships	A, C, D
8	Vectors and matrices	Algebra / geometry	Form	A, B, C
9	Dispersion, correlation and regression	Statistics	Relationships	A, C, D
10	Conditional probability and counting	Probability	Logic	A, B, C, D
11	Capstone modelling project	Integrated	Relationships	B, C, D
12	Cumulative revision and DP transition	Integrated	Logic	A, B, C, D

Assumed from Year 4

Rational exponents and surds; factorisation and algebraic fractions; linear and quadratic equations; inequalities and simultaneous equations; function notation and coordinate geometry; quadratic graphs and the discriminant; linear and quadratic sequences; percentage change and compound interest; congruence and similarity; right-triangle trigonometry; surface area and volume; histograms, quartiles and box plots; combined-event probability.

Taught in depth in Year 5

Polynomial division and the factor theorem; composite and inverse functions; exponential and logarithmic functions; arithmetic and geometric series, convergence and finance; quadratic inequalities and linear programming; circle theorems and formal proof; sine and cosine rules and 3D trigonometry; vectors in two and three dimensions; matrices, determinants and transformations; standard deviation; correlation, regression and critical interpretation; conditional probability, independence and counting.

Reserved for the Diploma Programme

Calculus: limits, differentiation, integration and their applications. Complex numbers. Proof by induction and formal set theory. Binomial expansion for non-integer powers. Trigonometric identities and equations beyond the sine and cosine rules. Scalar product and vector equations of lines and planes. Probability distributions, expectation and variance. Hypothesis testing and confidence intervals.

How the progression works

Year 5 moves from *method* to *judgement*: students choose a representation, justify why it is valid, and evaluate the limits of a model. The capstone and revision chapter make that shift explicit without pretending that school mathematics ends here.

PREREQUISITES

What Year 5 *assumes*

Year 5 does not reteach Year 4. The list below is the minimum assumed knowledge. Each chapter begins with a ten-minute diagnostic on the prerequisites it actually uses; if you cannot complete it, the chapter names the Year 4 section to revisit. A gap in one line does not block the whole course, but it will surface quickly in the exercises.

Number and algebra

Index laws with integer and rational exponents; simplification of surds and rationalisation of simple denominators; expansion of products, including $(a \pm b)^2$ and difference of two squares; factorisation by common factor, grouping, trinomials and the difference of squares; manipulation of algebraic fractions with monomial and simple binomial denominators; solving linear equations and inequalities, including those with fractional coefficients; solving quadratic equations by factorisation, completing the square and the formula; the discriminant and its interpretation; solving pairs of linear equations by substitution and elimination; rearranging formulae.

Functions and graphs

Function notation and substitution; domain and range of simple functions; plotting straight lines from equations and from a table; gradient, intercepts, midpoint and distance; parallel and perpendicular lines; recognising and sketching parabolas in completed-square form, including the vertex, axis and intercepts; interpreting and constructing linear models in context.

Sequences and finance

Generating terms from a term-to-term or position-to-term rule; recognising arithmetic and simple quadratic sequences; percentage increase and decrease; the multiplier method; compound interest and depreciation over a whole number of periods.

Geometry and trigonometry

Angle facts for straight lines, triangles, parallel lines and polygons; congruence criteria; similar triangles and the ratio of corresponding sides; enlargement and scale factors for length, area and volume; Pythagoras' theorem in two and three dimensions; the trigonometric ratios sine, cosine and tangent in right triangles; solving right triangles; angles of elevation and depression; circumference, area, surface area and volume of standard solids.

Statistics and probability

Constructing and reading frequency tables, bar charts, histograms with equal and unequal class widths, and box plots; median, quartiles, interquartile range and mode from raw data and from tables; the idea of an outlier; sample space, equally likely outcomes and the probability scale; the addition rule for mutually exclusive events; combined events represented in lists, tables, Venn diagrams and tree diagrams; the multiplication rule for independent events.

If something is missing

Use the diagnostic at the start of the relevant chapter. The chapter names the Year 4 section, and the Year 4 answer key contains fully worked solutions. Do not attempt to memorise a formula without being able to derive or explain it: Year 5 assessments ask you to justify, not merely reproduce.

> **A NOTE ON DEPTH**
>
> Year 4 introduced these ideas in familiar settings. Year 5 expects you to use them in unfamiliar ones, to choose between methods, and to explain why a method is valid. Fluent recall is necessary but not sufficient.

TRANSITION

From MYP *to the DP*

The Diploma Programme is a different qualification with its own subject guides, assessment objectives and examination conventions. This book prepares you for it in the sense that it builds fluent algebra, secure reasoning and the habit of justifying claims. It does not teach any DP syllabus, and no textbook can guarantee success in a particular DP course. Course selection, prior attainment and the school's own placement decisions all matter.

What changes

Pace and abstraction. DP courses move faster and expect you to handle general symbols, not only numerical examples. **Calculus.** Differentiation and integration are central to DP Mathematics and are not taught in this book. **Formal proof.** DP work expects structured argument and precise notation more consistently than MYP tasks do.

Technology. A graphing calculator is used in DP examinations. You should be able to use one, but also to explain what it is doing and when it misleads.

What stays the same

The modelling cycle, the demand for interpretation and evaluation, and the expectation that an answer is checked for reasonableness. These are MYP habits that DP examiners reward.

A realistic readiness profile

Before a DP Mathematics course, you should be able to: manipulate algebraic fractions without prompting; solve quadratic, exponential and trigonometric equations; use the sine and cosine rules confidently; interpret a scatter graph and a conditional probability; and write a short proof with a stated reason for each step. Chapter 12 checks these explicitly.

Course pathways

DP Mathematics is offered at Standard and Higher Level, in courses with different emphases on analysis and on applications. Requirements vary by school and by course. Ask your teacher which pathway fits your attainment and interests; this book does not rank or recommend one.

AN HONEST STATEMENT

Completing this coursebook is evidence of sustained study, not a certification of readiness. The final chapter asks you to reflect on your own evidence: what you can do without help, what you still need to repair, and what you have not yet met.

MYP and Diploma Programme are trademarks of the International Baccalaureate Organization. This publication is independent and is not endorsed by, affiliated with, or sponsored by the IB.

STRUCTURE BENEATH THE EXPRESSION

Polynomials and Fractions

01

FORM · EQUIVALENCE · GENERALISATION

Division, remainders and factors — reading the structure of a polynomial before you solve it, and keeping algebraic fractions exact.

STATEMENT OF INQUIRY

The structure of a polynomial — its factors, roots and remainders — determines both what it can be simplified to and what its graph will do.

KEY CONCEPT

Form

RELATED CONCEPTS

Equivalence · Generalisation · Simplification

GLOBAL CONTEXT

Personal and cultural expression — the language of algebra as a way of making structure visible

FACTUAL

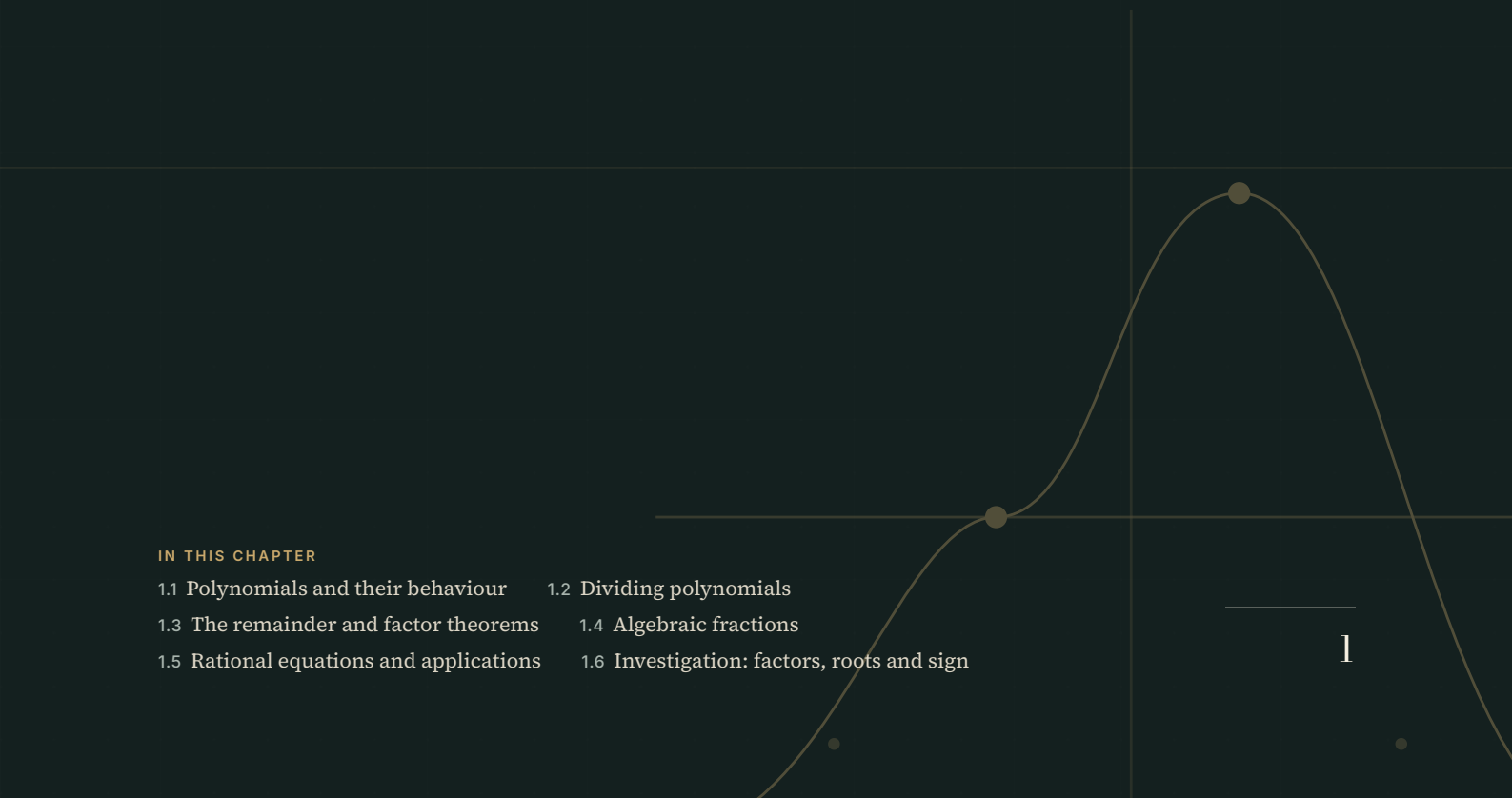
How do we divide one polynomial by another, and what does the remainder mean?

CONCEPTUAL

Why does a factor correspond to a root, and why does multiplicity change the graph?

DEBATABLE

Is an algebraic fraction "simplified" when it is shorter, or when its structure is clearer?



IN THIS CHAPTER

1.1 Polynomials and their behaviour
1.2 Dividing polynomials
1.3 The remainder and factor theorems
1.4 Algebraic fractions
1.5 Rational equations and applications
1.6 Investigation: factors, roots and sign

OPENING PROBLEM · THE BOX THAT MUST HOLD ITS SHAPE

Cutting a corner from a sheet of card

A designer starts with a rectangular sheet of card 24 cm by 18 cm. From each corner she cuts out a square of side x cm, then folds up the sides to make an open box.

The volume of the box is a function of the one number she chooses. **What size of square gives the largest box — and can a polynomial tell you before you build it?**

1. Write expressions for the length, width and height of the folded box in terms of x.
2. Write the volume $V(x)$ as a polynomial in x. State the degree and the values of x that make physical sense.
3. Evaluate V for $x = 1, 2, 3, 4, 5, 6, 7, 8$ and locate the largest value from your table.
4. Explain why the polynomial is only a model: name one physical feature of a real box that the volume formula ignores.

This chapter gives you the tools to answer the problem structurally: how a polynomial is built from its factors, how division reveals a remainder, and how a root announces itself in a graph. Those same tools let you simplify algebraic fractions without losing information.

CH 01

1.0 Learning intentions

ORIENTATION

- Recognise a polynomial, state its degree and describe its end behaviour and roots from its factored form.
- Divide polynomials by linear expressions using long division, and interpret the quotient and remainder.
- Apply the remainder and factor theorems to find remainders, test factors and factorise cubics.
- Simplify, combine and divide algebraic fractions, stating the domain restrictions that keep them equivalent.
- Solve rational equations, identify extraneous roots, and model a situation with a rational equation.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

state the degree, leading coefficient and constant term of a polynomial; divide a cubic or quartic by a linear factor and write the result as quotient plus remainder over divisor; use $P(a)$ to find a remainder and to test whether $(x - a)$ is a factor; factorise a cubic completely once one root is known; simplify an algebraic fraction by factorising and state the excluded values; add, subtract, multiply and divide algebraic fractions; solve a rational equation and reject any root that makes a denominator zero; translate a rate or sharing problem into a rational equation and interpret the solution.

ATL SKILLS · THINKING AND COMMUNICATION

Critical thinking: before dividing, look for structure — a common factor, a known root, a difference of squares. The fastest division is the one you do not need to perform. **Communication:** when you cancel a factor in a fraction, write the excluded value beside the simplified expression; the two expressions are equal only where the original was defined. **Transfer:** the factor theorem is a test, not a guess; use it to turn trial and error into a short list of candidates.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 1.1.

1. Expand $(x+4)(x-3)$ and $(2x-1)(x+5)$.
2. Factorise $x^2-7x+12$ and x^2-25.
3. Solve $x^2-5x+6=0$.
4. Simplify $\dfrac{6x^3}{2x}$ and $\dfrac{x^2-4}{x-2}$.
5. If $f(x)=2x^2-3x+1$, find $f(2)$ and $f(-1)$.
6. Write $x^3\cdot x^4$ and $\dfrac{x^5}{x^2}$ as powers of x.

REFRESHER · EXPANDING, FACTORISING, SUBSTITUTING

Expanding multiplies every term of one bracket by every term of the other. Factorising reverses this: remove the highest common factor, then look for a pattern. The difference of two squares is $a^2-b^2=(a-b)(a+b)$; a monic quadratic x^2+bx+c factorises as $(x+p)(x+q)$ with $p+q=b$ and $pq=c$.

Substitution replaces every x by the given value, keeping the signs. For $f(x)=2x^2-3x+1$, $f(-1)=2(1)+3+1=6$. A root of a polynomial is a value a for which $P(a)=0$.

CONCEPT

1.1 Polynomials and their behaviour

A polynomial is a sum of terms of the form ax^n, where n is a whole number and a is a constant. The word comes from *poly* (many) and *nomos* (term), but the defining feature is not the number of terms — it is that every power of the variable is a non-negative integer. That restriction is what makes polynomials predictable: they can be added, multiplied and divided in a controlled way, and their graphs have no breaks.

The **degree** is the highest power present. The **leading coefficient** is the coefficient of that highest power, and the **constant term** is the term with no variable. A polynomial of degree 1 is linear, degree 2 quadratic, degree 3 cubic, degree 4 quartic. The zero polynomial is usually excluded, because every power "is present" with coefficient zero.

DEFINITIONS

Polynomial, degree and root

A polynomial in x has the form

$$P(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0,\qquad a_n\neq 0,$$

where each a_i is a real number and n is a non-negative integer. The degree is n, the leading coefficient is a_n, and a_0 is the constant term.

A **root** (or zero) of P is a number a with $P(a)=0$. A **factor** of P is a polynomial that divides P exactly, with zero remainder.

EXAMPLE **1.1** READING A POLYNOMIAL

QUESTION

For $P(x) = 4x^5 - 2x^3 + x - 7$, state (a) the degree, (b) the leading coefficient, (c) the constant term, and (d) the coefficient of x^3.

SOLUTION

a The highest power is 5, so the degree is 5.

b The coefficient of x^5 is 4.

c The term without x is -7.

d The x^3 term is $-2x^3$, so the coefficient is -2. Note that there is no x^4 or x^2 term; their coefficients are 0.

> **Missing terms matter:** when dividing, write the polynomial with zero placeholders for missing powers. Omitting a gap is the most common cause of a misaligned long division.

Multiplying polynomials produces a new polynomial whose degree is the sum of the degrees. This is easy to see: the highest terms multiply to give the highest term of the product. For example, a cubic times a quadratic gives a degree 5 polynomial. The leading coefficient of the product is the product of the leading coefficients.

EXAMPLE **1.2** EXPANDING AND PREDICTING THE DEGREE

QUESTION

Expand $(2x - 3)(x^2 + 4x - 1)$ and state the degree of the result.

SOLUTION

1 Multiply each term of the first bracket by each term of the second:

$$2x(x^2 + 4x - 1) - 3(x^2 + 4x - 1).$$

2 Expand: $2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3$.

3 Collect like terms:

$$2x^3 + 5x^2 - 14x + 3.$$

4 The highest term is $2x^3$, so the degree is 3. The prediction from the factors is $1 + 2 = 3$, which agrees.

Check: substitute $x = 1$. The original is $(2 - 3)(1 + 4 - 1) = (-1)(4) = -4$. The expansion gives $2 + 5 - 14 + 3 = -4$. ✓

The factored form of a polynomial is often more informative than the expanded form. If

$$P(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots,$$

then the roots are $r_1, r_2, \ldots$, each raised to a **multiplicity** m_i. Near a root of odd multiplicity the graph crosses the axis; near a root of even multiplicity it touches the axis and turns back. This single fact explains most of the shape of a polynomial graph.

For large values of $|x|$, only the leading term matters. A polynomial of even degree with positive leading coefficient rises at both ends; with negative leading coefficient it falls at both ends. An odd-degree polynomial always has opposite behaviour at the two ends, so it must cross the axis at least once. That is why every cubic has at least one real root.

KEY IDEA · WHAT THE FACTORED FORM TELLS YOU

For $P(x) = a(x - r_1)^{m_1} \cdots (x - r_k)^{m_k}$: the roots are the r_i; the multiplicity m_i is odd for a crossing and even for a touch; the sign of a and the parity of the total degree determine the end behaviour; and the y-intercept is $P(0)$.

EXAMPLE **1.3** SKETCHING FROM STRUCTURE

QUESTION

The polynomial is $P(x) = -(x + 2)(x - 1)^2(x - 3)$. State the roots and their multiplicities, describe the behaviour at each root, and describe the end behaviour.

SOLUTION

1 Write each factor in the form $(x - r)$: $x + 2 = x - (-2)$. The roots are -2 (multiplicity 1), 1 (multiplicity 2) and 3 (multiplicity 1).

2 At $x = -2$ and $x = 3$ the multiplicity is odd, so the graph crosses the axis. At $x = 1$ the multiplicity is even, so the graph touches the axis and turns.

3 The degree is $1 + 2 + 1 = 4$, which is even, and the leading coefficient is -1, which is negative. So both ends go down: as $x \to \pm\infty$, $P(x) \to -\infty$.

4 The y-intercept is $P(0) = -(2)(1)(-3) = 6$.

Reasoning: the sign of P between roots can be found by testing one value in each interval. With four simple intervals, the signs alternate except at the double root, where the sign is the same on both sides.

COMMON MISCONCEPTIONS

"Every cubic has three roots." A cubic has three roots only if complex roots are counted. Over the reals it may have one root and a repeated root, or one root and a quadratic factor with no real roots.

"$(x - 1)^2$ means the root is -1." The root comes from setting the factor equal to zero: $x - 1 = 0$ gives $x = 1$. The exponent is the multiplicity, not part of the root.

"A double root means the graph is flat everywhere." It means the graph touches the axis at that point. Away from the root the graph is not flat.

"Degree 4 means four x-intercepts." A quartic can have 0, 1, 2, 3 or 4 real intercepts depending on multiplicity and complex factors.

GUIDED PRACTICE · COMPLETE THE REASONING

For $Q(x) = 2(x - 3)(x + 1)^3$: the roots are _____ with multiplicity ____, and _____ with multiplicity ____.

At $x = -1$ the graph _____ the axis because the multiplicity is _____.

The degree is ____, which is _____ (even/odd), and the leading coefficient is ____, so as $x \to +\infty$, $Q(x) \to$ _____.

The y-intercept is $Q(0) =$ _____.

EXERCISE 1A

POLYNOMIALS · DEGREE · STRUCTURE

FLUENCY

1 State the degree and leading coefficient of each polynomial.
   a. $3x^4 - 2x + 5$
   b. $-x^3 + 7x^2 - x$
   c. $8 - 2x^5$
   d. $(x - 2)(x + 3)(x - 1)$

2 Expand and simplify.
   a. $(x + 5)(x^2 - 2x + 3)$
   b. $(2x - 1)(x^2 + x - 4)$
   c. $(x - 3)^2(x + 1)$
   d. $(x + 2)(x - 2)(x^2 + 4)$

3 Write each polynomial in factored form.
   a. $x^3 - 4x$
   b. $x^3 - 9x^2 + 20x$
   c. $2x^3 - 8x$
   d. $x^4 - 16$

4 For $P(x) = (x - 1)(x + 2)(x - 4)$, find $P(0)$, $P(1)$ and $P(-2)$.

5 State the roots and multiplicities for each.
   a. $y = (x - 5)^2(x + 1)$
   b. $y = x(x - 2)^3$
   c. $y = (x + 4)(x - 1)^2(x - 3)$

6 Describe the end behaviour of each.
   a. $y = 3x^4 - 2x$
   b. $y = -2x^3 + x$
   c. $y = x^5 - 4x^2$
   d. $y = -x^2 + 6$

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes that the roots of $y = (x + 3)^2(x - 2)$ are 3 and -2, each with multiplicity 1. Identify both errors and correct them.

8 Explain why a polynomial of odd degree with real coefficients must have at least one real root. Use end behaviour in your answer.

9 A quartic has roots at $x = -1$ (double), $x = 2$ and $x = 5$, and passes through $(0, 20)$. Find its equation in factored form.

10 Sketch $y = (x + 1)^2(x - 2)(x - 4)$. Label the intercepts and state the sign of y on each interval between roots.

11 Two polynomials have the same three roots but different leading coefficients. Describe how their graphs differ and how they are alike.

12 Explain why the graph of $y = x^3$ is flat at the origin even though the root 0 has multiplicity 3, not 2.

APPLICATION AND EXTENSION

13 A cuboid has dimensions $(x + 1)$, $(x + 3)$ and $(x - 2)$ centimetres. Write its volume as an expanded polynomial and state the values of x for which the dimensions are all positive.

14 The graph of a cubic crosses the x-axis at $x = -2$ and touches it at $x = 1$. It passes through $(0, -4)$. Find its equation.

15 **Extension.** A polynomial has degree 4, a double root at $x = 2$, and no other real roots. Explain why its graph cannot cross the x-axis anywhere else, and sketch two possible shapes with different leading coefficients.

PROCEDURE AND MEANING

1.2 Dividing polynomials

Dividing polynomials works like dividing numbers, with powers of x playing the role of place value. The aim is to write

$$P(x) = D(x)Q(x) + R(x),$$

where Q is the quotient, R is the remainder, and the degree of R is less than the degree of D. When $D(x) = x - a$, the remainder is a constant, and that constant turns out to be $P(a)$.

The procedure is a loop: divide the leading term of what remains by the leading term of the divisor, multiply the divisor by the result, subtract, and repeat. Each pass removes the highest remaining power, so the process must terminate.

DIVISION IDENTITY

For polynomials P and $D \neq 0$, there are unique polynomials Q and R with

$$\deg R < \deg D \quad \text{and} \quad P(x) = D(x)Q(x) + R(x).$$

If $R = 0$, then D is a factor of P.

EXAMPLE **1.4** LONG DIVISION BY A LINEAR DIVISOR

QUESTION

Divide $2x^3 + 5x^2 - 14x + 3$ by $x + 3$, and write the result in the form $P(x) = D(x)Q(x) + R$.

SOLUTION

1. Divide the leading term: $2x^3 \div x = 2x^2$. Multiply: $2x^2(x + 3) = 2x^3 + 6x^2$. Subtract: $(2x^3 + 5x^2) - (2x^3 + 6x^2) = -x^2$.
2. Bring down $-14x$. Divide: $-x^2 \div x = -x$. Multiply: $-x(x + 3) = -x^2 - 3x$. Subtract: $(-x^2 - 14x) - (-x^2 - 3x) = -11x$.
3. Bring down $+3$. Divide: $-11x \div x = -11$. Multiply: $-11(x + 3) = -11x - 33$. Subtract: $(-11x + 3) - (-11x - 33) = 36$.
4. The remainder is 36, so

$$2x^3 + 5x^2 - 14x + 3 = (x + 3)(2x^2 - x - 11) + 36.$$

Check at $x = -3$: the first term vanishes, so the value should equal the remainder. $P(-3) = 2(-27) + 5(9) + 42 + 3 = -54 + 45 + 42 + 3 = 36$. ✓

EXAMPLE **1.5** DIVISION WITH A MISSING TERM

QUESTION

Divide $x^4 - 1$ by $x - 2$.

SOLUTION

1. Write the dividend with zero placeholders: $x^4 + 0x^3 + 0x^2 + 0x - 1$.
2. $x^4 \div x = x^3$. Multiply: $x^3(x - 2) = x^4 - 2x^3$. Subtract: $2x^3$.
3. $2x^3 \div x = 2x^2$. Multiply and subtract: $2x^3 - 2x^2(x - 2)$ leaves $4x^2$.
4. $4x^2 \div x = 4x$. Subtract $4x(x - 2) = 4x^2 - 8x$, leaving $8x$.
5. $8x \div x = 8$. Subtract $8(x - 2) = 8x - 16$, leaving 15.
6. So $x^4 - 1 = (x - 2)(x^3 + 2x^2 + 4x + 8) + 15$.

> **Why the placeholders:** without them, the x^2 and x columns shift and the subtraction aligns the wrong powers. Writing zeros keeps every power in its own column.

When the divisor is linear, there is a shorter arrangement sometimes called **synthetic division**. Write the coefficients of the dividend in order, bring down the first, multiply by the root a, add, and repeat. The final number is the remainder and the preceding numbers are the coefficients of the quotient, one degree lower. The method is not magic: it is long division with the powers of x omitted.

EXAMPLE **1.6** SYNTHETIC DIVISION

QUESTION

Use synthetic division to divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$.

SOLUTION

1 Write the coefficients $2, -3, -11, 6$ and the root 3.

2 Bring down 2. Multiply by 3 to get 6; add to -3 to get 3. Multiply 3 by 3 to get 9; add to -11 to get -2. Multiply -2 by 3 to get -6; add to 6 to get 0.

3 The quotient coefficients are $2, 3, -2$ and the remainder is 0:

$$2x^3 - 3x^2 - 11x + 6 = (x - 3)(2x^2 + 3x - 2).$$

4 Factor the quadratic: $2x^2 + 3x - 2 = (2x - 1)(x + 2)$. So $P(x) = (x - 3)(2x - 1)(x + 2)$.

Check: $P(3) = 54 - 27 - 33 + 6 = 0$. ✓ The roots are $3, \frac{1}{2}, -2$.

COMMON MISCONCEPTIONS

"Dividing by $x + 3$ means using $x = 3$ in synthetic division." The divisor $x + 3$ is $x - (-3)$, so the number to use is -3.

"The remainder is the last coefficient." The last number in the synthetic array is the remainder; the others are coefficients. Mixing them up shifts the whole quotient by one degree.

"Polynomial division always gives a polynomial quotient." The quotient is a polynomial only when the remainder is zero. Otherwise the exact result is $Q(x) + \dfrac{R}{D(x)}$, which is not a polynomial.

"You can stop when the remainder looks small." You stop when the degree of the remainder is less than the degree of the divisor — not when the number looks small.

GUIDED PRACTICE · COMPLETE THE REASONING

Divide $x^3 + 2x^2 - 5x - 6$ by $x - 2$. Synthetic division uses the root _____ and the coefficients _____.

Bring down ____; multiply by 2 and add to 2 to get ____; multiply by 2 and add to -5 to get ____; multiply by 2 and add to -6 to get ____.

The quotient is _____ and the remainder is ____. Since the remainder is _____, $x - 2$ is / is not a factor.

EXERCISE 1B

LONG AND SYNTHETIC DIVISION

FLUENCY

1 Divide and state the quotient and remainder.

a. $(x^2 + 5x + 6) \div (x + 2)$

b. $(x^3 - 8) \div (x - 2)$

c. $(2x^3 + x^2 - 8x + 5) \div (x + 3)$

d. $(x^4 - 3x^2 + 1) \div (x - 1)$

2 Use synthetic division to divide.

a. $x^3 - 4x^2 + x + 6$ by $x - 2$

b. $2x^3 + 3x^2 - 11x - 6$ by $x + 3$

c. $x^4 - 1$ by $x + 2$

d. $3x^3 - 7x + 4$ by $x - 1$

3 Write each division in the form $P(x) = D(x)Q(x) + R$.

a. $(x^3 + 1) \div (x + 1)$

b. $(x^3 + 1) \div (x - 1)$

c. $(6x^3 - 5x^2 + 4) \div (2x + 1)$

UNDERSTANDING AND REASONING

EXERCISE 1B · CONTINUED

LONG AND SYNTHETIC DIVISION

4 **Error analysis.** A student divides $x^3 - 2x + 4$ by $x - 1$ and writes the quotient as $x^2 - x - 1$ with remainder 3. Check by substitution and identify the error.

5 Without dividing, explain why $x - 4$ cannot be a factor of $x^3 + 2x^2 - 3x + 5$.

6 The division of $P(x)$ by $x - 2$ gives quotient $x^2 + 3x - 1$ and remainder 5. Write $P(x)$ and find $P(2)$.

7 Explain why the remainder on division by a linear divisor must be a constant.

8 When $x^3 + kx + 6$ is divided by $x - 3$, the remainder is 0. Find k.

APPLICATION AND EXTENSION

9 A polynomial P satisfies $P(x) = (x - 1)(x^2 + 4x + c) + 7$. Given that $P(1) = 7$ and $P(2) = 17$, find c.

10 Divide $x^4 + 4x^3 - 7x^2 - 22x + 24$ by $x - 2$, then factor the quotient completely and state all four roots of P.

11 **Extension.** The polynomial $x^3 + ax^2 + bx + c$ has remainder 0 on division by $x - 1$, remainder 6 on division by $x - 2$, and remainder 0 on division by $x + 1$. Find a, b, c.

THEOREM AND USE

1.3 The remainder and factor theorems

The division identity becomes especially powerful when the divisor is linear. If $P(x) = (x - a)Q(x) + R$, substituting $x = a$ makes the first term zero, so $P(a) = R$. The remainder is not something you have to compute by division: it is the value of the polynomial at the root of the divisor.

The **factor theorem** is the special case $R = 0$: $(x - a)$ is a factor exactly when $P(a) = 0$. Together with the rational root theorem — which limits the candidates for a rational root to factors of the constant term divided by factors of the leading coefficient — this turns "guess a root" into a systematic search.

REMAINDER AND FACTOR THEOREMS

Remainder theorem. When $P(x)$ is divided by $x - a$, the remainder is $P(a)$.
Factor theorem. $(x - a)$ is a factor of $P(x)$ if and only if $P(a) = 0$.
Rational root theorem. If a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term and q divides the leading coefficient.

EXAMPLE 1.7 FINDING A REMAINDER WITHOUT DIVIDING

QUESTION

Find the remainder when $P(x) = 3x^3 - 4x^2 + 2x - 7$ is divided by (a) $x - 2$ and (b) $x + 1$.

SOLUTION

a The divisor $x - 2$ has root 2, so the remainder is $P(2) = 3(8) - 4(4) + 4 - 7 = 24 - 16 + 4 - 7 = 5$.

b The divisor $x + 1$ has root -1, so the remainder is $P(-1) = -3 - 4 - 2 - 7 = -16$.

> **Sign care:** $x + 1 = x - (-1)$, so the value is -1. A large fraction of errors in this section come from using 1 instead of -1.

EXAMPLE **1.8** TESTING AND USING A FACTOR

QUESTION

Show that $x - 2$ is a factor of $P(x) = x^3 - 4x^2 + x + 6$, and hence factorise P completely.

SOLUTION

1 Evaluate: $P(2) = 8 - 16 + 2 + 6 = 0$. By the factor theorem, $x - 2$ is a factor.

2 Divide to find the other factor: $P(x) = (x - 2)(x^2 - 2x - 3)$.

3 Factor the quadratic: $x^2 - 2x - 3 = (x - 3)(x + 1)$.

4 So $P(x) = (x - 2)(x - 3)(x + 1)$. The roots are $2, 3, -1$.

Check: expand $(x - 2)(x - 3)(x + 1)$. First $(x - 2)(x - 3) = x^2 - 5x + 6$. Multiply by $x + 1$: $x^3 - 4x^2 + x + 6$. ✓

EXAMPLE **1.9** FINDING AN UNKNOWN COEFFICIENT

QUESTION

The polynomial $P(x) = 2x^3 + kx^2 - 5x + 3$ has a factor $x + 1$. Find k and factorise P completely.

SOLUTION

1 By the factor theorem, $P(-1) = 0$:

$$-2 + k + 5 + 3 = 0 \Rightarrow k = -6.$$

2 So $P(x) = 2x^3 - 6x^2 - 5x + 3$. Divide by $x + 1$: the quotient is $2x^2 - 8x + 3$, since $2x^3 - 6x^2 - 5x + 3 = (x + 1)(2x^2 - 8x + 3)$.

3 The quadratic $2x^2 - 8x + 3$ does not factor over the integers (its discriminant is $64 - 24 = 40$, not a square). So the complete factorisation over the rationals is

$$P(x) = (x + 1)(2x^2 - 8x + 3).$$

Interpretation: "factorise completely" depends on the number system. Over the reals the quadratic can be written with surds; over the rationals it stays as it is.

COMMON MISCONCEPTIONS

"If $P(a) = 0$ then $x + a$ is a factor." The factor is $x - a$. For $a = 3$, the factor is $x - 3$.

"The remainder theorem only works for cubics." It works for any polynomial divided by any linear expression $x - a$, whatever the degree.

"Finding one root means you have factorised completely." One linear factor leaves a quotient of degree $n - 1$ that must still be factorised.

"A polynomial with no rational root has no factors." It may factor into quadratics with irrational or complex coefficients, or into a linear factor with an irrational root.

GUIDED PRACTICE · COMPLETE THE REASONING

For $P(x) = x^3 + 3x^2 - 4$: $P(1) =$ _____, so $x - 1$ is / is not a factor. $P(-2) =$ _____, so $x + 2$ is / is not a factor.

Divide by $x + 2$: the quotient is _____, and $P(x) =$ _____.

Factor the quotient: _____. The three roots are _____.

EXERCISE 1C

REMAINDER AND FACTOR THEOREMS

FLUENCY

1 Use the remainder theorem to find the remainder.
   a. $x^3 - 2x^2 + 5$ divided by $x - 1$
   b. $2x^3 + x - 7$ divided by $x + 2$
   c. $x^4 - 3x + 1$ divided by $x - 2$
   d. $5x^3 - 2x^2 + x - 4$ divided by $2x - 1$

2 Determine whether the given linear expression is a factor.
   a. $x - 3$, $P(x) = x^3 - 2x^2 - 5x + 6$
   b. $x + 1$, $P(x) = 2x^3 + 3x^2 - 2x - 3$
   c. $x - 2$, $P(x) = x^4 - 16$
   d. $x + 3$, $P(x) = x^3 + 27$

3 Factorise completely.
   a. $x^3 - x^2 - 14x + 24$
   b. $x^3 + 2x^2 - 5x - 6$
   c. $2x^3 - 5x^2 - 4x + 3$
   d. $x^3 - 6x^2 + 11x - 6$

UNDERSTANDING AND REASONING

4 **Error analysis.** A student claims $x - 1$ is a factor of $x^4 + x^3 - x - 2$ because $P(1) = 0$. Evaluate $P(1)$ and correct the conclusion.

5 Find k given that $x - 2$ is a factor of $x^3 + kx^2 - 3x + 2$.

6 When $P(x) = x^3 + ax^2 - 4x + b$ is divided by $x - 1$ the remainder is 2, and when divided by $x + 2$ the remainder is -4. Find a and b.

7 Explain the connection between the factor theorem and the statement "a root of P is a solution of $P(x) = 0$".

8 A cubic has roots $1, 2, -3$ and leading coefficient 2. Write it in expanded form and verify the factor theorem for $x = 2$.

9 Show that $x - 1$ is a factor of $x^n - 1$ for every positive integer n, and explain why $x + 1$ is a factor of $x^n - 1$ only when n is even.

APPLICATION AND EXTENSION

10 The volume of a cuboid is $V(x) = x^3 + 6x^2 + 11x + 6$ cm^3 and its height is $x + 1$ cm. Find expressions for the length and width, given that the length exceeds the width by 1 cm.

11 A monic polynomial P of degree 3 has $P(1) = P(-2) = 0$ and $P(0) = -8$. Find P and its third root.

12 **Extension.** Prove that if $x - a$ is a factor of $P(x)$ and also a factor of $Q(x)$, then it is a factor of $P(x) + Q(x)$. Does the converse hold? Give a counterexample or a proof.

EQUIVALENCE

1.4 Algebraic fractions

An algebraic fraction is a quotient of polynomials. The rules are the same as for numerical fractions, but the domain matters: a fraction is undefined wherever its denominator is zero, and simplifying cannot remove that restriction. The simplified expression is equal to the original only on the original domain.

To simplify, factorise the numerator and denominator completely, then cancel common factors. Cancelling a term rather than a factor is the central error: $\dfrac{x+2}{x+3}$ does not simplify to $\dfrac{2}{3}$, because x is not a factor of either sum.

OPERATIONS AND DOMAINS

For polynomials A, B, C, D with $B, D \neq 0$:

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}, \qquad \frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C} \ (C \neq 0),$$
$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}.$$

The **excluded values** are the values of x making any original denominator zero. They must be stated even after cancellation.

EXAMPLE **1.10** SIMPLIFYING BY FACTORISING

QUESTION

Simplify $\dfrac{x^2 - 9}{x^2 + x - 12}$ and state the excluded values.

SOLUTION

1 Factorise the numerator: $x^2 - 9 = (x - 3)(x + 3)$.

2 Factorise the denominator: $x^2 + x - 12 = (x + 4)(x - 3)$.

3 Cancel the common factor $x - 3$:

$$\frac{(x - 3)(x + 3)}{(x - 3)(x + 4)} = \frac{x + 3}{x + 4}.$$

4 The original denominator is zero at $x = -4$ and $x = 3$. So the simplification holds for $x \neq -4,\ 3$.

Why state $x \neq 3$? At $x = 3$ the original fraction is $\frac{0}{0}$, which is undefined. The simplified expression would give $\frac{6}{7}$, a value the original never takes.

EXAMPLE **1.11** MULTIPLYING AND DIVIDING

QUESTION

Simplify $\dfrac{x^2 - 4}{x^2 + 5x + 6} \div \dfrac{x - 2}{x + 3}$.

SOLUTION

1 Factorise: $x^2 - 4 = (x - 2)(x + 2)$ and $x^2 + 5x + 6 = (x + 2)(x + 3)$.

2 Invert the divisor and multiply:

$$\frac{(x - 2)(x + 2)}{(x + 2)(x + 3)} \cdot \frac{x + 3}{x - 2}.$$

3 Cancel $(x + 2)$, $(x + 3)$ and $(x - 2)$: the result is 1.

4 Excluded values: $x \neq -3, -2, 2$. The expression equals 1 wherever it is defined.

Interpretation: the two fractions are reciprocals of the same expression, so their quotient is 1. The domain is still restricted.

EXAMPLE **1.12** ADDING WITH DIFFERENT DENOMINATORS

QUESTION

Write $\dfrac{3}{x+1}+\dfrac{2}{x-2}$ as a single fraction.

SOLUTION

1 The common denominator is $(x+1)(x-2)$; the factors share no common factor, so this is the lowest common denominator.

2 Rewrite each fraction:

$$\frac{3(x-2)+2(x+1)}{(x+1)(x-2)}.$$

3 Expand the numerator: $3x-6+2x+2=5x-4$.

4 So the sum is $\dfrac{5x-4}{(x+1)(x-2)}$, with $x\neq -1, 2$.

> **Do not expand the denominator** unless the question asks for a quadratic form. The factorised denominator shows the restrictions and is easier to compare with other expressions.

COMMON MISCONCEPTIONS

"Cancel the x's." Terms cancel only as factors. $\dfrac{x+2}{x+3}\neq\dfrac{2}{3}$.

"Add the numerators and add the denominators." Fractions must share a denominator before their numerators can be added.

"Once it is simplified, the restriction disappears." The restriction belongs to the original expression and must be stated.

"The common denominator is always the product." The product works, but it is not always lowest; factor first and use the highest power of each factor.

GUIDED PRACTICE · COMPLETE THE REASONING

Simplify $\dfrac{x^2-x-6}{x^2-9}$. Numerator: _____. Denominator: _____.

The common factor is _____, so the simplified form is _____, with $x\neq$ _____.

Write $\dfrac{1}{x-3}-\dfrac{2}{x+1}$ as a single fraction: the common denominator is _____, and the numerator simplifies to _____.

EXERCISE 1D ALGEBRAIC FRACTIONS

FLUENCY

1 Simplify and state the excluded values.

a. $\dfrac{6x^2}{9x}$

b. $\dfrac{x^2-16}{x+4}$

c. $\dfrac{x^2+2x}{x^2+5x+6}$

d. $\dfrac{2x^2-8}{x^2-2x}$

2 Write as a single fraction.

a. $\dfrac{1}{x}+\dfrac{1}{x+2}$

b. $\dfrac{3}{x-1}-\dfrac{1}{x}$

c. $\dfrac{2}{x+3}+\dfrac{5}{x-2}$

d. $\dfrac{x}{x-4}-\dfrac{2}{x+1}$

EXERCISE 1D · CONTINUED ALGEBRAIC FRACTIONS

3 Multiply or divide and simplify.

a. $\dfrac{x^2-1}{x+2}\cdot\dfrac{x^2-4}{x-1}$

b. $\dfrac{x^2+3x}{x^2-9}\div\dfrac{x}{x-3}$

c. $\dfrac{4x^2}{x^2-1}\cdot\dfrac{x-1}{2x}$

d. $\dfrac{x^2-5x+6}{x^2-4}\div\dfrac{x-3}{x+2}$

UNDERSTANDING AND REASONING

4 **Error analysis.** A student simplifies $\dfrac{x^2+5x}{x}$ to $x+5$ and says the expression is defined for all x. Identify the omitted restriction and explain why it matters.

5 Explain why $\dfrac{a}{b}+\dfrac{c}{d}=\dfrac{ad+bc}{bd}$ is valid for all values where the denominators are non-zero.

6 Show that $\dfrac{x^2-1}{x-1}=x+1$ for $x\neq 1$, and explain what happens at $x=1$.

7 Simplify $\dfrac{1}{x-1}-\dfrac{1}{x+1}$, then evaluate the result at $x=0$ and at $x=10$. Compare with evaluating the original expression at the same values.

8 Write $\dfrac{2}{x^2-1}$ as a difference of two fractions of the form $\dfrac{A}{x-1}+\dfrac{B}{x+1}$, and find A and B.

APPLICATION AND EXTENSION

9 A rectangle has length $\dfrac{x+3}{x-2}$ and width $\dfrac{x-2}{x+1}$. Write its area as a simplified fraction and state the restrictions.

10 The resistance R of two resistors R_1 and R_2 in parallel satisfies $\dfrac{1}{R}=\dfrac{1}{R_1}+\dfrac{1}{R_2}$. Make R the subject.

11 **Extension.** Simplify $\dfrac{1}{x}+\dfrac{2}{x^2}+\dfrac{3}{x^3}$, and explain how the result shows that the expression is undefined at $x=0$ but defined for all other values.

SOLVING AND MODELLING

1.5 Rational equations and applications

A rational equation contains an algebraic fraction. The standard method multiplies every term by a common denominator, turning the equation into a polynomial. That step is valid only where the multiplier is non-zero, so any solution that makes an original denominator zero must be rejected as **extraneous**.

Rational equations arise naturally in rate problems (speed, work, flow), in sharing problems, and in any situation where two quantities are inversely related. The modelling step is usually the hardest: deciding which quantity is the variable and which relationships constrain it.

EXAMPLE **1.13** SOLVING A RATIONAL EQUATION

QUESTION

Solve $\dfrac{5}{x-1} - \dfrac{3}{x+1} = \dfrac{2}{x^2-1}$.

SOLUTION

1 Factorise the last denominator: $x^2 - 1 = (x-1)(x+1)$. The restrictions are $x \neq 1, -1$.

2 Multiply every term by $(x-1)(x+1)$:

$$5(x+1) - 3(x-1) = 2.$$

3 Expand and solve: $5x + 5 - 3x + 3 = 2$, so $2x + 8 = 2$ and $x = -3$.

4 Check the restriction: $-3 \neq \pm 1$. Substitute to verify: $\frac{5}{-4} - \frac{3}{-2} = -1.25 + 1.5 = 0.25$, and $\frac{2}{8} = 0.25$. ✓

Check every time: substitution catches both arithmetic slips and extraneous roots. It is not optional in an assessment.

EXAMPLE **1.14** AN EXTRANEOUS ROOT

QUESTION

Solve $\dfrac{1}{x-2} + \dfrac{1}{x+2} = \dfrac{4}{x^2-4}$.

SOLUTION

1 The denominator $x^2 - 4 = (x-2)(x+2)$, so $x \neq \pm 2$.

2 Multiply by $(x-2)(x+2)$:

$$(x+2) + (x-2) = 4, \text{ so } 2x = 4 \text{ and } x = 2.$$

3 But $x = 2$ makes two original denominators zero. It is extraneous and must be rejected.

4 The equation has no solution.

Why this happens: the left side simplifies to $\frac{2x}{x^2-4}$. Setting it equal to $\frac{4}{x^2-4}$ gives $2x = 4$ only after multiplying by an expression that is zero at $x = 2$.

EXAMPLE **1.15** MODELLING WITH A RATIONAL EQUATION

QUESTION

A cyclist and a walker both travel a 24 km route. The cyclist's average speed is 6 km/h greater than the walker's, and the cyclist takes 2 hours less. Find both speeds.

SOLUTION

1 Let the walker's speed be v km/h. Then the cyclist's speed is $v + 6$.

2 Time is distance divided by speed:

$$\frac{24}{v} - \frac{24}{v+6} = 2, \qquad v > 0.$$

3 Multiply by $v(v+6)$:

$$24(v+6) - 24v = 2v(v+6).$$

This simplifies to $144 = 2v^2 + 12v$, or $v^2 + 6v - 72 = 0$.

4 Factorise: $(v+12)(v-6) = 0$, so $v = 6$ or $v = -12$. Speed must be positive, so $v = 6$.

5 The walker travels at 6 km/h and the cyclist at 12 km/h. Times are 4 h and 2 h, a difference of 2 h. ✓

Modelling note: the algebraic solution $v = -12$ is not "wrong" — it is a solution of the equation but not of the situation. Stating why you reject it is part of the reasoning.

COMMON MISCONCEPTIONS

"Multiply by the denominator and ignore the restrictions." Multiplying by zero produces equations that are not equivalent. Always record the restrictions first.

"Every solution of the polynomial is a solution of the rational equation." Check each against the original denominators.

"Cross-multiplying works for sums." Cross-multiplication applies to a single fraction equal to a single fraction. For sums, use a common denominator.

"A negative time or speed is a valid answer." In context, reject values that the physical situation cannot take, and say why.

GUIDED PRACTICE · COMPLETE THE REASONING

Solve $\frac{2}{x-3} = \frac{3}{x+1}$. Restrictions: $x \neq$ _____. Cross-multiply: _____.

Solve the linear equation: $x =$ _____. Check against the restrictions: _____.

A tap fills a tank in x hours and a second tap in $x + 2$ hours. Together they fill it in _____ hours, so the equation is _____.

EXERCISE 1E

RATIONAL EQUATIONS AND MODELLING

FLUENCY

1 Solve and check.

a. $\frac{3}{x} = \frac{6}{x+2}$

b. $\frac{5}{x-1} = 2$

c. $\frac{x}{x+2} = 3$

d. $\frac{4}{x-3} = \frac{6}{x+1}$

2 Solve.

a. $\frac{1}{x} + \frac{1}{x+1} = 1$

b. $\frac{2}{x-1} - \frac{1}{x+1} = 1$

c. $\frac{x}{x-2} - \frac{4}{x^2-4} = 0$

d. $\frac{3}{x+2} + \frac{2}{x-1} = 2$

EXERCISE 1E · CONTINUED RATIONAL EQUATIONS AND MODELLING

3 State whether each equation has no solution, one solution or two solutions, and solve it.

a. $\dfrac{x}{x-3} = \dfrac{3}{x-3}$

b. $\dfrac{1}{x-2} + \dfrac{1}{x+2} = \dfrac{3}{x^2-4}$

c. $\dfrac{3}{x} + \dfrac{1}{x-2} = 1$

UNDERSTANDING AND REASONING

4 **Error analysis.** A student solves $\dfrac{1}{x-1} = 2$ by writing $1 = 2(x-1)$, then $x = 1.5$, and rejects it because $x \neq 1$. Is the rejection correct? Explain.

5 Explain why multiplying both sides of a rational equation by $x - 2$ can introduce a solution that the original equation does not have.

6 A number is increased by 3 and its reciprocal is taken; the result is $\frac{1}{2}$. Find the number.

7 The reciprocal of a positive integer is added to twice the reciprocal of the next integer, giving $\frac{7}{6}$. Find the integer.

8 Show that $\dfrac{1}{x-a} - \dfrac{1}{x+a} = \dfrac{2a}{x^2-a^2}$ and use it to solve $\dfrac{1}{x-3} - \dfrac{1}{x+3} = \dfrac{3}{x^2-9}$.

APPLICATION AND EXTENSION

9 A boat travels 30 km downstream in the same time it takes to travel 18 km upstream. The current is 3 km/h. Find the boat's speed in still water.

10 Two workers can complete a job together in 4 hours. One worker alone takes 6 hours longer than the other. Find the time each takes alone.

11 **Extension.** A rectangular garden has area 60 m^2. If its length is increased by 2 m and its width decreased by 1 m, the area is unchanged. Find the dimensions, forming and solving a rational equation.

PATTERN AND GENERALISATION

1.6 Investigation: factors, roots and sign

The factor theorem tells you where a polynomial is zero. The sign of the polynomial between those zeros is determined by the multiplicities. In this investigation you will build a sign table, predict a graph, and then explain why the pattern must hold for every polynomial.

INVESTIGATION 1 · SIGN CHANGES AND MULTIPLICITY

Why does a graph touch instead of cross?

Consider $P(x) = (x-1)^2(x+2)$.

1. State the roots and their multiplicities. Expand P and check its degree.
2. Complete a sign table for the intervals $x < -2$, $-2 < x < 1$ and $x > 1$, testing $x = -3, 0, 2$.
3. Sketch the graph using the sign table, the roots and the end behaviour.
4. Now consider $Q(x) = (x-1)^3(x+2)$. Repeat the table and sketch. What changes at $x = 1$?
5. Compare P and Q on the interval $0.9 < x < 1.1$. Explain why one graph stays on one side of the axis and the other crosses.
6. Generalise: for a root r of multiplicity m, when does the sign change and when does it not? Justify using the factor $(x-r)^m$.
7. Test your rule on $R(x) = -(x+3)(x-2)^4(x-5)^2$. Predict the sign on every interval before sketching.

Generalise. For a polynomial with real roots, the sign can change at a root only if the multiplicity is odd. Write a rule that gives the number of sign changes across the whole real line in terms of the number of odd-multiplicity roots, and explain why even-multiplicity roots never change the sign.

MODELLING · CRITERION D

The open box, revisited

Return to the opening problem: a 24 cm × 18 cm sheet, squares of side x cut from each corner.

1. **Model.** Write $V(x) = x(24-2x)(18-2x)$ and expand it. State the domain $0 < x < 9$.
2. **Calculate.** Evaluate V at $x = 1, 2, \ldots, 8$ to 1 decimal place. Identify the maximum from the table.
3. **Interpret.** What does the polynomial predict at $x = 0$ and at $x = 9$? Why are these values not useful box sizes?
4. **Evaluate.** State two assumptions of the model and explain how each would change the answer in a real workshop.

CUMULATIVE

1.7 Mixed practice

Attempt all questions without notes. Questions are not arranged by section; part of the purpose is to decide which method applies.

MIXED PRACTICE 1 POLYNOMIALS, DIVISION, FRACTIONS AND EQUATIONS

FLUENCY AND UNDERSTANDING

1. Expand $(x-2)(x^2+3x-1)$ and state the degree of the result.
2. Factorise completely: (a) $x^3 - 7x^2 + 12x$; (b) $x^3 + 8$; (c) $2x^3 - x^2 - 8x + 4$.
3. Find the remainder when $x^4 - 2x^3 + x - 5$ is divided by $x+2$.
4. Given $P(x) = x^3 - 3x^2 - 10x + 24$, show that $x-2$ is a factor and factorise P completely.
5. Simplify $\dfrac{x^2-2x-15}{x^2-9}$ and state the excluded values.
6. Write $\dfrac{4}{x-2} + \dfrac{3}{x+2}$ as a single fraction.

MIXED PRACTICE 1 · CONTINUED

POLYNOMIALS, DIVISION, FRACTIONS AND EQUATIONS

7 Solve $\dfrac{3}{x+1} = \dfrac{2}{x-2}$.

8 Solve $x^3 - 4x^2 + x + 6 = 0$ given that $x = 3$ is a root.

9 The graph of $y = (x+1)(x-3)^2$ touches or crosses the axis at which points? Justify.

10 Find k if $x + 2$ is a factor of $2x^3 + kx^2 - 5x + 6$.

11 Simplify $\dfrac{x^2-1}{x^2+2x+1} \cdot \dfrac{x+1}{x-1}$.

12 Explain why $\dfrac{x^2+4}{x}$ is undefined at $x = 0$ even though the numerator is never zero.

REASONING AND APPLICATION

13 A cubic has roots -1, 2 and 4, and passes through $(0, -16)$. Find its equation in expanded form.

14 The polynomial $P(x) = x^3 + ax^2 + bx - 6$ has factors $x - 1$ and $x + 2$. Find a and b, and the third factor.

15 Solve $\dfrac{2}{x-1} - \dfrac{3}{x+2} = \dfrac{1}{x^2+x-2}$.

16 A rectangle has perimeter 28 cm and area 45 cm^2. Form a quadratic equation and solve it to find the dimensions.

17 The volume of a cuboid is $x^3 + 9x^2 + 26x + 24$. Given that the height is $x + 2$, find expressions for the other two dimensions.

18 A train travels 240 km at a constant speed. If it travelled 20 km/h faster, the journey would take 1 hour less. Find the speed.

19 Show that $\dfrac{1}{x-1} + \dfrac{1}{x+1} = \dfrac{2x}{x^2-1}$ and state the values for which the identity fails.

20 The graph of $y = ax^3 + bx^2 + cx + d$ crosses the axis at -2, touches at 1, and passes through $(0, 18)$. Find a, b, c, d.

21 **Extension.** Prove that if a polynomial has integer coefficients and a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term.

22 **Extension.** The equation $\dfrac{x}{x-a} + \dfrac{x}{x-b} = 2$ has a solution $x = 0$ for all $a \neq b$. Find the other solution in terms of a and b, and state when it is extraneous.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Polynomial structure and division

Instructions. Answer all questions. Show sufficient working for the marks indicated. A calculator is not required.

1 For $P(x) = 2x^4 - 3x^3 + 5$, state the degree, the leading coefficient and the constant term. [3]

2 Expand $(2x - 1)(x^2 - 3x + 4)$. [3]

3 Divide $x^3 + 2x^2 - 7x - 10$ by $x - 2$, stating the quotient and remainder. [4]

4 Use the factor theorem to show that $x + 3$ is a factor of $x^3 + 2x^2 - 5x - 6$, and factorise completely. [5]

5 Simplify $\dfrac{x^2 - 4x}{x^2 - 16}$ and state the excluded values. [3]

6 Solve $\dfrac{2}{x-2} = \dfrac{5}{x+1}$. Check for extraneous roots. [2]

Evidence: correct statements, full expansion, aligned division, a factor-theorem evaluation, a correct restriction, and a checked solution.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Remainders that follow a pattern

Instructions. Work independently. Record results in a table and justify each generalisation.

For $n = 1, 2, 3, 4, 5$, let R_n be the remainder when $x^n - 1$ is divided by $x - 2$.

1. Calculate R_1, R_2, R_3, R_4, R_5. [5]
2. Describe the pattern in the sequence R_n. [3]
3. Predict R_6 and R_7, and verify one of them by division. [4]
4. Find a general formula for R_n in terms of n. Justify it using the remainder theorem. [6]
5. For which n is $x - 2$ a factor of $x^n - 1$? Explain using your formula and the factor theorem. [3]
6. Repeat the investigation for division by $x + 1$ and compare the two patterns. [3]

Evidence: a complete table, a described pattern, a prediction with verification, a justified general formula, and a factor interpretation.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining the factor theorem

Instructions. Write a 300–400 word explanation for a student who can divide polynomials but does not understand why the factor theorem works. Your explanation must: (i) state the division identity; (ii) show why substituting $x = a$ gives the remainder; (iii) explain the special case when the remainder is zero; (iv) give one worked example and one non-example. Use correct notation and a clear structure with a short introduction and conclusion.

Evidence: a labelled division identity, a substitution argument, a correct example, a clearly distinguished non-example, and coherent paragraphs.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Designing the strongest box

Instructions. A manufacturer cuts squares of side x cm from a $30\text{ cm} \times 20\text{ cm}$ sheet to make an open box. The box must have volume at least 800 cm^3.

1. Write $V(x)$ as a polynomial and state the domain. [3]
2. Evaluate V for $x = 1, 2, \ldots, 9$ and identify the maximum. [4]
3. Find the range of x for which $V(x) \geq 800$, using a table or graph. [6]
4. State two assumptions of the model and explain how each affects the recommended size. [5]
5. Recommend a value of x for manufacture and justify it using your results. [6]

Evidence: a correct polynomial, a complete table, an identified interval, explicit assumptions, and a justified recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A polynomial is a finite sum of terms ax^n with n a non-negative integer. Its degree is the highest power; its leading coefficient governs end behaviour.
- In factored form $a(x-r_1)^{m_1}\cdots$, the roots are the r_i. Odd multiplicity means the graph crosses; even multiplicity means it touches.
- Division gives $P(x) = D(x)Q(x) + R(x)$ with $\deg R < \deg D$. For a linear divisor $x - a$, the remainder is the constant $P(a)$.
- The factor theorem: $x - a$ is a factor exactly when $P(a) = 0$. Rational roots, if any, are $\frac{p}{q}$ with p dividing the constant term and q the leading coefficient.
- Algebraic fractions are simplified by factorising and cancelling factors, never terms. Excluded values remain even after cancellation.
- Rational equations are solved by multiplying by a common denominator; any value making an original denominator zero must be rejected.

$$P(x) = D(x)Q(x) + R(x), \qquad R = P(a) \text{ for } D(x) = x - a, \qquad (x - a) \mid P \iff P(a) = 0$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- I can state the degree, leading coefficient and constant term of a polynomial.
- I can expand a product of polynomials and predict its degree.
- I can describe roots, multiplicities and end behaviour from a factored form.
- I can divide a polynomial by a linear expression, using placeholders for missing terms.
- I can use synthetic division and interpret the final number as the remainder.
- I can use the remainder theorem without dividing.
- I can test a factor and factorise a cubic completely.
- I can find an unknown coefficient using the factor theorem.
- I can simplify an algebraic fraction and state the excluded values.
- I can add, subtract, multiply and divide algebraic fractions.
- I can solve a rational equation and reject extraneous roots.
- I can model a rate or sharing problem with a rational equation.
- I can explain why a graph touches or crosses at a root.
- I can evaluate a polynomial model's assumptions as well as its arithmetic.

REFLECTION

Write short answers in your notebook. Use evidence from your work, not general statements.

1. Which did you find more useful in this chapter: expanding or factorising? Give a question where the form you chose made the solution shorter.
2. Describe one error you made and the reasoning that corrected it. What will you check first next time?
3. The statement of inquiry claims that structure determines what can be simplified and what a graph will do. Use one example from this chapter to support the claim and one example that complicates it.
4. What question about polynomials or algebraic fractions do you still want to answer?

CHANGE OF VIEW, PRESERVATION OF STRUCTURE

Functions and Their Maps

02

Moving, stretching and reversing a graph — and asking which changes can be undone exactly.

RELATIONSHIPS · CHANGE · REPRESENTATION

STATEMENT OF INQUIRY

A function is a rule that pairs each input with exactly one output; transforming, composing and inverting are operations on the rule itself, and each leaves a trace on the graph.

KEY CONCEPT
Relationships

RELATED CONCEPTS
Change · Representation · Patterns

GLOBAL CONTEXT
Scientific and technical innovation — models that translate a measurement into a prediction

FACTUAL
What information does the notation $f(x)$ contain, and what does it leave out?

CONCEPTUAL
Why can some functions be reversed exactly while others cannot?

DEBATABLE
When a model is transformed to fit data, has the mathematics improved the model or merely disguised it?

IN THIS CHAPTER

2.1 Domain, range and function notation
2.2 Transformations of graphs
2.3 Composite functions
2.4 Inverse functions
2.5 Modelling with transformations
2.6 Investigation: when is a function invertible?

OPENING PROBLEM · THE SHADOW THAT SHRINKS

A lamp, a ball and a wall

A small lamp sits on the floor 2 m from a vertical wall. A ball of radius 0.5 m is placed between them, its centre x metres from the wall. The ball's shadow on the wall is a circle whose radius r depends on x.

Moving the ball changes the shadow in a way that can be described by transforming a simple graph. **Can you predict the shadow's size without measuring it — and can you work backwards from a shadow to the ball's position?**

1. Sketch the side view. Use similar triangles to write the shadow radius r as a function of the distance x.
2. State the domain of your function, in terms of what is physically possible.
3. Describe how the graph of $r(x)$ is obtained from the graph of $y = \dfrac{1}{x}$.
4. Given a shadow radius, can you always recover x? Explain what this says about reversing the rule.

A function is more than a formula: it is a set of allowed inputs, a rule, and a set of outputs. This chapter separates those three ideas, then shows how the graph responds when the rule is shifted, stretched, composed with another rule, or reversed.

ORIENTATION

2.0 Learning intentions

- State the domain and range of a function from its rule or graph, using set and interval notation.
- Apply translations, reflections and stretches to a parent graph and describe a transformation from an equation.
- Form and evaluate composite functions, and determine the domain of a composite.
- Decide whether a function is one-to-one, find its inverse algebraically, and relate the graphs of f and f^{-1}.
- Fit a transformed parent function to a real context and interpret the parameters.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

identify excluded inputs and state domain and range in interval notation; evaluate and solve simple equations involving function notation; sketch $y = f(x) + a$, $y = f(x - a)$, $y = af(x)$, $y = f(bx)$ and combinations, and describe the order of transformations; form $f(g(x))$ and $g(f(x))$ and state their domains; explain why composition is not usually commutative; test one-to-one behaviour with the horizontal line test; find an inverse function and state its domain; recognise that the graph of f^{-1} is the reflection of f in $y = x$; restrict a domain so that a function becomes invertible; build and criticise a transformed model of a real situation.

ATL SKILLS · THINKING AND TRANSFER

Critical thinking: when a graph is transformed, decide first which quantity changed — the input or the output. Horizontal changes are the ones that surprise students because they act in the opposite direction to the sign. **Transfer:** the idea of an inverse is the same idea as "undoing" a process: put on socks then shoes; remove shoes then socks. Composition records the order. **Communication:** a function is not fully specified until its domain is given; state it explicitly whenever the domain is restricted.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 2.1.

1. If $f(x) = 3x - 2$, find $f(0)$, $f(4)$ and $f(-1)$.
2. Sketch the graphs of $y = x^2$, $y = x^3$ and $y = \dfrac{1}{x}$.
3. Solve $2x - 5 = 11$ and $x^2 = 16$.
4. Write the set of values $x \geq -2$ in interval notation.
5. For what values of x is $\sqrt{x - 3}$ defined?
6. Make x the subject of $y = 2x + 7$.

REFRESHER · GRAPHS, INEQUALITIES AND REARRANGING

The graph of $y = x^2$ is a parabola with its lowest point at the origin; $y = x^3$ rises from lower left to upper right and has a flat point at the origin; $y = \dfrac{1}{x}$ has two branches and is undefined at $x = 0$. A square root is defined only for a non-negative radicand, so $\sqrt{x - 3}$ requires $x \geq 3$. Rearranging a formula uses inverse operations in reverse order: from $y = 2x + 7$, subtract 7 then divide by 2.

Interval notation: $a \leq x < b$ is written $[a, b)$, and $x \geq a$ is $[a, \infty)$. A round bracket excludes the endpoint, a square bracket includes it; ∞ is never included.

CONCEPT

2.1 Domain, range and function notation

A function is a rule that assigns to each element of one set exactly one element of another. The first set is the **domain**; the set of values that actually occur is the **range**. The phrase "exactly one" is the whole content of the definition: a rule that offers two outputs for one input is not a function.

Notation matters because it records the three parts. Writing $f(x) = 2x + 1$ names the function f, the input variable x, and the rule. Writing $f(3) = 7$ states that the input 3 produces the output 7. It does *not* mean "f multiplied by 3", and it does not tell you the domain unless one is stated. In this course, a function given by a formula without a stated domain has the **natural domain**: all real numbers for which the formula produces a real value.

DEFINITIONS

Function, domain, range, one-to-one

A function f from a set A to a set B assigns to every $x \in A$ exactly one value $f(x) \in B$. The set A is the domain; the set $\{f(x) : x \in A\}$ is the range. A function is **one-to-one** (injective) if different inputs always give different outputs: $f(a) = f(b)$ implies $a = b$.

A function is **increasing** on an interval if $a < b$ implies $f(a) < f(b)$ there, and **decreasing** if $a < b$ implies $f(a) > f(b)$. A function that is always increasing or always decreasing on its domain is called monotonic.

EXAMPLE 2.1 FINDING A NATURAL DOMAIN AND RANGE

For each function, state the natural domain and the range. (a) $f(x) = \dfrac{1}{x-2}$ (b) $g(x) = \sqrt{4-x}$ (c) $h(x) = x^2 - 4x + 3$

SOLUTION

(a) The denominator cannot be zero, so $x \neq 2$. The domain is $\mathbb{R} \setminus \{2\}$, or $(-\infty, 2) \cup (2, \infty)$. As x approaches 2 the fraction grows without bound in size, and as x grows large the fraction approaches 0 but never equals it. Hence the range is $\mathbb{R} \setminus \{0\}$.

(b) The radicand must be non-negative: $4 - x \geq 0$, so $x \leq 4$. Domain $(-\infty, 4]$. A square root is never negative, and it takes every value from 0 upward as x falls from 4, so the range is $[0, \infty)$.

(c) A quadratic is defined for all real x. Complete the square: $x^2 - 4x + 3 = (x-2)^2 - 1$. Since $(x-2)^2 \geq 0$, the output is at least -1, and every value from -1 upward occurs. Domain $\mathbb{R}$; range $[-1, \infty)$.

> The range is a property of the graph, not of the formula alone. Restricting the domain changes the range: h on $[0, 1]$ has range $[0, 3]$.

EXAMPLE 2.2 WORKING WITH FUNCTION NOTATION

Let $f(x) = x^2 - 3x$ and $g(x) = 2x + 1$. Find (a) $f(-2)$, (b) $g(f(1))$, (c) the values of x for which $f(x) = g(x)$.

SOLUTION

(a) $f(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$.

(b) Work from the inside: $f(1) = 1 - 3 = -2$, so $g(f(1)) = g(-2) = 2(-2) + 1 = -3$.

(c) $x^2 - 3x = 2x + 1$, so $x^2 - 5x - 1 = 0$. The quadratic formula gives

$$x = \frac{5 \pm \sqrt{25 + 4}}{2} = \frac{5 \pm \sqrt{29}}{2}.$$

Numerically $x \approx 5.19$ or $x \approx -0.19$. Both are in the natural domain, so both are solutions.

MISCONCEPTION

"The domain of $\dfrac{x^2 - 4}{x - 2}$ is all real numbers because it simplifies to $x + 2$." The simplification is valid only where the original is defined. At $x = 2$ the original expression is $\dfrac{0}{0}$, which is undefined; the simplified expression happens to equal 4 there, but that value is not in the domain of the original function. Always state restrictions before cancelling.

GUIDED PRACTICE

1. For $f(x) = 5 - 2x$, find $f(0)$ and the x for which $f(x) = 0$. (Answer: 5; $x = 2.5$.)
2. State the natural domain of $\sqrt{2x - 6}$. (Answer: $x \geq 3$.)
3. The range of $y = (x+1)^2 - 5$ is $[k, \infty)$. Find k. (Answer: -5.)
4. Explain why $y^2 = x$ does not define y as a function of x for all $x \geq 0$. (Two outputs $\pm\sqrt{x}$ for each positive x.)

EXERCISE 2A NOTATION, DOMAIN AND RANGE

FLUENCY

EXERCISE 2A · CONTINUED NOTATION, DOMAIN AND RANGE

3 State the natural domain of each.

a. $\dfrac{3}{x+4}$

b. $\sqrt{x+5}$

c. $\dfrac{1}{x^2-9}$

d. $\sqrt{7-2x}$

e. $\dfrac{x}{x^2+1}$

4 State the range of each, given its natural domain.

a. $y = x^2 + 1$

b. $y = 3 - \sqrt{x}$

c. $y = \dfrac{2}{x}$

d. $y = (x-3)^2$

5 Find the value(s) of x for which $f(x) = 0$.

a. $f(x) = 3x - 12$

b. $f(x) = x^2 - 5x + 6$

c. $f(x) = \dfrac{x-1}{x+2}$

6 Write each set in interval notation.

a. $-3 < x \leq 4$

b. $x < 1$

c. $x \neq 5$

d. $2 \leq x < 7$

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes: "For $f(x) = \sqrt{x-1}$, the domain is $x \geq 1$ and the range is $y \geq 1$." Identify the error in the range and correct it.

8 Explain, using the definition, why the circle $x^2 + y^2 = 9$ is not the graph of a function $y = f(x)$.

9 The function $f(x) = x^2 - 6x + 5$ is restricted to $x \geq 3$. Find its range on this restricted domain.

10 A function has domain $[-2, 4]$ and range $[1, 9]$. State the domain and range of $y = f(x-1) + 3$.

11 Give an example of a function whose natural domain is all real numbers but whose range is not all real numbers. Justify both claims.

12 Explain why $f(a) = f(b)$ does not always imply $a = b$, using $f(x) = x^2$ as an example. What extra condition would make the implication true?

APPLICATION AND EXTENSION

13 A taxi charges a fixed 3.50 plus 1.20 per kilometre. Write the fare $F(d)$ as a function of distance and state a sensible domain and range for a city journey of at most 30 km.

14 The height of a ball above the ground is $h(t) = 1 + 8t - 5t^2$ metres after t seconds. State a sensible domain for the physical situation and find the maximum height by completing the square.

15 **Extension.** The function $f(x) = \dfrac{ax+b}{cx+d}$ is defined for $x \neq -\dfrac{d}{c}$. Show that the range excludes $y = \dfrac{a}{c}$, and explain what happens when $a = 0$.

CONCEPT AND METHOD

2.2 Transformations of graphs

Every transformation in this section starts from a **parent graph**, such as $y = x^2$, $y = x^3$, $y = |x|$, $y = \sqrt{x}$ or $y = \dfrac{1}{x}$. A transformation changes the parent by a rule applied to the input or to the output. The distinction is decisive: **outside** changes act on the output and behave as they read; **inside** changes act on the input and behave in the opposite direction.

For a point (x, y) on the parent graph, the transformed point is found by asking what input the new function needs and what output it produces. If $y = f(x-3)$, the new graph at $x = 5$ uses the parent value at $x = 2$: every point moves 3 to the right. If $y = f(x) + 3$, every output is 3 larger: every point moves 3 up.

KEY IDEA · TRANSFORMATION SUMMARY

Starting from $y = f(x)$: $y = f(x) + a$ translates a units up; $y = f(x-a)$ translates a units right; $y = -f(x)$ reflects in the x-axis; $y = f(-x)$ reflects in the y-axis; $y = af(x)$ stretches vertically by factor a (a compression if $0 < a < 1$); $y = f(bx)$ stretches horizontally by factor $\dfrac{1}{b}$ (a compression if $b > 1$).

EXAMPLE **2.3** DESCRIBING A TRANSFORMATION FROM ITS EQUATION

Describe how the graph of each function is obtained from a parent graph. State the coordinates of the image of the origin or of the parent's key point. (a) $y = (x+2)^2 - 5$ (b) $y = -2\sqrt{x}$ (c) $y = \dfrac{1}{2x-4}$

SOLUTION

(a) Parent $y = x^2$. Inside, $x + 2 = x - (-2)$: translate 2 left. Outside, -5: translate 5 down. The vertex $(0, 0)$ moves to $(-2, -5)$.

(b) Parent $y = \sqrt{x}$. The factor -2 is outside: stretch vertically by 2 and reflect in the x-axis. The point $(0, 0)$ stays fixed; $(1, 1)$ moves to $(1, -2)$; $(4, 2)$ moves to $(4, -4)$.

(c) Write the denominator as $2(x-2)$, so $y = \dfrac{1}{2(x-2)} = \dfrac{1}{2} \cdot \dfrac{1}{x-2}$. Parent $y = \dfrac{1}{x}$: translate 2 right, then compress vertically by factor $\dfrac{1}{2}$. The asymptotes move from $x = 0, y = 0$ to $x = 2, y = 0$.

> Rewriting $2x - 4$ as $2(x-2)$ exposes both transformations. Without it, the horizontal compression and shift are easy to confuse.

EXAMPLE **2.4** COMBINING TRANSFORMATIONS IN ORDER

The graph of $y = f(x)$ passes through $(2, 5)$ and has a minimum there. Find the image of this point and describe the shape change for (a) $y = 3f(x-1) + 4$ and (b) $y = f(2x - 6)$.

SOLUTION

(a) The inside $x - 1$ moves the point 1 right; the outside $3(\cdot) + 4$ triples the output and then adds 4. The image is $(2 + 1,\ 3 \cdot 5 + 4) = (3, 19)$. The minimum remains a minimum; the graph is three times as steep vertically.

(b) Write $2x - 6 = 2(x - 3)$. The horizontal compression by factor $\dfrac{1}{2}$ comes first, then a translation 3 right. The point $(2, 5)$ moves to $(3 + 1,\ 5) = (4, 5)$: half the horizontal distance from the axis $x = 0$, then 3 right. Equivalently, translate 3 right and then compress by $\dfrac{1}{2}$. The minimum remains at height 5.

> The two descriptions in (b) look different but give the same point: $x \mapsto \frac{1}{2}x + 3$ has inverse $x \mapsto 2(x-3)$. What matters is that the horizontal operations are applied to the input, not to the output.

MISCONCEPTION

"The graph of $y = f(x+4)$ moves 4 units left because the sign is positive, and $y = f(2x)$ stretches horizontally by 2." Both halves are wrong. $x + 4$ means the input is 4 *larger* than the parent input at the same output, so the feature occurs 4 units earlier: left. And $f(2x)$ reaches the parent value twice as fast, so the graph is compressed by $\dfrac{1}{2}$, not stretched by 2. Test with one point: if $f(2) = 10$, then $f(2x) = 10$ when $x = 1$.

GUIDED PRACTICE

1. Describe the transformation $y = |x - 3| + 2$ from $y = |x|$. (3 right, 2 up.)
2. The point $(4, 1)$ lies on $y = f(x)$. Find its image on $y = f(x+1) - 3$. $(3, -2)$.)
3. Write $y = \sqrt{3x + 6}$ in the form $y = \sqrt{b(x-h)}$. $y = \sqrt{3(x+2)}$.)
4. Sketch $y = -f(2x)$ for $f(x) = x^2$, labelling three points. $(1, 1) \mapsto (0.5, -1)$; $(2, 4) \mapsto (1, -4)$.)

EXERCISE 2B

TRANSLATIONS, REFLECTIONS AND STRETCHES

FLUENCY

1 Describe each transformation of $y = f(x)$ in words.

a. $y = f(x) - 7$

b. $y = f(x + 3)$

c. $y = 4f(x)$

d. $y = f(-x)$

e. $y = f\left(\frac{1}{3}x\right)$

f. $y = -f(x) + 2$

2 The point $(6, -2)$ is on $y = f(x)$. Find its image under each transformation in Question 1.

3 Describe how each graph is obtained from its parent.

a. $y = (x - 4)^2 + 1$

b. $y = |x + 2| - 5$

c. $y = \dfrac{3}{x}$

d. $y = \sqrt{x - 1} + 3$

e. $y = -\dfrac{1}{2}x^3$

4 State the vertex or key point of each.

a. $y = (x + 6)^2 - 11$

b. $y = -3|x - 2| + 7$

c. $y = \sqrt{x + 8}$

d. $y = \dfrac{1}{x - 5} + 2$

5 Write an equation for each description.

a. $y = x^2$ shifted 3 left and 4 down

b. $y = |x|$ reflected in the x-axis and shifted 2 right

c. $y = \sqrt{x}$ stretched vertically by 3 and shifted 1 up

d. $y = \dfrac{1}{x}$ compressed horizontally by factor $\frac{1}{2}$

UNDERSTANDING AND REASONING

6 **Error analysis.** A student says $y = f(2x - 6)$ is obtained by compressing $y = f(x)$ horizontally by factor $\frac{1}{2}$ and then shifting 6 right. Test the claim on $f(x) = x^2$ using the point $(1, 1)$, and state the correct description.

7 Explain why the order of a horizontal translation and a horizontal stretch changes the result, while the order of a vertical translation and a vertical stretch does not. Use a point to support your explanation.

8 The graph of $y = f(x)$ has a maximum at $(-3, 8)$. Find the corresponding turning point of $y = -2f(x + 1) - 3$ and state whether it is a maximum or a minimum. Explain your choice.

9 Sketch $y = (x - 2)^2 - 3$ and $y = -(x - 2)^2 + 3$ on the same axes. Describe how the second is obtained from the first.

10 A curve is symmetric about the y-axis. Which of the transformations in Question 1 preserve that symmetry? Justify each answer.

11 Explain why $y = f(x - 2) + f(2 - x)$ is symmetric about $x = 2$ for any function f.

APPLICATION AND EXTENSION

12 The profile of a tunnel is modelled by $y = 6 - 0.5x^2$ for $-2\sqrt{3} \le x \le 2\sqrt{3}$, where y is height in metres. A second tunnel has the same shape, is twice as wide, and is 2 m taller. Write its equation and state its domain.

13 The graph of $y = \sqrt{x}$ is transformed to pass through $(3, 5)$ with its endpoint there. Write a possible equation and explain whether it is unique.

14 **Extension.** Show that applying $y = f(x - h)$ then $y = f(bx)$ is not the same as applying $y = f(bx)$ then $y = f(x - h)$. Find the single transformation that is equivalent to each order, in the form $y = f(bx + c)$.

CONCEPT AND METHOD

2.3 Composite functions

A composite function applies one rule after another. If g sends x to $g(x)$, and f then acts on that output, the result is $f(g(x))$, written $fg(x)$ or $(f \circ g)(x)$. The order is fixed: the function nearest the input acts first. Composition is not multiplication, and $fg(x)$ is not $f(x) \cdot g(x)$.

The domain of a composite is smaller than either domain in general. For $f(g(x))$ to exist, x must be in the domain of g, and $g(x)$ must be in the domain of f. Both conditions must be checked. This is why a composite can have a restricted domain even when both original functions have natural domain $\mathbb{R}$.

DEFINITION

Composite function

For functions f and g, the composite $f \circ g$ is defined by

$$(f \circ g)(x) = f(g(x)).$$

Its domain is the set of x in the domain of g for which $g(x)$ lies in the domain of f.

EXAMPLE 2.5 FORMING AND EVALUATING COMPOSITES

Let $f(x) = x^2 - 1$ and $g(x) = 3x + 2$. Find $f(g(x))$, $g(f(x))$, $f(g(1))$ and $g(f(1))$. Comment on the results.

SOLUTION

$f(g(x)) = (3x + 2)^2 - 1 = 9x^2 + 12x + 4 - 1 = 9x^2 + 12x + 3.$

$g(f(x)) = 3(x^2 - 1) + 2 = 3x^2 - 3 + 2 = 3x^2 - 1.$

$f(g(1)) = f(5) = 24.$ $g(f(1)) = g(0) = 2.$

> The two composites are different functions: composition is not commutative. Checking with a number is a fast way to catch an order error.

EXAMPLE 2.6 DOMAIN OF A COMPOSITE

Find the domain of each composite. (a) $f(x) = \sqrt{x}$, $g(x) = x - 5$, find $f(g(x))$. (b) $f(x) = \dfrac{1}{x - 2}$, $g(x) = x^2$, find $f(g(x))$.

SOLUTION

(a) $f(g(x)) = \sqrt{x - 5}$. The inner function is defined for all x, but the outer function needs a non-negative input: $x - 5 \geq 0$, so $x \geq 5$. Domain $[5, \infty)$.

(b) $f(g(x)) = \dfrac{1}{x^2 - 2}$. The inner function is defined for all x; the outer function requires $x^2 \neq 2$. Domain $\mathbb{R} \setminus \{-\sqrt{2}, \sqrt{2}\}$.

> Simplifying a composite can hide a restriction. In (b), do not cancel or rewrite the denominator without first recording $x^2 - 2 \neq 0$.

EXAMPLE **2.7** DECOMPOSING A FUNCTION

Write $h(x) = \sqrt{5x - 3}$ as a composite of two simpler functions, and $k(x) = (2x + 1)^3 - 4$ as a composite of three.

SOLUTION

(a) Let $g(x) = 5x - 3$ and $f(x) = \sqrt{x}$. Then $f(g(x)) = \sqrt{5x - 3} = h(x)$.

(b) Let $g(x) = 2x + 1$, $f(x) = x^3$ and $j(x) = x - 4$. Then $j(f(g(x))) = j((2x + 1)^3) = (2x + 1)^3 - 4 = k(x)$.

A decomposition is not unique: h could also be written with $g(x) = x$ and $f(x) = \sqrt{5x - 3}$. The useful decomposition exposes the order in which the operations are performed.

MISCONCEPTION

"$fg(x)$ means $f(x) \times g(x)$, and $f(g(x))$ has the same domain as g." The notation fg in this course means composition, while the product is written $f(x)g(x)$. And the domain of $f \circ g$ is the set of inputs for which *both* stages work; it can exclude values allowed by g because their images are not allowed by f.

GUIDED PRACTICE

1. For $f(x) = 2x + 1$, $g(x) = x^2$, find $f(g(3))$ and $g(f(3))$. (19; 49.)
2. Find the domain of $f(g(x))$ when $f(x) = \dfrac{1}{x}$, $g(x) = x - 4$. ($x \neq 4$.)
3. Decompose $y = (x - 7)^2 + 3$ into $f(g(x))$. ($g = x - 7$, $f = u^2 + 3$.)
4. If $f(g(x)) = x$ for all x, what does that say about the relationship between f and g? They undo each other; f is the inverse of g.)

EXERCISE 2C

COMPOSITION AND DOMAIN

FLUENCY

1 For $f(x) = x + 3$ and $g(x) = 2x - 1$, find $f(g(x))$, $g(f(x))$, $f(g(2))$ and $g(f(2))$.

2 For $f(x) = x^2$ and $g(x) = x - 4$, find $f(g(x))$, $g(f(x))$, $f(g(-1))$ and $g(f(-1))$.

3 For $f(x) = \sqrt{x}$ and $g(x) = x + 9$, find $f(g(x))$ and its domain.

4 For $f(x) = \dfrac{1}{x}$ and $g(x) = x - 3$, find $f(g(x))$ and its domain.

5 Evaluate $f(f(2))$ for (a) $f(x) = x^2 - 1$; (b) $f(x) = \dfrac{1}{x + 1}$.

6 Decompose each into two functions $f(g(x))$.

a. $\sqrt{4x + 1}$
b. $(x - 5)^2$
c. $\dfrac{1}{2x + 3}$
d. $|3x - 2|$

UNDERSTANDING AND REASONING

7 **Error analysis.** Given $f(x) = x^2$, $g(x) = x - 1$, a student writes $f(g(x)) = x^2 - 1$. Identify the error and write the correct composite.

8 Show that $f(g(x))$ and $g(f(x))$ are different for $f(x) = x + 1$, $g(x) = x^2$, by comparing both at $x = 2$.

9 Find the domain of $f(g(x))$ for $f(x) = \sqrt{x - 2}$, $g(x) = \dfrac{1}{x}$. Solve the inequality carefully.

10 If $f(x) = 2x + 3$, find $f(f(x))$ and solve $f(f(x)) = 0$.

11 Explain why the domain of $f \circ g$ can never be larger than the domain of g.

12 For $f(x) = x^2$ and $g(x) = \sqrt{x}$, state the domain of $f(g(x))$ and of $g(f(x))$. Explain why they differ.

EXERCISE 2C · CONTINUED

COMPOSITION AND DOMAIN

APPLICATION AND EXTENSION

13 A shop applies a 20% discount, then adds 15% tax. Write the final price as a composite of two functions of the original price, and evaluate it for a 240 item. Does the order matter? Justify numerically.

14 A car's fuel efficiency $E(v)$ in km per litre depends on speed, and the fuel used for a 300 km journey is $F(E) = \dfrac{300}{E}$. Write $F(E(v))$ and explain what the composite represents.

15 **Extension.** Find all functions f of the form $f(x) = ax + b$ satisfying $f(f(x)) = 9x + 4$.

CONCEPT AND METHOD

2.4 Inverse functions

An inverse function reverses a function. If f sends 3 to 11, then f^{-1} sends 11 back to 3. This is possible only when no two inputs share an output: otherwise the reverse rule would not know which input to return. A function with this property is one-to-one, and its graph passes the **horizontal line test** — no horizontal line meets the graph more than once.

The graph of f^{-1} is the reflection of the graph of f in the line $y = x$. The reason is that a point (a, b) on f means $f(a) = b$, which means $f^{-1}(b) = a$, so (b, a) lies on f^{-1}. The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f. The notation f^{-1} does not mean $\dfrac{1}{f}$.

DEFINITION

Inverse function

If f is one-to-one on its domain, its inverse f^{-1} is defined by

$$f^{-1}(y) = x \quad \Longleftrightarrow \quad f(x) = y.$$

Equivalently, $f^{-1}(f(x)) = x$ for every x in the domain of f, and $f(f^{-1}(y)) = y$ for every y in the range of f.

EXAMPLE 2.8 FINDING AN INVERSE ALGEBRAICALLY

Find the inverse of $f(x) = \dfrac{3x + 1}{x - 2}$, stating its domain.

SOLUTION

Write $y = \dfrac{3x + 1}{x - 2}$ and solve for x in terms of y. Multiply: $y(x - 2) = 3x + 1$, so $xy - 2y = 3x + 1$. Collect the x terms: $xy - 3x = 2y + 1$, hence $x(y - 3) = 2y + 1$ and

$$x = \frac{2y + 1}{y - 3}.$$

Interchanging the names gives $f^{-1}(x) = \dfrac{2x + 1}{x - 3}$, defined for $x \neq 3$. The original f is defined for $x \neq 2$ and its range excludes $y = 3$, which matches the domain restriction of the inverse.

Check. $f(0) = \dfrac{1}{-2} = -\dfrac{1}{2}$ and $f^{-1}\left(-\dfrac{1}{2}\right) = \dfrac{0}{-\frac{7}{2}} = 0$. Also $f^{-1}(f(x)) = x$ can be verified by substitution.

EXAMPLE **2.9** RESTRICTING THE DOMAIN

The function $f(x) = (x-2)^2$ is not one-to-one on $\mathbb{R}$. Restrict its domain so that it becomes invertible, and find the inverse on that restriction.

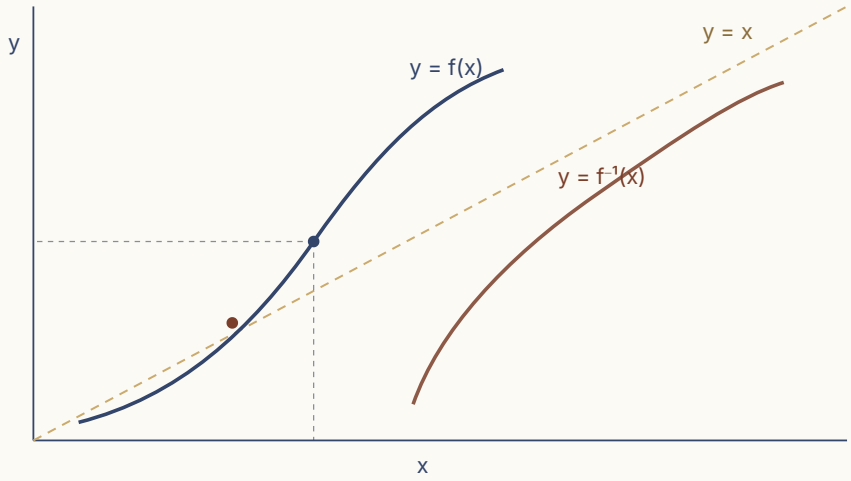
SOLUTION

The parabola has a minimum at $x = 2$, so it is decreasing on $(-\infty, 2]$ and increasing on $[2, \infty)$. Choose the restriction $x \geq 2$. Then $y = (x-2)^2$ with $y \geq 0$, and since $x - 2 \geq 0$,

$$x - 2 = \sqrt{y}, \qquad x = 2 + \sqrt{y}.$$

Thus $f^{-1}(x) = 2 + \sqrt{x}$ for $x \geq 0$, with range $[2, \infty)$. The other choice, $x \leq 2$, gives $f^{-1}(x) = 2 - \sqrt{x}$.

> The restriction must be stated as part of the answer. Without it, "the inverse of x^2" is not a function.



Reflecting a point in $y = x$ swaps its coordinates. The curve and its inverse meet on the line $y = x$ when $f(x) = x$.

MISCONCEPTION

"To find an inverse, put a -1 on the function: $f^{-1}(x) = \dfrac{1}{f(x)}$." The notation f^{-1} denotes the inverse function, not a reciprocal. For $f(x) = 2x$, the inverse is $f^{-1}(x) = \dfrac{x}{2}$, while the reciprocal is $\dfrac{1}{2x}$. Only for the special function $f(x) = \dfrac{1}{x}$ do the two coincide.

GUIDED PRACTICE

1. Find the inverse of $f(x) = 5x - 4$. ($f^{-1}(x) = \dfrac{x+4}{5}$.)
2. Find the inverse of $g(x) = x^3 + 1$. ($g^{-1}(x) = \sqrt[3]{x-1}$.)
3. Explain why $h(x) = |x|$ has no inverse on $\mathbb{R}$. It is not one-to-one: $h(2) = h(-2)$.)
4. The point $(4, 7)$ is on f. State a point on f^{-1}. ($(7, 4)$.)

EXERCISE 2D

INVERSES AND RESTRICTED DOMAINS

FLUENCY

1 Find the inverse of each linear function.

a. $f(x) = 3x + 2$

b. $f(x) = 7 - x$

c. $f(x) = \dfrac{x - 5}{2}$

d. $f(x) = -\dfrac{2}{3}x + 4$

2 Find the inverse of each, stating its domain.

a. $f(x) = x^3$

b. $f(x) = \sqrt{x + 4}$

c. $f(x) = \dfrac{1}{x - 1}$

d. $f(x) = 2 + \sqrt{x}$

3 Each point lies on a one-to-one function f. State the corresponding point on f^{-1}.

a. $(2, 9)$

b. $(-3, 5)$

c. $(0, -1)$

d. (a, b)

4 Show that $f(x) = \dfrac{1}{x}$ is its own inverse.

5 Find $f^{-1}(3)$ for $f(x) = 2x^2 + 1$, $x \geq 0$.

UNDERSTANDING AND REASONING

6 **Error analysis.** A student finds the inverse of $f(x) = (x - 1)^2$ as $f^{-1}(x) = 1 + \sqrt{x}$ without restricting the domain. Explain why the answer is incomplete.

7 Use the horizontal line test to decide which functions are one-to-one on their natural domain.

a. $y = x^3$

b. $y = x^2 - 4$

c. $y = \dfrac{1}{x}$

d. $y = |x - 2|$

e. $y = \sqrt{4 - x}$

8 Explain why the domain of f^{-1} equals the range of f. Use the reflection in $y = x$ in your explanation.

9 Find the inverse of $f(x) = \dfrac{x + 3}{x - 1}$ and state its domain. Verify with one numerical check.

10 The graph of $y = f(x)$ has a y-intercept at $(0, 5)$ and an x-intercept at $(4, 0)$. State the intercepts of $y = f^{-1}(x)$.

11 Show algebraically that $f(x) = 3x - 7$ is one-to-one by assuming $f(a) = f(b)$ and deducing $a = b$.

APPLICATION AND EXTENSION

12 A currency conversion is $C(d) = 0.92d$, converting dollars to euros. Find the inverse and explain what it computes.

13 The speed of a wave in shallow water is $v = \sqrt{9.8d}$ m/s at depth d metres. Make d the subject and find the depth needed for a speed of 7 m/s.

14 **Extension.** The function $f(x) = x^2 - 6x + 5$ is restricted to $x \geq 3$. Find $f^{-1}(x)$, state its domain and range, and explain why the restriction $x \leq 3$ would produce a different inverse.

2.5 Modelling with transformations

APPLICATION

A parent graph becomes a model when its parameters are chosen to match measurements. The general form $y = a\,f(b(x - h)) + k$ contains four adjustments: a vertical stretch a, a horizontal stretch $\dfrac{1}{b}$, a horizontal shift h and a vertical shift k. Each parameter has a meaning in the context — an initial value, a rate, a turning point, a baseline — and that meaning is what makes the model interpretable rather than merely fitted.

The order of work is: choose a parent whose shape matches the phenomenon; use the key point (vertex, endpoint, asymptote) to fix h and k; use one further measurement to fix the stretch; then check the model against a measurement not used in the fitting.

EXAMPLE 2.10 FITTING A PARABOLIC ARCH

A footbridge has a parabolic arch. The arch is 6 m wide at ground level and rises 2.5 m at the centre. Write its height y as a function of the horizontal distance x from the left end, and find the height 1 m from the left end.

SOLUTION

Take the parent $y = x^2$ and place the vertex at the centre. The centre is $x = 3$, and the top is 2.5, so the model has the form $y = a(x - 3)^2 + 2.5$.

At the left end, $x = 0$, the height is 0: $0 = a(0 - 3)^2 + 2.5$, so $9a = -2.5$ and $a = -\dfrac{5}{18} \approx -0.278$.

Hence

$$y = -\frac{5}{18}(x - 3)^2 + 2.5, \qquad 0 \le x \le 6.$$

At $x = 1$: $y = -\dfrac{5}{18}(4) + 2.5 = -1.111 + 2.5 = 1.389$ m (3 s.f.).

> The negative a reflects the parabola, because the arch opens downward. The domain is part of the model: outside $0 \le x \le 6$ the formula describes no bridge.

EXAMPLE 2.11 A SQUARE-ROOT MODEL AND ITS INVERSE

The time T seconds for a pendulum of length L metres to complete one swing is measured and modelled by $T = k\sqrt{L}$. A pendulum of length 1.2 m takes 2.2 s. Find k, predict the time for 3 m, and find the length needed for a period of 3 s.

SOLUTION

From $2.2 = k\sqrt{1.2}$, $k = \dfrac{2.2}{\sqrt{1.2}} = 2.008\ldots \approx 2.01$. The model is $T = 2.01\sqrt{L}$.

For $L = 3$: $T = 2.01\sqrt{3} = 3.48$ s (3 s.f.). For $T = 3$: $\sqrt{L} = \dfrac{3}{2.01} = 1.493$, so $L = 2.23$ m (3 s.f.).

The second calculation uses the inverse function $L = \left(\dfrac{T}{k}\right)^2$.

> The physical pendulum formula has $k = \dfrac{2\pi}{\sqrt{g}} \approx 2.006$ for small swings, so the fitted value is close to the theoretical one. Agreement supports the model but does not prove it.

MISCONCEPTION

"A good fit means the model is true." A transformed parent function can fit three points exactly and still be wrong between them. The shape was chosen in advance; the data only fixed its parameters. Always test the model on a point not used to fit it, and state the range of inputs over which it is plausible.

GUIDED PRACTICE

1. A parabola has vertex $(2, -1)$ and passes through $(0, 3)$. Find its equation. $y = (x - 2)^2 - 1$.)

2. The graph of $y = \sqrt{x}$ is shifted 4 right and 2 up. Write the equation and state its domain and range. $y = \sqrt{x - 4} + 2$; $x \ge 4$, $y \ge 2$.)

3. A model $y = a(x - h)^2 + k$ has $a < 0$. What does that tell you about the maximum or minimum? Maximum at (h, k).)

4. Why is a restricted domain necessary when a model is inverted? The parent may not be one-to-one; the context selects one branch.)

EXERCISE 2E

BUILDING AND USING TRANSFORMED MODELS

FLUENCY

1 Write the equation of each parabola.
   a. Vertex $(0, 0)$, passing through $(2, 8)$
   b. Vertex $(1, -3)$, passing through $(3, 5)$
   c. Vertex $(-2, 4)$, passing through $(0, 0)$

2 A curve $y = a\sqrt{x - h} + k$ has its endpoint at $(2, 1)$ and passes through $(6, 5)$. Find a, h, k.

3 The graph of $y = \dfrac{1}{x}$ is shifted 3 right and 1 down. Write the equation and state the asymptotes.

4 A function is modelled by $y = 4 - 2(x - 1)^2$. State its vertex, axis of symmetry and maximum value.

UNDERSTANDING AND REASONING

5 **Error analysis.** A student models a cable hanging between two towers with $y = a(x - h)^2 + k$ and $a > 0$, obtaining a curve that opens upward with the lowest point between the towers. Explain why $a > 0$ is correct here, and why the same sign would be wrong for a bridge arch.

6 A model predicts a quantity for $x = 10$ using data from $x = 1$ to 5. Explain why this is extrapolation and why the prediction is less reliable than an interpolation.

7 The shadow radius in the opening problem is $r(x) = \dfrac{0.5x}{2 - x}$ for a ball of radius 0.5 m. Describe how this graph relates to $y = \dfrac{1}{x}$, and state its domain.

8 A transformed square-root model has domain $x \geq 2$ and range $y \leq 7$. Write one possible equation.

9 Explain how the inverse of a model can be useful even when the model itself is only approximate.

APPLICATION AND EXTENSION

10 The height of a seedling is modelled by $h(t) = 2 + 0.8\sqrt{t}$ cm after t weeks. Find its height at $t = 0$, 4 and 16, and the time at which it reaches 6 cm.

11 A suspension bridge cable takes the shape $y = 0.02(x - 40)^2 + 3$ for $0 \leq x \leq 80$, where y is height in metres above the road. Find the height at the towers and at the lowest point.

12 **Extension.** A model has the form $y = a\,f(b(x - h)) + k$. Explain why the graph is unchanged if a is replaced by $2a$, k by $2k$, and the entire graph is then stretched vertically by $\dfrac{1}{2}$. What does this imply about fitting a model to data?

PATTERN AND GENERALISATION

2.6 Investigation: when is a function invertible?

An inverse exists when the rule can be reversed without ambiguity. This investigation asks what property of a graph guarantees that, how a restriction can create it, and why the reflection in $y = x$ preserves the essential information.

INVESTIGATION 2 · ONE-TO-ONE AND REVERSIBLE

Reading invertibility from a graph

1. Sketch $y = x^2$, $y = x^3$, $y = |x|$ and $y = \dfrac{1}{x}$ on separate axes. For each, state whether it passes the horizontal line test.
2. For the graphs that fail, identify the largest interval containing $x = 0$ on which the function is one-to-one. Write the restricted function and its inverse.
3. Copy and complete: a function is invertible exactly when every horizontal line meets its graph at most once, because …
4. Consider $f(x) = x^2 - 4x$. Complete the square and state the two natural restrictions that make it invertible. Find both inverses and their domains.
5. For $f(x) = \dfrac{2x + 1}{x - 1}$, calculate $f(f(x))$. What do you notice? Find f^{-1} and compare.
6. Sketch $y = f(x)$ and $y = f^{-1}(x)$ for $f(x) = 2x + 1$. Mark the intersection with $y = x$. Explain why any intersection of a function and its inverse must lie on $y = x$ or be paired with another intersection.
7. Generalise: if f is increasing on its whole domain, must it be invertible? What if it is decreasing? Justify using two inputs $a < b$.

Generalise. A strictly monotonic function is invertible. Prove that if f is strictly increasing and $f(a) = f(b)$, then $a = b$. Then explain why a function that is increasing on one interval and decreasing on another can fail to be invertible, even though it is one-to-one on each interval separately.

MODELLING · CRITERION D

The shrinking shadow

Return to the lamp, ball and wall. The lamp is 2 m from the wall, the ball has radius 0.5 m, and its centre is x metres from the wall.

1. **Model.** Using similar triangles, show that the shadow radius is $r(x) = \dfrac{0.5x}{2 - x}$. State the domain.
2. **Transform.** Rewrite $r(x)$ in the form $a + \dfrac{b}{2 - x}$ and describe the transformation from $y = \dfrac{1}{x}$.
3. **Invert.** Find x in terms of r. Explain why the inverse is valid on the physical domain.
4. **Evaluate.** State two assumptions of the model — for example about the lamp and the ball — and describe how each would affect the shadow in reality.

CUMULATIVE

2.7 Mixed practice

Attempt all questions without notes. Decide which representation — formula, graph or table — makes each question shortest.

MIXED PRACTICE 2

FUNCTIONS, TRANSFORMATIONS, COMPOSITION AND INVERSES

FLUENCY AND UNDERSTANDING

1 For $f(x) = \dfrac{6}{x-2}$, state the domain and range, and find $f(5)$ and $f(-1)$.

2 Describe the transformation of $y = x^2$ that produces $y = -2(x+3)^2 + 4$.

3 The point $(-2, 6)$ lies on $y = f(x)$. Find its image on $y = f(3x) + 1$.

4 For $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, find $f(g(x))$ and $g(f(x))$.

5 Find the domain of $\sqrt{5 - 2x}$.

6 Find the inverse of $f(x) = \dfrac{4x - 1}{2}$.

7 State whether $y = x^2 - 8x + 7$ is one-to-one. Justify, then restrict the domain to make it invertible.

8 Find $f^{-1}(x)$ for $f(x) = x^2 + 2$ on $x \geq 0$.

9 Sketch $y = \dfrac{1}{x+2}$ and state its asymptotes.

10 If $f(x) = 3x + 5$, solve $f^{-1}(x) = 4$.

11 Write $y = \sqrt{9 - 3x}$ in the form $y = \sqrt{b(h - x)}$, then describe the transformations.

12 Explain why $f(g(x))$ and $g(f(x))$ need not have the same domain.

REASONING AND APPLICATION

13 A parabola has vertex $(-1, 2)$ and passes through $(3, -6)$. Find its equation in the form $y = a(x - h)^2 + k$.

14 The graph of $y = f(x)$ has a maximum at $(4, 10)$. Find the corresponding turning point of $y = -f(2x - 6) + 1$ and state whether it is a maximum or a minimum.

15 Find the inverse of $f(x) = \dfrac{2x + 5}{3 - x}$ and state its domain.

16 A company's profit is $P(n) = 120n - 2n^2$ for $0 \leq n \leq 60$. Complete the square, state the maximum profit, and find the number of units that gives it.

17 The stopping distance of a car is modelled by $d = kv^2$. At 40 km/h the distance is 16 m. Find k, the distance at 60 km/h, and the speed for a 50 m stop.

18 Show that $f(x) = \dfrac{x}{x - 1}$ satisfies $f(f(x)) = x$ by computing $f(f(x))$ and simplifying. What does the result imply about f^{-1}?

19 Explain why the graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$, using the definition of inverse.

20 A rectangle has perimeter 40 cm. Write its area as a function of one side, state the domain, and find the maximum area.

21 **Extension.** Prove that if f and g are both one-to-one, then $f \circ g$ is one-to-one.

22 **Extension.** Find all values of k for which $f(x) = x^2 + kx + 4$ is not one-to-one on $\mathbb{R}$, and explain your answer by completing the square.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 20 marks

Functions and their properties

Instructions. Answer all questions. Show sufficient working. A calculator is permitted but not required.

1 For $f(x) = x^2 - 6x + 5$, find the domain, the range and the coordinates of the vertex. [4]

2 Describe the transformation that maps $y = |x|$ to $y = -2|x - 3| + 1$. [3]

3 For $f(x) = 3x - 2$ and $g(x) = x^2 + 1$, find $f(g(x))$ and $g(f(x))$. [4]

4 Find the inverse of $f(x) = \dfrac{x + 4}{2x - 3}$, stating its domain. [5]

5 Explain why $y = x^2 + 4$ has no inverse on $\mathbb{R}$, and find the inverse of the restriction $x \leq 0$. [4]

Evidence: correct domain and range, a complete transformation description, both composites, a correct algebraic inverse with restriction, and a justified domain choice.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Iterating a function

Instructions. Work independently. Use a table to record results and justify each generalisation.

Let $f(x) = 2x + 1$. Define $f^{(1)}(x) = f(x)$, $f^{(2)}(x) = f(f(x))$, and so on.

1. Calculate $f^{(2)}(x)$, $f^{(3)}(x)$ and $f^{(4)}(x)$. [6]
2. Describe the pattern in the coefficients of x and the constant terms. [4]
3. Predict $f^{(n)}(x)$ and test your formula for $n = 5$. [5]
4. Find the value of n for which the constant term first exceeds 1000. [4]
5. Repeat for $g(x) = 3x$. How does the pattern differ, and why? [3]
6. Explain what iteration has to do with inverse functions. [2]

Evidence: correct iterates, a described pattern, a verified general formula, a solved inequality, and a comparison with the multiplier case.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining the order of transformations

Instructions. Write a 300–400 word explanation for a classmate who believes that $y = f(2x - 6)$ is obtained by compressing $y = f(x)$ by $\frac{1}{2}$ and then shifting 6 right. Your explanation must: (i) state the correct order; (ii) use one numerical point to show the difference; (iii) explain why horizontal changes behave in the opposite direction to their sign; (iv) give a general rule for reading $y = af(b(x - h)) + k$. Use correct notation and a diagram or table where it helps.

Evidence: a correct order, a numerical counterexample, a causal explanation of the sign reversal, and a clear general rule.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Modelling a theme-park ride

Instructions. A drop-tower ride lifts passengers to a height of 60 m, holds them, then releases them. A simplified model of the fall is $h(t) = 60 - 5t^2$ for $0 \le t \le 2\sqrt{3}$, where t is the number of seconds since release and h is the height in metres.

1. State the height at $t = 0$, the maximum height and the time at which it occurs. [3]
2. Find the time at which the height is 40 m, and interpret both algebraic solutions. [5]
3. Find the inverse function and explain what it calculates. Why must a domain be chosen? [6]
4. State two assumptions of the model and explain how each would alter the real motion. [5]
5. Evaluate the model: is a quadratic a reasonable description of a drop tower? Justify using the shape and the physical situation. [5]

Evidence: correct key values, a solved quadratic with interpretation, a correct inverse with a restricted domain, explicit assumptions, and a critical evaluation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A function assigns exactly one output to each input. Its domain is the set of allowed inputs; its range is the set of outputs that occur.
- Outside a graph, a in $af(x)+k$ stretches vertically and shifts up; inside, b in $f(b(x-h))$ compresses horizontally by $\frac{1}{b}$ and shifts right by h.
- Composition $f(g(x))$ applies g first. Its domain requires x in the domain of g and $g(x)$ in the domain of f.
- A function is invertible exactly when it is one-to-one; its graph then passes the horizontal line test. The inverse is found by solving $y=f(x)$ for x and interchanging the variables.
- The graphs of f and f^{-1} are reflections in $y=x$; the domain of f^{-1} is the range of f.
- Restricting the domain of a many-to-one function to a monotonic interval makes it invertible, but the restriction must be stated.
- A transformed parent function becomes a model when its parameters are interpreted in context and tested beyond the fitting data.

$$f(g(x))\neq g(f(x)),\qquad f^{-1}(f(x))=x,\qquad f(f^{-1}(x))=x,\qquad y=f(b(x-h))+k$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can state a natural domain and range using interval notation.
- I can evaluate a function and solve $f(x)=k$.
- I can describe and sketch translations, reflections and stretches.
- I can read a combined transformation from $y=af(b(x-h))+k$.
- I can form $f(g(x))$ and $g(f(x))$ and show they differ.
- I can determine the domain of a composite function.
- I can test one-to-one behaviour with the horizontal line test.
- I can find an inverse algebraically and state its domain.
- I can restrict a domain to create an inverse.
- I can explain the reflection of a graph in $y=x$.
- I can fit a transformed parent function to data and interpret its parameters.
- I can evaluate the assumptions of a model.
- I can explain why composition is not commutative.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which transformation gave you the most difficulty, and what rule or test now keeps you from reversing it?
2. Describe one error you made in this chapter and the reasoning that corrected it.
3. The statement of inquiry says that transforming, composing and inverting are operations on the rule itself. Use one example to show what the graph “records” about the operation.
4. Why is the domain not optional when a function is inverted or composed? Give an example where ignoring it produces a wrong answer.
5. What question about functions do you still want to answer?

GROWTH AT A CONSTANT RATIO

Exponential Change

03

CHANGE · EQUIVALENCE · MODELLING

When a quantity multiplies by the same factor in every interval, its graph bends away from every straight line — and logarithms let us measure that bend.

STATEMENT OF INQUIRY

Exponential change is repeated multiplication; logarithms are the inverse language that turns multiplication into addition and makes the rate visible.

KEY CONCEPT

Change

RELATED CONCEPTS

Equivalence · Patterns · Representation

GLOBAL CONTEXT

Scientific and technical innovation — populations, medicines, sound and seismic energy

FACTUAL

What distinguishes exponential change from linear or quadratic change?

CONCEPTUAL

Why is a logarithm the inverse of an exponential, and why does that make it useful for solving for an unknown exponent?

DEBATABLE

When a quantity grows exponentially, is a logarithmic scale a more honest way to present it?



IN THIS CHAPTER

3.1 Exponential growth and decay
3.2 The number e and continuous change
3.3 Logarithms and their laws
3.4 Solving exponential equations
3.5 Modelling with logarithms
3.6 Investigation: how many digits?

OPENING PROBLEM · THE COOLING CUP

How long until the coffee is drinkable?

A cup of coffee is poured at 95°C in a room kept at 20°C. Its temperature is recorded every minute. The readings are close to the model

$$T(t) = 20 + 75(0.88)^t,$$

where t is in minutes and T is in degrees Celsius.

The drink is comfortable at 60°C and too cold below 30°C. **Can the model tell you the window of time in which the coffee is pleasant — and can it tell you how fast the coffee is losing heat at the start?**

1. Evaluate $T(0)$, $T(5)$ and $T(15)$. What is the temperature of the room in the model?
2. Explain why the factor 0.88 produces cooling rather than warming, and why the graph approaches 20°C but never goes below it.
3. Estimate from a table when the coffee first falls below 60°C and when it falls below 30°C.
4. Name one feature of real cooling that this model ignores. Does that make the model useless? Explain.

The model in the opening problem is exponential: each minute the temperature difference from the room is multiplied by the same factor. This chapter develops the algebra of such models, introduces the number e and the logarithm, and shows how to solve for a time that appears in an exponent.

ORIENTATION

3.0 Learning intentions

- Recognise exponential growth and decay from a table, graph or rule, and interpret the base as a growth or decay factor.
- Use the number e and continuous models $y = Ae^{kt}$, relating k to a doubling or half-life.
- Convert between exponential and logarithmic form and apply the laws of logarithms.
- Solve exponential and logarithmic equations, checking for extraneous solutions.
- Use logarithmic scales and linearised graphs to interpret data and evaluate a model.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

identify an exponential relationship and state its growth or decay factor; sketch $y = a^x$ and transformations of it, including the asymptote and intercepts; calculate doubling times and half-lives; use e and continuous growth models; convert $b^y = x$ to $\log_b x = y$ and back; apply the product, quotient and power laws and the change-of-base formula; solve $a^x = b$ using logarithms; solve equations involving $\ln x$ and reject values outside the domain; interpret pH, decibel and earthquake scales; use a log–linear graph to test an exponential model; explain the assumptions and limitations of a growth model.

ATL SKILLS · THINKING AND RESEARCH

Critical thinking: exponential statements are easy to misread. "Grows by 7% each year" is a multiplier of 1.07, not an addition of 7. "Halves every 6 hours" is a multiplier of $\frac{1}{2}$ per 6 hours, not per hour. Translate the sentence into a multiplier before writing any formula. **Research:** population, finance and science data are often presented on logarithmic axes; read the axis label before interpreting the slope.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 3.1.

1. Evaluate 2^5, 3^4, 5^0 and 2^{-3}.
2. Simplify $x^3 \cdot x^4$, $\dfrac{x^7}{x^3}$ and $(x^2)^5$.
3. Solve $2^x = 32$.
4. Increase 200 by 7%, then decrease the result by 7%.
5. Write 0.85 as a percentage decrease.
6. Make x the subject of $y = 3(2)^x$.

REFRESHER · INDICES AND PERCENTAGE MULTIPLIERS

For a positive base b, $b^m b^n = b^{m+n}$, $\dfrac{b^m}{b^n} = b^{m-n}$, $(b^m)^n = b^{mn}$, $b^0 = 1$ and $b^{-n} = \dfrac{1}{b^n}$. A repeated percentage change of $r\%$ per period is a multiplier of $1 + \dfrac{r}{100}$; a decrease of $r\%$ is $1 - \dfrac{r}{100}$. Repeating it n times gives a power of that multiplier, not a multiple of it.

To undo a power, you need a root only if the exponent is known. The equation $2^x = 32$ is solved by writing $32 = 2^5$, so $x = 5$. When the right-hand side is not a power of the base — for example $2^x = 30$ — a new tool is needed. That tool is the logarithm.

CONCEPT

3.1 Exponential growth and decay

An exponential function has the variable in the exponent: $y = ab^x$, where $a \neq 0$, $b > 0$ and $b \neq 1$. The constant a is the value when $x = 0$, because $b^0 = 1$. The base b is the **growth factor**: each time x increases by 1, the output is multiplied by b. If $b > 1$ the quantity grows; if $0 < b < 1$ it decays. The percentage rate is $b - 1$, expressed as a percentage: $b = 1.07$ is a 7% increase per unit, $b = 0.88$ is a 12% decrease.

The shape is distinctive. For $b > 1$, the graph rises increasingly steeply, passes through $(0, a)$ and approaches the x-axis as an asymptote on the left. For $0 < b < 1$, it falls, approaching the x-axis on the right. In both cases the range is $(0, \infty)$ when $a > 0$, and the graph never touches the axis. Exponential change is not linear (constant difference) and not quadratic (constant second difference); it has a constant *ratio* between successive values.

DEFINITIONS

Exponential function, growth factor, doubling time, half-life

An exponential function has the form $f(x) = ab^x$ with $a \neq 0$, $b > 0$, $b \neq 1$. The base b is the growth factor per unit increase in x. If $b > 1$, f models growth; if $0 < b < 1$, decay.

The **doubling time** is the increase in x needed to multiply the output by 2. The **half-life** is the increase in x needed to multiply it by $\frac{1}{2}$. Both are independent of the starting value a.

EXAMPLE 3.1 READING GROWTH AND DECAY FROM A RULE

For each function, state the initial value, the growth or decay factor, the percentage rate per unit, and whether the graph rises or falls. (a) $y = 400(1.06)^x$ (b) $y = 250(0.72)^x$ (c) $y = 5(3)^{-x}$

SOLUTION

(a) Initial value 400; factor 1.06; rate $1.06 - 1 = 0.06 = 6\%$ increase per unit; rising.

(b) Initial value 250; factor 0.72; rate $0.72 - 1 = -0.28$, a 28% decrease per unit; falling.

(c) Rewrite using $3^{-x} = (3^{-1})^x = \left(\frac{1}{3}\right)^x$. Initial value 5; factor $\frac{1}{3} \approx 0.333$; rate $-\frac{2}{3} \approx -66.7\%$ per unit; falling. Writing a negative exponent as a reciprocal base makes the direction obvious.

> The initial value is the coefficient, not the base. At $x = 0$, $y = a$ for every exponential ab^x.

EXAMPLE 3.2 CONSTRUCTING A MODEL FROM A DESCRIPTION

A colony of bacteria doubles every 3 hours. It starts with 800 bacteria. (a) Write the population after t hours. (b) Find the population after 12 hours. (c) Find the hourly growth factor to 3 significant figures.

SOLUTION

(a) The population is multiplied by 2 every 3 hours, so after t hours it has been multiplied by $2^{t/3}$:

$$P(t) = 800 \cdot 2^{t/3}.$$

(b) $P(12) = 800 \cdot 2^4 = 800 \cdot 16 = 12\,800$.

(c) We need b with $b^3 = 2$, so $b = 2^{1/3} = 1.2599\ldots \approx 1.26$. The equivalent model is $P(t) = 800(1.26)^t$, a 26% increase per hour.

> The two forms $800 \cdot 2^{t/3}$ and $800(1.26)^t$ are the same function, rounded. The first preserves the exact doubling time; the second makes the hourly rate explicit.

EXAMPLE 3.3 DEPRECIATION AND TRANSFORMATIONS

A delivery van costs $24\,000$. It loses 15% of its value each year. (a) Write its value after t years. (b) Find its value after 5 years. (c) Sketch $y = 3(0.5)^x + 2$, stating the asymptote and intercept.

SOLUTION

(a) A 15% loss means a multiplier of $1 - 0.15 = 0.85$ each year, so $V(t) = 24\,000(0.85)^t$.

(b) $V(5) = 24\,000(0.85)^5 = 24\,000(0.4437\ldots) = 10\,649$ (to the nearest unit).

(c) Start from $y = (0.5)^x$, a decreasing exponential through $(0, 1)$ with asymptote $y = 0$. The factor 3 stretches vertically; the $+2$ shifts the asymptote to $y = 2$. The intercept is $3(1) + 2 = 5$. The graph falls from $(0, 5)$ toward $y = 2$, never reaching it.

> A vertical shift changes the asymptote and therefore the range. Here the range is $(2, \infty)$, not $(0, \infty)$.

MISCONCEPTION

"A 7% increase each year for 10 years is a 70% increase in total." Repeated percentage change compounds: the multiplier is $1.07^{10} = 1.967$, a 96.7% increase. The linear answer 70% would be correct only if the increase were 7% of the *original* value each year, which is not what "7% per year" means in finance or population growth.

GUIDED PRACTICE

1. State the growth factor and rate: $y = 80(1.12)^x$, $y = 50(0.94)^x$. (1.12, +12%; 0.94, −6%.)

2. A quantity halves every 4 days. Write it as $Q = Q_0 b^t$ with t in days. $Q = Q_0(0.5)^{t/4}$.)

3. Evaluate $T(10)$ for $T(t) = 20 + 75(0.88)^t$. $20 + 75(0.2785) = 40.9°\text{C}$.)

4. Why can an exponential graph never cross the x-axis? A positive base to any power is positive; the horizontal asymptote is $y = 0$.)

EXERCISE 3A

GROWTH FACTORS, GRAPHS AND MODELS

FLUENCY

1 State the initial value and growth factor, then classify as growth or decay.
   a. $y = 15(1.3)^x$
   b. $y = 200(0.82)^x$
   c. $y = 4(2.5)^x$
   d. $y = 0.5(0.95)^x$
   e. $y = 7(4)^{-x}$

2 Evaluate each function at the given value.
   a. $y = 3(2)^x$ at $x = 4$
   b. $y = 100(0.9)^x$ at $x = 3$
   c. $y = 5(1.5)^x$ at $x = -2$
   d. $y = 2(0.5)^x$ at $x = 5$

3 Write the multiplier for each change.
   a. Increase of 8%
   b. Decrease of 12%
   c. Increase of 0.5%
   d. Decrease of 33%

4 A population of 12 000 grows by 4% each year. Write its size after t years and find the size after 6 years.

5 A radioactive sample has a half-life of 8 days. What fraction remains after 24 days? After 40 days?

6 Sketch $y = 2^x$ and $y = 2^{-x}$ on the same axes. State the asymptote and the y-intercept of each.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes that $y = 500(0.7)^x$ decreases by 7% per unit. Identify the error and state the correct percentage decrease.

8 Explain why $y = 3(2)^x$ and $y = 3(0.5)^{-x}$ have the same graph.

9 A table shows values 5, 10, 20, 40, 80 at equal intervals. Explain why the relationship is exponential, and find the factor and a possible formula.

10 For $y = ab^x$, describe the effect on the graph of (a) doubling a; (b) replacing b by b^2; (c) adding 4 to the whole function.

11 Explain why an exponential function with $b > 1$ eventually exceeds any quadratic, no matter how large its leading coefficient. Use a table of values to support your explanation.

12 The graph of $y = 4(0.8)^x$ is reflected in the y-axis. Write the new equation and describe the result.

APPLICATION AND EXTENSION

13 A car worth 32 000 depreciates by 18% in its first year and 12% in each later year. Find its value after 4 years.

14 An antibiotic dose of 500 mg has a half-life of 5 hours. Find the amount remaining after 18 hours and the first whole hour at which less than 40 mg remains.

15 **Extension.** A quantity grows by 20% every 5 years. Show that its annual growth factor is $1.2^{1/5}$, and find the annual percentage rate to 2 decimal places.

CONCEPT AND NUMBER

3.2 The number e and continuous change

Compound interest can be calculated more and more often: yearly, monthly, daily, every second. If an annual rate of 100% is split into n equal compounding periods, the year-end multiplier is $\left(1+\dfrac{1}{n}\right)^{n}$. The table shows what happens as n grows.

The values approach a fixed number, approximately 2.718281828, called e. It is the natural base for continuous growth: an annual rate r, compounded continuously, gives the multiplier e^{r}. More generally, a quantity that changes continuously at a constant proportional rate satisfies $y = Ae^{kt}$, where k is the continuous rate and A is the initial value. If $k > 0$ the quantity grows; if $k < 0$ it decays.

n	1	10	100	1 000	10 000	100 000
$\left(1+\frac{1}{n}\right)^{n}$	2	2.5937	2.7048	2.7169	2.71815	2.71827

DEFINITION

The exponential number and continuous change

The number e is the limiting value

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} \approx 2.71828.$$

A quantity modelled by $y = Ae^{kt}$ changes at a constant proportional rate k per unit of time. Its doubling time is $\dfrac{\ln 2}{k}$ for $k > 0$; its half-life is $\dfrac{\ln 2}{-k}$ for $k < 0$.

EXAMPLE 3.4 CONTINUOUS GROWTH AND DOUBLING TIME

A population is modelled by $P(t) = 1500e^{0.023t}$, where t is in years. (a) State the initial population and the continuous growth rate. (b) Find the population after 20 years. (c) Find the doubling time.

SOLUTION

(a) $P(0) = 1500$. The continuous rate is $k = 0.023$, or 2.3% per year.

(b) $P(20) = 1500e^{0.46} = 1500(1.5841) = 2376$ (nearest whole number).

(c) Solve $e^{0.023t} = 2$: $0.023t = \ln 2$, so

$$t = \frac{\ln 2}{0.023} = 30.1 \text{ years (3 s.f.)}.$$

The population doubles roughly every 30 years.

> The annual multiplier is $e^{0.023} = 1.0233$, a 2.33% increase per year. The continuous rate 2.3% and the annual rate 2.33% describe the same growth over a year; they are not interchangeable numbers.

EXAMPLE 3.5 COMPARING COMPOUNDING METHODS

2000 is invested at 5% per year for 8 years. Compare the final amount when interest is compounded (a) quarterly, (b) continuously.

SOLUTION

(a) Quarterly rate $0.05/4 = 0.0125$, with $4 \times 8 = 32$ periods:

$$A = 2000(1.0125)^{32} = 2000(1.48813) = 2976.26.$$

(b) Continuously: $A = 2000e^{0.05\times 8} = 2000e^{0.4} = 2000(1.49182) = 2983.65.$

Continuous compounding gives 7.39 more over 8 years. The difference is small because the rate is small and the term is short; it grows with the rate and the time.

> Both answers must be rounded at the end. Rounding the multiplier to 1.49 early would give 2980 and lose the comparison.

EXAMPLE 3.6 DECAY WITH A HALF-LIFE

Carbon-14 has a half-life of about 5730 years. A bone contains 62% of the carbon-14 expected in a living organism. Estimate its age.

SOLUTION

Use $N = N_0\left(\tfrac{1}{2}\right)^{t/5730}$ with $N/N_0 = 0.62$:

$$0.62 = \left(\tfrac{1}{2}\right)^{t/5730}.$$

Take logarithms: $\ln 0.62 = \dfrac{t}{5730}\ln 0.5$, so

$$t = 5730 \cdot \frac{\ln 0.62}{\ln 0.5} = 5730 \cdot \frac{-0.4780}{-0.6931} = 3951 \approx 3950 \text{ years}.$$

The bone is about 3950 years old, to 3 significant figures.

> The continuous form gives the same answer: $k = \dfrac{\ln 0.5}{5730}$ and $t = \dfrac{\ln 0.62}{k}$. Either form is acceptable if the half-life is stated.

MISCONCEPTION

"e is just a button on the calculator, so it can be replaced by 2.7." Rounding e to 2.7 changes a continuous growth rate substantially over time: $e^{0.05} = 1.05127$, but $2.7^{0.05} = 1.05016$. Use the calculator's e value, and round only the final answer. Also, e^{kt} is not the same as $(e^k)^t$ computed with a rounded e^k; keep the exponent exact until the end.

GUIDED PRACTICE

1. State the continuous rate and doubling time for $P = 80e^{0.06t}$. (6%; 11.6.)
2. Find $1000e^{0.03\times 5}$. (1161.83.)
3. A sample decays by 2% per year continuously. Write its model. ($A = A_0e^{-0.02t}$.)
4. Explain the difference between 5% compounded yearly and 5% continuous. Multipliers 1.05 vs $e^{0.05} = 1.0513$.)

EXERCISE 3B

THE NUMBER E AND CONTINUOUS MODELS

FLUENCY

1 Evaluate to 4 significant figures.
   a. e^2
   b. e^{-1}
   c. $e^{0.5}$
   d. e^0

2 Write each as a continuous model $y = Ae^{kt}$.
   a. Initial 200, grows 3% per year continuously
   b. Initial 50, decays 4% per year continuously
   c. Initial 8, grows 1.5% per month continuously

3 Find the doubling time for each continuous growth rate.
   a. $k = 0.02$
   b. $k = 0.07$
   c. $k = 0.12$

4 Find the half-life for each continuous decay rate.
   a. $k = -0.05$
   b. $k = -0.2$
   c. $k = -0.011$

5 1500 is invested at 4% continuous. Find its value after 10 years.

6 10 000 is invested at 3.5% compounded monthly. Find its value after 6 years.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student calculates $2000e^{0.05\times 8}$ as $2000(2.7)^{0.4}$. Explain the error and give the correct value.

8 Show that $P = P_0e^{kt}$ has the same doubling time for every starting value P_0.

9 A model $N = 500e^{0.08t}$ is claimed to grow by 8% per year. Is the claim exactly true? Explain, and give the exact annual percentage rate.

10 Explain why continuous compounding is a limit of more frequent compounding, not a different kind of interest.

11 A sample loses 30% of its mass in 12 years. Find its continuous decay rate and its half-life.

12 Compare $y = e^{0.1x}$ and $y = 2^{x/7}$. Which grows faster for large x? Convert one to the other's base to justify.

APPLICATION AND EXTENSION

13 A lake is stocked with 400 fish. The population is modelled by $P = 400e^{0.18t}$, t in years, until it reaches carrying capacity. Find the population after 3 years and the time to reach 1000.

14 A patient receives 80 mg of a drug that is eliminated continuously at 18% per hour. Find the amount after 4 hours and the time at which 10 mg remains.

15 **Extension.** Show that if a quantity doubles in time T, then $k = \dfrac{\ln 2}{T}$, and use this to write $y = Ae^{kt}$ as $y = A\,2^{t/T}$.

METHOD

3.3 Logarithms and their laws

A logarithm answers the question "to what power must the base be raised?". The statement $\log_b x = y$ means exactly $b^y = x$. The logarithm is therefore the inverse operation of exponentiation, and the graph of $y = \log_b x$ is the reflection of $y = b^x$ in the line $y = x$.

Because a logarithm is an exponent, the laws of logarithms are the laws of indices rewritten. Multiplying powers adds exponents, so the log of a product is the sum of the logs; dividing subtracts; raising to a power multiplies the exponent by that power. These laws hold for positive arguments and a positive base other than 1.

DEFINITION AND LAWS

Logarithm

For $b > 0$, $b \neq 1$ and $x > 0$,

$$\log_b x = y \quad \Longleftrightarrow \quad b^y = x.$$

The domain of $\log_b x$ is $x > 0$; its range is $\mathbb{R}$.

Laws (for $x, y > 0$): $\log_b(xy) = \log_b x + \log_b y$; $\log_b\left(\dfrac{x}{y}\right) = \log_b x - \log_b y$; $\log_b(x^n) = n \log_b x$; $\log_b b = 1$; $\log_b 1 = 0$; $\log_b\left(\dfrac{1}{x}\right) = -\log_b x$.

Change of base: $\log_b x = \dfrac{\log_c x}{\log_c b}$, for any valid base c.

EXAMPLE 3.7 EVALUATING LOGARITHMS

Evaluate without a calculator: (a) $\log_2 32$, (b) $\log_3 81$, (c) $\log_5\left(\dfrac{1}{25}\right)$, (d) $\log_9 3$, (e) $\log_7 1$.

SOLUTION

(a) $2^5 = 32$, so $\log_2 32 = 5$. (b) $3^4 = 81$, so $\log_3 81 = 4$. (c) $\dfrac{1}{25} = 5^{-2}$, so $\log_5\left(\dfrac{1}{25}\right) = -2$. (d) $9^{1/2} = 3$, so $\log_9 3 = \dfrac{1}{2}$. (e) $7^0 = 1$, so $\log_7 1 = 0$.

> Every evaluation is a rewriting of the argument as a power of the base. If that is not possible mentally, use change of base.

EXAMPLE 3.8 USING THE LAWS

Simplify: (a) $\log_2 48 - \log_2 3$, (b) $2\log 5 + \log 4$, (c) $\dfrac{\log_a 8}{\log_a 2}$.

SOLUTION

(a) $\log_2 48 - \log_2 3 = \log_2\left(\dfrac{48}{3}\right) = \log_2 16 = 4$.

(b) $2\log 5 + \log 4 = \log(5^2) + \log 4 = \log 25 + \log 4 = \log 100 = 2$.

(c) $\dfrac{\log_a 8}{\log_a 2} = \log_2 8 = 3$, by change of base. Equivalently, $8 = 2^3$, so the ratio of logs is 3.

> The bases must match before the product and quotient laws can be applied. $\log_2 8 + \log_3 9$ does not simplify to $\log 72$.

EXAMPLE 3.9 CHANGE OF BASE

Evaluate $\log_7 20$ to 4 significant figures, and solve $\log_4 x = 2.5$.

SOLUTION

Using natural logarithms, $\log_7 20 = \dfrac{\ln 20}{\ln 7} = \dfrac{2.9957}{1.9459} = 1.5395$. Check: $7^{1.5395} = 20.00$.

$\log_4 x = 2.5$ means $x = 4^{2.5} = 4^2 \cdot 4^{0.5} = 16 \cdot 2 = 32$.

> Common logarithms give the same value: $\log_7 20 = \dfrac{\log 20}{\log 7}$. Any valid base may be used; natural logarithms are convenient because the button is on every scientific calculator.

MISCONCEPTION

"$\log(x+y) = \log x + \log y$." The product law is for multiplication, not addition. For example, $\log(2+8) = \log 10 = 1$, while $\log 2 + \log 8 = 0.301 + 0.903 = 1.204$. There is no general law for the logarithm of a sum; factorise the argument first if possible.

GUIDED PRACTICE

1. Write $3^4 = 81$ in logarithmic form. $\log_3 81 = 4$.)
2. Evaluate $\log_2 64$, $\log_4 1$, $\log_6 6$. (6, 0, 1.)
3. Simplify $\log_3 6 + \log_3 4.5$. $\log_3 27 = 3$.)
4. Use change of base to evaluate $\log_5 40$. (2.292.)

EXERCISE 3C

LOGARITHMS, LAWS AND CHANGE OF BASE

FLUENCY

1 Convert each to logarithmic form.

a. $2^6 = 64$

b. $10^{-2} = 0.01$

c. $9^{1/2} = 3$

d. $e^0 = 1$

2 Convert each to exponential form.

a. $\log_3 27 = 3$

b. $\log_{10} 1000 = 3$

c. $\log_2\left(\frac{1}{8}\right) = -3$

d. $\ln 1 = 0$

3 Evaluate without a calculator.

a. $\log_4 16$

b. $\log_5 125$

c. $\log_2\left(\frac{1}{16}\right)$

d. $\log_{27} 3$

e. $\log_8 1$

4 Simplify using the laws.

a. $\log_2 5 + \log_2 7$

b. $\log_3 54 - \log_3 2$

c. $3\log_2 2$

d. $\log 4 + \log 25$

5 Use change of base to evaluate to 4 significant figures.

a. $\log_3 10$

b. $\log_6 200$

c. $\log_{12} 5$

6 Solve for x: (a) $\log_2 x = 5$; (b) $\log_5 x = 0$; (c) $\log_x 49 = 2$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes $\log_2 8 = \log_2 2 + \log_2 4 = 1 + 2 = 3$ but then claims the method works for $\log_2 10 = \log_2 5 + \log_2 5$. Explain both what is right and what is wrong.

8 Explain why $\log_b 0$ is undefined for every base b.

9 Show that $\log_b\left(\frac{1}{x}\right) = -\log_b x$ using the laws.

10 Explain why the graph of $y = \log_2 x$ has a vertical asymptote at $x = 0$ and passes through $(1, 0)$ and $(2, 1)$.

11 Prove that $\log_b x = \dfrac{\log_c x}{\log_c b}$ by letting $y = \log_b x$ and taking $\log_c$ of both sides of $b^y = x$.

12 Which is larger, $\log_2 7$ or $\log_3 20$? Justify without a calculator by comparing powers.

APPLICATION AND EXTENSION

13 The loudness of a sound in decibels is $L = 10\log_{10}\left(\dfrac{I}{I_0}\right)$, where $I_0 = 10^{-12}$ W/m^2. Find the loudness of a sound of intensity 10^{-5} W/m^2.

14 The pH of a solution is $\text{pH} = -\log_{10}[\text{H}^+]$. Find the pH when $[\text{H}^+] = 2.5 \times 10^{-4}$, and the hydrogen ion concentration when pH = 8.3.

15 **Extension.** Show that $\log_a b \cdot \log_b c \cdot \log_c a = 1$ for positive $a, b, c \neq 1$.

3.4 Solving exponential equations

METHOD

An exponential equation has the unknown in an exponent. Two strategies cover most cases. If both sides can be written as powers of the same base, equate the exponents. If not, take logarithms of both sides; the power law then brings the unknown down. For equations that mix a power with its square — for example 4^x and 2^x — substitute a new variable and solve the resulting quadratic.

Logarithmic equations require a domain check. The argument of every logarithm must be positive; a candidate solution that makes an argument zero or negative is not valid, even if it satisfies the algebra after combining logs.

EXAMPLE **3.10** TAKING LOGARITHMS

Solve, giving answers to 3 significant figures: (a) $3^x = 20$, (b) $5^{2x-1} = 40$, (c) $2^{x+1} = 3^x$.

SOLUTION

(a) Take natural logs: $x \ln 3 = \ln 20$, so $x = \dfrac{\ln 20}{\ln 3} = \dfrac{2.9957}{1.0986} = 2.727\ldots \approx 2.73$.

(b) $(2x-1)\ln 5 = \ln 40$, so $2x - 1 = \dfrac{3.6889}{1.6094} = 2.2921$, giving $x = \dfrac{3.2921}{2} = 1.646\ldots \approx 1.65$.

(c) $(x+1)\ln 2 = x \ln 3$, so $x \ln 2 + \ln 2 = x \ln 3$, hence $x(\ln 3 - \ln 2) = \ln 2$ and

$$x = \frac{\ln 2}{\ln 3 - \ln 2} = \frac{0.6931}{0.4055} = 1.709\ldots \approx 1.71.$$

Check: $2^{2.709} \approx 6.54$ and $3^{1.709} \approx 6.54$.

> In (c) the unknown appears on both sides. Collect the x-terms before dividing; dividing too early loses the exact form.

EXAMPLE **3.11** A HIDDEN QUADRATIC

Solve $e^{2x} - 5e^x + 6 = 0$.

SOLUTION

Let $u = e^x$, so $e^{2x} = u^2$. The equation becomes $u^2 - 5u + 6 = 0$, so $(u-2)(u-3) = 0$, giving $u = 2$ or $u = 3$.

Since $u = e^x > 0$, both values are admissible: $e^x = 2$ gives $x = \ln 2 \approx 0.693$, and $e^x = 3$ gives $x = \ln 3 \approx 1.099$.

> If the quadratic gave a negative root, that root would be rejected because e^x is always positive.

EXAMPLE **3.12** A LOGARITHMIC EQUATION WITH A CHECK

Solve $\log_2(x-1)+\log_2(x+3)=3$.

SOLUTION

Domain: $x-1>0$ and $x+3>0$, so $x>1$. Combine the logs:

$$\log_2[(x-1)(x+3)]=3 \implies (x-1)(x+3)=2^3=8.$$

Expand: $x^2+2x-3=8$, so $x^2+2x-11=0$. The quadratic formula gives

$$x=\frac{-2\pm\sqrt{4+44}}{2}=-1\pm 2\sqrt{3}.$$

The two values are approximately 2.464 and -4.464. Only $x=-1+2\sqrt{3}\approx 2.46$ satisfies $x>1$; the other is rejected.

> The rejected value is not a rounding error: it makes both original arguments negative, so the original equation is undefined there.

MISCONCEPTION

"$\log(x-1)+\log(x+3)=\log[(x-1)+(x+3)]$." The sum of logarithms corresponds to a product of arguments, not a sum. Also, after solving, students often forget to check the domain and accept an extraneous root. Both errors are avoidable by writing the domain restriction *before* combining the logarithms.

GUIDED PRACTICE

1. Solve $2^x=50$. (5.644.)
2. Solve $4^x=8^{x-1}$. ($2^{2x}=2^{3x-3}$, $x=3$.)
3. Solve $e^{2x}-4e^x=0$. ($e^x=4$, $x=\ln 4$.)
4. Solve $\log x+\log(x-3)=1$. ($x=5$; reject -2.)

EXERCISE 3D

EXPONENTIAL AND LOGARITHMIC EQUATIONS

FLUENCY

1 Solve by writing both sides as powers of the same base.
   a. $2^x=64$
   b. $3^{x+1}=81$
   c. $5^{2x}=125$
   d. $9^x=27$

2 Solve to 3 significant figures using logarithms.
   a. $2^x=30$
   b. $5^x=100$
   c. $7^{x-2}=50$
   d. $1.08^x=2$

3 Solve each equation.
   a. $e^x=12$
   b. $e^{2x}=5$
   c. $e^{-x}=0.4$
   d. $10^{x+1}=250$

4 Solve the logarithmic equations.
   a. $\log_2 x=4$
   b. $\log_3(x+2)=2$
   c. $\ln x=1.5$
   d. $\log_5(2x-1)=2$

5 Solve $3^{x+1}=4^x$, giving the exact answer and a decimal approximation.

6 Solve $2^{2x}-6\cdot 2^x+8=0$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student solves $\log_2 x+\log_2(x-2)=3$ and obtains $x=4$ and $x=-2$, accepting both. Explain which solution is invalid and why.

8 Explain why taking logarithms of both sides of an equation is a valid operation, but squaring both sides is not always reversible.

9 Solve $e^{2x}-7e^x+10=0$ and explain why both roots are admissible.

10 Solve $\log(x+1)-\log(x-1)=1$ and state the domain before solving.

EXERCISE 3D · CONTINUED EXPONENTIAL AND LOGARITHMIC EQUATIONS

11 Find the exact value of x for which $2^x = 3^{x-1}$.

12 Explain why an equation $a^x = b$ with $a > 0$, $a \neq 1$, $b > 0$ always has exactly one solution.

APPLICATION AND EXTENSION

13 An investment of 5000 grows at 4.5% per year. Find the number of whole years required for it to exceed 8000.

14 A population model is $P = 1200e^{0.028t}$. Find the time for the population to reach 2000, and the time to double.

15 **Extension.** Solve $4^x - 3 \cdot 2^{x+1} + 8 = 0$ by writing $4^x = (2^x)^2$ and $2^{x+1} = 2 \cdot 2^x$.

APPLICATION

3.5 Modelling with logarithms

Many quantities span such a wide range that a linear scale is unreadable. Sound intensity from a whisper to a jet engine varies by a factor of 10^{12}; hydrogen ion concentration varies by factors of millions. A **logarithmic scale** compresses these ranges by turning equal *ratios* into equal *distances*. The decibel, pH and earthquake magnitude scales are all logarithmic.

Logarithms also turn an exponential model into a straight line. If $y = ab^x$, then $\log y = \log a + x \log b$. A graph of $\log y$ against x is therefore linear, with gradient $\log b$ and intercept $\log a$. This is how an exponential relationship can be tested from data: plot the logarithms and look for a straight line.

KEY IDEA · LOGARITHMIC SCALES AND LINEARISATION

On a logarithmic scale, a fixed distance represents a fixed *multiplicative* factor. For $y = ab^x$, $\log y = \log a + x \log b$: the graph of $\log y$ against x is a straight line with gradient $\log b$. If the gradient is positive, the data grow exponentially; if negative, they decay.

EXAMPLE 3.13 DECIBELS AND PH

(a) Find the loudness in decibels of a sound of intensity $3.2 \times 10^{-6}\ \text{W/m}^2$. (b) Find the hydrogen ion concentration of a solution of pH 4.7.

SOLUTION

(a) $L = 10 \log_{10}\left(\dfrac{3.2 \times 10^{-6}}{10^{-12}}\right) = 10 \log_{10}(3.2 \times 10^{6}) = 10(6.505) = 65.05 \approx 65\ \text{dB}.$

(b) $\text{pH} = -\log_{10}[\text{H}^+]$, so $\log_{10}[\text{H}^+] = -4.7$ and

$$[\text{H}^+] = 10^{-4.7} = 2.0 \times 10^{-5}\ \text{mol/L}\ (2\ \text{s.f.}).$$

> A change of 1 pH unit is a factor of 10 in concentration, not an addition. A solution of pH 3 is ten times more acidic than one of pH 4.

EXAMPLE 3.14 TESTING AN EXPONENTIAL MODEL

The table shows a quantity y at equal intervals of x. Show that an exponential model is reasonable, and estimate a and b in $y = ab^x$.

SOLUTION

The table is:

x	0	1	2	3	4
y	3	4.5	6.8	10.2	15.3

The ratios are $4.5/3 = 1.50$, $6.8/4.5 = 1.51$, $10.2/6.8 = 1.50$, $15.3/10.2 = 1.50$. A constant ratio indicates exponential growth with $b \approx 1.50$. The initial value is $a = 3$. Hence $y \approx 3(1.50)^x$.

Taking logs gives $\log y = \log 3 + x \log 1.5$, a straight line with intercept 0.477 and gradient 0.176. Plotting the five log values would show that they lie close to that line; the small deviations are consistent with rounding in the data.

> A constant ratio is the table signature of an exponential; a constant difference would indicate a linear relationship.

EXAMPLE 3.15 EARTHQUAKE MAGNITUDES

The magnitude scale is $M = \log_{10}\left(\dfrac{A}{A_0}\right)$. How many times greater is the amplitude of a magnitude 7.2 earthquake than that of a magnitude 5.2 earthquake?

SOLUTION

Let A_1 and A_2 be the amplitudes. Then

$$M_1 - M_2 = \log_{10}\left(\frac{A_1}{A_0}\right) - \log_{10}\left(\frac{A_2}{A_0}\right) = \log_{10}\left(\frac{A_1}{A_2}\right).$$

So $\log_{10}\left(\dfrac{A_1}{A_2}\right) = 2.0$, giving $\dfrac{A_1}{A_2} = 10^2 = 100$. The amplitude is 100 times greater.

> The energy released scales differently from the amplitude; the amplitude ratio is what the magnitude difference of 2 measures directly. Do not claim the energy is 100 times greater without the appropriate energy relation.

MISCONCEPTION

"A magnitude 6 earthquake is twice as strong as a magnitude 3." The scale is logarithmic: a difference of 3 in magnitude corresponds to a factor of $10^3 = 1000$ in amplitude. Likewise, "90 dB is 30% louder than 60 dB" is wrong: the intensity ratio is $10^{(90-60)/10} = 10^3 = 1000$.

GUIDED PRACTICE

1. Find the pH of a solution with $[\mathrm{H}^+] = 10^{-3.2}$. (3.2.)
2. How many times more intense is 80 dB than 50 dB? ($10^3 = 1000$.)
3. For data with a constant ratio of 0.8 per step, is the model growth or decay? (Decay, 20% per step.)
4. In $\log y = \log a + x \log b$, what does the gradient represent? ($\log b$.)

EXERCISE 3E

LOGARITHMIC SCALES AND EXPONENTIAL DATA

FLUENCY

1 Find the decibel level for each intensity.
   a. $10^{-9}\ \text{W/m}^2$
   b. $10^{-4}\ \text{W/m}^2$
   c. $2 \times 10^{-6}\ \text{W/m}^2$

2 Find the hydrogen ion concentration for each pH.
   a. pH 2
   b. pH 7.4
   c. pH 11.5

3 How many times greater is the amplitude of a magnitude 8 earthquake than a magnitude 5?

4 For each table, state whether the relationship is linear or exponential, and give the constant difference or ratio.
   a. x: 0,1,2,3; y: 10,13,16,19
   b. x: 0,1,2,3; y: 10,15,22.5,33.75

5 For $y = 5(1.25)^x$, write $\log y$ in terms of x. State the gradient and intercept.

6 An exponential model passes through $(0, 8)$ and $(3, 27)$. Find a and b.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says a sound of 100 dB is "twice as loud" as 50 dB. Correct the statement and explain the scale.

8 Explain why a logarithmic scale cannot include zero on the axis.

9 The graph of $\log_{10} y$ against x is a straight line with gradient -0.3. Describe the original relationship and find the percentage change per unit increase in x.

10 Data for a quantity are plotted on semilog paper and lie on a straight line. What does this tell you about the data? What would a curve tell you?

11 Explain why the pH scale uses a negative logarithm rather than a positive one.

12 A student claims that since $\log 1000 = 3$, a quantity that grows by a factor of 1000 has "grown by 3". Explain what the 3 measures and what it does not.

APPLICATION AND EXTENSION

13 A population of 50 rabbits grows to 320 in 6 years. Assuming exponential growth, find the annual factor and the continuous rate.

14 The table gives the activity of a radioactive sample.

t (h)	0	2	4	6	8
A (Bq)	800	566	400	283	200

Show that the data are exponential, estimate the half-life, and write a model.

15 **Extension.** Show that if $y = ab^x$, then a graph of $\log_{10} y$ against x has gradient $\log_{10} b$. Use this to explain how a straight-line graph can distinguish exponential growth from quadratic growth.

PATTERN AND GENERALISATION

3.6 Investigation: how many digits?

Logarithms turn a question about the size of a number into a question about its exponent. In this investigation you will use that idea to count digits, predict leading digits, and explain a surprising pattern in the first digits of powers.

INVESTIGATION 3 · THE FIRST DIGITS OF POWERS

Counting without multiplying

1. Explain why a positive integer N with $k+1$ digits satisfies $10^k \le N < 10^{k+1}$, and hence why the number of digits is $\lfloor \log_{10} N \rfloor + 1$.
2. Use the rule to find the number of digits of 2^{10}, 2^{50} and 2^{100}. Check the first two by direct calculation.
3. Find the number of digits of 3^{200} and 7^{100}.
4. The leading digit of N is determined by the fractional part of $\log_{10} N$. For 2^n, compute the leading digit for $n = 1, 2, \ldots, 20$ and record the results in a table.
5. Count how often each leading digit 1–9 occurs in your table. Is the distribution uniform? Extend to $n = 1, \ldots, 60$ if you can.
6. Explain why 2^n has leading digit 1 exactly when the fractional part of $n \log_{10} 2$ lies between 0 and $\log_{10} 2$. Use this to predict the proportion of powers of 2 that begin with 1.
7. Generalise: describe how logarithms convert a multiplicative question (how large?) into an additive one (how many powers of 10?).

Generalise. For a positive number N, the integer part of $\log_{10} N$ counts the complete powers of 10 below N, and the fractional part fixes the leading digits. Write a rule for the number of digits of a^n and explain why doubling the exponent does not simply double the number of digits.

MODELLING · CRITERION D

Cooling, but how fast?

Return to the coffee model $T(t) = 20 + 75(0.88)^t$.

1. **Solve.** Find algebraically the time at which $T = 60$ and the time at which $T = 30$. Compare with your table estimates.
2. **Interpret.** Rewrite the model as $T - 20 = 75(0.88)^t$. What does $T - 20$ represent, and why is it the quantity that decays exponentially?
3. **Compare.** Find the half-life of the temperature difference. Does it depend on the starting temperature?
4. **Evaluate.** State two assumptions of the model and explain how each would affect the predicted drinking window in a real kitchen.

CUMULATIVE

3.7 Mixed practice

Attempt all questions without notes. Decide first whether the unknown is a base, an exponent or a logarithm.

MIXED PRACTICE 3 EXPONENTIALS, LOGARITHMS AND MODELS

FLUENCY AND UNDERSTANDING

1. State the growth factor and percentage rate for $y = 60(1.045)^x$ and $y = 90(0.93)^x$.
2. Evaluate $4(3)^2$, 2^{-4}, $e^{0.2}$ and $\ln 5$.
3. Write $5^3 = 125$ in logarithmic form and $\log_4 64 = 3$ in exponential form.
4. Simplify $\log_2 12 + \log_2 \left(\frac{4}{3}\right)$.
5. Solve $2^{3x-1} = 32$.
6. Solve $7^x = 250$ to 3 significant figures.

MIXED PRACTICE 3 · CONTINUED

EXPONENTIALS, LOGARITHMS AND MODELS

7 Solve $\ln(2x - 1) = 2$.

8 Solve $e^{2x} - 3e^{x} - 4 = 0$.

9 A quantity halves every 12 years. Write a model and find the continuous decay rate.

10 Find the pH of a solution with $[\mathrm{H}^+] = 4.0 \times 10^{-6}$.

11 How many digits does 6^{80} have?

12 Explain why $y = 2^{x}$ has no x-intercept.

REASONING AND APPLICATION

13 An investment doubles in 9 years. Find the annual percentage rate, assuming annual compounding.

14 A car worth 28 000 loses 20% of its value in the first year and 10% each year afterwards. Find its value after 5 years.

15 Solve $\log_3(x + 4) + \log_3(x - 2) = 2$, checking the domain.

16 The intensity of a sound is doubled. By how many decibels does the level increase? Justify using the definition.

17 The population of a town was 45 000 in 2015 and 52 000 in 2025. Assuming exponential growth, find the annual rate and predict the population in 2035.

18 A radioactive sample has a half-life of 30 years. Find the time for its activity to fall to 20% of its initial value.

19 Show that $\log_2 3 \cdot \log_3 8 = 3$.

20 Solve $3^{2x} - 10 \cdot 3^{x} + 9 = 0$.

21 **Extension.** Prove that if $\log_a x = \log_b x$ for all $x > 0$, then $a = b$.

22 **Extension.** The model $N = N_0 e^{kt}$ has $N = 2N_0$ at $t = 5$. Find k exactly, and the time at which $N = 3N_0$.

CRITERION A · KNOWING AND UNDERSTANDING

35 minutes · 20 marks

Exponentials and logarithms

Instructions. Answer all questions. Show sufficient working. Give non-exact answers to 3 significant figures.

1 For $y = 250(0.86)^{x}$, state the initial value, the decay factor and the percentage decrease per unit. [3]

2 Sketch $y = 2^{x}$ and $y = \log_2 x$ on the same axes, stating the asymptote and two key points for each. [4]

3 Evaluate $\log_3 81$, $\log_5\left(\frac{1}{125}\right)$ and $\log_7 1$. [3]

4 Solve $5^{2x+1} = 100$. [3]

5 Solve $\log_2(x - 3) + \log_2 x = 2$, checking for extraneous roots. [4]

6 A population model is $P = 800e^{0.035t}$. Find the initial population, the doubling time and the population at $t = 12$. [3]

Evidence: correct parameters, accurate graphs with asymptotes, exact logarithmic evaluations, a checked solution, and correct use of e.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Powers, digits and leading digits

Instructions. Work independently. Record results in tables and justify every generalisation.

1. Complete a table of $\log_{10}(2^n)$, the number of digits, and the leading digit for $n = 1, \dots, 12$. [6]
2. Describe how the number of digits changes as n increases by 1. Explain why it sometimes stays the same. [4]
3. Find a formula for the number of digits of 2^n, and use it to find the number of digits of 2^{200}. [5]
4. Investigate the leading digits for $n = 1, \dots, 40$: which digit occurs most often? Is the pattern uniform? [5]
5. Explain, using the fractional part of $n \log_{10} 2$, why a leading digit of 1 is more likely than a leading digit of 9. [4]

Evidence: complete tables, a justified digit formula, a described and verified distribution, and a correct fractional-part explanation.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Why logarithms solve exponential equations

Instructions. Write a 300–400 word explanation for a student who can solve $2^x = 32$ but not $2^x = 30$. Your explanation must: (i) define a logarithm as an inverse operation; (ii) show the steps that isolate x; (iii) explain why the same method works for any positive base; (iv) include one worked example and one common error. Use correct notation and a clear structure.

Evidence: a correct definition, a complete solution, a general justification, and a clearly diagnosed error.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

A cooling drink and a warming room

Instructions. A café models the temperature of a drink by $T(t) = 21 + 74(0.91)^t$, where t is in minutes and T in $^\circ\mathrm{C}$. The drink is served at $95^\circ\mathrm{C}$; customers prefer it between $55^\circ\mathrm{C}$ and $35^\circ\mathrm{C}$.

1. State the room temperature and the initial temperature according to the model. [2]
2. Find the time at which the drink reaches $55^\circ\mathrm{C}$ and the time at which it reaches $35^\circ\mathrm{C}$. [6]
3. Find the half-life of the temperature difference $T - 21$, and interpret it in context. [5]
4. State two assumptions of the model and explain how each would change the predicted window. [5]
5. The café wants to extend the window by 5 minutes. Suggest one practical change and evaluate whether the model supports it. [6]

Evidence: correct temperatures, both solved times, a correct half-life, explicit assumptions, and an evaluated recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- An exponential function $y = ab^x$ multiplies by b for each unit increase in x. The coefficient a is the value at $x = 0$.
- Growth factors above 1 produce growth; factors between 0 and 1 produce decay. The percentage rate is $b - 1$, expressed as a percentage.
- The number $e \approx 2.71828$ arises as the limit of $\left(1 + \frac{1}{n}\right)^n$. Continuous change is modelled by $y = Ae^{kt}$.
- A logarithm is an exponent: $\log_b x = y$ means $b^y = x$. Its domain is $x > 0$.
- The laws of logarithms convert products into sums, quotients into differences and powers into multiples. Change of base allows any calculator base.
- Exponential equations are solved by equating exponents when bases match, or by taking logarithms when they do not.
- Logarithmic equations must be checked against the domain; candidates that make an argument non-positive are rejected.
- Logarithmic scales compare ratios, not differences; $\log y = \log a + x \log b$ linearises an exponential model.

$$b^y = x \iff \log_b x = y, \qquad \log_b(xy) = \log_b x + \log_b y, \qquad \log_b(x^n) = n \log_b x, \qquad T_{1/2} = \frac{\ln 2}{-k}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can identify an exponential relationship from a table, graph or rule.
- I can state the growth factor and percentage rate.
- I can construct an exponential model from a description.
- I can find doubling times and half-lives.
- I can use e and continuous growth models.
- I can convert between exponential and logarithmic form.
- I can apply the product, quotient and power laws.
- I can use change of base.
- I can solve $a^x = b$ using logarithms.
- I can solve a hidden quadratic in e^x or 2^x.
- I can solve a logarithmic equation and reject extraneous roots.
- I can interpret pH, decibel and magnitude scales.
- I can use a log–linear graph to test an exponential model.
- I can evaluate the assumptions of an exponential model.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which representation — table, graph, formula or logarithm — helped you most in this chapter? Give a question where it made the solution shorter.
2. Describe one error you made and the reasoning that corrected it. What will you check first next time?
3. The statement of inquiry calls logarithms the “inverse language” of exponential change. Use an example to explain what that means.
4. Why must the domain be checked when solving a logarithmic equation? Give an example of a rejected root.
5. What question about exponential or logarithmic change do you still want to answer?

PATTERN, RULE, TOTAL

Sequences and Series

04

FORM · PATTERNS · SIMPLIFICATION

From a list of numbers to a rule you can trust — and from a rule to a total you can predict.

STATEMENT OF INQUIRY

A sequence is form made visible: one rule generates every term, and the sum of those terms simplifies to a formula whose behaviour — finite, unbounded or convergent — is decided by a single number.

KEY CONCEPT

Form

RELATED CONCEPTS

Patterns · Generalisation · Simplification

GLOBAL CONTEXT

Fairness and development — saving, loans and the long-term decisions that shape opportunity

FACTUAL

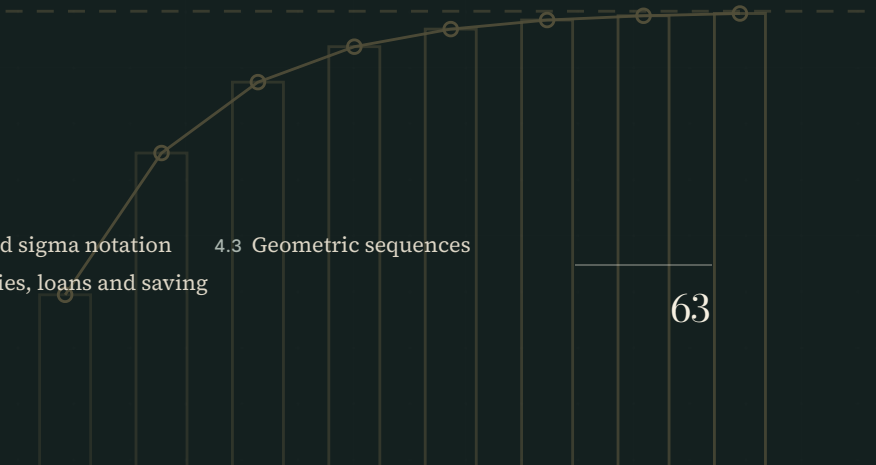
What makes a sequence arithmetic or geometric, and how is each rule written?

CONCEPTUAL

How can adding infinitely many positive numbers produce a finite total?

DEBATABLE

When a loan is advertised at an attractive rate, whose interests does the advertisement serve?



IN THIS CHAPTER

4.1 Arithmetic sequences
4.2 Arithmetic series and sigma notation
4.3 Geometric sequences
4.4 Geometric series and convergence
4.5 Annuities, loans and saving
4.6 Investigation: infinite sums

OPENING PROBLEM · THE THIRTY-DAY CONTRACT

Two offers and a signature

A small firm offers you a thirty-day contract. Option A pays a flat 1000 units each day. Option B pays 1 unit on the first day, 2 on the second, 4 on the third, doubling every day. A third option, C, pays 10 units on the first day and then 10 more than the previous day, every day.

Option B looks like an insult on day one and Option A looks like a safe salary. **Which contract would you sign — and which quantity, a single day's pay or the thirty-day total, actually decides the answer?**

1. Write Option B's daily pay as a sequence and list its first ten terms. Add the ten terms by hand.
2. Write a rule for the daily pay under each option. Use the rules to find the thirty-day total for A and for C.
3. Use a table or a spreadsheet to find the first day on which B's daily pay exceeds A's, and the first day on which B's running total exceeds 30 000.
4. Choose an option and justify the choice. Explain why the comparison cannot be made from the first day's pay alone.

A sequence is an ordered list produced by a rule; a series is the total of that list. The two ideas separate the question "how much next?" from the question "how much altogether?", and both answers can be written in closed form. This chapter builds those formulas, then uses them to decide what a savings plan, a depreciating machine and a loan really cost.

CH 04

4.0 Learning intentions

ORIENTATION

- Generate terms of arithmetic and geometric sequences from recursive and closed forms, and move fluently between the two.
- Recover a and d, or a and r, from information about particular terms, and decide whether a given value belongs to a sequence.
- Read and write sigma notation, evaluate an arithmetic series and solve for the number of terms, including cases that lead to a quadratic equation.
- Decide whether a geometric series converges, find its sum to infinity, and express a recurring decimal as a fraction.
- Model compound interest, depreciation, regular saving and loan repayment with recurrence relations and geometric sums, and evaluate the assumptions behind the model.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

write the recursive and closed forms of an arithmetic or geometric sequence; use $u_n = a + (n-1)d$ and $u_n = ar^{n-1}$ confidently, including a negative or fractional ratio; find a and d from two terms and explain why an even index gap gives two values of r; count terms and decide whether a value belongs; use $\sum$ notation and count the terms of a sum; prove and apply $S_n = \frac{n}{2}(a+l)$ and $S_n = \frac{a(1-r^n)}{1-r}$; solve $S_n = k$ for n, rejecting impossible roots; state and justify the condition $|r| < 1$ and evaluate $S_\infty = \frac{a}{1-r}$; convert a recurring decimal to a fraction; build a recurrence for a saving or loan balance and solve it with a geometric series; compare two financial offers on a common basis and criticise the assumptions of the comparison.

ATL SKILLS · THINKING, TRANSFER AND COMMUNICATION

Critical thinking: classify the sequence before choosing a formula — is the step a difference or a ratio? A numerical check on the second and third terms settles it. **Transfer:** the pairing argument that sums $1 + 2 + \cdots + 100$ sums the rows of a stadium; the geometric sum that prices an annuity describes a bouncing ball. **Communication:** a recurrence is complete only with its starting value, and a model is unfinished until its assumptions are stated.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 4.1.

1. Write the next three terms of $5,\ 9,\ 13,\ 17,\ \ldots$ and state the rule you used.
2. Solve $3n - 7 = 41$ and $2^n = 64$.
3. Evaluate 2^5, $\left(\frac{1}{2}\right)^3$, 1.05^2 and 5^0.
4. Factorise $x^2 + 3x - 40$ and hence solve $x^2 + 3x - 40 = 0$.
5. Write 0.75 as a fraction in lowest terms, and $\frac{3}{8}$ as a decimal.
6. Make l the subject of $S = \frac{n}{2}(a + l)$.

REFRESHER · INDICES, QUADRATICS AND LOGARITHMS

An index counts repeated multiplication: $2^5 = 32$, $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, and $a^0 = 1$ for any non-zero a. The laws $a^m a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$ and $(a^m)^n = a^{mn}$ are used constantly with geometric sequences. A growth of 5 per cent is multiplication by 1.05; a fall of 12 per cent is multiplication by 0.88, not by -0.12.

A quadratic such as $2n^2 + n - 465 = 0$ is solved by factorising when the numbers allow it, and otherwise by the formula $n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. In this chapter n must be a positive integer, so a negative or fractional root is rejected even when the algebra is correct.

Logarithms, developed in Chapter 3, undo powers: $\log_b x = y$ means $b^y = x$, and $\log_b(x^k) = k \log_b x$. To solve $1.05^n = 1.5$, take logs of both sides: $n = \frac{\log 1.5}{\log 1.05} \approx 8.31$. Since n counts years, the answer is the next whole number, 9.

CONCEPT AND METHOD

4.1 Arithmetic sequences

A sequence is an ordered list of numbers, and each number in it is a term. The position of a term is its index: the term in position n is written u_n, with u_1 the first term. Order matters. The sequences $2,\ 5,\ 8$ and $8,\ 5,\ 2$ contain the same numbers and are different sequences, because the rule that produces them reads in a direction.

A sequence can be described in three ways. A **list** shows the terms. A **recursive** rule says how to get the next term from the previous one, and must include a starting value. A **closed form** gives u_n directly as a formula in n, so any term can be computed without computing the ones before it. The closed form is usually the more powerful of the two, but the recursive form often shows the structure of a problem more honestly.

An arithmetic sequence changes by the same amount at every step. The amount is the common difference $d = u_{n+1} - u_n$. Adding d repeatedly is the same as multiplying d by the number of steps, so

$$u_n = a + (n - 1)d,$$

where $a = u_1$. Expanded, this is $u_n = dn + (a - d)$, a linear expression in n. The graph of u_n against n is therefore a set of points lying on a straight line of gradient d: an arithmetic sequence is the discrete version of constant growth.

DEFINITIONS

Sequence, arithmetic sequence, common difference

A **sequence** is a function whose domain is a set of consecutive positive integers; u_n is the nth term and n is its index. A sequence is **arithmetic** when consecutive terms differ by a constant:

$$u_{n+1} - u_n = d \quad \text{for every } n.$$

Its recursive form is $u_1 = a,\ u_{n+1} = u_n + d$; its closed form is $u_n = a + (n-1)d$.

The **common difference** may be positive, negative or zero. If $d > 0$ the sequence is increasing; if $d < 0$ it is decreasing; if $d = 0$ every term equals a. Three numbers $a,\ b,\ c$ are consecutive terms of an arithmetic sequence exactly when $b = \frac{a+c}{2}$, the **arithmetic mean** of a and c.

EXAMPLE **4.1** FROM A LIST TO A RULE, AND BACK AGAIN

The sequence begins $7,\ 11,\ 15,\ 19,\ \ldots$ (a) Find d and write the recursive and closed forms. (b) Find u_{20}. (c) Which term equals 143? (d) Is 200 a term of the sequence?

SOLUTION

(a) $11 - 7 = 4$ and $15 - 11 = 4$, so $d = 4$. Recursive: $u_1 = 7,\ u_{n+1} = u_n + 4$. Closed: $u_n = 7 + 4(n-1) = 4n + 3$.

(b) $u_{20} = 4(20) + 3 = 83$. The recursive route would need nineteen additions; the closed form needs one substitution. That is the practical reason for finding it.

(c) Solve $4n + 3 = 143$: $4n = 140$, so $n = 35$. The 35th term is 143.

(d) Solve $4n + 3 = 200$: $n = 49.25$. Since n must be a positive integer, 200 is not a term. The value lies between $u_{49} = 199$ and $u_{50} = 203$; a sequence with step 4 can never land on 200 because every term is one more than a multiple of 4.

EXAMPLE **4.2** RECOVERING THE SEQUENCE FROM TWO TERMS

An arithmetic sequence has $u_5 = 23$ and $u_{12} = 58$. (a) Find a and d. (b) How many terms are there in the sequence $3,\ 8,\ 13,\ \ldots,\ 98$?

SOLUTION

(a) From $u_{12} = u_5 + 7d$, $58 = 23 + 7d$, so $7d = 35$ and $d = 5$. Then $u_5 = a + 4d$ gives $23 = a + 20$, so $a = 3$. The closed form is $u_n = 3 + 5(n-1) = 5n - 2$.

(b) The sequence has $a = 3$ and $d = 5$, so $u_n = 5n - 2$. Setting $5n - 2 = 98$ gives $n = 20$. There are 20 terms. Equivalently,

$$n = \frac{l - a}{d} + 1 = \frac{98 - 3}{5} + 1 = 20.$$

The "$+1$" counts the first term, which the division alone omits.

EXAMPLE **4.3** A STADIUM STAND

A stand has 24 seats in the first row and 2 more seats in each row behind it. (a) How many seats are in row 30? (b) Which row has 70 seats? (c) A safety rule allows at most 75 seats in any row. How many rows can the stand have? (d) Can any row contain exactly 81 seats?

SOLUTION

Model the rows with $a = 24$ and $d = 2$, so $u_n = 24 + 2(n-1) = 2n + 22$.

(a) $u_{30} = 2(30) + 22 = 82$ seats.

(b) $2n + 22 = 70$ gives $n = 24$. Row 24 has 70 seats.

(c) Require $2n + 22 \leq 75$, so $n \leq 26.5$. Row 26 has 74 seats and row 27 would have 76, so at most 26 rows are allowed.

(d) $2n + 22 = 81$ gives $n = 29.5$, not an integer. Every row has an even number of seats, so 81 is impossible. The model itself, not the arithmetic, rules it out.

> The inequality in (c) answers a design question; the equation in (d) answers a membership question. Read which one the context demands.

MISCONCEPTION

"If the first term is 7 and the difference is 4, then $u_n = 7 + 4n$." Substituting $n = 1$ gives 11, not 7. The correct form is $7 + 4(n-1) = 4n + 3$: there are $n - 1$ steps from the first term to the nth, because the first term is the starting point. A second error hides nearby: d is found by subtracting consecutive terms, never by dividing the last term by the number of terms.

GUIDED PRACTICE

1. For $5,\ 12,\ 19,\ \ldots$, state d and find u_{10}. (Answer: $d = 7$; $u_{10} = 68$.)
2. Write the closed form of $30,\ 26,\ 22,\ \ldots$. (Answer: $u_n = 34 - 4n$.)
3. How many terms are there in $4,\ 11,\ 18,\ \ldots,\ 95$? (Answer: 14.)
4. An arithmetic sequence has $u_3 = 9$ and $u_7 = 25$. Find a and d. (Answer: $d = 4$, $a = 1$.)

EXERCISE 4A ARITHMETIC SEQUENCES

FLUENCY

1 State the common difference and write the next two terms.

a. $4,\ 11,\ 18,\ \ldots$

b. $30,\ 26,\ 22,\ \ldots$

c. $2.5,\ 4.0,\ 5.5,\ \ldots$

d. $-8,\ -5,\ -2,\ \ldots$

2 Write a closed form for u_n.

a. $3,\ 7,\ 11,\ \ldots$

b. $20,\ 17,\ 14,\ \ldots$

c. $-4,\ 1,\ 6,\ \ldots$

d. $9,\ 9,\ 9,\ \ldots$

3 For $u_n = 6n - 5$, find u_1, u_{10} and u_{50}. Which term equals 61?

4 The sequence $100,\ 93,\ 86,\ \ldots$ continues. Find u_{15} and the first negative term.

5 Find a and d, then write u_n.

a. $u_1 = 7,\ u_8 = 35$

b. $u_4 = 11,\ u_9 = 31$

c. $u_3 = 2,\ u_{10} = -19$

6 Count the terms.

a. $5,\ 9,\ 13,\ \ldots,\ 85$

b. $12,\ 19,\ 26,\ \ldots,\ 96$

c. $60,\ 55,\ 50,\ \ldots,\ 5$

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes that the sequence $3,\ 7,\ 11,\ \ldots$ has $u_n = 3 + 4n$. Show by substituting $n = 1$ that this is wrong, and write the correct rule.

8 Show that 175 is not a term of $8,\ 15,\ 22,\ \ldots$. Find the largest term of this sequence that is less than 175.

EXERCISE 4A · CONTINUED ARITHMETIC SEQUENCES

9 The sequence $u_n = 3n - 2$ is arithmetic. State d. Explain why $v_n = 2^n$ is not arithmetic by computing $v_2 - v_1$ and $v_3 - v_2$.

10 An arithmetic sequence has $u_3 = 14$ and $u_{11} = 46$. Find a, d and u_{20}.

11 Explain why an arithmetic sequence with $d \neq 0$ can never contain the same value twice. What changes when $d = 0$?

12 For $u_n = 5 - 2n$, decide whether the sequence is increasing or decreasing, and find the largest n for which $u_n > 0$. Justify the inequality step.

APPLICATION AND EXTENSION

13 A cinema has 18 seats in the first row and 3 more in each row behind, for 20 rows. How many seats are in the last row? Which row has 54 seats? Could a row contain 100 seats?

14 A stack of pipes has 20 pipes in the bottom row and one fewer in each row above. How many pipes are in the twelfth row? Which row has 8 pipes? If every row must contain at least 4 pipes, what is the greatest possible number of rows, and how many pipes does that stack hold?

15 **Extension.** An arithmetic sequence has $u_1 = 3$ and $u_9 = 3u_4$. Find d, and prove that for any arithmetic sequence $u_m + u_n = u_p + u_q$ whenever $m + n = p + q$.

CONCEPT, NOTATION AND METHOD

4.2 Arithmetic series and sigma notation

Adding the terms of a sequence produces a series. The sum of the first n terms is written S_n, so $S_n = u_1 + u_2 + \cdots + u_n$. An arithmetic series is the sum of an arithmetic sequence. The distinction between a term and a sum is the distinction between one row of a stadium and the whole stand.

No formula here is magic. The sum of an arithmetic series is found by writing the sum forwards and backwards and adding the two lines:

$$S_n = a + (a + d) + (a + 2d) + \cdots + l,$$
$$S_n = l + (l - d) + (l - 2d) + \cdots + a.$$

Each of the n columns adds to $a + l$, because moving one step forward in the first line is exactly balanced by moving one step back in the second. Adding the columns gives $2S_n = n(a + l)$, and halving gives the formula. This is the pairing argument attributed to Gauss, who is said to have summed $1 + 2 + \cdots + 100$ as fifty pairs of 101.

FORMULAS AND NOTATION

The two forms of the arithmetic sum

For an arithmetic sequence with first term a, common difference d, last term l and n terms,

$$S_n = \frac{n}{2}(a + l) = \frac{n}{2}\left(2a + (n - 1)d\right).$$

The first form is convenient when the last term is known, the second when n is known. The sum $u_1 + u_2 + \cdots + u_n$ is written $\sum_{k=1}^{n} u_k$; the index k is a dummy, so $\sum_{k=1}^{n} u_k$ and $\sum_{r=1}^{n} u_r$ mean the same. The number of terms is $(\text{upper limit}) - (\text{lower limit}) + 1$.

Shifting the index does not change the sum: replacing k by $k + 1$ lowers every starting value by one, so $\sum_{k=1}^{n} (k + 1)^2$ equals $\sum_{j=2}^{n+1} j^2$.

EXAMPLE **4.4** SUMMING AN ARITHMETIC SERIES

Find $7 + 11 + 15 + \cdots + 99$.

SOLUTION

The sequence has $a = 7$ and $d = 4$. First find how many terms there are: $7 + 4(n-1) = 99$ gives $4n + 3 = 99$, so $n = 24$. Then

$$S_{24} = \frac{24}{2}(7 + 99) = 12 \times 106 = 1272.$$

Check by the second form: $S_{24} = \frac{24}{2}\left(2(7) + 23(4)\right) = 12(14 + 92) = 12 \times 106 = 1272$. The two forms agree because they are the same statement written twice.

> Always count the terms before summing. Using $n = 99$ here would be the single most common error in the topic.

EXAMPLE **4.5** READING AND REWRITING SIGMA NOTATION

(a) Evaluate $\displaystyle\sum_{k=1}^{20}(3k + 2)$. (b) Evaluate $\displaystyle\sum_{r=3}^{15}(2r - 1)$. (c) Show that $\displaystyle\sum_{k=1}^{n}(4k - 1) = 2n^2 + n$.

SOLUTION

(a) The terms run from $3(1) + 2 = 5$ to $3(20) + 2 = 62$, and there are 20 of them. So $S_{20} = \frac{20}{2}(5 + 62) = 10 \times 67 = 670$.

(b) The limits are 3 and 15, so the number of terms is $15 - 3 + 1 = 13$, not 15. The first term is $2(3) - 1 = 5$ and the last is $2(15) - 1 = 29$. Hence $S_{13} = \frac{13}{2}(5 + 29) = 13 \times 17 = 221$.

(c) The first term is 3 and the nth is $4n - 1$, so

$$S_n = \frac{n}{2}\left(3 + (4n - 1)\right) = \frac{n}{2}(4n + 2) = n(2n + 1) = 2n^2 + n.$$

Substituting $n = 1$ and $n = 2$ checks the formula: 3 and $3 + 7 = 10$, and $2(1)^2 + 1 = 3$, $2(4) + 2 = 10$.

EXAMPLE **4.6** SOLVING FOR THE NUMBER OF TERMS

(a) The series $3 + 7 + 11 + \cdots$ has sum 465. Find the number of terms. (b) A pile of logs has 24 logs in the bottom row and one fewer in each row above. The pile contains 272 logs. How many rows are there?

SOLUTION

(a) Here $a = 3$, $d = 4$, so $S_n = \frac{n}{2}\left(6 + 4(n-1)\right) = \frac{n}{2}(4n+2) = n(2n+1)$. Setting $n(2n+1) = 465$:

$$2n^2 + n - 465 = 0, \qquad n = \frac{-1 + \sqrt{1 + 3720}}{4} = \frac{-1 + 61}{4} = 15.$$

The negative root is rejected because n counts terms. The sum of the first 15 terms is 465.

(b) With n rows, the top row holds $24 - (n-1) = 25 - n$ logs, so

$$S_n = \frac{n}{2}\left(24 + (25 - n)\right) = \frac{n(49 - n)}{2} = 272.$$

Hence $n^2 - 49n + 544 = 0$. The discriminant is $49^2 - 4(544) = 2401 - 2176 = 225 = 15^2$, so

$$n = \frac{49 \pm 15}{2} = 32 \text{ or } 17.$$

A pile of 32 rows would have a top row of $25 - 32 = -7$ logs, which is impossible, so the pile has 17 rows and the top row holds 8 logs. Check: $\frac{17}{2}(24 + 8) = 17 \times 16 = 272$.

> Both roots satisfy the algebra because the formula happily continues the sequence below zero. Only the context can reject $n = 32$. Always test a root against the meaning of the variables.

MISCONCEPTION

"$\sum_{k=3}^{12} u_k$ has $12 - 3 = 9$ terms, and S_n means the nth term." Both statements confuse a position with a count. From $k = 3$ to $k = 12$ inclusive there are 10 terms, because both endpoints are included; the subtraction counts the gaps, and there is always one more term than gap. And S_n is a sum of n terms, while u_n is a single term: $S_5 = u_1 + u_2 + u_3 + u_4 + u_5$, which is generally much larger than u_5.

GUIDED PRACTICE

1. Evaluate $4 + 9 + 14 + \cdots + 64$. (Answer: $n = 13$, $S = 442$.)
2. How many terms are in $\sum_{j=5}^{25}(3j)$? (Answer: 21.)
3. Find S_{10} for $a = 6$, $d = 3$. (Answer: 195.)
4. A series has $S_n = n^2 + 2n$. Find S_4 and u_4. (Answer: 24; $u_4 = S_4 - S_3 = 24 - 15 = 9$.)

EXERCISE 4B ARITHMETIC SERIES AND SIGMA NOTATION

FLUENCY

1 Evaluate each sum.

a. $2 + 5 + 8 + \cdots + 59$

b. $100 + 95 + 90 + \cdots + 5$

c. $1 + 2 + 3 + \cdots + 100$

d. $12 + 15 + 18 + \cdots + 99$

EXERCISE 4B · CONTINUED ARITHMETIC SERIES AND SIGMA NOTATION

2 Evaluate each sum.

a. $\sum_{k=1}^{10}(2k+3)$

b. $\sum_{k=1}^{12}(5k-1)$

c. $\sum_{r=4}^{20}3r$

d. $\sum_{j=1}^{8}(40-3j)$

3 Write each series in sigma notation.

a. $3+6+9+\cdots+60$

b. $5+9+13+\cdots+41$

c. $1+4+7+\cdots+100$

4 Find n.

a. $\sum_{k=1}^{n}(2k-1)=900$

b. $\sum_{k=1}^{n}(3k+1)=175$

5 For $u_n=4n-3$, find u_1, u_{15} and S_{15}.

6 An arithmetic series has $a=6$, $d=5$ and $n=18$. Find S_{18}.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student calculates $5+8+11+\cdots+47$ as $\frac{47}{2}(5+47)$, saying there are 47 terms. Identify both errors and give the correct sum.

8 Show that $\sum_{k=1}^{n}(4k+1)=n(2n+3)$, and use the result to find the number of terms when the sum equals 275.

9 Explain in your own words why pairing the first and last terms works. Use the pairing to find $1+2+\cdots+60$.

10 The sum of the first n terms of a series is $S_n=n^2+3n$. Find u_1, u_2 and d, and confirm your closed form for u_n by re-summing.

11 A series has $S_n=5n-n^2$. Find u_1, u_2 and u_3. Explain why $S_3=S_2$, and what that says about the third term.

12 In $S_n=\frac{n}{2}(a+l)$, explain why $a+l$ is always even when n is odd for an integer arithmetic sequence. Why does that guarantee an integer sum?

APPLICATION AND EXTENSION

13 A theatre has 26 rows, with 18 seats in the first row and 2 more in each successive row. Find the total number of seats.

14 A savings club asks members to deposit 80 in the first month and 15 more each month for 36 months. Find the total deposited, and the size of the final deposit.

15 **Extension.** The sum of the first n terms of an arithmetic series is $3n^2+2n$. Find the 10th term, the first term and the common difference, and verify that your values reproduce the sum formula.

CONCEPT AND METHOD

4.3 Geometric sequences

A geometric sequence multiplies by the same number at every step instead of adding one. That number is the common ratio r, and it may be found by dividing any term by the one before it. Because repeated multiplication is a power,

$$u_n = ar^{n-1},$$

where $a=u_1$. The recursive form is $u_1=a,\ u_{n+1}=ru_n$.

The ratio is a **multiplier**, and multipliers encode percentage change: a growth of 7 per cent multiplies by 1.07, a fall of 12 per cent multiplies by 0.88. A ratio above 1 gives growth, one between 0 and 1 gives decay towards zero, and a negative ratio gives terms that alternate in sign. Geometric sequences are therefore the natural language of compound interest, depreciation and half-life.

Two terms determine r only up to sign when the index gap is even. Dividing u_m by u_n cancels a and leaves $r^{m-n} = u_m / u_n$. An odd gap gives a unique r; an even gap allows both r and $-r$, and both must be reported unless the context excludes one.

DEFINITIONS

Geometric sequence, common ratio, geometric mean

A sequence is **geometric** when the ratio of consecutive terms is constant:

$$\frac{u_{n+1}}{u_n} = r \quad \text{for every } n, \qquad u_n = a r^{n-1}.$$

Three positive numbers a, b, c are consecutive terms of a geometric sequence exactly when $b^2 = ac$, that is $b = \sqrt{ac}$. The value b is the **geometric mean** of a and c. Unlike the arithmetic mean, it is defined by a product, which is why it describes proportional rather than additive change.

EXAMPLE **4.7** GENERATING AND LOCATING TERMS

For the sequence $3,\ 6,\ 12,\ 24,\ \ldots$ (a) state r and the closed form; (b) find u_{10}; (c) which term equals 768?

SOLUTION

(a) $6 \div 3 = 2$ and $12 \div 6 = 2$, so $r = 2$ and $a = 3$. Hence $u_n = 3 \cdot 2^{n-1}$.

(b) $u_{10} = 3 \cdot 2^{9} = 3 \times 512 = 1536$. Note the exponent: the tenth term uses 2^9, because the first term already contains one factor of 3 and no factor of 2.

(c) Solve $3 \cdot 2^{n-1} = 768$, so $2^{n-1} = 256 = 2^8$ and $n = 9$. The ninth term is 768. Because both sides are powers of 2, no logarithms are needed; matching exponents is enough. A check ($u_9 = 768$, $u_{10} = 1536$) confirms the doubling.

EXAMPLE 4.8 TWO TERMS, ONE SEQUENCE OR TWO?

(a) A geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find a and r, and the first term that exceeds $10\,000$. (b) A different geometric sequence has $u_2 = 6$ and $u_4 = 54$. Explain why these data do not determine a single sequence.

SOLUTION

(a) Divide: $\frac{u_5}{u_2} = \frac{ar^4}{ar} = r^3 = \frac{324}{12} = 27$. Hence $r = 3$, and $u_2 = ar = 12$ gives $a = 4$. The closed form is $u_n = 4 \cdot 3^{n-1}$.

Require $4 \cdot 3^{n-1} > 10\,000$, so $3^{n-1} > 2500$. Taking logarithms uses the law $\log(a^k) = k \log a$:

$$n - 1 > \frac{\log 2500}{\log 3} \approx \frac{3.398}{0.477} \approx 7.12.$$

Since $n - 1$ is an integer, $n - 1 = 8$ and $n = 9$. The ninth term is $4 \cdot 3^8 = 26\,244$; the eighth is 8748, which confirms that 9 is the first index that works.

(b) Dividing gives $r^2 = \frac{54}{6} = 9$, so $r = 3$ or $r = -3$. Since $u_2 = ar = 6$, the two possibilities are $a = 2$ with $r = 3$ and $a = -2$ with $r = -3$:

$$2,\ 6,\ 18,\ 54,\ \ldots \qquad \text{or} \qquad -2,\ 6,\ -18,\ 54,\ \ldots$$

Both fit the two given terms exactly. An even index gap always leaves this ambiguity, because squaring destroys the sign of r. Only extra information — all terms positive, or the terms alternate — selects one case. In both sequences the geometric mean of 6 and 54, namely $\sqrt{324} = 18$, is the middle term.

> Logarithms give the boundary; the context gives the integer. Rounding 7.12 down to 7 would give a term below $10\,000$.

EXAMPLE 4.9 DEPRECIATION AS A GEOMETRIC SEQUENCE

A delivery van costs $24\,000$ and loses 15 per cent of its value each year. (a) Write a model for its value after n years. (b) Find its value after 4 years. (c) When does the value first fall below 8000?

SOLUTION

(a) A fall of 15 per cent leaves 85 per cent, so the multiplier is 0.85. The values form a geometric sequence with $a = 24\,000$ and $r = 0.85$:

$$V_n = 24\,000(0.85)^n.$$

The index starts at 0, because V_0 is the value when the van is new. Keeping n as the number of years since purchase but writing $V_n = 24\,000(0.85)^{n-1}$ would price the new van at $24\,000 \div 0.85 = 28\,235$, which is wrong.

(b) $V_4 = 24\,000(0.85)^4 = 24\,000 \times 0.5220 = 12\,528$ (to the nearest unit).

(c) Require $24\,000(0.85)^n < 8000$, so $0.85^n < \frac{1}{3}$. Taking logarithms:

$$n > \frac{\log(1/3)}{\log 0.85} \approx \frac{-0.4771}{-0.0706} \approx 6.76.$$

Hence $n = 7$, and $V_7 = 24\,000(0.85)^7 \approx 7694$. The value is 9052 after six years, so the seventh year is the first below 8000.

> Dividing two negative logarithms gives a positive result. Check the direction of the inequality: a decay factor below 1 makes the value decrease as n grows.

MISCONCEPTION

"A geometric sequence always grows faster than an arithmetic one, and $u_n = ar^n$." The comparison depends on where you look: $100, 105, 110, \ldots$ (arithmetic, step 5) leads $100, 103, 106.09, \ldots$ (geometric, ratio 1.03) for many terms, and the geometric sequence overtakes only later. Growth is eventual, not immediate. The second error is the index: ar^n makes the first term ar, so the correct closed form is ar^{n-1}. A ratio below 1 also does not make the terms negative; it makes them smaller and positive, approaching zero without reaching it.

GUIDED PRACTICE

1. For $2, 10, 50, \ldots$, state r and find u_6. (Answer: $r = 5$; $u_6 = 6250$.)
2. Write the closed form of $81, 27, 9, \ldots$. (Answer: $u_n = 81\left(\frac{1}{3}\right)^{n-1}$.)
3. A sequence has $u_2 = 10$ and $u_4 = 40$. Find both possible values of r. (Answer: $r = \pm 2$.)
4. A price rises by 4 per cent each year. Write the multiplier and the value of a 250 item after 3 years. (Answer: 1.04; $250(1.04)^3 = 281.22$, or 281 to the nearest unit.)

EXERCISE 4C

GEOMETRIC SEQUENCES

FLUENCY

1 State the common ratio and the next two terms.
a. $2, 10, 50, \ldots$
b. $81, 27, 9, \ldots$
c. $5, -10, 20, \ldots$
d. $0.4, 0.8, 1.6, \ldots$

2 Write a closed form for u_n.
a. $4, 12, 36, \ldots$
b. $64, 32, 16, \ldots$
c. $-3, 6, -12, \ldots$
d. $0.5, 0.05, 0.005, \ldots$

3 For $u_n = 5 \cdot 2^{n-1}$, find u_1, u_6 and u_{10}, and the first term that exceeds 5000.

4 Find a and r.
a. $u_1 = 3,\ u_4 = 24$
b. $u_2 = 10,\ u_5 = 80$
c. $u_3 = 12,\ u_6 = -96$

5 A geometric sequence has $u_3 = 8$ and $u_5 = 32$. Find both possible sequences.

6 Write the multiplier for each change.
a. an increase of 7 per cent
b. a decrease of 12 per cent
c. trebling
d. halving
e. an increase of 0.5 per cent

UNDERSTANDING AND REASONING

7 **Error analysis.** For $3, 6, 12, \ldots$, a student writes $u_{10} = 3 \cdot 2^{10} = 3072$. Identify the error and give the correct value.

8 2000 is invested at 4.5 per cent per year. Write a model for the value after n years, find the value after 6 years, and find the first year in which the value exceeds 3000.

9 Explain why a geometric sequence with $-1 < r < 0$ alternates in sign while its terms shrink. What happens when $r = 1$, and when $r = -1$?

10 Insert two numbers between 5 and 40 so that all four numbers form a geometric sequence with positive terms.

11 A culture doubles every 4 hours. Find the factor by which it grows in one hour, and the factor over 24 hours. Explain why the hourly factor is not 2.

12 For $u_n = 3(1.2)^{n-1}$, find the first term that exceeds 100. Use logarithms and state the integer clearly.

APPLICATION AND EXTENSION

13 A machine costs $45\,000$ and depreciates by 18 per cent per year on a reducing balance. Find its value after 5 years and the first year in which it is worth less than $10\,000$.

14 A town of $12\,000$ people grows by 2.5 per cent per year. Find the population after 10 years and the first year in which it doubles.

EXERCISE 4C · CONTINUED GEOMETRIC SEQUENCES

15 **Extension.** A geometric sequence satisfies $u_1 + u_2 = 12$ and $u_3 + u_4 = 27$. Find all possible values of r and a, and explain why there are two answers.

CONCEPT AND METHOD

4.4 Geometric series and convergence

Adding the terms of a geometric sequence gives a geometric series. Its sum is found by a trick that cancels almost everything: multiply S_n by r, subtract, and only two terms survive.

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1},$$
$$rS_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n.$$

Subtracting the second line from the first gives $(1 - r)S_n = a - ar^n$, so for $r \neq 1$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}.$$

When $r = 1$ the division is impossible, and the series is simply n copies of a: $S_n = na$.

The formula contains the whole story of convergence. If $|r| < 1$, then r^n shrinks towards zero as n grows, so S_n approaches the fixed value $\frac{a}{1-r}$: the series converges, and that limit is the sum to infinity S_∞. If $|r| > 1$ the powers grow without bound, the partial sums do the same, and the series diverges; no finite total exists.

Two warnings follow. First, substituting $|r| \geq 1$ into $S_\infty = \frac{a}{1-r}$ produces a number but not a sum: for $1 + 2 + 4 + \cdots$ it gives -1, which is false because every partial sum is positive. Second, terms that tend to zero are not enough to guarantee convergence; Section 4.6 exhibits a series whose terms shrink to zero yet whose total grows without limit.

KEY IDEA · FINITE AND INFINITE GEOMETRIC SUMS

For $r \neq 1$, $S_n = \frac{a(1-r^n)}{1-r}$. If $|r| < 1$ the series converges and $S_\infty = \frac{a}{1-r}$. If $|r| \geq 1$ the series diverges, with the single exception $a = 0$. A series with $r = -0.8$ converges, because convergence depends on $|r|$, not on the sign of r.

DEFINITIONS

Geometric series, convergence, sum to infinity

A **geometric series** is the sum of the terms of a geometric sequence: $S_n = a + ar + \cdots + ar^{n-1}$. The series **converges** if the partial sums S_n approach a finite limit as n increases without bound; that limit is the **sum to infinity**. Otherwise the series **diverges**. For a geometric series with $a \neq 0$, convergence occurs exactly when $|r| < 1$.

EXAMPLE 4.10 FINITE SUMS, INCLUDING THE CASE $r = 1$

Find (a) the sum of the first 8 terms of $2 + 6 + 18 + \cdots$; (b) the sum of the first 10 terms of $80 + 40 + 20 + \cdots$; (c) $7 + 7 + \cdots + 7$ with 12 terms.

SOLUTION

(a) $a = 2, r = 3, n = 8$:

$$S_8 = \frac{2(3^8 - 1)}{3 - 1} = 3^8 - 1 = 6560.$$

The cancellation is visible here: the factor 2 and the denominator 2 disappear.

(b) $a = 80, r = 0.5, n = 10$:

$$S_{10} = \frac{80\left(1 - 0.5^{10}\right)}{1 - 0.5} = 160\left(1 - \frac{1}{1024}\right) = 159.84375.$$

The sum is already close to 160, which is the value the series is heading towards.

(c) Here $r = 1$, so the formula would divide by zero. The sum is $12 \times 7 = 84$, and in general $S_n = na$ when $r = 1$.

EXAMPLE 4.11 SUMS TO INFINITY, AND A SERIES THAT HAS NONE

(a) Find S_∞ for $80 + 40 + 20 + \cdots$. (b) Find S_∞ for $12 - 6 + 3 - \cdots$. (c) Explain why $1 + 2 + 4 + \cdots$ has no sum, even though the formula produces the number -1.

SOLUTION

(a) $a = 80$ and $r = 0.5$, and $|r| < 1$, so

$$S_\infty = \frac{80}{1 - 0.5} = 160.$$

The partial sums $80,\ 120,\ 140,\ 150,\ 155,\ \ldots$ climb towards 160 without passing it.

(b) $a = 12$ and $r = -0.5$. Since $|-0.5| < 1$ the series converges:

$$S_\infty = \frac{12}{1 - (-0.5)} = \frac{12}{1.5} = 8.$$

The partial sums are $12,\ 6,\ 9,\ 7.5,\ 8.25,\ \ldots$, alternating above and below 8 while closing in on it.

(c) Here $r = 2$, so $|r| > 1$ and the series diverges. The partial sums are $1,\ 3,\ 7,\ 15,\ 31,\ \ldots$, doubling and adding one each time, and they exceed any bound. Substituting into $\frac{a}{1-r}$ gives $\frac{1}{-1} = -1$, a number produced by algebra outside its domain of validity. A formula without its condition is not a result.

> An alternating series can converge: the sign of r is irrelevant, only its size matters.

EXAMPLE **4.12** RECURRING DECIMALS AS GEOMETRIC SERIES

Express as fractions in lowest terms: (a) $0.272727\ldots$ (b) $0.8333\ldots$ (c) $0.999\ldots$

SOLUTION

(a) $0.272727\ldots = 0.27 + 0.0027 + 0.000027 + \cdots$, a geometric series with $a = 0.27$ and $r = 0.01$:

$$S_\infty = \frac{0.27}{1 - 0.01} = \frac{0.27}{0.99} = \frac{27}{99} = \frac{3}{11}.$$

(b) Only the tail recurs, so split off the non-recurring digit:

$$0.8333\ldots = 0.8 + 0.03 + 0.003 + \cdots = \frac{4}{5} + \frac{0.03}{1 - 0.1} = \frac{4}{5} + \frac{1}{30} = \frac{5}{6}.$$

(c) $0.999\ldots = \frac{0.9}{1-0.1} = 1$. The difference $1 - 0.999\ldots$ is smaller than every positive number, so it is zero. The decimal $0.999\ldots$ and the number 1 are two names for the same value, not two values that are close.

> The method is always the same: write the decimal as a first term plus a geometric tail, identify r as a power of 10, and sum.

MISCONCEPTION

"An infinite series always has an infinite sum, and $S_\infty = \frac{a}{1-r}$ can be used for any r." An infinite sum is a limit of finite sums, and a limit may be finite: $0.999\ldots = 1$ is an infinite sum with a finite value. The formula is conditional. It was derived by dividing by $1 - r$, legitimate only when $r \neq 1$, and the step $r^n \to 0$ holds only when $|r| < 1$. Outside those conditions, $\frac{a}{1-r}$ is arithmetic without meaning.

GUIDED PRACTICE

1. Find S_6 for $3 + 6 + 12 + \cdots$. (Answer: $3(2^6 - 1) = 189$.)
2. Find S_∞ for $18 + 6 + 2 + \cdots$. (Answer: $18 \div \frac{2}{3} = 27$.)
3. Does $4 - 6 + 9 - \cdots$ converge? (Answer: No; $r = -1.5$ and $|r| > 1$.)
4. Write $0.444\ldots$ as a fraction. (Answer: $\frac{4}{9}$.)

EXERCISE 4D

GEOMETRIC SERIES AND CONVERGENCE

FLUENCY

1 Find the sum of the stated number of terms.

a. $3 + 6 + 12 + \cdots$, 8 terms
b. $2 + 6 + 18 + \cdots$, 6 terms
c. $100 + 50 + 25 + \cdots$, 8 terms
d. $1 - 3 + 9 - \cdots$, 7 terms

2 Find the sum to infinity where it exists.

a. $12 + 6 + 3 + \cdots$
b. $1 + \frac{1}{3} + \frac{1}{9} + \cdots$
c. $8 - 4 + 2 - \cdots$
d. $5 + \frac{5}{4} + \frac{5}{16} + \cdots$

3 Write each recurring decimal as a fraction.

a. $0.777\ldots$
b. $0.363636\ldots$
c. $0.1666\ldots$
d. $0.41666\ldots$

4 For each ratio, state whether the geometric series with that ratio converges, and give its sum to infinity when it exists: (a) $r = 1.03$ (b) $r = -0.6$ with $a = 10$ (c) $r = 5$ (d) $r = \frac{2}{3}$ with $a = 9$ (e) $r = -1$ (f) $r = 0.999$ with $a = 2$.

5 A geometric series has $S_\infty = 50$ and $a = 20$. Find r. Repeat for $S_\infty = 36$ and $r = 0.25$.

6 For $a = 4$ and $r = 1.5$, find S_{10}.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes $1 + 2 + 4 + 8 + \cdots = \frac{1}{1-2} = -1$. Explain what is wrong, and support your explanation with three partial sums.

8 Derive $S_n = \frac{a(1-r^n)}{1-r}$ by subtracting rS_n from S_n. Then explain which step fails when $r = 1$, and what S_n equals in that case.

EXERCISE 4D · CONTINUED

9 Explain why $S_\infty = \frac{a}{1-r}$ is described as a limit rather than a sum of infinitely many additions.

10 A geometric series has sum to infinity 16 and first term 4. Find r and the third term.

11 For $a = 8$ and $r = -0.5$, find S_4 and S_∞, and describe how the partial sums behave.

12 For $a > 0$ and $0 < r < 1$, prove that $S_n < S_\infty$ for every n, and explain why the partial sums increase.

APPLICATION AND EXTENSION

13 **The thirty-day contract.** Return to the opening problem. (a) Show that Option B pays 536 870 912 on day 30 and 1 073 741 823 in total. (b) Show that Option A pays 30 000 and Option C pays 4650 over the thirty days. (c) Find the first day on which B's daily pay exceeds A's, and the first day on which B's running total exceeds 30 000. (d) State which option you would sign and justify the choice using both a term and a sum.

14 A ball is dropped from 2 m and rebounds to 0.7 of its previous height after each bounce. Find the total vertical distance travelled before it comes to rest.

15 **Extension.** Find every value of x for which $1 + 2x + 4x^2 + \cdots = 3$, stating the condition on x that makes the sum to infinity valid.

APPLICATION

4.5 Annuities, loans and saving

Financial mathematics is the arithmetic of a recurrence relation. A balance that earns interest and receives a deposit obeys $B_{n+1} = RB_n + d$, where $R = 1 + i$ is the annual or monthly multiplier and d is the deposit made at the end of the period. A loan obeys the same rule with a subtraction: $B_{n+1} = RB_n - p$, where p is the repayment. Each step is one application of the recurrence; unrolling the steps produces a geometric series.

With no further deposits the recurrence unrolls to $B_n = B_0 R^n$, the compound interest formula; depreciation is the same formula with $R < 1$. With regular deposits, each deposit earns interest for a different length of time: the first for $n - 1$ periods, the last for none. Their values at the end form a geometric series with first term d, ratio R and n terms, so

$$B_n = \frac{d(R^n - 1)}{R - 1} = \frac{d\left((1 + i)^n - 1\right)}{i}.$$

This is the value of an annuity. A deposit made at the start of each period earns one extra period, so the total is multiplied by R.

For a loan of L repaid in n equal instalments, the balance after n payments must be zero. Setting $LR^n - p\frac{R^n - 1}{R - 1} = 0$ and solving for p:

$$p = \frac{LiR^n}{R^n - 1} = \frac{Li}{1 - R^{-n}}.$$

The total repaid is np, so the total interest is $np - L$. Because p is fixed while the balance falls, the interest portion of each payment shrinks and the capital portion grows — a fact a spreadsheet reveals at once.

KEY IDEA · FOUR FINANCIAL FORMULAS FROM ONE RECURRENCE

Growth $B_n = B_0(1 + i)^n$; decay $B_n = B_0(1 - i)^n$; deposits $B_n = \frac{d((1+i)^n - 1)}{i}$; loan instalment $p = \frac{Li}{1 - (1+i)^{-n}}$. The **effective annual rate** of a nominal rate i compounded monthly is $(1 + i/12)^{12} - 1$, which puts offers with different compounding intervals on one basis.

EXAMPLE **4.13** COMPOUND GROWTH AND THE TIME TO A TARGET

4000 is invested at 5 per cent per year, compounded annually. (a) Find its value after 6 years. (b) Find the first year at the end of which the value exceeds 6000.

SOLUTION

(a) $R = 1.05$, so $B_6 = 4000(1.05)^6 = 4000 \times 1.3401 = 5360.38$, to the nearest hundredth.

(b) Require $4000(1.05)^n > 6000$, so $1.05^n > 1.5$ and

$$n > \frac{\log 1.5}{\log 1.05} \approx 8.31.$$

The first whole year is $n = 9$, when the value is 6205.31; at $n = 8$ it is 5909.82, just short. "Compounded annually" means interest is added once per year to the whole balance, including earlier interest; simple interest would add 200 a year and reach only 5200.

EXAMPLE **4.14** REGULAR DEPOSITS FORM A GEOMETRIC SERIES

200 is deposited at the end of each year for 10 years into an account paying 4 per cent per year. (a) Find the balance at the end of the tenth year. (b) How much of it is interest? (c) What would the balance be if each deposit were made at the start of the year?

SOLUTION

(a) The first deposit earns interest for 9 years and the last for none, so the balance is

$$200\left(1.04^9 + 1.04^8 + \cdots + 1.04 + 1\right) = \frac{200\left(1.04^{10} - 1\right)}{0.04} = 2401.22.$$

Notice the order of the series: writing it from the last deposit to the first makes the first term 200 and the ratio 1.04.

(b) The deposits total 2000, so the interest is $2401.22 - 2000 = 401.22$.

(c) Each deposit now earns one more year, so every term is multiplied by 1.04 and the balance is $2401.22 \times 1.04 = 2497.27$. The timing of the payments, not their size, accounts for the difference of 96.05.

> The exponent on the last deposit is 0, not 1. A deposit made at the end of the final year cannot earn interest in that year.

EXAMPLE **4.15** LOAN REPAYMENTS AND COMPARING OFFERS

(a) A loan of 8000 at 6 per cent per year is repaid in five equal annual instalments. Find the instalment and the total interest. (b) Find the instalment if the term is four years instead. (c) A bank offers 5.2 per cent compounded annually; another offers 5.0 per cent compounded monthly. Which is cheaper?

SOLUTION

(a) $i = 0.06$ and $n = 5$:

$$p = \frac{8000(0.06)}{1 - 1.06^{-5}} = \frac{480}{1 - 0.747258} = 1899.17.$$

Five payments total 9495.85, so the interest is 1495.85. A forward check confirms the balance: after one year $8000(1.06) - 1899.17 = 6580.83$, and after five years the balance is zero to the nearest unit.

(b) With $n = 4$, $p = \frac{480}{1-1.06^{-4}} = 2308.73$. The total is 9234.92 and the interest 1234.92. A shorter term raises each payment by 409.56 but saves 260.93 in interest.

(c) Compare effective annual rates. The first is 5.2 per cent. The second is

$$\left(1 + \frac{0.05}{12}\right)^{12} - 1 = 0.05116\ldots,$$

about 5.12 per cent. The monthly offer is cheaper, despite the larger number of compounding steps. Comparing nominal rates directly would have suggested the opposite.

> The effective rate is a comparison tool, not a promise: it assumes the rate stays fixed and ignores fees, which can change the ranking of two offers.

MISCONCEPTION

"Twelve payments of 100 at 6 per cent per year produce 1200 plus 6 per cent, namely 1272." Only the earliest deposits earn a full year of interest; the last earns none. If the 6 per cent is a nominal annual rate compounded monthly, the rate per month is 0.5 per cent and the correct total is $100\left(\frac{1.005^{12}-1}{0.005}\right) = 1233.56$ — the deposits plus 33.56 of interest, not 72. A second error is to charge interest on the original loan while ignoring that the balance is falling; interest is charged on what is still owed, which is exactly what the recurrence records.

GUIDED PRACTICE

1. Find the value of 3000 after 5 years at 4 per cent compounded annually. (Answer: 3649.96.)
2. Find the balance after three annual deposits of 500 at 5 per cent. (Answer: 1576.25.)
3. Find the annual instalment on 2000 at 10 per cent over two years. (Answer: 1152.38.)
4. Find the effective annual rate of 6 per cent compounded monthly. (Answer: $1.005^{12} - 1 = 6.17$ per cent.)

EXERCISE 4E

SAVING, BORROWING AND COMPARING

FLUENCY

1 Find the value of 2500 after 5 years at 3.5 per cent compounded annually.

2 A machine costing $12\,000$ depreciates by 20 per cent per year on a reducing balance. Find its value after 4 years.

3 Find the balance after 8 annual deposits of 150 at 3 per cent, made at the end of each year.

4 Find the annual instalment on a loan of 6000 at 7 per cent repaid over four years, and the total interest.

EXERCISE 4E · CONTINUED SAVING, BORROWING AND COMPARING

5 1000 is invested at 4 per cent, and 100 is added at the end of each year. Find the balance after three years by iterating the recurrence, then check with the annuity formula.

6 Find the effective annual rate of 4.8 per cent compounded monthly.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student claims that saving 100 a month for a year at 5 per cent per year gives $1200 \times 1.05 = 1260$. Explain why this overstates the interest, and identify which deposit earns the most.

8 Explain how the annuity formula arises from a geometric series. For 200 deposited annually for 10 years at 4 per cent, state a, r and n of that series.

9 For the loan in Example 4.15, calculate the interest charged in the first year and in the fifth year. Explain the pattern.

10 Which is cheaper: 3.9 per cent compounded annually, or 3.8 per cent compounded monthly? Support your answer with effective rates.

11 Show that making each deposit at the start of the year multiplies the final balance by $1 + i$. Evaluate the difference for 100 a year for five years at 5 per cent.

12 A loan can be taken over three years or over five years at the same rate. Explain, using the formula, why the longer term lowers each payment but raises the total interest.

APPLICATION AND EXTENSION

13 A saver needs 15 000 in six years and can earn 4.5 per cent per year. Find the equal annual deposit, made at the end of each year, that reaches the target.

14 A loan of 12 000 is offered at 6.0 per cent or at 5.4 per cent, both repaid in three equal annual instalments. Find both instalments and both totals, and state the saving from the lower rate.

15 **Extension.** A credit card charges 1.5 per cent per month on the outstanding balance. A balance of 1200 is repaid at 150 per month. Find the balance after three months and the number of months needed to clear the debt, and explain why the final payment is smaller than 150.

PATTERN AND GENERALISATION

4.6 Investigation: infinite sums

An infinite series is not a single act of addition, which no one can finish. It is a sequence of totals, and the question is whether those totals settle down. This investigation looks at sums that settle, at one that famously does not, and at the reasoning that separates the two.

INVESTIGATION 4 · WHAT DOES AN INFINITE SUM MEAN?

Totals that settle and totals that do not

1. Copy and complete the table for $S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}$. Give each total as a fraction with denominator 2^n and as a decimal.

n	1	2	3	4	5	6
S_n	0.5	0.75				

   Predict S_{10} and S_∞. Explain how the gap to 1 behaves at each step.

2. Draw a square of side 1. Shade half of it, then half of what remains, then half of the new remainder, and so on. Explain why the shaded area is 1 in the limit, and why no finite number of steps ever shades the whole square.

3. Write $0.999\ldots$ as a geometric series and sum it. Separately, note that $\frac{1}{3} = 0.333\ldots$ and multiply both sides by 3. Do the two arguments agree? What does that tell you about the notation $0.999\ldots$?

4. Zeno's runner covers 100 m, then 50 m, then 25 m, and so on. Sum the distances. If each stage takes half the time of the one before, what total time does the runner take, and what is paradoxical about the description?

5. Now sum the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$. Compute the partial sums after 10, 100 and 1000 terms (about 2.93, 5.19 and 7.49). Group the terms as $\left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots$ and show that each group exceeds $\frac{1}{2}$. What follows?

6. For $r = 0.5$, compute $S_n = \frac{a(1-r^n)}{1-r}$ with $a = 1$ for $n = 5$ and $n = 10$, and compare with $\frac{1}{1-r} = 2$. Repeat for $r = 0.9$ and for $r = 1.1$. Which case approaches a limit, and which does not?

7. Write a sentence that begins: "An infinite geometric series converges exactly when …", and justify it by describing what happens to r^n in each case, including $r = -1$.

Generalise. For a geometric series, $S_n = \frac{a(1-r^n)}{1-r}$. If $|r| < 1$, the power r^n becomes arbitrarily small, so S_n differs from $\frac{a}{1-r}$ by $\frac{ar^n}{1-r}$, a quantity that tends to zero; hence $S_\infty = \frac{a}{1-r}$. If $|r| > 1$, the powers grow and no limit exists; if $r = 1$, the sum is na and grows without bound when $a \neq 0$. Then explain why the harmonic series shows that "terms tend to zero" is necessary but not sufficient for convergence: its terms do tend to zero, yet its partial sums pass every bound.

MODELLING · CRITERION D

Designing a repayment plan

A small enterprise borrows $15\,000$ to buy equipment. The lender charges 0.6 per cent per month on the outstanding balance, and the loan is repaid in equal monthly instalments over three years.

1. **Model.** Write the recurrence $B_{n+1} = 1.006B_n - p$ with $B_0 = 15\,000$, and solve $B_{36} = 0$ to find the monthly instalment. State the total repaid and the total interest.
2. **Compare.** Repeat for a four-year term. Explain why the shorter term costs more each month but less overall, and by how much.
3. **Test a change.** Suppose the enterprise can pay 50 more each month. Use logarithms to find the new number of payments, and estimate the interest saved.
4. **Evaluate.** State three assumptions of the model — about the rate, the payments and the costs — and explain how a change in each would alter the plan and the advice you would give.

CUMULATIVE

4.7 Mixed practice

Attempt all questions without notes. Decide first whether the question is about a term, a finite sum or a limit; that choice determines the method.

MIXED PRACTICE 4

SEQUENCES, SERIES AND FINANCIAL MODELS

FLUENCY AND UNDERSTANDING

1. For $6,\ 10,\ 14,\ \ldots$, write u_n, find u_{25}, and find which term equals 142.
2. For $81,\ 54,\ 36,\ \ldots$, state r, write u_n and find u_6.
3. Evaluate $\sum_{k=1}^{12}(5k-2)$.
4. Find S_∞ for $18 + 12 + 8 + \cdots$.
5. Write $0.727272\ldots$ as a fraction in lowest terms.
6. Insert three numbers between 2 and 162 so that all five form a geometric sequence with positive terms.
7. An arithmetic sequence has $u_4 = 17$ and $u_{10} = 41$. Find a, d and u_{20}.
8. A geometric sequence has $u_2 = 20$ and $u_5 = 160$. Find r and a.
9. Find n if $\sum_{k=1}^{n}(3k-1) = 155$.
10. Find the balance after 24 monthly deposits of 120 at 0.4 per cent per month, made at the end of each month.
11. State which series converge, giving a reason: (a) $5 + 1 + 0.2 + \cdots$ (b) $3 - 3 + 3 - 3 + \cdots$ (c) $7 + 7 + 7 + \cdots$ (d) $2 + 3 + 4.5 + \cdots$.

REASONING AND APPLICATION

12. **Error analysis.** A student sums $4 + 9 + 14 + \cdots + 94$ as $\frac{94}{2}(4 + 94)$, saying there are 94 terms. Find the correct number of terms and the correct sum, and identify the faulty reasoning.
13. A ball is dropped from 5 m and rebounds to 0.8 of its previous height. Find the total distance travelled before it rests.
14. Prove that in any arithmetic sequence $u_1 + u_n = u_2 + u_{n-1}$, and use the result to explain why Gauss pairing gives the sum of the series.
15. A salary is £$32\,000$ in the first year and rises by 3 per cent each year after that. Find the salary in the tenth year and the total earned over the ten years.

MIXED PRACTICE 4 · CONTINUED

16 A loan of 5000 is repaid in 12 equal monthly instalments at 0.8 per cent per month. Find the instalment and the total interest.

17 A geometric series has $S_\infty = 24$ and $r = 0.25$. Find a and u_3.

18 Sequence A is $100, 90, 81, \ldots$ and sequence B is $100, 95, 90, \ldots$. Compare their 10th terms, and find the first term of A that is less than 1.

19 **Extension.** A geometric series has $S_\infty = 9$ and the sum of its first two terms is 5. Find all possible values of r and a.

20 **Extension.** An arithmetic sequence has first term 5, and the sum of its first 8 terms equals the sum of the next 4 terms. Find d and verify your answer by direct addition.

21 On day 1, three people hear a rumour. Each day, the number of *new* people told is double the number told on the previous day. (a) How many new people are told on day 6? (b) How many people in total have heard it by the end of day 6? (c) The town has 5000 people. On which day does the cumulative total first exceed 5000?

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Sequences and series

Instructions. Answer all questions. Show sufficient working. A calculator is permitted; give monetary answers to two decimal places where appropriate.

1 For the sequence $11, 15, 19, \ldots$: (a) state d; (b) write u_n; (c) find u_{30}; (d) decide whether 203 is a term. [5]

2 For $6, 18, 54, \ldots$: (a) state r and write u_n; (b) find u_6; (c) find the first term that exceeds 5000. [5]

3 Evaluate (a) $\displaystyle\sum_{k=1}^{15}(4k-3)$ and (b) $24 + 12 + 6 + \cdots$ to infinity. [4]

4 2000 is invested at 4 per cent per year, and 300 is added at the end of each year. Find the balance after three years. [3]

5 (a) Explain why $1 + 3 + 9 + \cdots$ has no finite sum, using three partial sums. (b) Find $1 - \frac{1}{3} + \frac{1}{9} - \cdots$ to infinity. [3]

Evidence: correct common difference and closed form; a justified membership decision; correct ratio, term and inequality solution; accurate evaluation of a finite and an infinite sum; a balance built from both a power and an annuity; a divergence argument supported by partial sums, and a convergent alternating sum.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

A tower of cans

Instructions. Work independently. Record results in a table, describe each pattern in words before writing algebra, and justify every general formula.

A display tower is built from cans. The top row holds 2 cans, and each row below holds 3 more cans than the row above it.

1 Copy and complete the table for $n = 1$ to 6, where n is the number of rows. [6]

Rows, n	1	2	3	4	5	6
Cans in row n	2	5				
Total cans, T_n	2	7				

2 Describe the pattern in the row totals and the pattern in the first differences of T_n. Write formulas for the number of cans in row n and for T_n, and test both with $n = 6$. [6]

3 Prove your formula for T_n by pairing the top and bottom rows. Show that the first differences of T_n reproduce the row totals. [5]

4 A tower contains exactly 100 cans. Find n and the number of cans in the bottom row, and explain why your answer is the only possible one. [4]

5 A second design has 3 cans in the top row and 4 more in each row below. Write its row and total formulas, and find the number of rows in a tower of 105 cans. [3]

Evidence: a complete and correct table; patterns described in words; verified formulas for the row and the total; a pairing proof and a difference check; a solved quadratic with the negative root rejected; a second model handled independently.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining an infinite sum

Instructions. Write a 300–400 word explanation for a classmate who believes that "adding forever must give infinity" and that $0.999\ldots$ is "just less than 1". Your explanation must:

1 define a partial sum and say what it means for a series to converge; [3]

2 use the series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ with a table of at least four partial sums to show convergence; [4]

3 derive $S_n = \frac{a(1-r^n)}{1-r}$ and explain why it approaches $\frac{a}{1-r}$ when $|r| < 1$; [5]

4 explain why the same formula cannot be used when $|r| \geq 1$, with a counterexample; [4]

5 use a geometric series to show that $0.999\ldots = 1$; [2]

6 use correct notation throughout, with a diagram or table where it helps. [2]

Evidence: a clear definition of a limit of partial sums; a correct table; a derivation with the condition stated; a counterexample that exposes the failure of the formula; a correct recurring-decimal argument; coherent structure and precise notation.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Save now or borrow now?

Instructions. A community group needs equipment costing 9500. It has two options. **Option S:** save a fixed amount at the end of each month for 36 months in an account paying 3.6 per cent per year compounded monthly, so that the balance reaches 9500. **Option B:** borrow 9500 now at 7.2 per cent per year compounded monthly and repay it in 36 equal monthly instalments. Assume in both cases that the rate stays fixed and that no fees are charged.

1. Model Option S and find the monthly deposit, the total deposited and the interest earned. [5]
2. Model Option B and find the monthly instalment, the total repaid and the interest paid. [5]
3. Compare the two options in terms of monthly cost, total cost and timing. State one advantage of each. [5]
4. State three assumptions of your models and explain how the comparison would change if each failed. [5]
5. Recommend an option for a group that can afford 300 per month, and evaluate the limitations of your recommendation. [4]

Evidence: correct use of the annuity and loan formulas with the monthly rate; accurate totals and interest; a comparison on a common basis; explicit and relevant assumptions; a recommendation that follows from the mathematics and acknowledges its limits.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- An arithmetic sequence adds a constant difference: $u_n = a + (n-1)d$. Its graph is a set of points on a line of gradient d.
- A geometric sequence multiplies by a constant ratio: $u_n = ar^{n-1}$. The ratio is a percentage multiplier, so it models growth and decay.
- An arithmetic series sums by pairing first and last: $S_n = \frac{n}{2}(a+l) = \frac{n}{2}\left(2a + (n-1)d\right)$.
- A geometric series sums by subtracting rS_n from S_n: $S_n = \frac{a(1-r^n)}{1-r}$ for $r \neq 1$.
- An infinite geometric series converges exactly when $|r| < 1$, with $S_\infty = \frac{a}{1-r}$; otherwise it diverges and the formula is meaningless.
- Recurring decimals are geometric series with ratio a power of 0.1, which is why every recurring decimal is rational.
- A recurrence such as $B_{n+1} = RB_n \pm d$ unrolls into a geometric series; this gives the formulas for compound interest, annuities and loan instalments, and allows offers to be compared on an effective annual basis.

$$u_n = a + (n-1)d, \qquad S_n = \frac{n}{2}(a+l), \qquad u_n = ar^{n-1}, \qquad S_n = \frac{a(1-r^n)}{1-r}, \qquad S_\infty = \frac{a}{1-r} \ \ (|r| < 1)$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can identify an arithmetic or geometric sequence from three terms.
- I can write both recursive and closed forms and move between them.
- I can find a and d, or a and r, from two given terms.
- I can count the terms of a finite sequence or a sigma expression.
- I can decide whether a given value belongs to a sequence.
- I can use $\sum$ notation and shift the index without changing the sum.
- I can derive and apply the arithmetic sum formula.
- I can solve $S_n = k$ for n and reject impossible roots.
- I can derive and apply the finite geometric sum formula.
- I can state and justify the condition for convergence.
- I can find a sum to infinity and convert a recurring decimal.
- I can build a recurrence for a saving or loan balance.
- I can compare two financial offers using effective annual rates.
- I can state the assumptions of a financial model and evaluate them.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. The statement of inquiry claims that one number decides the behaviour of a geometric series. Which number, and how does it decide between a finite total and unlimited growth?
2. Which did you find harder — finding a term or finding a total? What is the conceptual difference between the two questions?
3. Describe one error you made in this chapter and the check that now prevents it.
4. In the opening problem, Option B looked absurd on day one. What does that example teach about judging a process from its first step?
5. The global context is fairness and development. How could an understanding of compound interest change a decision that affects a family or a community over ten years?

CONSTRAINTS, CORNERS AND THE BEST CHOICE

Inequalities and Decisions

05

RELATIONSHIPS · CHANGE · SYSTEM

Describing what is allowed, then choosing the best option inside it — with algebra, graphs and a healthy suspicion of the answer.

STATEMENT OF INQUIRY

A system of inequalities describes what is possible; a linear objective turns possibility into choice. Solving a quadratic inequality and optimising under linear constraints are the same habit: compare, bound, and justify.

KEY CONCEPT

Relationships

RELATED CONCEPTS

Change · Equivalence · System

GLOBAL CONTEXT

Globalization and sustainability — production planning under resource limits

FACTUAL

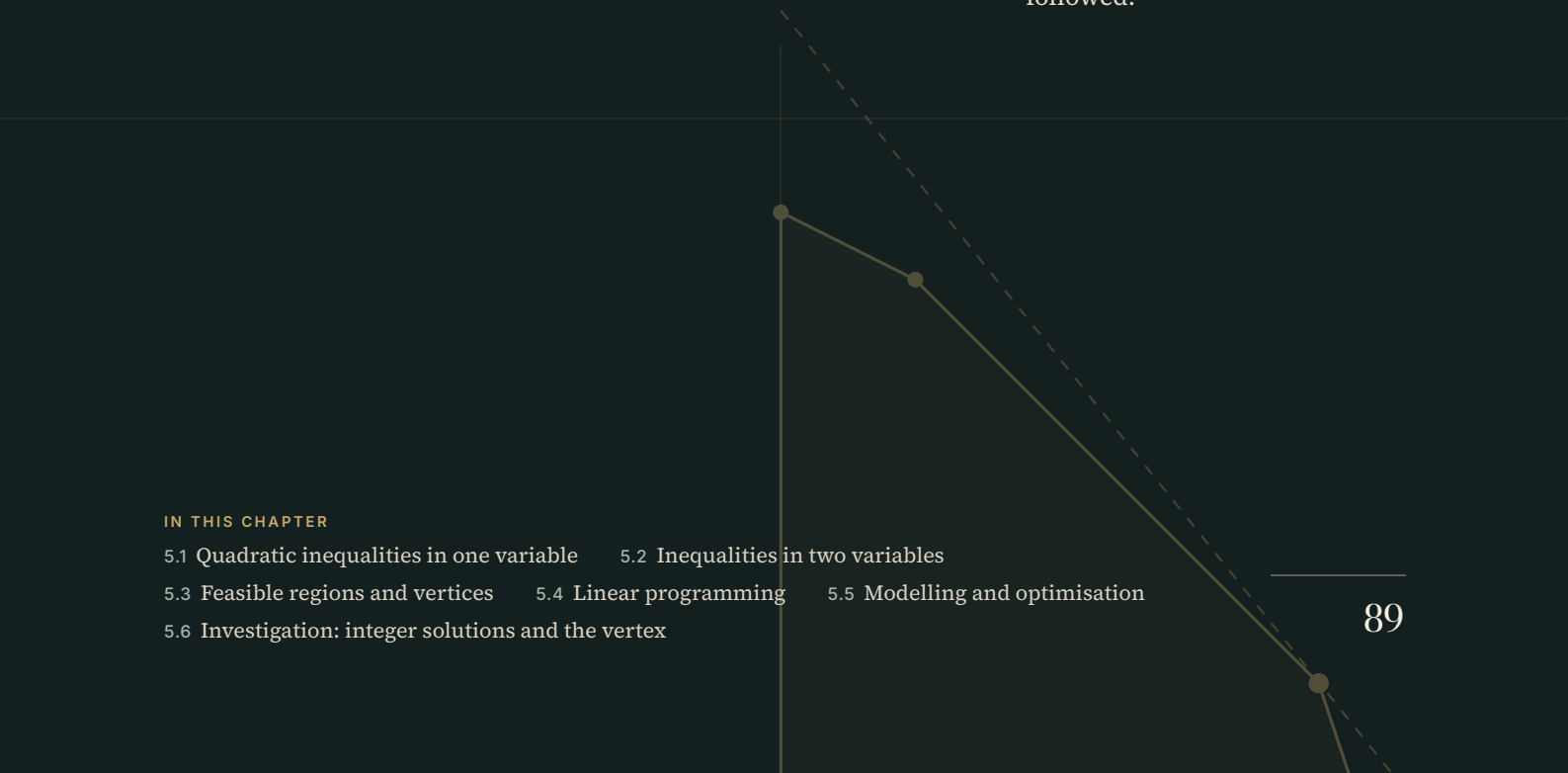
What is the difference between the solution of an equation and the solution of an inequality?

CONCEPTUAL

Why does the best value of a linear objective occur at a corner of a bounded feasible region?

DEBATABLE

When a model recommends a plan that is mathematically optimal but socially awkward, should the recommendation be followed?



IN THIS CHAPTER

5.1 Quadratic inequalities in one variable
5.2 Inequalities in two variables
5.3 Feasible regions and vertices
5.4 Linear programming
5.5 Modelling and optimisation
5.6 Investigation: integer solutions and the vertex

OPENING PROBLEM · TWO PRODUCTS, THREE LIMITS

The Dovetail Workshop

Dovetail makes desks and shelves from reclaimed timber. A desk takes 3 hours of assembly and 1 hour of finishing; a shelf takes 1 hour of assembly and 2 hours of finishing. Next week there are 54 assembly hours, 40 finishing hours, and storage for 22 finished items. A desk contributes £50 profit and a shelf £40.

The workshop plans 10 desks and 8 shelves. **Is that the best use of its limits — and which limit should it pay to expand?**

1. Write five inequalities for x desks and y shelves, and sketch the plans they allow.
2. Check the current plan against every limit and calculate its profit.
3. Find a plan earning more than £900. Is there a single best plan?
4. Four more assembly hours or four more storage units: which is worth more? Decide with evidence.

An inequality does not name a number; it names a set. This chapter learns to describe those sets precisely — on a number line, in the plane, and inside a workshop — and then to choose the best point in the set.

5.0 ORIENTATION Learning intentions

- Solve quadratic inequalities using factorisation, critical values and sign diagrams.
- Relate an inequality to the parts of a parabola, including repeated and no real roots.
- Represent linear inequalities as half-planes and combine them into systems.
- Find a feasible region and its vertices, and use them to optimise a linear objective.
- Build, interpret and criticise a linear programming model, including whole-number and sensitivity questions.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

factorise a quadratic, place its critical values on a number line and solve all four forms, including repeated and no real roots; shade a half-plane and translate words into inequalities; find the vertices of a feasible region; evaluate a linear objective at every vertex and recognise when no maximum exists; check whole-number solutions, interpret slack, test a change in one constraint and state the assumptions behind a recommendation.

ATL SKILLS · THINKING, TRANSFER AND COMMUNICATION

Critical thinking: an inequality is a claim about a set, so one counterexample disproves it but no number of examples proves it. **Transfer:** the sign diagram, the half-plane and the feasible region are one idea in different dimensions. **Communication:** a recommendation needs its variables, units, binding limits and assumptions, not just a number.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 5.1.

1. Factorise $x^2 - 5x + 6$ and $x^2 - 9$.
2. Solve $x^2 - 5x + 6 = 0$ and $2x^2 + x - 3 = 0$.
3. Sketch $y = x^2 - 4x + 3$, marking both intercepts.
4. Solve $3x - 7 \leq 8$ and $-2x < 6$.
5. Write $-1 \leq x < 4$ in interval notation.
6. Find the equation of the line through $(4, 0)$ and $(2, 3)$.

REFRESHER · FACTORISING, LINES AND INEQUALITY SIGNS

A quadratic factorises when two numbers multiply to the constant term and add to the coefficient of x: $x^2 - 5x + 6 = (x - 2)(x - 3)$. The quadratic formula gives the roots when factorising fails:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant $b^2 - 4ac$ decides how many real roots exist: positive gives two, zero gives one repeated root, negative gives none.

Multiplying or dividing an inequality by a negative number reverses its sign, so $-2x < 6$ becomes $x > -3$. A line is quickest to draw from its intercepts: $3x + 2y = 12$ passes through $(4, 0)$ and $(0, 6)$. Interval notation includes an endpoint with a square bracket and excludes it with a round bracket: $-1 \leq x < 4$ is $[-1, 4)$.

CONCEPT AND METHOD

5.1 Quadratic inequalities in one variable

An equation asks where two quantities are equal; a quadratic inequality asks where one is larger. The answer to $x^2 - x - 6 = 0$ is a pair of numbers, but the answer to $x^2 - x - 6 > 0$ is a set of intervals. Most errors here come from solving the equation and reporting only its roots.

The method has three stages: rearrange to zero and factorise; mark the critical values where each factor vanishes; determine the sign of the product on each interval. The sign can change only at a critical value, so testing one number per interval is enough. On the graph, $ax^2 + bx + c > 0$ asks where the parabola is above the axis.

DEFINITIONS

Quadratic inequality, critical values, sign diagram

A **quadratic inequality** has the form $ax^2 + bx + c > 0$, ≥ 0, < 0 or ≤ 0, with $a \neq 0$. Its **critical values** are the real solutions of $ax^2 + bx + c = 0$. A **sign diagram** marks the critical values on a number line and records the sign on each interval.

The discriminant is $\Delta = b^2 - 4ac$: positive means two critical values and alternating signs; zero means one repeated value and no change of sign; negative means no critical values and a constant sign.

EXAMPLE **5.1** A FACTORISED QUADRATIC

Solve $x^2 - x - 6 > 0$.

SOLUTION

$x^2 - x - 6 = (x - 3)(x + 2)$, so the critical values are 3 and -2. Test one value in each interval: at $x = -3$, $(-6)(-1) = 6 > 0$; at $x = 0$, $(-3)(2) = -6 < 0$; at $x = 4$, $(1)(6) = 6 > 0$. The product is positive outside the critical values and negative between them.

The inequality is strict, so the roots are excluded:

$$x < -2 \quad \text{or} \quad x > 3, \qquad (-\infty, -2) \cup (3, \infty).$$

The parabola opens upward and crosses at -2 and 3, so it lies above the axis exactly on these two intervals.

> "Or" is essential: no number is both less than -2 and greater than 3, so the solution is a union of two intervals.

EXAMPLE **5.2** REARRANGING FIRST

Solve $2x^2 + 5x \leq 12$.

SOLUTION

Rearrange: $2x^2 + 5x - 12 \leq 0$. Two numbers with product $2(-12) = -24$ and sum 5 are 8 and -3, so

$$2x^2 + 8x - 3x - 12 = 2x(x + 4) - 3(x + 4) = (2x - 3)(x + 4).$$

The critical values are $\dfrac{3}{2}$ and -4. Since $a = 2 > 0$, the expression is negative between the roots. Equality is allowed, so

$$-4 \leq x \leq \frac{3}{2}, \qquad \left[-4, \frac{3}{2}\right].$$

Check. At $x = 0$, $-12 \leq 0$; at $x = \dfrac{3}{2}$, $2\left(\dfrac{9}{4}\right) + \dfrac{15}{2} = 12$. Both hold, and $x = 3$ gives $33 > 12$ as expected.

> Do not divide by x: it may be zero or negative, and the division would lose the solution $x = 0$ or reverse the inequality.

EXAMPLE **5.3** TOUCHING THE AXIS, OR MISSING IT

Solve (a) $x^2 - 6x + 9 \geq 0$, (b) $x^2 - 6x + 9 > 0$, (c) $x^2 + 4x + 7 > 0$, (d) $x^2 + 4x + 7 \leq 0$.

SOLUTION

(a) $x^2 - 6x + 9 = (x - 3)^2 \geq 0$ for every real x, so the solution is $\mathbb{R}$.

(b) The square is zero only at $x = 3$, so the solution is $\mathbb{R} \setminus \{3\}$.

(c) $x^2 + 4x + 7 = (x + 2)^2 + 3 \geq 3 > 0$ for every x. The discriminant is $16 - 28 = -12 < 0$, so there are no real roots and the sign is constant; since $a > 0$, it is always positive.

(d) The smallest value is 3, so the expression is never ≤ 0. The solution set is empty, $\varnothing$.

> A repeated root is one critical value, and the sign is the same on both sides. "No real roots" does not mean "no solution": the inequality may hold for every x.

MISCONCEPTION

"$x^2 > 9$, so $x > 3$." Taking a square root is reversible only when both sides are known to be non-negative; here $x = -5$ also works. The correct step is $(x-3)(x+3) > 0$, giving $x < -3$ or $x > 3$. A related error is to assume the solution is always "between the roots", ignoring the direction of the inequality.

GUIDED PRACTICE

1. Solve $(x-4)(x+1) < 0$. (Answer: $-1 < x < 4$.)
2. Solve $x^2 \geq 25$. (Answer: $x \leq -5$ or $x \geq 5$.)
3. For which x is $x^2 + 2x + 1 \leq 0$? (Answer: $x = -1$ only.)
4. Where does $y = x^2 - 2x - 8$ lie below the axis? (Answer: $-2 < x < 4$.)

EXERCISE 5A

QUADRATIC INEQUALITIES IN ONE VARIABLE

FLUENCY

1 Solve.
a. $(x-2)(x-6) > 0$
b. $(x+5)(x+1) \leq 0$
c. $x(x+3) \geq 0$

2 Solve.
a. $x^2 - 16 < 0$
b. $x^2 - x - 12 \geq 0$
c. $2x^2 + 7x + 3 \leq 0$

3 Rearrange, then solve.
a. $x^2 < 5x$
b. $x^2 + 3x \geq 10$
c. $4x^2 > 9$

4 Write the solutions to Question 1 in interval notation.

5 Solve.
a. $(x-3)^2 \leq 0$
b. $(x+2)^2 > 0$
c. $(x-1)^2 < 0$

6 Classify each expression as always positive, always negative, never negative, or changing sign.
a. $x^2 - 2x + 5$
b. $-2x^2 + 4x - 5$
c. $9x^2 - 6x + 1$

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes "$x^2 > 9$, so $x > 3$". Find a value satisfying the inequality but not the answer, and write the correct solution.

8 Complete the square to explain why $x^2 + 6x + 10 > 0$ for every real x.

9 The graph of $y = x^2 - 4x - 5$ crosses the axis at $x = -1$ and $x = 5$. State where it is below the axis and where it is on or above it.

10 Find k for which $x^2 + kx + 25 = 0$ has a repeated root. For each, solve $x^2 + kx + 25 \geq 0$.

11 Prove that $x^2 - 2x + 3 > 0$ for all real x.

12 A student claims "if $x^2 > y^2$ then $x > y$". Disprove it with a numerical example and explain why the implication fails.

APPLICATION AND EXTENSION

13 A ball has height $h = 20t - 5t^2$ metres after t seconds. For how long is it more than 15 m above the ground? Interpret your answer.

14 A factory's profit, in hundreds of pounds, is $P = -2x^2 + 14x - 20$ for x hundred units. For what production levels is profit positive? Find the maximum profit.

15 **Extension.** Find all m for which $x^2 + mx + 4 > 0$ for every real x. Describe the cases $m = 4$ and $m = -4$.

CONCEPT AND REPRESENTATION

5.2 Inequalities in two variables

An equation such as $2x + 3y = 12$ describes a line. The inequality $2x + 3y \leq 12$ describes everything on one side of it. The line is the **boundary** and each side is a half-plane. A solid boundary includes the line; a dashed boundary excludes it.

To choose the side, test a point off the line: if it satisfies the inequality, shade its side; if not, shade the other. A system is solved by shading every half-plane and keeping the overlap.

DEFINITIONS

Linear inequality, boundary line, half-plane

A **linear inequality in two variables** has the form $ax + by \leq c$ (or $\geq, <, >$), with a, b not both zero. Its **boundary line** is $ax + by = c$, which divides the plane into two **half-planes**. The solution is one half-plane, including the boundary for $\leq$ and $\geq$ and excluding it for $<$ and $>$. The solution of a **system** is the intersection of the solution sets.

EXAMPLE **5.4** DRAWING A HALF-PLANE

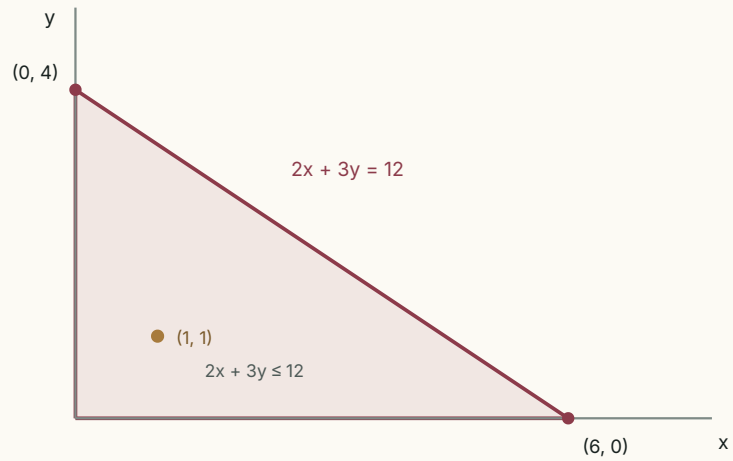
Draw $2x + 3y \leq 12$. Do $(1, 1)$, $(5, 1)$ and $(0, 4)$ belong to it?

SOLUTION

The boundary $2x + 3y = 12$ has intercepts $(6, 0)$ and $(0, 4)$. The inequality is non-strict, so the line is solid. Testing the origin gives $0 \leq 12$, true, so shade the side containing the origin.

Now substitute: $(1, 1)$ gives $5 \leq 12$, inside; $(5, 1)$ gives $13 > 12$, outside; $(0, 4)$ gives $12 \leq 12$, on the boundary and therefore included.

> If a boundary passes through the origin, $0 \leq 0$ is true but uninformative, because the origin is on the line. Test a different point.



EXAMPLE **5.5** FROM WORDS TO A SYSTEM

A van carries x small and y large crates. It holds at most 60 crates in total. A large crate takes twice the space of a small one, and the load limit is 90 small-crate equivalents. Insurance requires at least 10 large crates, and the depot supplies at most 40 small crates. Write the system.

SOLUTION

Translate each sentence:

1. Total: $x + y \leq 60$.
2. Load: $x + 2y \leq 90$.
3. Insurance: $y \geq 10$.
4. Supply: $x \leq 40$.
5. Common sense: $x \geq 0, y \geq 0$.

So $x + y \leq 60$, $x + 2y \leq 90$, $y \geq 10$ and $0 \leq x \leq 40$. The point $(30, 20)$ works: $50 \leq 60$, $70 \leq 90$, $20 \geq 10$, $30 \leq 40$. The point $(45, 20)$ fails only $x \leq 40$.

> "At least" gives $\geq$; "at most" gives $\leq$. Underline those words before writing symbols.

MISCONCEPTION

"$\leq$ always means shade below the line." It does not: in $x \leq 4$ there is no "below", and in $-y \leq 2x$ the solution is above. Always test a point. Testing the origin when the boundary passes through it is a second trap: $0 \leq 0$ is true whether the origin is inside or on the edge.

GUIDED PRACTICE

1. Is $(2, 5)$ a solution of $3x + y \leq 12$? (Yes, $11 \leq 12$.)
2. Sketch $y \geq x - 2$. Solid or dashed? (Solid; shade above.)
3. Write an inequality for the half-plane above the line through $(0, 3)$ and $(6, 0)$. ($x + 2y \geq 6$.)
4. Why is $x \geq 0$ almost always included in a quantity model? (Negative quantities are meaningless.)

EXERCISE 5B

INEQUALITIES IN TWO VARIABLES

FLUENCY

1. Test $(0, 0)$, $(5, 2)$, $(2, 6)$, $(6, 0)$ in $3x + 2y \leq 18$: inside, outside or on the boundary?
2. Sketch the region for each.
   a. $x + y \leq 5$
   b. $y \geq 2x - 1$
   c. $x \leq 4$
   d. $2x - y > 3$
3. Write inequalities for: "x is at least 3, y is non-negative, the total is at most 20."
4. Does $(3, 4)$ satisfy $x \geq 0$, $y \geq 0$, $x + y \leq 8$, $y \leq 2x$? Show substitutions.
5. A line passes through $(4, 0)$ and $(0, 6)$. Write the inequality for the half-plane containing the origin.
6. For $y < 2x + 1$, is the boundary included? Does $(0, 0)$ satisfy it?

UNDERSTANDING AND REASONING

7. **Error analysis.** A student shades above $2x + y = 10$ for $2x + y \leq 10$. Use $(0, 0)$ to explain the error and state the correct side.
8. Explain why $x \geq 0$ and $x \leq 0$ together describe a line, not a two-dimensional region.
9. Write four inequalities for the rectangle $1 \leq x \leq 4$, $2 \leq y \leq 6$.
10. For $x + y \geq 6$, $y \leq 8$, $x \geq 0$, decide whether $(2, 2)$ and $(10, 0)$ belong. Explain why the region is unbounded.

EXERCISE 5B · CONTINUED INEQUALITIES IN TWO VARIABLES

11 "At least twice as many adults as children": with a adults and c children, write the inequality. Why are $a \geq 0$, $c \geq 0$ stated separately?

12 Write a system for the region with vertices $(0,0)$, $(6,0)$, $(4,3)$, $(0,5)$.

APPLICATION AND EXTENSION

13 A taxi firm has x saloons and y minibuses. A minibus takes 3 garage spaces, a saloon 2, and there are 24 spaces. There are at most 10 drivers and at least 2 minibuses. Write the system and test $(4,4)$.

14 A snack mix uses x kg of A at £4/kg and y kg of B at £6/kg. It must weigh at least 5 kg, contain at most 3 kg of B, and cost at most £48. Write the constraints.

15 **Extension.** Explain why $x + y \leq 4$, $x + y \geq 6$, $x \geq 0$, $y \geq 0$ has no solution. What does that mean for the situation modelled?

REPRESENTATION AND STRUCTURE

5.3 Feasible regions and vertices

Each inequality shades a half-plane; the plans satisfying all of them at once form the overlap. That overlap is the feasible region, and its corners carry the answers. A corner is a vertex: a point where two boundary lines meet and the region changes direction. Not every intersection of two boundary lines qualifies — the point must also satisfy every other inequality.

An unbounded region may still have vertices, but its boundary includes rays. When the variables count objects, only lattice points are available, so counting them is part of describing the region.

DEFINITIONS

Feasible region, vertex, bounded region

The **feasible region** of a system of linear inequalities is the set of points that satisfies all of them simultaneously. A **vertex** is a point of the feasible region where two boundary lines meet. The region is **bounded** if it lies inside a circle of finite radius; otherwise it is **unbounded**. A **lattice point** has integer coordinates.

EXAMPLE 5.6 THE CORNERS OF THE WORKSHOP

Find the vertices of the Dovetail feasible region:

$$3x + y \leq 54, \qquad x + 2y \leq 40, \qquad x + y \leq 22, \qquad x \geq 0, \quad y \geq 0.$$

SOLUTION

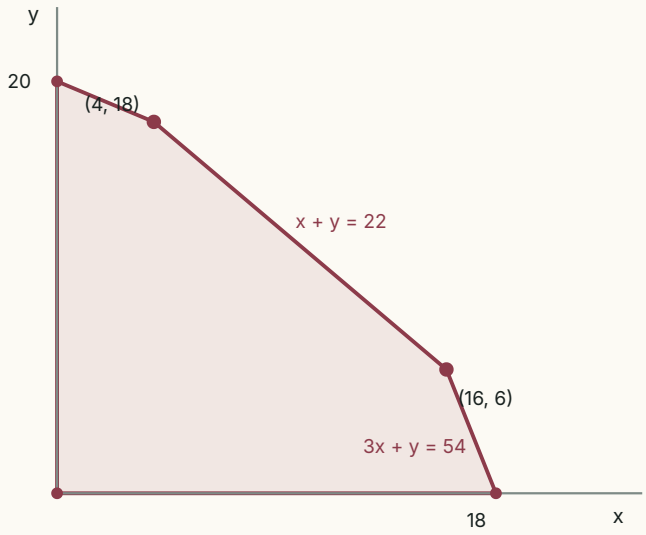
On the axes, $3x + y = 54$ gives $(18, 0)$ and $x + 2y = 40$ gives $(0, 20)$; the origin is also a corner. Now solve pairs of boundary lines.

$3x + y = 54$ with $x + y = 22$: subtracting gives $2x = 32$, so $x = 16$ and $y = 6$. $x + y = 22$ with $x + 2y = 40$: subtracting gives $y = 18$, so $x = 4$. The third pair, $3x + y = 54$ with $x + 2y = 40$, gives $x = 13.6$, $y = 13.2$, but $13.6 + 13.2 = 26.8 > 22$, so that point is outside the region.

Check: $(16, 6)$ uses $54, 28, 22$; $(4, 18)$ uses $30, 40, 22$; both are feasible. The vertices are

$$(0,0), \quad (18,0), \quad (16,6), \quad (4,18), \quad (0,20).$$

> Solving equations finds where lines cross; the inequalities decide which crossings are corners. Substituting is not optional.



Five vertices; the crossing at $(13.6, 13.2)$ lies outside the region.

EXAMPLE **5.7** COUNTING WHOLE-NUMBER PLANS

How many lattice points satisfy $x + 2y \le 7$, $x \ge 0$, $y \ge 0$?

SOLUTION

Work along each horizontal row, because y is restricted more tightly than x. For a fixed whole number y, the inequality gives $x \le 7 - 2y$, so x runs from 0 to $7 - 2y$.

y	0	1	2	3
largest x	7	5	3	1
points in row	8	6	4	2

The total is $8 + 6 + 4 + 2 = 20$. The vertices of the region are $(0, 0)$, $(7, 0)$ and $(0, 3.5)$; the last is not available, which is exactly why counting is not the same as measuring area.

> Row-by-row counting avoids both double counting and the temptation to round a boundary value upward.

MISCONCEPTION

"Every crossing of two boundary lines is a corner." The lines $3x + y = 54$ and $x + 2y = 40$ cross at $(13.6, 13.2)$, which breaks the storage limit $x + y \le 22$. A second error is to forget the axes: $(18, 0)$ and $(0, 20)$ are vertices created by $x = 0$ or $y = 0$, not by two "interesting" lines.

GUIDED PRACTICE

1. Find the vertices of $x \ge 0$, $y \ge 0$, $x + y \le 5$. (Answer: $(0, 0)$, $(5, 0)$, $(0, 5)$.)
2. Which of $(3, 3)$ and $(1, 5)$ satisfy $x + y \le 6$ and $y \le 4$? (Answer: $(3, 3)$ only.)
3. How many lattice points satisfy $x + y \le 3$, $x \ge 0$, $y \ge 0$? (Answer: 10.)
4. Is $x \ge 0$, $y \ge 0$, $x + 2y \ge 6$ bounded? (Answer: No.)

EXERCISE 5C

FEASIBLE REGIONS AND VERTICES

FLUENCY

1 Find the vertices of each region.
   a. $x \geq 0, y \geq 0, x + y \leq 6$
   b. $x \geq 0, y \geq 0, 2x + y \leq 8, x + 2y \leq 8$

2 Which of $(0, 0)$, $(4, 2)$, $(6, 0)$, $(2, 5)$ are vertices of $x \geq 0, y \geq 0, x + y \leq 6, y \leq 4$?

3 Classify each region as bounded or unbounded.
   a. $x \geq 1, y \geq 1, x + y \leq 8$
   b. $x \geq 0, y \geq 0, x + y \geq 4$
   c. $2 \leq x \leq 6, 1 \leq y \leq 5$

4 Find where $2x + 3y = 12$ and $x - y = 1$ meet.

5 Find the vertices of $x \geq 2, y \geq 0, x + y \leq 9$.

6 Count the lattice points with $x + 2y \leq 8, x \geq 0, y \geq 0$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student lists $(13.6, 13.2)$ as a vertex of the Dovetail region. Identify the inequality it fails and state what should happen to it.

8 Explain why $(0, 20)$ is a vertex of the Dovetail region even though it is not the crossing of two of the three "resource" lines.

9 Sketch $x \geq 0, y \geq 1, x + y \geq 4$. Give its vertices and explain why it has no corner at the top right.

10 How many lattice points satisfy $x + 2y \leq 6, x \geq 0, y \geq 0$? Compare with the area of the triangle, and explain the difference.

11 A region has vertices $(1, 1)$, $(6, 1)$, $(1, 6)$. Write three inequalities that define it.

12 Show that adding $x + y \leq 22$ to the Dovetail system removes the crossing $(13.6, 13.2)$ and creates the vertex $(16, 6)$.

APPLICATION AND EXTENSION

13 A stall has x small tables and y large ones, with $x + y \leq 12$, $2x + y \leq 18$, $x \geq 0$, $y \geq 0$. Find the vertices and sketch the region.

14 A workshop makes x chairs and y tables, with $x + y \leq 10$, $2x + y \leq 16$, $x \geq 2$, $y \geq 0$. Find the vertices and describe the shape of the region.

15 **Extension.** Explain why a bounded feasible region must have at least three vertices, while an unbounded one may have only two.

METHOD AND PROOF

5.4 Linear programming

A feasible region lists what is possible; an objective function measures how good each possibility is. For Dovetail, $P = 50x + 40y$. The level lines $P = c$ are parallel straight lines, and increasing c slides the line across the plane. The best value is reached where the line last touches the region: at a vertex, or along a whole edge when the level line is parallel to that edge.

This is the corner-point principle: a linear expression cannot be largest in the interior, because from an interior point a small step in the increasing direction stays feasible. The maximum therefore lies on the boundary, and along an edge P changes steadily, so its largest value on that edge is at an end.

DEFINITIONS

Linear programming, objective, constraints

Linear programming is the problem of maximising or minimising a linear **objective function** subject to linear **constraints**. The constraints define the feasible region; the objective is a separate expression that is not part of the region. A **level line** is the set of points giving one fixed value of the objective.

KEY IDEA · THE CORNER-POINT PRINCIPLE

If a linear objective is defined on a non-empty bounded feasible region, its maximum and minimum both occur at vertices. Evaluate the objective at every vertex and compare: the largest value is the maximum. If two adjacent vertices give the same value, every point on the edge between them is equally good. On an unbounded region a maximum may fail to exist, so the vertices alone are not enough — the behaviour along the rays must also be checked.

EXAMPLE 5.8 MAXIMISING THE WORKSHOP PROFIT

For the Dovetail region, maximise $P = 50x + 40y$.

SOLUTION

The region is bounded, so by the corner-point principle it is enough to test the five vertices.

VERTEX	$(0,0)$	$(18,0)$	$(16,6)$	$(4,18)$	$(0,20)$
P	0	900	1040	920	800

The largest value is 1040, at $x = 16$, $y = 6$: 16 desks and 6 shelves earn £1040. Assembly and storage are exactly full, while finishing uses 28 of its 40 hours.

Level-line check. $50x + 40y = 1040$ is $5x + 4y = 104$, with intercepts $(20.8, 0)$ and $(0, 26)$. Every other vertex lies below this line: at $(18, 0)$, $5(18) + 4(0) = 90 < 104$.

> The optimum need not use every resource. Twelve spare finishing hours are not waste; they are evidence that finishing is not the bottleneck.

EXAMPLE 5.9 WHEN THERE IS NO MAXIMUM

Find the maximum and minimum of $P = 2x + y$ subject to $x \geq 0$, $y \geq 1$, $x + y \geq 4$.

SOLUTION

The lines meet at $(0, 1)$, $(3, 1)$ and $(0, 4)$, but $(0, 1)$ gives $0 + 1 < 4$, so it is not feasible. The region is unbounded, above and right of the line through $(0, 4)$ and $(3, 1)$, with vertices $(0, 4)$ and $(3, 1)$.

At the vertices, $P = 4$ and $P = 7$. The minimum is 4, at $(0, 4)$. There is no maximum: for every $t \geq 3$, the point $(t, 1)$ is feasible and gives $P = 2t + 1$, which grows without limit.

> The corner-point principle is a statement about bounded regions. On this region the vertices give the minimum but say nothing about a maximum, because the region escapes along a ray where P increases.

EXAMPLE 5.10 WHOLE NUMBERS ONLY

Maximise $P = 4x + 5y$ subject to $3x + y \leq 10$, $x + 2y \leq 9$, $x \geq 0$, $y \geq 0$, where x and y are integers.

SOLUTION

First solve the continuous problem. The vertices are $(0, 0)$, $\left(\frac{10}{3}, 0\right)$, $(0, 4.5)$ and the crossing of $3x + y = 10$ with $x + 2y = 9$: substituting $y = 10 - 3x$ gives $x + 20 - 6x = 9$, so $x = 2.2$ and $y = 3.4$. The continuous maximum is at $(2.2, 3.4)$, with $P = 8.8 + 17 = 25.8$.

That point is not available. Rounding to $(2, 3)$ gives $8 + 15 = 23$, but a neighbour is better: $(1, 4)$ gives $4 + 20 = 24$, and it is feasible because $3 + 4 = 7 \leq 10$ and $1 + 8 = 9 \leq 9$. The point $(2, 4)$ would give 28 but fails $x + 2y \leq 9$. The best lattice value is 24, at $(1, 4)$.

> Rounding the continuous answer is not a method: it can be infeasible, and even when it is feasible a different

MISCONCEPTION

"The best vertex is the one furthest up and to the right." Position says nothing about the objective: with $P = 50x + 40y$, the vertex $(0, 20)$ is higher than $(16, 6)$ but earns £240 less. Equally, "round the continuous optimum" fails in Example 5.10, where $(2, 3)$ scores 23 but $(1, 4)$ scores 24.

GUIDED PRACTICE

1. Maximise $P = 3x + y$ on the vertices $(0, 0)$, $(5, 0)$, $(4, 2)$, $(0, 4)$. (Answer: 15 at $(5, 0)$.)
2. Minimise $P = x + 2y$ on $(1, 1)$, $(6, 1)$, $(2, 5)$. (Answer: 3 at $(1, 1)$.)
3. Why must every vertex be tested? (The level line can last touch at any of them.)
4. Maximise $P = 4x + 5y$ on $(0, 0)$, $(6, 0)$, $(2, 4)$, $(0, 5)$. (Answer: 28 at $(2, 4)$.)

EXERCISE 5D

LINEAR PROGRAMMING

FLUENCY

1 Evaluate $P = 2x + 3y$ at $(0, 0)$, $(4, 0)$, $(3, 2)$, $(0, 5)$ and state both extreme values.

2 The region with vertices $(0, 0)$, $(6, 0)$, $(4, 4)$, $(0, 3)$ is bounded. Maximise $P = 5x + 4y$.

3 For the same region, minimise $P = x + y$.

4 Find the vertices of $x \geq 0$, $y \geq 0$, $x + y \leq 8$, $x + 3y \leq 18$, then maximise $P = 3x + 2y$.

5 For $P = 4x + 3y$, find the intercepts of the level line $P = 12$ and state which side of it increases P.

6 Maximise $P = x + 3y$ on the vertices $(0, 0)$, $(10, 0)$, $(6, 4)$, $(0, 6)$. What is unusual about the answer?

UNDERSTANDING AND REASONING

7 **Error analysis.** A student tests only $(0, 0)$ and $(10, 0)$ for $P = x + 3y$ on the region of Question 6 and reports a maximum of 10. Explain the error.

8 Explain why a level line parallel to an edge of the feasible region produces two optimal vertices rather than one.

9 Explain why the corner-point principle cannot promise a maximum on the region in Example 5.9, even though the region has vertices.

10 For the region with vertices $(0, 0)$, $(8, 0)$, $(6, 4)$, $(0, 10)$, find k so that $(6, 4)$ and $(0, 10)$ give equal values of $P = x + ky$. For which k is $(0, 10)$ the unique maximum?

11 Maximise $P = 5x + 2y$ subject to $3x + y \leq 10$, $x + 2y \leq 9$, $x, y \geq 0$ integers.

12 At $(16, 6)$ and at $(4, 18)$, calculate the slack in each Dovetail constraint. Which resources are fully used at each plan?

APPLICATION AND EXTENSION

13 A print enterprise makes x booklets and y posters, with $2x + y \leq 44$, $x + 2y \leq 46$, $x + y \leq 28$. Profit is £5 per booklet and £6 per poster. Find the vertices and the maximum profit.

14 A market garden plants x beds of herbs and y beds of salad, with $x + y \leq 20$, $3x + 2y \leq 60$, $x + 2y \leq 30$. Returns are £20 and £30 per bed. Find the best planting and its return.

15 **Extension.** A courier uses x vans and y lorries, with $x + 2y \geq 12$, $x + y \leq 10$, $y \geq 2$. Costs are £40 and £90. Minimise cost and explain why the cheapest fleet is not the one with the fewest vehicles.

MODELLING

5.5 Modelling and optimisation

Real questions arrive as sentences. Building a model means choosing variables, writing constraints and an objective, and stating the assumptions that make the arithmetic possible: prices stay constant, every item sells, units are whole.

Sensitivity analysis asks how the answer responds when one number changes. It turns an answer into advice, because a manager asks not only "what is best?" but "what is worth paying for?".

DEFINITIONS

Model, assumption, sensitivity analysis

A **mathematical model** is a simplified description of a situation using variables, constraints and an objective. An **assumption** is a condition accepted as true in order to build the model, although it may fail in reality. **Sensitivity analysis** examines how the optimum changes when a constraint or coefficient is altered. The **slack** in a constraint is the amount by which it is not fully used at a given point.

EXAMPLE 5.11 A PRINT STUDIO

A studio makes x posters and y banners. A poster uses 2 units of ink and 1 press hour; a banner uses 1 unit and 1 hour. Weekly limits are 60 ink units and 40 press hours, and at most 24 banners can be sold. Existing orders require at least 8 posters. Profit is £12 per poster and £20 per banner. Maximise profit.

SOLUTION

The model is $P = 12x + 20y$ subject to

$$2x + y \leq 60, \qquad x + y \leq 40, \qquad y \leq 24, \qquad x \geq 8, \quad y \geq 0.$$

The vertices are $(8, 0)$, $(30, 0)$, $(20, 20)$, $(16, 24)$ and $(8, 24)$. For example, $2x + y = 60$ with $x + y = 40$ gives $(20, 20)$; with $y = 24$ the ink line gives $x = 18$, but $18 + 24 > 40$, so the corner there is $(16, 24)$.

VERTEX	$(8, 0)$	$(30, 0)$	$(20, 20)$	$(16, 24)$	$(8, 24)$
P	96	360	640	672	576

The maximum is £672 at 16 posters and 24 banners. Ink use is $32 + 24 = 56$, leaving 4 units spare; press time and the banner limit are both exactly full.

> Four spare ink units are not a reason to buy more ink: the press and the market for banners are what limit the profit.

EXAMPLE 5.12 WHICH LIMIT IS WORTH EXPANDING?

Dovetail can add either 4 hours of assembly or 4 units of storage. Compare the two options.

SOLUTION

Storage. Replace $x + y \leq 22$ by $x + y \leq 26$. The optimum is now where $3x + y = 54$ meets $x + y = 26$: $2x = 28$, so $x = 14$, $y = 12$. Finishing needs $14 + 24 = 38 \leq 40$. Profit is £1180, from $50(14) + 40(12) = 700 + 480 = 1180$, an increase of £140.

Assembly. Replace $3x + y \leq 54$ by $3x + y \leq 58$. The optimum is where $3x + y = 58$ meets $x + y = 22$: $2x = 36$, so $x = 18$, $y = 4$. Finishing needs $18 + 8 = 26 \leq 40$. Profit is £1060, from $50(18) + 40(4) = 900 + 160 = 1060$, an increase of only £20.

Storage is worth £35 per extra unit ($140 \div 4$); assembly is worth £5 per extra unit ($20 \div 4$). Storage is the better investment, provided the workshop can really store the extra items and still sell them.

> Both changes keep the same binding constraints; a large enough change would make another constraint bind and alter the rate.

MISCONCEPTION

"The optimum is the decision." The optimum is a recommendation conditional on the model. It assumes constant prices, unlimited demand and perfectly divisible time. A manager who expands storage to £1180 without checking whether 26 items can be sold has optimised the arithmetic, not the business.

GUIDED PRACTICE

1. In Example 5.11, how many ink units are spare at the optimum? (Answer: 4.)
2. Which constraint in Example 5.11 would you relax first, and why? (The banner limit: it is binding and each banner earns £20.)
3. If the banner limit rose to 26, which vertex would become optimal? (Answer: $(14, 26)$, profit £688.)
4. Name one assumption behind the profit figures. (Prices and costs stay constant at every scale.)

EXERCISE 5E

MODELLING AND OPTIMISATION

FLUENCY

1 A café sells x coffees at £2.20 profit and y cakes at £1.80. It can make at most 120 coffees and 80 cakes, and at most 160 items in total. Write the objective and the constraints.

2 Find the vertices of the region in Question 1 and the maximum profit.

3 A region has vertices $(0, 0)$, $(9, 0)$, $(6, 3)$, $(0, 6)$. Maximise $P = 3x + 4y$.

4 For $x + y \leq 12$, $2x + y \leq 20$, $x \geq 0$, $y \geq 0$, calculate the slack in each constraint at $(6, 4)$. Is $(6, 4)$ a vertex?

5 Write the objective for x chairs at £35 profit and y tables at £80.

6 State two assumptions made when a bakery models profit per loaf as constant.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says the best plan must use every resource completely. Use the Dovetail optimum to show why this is false.

8 Explain why increasing a resource that is not fully used at the optimum changes nothing, while reducing it can eventually change the optimum.

9 Dovetail's finishing limit rises from 40 to 50 hours. Explain, using a relaxed problem, why the optimum does not change.

10 In Example 5.11 the ink limit is not binding. Should the studio buy more ink? Justify with the slacks.

11 Give one reason why profit per poster might fall as output rises, and explain how that would weaken the model.

12 A continuous optimum is $(10.4, 6.2)$. Describe two legitimate ways to produce a practical plan, and say why neither is guaranteed to be best.

APPLICATION AND EXTENSION

13 A studio makes x posters and y banners with $3x + y \leq 48$, $x + y \leq 30$, $y \leq 18$, $x \geq 6$. Profit is £15 per poster and £22 per banner. Find the vertices and the maximum profit.

14 The studio can add 4 banners of storage ($y \leq 22$) or 2 units of total capacity ($x + y \leq 32$). Evaluate both and recommend one.

15 **Extension.** Remove the banner limit entirely and re-solve. Explain why a maximum still exists, and comment on whether the limit should be removed.

PATTERN AND GENERALISATION

5.6 Investigation: integer solutions and the vertex

Example 5.10 showed that the continuous optimum and the whole-number optimum need not agree. This investigation measures the gap, tests whether rounding is ever safe, and looks for the structure behind the numbers.

INVESTIGATION 5 · WHEN THE CORNER IS NOT WHOLE

Hunting the best lattice point

1. For R: $3x + y \le 10$, $x + 2y \le 9$, $x, y \ge 0$, find the vertices and the continuous maximum of $P = 4x + 5y$.
2. List every lattice point of R with $x = 0, 1, 2, 3$. Evaluate P at each and record the best.
3. Repeat for $P = 5x + 2y$ and $P = 2x + 3y$, and copy the table.

OBJECTIVE	CONTINUOUS MAXIMUM	INTEGER MAXIMUM	GAP
$4x + 5y$	25.8	24	1.8
$5x + 2y$	17.8	17	0.8
$2x + 3y$	14.6	14	0.6

4. Replace $3x + y \le 10$ by $3x + y \le 11$. Find the new fractional vertex, the continuous maximum of $4x + 5y$, and the best lattice point. Does the gap grow or shrink?
5. Which lattice point wins for each objective? Mark them on one sketch and describe where they sit relative to the fractional vertex.
6. Test the rule "round each coordinate to the nearest integer" on all four cases. In how many does it give the best lattice point — or even a feasible one?
7. Try $2x + y \le 10$, $x + 2y \le 10$, $x, y \ge 0$ with $P = 7x + 4y$. Find both maxima. Is the vertex integral?

Generalise. If a vertex is integral, the continuous and integer optima coincide there. If not, the integer optimum is a lattice point on or near the boundary close to that vertex, and rounding is not a method: it may leave the region, and another neighbour may score higher. State which of your results support each sentence, then write a rule for how many lattice points are worth checking and test it on the bike workshop below.

MODELLING · CRITERION D

The bike workshop

A workshop assembles x bicycles and y e-bikes each week. A bicycle needs 1 frame and 2 assembly hours; an e-bike needs 1 frame, 3 hours and 1 motor. There are 40 frames, 90 hours and 12 motors. Contribution is £80 per bicycle and £150 per e-bike.

1. **Model.** Write the constraints and the objective. Explain why both $x + y \le 40$ and $y \le 12$ are needed.
2. **Solve.** Find the vertices and the best weekly plan, and state the slack in each resource.
3. **Test.** Three more motors or six more assembly hours: evaluate both changes, allowing only whole bikes.
4. **Evaluate.** State two assumptions and one reason the recommendation might fail in practice.

CUMULATIVE

5.7 Mixed practice

Attempt all questions without notes. Decide first whether the question is about a set on a line, a set in a plane, or a best point in a set.

MIXED PRACTICE 5

INEQUALITIES, REGIONS AND OPTIMISATION

FLUENCY AND UNDERSTANDING

1 Solve $x^2 - 5x - 14 \leq 0$.

2 Solve $3x^2 - 5x - 2 > 0$.

3 Solve $x^2 + 6x + 10 > 0$.

4 Solve $(x - 5)^2 < 0$.

5 Write $-3 < x \leq 2$ in interval notation.

6 Shade $3x + 2y \geq 12$. Is $(2, 3)$ inside, outside or on the boundary?

7 Find the vertices of $x \geq 0$, $y \geq 0$, $x + y \leq 10$, $2x + y \leq 16$.

8 Maximise $P = 4x + 3y$ on the vertices from Question 7.

9 Calculate the slack in each constraint of Question 7 at $(6, 4)$.

10 Explain why $(0, 10)$ is not optimal in Question 8 although it has the largest y.

11 Maximise $P = 3x + 7y$ for $2x + y \leq 9$, $x + 2y \leq 9$, $x, y \geq 0$ integers.

12 Dovetail's storage limit falls from 22 to 20. Find the new vertices and the new maximum profit.

REASONING AND APPLICATION

13 A workshop makes x chairs and y tables with $2x + y \leq 30$, $x + 2y \leq 36$, $x + y \leq 20$. Profit is £4 and £5. Find the best plan. What happens if the total-capacity limit rises to 22?

14 A bakery makes x loaves and y cakes with $2x + y \leq 90$, $x + 2y \leq 80$, $x + y \leq 55$. Profit is £1.20 and £2.00. Find the vertices and the maximum profit, and name the binding limits.

15 Using the region with vertices $(0, 0)$, $(10, 0)$, $(4, 6)$, $(0, 8)$, explain with numbers how raising the coefficient of y in $P = x + ky$ can change the optimal vertex.

16 A continuous optimum is $(7.6, 2.8)$. List four lattice points worth checking and explain why the list is not arbitrary.

17 Maximise $P = 5x + 4y$ subject to $3x + 2y \leq 24$, $x + 2y \leq 16$, $x \geq 2$, $y \geq 0$.

18 A linear programme recommends producing none of product A. Give two reasons why a manager might still produce some.

19 **Extension.** Show that if two vertices A and B give the same value of a linear objective, then every point on the segment AB gives that value too.

20 **Extension.** A student claims that adding a constraint can never increase the maximum of a linear programme. Is the claim true or false? Justify with an example or a proof.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Inequalities and feasible regions

Instructions. Answer all questions. Show sufficient working. A calculator is permitted but not required.

1 Solve. **(a)** $x^2 - 7x + 10 > 0$ **(b)** $2x^2 + x - 6 \leq 0$ **(c)** $x^2 + 2x + 5 < 0$ [6]

2 Solve. **(a)** $(x - 4)^2 \geq 0$ **(b)** $x^2 - 4x + 4 > 0$ **(c)** Use the discriminant to explain why $x^2 - 3x + 7 > 0$ for every real x. [4]

3 Draw $2x - y \leq 4$ and test $(3, 2)$, $(0, -1)$ and $(5, 1)$. [4]

4 Find the vertices of $x \geq 0$, $y \geq 0$, $x + y \leq 9$, $2x + y \leq 14$. [3]

5 Explain why no point satisfies both $x + y \leq 3$ and $x + y \geq 5$. What would this mean for a production model? [3]

Evidence: correct critical values and interval notation, repeated and no real roots handled, a shaded half-plane with three tests, all four vertices, and a clear interpretation of an empty region.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

How the objective chooses a corner

Instructions. Work independently. Use a table, show each substitution, and justify every generalisation.

A feasible region has vertices $A(0,0)$, $B(10,0)$, $C(4,6)$ and $D(0,8)$. For a constant $k \geq 0$, let $P = x + ky$.

1. Complete a table for $k = 0.5, 1, 1.5, 2, 3$, giving P at each vertex and naming the optimal vertex or vertices. [5]
2. Describe the pattern in the winning vertex as k increases. [4]
3. Find exactly the two values of k at which the winner changes, and state which vertices tie at each. [5]
4. Explain your values using the gradients of BC and CD, and the gradient of the level line $P = c$. [4]
5. Predict the optimal vertex for $k = 10$ and for $k = 0.1$, justifying each prediction. [3]
6. At a transition, explain why every point of an edge is optimal, not only its endpoints. [3]

Evidence: a complete table, a described pattern, both transition values derived, gradients linked to parallel level lines, and a justified extrapolation.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Why the best point is a corner

Instructions. Write a 350–450 word explanation for a classmate who believes that a linear programme is solved by testing the point with the largest coordinates. Your explanation must: **(i)** define feasible region, constraint and objective [4]; **(ii)** use the Dovetail model to give a numerical counterexample to the classmate's rule [5]; **(iii)** use a level line to explain why the optimum lies on the boundary and at a vertex [5]; **(iv)** explain what changes when the region is unbounded [3]; **(v)** name one assumption of the model and one limitation of the recommendation [3]. Use correct notation and a sketch where it helps.

Evidence: precise definitions, a counterexample with values, a causal level-line argument, a correct statement about unbounded regions, and context-specific assumptions and limitations.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The school canteen

Instructions. A canteen serves x vegetarian and y chicken meals each day. The kitchen can produce at most 28 meals, the ovens hold $x + 2y \leq 44$ tray-units, and the fridge holds at most 18 chicken meals. Each vegetarian meal contributes £1.50 and each chicken meal £2.00.

1. Write the constraints and the objective, stating the units of each. [4]
2. Draw the feasible region, find all vertices, and identify the binding constraints at each. [6]
3. Find the optimal daily plan and its contribution. [4]
4. The canteen can add either 4 meals of kitchen capacity or 6 tray-units of oven space. Evaluate both, and recommend one with evidence. [6]
5. State two assumptions of the model and evaluate whether the recommendation would survive a change in either. [4]

Evidence: a correct model with units, an accurate region and vertices, correct evaluation at every vertex, a quantified comparison, and context-specific assumptions and evaluation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A quadratic inequality is solved by rearranging to zero, factorising, marking the critical values and testing the sign on each interval.
- Strict inequalities exclude the critical values; the solution may be one interval, two intervals, all reals, all reals except a point, or empty.
- The discriminant decides the shape: two roots give alternating signs, a repeated root gives no change, and a negative discriminant gives a constant sign.
- A linear inequality describes a half-plane; use a solid boundary for $\leq$ and $\geq$, a dashed one for $<$ and $>$, and a test point to choose the side.
- The feasible region is the intersection of the half-planes. Its vertices must satisfy every constraint, and only lattice points are available when the variables count objects.
- On a bounded region a linear objective is extreme at a vertex, and a level line parallel to an edge makes the whole edge optimal. On an unbounded region a maximum may not exist.
- Slack shows which resources are fully used; sensitivity analysis values a change in one constraint; integer answers must be searched, not rounded.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \qquad \Delta = b^2 - 4ac, \qquad P = ax + by, \qquad \text{optimum at a vertex}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can solve a factorised quadratic inequality with a sign diagram.
- I can rearrange an inequality before solving it.
- I can handle repeated roots and a negative discriminant.
- I can write solution sets in interval notation.
- I can shade a half-plane using a test point.
- I can translate a word problem into a system of inequalities.
- I can find and check vertices, decide whether a region is bounded, and count lattice points.
- I can evaluate an objective at every vertex and justify the optimum.
- I can recognise a tie and an unbounded region with no maximum.
- I can find an integer optimum and explain why rounding fails.
- I can interpret slack and compare two constraint changes.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which step of solving a quadratic inequality caused you most difficulty, and what check now catches the error?
2. Describe one mistake you made with a half-plane or a vertex and the reasoning that corrected it.
3. The statement of inquiry says that constraints describe what is possible and the objective turns possibility into choice. Use one model from this chapter to show what that means.
4. Why is it dangerous to report a rounded continuous optimum as a practical plan? Give an example with values.
5. What question about optimisation do you still want to answer?

STRUCTURE, SYMMETRY AND CERTAINTY

Circles and Proof

06

LOGIC · SPACE · JUSTIFICATION

Why a circle forces angles to be equal — and how a chain of reasons turns a drawing into a guarantee.

STATEMENT OF INQUIRY

A circle conceals equal angles and fixed distances; a few theorems, proved from radii and triangles, let us deduce measurements and justify every claim.

KEY CONCEPT
Logic

RELATED CONCEPTS
Space · Generalisation · Justification

GLOBAL CONTEXT
Personal and cultural expression — the geometry of architecture and ornament

FACTUAL
Which angles in a circle are equal, and what determines their size?

CONCEPTUAL
Why do the circle theorems follow from the properties of isosceles triangles and radii?

DEBATABLE
When a diagram suggests a result, what more is needed before we may call it true?



IN THIS CHAPTER

6.1 Angles in circles
6.2 Cyclic quadrilaterals
6.3 Tangents and radii
6.4 Chords, secants and the alternate segment
6.5 Constructing and writing proofs
6.6 Investigation: circle theorems under transformation

OPENING PROBLEM · THE ARCH THAT MUST ADMIT A VAN

A circular gateway and a delivery

A community hall has a circular arch over its entrance. The opening is 8 m wide at ground level and rises 2.4 m at its highest point. A delivery van is 2.2 m wide and 2.3 m high and must pass through the centre of the opening.

The architect claims the van will fit because the arch is higher than the van at its centre. **Is the claim correct, and what measurement decides it?**

1. Draw the circular arc through the two ground points and the highest point. Mark the centre O, the chord joining the ground points, and the highest point.
2. Use the perpendicular from O to the chord to find the radius.
3. Find the height of the arch 1.1 m either side of the centre, where the van's top corners pass.
4. Decide whether the van fits. State one assumption of your model and explain how it could change the answer.

A circle is the simplest curved figure, yet it supports a remarkable web of results. This chapter builds them from two facts you already know — the radii of a circle are equal, and the angles of a triangle sum to 180° — and then teaches you to write the reasoning so that a reader can check it.

ORIENTATION

6.0 Learning intentions

- Apply and prove the angle-at-the-centre, same-segment and semicircle theorems, including the reflex case.
- Use the opposite-angle and exterior-angle properties of cyclic quadrilaterals, and recognise concyclic points.
- Use tangent–radius, equal-tangent and incircle relationships to justify measurements.
- Apply chord, intersecting-chord, alternate-segment and tangent–secant theorems, distinguishing each from its converse.
- Write a two-column or paragraph proof with a named reason for every step, and criticise invalid reasoning.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

prove the angle-at-the-centre theorem and handle reflex cases; use the same-segment, semicircle and cyclic-quadrilateral theorems and their valid converses; use tangent, chord, intersecting-chord, alternate-segment and tangent–secant relationships; write a proof with a reason for every step; and expose a false converse with a counterexample.

ATL SKILLS · CRITICAL THINKING AND COMMUNICATION

Critical thinking: a diagram shows what might be true, never what must be. Ask which step would fail in a different drawing. **Communication:** write for a sceptical reader; every line needs a reason checkable without the picture. **Transfer:** given, construction, congruent triangles, conclusion — the same pattern recurs in coordinate geometry, vectors and trigonometry.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 6.1.

1. Triangle ABC is isosceles with $AB = AC$. If $\angle B = 48°$, find $\angle A$.
2. In a right-angled triangle the hypotenuse is 13 and one leg is 5. Find the other leg.
3. Two angles lie on a straight line. One is $118°$. Find the other.
4. Solve $3x + 12 = 180$.
5. Two right-angled triangles have equal hypotenuses and one pair of equal legs. Which congruence test applies?
6. Describe a circle in words, and name the segment joining two points on it.

REFRESHER · ISOSCELES TRIANGLES, CONGRUENCE AND CIRCLE WORDS

In an isosceles triangle the angles opposite the equal sides are equal, so base angles $48°$ give vertex angle $84°$. Pythagoras gives the third side: with hypotenuse 13 and one leg 5, the other is 12. Triangles are congruent by SAS, SSS, AAS or RHS, and corresponding parts are then equal.

A **circle** is the set of points at a fixed distance from a fixed point. The fixed point is the centre, the distance is the radius, and a segment joining two points on the circle is a **chord**. A chord through the centre is a diameter.

CONCEPT AND PROOF

6.1 Angles in circles

An angle in a circle is formed by two chords meeting on the circumference, or by two radii meeting at the centre. The angle subtends the arc between its endpoints. Every result here compares angles that subtend the same arc from different places.

The proofs begin by joining the centre to points on the circumference. Two radii are equal, so each triangle formed is isosceles: the theorems follow from one equality and one angle sum, not from memorising a list.

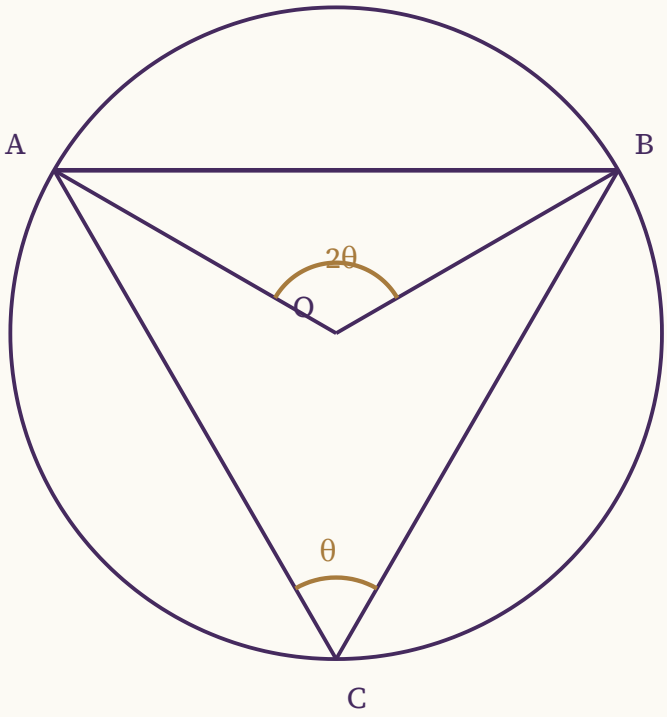
DEFINITIONS

Chord, arc, segment and concyclic points

A chord joins two points of a circle; a diameter is a chord through the centre. A chord divides the circle into a major arc and a minor arc, and the chord with either arc bounds a **segment**. Points on the same circle are concyclic. An angle **subtends** an arc when its sides pass through the arc's endpoints.

THEOREMS 6.1–6.3 · ANGLES STANDING ON THE SAME ARC

6.1 Angle at the centre. The angle subtended by an arc at the centre is twice the angle subtended by the same arc at any point on the remaining circumference. The central angle is the one standing on the arc not containing that point; it is reflex when the point lies on the minor arc.
6.2 Same segment. Angles subtended by the same arc at points on the same side of its chord are equal. Points on opposite sides give supplementary angles, not equal ones.
6.3 Semicircle. If AB is a diameter and C is another point on the circle, then $\angle ACB = 90°$.
Converse of 6.3 (true). If $\angle ACB = 90°$ for three points on a circle, then AB is a diameter. This converse is a separate statement and needs its own justification.



EXAMPLE **6.1** PROVING THE ANGLE-AT-THE-CENTRE THEOREM

In a circle with centre O, points A, B, C lie on the circumference and C is on the major arc AB. Prove that $\angle AOB = 2\angle ACB$.

SOLUTION

1. **Construction.** Join C to O and extend to meet the circle again at D. Write $\angle OCA = \alpha$ and $\angle OCB = \beta$, so $\angle ACB = \alpha + \beta$.
2. **Isosceles triangles.** $OA = OC$ (radii), so the base angles of $\triangle OAC$ are equal: $\angle OAC = \alpha$. Similarly, $\angle OBC = \beta$.
3. **Exterior angles.** An exterior angle equals the sum of the two interior opposite angles, so $\angle AOD = 2\alpha$ and $\angle BOD = 2\beta$.
4. **Add.** The ray OD lies inside $\angle AOB$, so $\angle AOB = 2\alpha + 2\beta = 2(\alpha + \beta) = 2\angle ACB$.

> **Reflex case.** If C is on the minor arc, the same argument gives the reflex angle at O equal to $2\angle ACB$; the ordinary angle is then $360^\circ - 2\angle ACB$.

If C and D lie on the same arc, both are half the same central angle, so they are equal (Theorem 6.2). If AB is a diameter, the central angle is 180°, so every angle in the semicircle is 90° (Theorem 6.3); conversely, a right angle at C makes A, O, B collinear.

EXAMPLE 6.2 USING THE CENTRE AND THE REFLEX ANGLE

In a circle with centre O, the smaller $\angle AOB = 128^\circ$. Point C lies on the major arc AB and point D on the minor arc. Find $\angle ACB$, the reflex angle AOB, and $\angle ADB$.

SOLUTION

C subtends the minor arc AB, whose central angle is 128°, so

$$\angle ACB = \tfrac{1}{2}(128^\circ) = 64^\circ.$$

The reflex angle at O is $360^\circ - 128^\circ = 232^\circ$. Point D subtends the major arc, so

$$\angle ADB = \tfrac{1}{2}(232^\circ) = 116^\circ.$$

MISCONCEPTION

"The central angle is always the one drawn inside the triangle, so it is less than 180°." When the circumference point lies on the minor arc, the relevant central angle is reflex. Always identify which arc the angle subtends before doubling or halving.

GUIDED PRACTICE

1. O is the centre and $\angle AOB = 150^\circ$; C is on the major arc. Find $\angle ACB$. (75°.)
2. $\angle ACB = 42^\circ$, with C on the major arc. Find $\angle AOB$. (84°.)
3. AB is a diameter and $\angle ABC = 37^\circ$. Find $\angle BAC$. (53°.)
4. C, D lie on the same side of chord AB, with $\angle ACB = 65^\circ$. Find $\angle ADB$, naming the theorem. (65°; same segment.)

EXERCISE 6A ANGLES AT THE CENTRE AND CIRCUMFERENCE

FLUENCY

1 O is the centre and $\angle AOB = 140^\circ$. Point C is on the major arc. Find $\angle ACB$.

2 $\angle ACB = 38^\circ$, with C on the major arc. Find $\angle AOB$ and the reflex angle AOB.

3 AB is a diameter and $\angle ABC = 52^\circ$. Find $\angle BAC$ and $\angle ACB$.

4 A, B, C, D are concyclic and C, D are on the same side of AB. Given $\angle ACB = 47^\circ$, find $\angle ADB$ and the central angle on the minor arc AB.

5 C lies on the minor arc AB and $\angle ACB = 125^\circ$. Find the reflex and smaller angles AOB.

6 In a circle with centre O, $\angle OAB = 35^\circ$. Find $\angle AOB$, then $\angle ACB$ for C on the major arc.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says $\angle AOB = 100^\circ$ and $\angle ACB = 130^\circ$ disprove the theorem because $100 \neq 2 \times 130$. Explain the error and identify the angle to double.

8 **Find the reason.** In the proof of Theorem 6.1, which reason justifies $OA = OC$, and which justifies $\angle AOD = 2\alpha$? Write both in full.

9 **Complete the proof.** Prove Theorem 6.2: if C, D are on the same arc AB, then $\angle ACB = \angle ADB$. Use Theorem 6.1 twice and name the central angle.

10 In a circle, $\angle ACB = 90^\circ$, $AC = 9$ cm and $BC = 12$ cm. Find AB and the radius; state which converse you used.

11 $\angle ACB = 3x + 10^\circ$ and $\angle ADB = 5x - 20^\circ$ are subtended by chord AB, with C, D on the same side. Find x and each angle.

12 Explain why an angle in the major segment is acute and one in the minor segment is obtuse, for a chord that is not a diameter.

APPLICATION AND EXTENSION

13 A circular table has diameter 1.2 m. A triangular brace ABC has AB as a diameter and $\angle ABC = 34^\circ$. Find the other two angles and explain why the brace is right-angled.

14 Survey points A, B are 40 m apart. From C, $\angle ACB = 30^\circ$. By finding the central angle of the circle through A, B, C, show that its radius is 40 m.

EXERCISE 6A · CONTINUED ANGLES AT THE CENTRE AND CIRCUMFERENCE

15 **Extension.** AB and CD are parallel chords of a circle, with A, B, C, D in order. Prove that $ABCD$ is an isosceles trapezium.

CONCEPT AND PROOF

6.2 Cyclic quadrilaterals

A quadrilateral whose vertices lie on a circle is cyclic; the circle is its circumcircle, and the vertices are named in order around it. The opposite-angle theorem follows from the angle-at-the-centre theorem, since each interior angle subtends the arc between the two non-adjacent vertices, and the two arcs concerned are complementary.

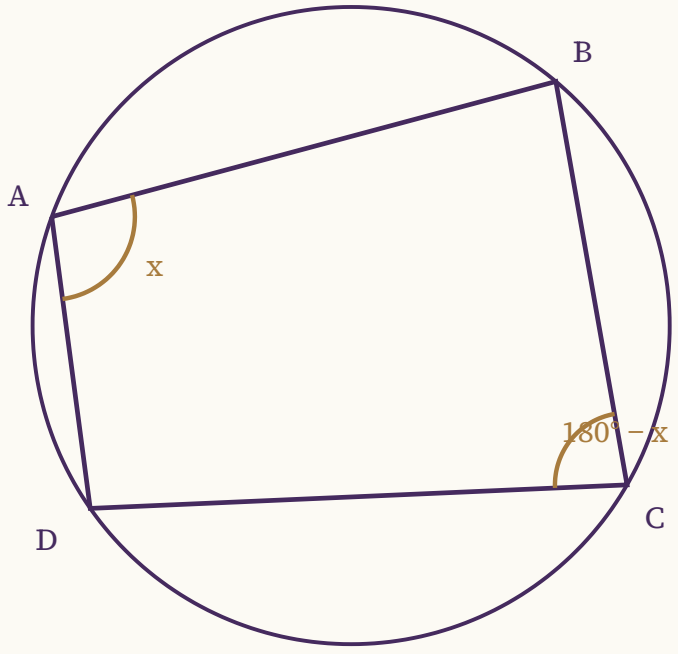
To recognise a cyclic quadrilateral, look for supplementary opposite angles, equal angles on the same side of a common segment, two right angles on the same segment, or an isosceles trapezium. Each test is a converse, so the configuration matters.

THEOREM 6.4 · OPPOSITE ANGLES OF A CYCLIC QUADRILATERAL

In a cyclic quadrilateral, each pair of opposite angles sums to 180°.

Proof. Let $ABCD$ be cyclic with centre O, and write $\angle DAB = \alpha$, $\angle BCD = \gamma$. These angles subtend arcs BCD and BAD, whose central angles are 2α and 2γ. The two arcs make the whole circle, so $2\alpha + 2\gamma = 360^\circ$, giving $\alpha + \gamma = 180^\circ$. The other pair is similar.

Converse of 6.4 (true). If a pair of opposite angles of a quadrilateral sums to 180°, the quadrilateral is cyclic. This converse lets you prove that four points lie on a circle; it is not a definition.



THEOREM 6.5 · EXTERIOR ANGLE OF A CYCLIC QUADRILATERAL

If one side is extended, the exterior angle formed equals the interior opposite angle. For example, if AB is extended beyond B to E, then $\angle CBE = \angle ADC$.

Proof. $\angle ABC + \angle CBE = 180^\circ$ (angles on a straight line) and $\angle ABC + \angle ADC = 180^\circ$ (opposite angles). Subtracting the common angle gives $\angle CBE = \angle ADC$. The converse is also true: if an exterior angle equals the interior opposite angle, the quadrilateral is cyclic.

EXAMPLE 6.3 INTERIOR AND EXTERIOR ANGLES TOGETHER

$ABCD$ is cyclic, with vertices in order. $\angle A = 2x + 10^\circ$, $\angle C = 3x + 20^\circ$, and AB is extended beyond B to E with $\angle CBE = 72^\circ$. Find x and all four interior angles.

SOLUTION

Opposite angles are supplementary:

$$(2x + 10) + (3x + 20) = 180, \qquad 5x + 30 = 180, \qquad x = 30.$$

Hence $\angle A = 70^\circ$ and $\angle C = 110^\circ$.

Angles on the straight line ABE give $\angle ABC = 180^\circ - 72^\circ = 108^\circ$. The exterior-angle theorem (or the opposite-angle theorem) then gives

$$\angle ADC = 72^\circ.$$

EXAMPLE 6.4 RECOGNISING A CYCLIC QUADRILATERAL FROM RIGHT ANGLES

Points C, D lie on the same side of AB, and $\angle ACB = \angle ADB = 90^\circ$. Prove that A, B, C, D are concyclic and that AB is a diameter. If $AB = 10$ cm, find the radius.

SOLUTION

In the circle through A, B, C, the angle at C is 90°, so the converse of the semicircle theorem makes AB a diameter. In the circle through A, B, D, the angle at D is 90°, so AB is also a diameter of that circle.

A diameter determines a unique circle, so the two circles coincide; hence D lies on the circle through A, B, C, and the four points are concyclic. The radius is 5 cm.

MISCONCEPTION

"A quadrilateral is cyclic if it has any pair of equal angles." Equal angles must be subtended by the same chord and lie on the same side of it. A kite has a pair of equal opposite angles but is not cyclic in general. Use supplementary opposite angles as the reliable test.

GUIDED PRACTICE

1. $ABCD$ is cyclic and $\angle A = 85^\circ$. Find $\angle C$. (95°.)
2. $ABCD$ is cyclic and $\angle B = 112^\circ$. Find $\angle D$. (68°.)
3. In cyclic $ABCD$, AD is extended beyond D to E and $\angle CDE = 75^\circ$. Find $\angle ABC$. (75°.)
4. A quadrilateral has angles $2x, 3x, 4x, 3x$ in order. Find x and decide whether it is cyclic. ($x = 30^\circ$; yes, opposite pairs sum to 180°.)

EXERCISE 6B

CYCLIC QUADRILATERALS AND THEIR CONVERSES

FLUENCY

1 $ABCD$ is cyclic in order. If $\angle A = 85^\circ$, find $\angle C$.

2 In cyclic $ABCD$, $\angle B = 112^\circ$. Find $\angle D$.

EXERCISE 6B · CONTINUED

CYCLIC QUADRILATERALS AND THEIR CONVERSES

3 In cyclic $ABCD$, AD is extended beyond D to E. If $\angle CDE = 75°$, find $\angle ABC$.

4 A quadrilateral has angles $2x, 3x, 4x, 3x$ in order. Find x, all four angles, and whether it is cyclic.

5 In cyclic $ABCD$, $\angle A = x + 20°$ and $\angle C = 2x - 5°$. Find x, $\angle A$ and $\angle C$.

6 $ABCD$ is an isosceles trapezium with $AB \parallel CD$ and $\angle A = 68°$. Find the other angles and explain why it is cyclic.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student claims any quadrilateral with a pair of equal opposite angles is cyclic. Give a counterexample and state the correct test.

8 **Find the reason.** In the proof of Theorem 6.4, why do the central angles add to $360°$, and which theorem writes each central angle as twice an interior angle?

9 **Complete the proof.** Prove the exterior-angle theorem for cyclic $ABCD$ with AB extended to E: use a straight-line angle and the opposite-angle theorem.

10 Must each quadrilateral be cyclic? (a) a rectangle; (b) a non-square rhombus; (c) a kite with two opposite right angles; (d) a non-rectangular parallelogram. Justify each.

11 In quadrilateral $ABCD$, $\angle A + \angle C = 180°$. Explain why D lies on the circle through A, B, C, referring to the uniqueness of the circle through three non-collinear points.

12 C, D lie on the same side of AB, and $\angle ADB = \angle ACB = 90°$. Prove A, B, C, D are concyclic and AB is a diameter. If $CD = 6$ cm and $AB = 10$ cm, find the distance from the centre to CD.

APPLICATION AND EXTENSION

13 Four boundary posts A, B, C, D lie on a circular lawn, with $\angle ABC = 96°$ and $\angle CDA = 84°$. Explain why they are concyclic. If $\angle CDA$ were $80°$ instead, would they still be concyclic? Explain.

14 A stained-glass panel is a cyclic quadrilateral with angles $95°, 80°, 100°, 85°$ in some order. Find a cyclic order and explain why order matters.

15 **Extension.** Prove that a parallelogram is cyclic if and only if it is a rectangle.

CONCEPT AND PROOF

6.3 Tangents and radii

A tangent touches a circle at exactly one point, the point of contact; a secant meets it twice. The tangent is the limiting position of a secant as its intersections merge. That image explains, but does not prove, the right angle between a tangent and the radius at the point of contact.

DEFINITIONS

Tangent, point of contact, secant and incircle

A **tangent** is a line meeting a circle at exactly one point, the **point of contact**. A **secant** meets it at two points. A **tangent segment** runs from an external point to the point of contact. An **incircle** of a triangle is tangent to all three sides.

THEOREM 6.6 · TANGENT AND RADIUS

A tangent is perpendicular to the radius at the point of contact.

Proof by contradiction. Suppose the tangent at T is not perpendicular to OT. Drop a perpendicular from O to the tangent at $N \neq T$. In right triangle OTN, hypotenuse OT is longer than ON, so N lies inside the circle. A line through an interior point meets the circle twice, contradicting the tangent meeting it only at T. Hence OT is perpendicular to the tangent.

Converse of 6.6 (true). If a line through T on a circle is perpendicular to OT, it is tangent at T: for any other point Q on the line, $OQ^2 = OT^2 + TQ^2 > OT^2$, so Q is outside the circle.

EXAMPLE **6.5** A TANGENT LENGTH AND AN ANGLE

PT is tangent to a circle of radius 8 cm at T, and $OP = 17$ cm. Find PT and $\angle TPO$ to 1 decimal place.

SOLUTION

By Theorem 6.6, $\angle OTP = 90°$. Pythagoras in $\triangle OTP$ gives

$$PT = \sqrt{17^2 - 8^2} = \sqrt{225} = 15 \text{ cm}.$$

Also $\sin \angle TPO = \dfrac{8}{17}$, so

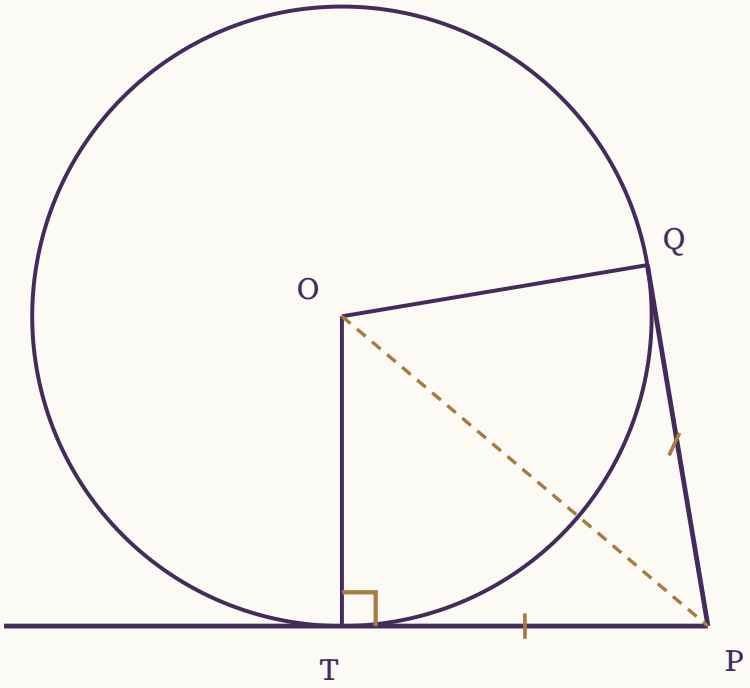
$$\angle TPO = \sin^{-1}\left(\frac{8}{17}\right) = 28.1° \text{ (1 d.p.)}.$$

THEOREM 6.7 · TWO TANGENTS FROM AN EXTERNAL POINT

From an external point P, the tangent segments PT and PQ are equal. The line OP bisects the angle between the tangents, $\angle TPQ$, and the angle between the radii, $\angle TOQ$.

Proof. $OT = OQ$ (radii), OP is common, and $\angle OTP = \angle OQP = 90°$ by Theorem 6.6. Triangles OTP and OQP are congruent by RHS. Hence $PT = PQ$, $\angle TPO = \angle QPO$ and $\angle TOP = \angle QOP$.

In an incircle, the two tangent segments from each vertex are equal. Writing them x, y, z gives three linear equations; and since joining the incentre to the vertices splits the triangle into three triangles of height r, the area is rs, where s is the semiperimeter.



EXAMPLE **6.6** TWO TANGENTS FROM ONE POINT

Tangents PT and PQ are drawn from an external point P. Given $\angle TPQ = 64°$, find $\angle TOQ$, $\angle TPO$ and $\angle TOP$. If $PT = 9$ cm, find PQ.

SOLUTION

The quadrilateral $OTPQ$ has right angles at T and Q, so

$$\angle TOQ = 360° - 90° - 90° - 64° = 116°.$$

By Theorem 6.7, OP bisects the angle between the tangents: $\angle TPO = 32°$. In right triangle OTP,

$$\angle TOP = 180° - 90° - 32° = 58°.$$

Also $PQ = PT = 9$ cm.

MISCONCEPTION

"The line from the external point to the centre is perpendicular to the tangent." It is the radius to the point of contact that is perpendicular to the tangent, not OP. In Figure 6.3, $OT \perp PT$, while OP bisects $\angle TPQ$. Using the wrong line gives the wrong right angle.

GUIDED PRACTICE

1. PT is tangent, $OT = 9$ and $OP = 15$. Find PT. (12.)
2. Two tangents from P each have length 7 cm. Find the other tangent. (7 cm.)
3. Two tangents from P make $70°$. Find the angle between the radii. (110°.)
4. A right-angled triangle has legs 6 and 8. Find its inradius. (2.)

EXERCISE 6C

TANGENTS, RADII AND INCIRCLES

FLUENCY

1 PT is tangent at T. Given $OP = 15$ and radius 9, find PT.

2 PT is tangent at T. Given $PT = 24$ and radius 7, find OP.

3 Tangents from P touch at T, Q, and $\angle TPQ = 70°$. Find $\angle TOQ$ and $\angle TPO$.

4 Tangents from P have $\angle TOP = 40°$. Find $\angle TPO$, $\angle TPQ$ and $\angle TOQ$.

5 PT is tangent at T, with $OT = 6$ and $PT = 8$. Find OP and $\angle OTP$.

6 A 5-12-13 triangle has an incircle. Find its radius.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes "since PT is tangent, $OP \perp PT$". Explain why this is wrong, name the correct perpendicular, and describe what OP does.

8 **Find the reason.** In the tangent–radius proof, why is $ON < OT$, and why does that put N inside the circle? Which assumption is contradicted?

9 **Complete the proof.** Prove that tangent segments PT and PQ are equal. Use the right angles, common side OP, equal radii, and name the congruence test.

10 Prove that OP bisects $\angle TOQ$ as well as $\angle TPQ$. Which pair of angles in the congruent triangles gives this?

11 A person stands on a 100 m cliff; Earth's radius is about 6400 km. Find the distance to the horizon along a tangent, to the nearest kilometre. (Convert 100 m to km first.)

12 A triangle has sides $13, 14, 15$ cm and area 84 cm^2. Find its inradius and the three tangent lengths from its vertices.

APPLICATION AND EXTENSION

13 A roundabout has radius 6 m. A straight path is tangent at T and passes through a gate P with $PT = 8$ m. How far is the gate from the centre?

14 A triangular garden has sides $5, 12, 13$ m. A circular fountain is tangent to all three sides. Find its radius and the area of the water surface, in terms of π.

EXERCISE 6C · CONTINUED TANGENTS, RADII AND INCIRCLES

15 **Extension.** From external point P, tangents PT and PQ touch a circle at T, Q. Prove that $\angle TPQ + \angle TOQ = 180°$, and that OP is the perpendicular bisector of TQ.

CONCEPT AND PROOF

6.4 Chords, secants and the alternate segment

Two results connect lengths rather than angles. Intersecting chords create four segments whose products are equal, and a tangent with a secant creates a length whose square equals a product. Both are proved by similar triangles, so the angle theorems of Section 6.1 keep returning.

Chords also give a practical way to find a radius. The perpendicular from the centre bisects a chord, producing a right triangle with the radius as hypotenuse. This is the calculation behind the opening problem.

THEOREMS 6.8–6.9 · CHORDS AND DISTANCE FROM THE CENTRE

6.8. The perpendicular from the centre to a chord bisects the chord. Conversely, the line from the centre to the midpoint of a chord is perpendicular to it, so the perpendicular bisector of a chord passes through the centre.

6.9. Equal chords are equidistant from the centre; conversely, chords equidistant from the centre are equal.

Proof of 6.8. Let $OM \perp AB$. Then $OA = OB$ (radii), OM is common, and $\angle OMA = \angle OMB = 90°$. Triangles OMA and OMB are congruent by RHS, so $AM = MB$. Equal chords have equal half-chords; comparing the resulting right triangles by RHS gives equal distances, proving 6.9.

EXAMPLE 6.7 CHORDS AND DISTANCES FROM THE CENTRE

A circle has radius 13 cm. (a) Find the distance from the centre to a chord of length 24 cm. (b) A second chord has length 10 cm; find its distance and state which chord is closer to the centre.

SOLUTION

(a) The perpendicular bisects the chord, so each half is 12 cm. The distance is

$$d = \sqrt{13^2 - 12^2} = 5 \text{ cm}.$$

(b) The half-chord is 5 cm, so the distance is $\sqrt{13^2 - 5^2} = 12$ cm. The 24 cm chord is closer to the centre.

THEOREM 6.10 · INTERSECTING CHORDS

If chords AB and CD intersect at P inside a circle, then

$$PA \cdot PB = PC \cdot PD.$$

Proof outline. $\angle APC = \angle DPB$ (vertically opposite) and $\angle PAC = \angle PDB$ (angles in the same segment on arc BC). Hence $\triangle PAC \sim \triangle PDB$ by AA, so $\dfrac{PA}{PD} = \dfrac{PC}{PB}$, which gives $PA \cdot PB = PC \cdot PD$.

EXAMPLE **6.8** INTERSECTING CHORDS: PROOF AND CALCULATION

Chords AB and CD intersect at P inside a circle. (a) Explain why $\triangle PAC \sim \triangle PDB$. (b) Given $PA = x$, $PB = x + 5$, $PC = 4$, $PD = 9$, find x.

SOLUTION

(a) $\angle APC = \angle DPB$ (vertically opposite) and $\angle PAC = \angle PDB$ (same segment, arc BC). Two pairs of equal angles give similarity.

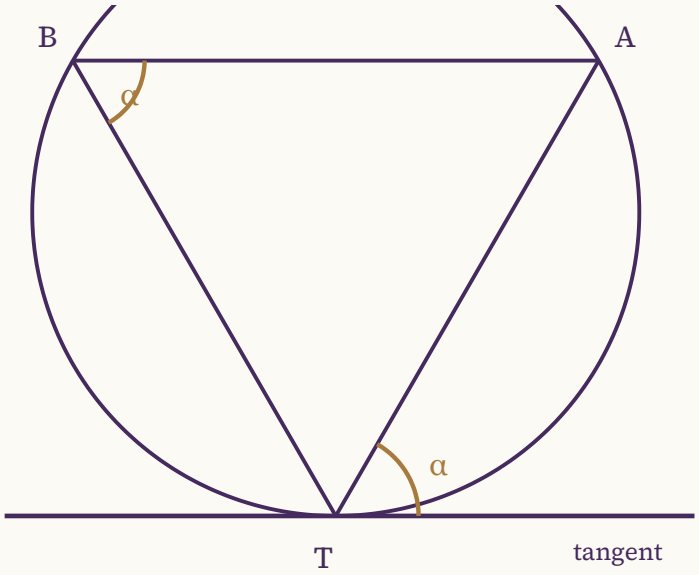
(b) By Theorem 6.10, $x(x + 5) = 36$, so $x^2 + 5x - 36 = 0$, i.e. $(x + 9)(x - 4) = 0$. A length is positive, so $x = 4$; hence $PA = 4$ and $PB = 9$.

THEOREM 6.11 · THE ALTERNATE SEGMENT

The angle between a tangent and a chord at the point of contact equals the angle in the alternate segment. If PT is tangent at T, then the angle between PT and TA equals $\angle TBA$, where B lies on the arc opposite the tangent ray.

Proof. Let $\angle OTA = \alpha$. Since $OT = OA$, triangle OTA is isosceles, so $\angle OAT = \alpha$ and $\angle TOA = 180^\circ - 2\alpha$. Thus $\angle TBA = \frac{1}{2}(180^\circ - 2\alpha) = 90^\circ - \alpha$. Since $OT \perp PT$, the tangent–chord angle is also $90^\circ - \alpha$. The obtuse angle on the other side belongs to the other segment, since angles on a straight line sum to 180°.

Converse of 6.11 (true). If a line through T makes an angle with chord TA equal to the angle in the alternate segment, the line is tangent at T.



THEOREM 6.12 · TANGENT AND SECANT

If PT is tangent at T and secant PAB meets the circle at A, B, with A between P and B, then

$$PT^2 = PA \cdot PB.$$

Proof. $\angle PTA = \angle TBA$ (alternate segment) and $\angle TPA = \angle TPB$ (same angle). Hence $\triangle PTA \sim \triangle PBT$ by AA, so $\dfrac{PT}{PB} = \dfrac{PA}{PT}$, giving $PT^2 = PA \cdot PB$.

EXAMPLE 6.9 THE ALTERNATE SEGMENT AND A TANGENT–SECANT CALCULATION

(a) A tangent at T makes 62° with chord TA. Find the angle in the alternate segment, the central angle on TA, and the angle between TA and radius OT. (b) From P, tangent $PT = 12$ cm and secant PAB has $PA = 8$ cm. Find PB and AB.

SOLUTION

(a) The alternate-segment angle is 62°, and the central angle is $2 \times 62^\circ = 124^\circ$. Since the radius is perpendicular to the tangent, the angle between TA and OT is $90^\circ - 62^\circ = 28^\circ$.

(b) $12^2 = 8 \times PB$, so $PB = 18$ cm and $AB = 18 - 8 = 10$ cm.

> **MISCONCEPTION**
>
> **"The tangent–chord angle equals the angle in the segment on the same side."** It equals the angle in the *alternate* segment, on the other side of the chord from the tangent ray. Choosing the same side gives the supplementary angle.

GUIDED PRACTICE

1. A 16 cm chord lies in a circle of radius 10 cm. Find its distance from the centre. (6 cm.)
2. Chords intersect at P, with $PA = 3$, $PB = 8$, $PC = 4$. Find PD. (6.)
3. A tangent–chord angle is 35°. Find the alternate-segment angle. (35°.)
4. PT is tangent, $PA = 4$ and $AB = 5$. Find PT. (6.)

EXERCISE 6D CHORDS, INTERSECTING CHORDS AND THE ALTERNATE SEGMENT

FLUENCY

1 A 24 cm chord lies in a circle of radius 13 cm. Find its distance from the centre.

2 A circle has radius 10 cm and a 16 cm chord. Find the distance from the centre to the chord.

3 Equal chords of length 18 cm lie in a circle of radius 15 cm. Find the distance of each from the centre.

4 Two chords are each 7 cm from the centre. One is 20 cm long. Find the other.

5 Chords AB, CD intersect at P. Given $PA = 3$, $PB = 8$, $PC = 4$, find PD.

6 PT is tangent and PAB is a secant with $PA = 4$, $PB = 9$. Find PT.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says every line from the centre to a chord bisects it. Explain why this is false, state the correct condition, and give the special case in which it is true.

8 **Find the reason.** In the chord-bisector proof, which sides and angles are equal, which congruence test is used, and why is the hypotenuse the radius?

9 **Complete the proof.** Prove that equal chords are equidistant from the centre, using $OM \perp AB$, $ON \perp CD$ and congruent right triangles.

10 A tangent–chord angle is 58°. Find the alternate-segment angle, the central angle on the chord, and the angle between the chord and the radius at contact.

11 PT is tangent, PAB is a secant, $PA = 4$ and $AB = 5$. Find PT.

12 In a circle of radius 13 cm, chords of length 24 cm and 10 cm are drawn. Find their distances from the centre and explain how chord length relates to distance.

APPLICATION AND EXTENSION

13 A circular table has radius 1 m. A chord strip is 1.6 m long. How far is it from the centre? A second strip is 1.2 m long; which is closer to the centre?

14 A circular arch is 8 m wide and rises 2.4 m at the centre. Find its radius. A van 2.2 m wide and 2.3 m high passes centrally; show that its top corners do not clear the arch, and state the vertical shortfall at each corner.

EXERCISE 6D · CONTINUED CHORDS, INTERSECTING CHORDS AND THE ALTERNATE SEGMENT

15 **Extension.** Prove the tangent–secant theorem: if PT is tangent and PAB is a secant, then $PT^2 = PA \cdot PB$. Use the alternate segment theorem and similar triangles.

METHOD

6.5 Constructing and writing proofs

A proof is a finite chain of statements from the given information to the required conclusion, each statement justified by a definition, a known theorem, a construction or a congruence. A theorem has a hypothesis (the "if" part) and a **conclusion** (the "then" part). The converse swaps them, and is a new statement that may be false.

Constructions such as joining two points, extending a line or dropping a perpendicular are allowed, but each must be described. They add no new assumptions. A two-column proof lists statements beside reasons; a paragraph proof writes the same chain in continuous prose. Both must name a reason for every step.

REASON BANK · ACCEPTABLE JUSTIFICATIONS

Given; **radii of the same circle**; **angles on a straight line**; **vertically opposite angles**; **triangle angle sum**; **isosceles triangle**; **congruence** (SAS, SSS, RHS, AAS); **tangent perpendicular to radius**; **angle at the centre** or **same segment**; **cyclic quadrilateral**. A reason is admissible only when its configuration is present.

EXAMPLE **6.10** A TWO-COLUMN PROOF

From an external point P, tangents PT and PQ touch a circle at T and Q. Prove that $\angle TPQ + \angle TOQ = 180^\circ$.

SOLUTION

STATEMENT	REASON
$OT \perp PT$, $OQ \perp PQ$, so $\angle OTP = \angle OQP = 90^\circ$	Tangent perpendicular to radius; definition of perpendicular
$\angle OTP + \angle TPQ + \angle PQO + \angle QOT = 360^\circ$	Angle sum of quadrilateral $OTPQ$
$90^\circ + \angle TPQ + 90^\circ + \angle TOQ = 360^\circ$	Substitution
$\angle TPQ + \angle TOQ = 180^\circ$	Subtract 180° from both sides

Every line states a fact and a reason. The proof does not depend on the diagram being to scale.

EXAMPLE **6.11** A THEOREM, ITS CONVERSE AND A COUNTEREXAMPLE

Consider the statement: "If a quadrilateral is a rectangle, then its diagonals are equal." (a) Is it true? (b) Write its converse and decide whether it is true. (c) Give a counterexample if it is false.

SOLUTION

(a) True. In a rectangle, the two diagonals are equal because each is the hypotenuse of a right triangle with the same two side lengths.

(b) Converse: "If a quadrilateral has equal diagonals, then it is a rectangle." This is false.

(c) An isosceles trapezium that is not a rectangle has equal diagonals but is not a rectangle. Its diagonals are equal by symmetry, yet its base angles are not $90°$.

> **Common logical errors:** assuming what is to be proved; treating a diagram or measurement as proof; using a theorem backwards without checking its converse; adding a construction point without saying how it is defined; and claiming a triangle is equilateral from two equal radii alone.

MISCONCEPTION

"The diagram shows it, so it is proved." A diagram can suggest a relationship, but it cannot establish that the relationship holds in every case. A proof must use stated properties, not appearance. This is why the reflex case of Theorem 6.1 needs a separate argument: the standard diagram does not show it.

GUIDED PRACTICE

1. Identify the hypothesis and conclusion of "If a quadrilateral is cyclic, then opposite angles sum to $180°$". (Hypothesis: cyclic; conclusion: opposite angles sum to $180°$.)
2. Write the converse of "If AB is a diameter, then $\angle ACB = 90°$" and state whether it is true. (If $\angle ACB = 90°$, then AB is a diameter; true.)
3. Name the reason for $OA = OB$. (Radii of the same circle.)
4. A proof begins "Triangle ABC is isosceles because it looks symmetrical". What is wrong? (Appearance is not a reason; equal sides or angles must be given or proved.)

EXERCISE 6E

PROOF, REASONS AND CONVERSES

FLUENCY

1 Identify the hypothesis and conclusion: (a) if a chord is a diameter, it is the longest chord; (b) if a quadrilateral is a rectangle, it is cyclic.

2 Write the converse of each statement in Question 1 and decide whether it is true.

3 Name the reason for each step: (a) $OA = OB$; (b) $\angle AOB = 2\angle ACB$; (c) $\angle ABC + \angle CBE = 180°$.

4 Write a two-column proof that an angle in a semicircle is $90°$, using Theorem 6.1.

5 Match each statement to a reason: (a) $\angle APC = \angle DPB$; (b) $OA = OB$; (c) $OT \perp PT$; (d) $\angle A + \angle C = 180°$; (e) $\angle ACB = \angle ADB$. Reasons: vertically opposite, radii, tangent–radius, cyclic quadrilateral, same segment.

6 Write a two-column proof that tangent segments from an external point are equal.

UNDERSTANDING AND REASONING

7 **Error analysis.** A proof of the angle-at-the-centre theorem begins "Since $\triangle OAC$ is isosceles, $\angle AOC = 2\angle OCA$". Identify two errors and correct them.

8 Explain why "the diagonals of a cyclic quadrilateral bisect each other" looks true in a rectangle but is false in general. Give a counterexample.

9 Give a counterexample to "if a quadrilateral has perpendicular diagonals, then it is cyclic". Is the converse true?

10 Prove in two columns that equal chords are equidistant from the centre.

EXERCISE 6E · CONTINUED PROOF, REASONS AND CONVERSES

11 Prove the converse: chords equidistant from the centre are equal.

12 Explain the difference between a theorem and its converse, using a true theorem whose converse is false.

APPLICATION AND EXTENSION

13 Write a paragraph proof that the angle at the centre is twice the angle at the circumference for the acute case. Include the construction and the reflex case in one sentence.

14 Write a paragraph proof that if a tangent at T is parallel to chord AB, then $TA = TB$.

15 **Extension.** Prove that a cyclic parallelogram is a rectangle. State clearly where the cyclic property and the parallelogram property are used.

PATTERN AND GENERALISATION

6.6 Investigation: circle theorems under transformation

A theorem covers every configuration satisfying its hypothesis. Here you vary a point on an arc, then reflect and rotate the figure, to decide which measurements are forced and which are accidents of one drawing.

INVESTIGATION 6 · WHAT STAYS THE SAME?

Moving, reflecting and rotating a circle theorem

1. Construct a circle with centre O and radius 5 cm. Draw a chord AB with $\angle AOB = 100°$. Place C on the major arc and measure $\angle ACB$.
2. Move C to five different positions on the major arc. Record $\angle ACB$, CA and CB in a table. Which quantities change and which stay the same?
3. Move C to the minor arc. Measure the new angle. How does it relate to the first angle, and which central angle explains it?
4. Place D on the same major arc as the original C. Compare $\angle ADB$ with the values from step 2. State a conjecture.
5. Reflect the figure in AB. Where does C go, what is the angle at its image, and what happens to the major and minor arcs?
6. Rotate the figure about O by $40°$. Measure the new angles and chord lengths. Which properties are invariant under this rotation?
7. Prove your conjecture from step 4 using two isosceles triangles. Identify exactly where the equality of radii and the exterior-angle property are used.

Generalise. For a fixed chord AB, every point on the same arc gives the same angle, half the central angle on the opposite arc. Moving the vertex to the other arc keeps the chord but replaces the angle by its supplement, because the two angles together subtend the whole circumference; rotation about the centre preserves both chord and angle. Explain why this shows the $2:1$ ratio belongs to the circle, not to one drawing.

MODELLING · CRITERION D

The arch, revisited

Return to the opening problem: a circular arch spans 8 m and rises 2.4 m.

1. **Model.** Let the half-span be $a = 4$ m and the rise be h. Use the chord-bisector theorem and Pythagoras to show that the radius is $r = \dfrac{a^2 + h^2}{2h}$. Evaluate it for $h = 2.4$.
2. **Calculate.** Find the height of the arch 1.1 m from the centre and the vertical shortfall for a 2.3 m van.
3. **Redesign.** Find the smallest rise, to the nearest 0.05 m, that gives 2.4 m clearance at 1.1 m from the centre.
4. **Evaluate.** State two assumptions of the model and explain how each could change the required rise.

CUMULATIVE

6.7 Mixed practice

MIXED PRACTICE 6

CIRCLE GEOMETRY AND GEOMETRIC PROOF

FLUENCY AND UNDERSTANDING

1. O is the centre and $\angle AOB = 76^\circ$. Find $\angle ACB$ for C on the major arc, and the reflex angle AOB.
2. In cyclic $ABCD$, $\angle A = 3x + 8^\circ$ and $\angle C = 5x + 12^\circ$. Find x, $\angle A$ and $\angle C$.
3. PT is tangent at T, with $OP = 25$ and radius 7. Find PT.
4. A 10 cm chord lies in a circle of radius 13 cm. Find its distance from the centre.
5. Chords AB, CD intersect at P, with $PA = 6$, $PB = 4$, $PC = 3$. Find PD.
6. A tangent–chord angle is 47°. Find the alternate-segment angle and the central angle on the chord.
7. Two tangents from P make $\angle TPQ = 58^\circ$. Find $\angle TOQ$ and $\angle TPO$.
8. AB is a diameter and $\angle BAC = 28^\circ$. Find $\angle ABC$.
9. In cyclic $ABCD$, the exterior angle at A is 118°. Find the interior opposite angle at C.
10. PT is tangent, PAB is a secant, $PT = 8$ and $PA = 4$. Find PB and AB.
11. A quadrilateral has angles $95^\circ, 85^\circ, 100^\circ, 80^\circ$. Give one order that is cyclic and one that is not, and justify both.

REASONING AND APPLICATION

12. **Complete the proof.** Prove that the angle at the centre is twice the angle at the circumference for the acute case. Include the construction, both isosceles triangles and the exterior-angle step.
13. **Error analysis.** C, D lie on opposite sides of AB and $\angle ACB = \angle ADB = 40^\circ$. A student concludes that A, B, C, D are concyclic. Explain why this is invalid and what extra condition would make it valid.
14. In a circle of radius 13 cm, a chord is 24 cm long. A second chord is 12 cm from the centre. Find the length of the second chord.
15. In cyclic $ABCD$, $\angle A : \angle B : \angle C = 2 : 3 : 4$. Find all four angles.
16. Points A, B are 12 km apart and $\angle ASB = 90^\circ$. Find the radius of the circle through A, B, S. If $\angle SAB = 35^\circ$, find SB to 3 significant figures.
17. A circular arch spans 10 m and rises 2.5 m. Find its radius and its height 2 m from the centre.
18. **Prove** the intersecting-chords theorem: if chords AB, CD meet at P inside a circle, then $PA \cdot PB = PC \cdot PD$.
19. A 5-12-13 triangle has an incircle. Find its radius and the three tangent lengths from its vertices.

MIXED PRACTICE 6 · CONTINUED

20 **Extension.** Two equal chords of a circle are drawn. Prove that the minor arcs they cut off are equal.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Angles, tangents and chords

Instructions. Answer all questions; show working. Calculators permitted.

1 In a circle with centre O, the smaller $\angle AOB = 118°$. Point C is on the major arc and D on the minor arc. Find $\angle ACB$, the reflex angle AOB, and $\angle ADB$. [4]

2 In cyclic $ABCD$, $\angle A = 4x - 10°$ and $\angle C = 2x + 40°$. Find x, $\angle A$ and $\angle C$. The exterior angle at B is $115°$; find $\angle D$. [4]

3 PT is tangent at T, with $OP = 13$ cm and radius 5 cm. Find PT and $\angle TPO$ to 1 decimal place. [4]

4 A 16 cm chord lies in a circle of radius 10 cm. Find its distance from the centre. Chords AB, CD intersect at P, with $PA = 6$, $PB = 8$ and $PC = 4$; find PD. [5]

5 Complete the proof that an angle in a semicircle is $90°$: state the central angle when AB is a diameter, apply Theorem 6.1, and explain why the converse is also true. [3]

Evidence: correct angle, reflex and tangent values; a solved equation; a chord distance and intersecting-chord product; a complete semicircle argument.

CRITERION B · INVESTIGATING PATTERNS — 40 minutes · 24 marks

The arc of constant angle

Instructions. Work independently. Use a table for measurements and justify every generalisation. Draw $AB = 6$ cm.

1 Construct five points C on one side of AB with $\angle ACB = 60°$. Record the perpendicular distance from each C to AB and the lengths CA, CB. [6]

2 Describe the locus of C. Is it a line, an arc or a circle? Explain why there are two symmetric loci, one on each side of AB. [4]

3 Find the radius and the centre of the arc for $\angle ACB = 60°$. Verify that your measured points lie on it. [5]

4 Repeat for $\angle ACB = 90°$. What happens to the centre and the radius? [3]

5 Predict the radius for $\angle ACB = 30°$, then test your prediction by construction. [3]

6 Generalise: express the radius in terms of AB and $\theta = \angle ACB$. Prove your formula using the isosceles triangle formed by the centre and the chord. [3]

Evidence: a systematic table, a described locus, a correct centre and radius, a tested prediction, and a justified formula.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining why the centre angle doubles

Instructions. Write a 300–400 word explanation for a student who knows isosceles base angles but has not met the angle-at-the-centre theorem. Your explanation must: (i) state the theorem and identify the arc; (ii) describe the construction of D; (iii) use two isosceles triangles; (iv) justify the exterior-angle step; (v) explain the reflex case. Include a labelled diagram and a conclusion on why the result is not merely a measurement.

Evidence: a clear statement with the correct arc, an accurate construction, two isosceles triangles, a valid exterior-angle argument, a separate reflex case, and suitable language.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

A circular stained-glass panel

Instructions. A designer is cutting a central quadrilateral pane from a circular window of radius 1.5 m. The four corners of the pane lie on the rim, and the sides are straight lead lines.

1. The proposed angles, in order, are $80°, 100°, 95°, 85°$. Show that this order cannot be cyclic, and give an order that can. [4]
2. Explain why a cyclic quadrilateral can be inscribed in the circular window, and why the order of the angles matters. [4]
3. A chord of the window is 2.4 m long; find its distance from the centre. A second chord is 1.8 m long; find its distance and state which is closer to the centre. [4]
4. State two assumptions of the design — for example lead width and angle accuracy — and explain how each affects the glass area. [6]
5. A second proposal lists the angles $70°, 110°, 70°, 110°$ in order. Evaluate both proposals for symmetry and for whether they can be cyclic, and recommend one with a reason. [6]

Evidence: correct cyclic and non-cyclic orders, a justified inscription argument, both chord distances, assumptions with consequences, and a critical recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- The angle at the centre is twice the angle at the circumference on the same arc; the reflex case uses the reflex central angle.
- Angles in the same segment are equal; an angle in a semicircle is $90°$, and conversely a right angle makes the hypotenuse a diameter.
- Opposite angles of a cyclic quadrilateral sum to $180°$; an exterior angle equals the interior opposite angle. The converses are valid tests for concyclic points.
- A tangent is perpendicular to the radius at the point of contact. Two tangents from an external point are equal, and the line to the centre bisects both relevant angles.
- The perpendicular from the centre bisects a chord; equal chords are equidistant from the centre. For intersecting chords, $PA \cdot PB = PC \cdot PD$.
- The tangent–chord angle equals the angle in the alternate segment. For a tangent and secant, $PT^2 = PA \cdot PB$.
- A proof is a chain of justified statements. A theorem and its converse are different implications; a diagram or measurement is not a proof.

$$\angle AOB = 2\angle ACB, \qquad \angle A + \angle C = 180°, \qquad PA \cdot PB = PC \cdot PD, \qquad PT^2 = PA \cdot PB$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can state and prove the angle-at-the-centre theorem.
- I can handle the reflex case correctly.
- I can use angles in the same segment and in a semicircle.
- I can prove and use the opposite-angle property of a cyclic quadrilateral.
- I can use the exterior-angle property and recognise cyclic quadrilaterals.
- I can prove that a tangent is perpendicular to the radius.
- I can use equal tangents and solve incircle problems.
- I can find chord lengths and distances from the centre.
- I can use the intersecting-chords theorem.
- I can use the alternate-segment and tangent–secant theorems.
- I can write a two-column or paragraph proof with reasons.
- I can distinguish a theorem from its converse and give a counterexample.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which circle theorem did you first treat as a fact to memorise, and how did the proof change your understanding of it?
2. Why is a construction line, such as joining the centre to a point, legitimate in a proof? What must you say when you add one?
3. Give an example of a statement that looked true in a diagram but was false. What convinced you?
4. Which real object near you could be modelled with a circle theorem, and which theorem would you use?

DISTANCE WITHOUT A TAPE MEASURE

Triangles in Space

07

RELATIONSHIPS · SPACE · MEASUREMENT

Two bearings and a baseline can locate a fire, a peak or a satellite — if you know which rule the triangle is asking for.

STATEMENT OF INQUIRY

Triangles that are not right-angled can still be solved: the sine and cosine rules turn measurements of angle and distance into positions, areas and three-dimensional structure.

KEY CONCEPT
Relationships

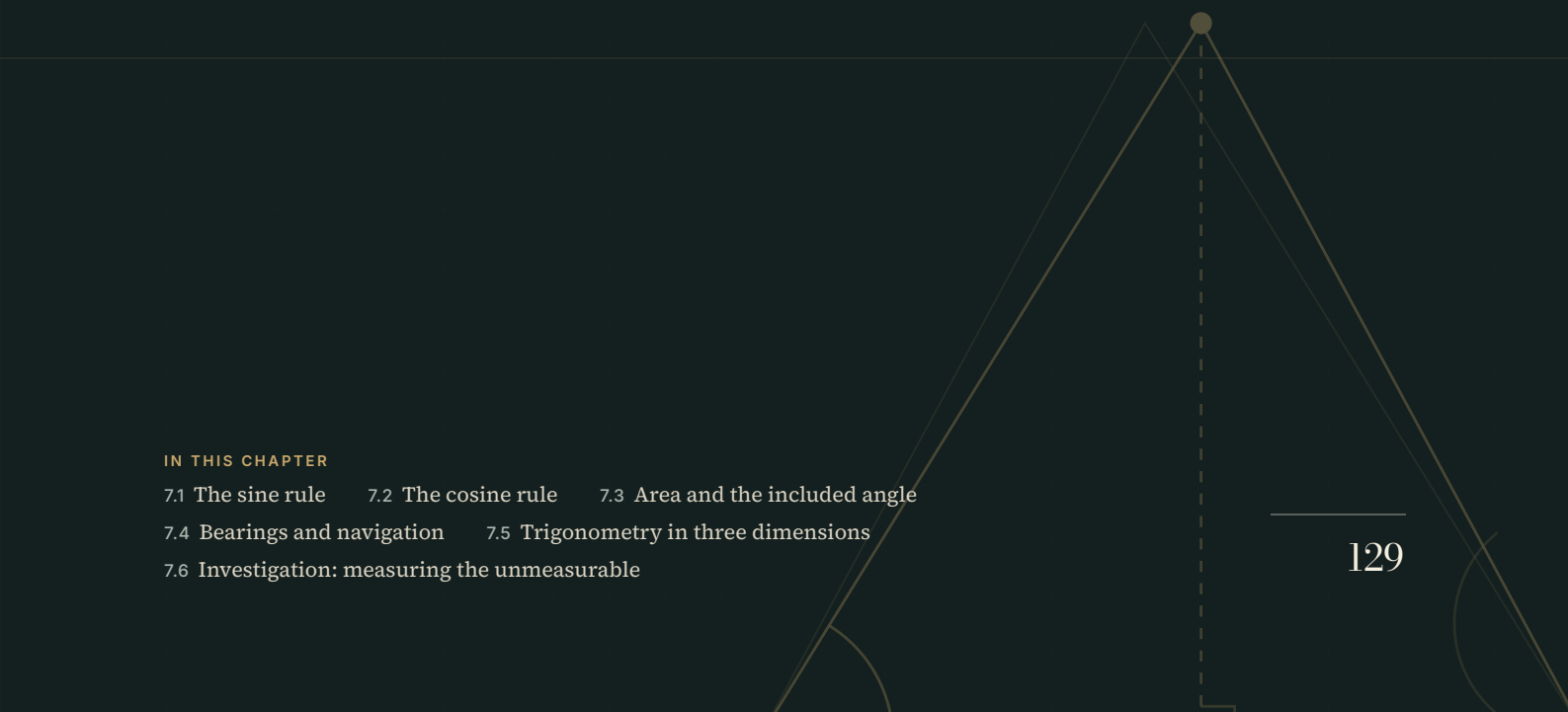
RELATED CONCEPTS
Space · Change · Measurement

GLOBAL CONTEXT
Orientation in space and time — navigation, surveying and the geometry of place

FACTUAL
What is the minimum information needed to determine every side and angle of a triangle?

CONCEPTUAL
Why do two different rules exist, and how does each one encode the geometry of a triangle?

DEBATABLE
When a measurement cannot be taken directly, how much should we trust a value calculated from angles?



IN THIS CHAPTER

7.1 The sine rule
7.2 The cosine rule
7.3 Area and the included angle
7.4 Bearings and navigation
7.5 Trigonometry in three dimensions
7.6 Investigation: measuring the unmeasurable

OPENING PROBLEM · TWO BEARINGS AND A COLUMN OF SMOKE

The fire between the towers

Two fire lookout towers stand 8 km apart on a straight east–west firebreak. Tower A is west of tower B. On a dry afternoon the observer at A records a column of smoke on a bearing of 040°. At the same moment the observer at B records the same fire on a bearing of 330°.

Neither observer can walk to the fire, and the smoke is too thick to judge distance by eye. **Can the fire be located from the two bearings alone — and how far is it from each tower?**

1. Draw triangle ABF, with north marked at both towers. Find the two angles at A and B inside the triangle.
2. Use the sine rule to calculate AF and BF, then find the perpendicular distance from the fire to the firebreak.
3. Estimate the area of triangle ABF, the ground that both towers overlook.
4. Decide which tower is closer to the fire. Then repeat with bearings 042° and 328°, and describe how far the calculated position moves.

A right-angled triangle is a special case, not the general one. Most triangles that matter — a fire seen from two towers, a roof truss, the diagonal of a room — contain no right angle at all. This chapter builds two rules that solve every triangle, then uses them to measure distances that cannot be reached and angles that cannot be seen.

ORIENTATION

7.0 Learning intentions

- Derive the sine rule from an altitude, and use it to find a side or an angle, including the ambiguous case.
- Derive the cosine rule and use it when the given information is SAS or SSS.
- Calculate the area of a triangle from two sides and the included angle, and work backwards from area to angle.
- Interpret three-figure bearings, construct reliable diagrams and combine bearings with the sine and cosine rules.
- Find the angle between a line and a plane, and between two planes, by isolating right triangles in three dimensions.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

state the sine rule and identify a matching side–angle pair; use it to find a side or angle to 3 s.f.; recognise when SSA gives two triangles and state both solutions; derive and apply the cosine rule for SAS and SSS; choose the appropriate rule from the given information; explain why the cosine rule becomes Pythagoras at 90°; calculate area with $\frac{1}{2}ab\sin C$ and reverse the formula; use bearings and back bearings; solve a two-leg navigation problem with speed–distance–time; find line–plane and dihedral angles by isolating right triangles; keep exact surds where possible; and evaluate the assumptions and likely error in a calculated measurement.

ATL SKILLS · THINKING, TRANSFER AND COMMUNICATION

Critical thinking: before calculating, ask which angle lies between which two sides. Most errors in this chapter are diagram errors, not arithmetic errors. **Transfer:** the cosine rule is Pythagoras with a correction for the angle not being 90°; the same idea reappears with vectors and resolving forces. **Communication:** a bearing has three digits and is measured clockwise from north; a navigation diagram without north arrows is incomplete.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator except where stated. If you miss more than two, read the refresher box before Section 7.1.

1. A right-angled triangle has legs 5 cm and 12 cm. Find the hypotenuse.
2. In a right-angled triangle the hypotenuse is 10 cm and one angle is 35°. Find the opposite side.
3. Write down the exact values of $\sin 30^\circ$, $\cos 60^\circ$ and $\tan 45^\circ$.
4. Solve $\sin x = 0.6$ for $0^\circ < x < 90^\circ$.
5. Write the bearing $N30^\circ E$ as a three-figure bearing, and find the back bearing of 120°.
6. Use a calculator to evaluate $\dfrac{8\sin 50^\circ}{\sin 70^\circ}$ to 3 s.f.

REFRESHER · RIGHT TRIANGLES, EXACT VALUES AND BEARINGS

In a right-angled triangle, $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$ and $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$. Inverse functions undo them: if $\sin x = 0.6$ then $x = 36.9^\circ$ to 1 d.p. A calculator returns the acute value, but the obtuse supplement $180^\circ - x$ may also fit the triangle. That fact produces the ambiguous case in Section 7.1.

Know the exact values $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\cos 45^\circ = \frac{\sqrt{2}}{2}$ and $\cos 60^\circ = \frac{1}{2}$. A three-figure bearing is measured clockwise from north, so east is 090°, south 180° and west 270°. The back bearing differs by 180°. Round only at the end; carry 4 or 5 significant figures through intermediate steps.

CONCEPT AND DERIVATION

7.1 The sine rule

The sine rule connects each side of a triangle to the sine of the angle opposite it, and applies to every triangle. Side a is opposite angle A, side b opposite B, and side c opposite C. A side and its opposite angle form a **matching pair**, and every matching pair has the same ratio.

To see why, drop an altitude from C to AB, meeting it at D. In right triangle ACD, the altitude is $h = b\sin A$; in right triangle BCD, it is $h = a\sin B$. Equating gives $a\sin B = b\sin A$, and dividing by $\sin A\sin B$ gives $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$. A second altitude produces the third ratio. If the triangle is obtuse, the foot may lie outside the base, but the proof still works because $\sin(180^\circ - \theta) = \sin\theta$.

KEY IDEA · THE SINE RULE

In any triangle ABC, with sides opposite the angles as labelled,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Equivalently, $\dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c}$. Use the first form for a side and the second for an angle. You need one complete matching pair, plus one further side or angle.

EXAMPLE **7.1** FINDING A SIDE WITH THE SINE RULE

In triangle ABC, $A = 38°$, $B = 64°$ and $a = 9.4$ cm. Find b and c.

SOLUTION

The third angle is $C = 180° - 38° - 64° = 78°$. Pair b with B and c with C:

$$b = \frac{9.4 \sin 64°}{\sin 38°} = 13.72\ldots \text{ cm}, \qquad c = \frac{9.4 \sin 78°}{\sin 38°} = 14.93\ldots \text{ cm}.$$

Hence $b = 13.7$ cm and $c = 14.9$ cm (3 s.f.).

> The longest side, $c = 14.9$ cm, is opposite the largest angle, $C = 78°$ — a quick check that the labels have not been swapped.

EXAMPLE **7.2** THE AMBIGUOUS CASE: TWO TRIANGLES FROM THE SAME DATA

In triangle ABC, $A = 30°$, $a = 6$ cm and $b = 10$ cm. Solve the triangle, stating all possibilities.

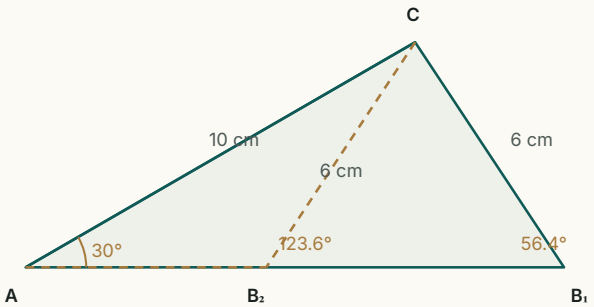
SOLUTION

Apply the sine rule to the pair a, A:

$$\sin B = \frac{b \sin A}{a} = \frac{10 \sin 30°}{6} = \frac{5}{6} = 0.8333\ldots$$

The inverse sine gives $56.4°$, but the supplement has the same sine, so $B = 56.4°$ or $123.6°$. Both fit here because A is acute and $a < b$. For each value, $C = 180° - 30° - B$ and $c = \dfrac{6 \sin C}{\sin 30°}$. **First triangle:** $B = 56.4°$, $C = 93.6°$, $c = 12.0$ cm.
Second triangle: $B = 123.6°$, $C = 26.4°$, $c = 5.34$ cm.

> These are two genuinely different triangles. Two solutions occur precisely when the given angle is acute and opposite the shorter given side; otherwise there is at most one triangle.



Two triangles with $A = 30°$, $a = BC = 6$ cm and $b = AC = 10$ cm. The circle centred at C with radius 6 cm meets the base line twice, at B_1 and B_2.

Exact values matter when the angles are $30°$, $45°$ or $60°$: if $A = 45°$, $B = 60°$ and $a = 6\sqrt{2}$, then $b = \dfrac{6\sqrt{2} \sin 60°}{\sin 45°} = 6\sqrt{3}$. Otherwise quote lengths and angles to 3 significant figures with units. Use the sine rule for ASA, AAS and SSA, where a complete side–angle pair exists; SAS and SSS belong to the cosine rule in Section 7.2.

MISCONCEPTION

"$\sin B = 0.8333$, so $B = 56.4°$ and the triangle is solved." The inverse sine returns one angle, but $\sin B = 0.8333$ has two solutions between $0°$ and $180°$: $56.4°$ and $123.6°$. Either may be used, provided the angles sum to $180°$. The ambiguous case is a real second triangle, not a calculator fault. Always test the supplement against the angle sum and side ordering.

GUIDED PRACTICE

1. In triangle ABC, $A = 52°$, $B = 73°$ and $a = 8.6$ cm. Find b. (Answer: $b = 10.4$ cm.)
2. In triangle ABC, $A = 36°$, $a = 5.0$ cm and $b = 7.0$ cm. Find both possible values of B. (Answer: $55.4°$ or $124.6°$.)
3. In triangle ABC, $A = 45°$, $B = 60°$ and $a = 6\sqrt{2}$ cm. Find b exactly. (Answer: $6\sqrt{3}$ cm.)
4. Why can the sine rule not begin a problem when two sides and the included angle are given? (No side is paired with its opposite angle; use the cosine rule.)

EXERCISE 7A

THE SINE RULE, INCLUDING THE AMBIGUOUS CASE

FLUENCY

1 In triangle ABC, $A = 47°$, $B = 63°$ and $a = 11.2$ cm. Find C, b and c.

2 Find the marked quantity. Give lengths to 3 s.f. and angles to 1 d.p.

a. $P = 35°$, $Q = 80°$, $p = 6.4$ cm; find q.

b. $X = 28°$, $x = 9.0$ cm, $y = 14.0$ cm; find the two possible values of Y.

c. $D = 112°$, $E = 34°$, $e = 8.5$ cm; find d.

3 In triangle ABC, $A = 60°$, $B = 45°$ and $a = 8\sqrt{3}$ cm. Find b and c in exact form.

4 Solve triangle ABC given $A = 39°$, $a = 7.4$ cm and $b = 10.2$ cm. Give both possible sets of remaining values.

5 Decide whether the sine rule can be used directly to begin each problem.

a. $A = 40°$, $B = 65°$, $c = 9$ cm

b. $b = 7$ cm, $c = 9$ cm, $A = 50°$

c. $a = 6$ cm, $b = 8$ cm, $A = 30°$

d. $a = 5$ cm, $b = 6$ cm, $c = 7$ cm

e. $A = 50°$, $B = 60°$, $a = 8$ cm

6 The shortest side of a triangle is 9.6 cm. Two of its angles are $58°$ and $44°$. Find the longest side.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes: "For $a = 7$, $b = 8$ and $A = 35°$, $\sin B = 0.6555$, so $B = 41.0°$." Explain why the answer is incomplete, and give the second possible value of B.

8 Derive the sine rule by dropping an altitude from B to AC. Let the foot be E, write two expressions for BE, and deduce $\dfrac{b}{\sin B} = \dfrac{c}{\sin C}$.

9 **Diagram-based.** The diagram in Example 7.2 shows two triangles with $a = 6$ cm, $b = 10$ cm and $A = 30°$. Identify which triangle has $B = 56.4°$ and which has $B = 123.6°$. Explain why both satisfy the data.

10 Determine how many triangles satisfy $a = 8$ cm, $b = 6$ cm and $A = 40°$. Justify your answer using the sine value and the angle sum.

11 In a triangle, $A = 30°$, $a = 5$ cm and $b = 9$ cm. Show that two triangles exist, and find the two possible values of B.

12 Explain why the sine rule alone cannot decide whether an angle is acute or obtuse. What additional check settles the question in a triangle?

APPLICATION AND EXTENSION

13 Two fire towers A and B are 6 km apart, with B due east of A. A fire F is seen on a bearing of $052°$ from A and $338°$ from B. Find AF, BF and the perpendicular distance from the fire to the line AB.

14 A kite lies in a vertical plane above level ground. From point A the angle of elevation of the kite is $38°$. From point B, 20 m farther away along the same horizontal line, the angle of elevation is $52°$. Find the height of the kite above the ground.

EXERCISE 7A · CONTINUED THE SINE RULE, INCLUDING THE AMBIGUOUS CASE

15 **Extension.** In triangle ABC, angle $A = 2B$. Use the sine rule and the double-angle identity $\sin 2B = 2\sin B\cos B$ to prove that $a = 2b\cos B$.

CONCEPT AND DERIVATION

7.2 The cosine rule

The sine rule needs a complete side–angle pair. When the information is two sides and the included angle (SAS), or three sides (SSS), no such pair exists. The cosine rule supplies the missing relationship; it is Pythagoras plus a correction recording how far the included angle is from a right angle.

Here is a coordinate proof. Place A at the origin and B at $(c, 0)$. Since $AC = b$ makes angle A with AB, C is $(b\cos A,\ b\sin A)$. The side a is the distance from B to C:

$$a^2 = (b\cos A - c)^2 + (b\sin A)^2 = b^2\cos^2 A - 2bc\cos A + c^2 + b^2\sin^2 A.$$

Since $\cos^2 A + \sin^2 A = 1$, this simplifies to $a^2 = b^2 + c^2 - 2bc\cos A$. The coordinates simply record the components of AC.

KEY IDEA · THE COSINE RULE

In any triangle ABC,

$$a^2 = b^2 + c^2 - 2bc\cos A,$$

with the letters cycled:

$$b^2 = c^2 + a^2 - 2ca\cos B, \qquad c^2 = a^2 + b^2 - 2ab\cos C.$$

Rearranged for an angle,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

The angle A must be the angle *between* sides b and c. When $A = 90^\circ$, $\cos A = 0$ and the rule becomes Pythagoras. When A is obtuse, $\cos A < 0$, so the correction term is added and $a^2 > b^2 + c^2$.

EXAMPLE **7.3** FINDING A SIDE WITH THE COSINE RULE

In triangle ABC, $b = 7.2$ cm, $c = 5.4$ cm and $A = 58^\circ$. Find a.

SOLUTION

The known angle A lies between the known sides, so

$$a^2 = 7.2^2 + 5.4^2 - 2(7.2)(5.4)\cos 58^\circ = 51.84 + 29.16 - 41.21 = 39.79\ldots$$

and $a = 6.31$ cm (3 s.f.).

> The correction is subtracted because 58° is acute. At 122° the cosine is negative, so the correction is added and a is larger.

EXAMPLE **7.4** FINDING AN ANGLE WITH THE COSINE RULE

A triangle has sides $a = 9$ cm, $b = 7$ cm and $c = 5$ cm. Find all three angles.

SOLUTION

The largest angle is opposite the largest side, so begin with A:

$$\cos A = \frac{7^2 + 5^2 - 9^2}{2(7)(5)} = \frac{49 + 25 - 81}{70} = -0.1.$$

Hence $A = 95.7°$ (1 d.p.). The negative cosine confirms that A is obtuse. For B,

$$\cos B = \frac{9^2 + 5^2 - 7^2}{2(9)(5)} = \frac{81 + 25 - 49}{90} = 0.6333\ldots,$$

so $B = 50.7°$ and $C = 180° - 95.7° - 50.7° = 33.6°$.

> Find the largest angle first. If the other two are then found by the sine rule, both are necessarily acute, so no ambiguity arises.

MISCONCEPTION

"The cosine rule is $a^2 = b^2 + c^2 + 2bc\cos A$." The correction is negative. For an acute angle this makes a^2 smaller than $b^2 + c^2$; for an obtuse angle the cosine is negative, so the correction becomes positive. A second error is using an angle that is not between the two known sides: in this form the angle must be A, opposite the side being found.

GUIDED PRACTICE

1. In triangle ABC, $b = 5$ cm, $c = 8$ cm and $A = 60°$. Find a. (Answer: $a = 7$ cm.)
2. A triangle has sides $a = 6$ cm, $b = 5$ cm and $c = 4$ cm. Find its largest angle. (Answer: $A = 82.8°$.)
3. Show that the cosine rule becomes Pythagoras at $A = 90°$. ($\cos 90° = 0$.)
4. In triangle ABC, $b = 9$ cm, $c = 6$ cm and $A = 120°$. Find a. (Answer: $a = 13.1$ cm.)

EXERCISE 7B

THE COSINE RULE AND THE CHOICE OF METHOD

FLUENCY

1 In triangle ABC, $b = 8.4$ cm, $c = 6.2$ cm and $A = 54°$. Find a.

2 Find the largest angle of the triangle with sides 11 cm, 9 cm and 6 cm.

3 Solve the triangle with sides 7 cm, 8 cm and 10 cm. Give all angles to 1 d.p.

4 In triangle ABC, $a = 5$ cm, $b = 12$ cm and $C = 90°$. Find c, and comment on the result.

5 State which rule you would use first, and why.

a. $b = 8$, $c = 11$, $A = 43°$
b. $a = 6$, $b = 7$, $c = 9$
c. $A = 52°$, $B = 61°$, $c = 14$
d. $a = 7$, $b = 10$, $A = 38°$
e. A right triangle with hypotenuse 13 and angle $28°$

6 Find all three angles of the triangle with sides 5 cm, 6 cm and 7 cm.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student uses $a^2 = b^2 + c^2 + 2bc\cos A$ for $b = 6$ cm, $c = 7$ cm and $A = 120°$. Explain how the sign error affects the answer, and calculate the correct value of a.

8 Derive $a^2 = b^2 + c^2 - 2bc\cos A$ by placing A at the origin, B at $(c, 0)$ and C at $(b\cos A, b\sin A)$.

9 Show that the cosine rule reduces to Pythagoras when $A = 90°$. Then explain why an obtuse angle makes a^2 greater than $b^2 + c^2$.

EXERCISE 7B · CONTINUED THE COSINE RULE AND THE CHOICE OF METHOD

10 **Diagram-based.** Two triangles each have sides 7 cm and 9 cm. In one the included angle is $30°$; in the other it is $150°$. Calculate the third side in each case, and explain the difference using the cosine rule.

11 A triangle has sides 4 cm, 5 cm and 8 cm. Decide whether it is acute, right-angled or obtuse, and find its largest angle.

12 Explain, in a short table or paragraph, how the given information determines whether you should use the sine rule, the cosine rule or right-triangle trigonometry.

APPLICATION AND EXTENSION

13 Two ships leave a port together. One sails 12 km on a bearing of $060°$; the other sails 18 km on a bearing of $145°$. Find the distance between the ships and the bearing of the second ship from the first.

14 A triangular field has sides 45 m, 60 m and 75 m. Show that it is right-angled, find its largest angle, and calculate its area.

15 **Extension.** In triangle ABC, prove that C is obtuse if and only if $a^2 + b^2 < c^2$. Explain why this gives a quick test for the type of triangle from its three sides.

CONCEPT AND APPLICATION

7.3 Area and the included angle

The familiar formula "half base times height" is still true, but the height is rarely given. The area formula in this section replaces the height with a side and the sine of the included angle. Take AC as the base, so its length is b. The perpendicular height from B to AC is $a\sin C$, because in the right triangle formed by dropping that perpendicular, a is the hypotenuse and C is the angle at C. Hence

$$\text{area} = \tfrac{1}{2} \times \text{base} \times \text{height} = \tfrac{1}{2}\, b\,(a\sin C) = \tfrac{1}{2}ab\sin C.$$

Choosing a different base produces the same formula with the letters cycled. The angle must be the one between the two sides used; if it is not, find it first or use a different pair.

KEY IDEA · AREA FROM TWO SIDES AND THE INCLUDED ANGLE

For any triangle ABC,

$$K = \tfrac{1}{2}ab\sin C = \tfrac{1}{2}bc\sin A = \tfrac{1}{2}ca\sin B.$$

For a parallelogram with adjacent sides a and b and included angle θ, the area is $ab\sin\theta$.

Rearranged, $\sin C = \dfrac{2K}{ab}$. Since $\sin C = \sin(180° - C)$, a given area and pair of sides usually allow two possible included angles, supplementary to each other.

EXAMPLE **7.5** AREA FROM TWO SIDES AND AN INCLUDED ANGLE

A triangular metal plate has sides 8.4 cm and 6.7 cm enclosing an angle of $52°$. Find its area.

SOLUTION

The given angle is between the two given sides, so

$$K = \tfrac{1}{2}(8.4)(6.7)\sin 52° = 28.14 \times 0.7880\ldots = 22.17\ldots \text{ cm}^2.$$

The area is 22.2 cm^2 (3 s.f.).

> The answer is an area, so the unit is squared. A quick check: at $90°$ the area would be 28.1 cm^2, and $52°$ is smaller, so the result should be a little less.

MISCONCEPTION

"The area is $\frac{1}{2}ab\sin A$, where A is any angle." The angle must lie *between* sides a and b; using a different angle gives the wrong area. Also remember the factor $\frac{1}{2}$: $ab\sin C$ is the parallelogram's area. When reversing the formula, do not discard the obtuse solution merely because a calculator returned an acute angle.

GUIDED PRACTICE

1. Find the area of a triangle with sides 6.4 cm and 9.2 cm enclosing $41°$. (Answer: 19.3 cm^2.)
2. Find the area of a triangle with sides 12 m and 10 m enclosing $115°$. (Answer: 54.4 m^2.)
3. A triangle has area 20 cm^2 and two sides 7 cm and 8 cm. Find the two possible included angles. (Answer: $45.6°$ or $134.4°$.)
4. Explain why a parallelogram's area is twice the area of the triangle formed by a diagonal. (The diagonal divides it into two congruent triangles.)

EXERCISE 7C

AREA, PARALLELOGRAMS AND WORKING BACKWARDS

FLUENCY

1 Find the area of a triangle with $a = 9.2$ cm, $b = 7.5$ cm and $C = 48°$.

2 Find the area of a triangle with $b = 12$ m, $c = 10$ m and $A = 115°$.

3 A triangle has area 24 cm^2 and two sides 8 cm and 7 cm. Find the two possible included angles.

4 Find the area of a parallelogram with adjacent sides 7.4 cm and 5.6 cm and included angle $62°$.

5 Find the area of each triangle.

a. Base 14 cm, perpendicular height 9 cm.

b. Sides 7 cm and 12 cm, included angle $30°$.

6 A triangle has area 36 cm^2 and two sides 9 cm and 10 cm. Find the possible included angles.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student calculates the area of a triangle using sides 6 cm and 9 cm and an angle of $40°$ that is opposite the 6 cm side. Explain why the calculation is wrong, and state what information would make the formula valid.

8 Derive $K = \frac{1}{2}ab\sin C$ by taking AC as the base and dropping a perpendicular from B. Draw and label the altitude.

9 Show that $\frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$. What does this imply about the three altitudes of a triangle?

10 **Diagram-based.** Two triangles share the same base and have their third vertices on a line parallel to the base. Explain, using $\frac{1}{2}ab\sin C$, why their areas are equal.

11 A triangle has area 20 cm^2 and two sides 6 cm and 9 cm. Find the possible included angles, and explain why there are two triangles.

12 A triangle has sides 13 m, 14 m and 15 m. Find its area by first calculating an angle, then check with Heron's formula. Which method is more direct, and why?

APPLICATION AND EXTENSION

13 A triangular garden has sides 18 m and 24 m enclosing $72°$. Turf costs £8.50 per square metre. Find the area and the cost of turfing the garden.

14 A parallelogram-shaped solar array has adjacent sides 65 m and 42 m and included angle $58°$. Find its area, and the number of complete 5 m^2 panels that fit if none may be cut.

15 **Extension.** A triangle has area K and two sides a and b. Show that there are generally two possible included angles, and that they are supplementary. For what value of K is there exactly one triangle?

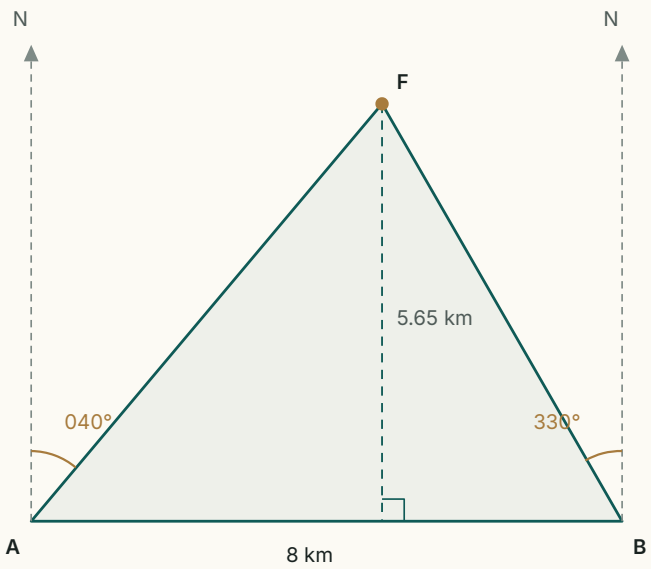
APPLICATION AND REPRESENTATION

7.4 Bearings and navigation

A three-figure bearing gives a direction as an angle measured clockwise from north and is written with three digits: 035°, 090°, 207°. North is 000°, east 090°, south 180° and west 270°. Every diagram should show a north arrow at each point where a direction is measured.

North lines at different places are parallel, so a bearing can be transferred by drawing a parallel line. The back bearing is the bearing of the original point seen from the destination; it differs by 180°. For example, the back bearing of 047° is 227°, and of 300° is 120°.

In a navigation problem, mark north at each vertex and identify the angles *inside* the triangle; these are usually differences of bearings. SAS calls for the cosine rule, and a side with its opposite angle calls for the sine rule. Distance, speed and time are linked by $d = st$, with consistent units.



The opening problem drawn to scale. The bearings are measured at A and B from parallel north lines; the angles inside the triangle are 50° at A and 60° at B.

EXAMPLE **7.6** A TWO-LEG SEA ROUTE

A ship sails 15 km from port P on a bearing of 070° to point Q, then 9 km on a bearing of 150° to point R. Find the distance PR and the bearing of R from P.

SOLUTION

At Q, the bearing back to P is $070^\circ + 180^\circ = 250^\circ$, so the angle inside the triangle is

$$\angle PQR = 250^\circ - 150^\circ = 100^\circ.$$

This is SAS, so

$$PR^2 = 15^2 + 9^2 - 2(15)(9)\cos 100^\circ = 225 + 81 + 46.9 = 352.9,$$

giving $PR = 18.8$ km (3 s.f.). By the sine rule,

$$\sin \angle QPR = \frac{9 \sin 100^\circ}{18.79} = 0.4717\ldots,$$

so $\angle QPR = 28.2^\circ$. Since R lies clockwise from Q, the bearing of R from P is $070^\circ + 28.2^\circ = 098^\circ$ (nearest degree).

> The angle at Q is not $150^\circ - 70^\circ = 80^\circ$. Bearings are measured from north at different points, so the back bearing 250° must be used at Q.

MISCONCEPTION

"The bearing of A from B is the same as the bearing of B from A." Bearings are directed: reversing the direction changes the bearing by $180°$. Do not measure anticlockwise from north, or clockwise from east. Finally, an angle inside a navigation triangle is often a *difference* of bearings, and one bearing may need converting to a back bearing first.

GUIDED PRACTICE

1. Write as three-figure bearings: $N40°E$, $S25°W$ and north-west. (Answer: $040°$, $205°$, $315°$.)

2. A walker goes 6 km on a bearing of $040°$, then 5 km on a bearing of $130°$. Find the distance and bearing from the start. (Answer: 7.81 km, $080°$.)

3. Why can a bearing be transferred between two points? (North lines are parallel, so corresponding angles are equal.)

4. From A, B is 12 km away on a bearing of $070°$. From B, C is 9 km away on a bearing of $150°$. Find AC and the bearing of C from A. (Answer: 16.2 km, $103°$.)

EXERCISE 7D

BEARINGS, BACK BEARINGS AND NAVIGATION

FLUENCY

1 Write each direction as a three-figure bearing: $N35°E$, $S20°W$, north-west, due east, $S40°W$.

2 Find the back bearing of each: $047°$, $128°$, $300°$, $355°$.

3 A walker goes 5 km on a bearing of $030°$, then 4 km on a bearing of $120°$. Find the distance and bearing from the start.

4 A ship sails 20 km on a bearing of $075°$, then 15 km on a bearing of $165°$. Find its distance and bearing from the starting point.

5 A point P is 12 km from a harbour on a bearing of $230°$. Find the bearing and distance of the harbour from P.

6 An aircraft flies at 240 km/h for 45 minutes on a bearing of $110°$. Find the distance flown and its displacement east and south of the start.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student draws a bearing of $070°$ as an angle $70°$ above the horizontal, measured anticlockwise from east. Explain what is wrong and how the diagram should be drawn.

8 Explain why north lines at different points on a map are parallel, and why this allows a bearing to be transferred from one vertex to another.

9 **Diagram-based.** Use the diagram above. Given the bearings $040°$ from A and $330°$ from B, and the fact that AB runs east–west, find the angles of triangle ABF at A, B and F.

10 The bearing of A from B is $050°$ and the bearing of A from C is $320°$. Explain how to find $\angle BAC$, and state whether the answer depends on the distances AB and AC.

11 A ship sails 10 km due north, then 10 km due east. Show that its bearing from the start is $045°$, not $090°$. Use a diagram in your explanation.

12 Two rescue stations report bearings of $040°$ and $310°$ to the same distress flare. Explain why the two rays might not intersect where expected if the stations are not on an east–west line, and why a third bearing would remove the uncertainty.

APPLICATION AND EXTENSION

13 Two ships leave a port at the same time. Ship P travels 24 km on a bearing of $055°$; ship Q travels 30 km on a bearing of $140°$. Find the distance PQ and the bearing of Q from P.

14 A yacht sails at 8 km/h for 3 hours on a bearing of $215°$, then at the same speed for 5 hours on a bearing of $125°$. Find its distance and bearing from the start, and the time it would take to return directly at 10 km/h.

15 **Extension.** Town B is 40 km from town A on a bearing of $060°$. Town C is 55 km from A on a bearing of $145°$. Find BC, the bearing of C from B, and the area of triangle ABC.

CONCEPT AND METHOD

7.5 Trigonometry in three dimensions

In three dimensions, an angle involving a line and a plane is defined by a projection. Drop a perpendicular from a point on the line to the plane, then join its foot to the point where the line meets the plane. The angle between the line and this projection is the required angle; it is the smallest angle the line makes with any line in the plane, and lies between 0° and 90°.

The angle between two planes is a dihedral angle. Find the line of intersection, then in each plane draw a line perpendicular to that intersection at the same point. The angle between those perpendiculars is the dihedral angle; a cross-section perpendicular to a roof ridge displays it directly.

Always reduce the problem to a right triangle: use Pythagoras and right-triangle trigonometry in a face or cross-section, and the sine or cosine rule in a sloping face. Keep exact surds until the final step, and name the triangle you are working in.

KEY IDEA · ANGLES AND LENGTHS IN A CUBOID

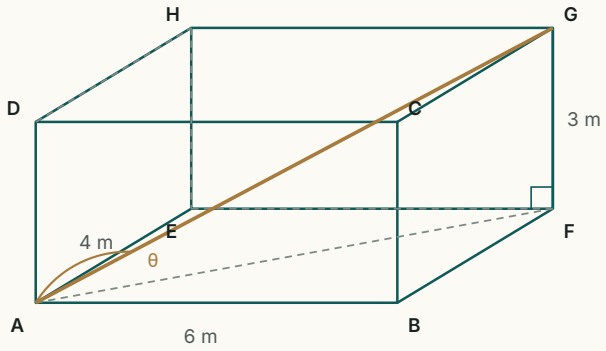
For a cuboid with length l, width w and height h, the space diagonal is

$$d = \sqrt{l^2 + w^2 + h^2}.$$

The angle θ between the space diagonal and the base is given by

$$\tan\theta = \frac{h}{\sqrt{l^2 + w^2}},$$

because the projection of the diagonal on the base is the base diagonal. Identify the perpendicular projection for a line–plane angle, and a perpendicular to the intersection for a dihedral angle.



The space diagonal AG projects onto the base diagonal AF. The right triangle AFG contains the height $FG = 3$ m, the projection AF and the diagonal AG, so the angle θ at A is the angle between the diagonal and the base.

EXAMPLE **7.7** A CUBOID DIAGONAL AND ITS ANGLE WITH THE BASE

A cuboid has length 6 m, width 4 m and height 3 m. Find the length of the space diagonal and the angle it makes with the base. Then find the angle it makes with the vertical face containing the 4 m and 3 m edges.

SOLUTION

The base diagonal is $\sqrt{6^2+4^2}=\sqrt{52}$. The space diagonal is the hypotenuse of the right triangle formed by this diagonal and the height:

$$AG=\sqrt{52+3^2}=\sqrt{61}=7.81 \text{ m } (3 \text{ s.f.}).$$

The angle θ with the base satisfies $\tan\theta=\dfrac{3}{\sqrt{52}}$, so $\theta=22.6^\circ$ (1 d.p.). For the angle with the 4 m by 3 m face, project AG onto that face: the projection has length $\sqrt{4^2+3^2}=5$ m, and the perpendicular is the 6 m edge. Thus

$$\tan\phi=\frac{6}{5}, \qquad \phi=50.2^\circ \text{ (1 d.p.)}.$$

> The angles differ because the projections differ: the base uses $\sqrt{52}$, the side face uses 5. Naming the projection is the key step.

EXAMPLE **7.8** A SQUARE PYRAMID

A square pyramid has a base of side 10 m and a vertical height of 12 m, with the apex above the centre of the base. Find the length of a sloping edge, the angle between a sloping edge and the base, and the dihedral angle between a triangular face and the base.

SOLUTION

The base half-diagonal is $5\sqrt{2}$ m, so a sloping edge satisfies

$$l^2=12^2+(5\sqrt{2})^2=144+50=194, \qquad l=13.9 \text{ m } (3 \text{ s.f.}).$$

The projection of a sloping edge is the half-diagonal, so

$$\tan\alpha=\frac{12}{5\sqrt{2}}, \qquad \alpha=59.5^\circ \text{ (1 d.p.)}.$$

For the dihedral angle, take a cross-section through the apex and the midpoint of a base side. The slant height is $\sqrt{12^2+5^2}=13$ m, and the horizontal distance is 5 m, so

$$\tan\beta=\frac{12}{5}, \qquad \beta=67.4^\circ \text{ (1 d.p.)}.$$

> The edge angle uses the half-diagonal $5\sqrt{2}$; the face angle uses the apothem 5. Mixing these projections is the most common 3D error.

EXAMPLE 7.9 A TENT: COSINE RULE IN A FACE, PYTHAGORAS IN A CROSS-SECTION

A ridge tent has a triangular cross-section with base 2.4 m and two equal sloping sides of 2.0 m. The tent is 3.0 m long. Find the angle at the ridge, the height of the ridge, the area of canvas needed for the two sloping rectangular sides and two triangular ends (ignoring overlaps), and the angle a sloping side makes with the ground.

SOLUTION

In the triangular end, let the ridge angle be C. By the cosine rule,

$$\cos C = \frac{2.0^2 + 2.0^2 - 2.4^2}{2(2.0)(2.0)} = \frac{8 - 5.76}{8} = 0.28,$$

so $C = 73.7^\circ$ (1 d.p.). The height is $h = \sqrt{2.0^2 - 1.2^2} = 1.6$ m. Each sloping rectangular side has area $2.0 \times 3.0 = 6.0$ m^2, and each triangular end has area $\frac{1}{2}(2.4)(1.6) = 1.92$ m^2. The total canvas is

$$2(6.0) + 2(1.92) = 15.84 \approx 15.8 \text{ m}^2.$$

In the cross-section, the angle θ with the ground satisfies $\tan\theta = \dfrac{1.6}{1.2}$, so $\theta = 53.1^\circ$ (1 d.p.).

> The cosine rule found the ridge angle; Pythagoras found the height. The angle with the ground is not the ridge angle: it is measured in the cross-section, between one sloping side and the base.

MISCONCEPTION

"The angle between a line and a plane is the angle with any line drawn in the plane." It is the angle with the line's *projection*, which is the smallest such angle. Do not use a sloping edge as the perpendicular height: the height must be perpendicular to the base. For a dihedral angle, both lines must be perpendicular to the intersection; arbitrary sloping edges give a different angle.

GUIDED PRACTICE

1. A cuboid measures 5 m by 4 m by 3 m. Find its space diagonal and the angle it makes with the base. (Answer: 7.07 m, 25.1°.)

2. A square pyramid has base side 8 m and vertical height 6 m. Find a sloping edge and the angle it makes with the base. (Answer: 8.25 m, 46.7°.)

3. Distinguish a line–plane angle from a dihedral angle. (Line–plane uses the projection; plane–plane uses perpendiculars to the intersection.)

4. Why is a sloping edge not the perpendicular height of a pyramid? (It is measured along a face, not perpendicular to the base.)

EXERCISE 7E

LINES, PLANES AND THREE-DIMENSIONAL TRIANGLES

FLUENCY

1 A cuboid measures 8 m by 5 m by 4 m. Find its space diagonal and the angle it makes with the base.

2 A cuboid measures 10 m by 10 m by 6 m. Find its space diagonal and the angle it makes with the base.

3 A square pyramid has base side 12 m and vertical height 8 m. Find a sloping edge, the angle between that edge and the base, and the dihedral angle between a triangular face and the base.

4 A room measures 6 m by 4 m by 2.8 m. Find the length of the space diagonal and its angle with the floor.

5 A cuboid measures 5 m by 4 m by 3 m. Find the angle between its space diagonal and the face containing the 4 m and 3 m edges.

6 A wedge has a right-triangular cross-section with legs 3 m and 4 m, and length 6 m. Find the length of the longest edge on a sloping rectangular face.

UNDERSTANDING AND REASONING

EXERCISE 7E · CONTINUED

LINES, PLANES AND THREE-DIMENSIONAL TRIANGLES

7 **Error analysis.** A student finds the angle between a cuboid's space diagonal and the base by calculating $\tan\theta = \dfrac{\text{height}}{\text{space diagonal}}$. Explain why this is wrong, and state the correct denominator.

8 Explain, with a sketch, how to find the angle between a line and a plane using the projection of the line. Why is this the smallest angle between the line and any line in the plane?

9 **Diagram-based.** Use the cuboid in the diagram above. Find the angle between AG and the base $ABCD$, and the angle between AG and the face $BCGF$.

10 Two roof planes meet along a ridge. Explain how to identify the dihedral angle between them, and why a cross-section perpendicular to the ridge displays that angle directly.

11 A ladder 5 m long leans against a vertical wall with its foot 1.8 m from the wall. Find the angle it makes with the ground and the angle it makes with the wall.

12 Explain why a space diagonal is longer than each edge, and show that $\sqrt{l^2+w^2+h^2} < l+w+h$ for positive l, w, h.

APPLICATION AND EXTENSION

13 A house is 12 m long and 8 m wide. The eaves are 4 m above the ground and the ridge is 7 m above the ground, running centrally along the length. Find the angle each roof face makes with the horizontal, and the total area of the two roof faces.

14 A square pyramid has base side 230 m and vertical height 147 m. Find a sloping edge, the angle between that edge and the base, and the dihedral angle between a triangular face and the base.

15 **Extension.** In a cuboid, a space diagonal makes angles α, β and γ with the three edges that meet at one vertex. Show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

PATTERN AND GENERALISATION

7.6 Investigation: measuring the unmeasurable

Triangulation replaces a distance that cannot be measured with a baseline that can, together with angles measured from both ends. A clinometer measures an angle of elevation; parallax uses the apparent shift of an object when the viewpoint changes. This investigation asks how accurate such indirect measurements can be, and what assumptions they hide.

INVESTIGATION 7 · INDIRECT MEASUREMENT

How far is it, really?

1. **Build a clinometer.** Tape a drinking straw along the straight edge of a protractor and hang a weight from the centre. Sight the top of a building through the straw; the weighted string reads the angle of elevation. Measure the angle from P and the horizontal distance d to the wall, then calculate $h = d\tan\theta + e$, where e is the height of your eye.
2. **Repeat from farther away.** Move 20 m back along the same line and measure again. Compare the two heights and suggest two reasons for any difference.
3. **Triangulate across a river.** Choose a baseline $AB = 40$ m along one bank and a tree T opposite. Measure the bearings of T from A and B, convert them to the angles inside triangle ABT, and use the sine rule to find AT and BT. Check with a scale drawing.
4. **Use parallax.** Hold a finger at arm's length and look at it alternately with each eye. Estimate the apparent shift against a distant background. Explain, using similar triangles, why a larger shift means a nearer object, and estimate the distance to your finger.
5. **Quantify the error.** Take five readings of one angle. Find the range and mean, recalculate the height using the smallest and largest readings, and decide whether the angle or the distance contributes more uncertainty.
6. **List the assumptions.** For the building: level ground, a vertical wall, a horizontal baseline and no overhang. For the river: a straight bank and bearings from true north. Explain how each assumption would bias the answer.
7. **Return to the fire.** With the 8 km baseline and bearings $040°$ and $330°$, calculate AF and BF. Repeat with $042°$ and $328°$. How far does the calculated fire position move, and would that change which tower is sent?

Generalise. Indirect measurement converts an inaccessible length into a baseline and a set of angles. Its accuracy depends on the baseline relative to the distance, the precision of the angles, and the shape of the triangle. A long, narrow triangle is poorly conditioned because a small angular error moves the far vertex a long way. Using the sine rule, explain why an angle close to $0°$ or $180°$ makes the calculated side especially sensitive. Then state two rules for designing a reliable triangulation.

MODELLING · CRITERION D

The fire crew's decision

Return to the two towers: A and B are 8 km apart, with B due east of A. The fire F is on bearings $040°$ from A and $330°$ from B.

1. **Model.** Calculate AF, BF and the perpendicular distance from F to the firebreak. State which tower is closer and by how much.
2. **Logistics.** A crew at B can drive at 45 km/h; a helicopter at A flies at 160 km/h. Estimate the travel time from each tower and decide who responds first. State any assumptions about loading and terrain.
3. **Area.** Estimate the area of triangle ABF. A water bomber lays a retardant strip 0.8 km wide along BF; estimate the area treated.
4. **Evaluate.** Repeat with bearings $042°$ and $328°$. How far does the fire move? Discuss two assumptions and recommend how the crew should use the calculated position.

CUMULATIVE

7.7 Mixed practice

Attempt all questions without notes. Questions 1–10 practise selection and routine calculation; questions 11–20 require combination, proof or evaluation. Give lengths and angles to 3 s.f. unless an exact answer is requested.

MIXED PRACTICE 7

SINE RULE, COSINE RULE, AREA, BEARINGS AND 3D TRIGONOMETRY

FLUENCY AND UNDERSTANDING

1 In triangle ABC, $A = 36°$, $B = 81°$ and $a = 14.5$ cm. Find b.

2 In triangle ABC, $A = 32°$, $a = 6.5$ cm and $b = 9.0$ cm. Find both possible values of B.

3 In triangle ABC, $b = 9.5$ cm, $c = 7.2$ cm and $A = 63°$. Find a.

4 Find the largest angle of the triangle with sides 6 cm, 7 cm and 9 cm.

5 Find the area of a triangle with sides 11.4 cm and 8.6 cm enclosing $47°$.

6 A ship is 14 km from port on a bearing of $118°$. Write down the bearing of the port from the ship.

7 A cuboid measures 9 m by 6 m by 4 m. Find its space diagonal and the angle it makes with the base.

8 State which rule you would use first, and why.

a. $A = 40°$, $B = 60°$, $c = 12$ cm

b. $a = 5$ cm, $b = 7$ cm, $C = 110°$

c. $a = 8$ cm, $b = 9$ cm, $c = 10$ cm

d. $a = 7$ cm, $b = 9$ cm, $A = 25°$

9 In triangle ABC, $A = 45°$, $B = 60°$ and $a = 10\sqrt{2}$ cm. Find b exactly.

10 **Error analysis.** A student finds the area of a triangle with sides 8 cm and 11 cm using $\frac{1}{2}(8)(11)\sin 35°$, where $35°$ is opposite the 8 cm side. Explain the error and describe the correct method.

REASONING AND APPLICATION

11 Solve triangle ABC given $A = 48°$, $a = 9.5$ cm and $b = 7.2$ cm. Explain why only one triangle is possible.

12 A triangle has area 45 cm^2 and two sides 8.5 cm and 12 cm. Find the two possible included angles.

13 A ship sails 26 km on a bearing of $055°$, then 18 km on a bearing of $125°$. Find its distance and bearing from the start.

14 A roof is 10 m wide and 15 m long, with eaves 3.5 m and a central ridge 6.5 m above the floor. Find the angle of each face to the horizontal and the total area of the two faces.

15 A square pyramid has base side 16 m and vertical height 10 m. Find a sloping edge, its angle with the base, and the dihedral angle between a face and the base.

16 Prove $a^2 = b^2 + c^2 - 2bc\cos A$ by dropping the altitude from C to AB and applying Pythagoras to the two right triangles. Consider both acute and obtuse A.

17 Two towers 10 km apart on an east–west line sight a fire on bearings $048°$ from the western tower A and $322°$ from the eastern tower B. Find AF, BF and the area of ABF.

18 The opening fire is located from bearings $040°$ and $330°$. If each bearing could be wrong by $1.5°$, estimate how far the calculated position could move. Does the nearer tower change?

19 **Extension.** A parallelogram has adjacent sides a and b and acute included angle θ. Show that its longer diagonal has length $\sqrt{a^2 + b^2 + 2ab\cos\theta}$ and its shorter diagonal has length $\sqrt{a^2 + b^2 - 2ab\cos\theta}$.

20 **Extension.** A cube has edge 1. Show that the angle between a space diagonal and any edge it meets is $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$, and find the angle between a space diagonal and a face it meets.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 20 marks

Solving triangles

Instructions. Answer all questions. Show sufficient working. A calculator is permitted. Give lengths and angles to 3 s.f. unless an exact answer is requested.

1 In triangle ABC, $A = 41^\circ$, $B = 68^\circ$ and $c = 12.4$ cm. Find a and b. [4]

2 Solve triangle ABC given $a = 8.5$ cm, $b = 6.2$ cm and $C = 52^\circ$. [4]

3 A triangle has sides 9.6 cm and 12.4 cm enclosing 47°. Find its area, and find the obtuse angle between the same two sides that gives the same area. [3]

4 A cuboid measures 10 m by 6 m by 5 m. Find its space diagonal and the angle it makes with the base. [3]

5 A ship sails 18 km on a bearing of 065°, then 12 km on a bearing of 155°. Find its distance and bearing from the starting point. [4]

6 Explain, with a sketch, why SSA can produce two triangles. State the condition for two solutions. [2]

Evidence: a correct matching pair, a complete SAS solution, a correct area and supplementary angle, a cuboid diagonal with its projection, a two-leg bearing calculation, and a precise ambiguous-case condition.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

The ambiguous triangle

Instructions. Work independently. Record results in a table and justify each generalisation. Keep angles to 1 d.p. where needed.

In triangle ABC, angle $A = 30^\circ$ and side $b = 10$ cm are fixed. Side a is allowed to vary. The sine rule gives $\sin B = \dfrac{10 \sin 30^\circ}{a} = \dfrac{5}{a}$.

1 Complete a table for $a = 4, 5, 6, 7, 8, 9, 10, 12, 15$. For each value, calculate $\sin B$, the possible values of B, and the number of non-degenerate triangles. [6]

2 Describe how the number of triangles changes as a increases. State the threshold values of a at which the number changes. [4]

3 Explain why no triangle exists when $a < 5$, and why exactly one exists when $a = 5$. [5]

4 Generalise: for an acute angle A and a fixed side b, state the conditions on a for zero, one or two triangles. [4]

5 Test your rule at $a = 10$ and $a = 9.9$. Explain what happens at the boundary $a = b$, where one of the sine solutions becomes degenerate. [3]

6 Explain how a compass-and-straight-edge construction with centre C and radius a shows the same pattern. [2]

Evidence: a consistent table, a described threshold pattern, correct $a < 5$ and $a = 5$ cases, a justified general rule, a boundary discussion, and a construction-based explanation.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Why two rules?

Instructions. Write a 400–500 word explanation for a classmate who believes that the sine rule is enough for every triangle. Your explanation must: (i) derive the cosine rule from coordinates; (ii) show how it reduces to Pythagoras when $A = 90^\circ$; (iii) explain which information each rule requires; (iv) work one example in which the sine rule alone is ambiguous and the cosine rule resolves it; and (v) identify one error to avoid. Use a diagram and correct notation.

Evidence: a complete coordinate derivation, the limiting case, a comparison of required information, a valid worked example, an identified error, and communication suited to the audience.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

A coastal survey

Instructions. Two headlands A and B are 4.2 km apart, with B due east of A. A lighthouse L is sighted on a bearing of 062° from A and 318° from B. A buoy C is 2.8 km from A on a bearing of 145°. A patrol boat travels from A to L and then to C at an average speed of 25 km/h.

1. Draw a clear diagram and find the three angles of triangle ABL. [4]
2. Calculate AL and BL. [5]
3. Calculate LC and the bearing of C from L. [6]
4. Find the total distance travelled by the patrol boat and the time for the journey, to the nearest minute. [4]
5. Each bearing could be wrong by 1.5°. Estimate how far the calculated position of L could move, state two assumptions of your model, and recommend how the survey should be checked. [5]

Evidence: a diagram with north arrows, accurate triangle angles, correct sine-rule distances, a correctly oriented bearing for C, a consistent speed–distance–time calculation, and a critical evaluation of error and assumptions.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- The sine rule $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ applies to every triangle and needs one complete side–angle pair. Use it for ASA, AAS and SSA.
- SSA can produce two triangles when the given angle is acute and opposite the shorter given side; both solutions must be stated.
- The cosine rule $a^2 = b^2 + c^2 - 2bc\cos A$ handles SAS and SSS. It becomes Pythagoras at 90°, and an obtuse angle makes $a^2 > b^2 + c^2$.
- The area is $\frac{1}{2}ab\sin C$, where C is the included angle; a parallelogram with adjacent sides a, b has area $ab\sin C$. Reversing the formula can give two supplementary angles.
- Bearings are measured clockwise from north and written with three digits. The back bearing differs by 180°, and angles inside a navigation triangle are differences of bearings.
- In three dimensions, isolate a right triangle: project a line onto a plane for a line–plane angle, and use perpendiculars to the intersection for a dihedral angle.
- Indirect measurements depend on a baseline, precise angles and a well-conditioned triangle; a small angular error can move a distant point a long way.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \qquad a^2 = b^2 + c^2 - 2bc\cos A, \qquad K = \tfrac{1}{2}ab\sin C$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can state the sine rule and identify a matching side–angle pair.
- I can find a side or an angle with the sine rule to 3 s.f.
- I can recognise and solve the ambiguous SSA case.
- I can derive and apply the cosine rule for SAS and SSS.
- I can choose between the sine rule, cosine rule and right-triangle trigonometry.
- I can calculate area with $\frac{1}{2}ab\sin C$ and reverse it to find an angle.
- I can use bearings, back bearings and north lines correctly.
- I can solve a two-leg navigation problem and use $d = st$.
- I can find a cuboid's space diagonal and its angle with the base.
- I can find a line–plane angle using a projection.
- I can find a dihedral angle using a perpendicular cross-section.
- I can evaluate the assumptions and error in an indirect measurement.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which rule did you find easier to choose correctly, and what check now prevents you from choosing the wrong one?
2. Describe one diagram error you made in this chapter and how redrawing it changed your answer.
3. The statement of inquiry says that the rules turn measurements into positions. Use the fire problem to explain what the mathematics adds beyond the two bearings.
4. Why is the ambiguous case not an error in the mathematics? What does it reveal about the information in SSA?
5. When you calculate a distance you cannot measure, what would make you trust the result — and what would make you ask for a second measurement?

DIRECTION, ARRAY, TRANSFORMATION

Vectors and Matrices

08

FORM · SPACE · REPRESENTATION

Two ways of writing down movement — one arrow at a time, or the whole plane at once — and the number that decides whether a movement can be reversed.

STATEMENT OF INQUIRY

Vectors encode displacement by magnitude and direction, while matrices encode transformations of the plane as arrays of numbers; the determinant measures how a transformation stretches or collapses area, and decides whether it can be undone.

KEY CONCEPT

Form

RELATED CONCEPTS

Space · Representation · Systems

GLOBAL CONTEXT

Scientific and technical innovation — computer graphics, navigation and engineering

FACTUAL

What does a column vector record, and what does it deliberately leave out?

CONCEPTUAL

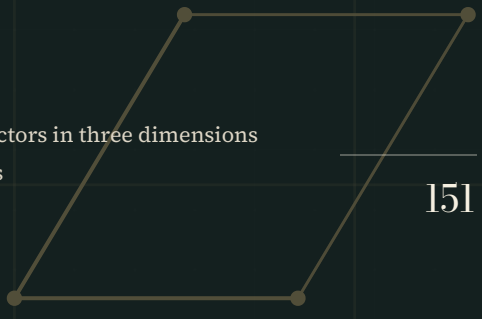
How can four numbers in a square array describe a rotation, a reflection or a stretch of the entire plane?

DEBATABLE

When a graphics engine transforms an object, has the object changed or only the coordinates used to describe it?

IN THIS CHAPTER

8.1 Vectors: notation and operations
8.2 Vectors and geometry
8.3 Vectors in three dimensions
8.4 Matrices and transformations
8.5 Determinants and inverse matrices
8.6 Investigation: area scale factors



OPENING PROBLEM · THE LOGO THAT COLLAPSED

Four corners, three edits, one stuck export

A studio stores its logo as a quadrilateral with corners $A(0,0)$, $B(3,0)$, $C(4,2)$ and $D(1,2)$, and animates it by applying a matrix to each corner's position vector. The approved lean is $L = \begin{pmatrix} 1 & 0.5 \\ 0 & 1 \end{pmatrix}$; the flip is $F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

On the final night an export setting applies $E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. The logo collapses to a straight line, and no slider brings it back.

The storyboard applies the flip first. Which product, LF or FL, matches it, and which matrix undoes the lean? And why is the export irreversible when the other two edits are not?

1. Write each corner as a position vector and calculate its image under L. Sketch both shapes on one grid.
2. Calculate LF and FL, apply each to $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$, and state which edit acts first in each case.
3. Find a matrix that reverses L, and use it to recover the corner that L sends to $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$.
4. Show that E sends every point onto the line $y = x$, and explain why no matrix can reverse this.

A vector is a movement written as a list of numbers. A matrix is a rule that moves every point at once. The two ideas meet in the equation $\vec{v}' = M\vec{v}$, and the number $\det M$ tells you whether that equation can be run backwards.

8.0 ORIENTATION Learning intentions

- Represent vectors in column and $\vec{i}, \vec{j}$ form; add, subtract and scale them.
- Find magnitudes and unit vectors; model displacement, velocity and force.
- Use position vectors to prove results about midpoints, parallel sides, collinearity and ratios.
- Extend vector methods to three dimensions.
- Represent transformations by 2×2 matrices, apply and combine them in order.
- Calculate determinants and inverses, solve 2×2 systems, and interpret a zero determinant.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

combine vectors in component form and interpret the result; find magnitudes and unit vectors; use $\overrightarrow{AB} = \vec{b} - \vec{a}$ for displacements, midpoints and distances; prove parallelism and collinearity; divide a segment in a ratio; work in three dimensions; state the order of a matrix and multiply matrices, remembering that AB means B first; write the standard transformation matrices; calculate determinants and inverses, verify $A^{-1}A = I$ and solve $A\vec{x} = \vec{b}$; explain a zero determinant as collapsed area and lost reversibility.

ATL SKILLS · THINKING AND TRANSFER

Critical thinking: test a doubtful matrix on $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$; those images determine it. **Transfer:** $\vec{b} - \vec{a}$ is "finish minus start"; the inverse is the "undo" of Chapter 2 as an array. **Communication:** distinguish position vectors from free displacements, and state the order in which a product acts.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box.

1. Solve $2x + y = 11$ and $x - y = 1$.
2. Find the hypotenuse of a right-angled triangle with legs 5 and 12.
3. Find the midpoint of $(-2, 3)$ and $(6, -5)$.
4. Make y the subject of $3x + 4y = 12$.
5. Reflect $(3, -2)$ in the x-axis and in $y = x$.
6. Evaluate $ad - bc$ when $a = 3, b = 1, c = 2, d = 4$.

REFRESHER · PYTHAGORAS, SIMULTANEOUS EQUATIONS AND COORDINATES

A segment's length comes from a right-angled triangle whose legs are the differences of the coordinates; for $(-2, 3)$ and $(6, -5)$ those differences are 8 and -8, so the midpoint is $(2, -1)$ and the length is $8\sqrt{2}$. Eliminate to solve: adding $2x + y = 11$ and $x - y = 1$ gives $3x = 12$, so $x = 4$ and $y = 3$. Reflections change or swap coordinates: in the x-axis $(x, y) \mapsto (x, -y)$, in the y-axis $(x, y) \mapsto (-x, y)$, in $y = x$ they swap.

CONCEPT AND METHOD

8.1 Vectors: notation and operations

A vector has both size and direction; a scalar has size only, like mass or time. On a grid a vector is a column of two numbers: $\begin{pmatrix}3\\-2\end{pmatrix}$ means 3 right and 2 down. The arrow may start anywhere; its length and direction identify it.

Addition is componentwise and has a picture: put the second arrow's tail at the first's head, and the sum runs from first tail to last head, the **triangle law**; the parallelogram gives the same sum. Subtraction adds the reversed arrow, so $\vec{a} - \vec{b}$ points from the head of $\vec{b}$ to the head of $\vec{a}$. A scalar multiple $k\vec{a}$ keeps direction when $k > 0$, reverses it when $k < 0$, and scales length by $|k|$.

The magnitude is $\left|\begin{pmatrix}x\\y\end{pmatrix}\right| = \sqrt{x^2 + y^2}$; dividing by it gives a unit vector. With $\vec{i} = \begin{pmatrix}1\\0\end{pmatrix}$ and $\vec{j} = \begin{pmatrix}0\\1\end{pmatrix}$, every vector is $x\vec{i} + y\vec{j}$. A vector tied to the origin is a position vector; a free one describes displacement, velocity or force.

DEFINITIONS

Components, equality, magnitude, unit vector

A vector is $\vec{a} = \begin{pmatrix} x \\ y \end{pmatrix} = x\vec{i} + y\vec{j}$. Two vectors are equal exactly when their components are equal. The **magnitude** is $|\vec{a}| = \sqrt{x^2 + y^2}$, and the unit vector in the direction of $\vec{a} \neq \vec{0}$ is $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$. For a scalar k, $k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$; $\vec{a}$ and $\vec{b}$ are parallel when one is a nonzero scalar multiple of the other.

EXAMPLE 8.1 COMBINING VECTORS AND FINDING A UNIT VECTOR

Let $\vec{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$. Find (a) $\vec{a} + \vec{b}$, (b) $3\vec{a} - 2\vec{b}$, (c) $|\vec{a}|$, (d) the unit vector in the direction of $\vec{a}$.

SOLUTION

(a) $\vec{a} + \vec{b} = \begin{pmatrix} 3 - 1 \\ -2 + 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

(b) $3\vec{a} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}$ and $2\vec{b} = \begin{pmatrix} -2 \\ 10 \end{pmatrix}$, so $3\vec{a} - 2\vec{b} = \begin{pmatrix} 11 \\ -16 \end{pmatrix}$.

(c) $|\vec{a}| = \sqrt{9 + 4} = \sqrt{13} \approx 3.61$.

(d) $\hat{a} = \dfrac{1}{\sqrt{13}}\begin{pmatrix} 3 \\ -2 \end{pmatrix} \approx \begin{pmatrix} 0.832 \\ -0.555 \end{pmatrix}$, and $0.832^2 + 0.555^2 \approx 1$.

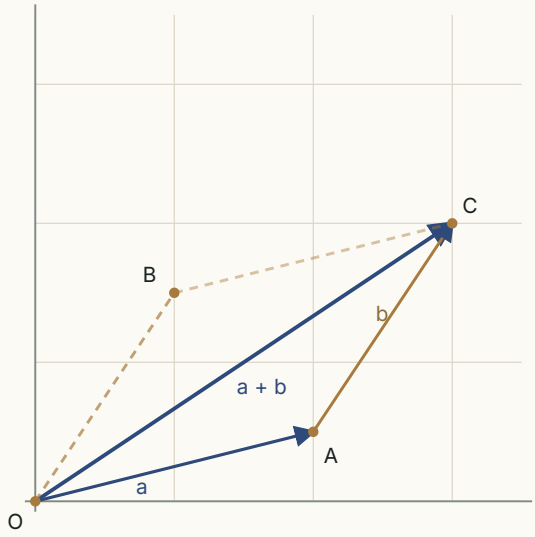
EXAMPLE 8.2 VELOCITY, WIND AND DISPLACEMENT

A drone flies with velocity $\vec{v} = 6\vec{i} - 3\vec{j}$ metres per second relative to the air, in a wind $\vec{w} = -\vec{i} + 2\vec{j}$. Find its ground velocity, its speed, and its displacement after 10 seconds.

SOLUTION

Velocities add as vectors: $\vec{v} + \vec{w} = \begin{pmatrix} 6 - 1 \\ -3 + 2 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$, so the ground velocity is $5\vec{i} - \vec{j}$.

The speed is $\sqrt{25 + 1} = \sqrt{26} \approx 5.10$ m/s. Displacement is velocity multiplied by time: $10\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 50 \\ -10 \end{pmatrix}$, so the drone is 50 m east and 10 m south of its start, at a distance of $10\sqrt{26} \approx 51.0$ m.



Scale: one grid square is 40 px, one unit. $\vec{OA} = \vec{a}$, $\vec{AC} = \vec{b}$ and $\vec{OC} = \vec{a} + \vec{b}$; the dashed arrows complete the parallelogram.

MISCONCEPTION

"$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$, and a vector is just the coordinates of a point." Take $\vec{a} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$: then $|\vec{a} + \vec{b}| = 5$ but $|\vec{a}| + |\vec{b}| = 7$. Lengths add only when the arrows point along the same line. And $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ names a movement, not the point $(3, 4)$; only with its tail at the origin does a vector coincide with a point.

GUIDED PRACTICE

1. For $\vec{a} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$, find $\vec{a} + \vec{b}$. (Answer: $\begin{pmatrix} -1 \\ -6 \end{pmatrix}$.)
2. Find $\left| \begin{pmatrix} 5 \\ -12 \end{pmatrix} \right|$. (Answer: 13.)
3. Find the unit vector in the direction of $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$. (Answer: $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$.)
4. For $\vec{a} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, find $2\vec{a} - 3\vec{b}$. (Answer: $\begin{pmatrix} -4 \\ 11 \end{pmatrix}$.)

EXERCISE 8A

COMPONENTS, MAGNITUDE AND UNIT VECTORS

FLUENCY

1 Write as a column vector: (a) 4 right, 3 up; (b) 2 left, 5 down; (c) 6 right, no vertical change; (d) 7 down, no horizontal change.

2 For $\vec{a} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ and $\vec{c} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, find (a) $\vec{a} + \vec{b}$, (b) $\vec{b} + \vec{c}$, (c) $\vec{a} - \vec{b}$, (d) $2\vec{a} + 3\vec{b}$.

3 Find the magnitude of each vector: $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $\begin{pmatrix} 5 \\ -12 \end{pmatrix}$, $\begin{pmatrix} -8 \\ 6 \end{pmatrix}$, $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

4 Find a unit vector in the direction of each of the first three vectors in Question 3.

5 Write in $\vec{i}, \vec{j}$ form: $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$, $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$.

6 Given $\vec{p} = 3\vec{i} - 4\vec{j}$ and $\vec{q} = -\vec{i} + 2\vec{j}$, find (a) $\vec{p} + \vec{q}$, (b) $\vec{p} - \vec{q}$, (c) $2\vec{p} - 3\vec{q}$, (d) $|\vec{p} + \vec{q}|$.

UNDERSTANDING AND REASONING

EXERCISE 8A · CONTINUED

COMPONENTS, MAGNITUDE AND UNIT VECTORS

7 **Error analysis.** A student writes $|(6,8)| = 6+8 = 14$ and $|(6,8)+(-6,-8)| = 28$. Identify both errors and give the correct values.

8 Using components, explain why $\overrightarrow{AB} = -\overrightarrow{BA}$ for any points A and B.

9 For $\vec{a} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, write two different vectors of length 10 parallel to $\vec{a}$, one in each direction.

10 The vector $\begin{pmatrix} x \\ 6 \end{pmatrix}$ has magnitude 10. Find the two possible values of x.

11 Explain why a vector cannot have a negative magnitude, even when both components are negative.

12 The resultant of $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\vec{u}$ is $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$. Find $\vec{u}$ and $|\vec{u}|$.

APPLICATION AND EXTENSION

13 A drone moves $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$, then $\begin{pmatrix} -3 \\ -4 \end{pmatrix}$, then $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$ metres. Find its final displacement and its distance from the start.

14 Forces $\begin{pmatrix} 8 \\ 3 \end{pmatrix}$ N and $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ N act on a particle. Find the resultant and its magnitude, and the force needed for equilibrium.

15 **Extension.** Prove that $|k\vec{a}| = |k|\,|\vec{a}|$ for any scalar k and vector $\vec{a}$. Hence explain why a nonzero vector has exactly two unit vectors parallel to it.

CONCEPT AND PROOF

8.2 Vectors and geometry

Geometry becomes arithmetic with position vectors. Fix an origin O and write $\vec{a} = \overrightarrow{OA}$: the position vector pins a point down, while the displacement from A to B is

$$\overrightarrow{AB} = \vec{b} - \vec{a},$$

"finish minus start", independent of O since moving the origin adds the same vector to $\vec{a}$ and to $\vec{b}$.

Hence the midpoint of AB is $\frac{1}{2}(\vec{a}+\vec{b})$ and $AB = |\vec{b}-\vec{a}|$. Parallel vectors are scalar multiples. A, B, C are collinear when $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel, so the vectors share A; parallel arrows alone may lie on different lines. To prove a theorem, translate the figure into equations — a parallelogram has $\overrightarrow{AB} = \overrightarrow{DC}$ — and manipulate them; no diagram is needed, though one suggests the equations.

DEFINITIONS

Position vector, midpoint, parallel, section formula

For A with position vector $\vec{a}$ and B with $\vec{b}$: $\overrightarrow{AB} = \vec{b} - \vec{a}$, the midpoint is $\frac{1}{2}(\vec{a}+\vec{b})$, and $AB = |\vec{b}-\vec{a}|$. Points A, B, C are collinear if $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel.

Section formula. If P divides AB internally with $AP : PB = m : n$, then

$$\vec{p} = \frac{n\vec{a} + m\vec{b}}{m+n}.$$

The weights are crossed, so the point nearer B carries more of $\vec{b}$.

EXAMPLE **8.3** COLLINEARITY AND DIVISION IN A RATIO

Show that $A(-1,1)$, $B(3,3)$ and $C(11,7)$ are collinear, find the ratio in which B divides AC, and confirm it with the section formula.

SOLUTION

$\overrightarrow{AB} = \begin{pmatrix}4\\2\end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix}12\\6\end{pmatrix} = 3\begin{pmatrix}4\\2\end{pmatrix}$. The vectors are parallel and share A, so the points are collinear.

$\overrightarrow{BC} = \begin{pmatrix}8\\4\end{pmatrix} = 2\overrightarrow{AB}$, so $AB : BC = 1 : 2$ and B divides AC internally in the ratio $1 : 2$.

Check with $m : n = 1 : 2$: $\vec{b} = \dfrac{2\vec{a} + \vec{c}}{3} = \dfrac{1}{3}\begin{pmatrix}-2+11\\2+7\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$. Note that $AB : AC = 1 : 3$, not $1 : 2$: a ratio of parts differs from a ratio to the whole.

EXAMPLE **8.4** PROVING A PROPERTY OF A PARALLELOGRAM

$ABCD$ is a parallelogram. Prove that its diagonals bisect each other.

SOLUTION

Let the position vectors of the vertices be $\vec{a}, \vec{b}, \vec{c}, \vec{d}$. Since AB is equal and parallel to DC,

$$\vec{b} - \vec{a} = \vec{c} - \vec{d}.$$

Rearranging gives $\vec{a} + \vec{c} = \vec{b} + \vec{d}$, and halving gives

$$\tfrac{1}{2}(\vec{a} + \vec{c}) = \tfrac{1}{2}(\vec{b} + \vec{d}).$$

The left side is the midpoint of AC, the right side the midpoint of BD. The midpoints have the same position vector, so they coincide: each diagonal passes through the midpoint of the other.

Converse. If the diagonals bisect each other, then $\vec{a} + \vec{c} = \vec{b} + \vec{d}$, so $\vec{b} - \vec{a} = \vec{c} - \vec{d}$ and AB is equal and parallel to DC: the quadrilateral is a parallelogram. No lengths or angles were used, so the result holds for every parallelogram.

MISCONCEPTION

"If $\overrightarrow{AB}$ is parallel to $\overrightarrow{CD}$ then A, B, C, D are collinear." Parallel arrows can lie on different lines; collinearity needs two vectors on the *same* line, which is why the test uses $\overrightarrow{AB}$ and $\overrightarrow{AC}$. A second slip is writing $\overrightarrow{AB} = \vec{a} + \vec{b}$: displacement is finish minus start.

GUIDED PRACTICE

1. For $A(2,1)$ and $B(6,4)$, find $\overrightarrow{AB}$ and $|\overrightarrow{AB}|$. (Answer: $\begin{pmatrix}4\\3\end{pmatrix}$; 5.)
2. Find the midpoint of the segment joining $(3,-1)$ and $(-5,7)$. (Answer: $(-1,3)$.)
3. P divides AB in the ratio $1 : 3$, with $A(0,0)$ and $B(8,12)$. Find P. (Answer: $(2,3)$.)
4. Are $\begin{pmatrix}2\\6\end{pmatrix}$ and $\begin{pmatrix}-3\\-9\end{pmatrix}$ parallel? Justify. (Yes: the second is -1.5 times the first.)

EXERCISE 8B

POSITION VECTORS, MIDPOINTS AND PROOF

FLUENCY

1 For $A(2,5)$, $B(-1,3)$ and $C(4,-2)$, find $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CA}$ and $|\overrightarrow{AB}|$.

2 Find the midpoint of each segment: (a) $A(1,3)$, $B(7,9)$; (b) $A(-4,2)$, $B(2,-8)$.

EXERCISE 8B · CONTINUED

POSITION VECTORS, MIDPOINTS AND PROOF

3 P divides AB internally in the ratio $2:1$, with $A(1,-2)$ and $B(7,4)$. Find the position vector of P.

4 For $A(1,1)$, $B(4,2)$, $C(5,5)$ and $D(2,4)$, show that $\overrightarrow{AB}=\overrightarrow{DC}$ and name the quadrilateral.

5 Decide whether each pair is parallel: (a) $\begin{pmatrix}2\\4\end{pmatrix}$, $\begin{pmatrix}1\\2\end{pmatrix}$; (b) $\begin{pmatrix}3\\-1\end{pmatrix}$, $\begin{pmatrix}-6\\2\end{pmatrix}$.

6 Find the distance between $A(1,3)$ and $B(5,6)$, and between $C(-2,4)$ and $D(3,-8)$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student shows that $\overrightarrow{AB}=2\overrightarrow{CD}$ and concludes that $ABCD$ is a parallelogram. Explain why this does not follow, and state what the equation does tell you.

8 Show that $A(-2,1)$, $B(1,5)$ and $C(7,13)$ are collinear, and state the ratio $AB:BC$.

9 Find the values of k for which $\begin{pmatrix}k\\3\end{pmatrix}$ and $\begin{pmatrix}4\\k-1\end{pmatrix}$ are parallel.

10 In parallelogram $OACB$, $\overrightarrow{OA}=\vec{a}$ and $\overrightarrow{OB}=\vec{b}$. M is the midpoint of AC and N the midpoint of BC. Show that $\overrightarrow{MN}$ is parallel to $\overrightarrow{AB}$ and half its length.

11 Prove the converse of Example 8.4: if the diagonals of quadrilateral $ABCD$ bisect each other, then $ABCD$ is a parallelogram.

12 For $A(1,1)$, $B(4,3)$ and $C(10,7)$, decide whether C lies on line AB; if so, state $AB:BC$.

APPLICATION AND EXTENSION

13 A field has corners $A(0,0)$, $B(60,0)$, $C(80,40)$ and $D(20,40)$, in metres. Show that it is a parallelogram and find the length of diagonal AC.

14 Tugs pull a ship with forces $\begin{pmatrix}12\\5\end{pmatrix}$ kN and $\begin{pmatrix}3\\9\end{pmatrix}$ kN. Find the magnitude of the resultant and the force a third tug must supply to hold the ship steady.

15 **Extension.** For $A(0,0)$, $B(4,1)$, $C(3,5)$ and $D(-1,4)$, show that all four sides have the same length and find where the diagonals meet. Explain why you must also know that the vertices are joined in order, or that the diagonals bisect each other, before concluding that the figure is a rhombus.

CONCEPT AND EXTENSION

8.3 Vectors in three dimensions

A point in space needs three numbers. With three mutually perpendicular axes meeting at O, every point is $P(x,y,z)$ and $\overrightarrow{OP}=\begin{pmatrix}x\\y\\z\end{pmatrix}$. The **coordinate planes** are $x=0$, $y=0$ and $z=0$: a point with one zero coordinate lies on one of them. Every formula of Sections 8.1 and 8.2 survives with one more term – $|\vec{v}|=\sqrt{x^2+y^2+z^2}$ applies Pythagoras twice, and displacement, midpoint and collinearity work as before. Only determinants and inverses remain two-dimensional.

DEFINITIONS

Vectors in space

A vector in three dimensions is $\vec{v}=\begin{pmatrix}x\\y\\z\end{pmatrix}=x\vec{i}+y\vec{j}+z\vec{k}$, with $|\vec{v}|=\sqrt{x^2+y^2+z^2}$. For $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$, $\overrightarrow{AB}=\begin{pmatrix}x_2-x_1\\y_2-y_1\\z_2-z_1\end{pmatrix}$ and the midpoint is $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\frac{z_1+z_2}{2}\right)$.

EXAMPLE 8.5 DISPLACEMENT, LENGTH AND MIDPOINT IN SPACE

For $A(2,-1,3)$ and $B(5,3,7)$, find $\overrightarrow{AB}$, $|\overrightarrow{AB}|$, the unit vector in that direction, and the midpoint of AB.

SOLUTION

$\overrightarrow{AB} = \begin{pmatrix}3\\4\\4\end{pmatrix}$, so $|\overrightarrow{AB}| = \sqrt{9+16+16} = \sqrt{41} \approx 6.40$, and the unit vector is $\dfrac{1}{\sqrt{41}}\begin{pmatrix}3\\4\\4\end{pmatrix} \approx \begin{pmatrix}0.469\\0.625\\0.625\end{pmatrix}$.

The midpoint is $\left(\frac{7}{2}, 1, 5\right) = (3.5, 1, 5)$, which agrees with $\begin{pmatrix}2\\-1\\3\end{pmatrix} + \frac{1}{2}\begin{pmatrix}3\\4\\4\end{pmatrix}$.

EXAMPLE 8.6 A CUBOID DIAGONAL AND A POINT BEYOND

(a) A cuboid has a corner at $O(0,0,0)$ and edges of lengths 4, 3 and 2 along the axes, so the opposite corner is $F(4,3,2)$. Find OF. (b) For $A(1,2,3)$ and $B(4,6,5)$, find C such that B is the midpoint of AC, and show that A, B, C are collinear.

SOLUTION

(a) $OF = \sqrt{16+9+4} = \sqrt{29} \approx 5.39$. The base diagonal is $\sqrt{16+9} = 5$, and the space diagonal is the hypotenuse of a right-angled triangle with legs 5 and 2. It is not $4+3+2=9$: the edges are perpendicular, so lengths combine by Pythagoras.

(b) From $\vec{b} = \frac{1}{2}(\vec{a}+\vec{c})$, $\vec{c} = 2\vec{b} - \vec{a} = \begin{pmatrix}8\\12\\10\end{pmatrix} - \begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}7\\10\\7\end{pmatrix}$. Then $\overrightarrow{AB} = \begin{pmatrix}3\\4\\2\end{pmatrix} = \overrightarrow{BC}$, so the points are collinear and $AC = 2\sqrt{29} \approx 10.77$.

MISCONCEPTION

"The length of $\begin{pmatrix}1\\2\\2\end{pmatrix}$ is $1+2+2=5$." Adding components gives the length of a staircase path, not the straight line; the magnitude is $\sqrt{1+4+4} = 3$. Also, $(3,0,2)$ is not the point $(3,2)$: the zero says it lies on the plane $y=0$.

GUIDED PRACTICE

1. Find $\left|\begin{pmatrix}2\\3\\6\end{pmatrix}\right|$. (Answer: 7.)
2. Find the midpoint of $A(1,-2,4)$ and $B(7,6,-2)$. (Answer: $(4,2,1)$.)
3. Write the displacement from $(5,1,0)$ to $(2,4,3)$. (Answer: $\begin{pmatrix}-3\\3\\3\end{pmatrix}$.)
4. On which coordinate plane does $(0,2,-5)$ lie? (The yz-plane, $x=0$.)

EXERCISE 8C POINTS, LENGTHS AND LINES IN SPACE

EXERCISE 8C · CONTINUED

POINTS, LENGTHS AND LINES IN SPACE

2 Find each magnitude: $\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\begin{pmatrix}1\\-2\\2\end{pmatrix}$, $\begin{pmatrix}4\\4\\7\end{pmatrix}$, $\begin{pmatrix}0\\3\\-4\end{pmatrix}$.

3 Find the midpoint of each pair in Question 1.

4 Find a unit vector in the direction of the first two vectors in Question 2.

5 For $\vec{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}-1\\4\\0\end{pmatrix}$, find (a) $\vec{a}+\vec{b}$, (b) $\vec{a}-\vec{b}$, (c) $3\vec{a}-2\vec{b}$, (d) $|\vec{a}+\vec{b}|$.

6 State which coordinate plane or axis contains each point: $(3,0,2)$, $(0,4,0)$, $(5,0,0)$, $(1,1,0)$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student calculates $\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\right| = 1+2+2=5$. Explain the error and give the correct value.

8 Show that $A(1,0,2)$, $B(3,2,4)$ and $C(6,5,7)$ are collinear, and state $AB:BC$.

9 $P(2,1,4)$ and $Q(5,4,z)$ satisfy $PQ=\sqrt{27}$. Find the two possible values of z.

10 A cuboid measures $2\times3\times6$. Show that its space diagonal is 7, and explain why the answer is not 11.

11 A square-based pyramid has base $A(0,0,0)$, $B(4,0,0)$, $C(4,4,0)$, $D(0,4,0)$ and apex $V(2,2,6)$. Find VA and the height VM, where M is the centre of the base, and verify that M is the midpoint of AC.

12 Explain why the midpoint formula needs no change when a third coordinate is added.

APPLICATION AND EXTENSION

13 A drone is at $(10,20,30)$ metres and moves with velocity $\begin{pmatrix}-2\\3\\-1\end{pmatrix}$ m/s. Find its position after 5 seconds and its distance from the origin then.

14 A vertical mast 12 m tall stands with its base at $(3,4,0)$. A guy wire runs from the top to the origin. Find its length.

15 **Extension.** For $A(1,2,3)$, $B(4,2,3)$ and $C(4,6,7)$, show that triangle ABC is right-angled at B by finding all three side lengths and using the converse of Pythagoras.

CONCEPT AND REPRESENTATION

8.4 Matrices and transformations

A matrix is a rectangular array of numbers, of order "rows × columns". Matrices of the same order are added entry by entry, and a scalar multiplies every entry. Multiplication is different: AB is defined only when the number of columns of A equals the number of rows of B, and the entry in row i, column j of AB is the sum of products from row i of A and column j of B.

The rule looks arbitrary until it is seen as composition. If B transforms a vector and then A transforms the result, the single matrix doing both is AB; because order matters, $AB\neq BA$ in general. A 2×2 matrix acts by $M\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}ax+by\\cx+dy\end{pmatrix}$, and its columns are the images of $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$, so a transformation is determined by where it sends the axes. The origin is always fixed, since $M\vec{0}=\vec{0}$.

DEFINITIONS AND STANDARD MATRICES

Order, identity, product, transformations

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$,

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

The **identity** $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ satisfies $AI = IA = A$, and $\vec{v} \mapsto M\vec{v}$ is a transformation fixing the origin.

Rotation $90°$ anticlockwise: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; $180°$: $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Reflection in the x-axis: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$; in the y-axis: $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$; in $y = x$: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Enlargement k: kI. Stretch parallel to the x-axis: $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$. A shear parallel to the x-axis, factor k, is $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, which fixes $(1, 0)$ and sends $(0, 1)$ to $(k, 1)$.

EXAMPLE 8.7 MATRIX ARITHMETIC AND THE FAILURE OF COMMUTATIVITY

Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 4 \\ -2 & 5 \end{pmatrix}$. Find $2A - 3B$, AB and BA, and compare.

SOLUTION

$2A = \begin{pmatrix} 4 & -2 \\ 6 & 0 \end{pmatrix}$ and $3B = \begin{pmatrix} 3 & 12 \\ -6 & 15 \end{pmatrix}$, so $2A - 3B = \begin{pmatrix} 1 & -14 \\ 12 & -15 \end{pmatrix}$.

For AB, row 1 times column 1 gives $2(1) + (-1)(-2) = 4$; continuing,

$$AB = \begin{pmatrix} 4 & 3 \\ 3 & 12 \end{pmatrix}, \qquad BA = \begin{pmatrix} 14 & -1 \\ 11 & 2 \end{pmatrix}.$$

The products differ, so matrix multiplication is not commutative.

> Both products exist here only because both matrices are 2×2; even then they need not agree.

EXAMPLE 8.8 COMPOSITE TRANSFORMATIONS AND THEIR ORDER

Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation $90°$ anticlockwise) and $S = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ (reflection in the y-axis). For $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, calculate $RS\vec{v}$ and $SR\vec{v}$, and describe each composite.

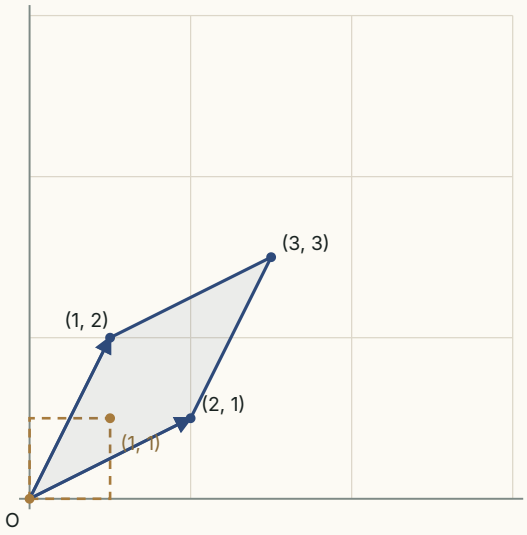
SOLUTION

RS means "apply S first, then R", since $RS\vec{v} = R(S\vec{v})$. Now $S\vec{v} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, then $R\begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

In the other order, $R\vec{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$, then $S\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. The matrices confirm this:

$$RS = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad SR = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

So RS is the reflection in $y = -x$, while SR is the reflection in $y = x$. Read a product from right to left, like the innermost bracket of a composite function.



Scale: one grid square is 46 px, one unit. The dashed unit square maps to the parallelogram with vertices $(0,0)$, $(2,1)$, $(3,3)$, $(1,2)$ under $\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$; its columns are the two arrows shown.

MISCONCEPTION

"Multiply matrices entry by entry, and AB means A acts first." For $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, the matching-entry product is $\begin{pmatrix}0 & 2\\ 3 & 0\end{pmatrix}$, but the true product is $\begin{pmatrix}2 & 1\\ 4 & 3\end{pmatrix}$. And $AB\vec{v} = A(B\vec{v})$, so B acts first: the matrix on the right is the inner operation.

GUIDED PRACTICE

1. State the order of $\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}$. (Answer: 2×3.)
2. For $A = \begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$ and $B = \begin{pmatrix}1 & -1\\ 2 & 0\end{pmatrix}$, find AB. (Answer: $\begin{pmatrix}2 & -2\\ 7 & -1\end{pmatrix}$.)
3. Find the image of $\begin{pmatrix}3\\ -4\end{pmatrix}$ under reflection in the y-axis. (Answer: $\begin{pmatrix}-3\\ -4\end{pmatrix}$.)
4. For $M = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$ and $N = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$, find MN and describe it. (Answer: $\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, reflection in $y = x$.)

EXERCISE 8D

MATRIX ALGEBRA AND TRANSFORMATIONS

FLUENCY

1 State the order of each matrix and the value of the named entry: (a) $\begin{pmatrix}2 & -1 & 3\\ 0 & 4 & 1\end{pmatrix}$, a_{13}; (b) $\begin{pmatrix}5\\ -2\end{pmatrix}$, a_{21}.

2 For $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}0 & 1\\ 2 & -1\end{pmatrix}$, find $A + B$, $A - B$, $3A$ and $2A - B$.

3 For the same matrices, find AB and BA. Are they equal?

4 Write the matrix for each transformation: (a) reflection in the x-axis; (b) rotation $180°$ about O; (c) enlargement of scale factor 4 about O.

5 Find the image of $\begin{pmatrix}2\\ -5\end{pmatrix}$ under each transformation in Question 4.

6 Find the image of $\begin{pmatrix}3\\ 1\end{pmatrix}$ under a rotation of $90°$ anticlockwise, and under a shear of factor 2 parallel to the x-axis.

UNDERSTANDING AND REASONING

EXERCISE 8D · CONTINUED

MATRIX ALGEBRA AND TRANSFORMATIONS

7 **Error analysis.** A student computes $\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}5&6\\7&8\end{pmatrix}=\begin{pmatrix}5&12\\21&32\end{pmatrix}$. Explain the error and give the correct product.

8 Explain why a 2×3 matrix cannot multiply another 2×3 matrix, but can multiply a 3×1 matrix. State the order of the result.

9 The matrix $\begin{pmatrix}0&1\\-1&0\end{pmatrix}$ represents a rotation. Describe it and explain how it relates to $\begin{pmatrix}0&-1\\1&0\end{pmatrix}$.

10 A transformation sends $(1,0)$ to $(3,2)$ and $(0,1)$ to $(-1,4)$. Write its matrix and find the image of $(1,1)$.

11 Show that reflecting in the x-axis and then in the y-axis is the same as rotating by $180°$, by multiplying the two matrices.

12 Explain why every 2×2 matrix transformation fixes the origin, and why a shape not containing the origin therefore also changes position.

APPLICATION AND EXTENSION

13 A logo has corners $(0,0)$, $(2,0)$, $(3,1)$ and $(1,1)$. Apply the shear $\begin{pmatrix}1&1\\0&1\end{pmatrix}$ to each corner and describe how the logo changes.

14 A robot arm moves a tool by $\begin{pmatrix}12\\-5\end{pmatrix}$ cm, then by $\begin{pmatrix}-4\\9\end{pmatrix}$ cm. Find the single displacement with the same effect and its length.

15 **Extension.** For $R=\begin{pmatrix}0&-1\\1&0\end{pmatrix}$ and $S=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$, show that $RS\neq SR$ by calculating both, and identify each composite as a reflection in a line through the origin.

CONCEPT AND METHOD

8.5 Determinants and inverse matrices

The determinant of $A=\begin{pmatrix}a&b\\c&d\end{pmatrix}$ is $\det A=ad-bc$: it measures how the transformation changes area, and its sign records whether the plane is turned over, as Section 8.6 shows. When $\det A\neq0$ there is an inverse with $A^{-1}A=AA^{-1}=I$:

$$A^{-1}=\frac{1}{ad-bc}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}.$$

When $\det A=0$ the matrix is singular: it collapses the plane onto a line or a point, and no inverse exists. Then $A\vec{x}=\vec{b}$ has no unique solution; otherwise $\vec{x}=A^{-1}\vec{b}$.

DEFINITIONS

Determinant, inverse, singular

For $A=\begin{pmatrix}a&b\\c&d\end{pmatrix}$, $\det A=ad-bc$. If $\det A\neq0$, then $A^{-1}=\dfrac{1}{\det A}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$; if $\det A=0$, A is singular and has no inverse. Also $\det I=1$ and $\det(AB)=\det A\cdot\det B$.

EXAMPLE **8.9** AN INVERSE, A CHECK AND A SYSTEM

For $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$, find $\det A$ and A^{-1}, verify that $AA^{-1} = I$, and hence solve $3x + y = 7$ and $2x + 4y = 8$.

SOLUTION

$\det A = 12 - 2 = 10$, so

$$A^{-1} = \frac{1}{10}\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 0.4 & -0.1 \\ -0.2 & 0.3 \end{pmatrix}.$$

Then $AA^{-1} = \begin{pmatrix} 1.2 - 0.2 & -0.3 + 0.3 \\ 0.8 - 0.8 & -0.2 + 1.2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, so the inverse is correct. The factor $\frac{1}{10}$ multiplies every entry.

Writing the system as $A\vec{x} = \vec{b}$ with $\vec{b} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$ gives $\vec{x} = A^{-1}\vec{b}$:

$$\vec{x} = \frac{1}{10}\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}\begin{pmatrix} 7 \\ 8 \end{pmatrix} = \frac{1}{10}\begin{pmatrix} 28 - 8 \\ -14 + 24 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Check. $3(2) + 1 = 7$ and $2(2) + 4(1) = 8$.

EXAMPLE **8.10** A SINGULAR MATRIX AND THE COLLAPSE OF AREA

For $B = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$, show that B has no inverse, describe its effect on the unit square, and explain what this implies for $2x + 4y = 3$ and $x + 2y = 1$.

SOLUTION

$\det B = 4 - 4 = 0$, so B is singular. Its columns $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ are parallel, so the unit square maps to a line segment on $y = \frac{1}{2}x$ and its area becomes zero.

For the equations, doubling $x + 2y = 1$ gives $2x + 4y = 2$, which contradicts $2x + 4y = 3$. There is no solution. If the right side were $2x + 4y = 2$, the equations would be identical and every point of $x + 2y = 1$ would be a solution: infinitely many. A zero determinant means "no unique solution", never "one solution".

MISCONCEPTION

"Invert $\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ by dividing by the determinant and keeping the entries." The formula rearranges and changes signs: $\frac{1}{10}\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}$. Equally damaging is continuing when $\det A = 0$: division by zero is undefined, and no algebra will recover the lost information.

GUIDED PRACTICE

1. Find the determinant of $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$. (Answer: -1.)

2. Find the inverse of $\begin{pmatrix} 4 & 1 \\ 3 & 1 \end{pmatrix}$. (Answer: $\begin{pmatrix} 1 & -1 \\ -3 & 4 \end{pmatrix}$.)

3. For which k is $\begin{pmatrix} k & 3 \\ 2 & k-1 \end{pmatrix}$ singular? (Answer: $k = 3$ or $k = -2$.)

4. A matrix has determinant -6. What is its area scale factor? (Answer: 6; the sign shows reversed orientation.)

EXERCISE 8E

DETERMINANTS, INVERSES AND SYSTEMS

FLUENCY

1 Find the determinant of each matrix.

a. $\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$

b. $\begin{pmatrix} 5 & -2 \\ 3 & 1 \end{pmatrix}$

c. $\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$

d. $\begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$

2 Find the inverse of each matrix in Question 1 that is invertible.

3 Verify that $A^{-1}A = I$ for $A = \begin{pmatrix} 4 & 1 \\ 3 & 1 \end{pmatrix}$.

4 Solve each system using an inverse matrix: (a) $4x + y = 9$, $3x + y = 7$; (b) $3x + 2y = 8$, $x + 4y = -2$.

5 Find the value(s) of k for which $\begin{pmatrix} k & 4 \\ 3 & k+1 \end{pmatrix}$ is singular.

6 State whether each matrix has an inverse, giving a reason: (a) $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$; (b) $\begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}$; (c) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes $A^{-1} = \frac{1}{7}\begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix}$ for $A = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix}$. Identify two errors and give the correct inverse.

8 Explain why a matrix with determinant zero cannot have an inverse, using the idea that its transformation loses information.

9 A matrix has determinant -3. Describe its effect on area and on orientation.

10 For $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$, calculate $\det A$, $\det B$, AB and $\det(AB)$. Comment.

11 The system $2x + 3y = 5$, $4x + 6y = 10$ has infinitely many solutions. Explain how the determinant reveals this, and describe the solution set.

12 Explain why $(AB)^{-1} = B^{-1}A^{-1}$ makes sense for transformations, and check it for the matrices in Question 10.

APPLICATION AND EXTENSION

13 A message is encoded by multiplying number pairs by $M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. The pair $(7, 4)$ is received. Decode it using M^{-1}.

14 A transformation maps a shape of area 24 cm^2 to one of area 60 cm^2. A possible matrix is $\begin{pmatrix} k & 0 \\ 0 & 2.5 \end{pmatrix}$. Find k and the determinant.

15 **Extension.** Show that if $\det A \neq 0$ then $\det(A^{-1}) = \dfrac{1}{\det A}$, and use this to find the determinant of the inverse of $\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ without computing the inverse.

PATTERN AND GENERALISATION

8.6 Investigation: area scale factors

The determinant is usually met as a formula. This investigation reverses the order: by measuring areas of images you will discover what the determinant measures, and why a zero determinant destroys reversibility.

INVESTIGATION 8 · MEASURING WHAT A MATRIX DOES TO AREA

From unit square to area scale factor

1. Copy the table. For each matrix A, calculate $\det A$ and the images of $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$; plot them, name each shape and find its area.
2. Complete the last two columns, then relate the image area to $\det A$ in one sentence.
3. Write a conjecture for the area scale factor of a 2×2 matrix, including negative determinants.
4. Test it on the triangle $(0,0)$, $(3,1)$, $(1,2)$, area 2.5: confirm this by a 3×2 rectangle minus three right-angled triangles. Apply $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find the image triangle and its area by the same method or the shoelace rule $\frac{1}{2}|x_1y_2 + x_2y_3 + x_3y_1 - x_2y_1 - x_3y_2 - x_1y_3|$.
5. Repeat with $C = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$. Compare the area ratio with $\det C$ and state whether orientation is preserved.
6. Examine the row with $\det A = 0$. Using the columns, explain why every image point lies on one line, and link this to Section 8.5.

MATRIX A	$\det A$	IMAGE VERTICES	AREA	AREA FACTOR	$\lvert\det A\rvert$
$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	1	$(0,0), (1,0), (1,1), (0,1)$	1	1	1
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	3	$(0,0), (2,1), (3,3), (1,2)$	3		
$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$	2	$(0,0), (1,1), (0,2), (-1,1)$	2		
$\begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$	-3	$(0,0), (3,0), (3,-1), (0,-1)$	3		
$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$	0	$(0,0), (1,2), (3,6), (2,4)$	0		

Generalise. For any 2×2 matrix A, every area is multiplied by $|\det A|$: positive determinants preserve orientation, negative ones reverse it, and zero destroys area and reversibility. Explain the rule for an axis-aligned rectangle with sides $p\vec{i}$ and $q\vec{j}$, then explain why it applies to every shape, since any shape can be divided into small squares.

MODELLING · CRITERION D

Restoring the exported logo

Return to the logo with corners $A(0,0)$, $B(3,0)$, $C(4,2)$, $D(1,2)$, the lean $L = \begin{pmatrix} 1 & 0.5 \\ 0 & 1 \end{pmatrix}$ and the export $E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

1. **Model.** Calculate the corner vectors after the lean, and show that the logo's area is unchanged. Explain why this agrees with $\det L$.
2. **Combine.** The storyboard applies the flip first and the lean second. Write the single matrix for this composite and find the final position of C.
3. **Invert.** Find L^{-1} and use it to recover a corner of the approved logo from the leaned corner $(5, 2)$. Check against the original coordinates.
4. **Evaluate.** Explain why E cannot be inverted, using both the determinant and the geometry of the export. State one assumption about screen coordinates and one way the model fails on a non-square pixel grid.

CUMULATIVE

8.7 Mixed practice

Attempt all questions without notes. Decide first whether a diagram, a component calculation or a matrix product is the shortest route.

MIXED PRACTICE 8

VECTORS, GEOMETRY, MATRICES AND INVERSES

FLUENCY AND UNDERSTANDING

1. For $\vec{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$, find $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$, $2\vec{a} + 3\vec{b}$ and $|\vec{a}|$.
2. Find a unit vector parallel to $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.
3. For $A(1,3)$ and $B(5,6)$, find $\overrightarrow{AB}$, $|\overrightarrow{AB}|$ and the midpoint of AB.
4. P divides AB in the ratio $3:2$, with $A(-1,4)$ and $B(9,-1)$. Find P.
5. Show that $A(0,1)$, $B(4,3)$ and $C(10,6)$ are collinear, and state $AB : BC$.
6. For $A(2,1,0)$ and $B(5,5,12)$, find $|\overrightarrow{AB}|$.
7. Find the midpoint of $(1,-2,4)$ and $(7,6,-2)$.
8. State the order of $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{pmatrix}$ and the value of a_{21}.
9. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$, find AB and BA.
10. Find the image of $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ under reflection in $y = x$, under a rotation of $90°$ anticlockwise, and under an enlargement of factor 3.

REASONING AND APPLICATION

11. Find the determinant and inverse of $\begin{pmatrix} 4 & 3 \\ 2 & 5 \end{pmatrix}$.
12. Solve $4x + 3y = 18$ and $2x + 5y = 16$ using an inverse matrix.
13. Find the values of k for which $\begin{pmatrix} k & 4 \\ 3 & k+1 \end{pmatrix}$ is singular.
14. For $P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, find PQ and QP, and describe each composite.
15. In parallelogram $ABCD$, X is the midpoint of AB and Y the midpoint of CD. Show that $AXYD$ is a parallelogram.
16. Show that $(0,0)$, $(5,0)$, $(8,4)$ and $(3,4)$ are the vertices of a rhombus, and find the lengths of its diagonals.

MIXED PRACTICE 8 · CONTINUED

17 A matrix sends $\begin{pmatrix}1\\0\end{pmatrix}$ to $\begin{pmatrix}3\\1\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ to $\begin{pmatrix}-1\\2\end{pmatrix}$. Write the matrix and find the image of $\begin{pmatrix}2\\3\end{pmatrix}$.

19 **Extension.** Prove that $\det(AB) = \det A \cdot \det B$ for $A = \begin{pmatrix}a & b\\c & d\end{pmatrix}$ and $B = \begin{pmatrix}e & f\\g & h\end{pmatrix}$ by expanding both sides.

18 A triangle of area $8\ \text{cm}^2$ is transformed by $\begin{pmatrix}2 & 0\\0 & 3\end{pmatrix}$. Find the area of the image, and explain your reasoning.

20 **Extension.** The matrix $\begin{pmatrix}2 & 1\\1 & 1\end{pmatrix}$ has integer entries and determinant 1. Find its inverse and explain why the inverse also has integer entries. Find one further 2×2 matrix with integer entries and determinant 1.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Vectors, matrix products and inverses

Instructions. Answer all questions. Show sufficient working. A calculator is permitted but not required.

1 For $\vec{a} = \begin{pmatrix}4\\-3\end{pmatrix}$ and $\vec{b} = \begin{pmatrix}-2\\5\end{pmatrix}$, find $2\vec{a} - \vec{b}$, $|\vec{a}|$ and the unit vector in the direction of $\vec{a}$. [4]

2 Show that $A(1, 2)$, $B(4, 6)$ and $C(10, 14)$ are collinear, and state the ratio $AB : BC$. [4]

3 For $A = \begin{pmatrix}2 & -1\\3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}0 & 5\\1 & -2\end{pmatrix}$, find AB and BA. [4]

4 Find the inverse of $\begin{pmatrix}5 & 2\\3 & 1\end{pmatrix}$, verify that $AA^{-1} = I$, and hence solve $5x + 2y = 1$ and $3x + y = 0$. [8]

Evidence: a correct linear combination and unit vector, a justified collinearity with ratio, both matrix products, a correct inverse with verification, and a solved system consistent with that inverse.

CRITERION B · INVESTIGATING PATTERNS — 40 minutes · 24 marks

Repeating a shear

Instructions. Work independently. Record results in a table and justify each generalisation.

Let $S = \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$, a shear parallel to the x-axis. Powers are defined by $S^2 = SS$, $S^3 = S^2S$, and so on.

1 Calculate S^2, S^3 and S^4, and record them in a table with the entries of the top row. [6]

2 Describe the pattern in the entries, and predict S^n. [4]

3 Verify your formula for $n = 6$, prove it by induction, and find S^{-1}; explain how it fits the pattern. [8]

4 Repeat for $R = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}$: calculate R^2, R^3, R^4 and describe the cycle geometrically. [4]

5 State a general rule for $\det(S^n)$ and for $\det(R^n)$, and explain both. [2]

Evidence: a systematic table, a described pattern, a verified and proved general formula, a correct inverse linked to S^n, a described rotation cycle, and a justified determinant rule.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining why order matters

Instructions. Write a 300–400 word explanation for a classmate who believes that AB always equals BA and that AB means "A first, then B". State the correct meaning of $AB\vec{v}$; give a numerical counterexample using a rotation and a reflection; explain why order changes the result, using the images of $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$; and state what this implies for solving $A\vec{x}=\vec{b}$ by inverse. Use correct notation and a diagram or table where it helps.

Evidence: a correct statement of order, a computed counterexample with matrices, a causal explanation using the images of the basis vectors, and a valid consequence for inverse methods.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Mapping a survey by drone

Instructions. Camera coordinates (x, y) map to map coordinates by $M=\begin{pmatrix}1.5 & 0.5\\0.5 & 1.0\end{pmatrix}$, in metres. The drone's airspeed is $\begin{pmatrix}4\\-3\end{pmatrix}$ m/s and the wind is $\begin{pmatrix}-1\\-1\end{pmatrix}$ m/s.

1. State the order of M, calculate $\det M$, and decide whether the mapping can be reversed. [3]
2. A feature is at $(150, 80)$ in camera coordinates. Find its map coordinates. [4]
3. A feature appears at $(300, 200)$ on the map. Find M^{-1}, recover its camera coordinates and verify the result. [5]
4. A rectangular field measures 200 m by 150 m in camera coordinates. Find its area on the map and explain the determinant's role. [4]
5. Find the drone's ground velocity and the time to cover 500 m over the ground. [3]
6. State two assumptions of the model and evaluate how well it describes a flight over hilly ground with a changing wind. [5]

Evidence: a correct determinant and decision, accurate forward and inverse mappings with a check, an area justified by the determinant, a velocity addition leading to a time, and explicit assumptions with critical evaluation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A vector records magnitude and direction; components add and scale entry by entry, matching the triangle and parallelogram laws.
- $|\vec{a}| = \sqrt{x^2 + y^2}$, and $\vec{a}/|\vec{a}|$ is the unit vector. In space, add a z-component and a z^2 term.
- With position vectors, $\overrightarrow{AB} = \vec{b} - \vec{a}$; the midpoint is $\frac{1}{2}(\vec{a} + \vec{b})$ and distance is $|\vec{b} - \vec{a}|$. Parallel vectors are scalar multiples; collinear points need a shared point; the section formula divides a segment in a ratio.
- An $m \times n$ matrix multiplies an $n \times p$ matrix to give an $m \times p$ matrix. Multiplication is not commutative, and AB means B acts first.
- Standard transformation matrices send the basis vectors to their columns; every such transformation fixes the origin.
- $\det A = ad - bc$ is the area scale factor. If $\det A = 0$ the matrix is singular and no inverse exists; otherwise A^{-1} solves $A\vec{x} = \vec{b}$ uniquely.

$$\overrightarrow{AB} = \vec{b} - \vec{a}, \qquad |\vec{v}| = \sqrt{x^2 + y^2 + z^2}, \qquad \det A = ad - bc, \qquad A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can add, subtract and scale vectors in component form.
- I can find a magnitude and a unit vector.
- I can move between column and $\vec{i}, \vec{j}$ notation.
- I can distinguish position vectors from free displacements.
- I can find a displacement, a midpoint and a distance.
- I can prove vectors parallel and points collinear.
- I can divide a segment in a given ratio.
- I can work with three-dimensional vectors.
- I can state the order of a matrix and multiply compatible matrices.
- I can explain why AB and BA differ, and why B acts first.
- I can write and apply matrices for standard transformations.
- I can calculate a determinant and an inverse, and verify the result.
- I can solve a 2×2 system by inverse and interpret $\det A = 0$.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which representation — arrow, column vector or matrix — made the hardest problem in this chapter easiest, and why?
2. Describe one error you made with signs or order, and the check that now prevents it.
3. The statement of inquiry links the determinant to whether a transformation can be undone. Use the singular matrix in Section 8.5 to explain that link in your own words.
4. Why does a proof with position vectors not need a diagram, even though a diagram usually suggests the proof?
5. What question about vectors or matrices do you still want to answer?

SAME AVERAGE, DIFFERENT STORY

Reading the Spread

09

RELATIONSHIPS · PATTERNS · REPRESENTATION

Measuring how far numbers scatter, how two quantities move together, and how far a straight line can be trusted.

STATEMENT OF INQUIRY

Two groups can share a centre and differ in every other respect; measuring dispersion and association describes those differences precisely, while the origin of the data limits what the description can prove.

KEY CONCEPT

Relationships

RELATED CONCEPTS

Patterns · Representation · Measurement

GLOBAL CONTEXT

Identities and relationships — how groups differ, and how claims about them are made

FACTUAL

What does a standard deviation measure that a mean does not?

CONCEPTUAL

Why can two variables be strongly associated without one causing the other?

DEBATABLE

When should a statistical model justify a decision about people?

IN THIS CHAPTER

9.1 Measuring spread
9.2 Standard deviation and grouped data
9.3 Comparing distributions
9.4 Scatter graphs and correlation
9.5 Regression and prediction
9.6 Investigation: correlation, causation and reliability

OPENING PROBLEM · SAME AVERAGE, DIFFERENT STORY

Two programmes, one stopwatch

A school trials two eight-week sprint programmes, with ten runners in each. Their final 100 m times, in seconds, are:

A: $13.1,\ 13.6,\ 14.0,\ 14.3,\ 14.5,\ 14.5,\ 14.7,\ 15.0,\ 15.4,\ 15.9$
B: $12.4,\ 13.0,\ 13.9,\ 14.2,\ 14.5,\ 14.5,\ 14.8,\ 15.1,\ 16.0,\ 16.6$

Both groups have mean 14.5 seconds. **Is the average enough to choose between them — and what would it take to claim that a programme caused the difference?**

1. Calculate the mean and median of each programme. What do they hide?
2. Calculate the range and interquartile range. Which programme is more consistent?
3. Write a recommendation using at least three summary measures as evidence.
4. List two pieces of evidence that would strengthen a causal claim, and one that would undermine it.

A mean is a single point pretending to speak for a group. This chapter supplies the language to disagree with it: quartiles, standard deviation, correlation and regression describe how values are arranged, how two variables move together, and how cautiously a line may predict.

ORIENTATION

9.0 Learning intentions

- Measure and interpret range, quartiles, interquartile range and standard deviation, including for grouped data.
- Construct and read five-number summaries, box plots and cumulative frequency graphs, and identify outliers.
- Compare distributions by centre, spread, shape and outliers, and justify a conclusion with evidence.
- Use scatter graphs and the correlation coefficient r to describe linear association.
- Fit and interpret a least-squares line, use residuals and r^2, and judge when prediction is trustworthy.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

find quartiles and the IQR using a stated convention; draw a box plot to scale; test an outlier and explain its effect; calculate variance and standard deviation from a list and from grouped data; interpret a standard deviation in context; compare distributions by centre, spread, shape and outliers; read a cumulative frequency graph; describe a scatter graph and calculate r; fit $y = a + bx$, interpret its parameters and distinguish interpolation from extrapolation; calculate residuals and r^2; and evaluate a claim using its sampling, provenance and causal language.

ATL SKILLS · CRITICAL THINKING AND COMMUNICATION

Critical thinking: every summary is a choice, and selecting one is an argument. **Research:** ask who collected the data, from whom, with what wording, and who paid. **Communication:** a comparative conclusion needs a direction, a size and a qualification.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box.

1. Find the mean, median and mode of $4,\ 7,\ 7,\ 9,\ 13$.
2. Find the median of $12,\ 15,\ 18,\ 21,\ 24,\ 26$.
3. Evaluate $\sum(x-5)^2$ for $x = 3,\ 5,\ 7,\ 9$.
4. A score of 72 is at the 80th percentile. Interpret this.
5. Plot $(1,3), (2,5), (3,4), (4,7)$ and describe the trend.
6. Solve $\dfrac{k}{4} = 0.35$.

REFRESHER · CENTRE, SIGMA NOTATION AND PERCENTAGES

The mean is the total divided by n. The median is the middle ordered value, or the mean of the two middle values. The range is the largest value minus the smallest. Sigma notation compresses a sum: $\sum(x-\bar{x})^2$ means subtract the mean from each value, square, then add. Squaring removes the sign, so a deviation of -3 contributes 9. A percentage is a fraction of 100: 0.35 is 35%, and 35% of 80 is 28.

On a scatter graph each point represents one individual measured on two variables. A rising trend is positive, a falling trend negative, and a formless cloud has no linear trend.

CONCEPT AND MEASURE

9.1 Measuring spread

Two groups can share a mean and differ utterly. In the opening problem one programme contains both the fastest and the slowest runner. A measure of dispersion describes how tightly values cluster. The simplest is the range, largest minus smallest, but it uses only two observations, so one unusual value can inflate it.

The **quartiles** divide the ordered data into four parts of roughly equal size. Q_1 has about one quarter of values at or below it, Q_3 about three quarters. The interquartile range $Q_3 - Q_1$ spans the middle half and ignores the extreme quarters. When n is odd, exclude the median and take the median of each half; when n is even, split into two equal halves. Software may use other conventions, so state yours.

A five-number summary records minimum, Q_1, median, Q_3 and maximum. Drawn to scale, these give a box plot: a box from Q_1 to Q_3, a line at the median, and whiskers to the most extreme non-outliers. The box shows the middle half; asymmetry in the whiskers or the median's position indicates skew.

An outlier is unusually far from the rest. The $1.5 \times \mathrm{IQR}$ rule flags values below $Q_1 - 1.5\,\mathrm{IQR}$ or above $Q_3 + 1.5\,\mathrm{IQR}$. It is a convention, not a law: a flagged value may be an error, a genuine extreme case, or evidence of two populations. Investigate before deleting.

DEFINITIONS

Range, quartiles, IQR, five-number summary, outlier, percentile

The **range** is maximum $-$ minimum. The **lower quartile** Q_1 and **upper quartile** Q_3 are the medians of the lower and upper halves. The **interquartile range** is $\mathrm{IQR} = Q_3 - Q_1$. The **five-number summary** is $(\min, Q_1, \text{median}, Q_3, \max)$. A value is a **potential outlier** if it lies below $Q_1 - 1.5\,\mathrm{IQR}$ or above $Q_3 + 1.5\,\mathrm{IQR}$.

The pth **percentile** has about $p\%$ of the data at or below it; quartiles are the 25th, 50th and 75th percentiles. Percentiles describe position, not quality.

EXAMPLE **9.1** QUARTILES, IQR AND AN OUTLIER

The waiting times, in minutes, for nine customers are $3,\ 5,\ 6,\ 8,\ 9,\ 11,\ 12,\ 15,\ 27$. Find the range, median, quartiles and IQR, and test 27 using the $1.5 \times \mathrm{IQR}$ rule.

SOLUTION

With $n = 9$, the median is the fifth value, 9. The lower half $3, 5, 6, 8$ has median 5.5; the upper half $11, 12, 15, 27$ has median 13.5. Hence $Q_1 = 5.5$, $Q_3 = 13.5$ and $\mathrm{IQR} = 8$. The range is $27 - 3 = 24$.

The fences are $5.5 - 12 = -6.5$ and $13.5 + 12 = 25.5$. Since $27 > 25.5$, 27 is a potential outlier. The other eight values lie between 3 and 15, so the range is driven by one customer; the IQR of 8 describes the typical spread far better.

> Without 27, the range falls to 12 and the IQR to 6. The median and IQR are **resistant**; the mean and range are not.

EXAMPLE **9.2** A FIVE-NUMBER SUMMARY

Find the five-number summary of $12,\ 14,\ 15,\ 17,\ 18,\ 19,\ 21,\ 23,\ 24,\ 26,\ 29,\ 31$, test for outliers, and describe the shape.

SOLUTION

With $n = 12$, the median is $\dfrac{19 + 21}{2} = 20$. The lower half has median $\dfrac{15 + 17}{2} = 16$; the upper half has median $\dfrac{24 + 26}{2} = 25$. Thus the summary is

$$12, \quad 16, \quad 20, \quad 25, \quad 31.$$

The fences are $16 - 13.5 = 2.5$ and $25 + 13.5 = 38.5$, so there are no outliers and the whiskers reach 12 and 31.

The upper whisker is 6 units long and the lower 4, and the mean 20.75 lies slightly above the median. The distribution is approximately symmetric with a mild positive skew.

> A box plot hides the sample size and any clusters inside a quartile; always state n.

MISCONCEPTION

"The lower quartile is one quarter of the way along the range." Quartiles are positions in the ordered data. For $3, 5, 6, 8, 9, 11, 12, 15, 27$, one quarter of the range is 6, which would give $Q_1 = 9$; the correct value is 5.5. The median is also not always a data value: with an even n it is usually the mean of two values.

GUIDED PRACTICE

1. Find the median and range of $7, 9, 9, 12, 15, 18, 21$. (12; 14.)
2. Find Q_1, Q_3 and IQR of $3, 4, 6, 8, 10, 12, 14, 16$. (5; 13; 8.)
3. Is 40 an outlier in $12, 15, 18, 20, 22, 25, 28, 40$? (No: fences 1.5 and 41.5.)
4. A value is at the 90th percentile of 200 observations. About how many are at or below it? (180.)

EXERCISE 9A

RANGE, QUARTILES, BOX PLOTS AND OUTLIERS

FLUENCY

1 Find the range, median, quartiles and IQR of (a) $4, 7, 9, 10, 12, 15, 18$; (b) $21, 24, 24, 27, 30, 33, 35, 41$.

2 Write the five-number summary of $5, 6, 8, 8, 10, 12, 13, 15, 22$.

3 Test 32 for being an outlier in $2, 5, 6, 7, 8, 9, 10, 11, 32$.

4 The visiting times, in minutes, of 50 museum visitors are: $0 < t \leq 10$: 6; $10 < t \leq 20$: 14; $20 < t \leq 30$: 18; $30 < t \leq 40$: 10; $40 < t \leq 50$: 2. Complete a cumulative frequency column and estimate the median.

5 A runner's time is at the 80th percentile of 150 runners, where smaller is better. Is she in the fastest or slowest fifth? Interpret.

6 The 25th percentile of clinic waiting times is 12 minutes. Interpret this for a patient.

UNDERSTANDING AND REASONING

7 **Error analysis.** For Example 9.2, a student takes Q_1 as the value in position $12 \div 4 = 3$, namely 15. Identify the error and give the correct value.

8 Each value is increased by 5. State which of range, median, IQR and mean change, with reasons.

9 Two sets both have range 20. Give an example where one has IQR 2 and the other IQR 15. Why is range alone insufficient?

10 A value is flagged as an outlier. Give two reasons it may still be correct, and say what you would check.

11 A box plot has a long right whisker and mean greater than median. Describe the shape and explain the mean's movement.

12 Why is a median estimated from a grouped frequency table only an estimate? Refer to class intervals.

APPLICATION AND EXTENSION

13 Depot P: $22, 25, 26, 28, 30, 31, 33, 35, 48$. Depot Q: $24, 27, 28, 29, 30, 31, 32, 33, 34$. Find five-number summaries and compare predictability. What does P's extreme value suggest?

14 A hospital reports median 18 min, IQR 7 min and mean 31 min. Explain how all three can be true, and advise a patient.

15 **Extension.** Every value is multiplied by 2 then increased by 3. Find the new median, quartiles and IQR in terms of m, q_1, q_3. Explain why adding 3 changes neither spread measure.

CONCEPT AND METHOD

9.2 Standard deviation and grouped data

The standard deviation uses every value. It answers: on average, how far is a value from the mean? Signed deviations add to zero, so we square them first; the mean of the squares is the variance, and its square root is the standard deviation.

For a data set of n values,

$$s^2 = \frac{\sum (x - \bar{x})^2}{n}, \qquad s = \sqrt{s^2}.$$

If the data are a **sample** used to estimate a larger population, many calculators divide by $n - 1$, giving a slightly larger *sample standard deviation*. For large n the difference is small; for $n = 5$ it is not. This course uses n for the data set being described; state clearly if you use $n - 1$. A standard deviation is non-negative, has the same units as the data, and is not a percentage. In many roughly symmetric distributions, about two thirds of values lie within one standard deviation of the mean.

For grouped data the individual values are lost. Replace each class by its midpoint and weight by the frequency:

$$\bar{x} \approx \frac{\sum f x}{n}, \qquad s^2 \approx \frac{\sum f x^2}{n} - \bar{x}^2.$$

The estimate assumes every value sits at its midpoint. Use equal-width classes with no gaps, usually five to twelve classes: too few hide the shape, too many leave noisy estimates.

DEFINITIONS

Deviation, variance, standard deviation

The **deviation** of x_i is $x_i - \bar{x}$. The **variance** is the mean squared deviation,

$$s^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2,$$

and the **standard deviation** is $s = \sqrt{s^2}$. For grouped data with midpoints x_i and frequencies f_i, use $s^2 \approx \dfrac{\sum f_i x_i^2}{n} - \bar{x}^2$, where $n = \sum f_i$. The **sample standard deviation** uses $n - 1$ in place of n.

EXAMPLE **9.3** CALCULATING A STANDARD DEVIATION

Find the variance and standard deviation of $5, 7, 8, 10, 12, 13, 15$, using denominator n. Then find the value using $n - 1$.

SOLUTION

The total is 70, so $\bar{x} = 10$. The squared deviations are

$$25,\ 9,\ 4,\ 0,\ 4,\ 9,\ 25,$$

with total 76. Hence

$$s^2 = \frac{76}{7} = 10.857\ldots, \qquad s = 3.295\ldots \approx 3.30.$$

With $n - 1 = 6$, $s_{n-1} = \sqrt{76/6} = 3.559\ldots \approx 3.56$. The two differ by about 8%, typical for a small set.

> The deviations sum to zero; if they do not, the mean in the table is wrong.

EXAMPLE 9.4 GROUPED MEAN AND STANDARD DEVIATION

Forty students recorded weekly study hours. Estimate the mean and standard deviation.

SOLUTION

HOURS	MIDPOINT x	f	fx	fx^2
$0 < h \le 4$	2	4	8	16
$4 < h \le 8$	6	11	66	396
$8 < h \le 12$	10	15	150	1500
$12 < h \le 16$	14	7	98	1372
$16 < h \le 20$	18	3	54	972
Total		40	376	4256

$$\bar{x} = \frac{376}{40} = 9.4, \qquad s^2 = \frac{4256}{40} - 9.4^2 = 18.04, \qquad s = 4.247\ldots \approx 4.25.$$

A typical student studied about 9.4 hours, give or take roughly 4.2. If students cluster near the top of each class, the true values would be slightly larger.

> In the formula, $\bar{x}^2$ is the square of the mean, not the mean of the squares.

MISCONCEPTION

"The standard deviation is the average distance from the mean, so average the deviations." The signed deviations always total zero, so their average is zero. The standard deviation is the square root of the *mean squared* deviation. Also check units before comparing: 2 cm and 2 mm are not the same spread.

GUIDED PRACTICE

1. Find the mean and standard deviation of $2, 4, 4, 6, 9$. (5; $s^2 = 5.6$, $s = 2.37$.)
2. Every value is increased by 10. What happens to mean and sd? (Mean +10; sd unchanged.)
3. Every value is doubled. What happens to sd? (It doubles.)
4. Midpoints $5, 15, 25$ have frequencies $4, 6, 10$. Estimate the mean. (18.)

EXERCISE 9B VARIANCE, STANDARD DEVIATION AND GROUPED ESTIMATES

FLUENCY

1 Find the mean, variance and standard deviation of $3, 5, 6, 8, 9, 11$.

2 Find the mean, variance and standard deviation of $6, 8, 10, 12, 14, 16, 18, 20$.

3 Estimate the mean and standard deviation of the ages: $0 < a \le 10$: 5; $10 < a \le 20$: 12; $20 < a \le 30$: 18; $30 < a \le 40$: 10; $40 < a \le 50$: 5.

4 Complete a deviation table for $4, 6, 9, 11, 14$ and find s.

5 A set of lengths has sd 4.2 cm. State it in millimetres and metres.

6 Estimate the mean mass: $10 < m \le 20$: 7; $20 < m \le 30$: 13; $30 < m \le 40$: 15; $40 < m \le 50$: 5.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student finds the deviations sum to zero and concludes $s = 0$. Explain the mistake and the correct procedure.

8 A set has sd 6. Find the new sd if every value is (a) increased by 10, (b) multiplied by 3, (c) divided by 2. Justify.

9 Why does a calculator offer both σx and sx? Which matches this section, and when is the other preferred?

10 Two classes both average 60; their sds are 3 and 12. Compare them in two sentences.

EXERCISE 9B · CONTINUED

VARIANCE, STANDARD DEVIATION AND GROUPED ESTIMATES

11 Why can a grouped estimate of the mean be close while the estimate of sd is less accurate?

12 A student compares sd of heights in cm with sd of masses in kg and says height is more variable. Explain the error.

APPLICATION AND EXTENSION

13 Machine A fills 500 g bags with sd 4 g; B fills 502 g with sd 1.5 g. Which is more consistent? Which risks underfilling? Recommend one and state the trade-off.

14 Route 1: mean 22 min, sd 5.1 min. Route 2: mean 26 min, sd 1.4 min. Advise a student who must arrive by a fixed time.

15 **Extension.** Show that for two values a, b, the sd using n is $\frac{|a-b|}{2}$. Why is this half the range?

REASONING AND COMMUNICATION

9.3 Comparing distributions

A comparison is an argument, and a mean alone is a weak premise. Describe **centre** (mean and median), **spread** (range, IQR, standard deviation), **shape** (symmetric or skewed) and **outliers**, then state what the combination implies. The same groups can support different recommendations depending on whether consistency or a high peak matters.

Parallel box plots place distributions on one scale; back-to-back stem plots and paired histograms compare shape while preserving more detail. Neither replaces the numbers: a box plot hides clusters and sample size, and a stem plot becomes unwieldy for large n. The standard deviation uses every value and is sensitive to extremes; the IQR describes the middle half and resists them. For skewed data or data with outliers, median and IQR usually give a fairer summary, while mean and sd may be pulled towards the tail.

Summary statistics can mislead even when correct. A mean salary can be inflated by one executive; a pass rate can exclude students who left; a graph can start its axis above zero so a small change looks dramatic. Ask what was measured, on whom, and what was left out.

DEFINITIONS

Centre, spread, shape, parallel box plot

The **centre** is described by mean and median; the **spread** by range, IQR and sd. A distribution is **symmetric** if its sides are mirror images, **positively skewed** with a longer right tail, and **negatively skewed** with a longer left tail. A **parallel box plot** draws two or more box plots on a common labelled scale.

A conclusion is **evidence-based** when it names the measure, its value, the direction of the difference and a qualification. “B is better” is an opinion; “B has median 64 versus 60 and IQR 8 versus 17, so it is higher and more consistent, although the sample is small” is an argument.

EXAMPLE **9.5** SAME MEAN, DIFFERENT SPREAD

Compare the two sprint programmes using centre, spread, shape and outliers. Recommend one for consistency and one for a chance of an exceptional time.

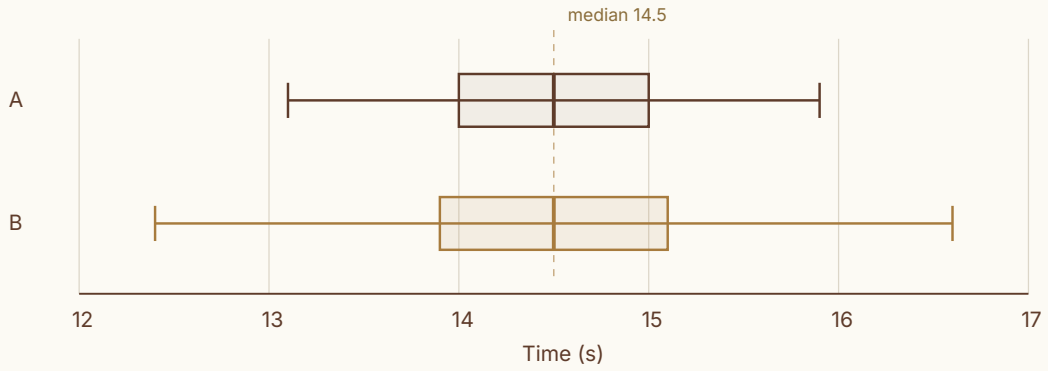
SOLUTION

Both sets have mean and median 14.5. For A: range 2.8, $Q_1 = 14.0$, $Q_3 = 15.0$, IQR 1.0, $s = 0.782$. For B: range 4.2, $Q_1 = 13.9$, $Q_3 = 15.1$, IQR 1.2, $s = 1.192$. B's sd is about 1.5 times A's, and its range is 1.5 times as large.

No value is an outlier: A's fences are 12.5 and 16.5, B's are 12.1 and 16.9. Both distributions are approximately symmetric, but B stretches further both ways, containing the fastest 12.4 and slowest 16.6.

Conclusion. The programmes have the same average, but A is more consistent. A team valuing predictability should choose A. A team wanting a chance of an exceptional time might choose B, accepting a greater chance of a slow time. With ten runners and no random assignment, neither comparison establishes causation.

> "Same mean" is not "same distribution"; spread, shape and context complete the argument.



Both programmes have median 14.5 s; B has a wider box and longer whiskers.

EXAMPLE **9.6** A TRUE AVERAGE THAT MISLEADS

Nine salaries, in thousands of pounds, are $18, 19, 20, 21, 22, 23, 24, 25, 90$. The company advertises "average salary £29,100". Is this accurate? Write a fairer summary.

SOLUTION

The total is 262, so the mean is $29.111\ldots$, which rounds to 29.1. The statement is arithmetically correct.

But the median is 22, $Q_1 = 19.5$, $Q_3 = 24.5$, IQR 5, and 90 is an outlier because the upper fence is 32. Every salary except the outlier lies between 18 and 25. The mean and sd (21.6) are dominated by the one exceptional value.

Fairer summary. "The median salary is £22,000; the middle half lies between £19,500 and £24,500." This describes a typical employee without hiding the outlier.

> "Average" is ambiguous; choosing mean or median is a rhetorical act as well as a mathematical one.

MISCONCEPTION

"If two groups have the same mean, their distributions are the same" or "the larger mean is always better." The opening means are identical while the spreads differ by a factor of about 1.5. A larger mean may come with much greater spread, so the "better" group may also be less reliable. State which feature matters, then support it with the appropriate measure.

GUIDED PRACTICE

1. A: mean 50, sd 2; B: mean 50, sd 5. Which is more consistent? (A.)
2. A box plot has median 20, $Q_1 = 17$, $Q_3 = 23$. Describe the middle half. (Between 17 and 23; IQR 6.)
3. Mean 50, median 70. What shape? (Negative skew.)
4. Why can a box plot hide an important feature? (It shows positions only; clusters, gaps and n are invisible.)

EXERCISE 9C

COMPARING CENTRES, SPREADS AND SHAPES

FLUENCY

1 Waiting times, in minutes, at two cafés. Café X: $6, 8, 9, 10, 11, 12, 14, 15, 25$. Café Y: $8, 9, 10, 10, 11, 11, 12, 12, 13$. Find median, quartiles and IQR, test outliers, and compare.

2 Group P: 21,24,27,30,33,36,42,48,51. Group Q: 22,25,31,34,38,40,45,52,56. Find the median and range of each.

3 Classes: 9P mean 61, median 60, IQR 14, sd 9; 9Q 64, 65, 7, 4; 9R 58, 62, 20, 13. Which is most consistent? Which has the highest typical score?

4 Explain the difference between sd 0 and range 0. Can one be zero while the other is not?

5 Which is likely negatively skewed: mean 45, median 50, or mean 50, median 45?

6 Match: (a) "middle half of salaries"; (b) "full span of temperatures"; (c) "uses every value, sensitive to extremes" — to range, IQR or sd.

UNDERSTANDING AND REASONING

7 **Error analysis.** "Team B is better because its highest score is 98 versus 91." Why is this invalid? Name two fairer summaries.

8 Construct two small sets with the same mean but different sds, and verify both.

9 A set has mean 20, median 19, range 30, IQR 8. A value 50 is added. Describe the likely effect on each.

10 Why are median and IQR preferred for skewed data? Use the salary example.

11 A test-score graph starts its vertical axis at 55. How can a 2-mark difference look dramatic? State the fair-graph principle.

12 Groups have sd 2 cm and 0.02 m. Which is more variable? Explain.

APPLICATION AND EXTENSION

13 A newspaper ranks schools by mean exam score and a small selective school tops the table. Give two reasons this may not measure teaching quality, and suggest a better comparison.

14 Waiting times: median 18, IQR 7, mean 31, one wait of 190 minutes. Write an accurate statement for patients and explain why the mean alone is unfair.

15 **Extension.** Why is sd more informative than range for monitoring production, but IQR preferable when genuine extreme values occur? Refer to one observation's effect.

CONCEPT AND REPRESENTATION

9.4 Scatter graphs and correlation

A scatter graph plots one point per individual, with one variable on each axis. The pattern describes the **association**: direction, strength, form and outliers. The horizontal variable is the explanatory variable and the vertical is the **response**. The names suggest explanation, but the choice is often a modelling decision; r is unchanged if the axes are swapped.

Direction is positive if the cloud rises, negative if it falls. Strength describes how closely points follow a pattern. Form may be linear or curved: a strong curve can have r near 0, because the correlation coefficient measures *linear* association only. One outlier can change r substantially.

Pearson's r lies between -1 and 1. Values of x and y that are both above their means, or both below, contribute positively; opposite movements contribute negatively. The formula standardises the contributions, so r has no units. Correlation is not causation, and r is not a percentage.

DEFINITIONS

Bivariate data, explanatory and response variables, correlation

Bivariate data are pairs measured on the same individuals. The **explanatory variable** is used to explain or predict; the **response** is explained or predicted. The **Pearson correlation coefficient** is

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \, \sum (y - \bar{y})^2}}.$$

The sign gives direction and $|r|$ gives linear strength; $r = 0$ means no linear association, not no relationship. **Correlation does not imply causation.**

EXAMPLE **9.7** DESCRIBING A SCATTER GRAPH

Nine students recorded weekly training hours x and improvement y, in seconds:

x	2	3	3	4	5	6	6	7	8
y	0.3	0.4	0.6	0.5	0.8	1.0	0.9	1.2	1.3

Identify the variables and describe the association.

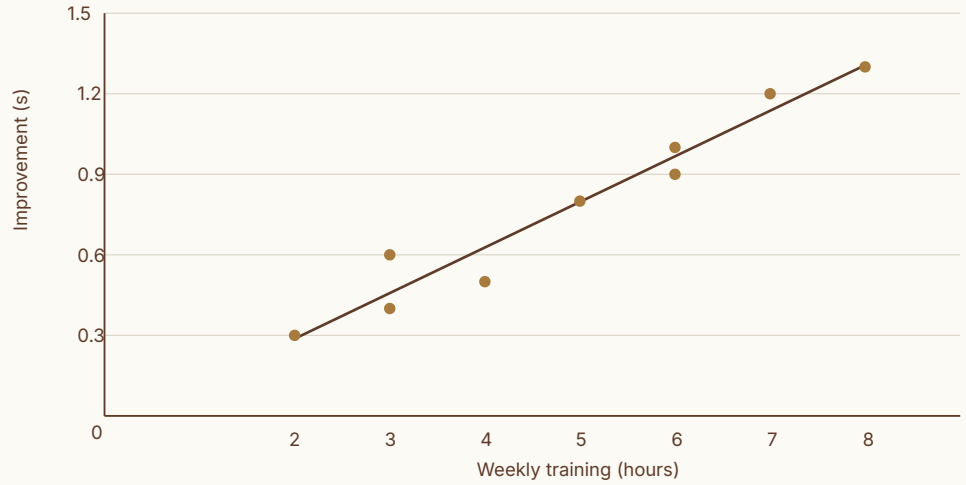
SOLUTION

Hours is explanatory and improvement response, because the question is whether more training is associated with greater improvement.

The association is **positive**, **strong** and approximately **linear**: the points form a rising band with no obvious curve. There are no points far from the pattern, although the two students training 3 hours improved by different amounts (0.4 and 0.6), as did the two training 6 hours (0.9 and 1.0). These are ordinary variation, not outliers.

Because improvements are small, a difference of 0.1 second matters in context. The scatter justifies fitting a line, but does not show that training caused the improvement.

> A description needs four words: direction, strength, form and outliers.



The points rise together around a straight line, fitted in Section 9.5.

EXAMPLE **9.8** CALCULATING r

Calculate r for $(1, 2), (2, 4), (3, 5), (4, 4), (5, 7), (6, 8), (7, 9)$ and interpret it.

SOLUTION

Here $n = 7$, $\bar{x} = 4$, $\bar{y} = \dfrac{39}{7} = 5.571\ldots$. The deviations give:

x	y	$x - \bar{x}$	$y - \bar{y}$	PRODUCT	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1	2	-3	-3.57	10.71	9	12.76
2	4	-2	-1.57	3.14	4	2.47
3	5	-1	-0.57	0.57	1	0.33
4	4	0	-1.57	0	0	2.47
5	7	1	1.43	1.43	1	2.04
6	8	2	2.43	4.86	4	5.90
7	9	3	3.43	10.29	9	11.76
				31.00	**28**	**37.71**

$$r = \frac{31}{\sqrt{28 \times 37.714\ldots}} = \frac{31}{\sqrt{1056}} = 0.9539\ldots \approx 0.954.$$

The value is positive and close to 1: a strong positive linear association. It is not perfect; the point $(4, 4)$ lies below the general rise.

> Use the unrounded mean $39/7$ in the squares, or the third decimal of r may change.

MISCONCEPTION

"$r = 0.8$ means 80% of the change in y is caused by x," or "$r = 1.2$ is very strong." Correlation cannot exceed 1 in absolute value; 1.2 signals an error. And r is not a percentage. Its square, r^2, is a proportion of variation, not of causation. A strong r can arise from a third variable, a curved relationship, or a restricted group.

GUIDED PRACTICE

1. Identify explanatory and response: "daily temperature and heating cost". (Temperature; cost.)
2. Describe $r = -0.85$. (Strong negative linear association.)
3. A scatter is a perfect downward line. State r. (-1.)
4. Why does $r = 0$ not imply no relationship? (The relationship may be curved.)

EXERCISE 9D

SCATTER GRAPHS, ASSOCIATION AND CORRELATION

FLUENCY

1 Identify explanatory and response: (a) hours studied and test score; (b) temperature and heating cost; (c) age and height of children; (d) advertising and sales.

2 Describe direction and strength: (a) 0.92; (b) -0.45; (c) 0.03; (d) -0.88.

3 Calculate r for $(2,5), (3,6), (4,6), (5,8), (6,9), (7,11)$.

4 Match $r = 0.95, -0.86, 0.12, -0.34$ to: weak negative; strong positive; almost none; strong negative.

5 A study reports $r = 0.68$ between sport and happiness. Interpret in one sentence, with a limitation.

6 A rising scatter has one point far below the band. Describe the association and the point's effect on r.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student reports $r = 1.15$ as "extremely strong". Identify both errors.

8 $r = 0.90$ between books in a home and reading scores. Why does this not show that buying books raises scores? Name two confounders.

9 Without calculating, how would removing a point far from a strong positive pattern affect r?

10 A correlation is calculated only for students scoring above 80. Why may it be smaller than for the whole year?

11 Decide whether causation is plausible and name a third variable: (a) ice-cream sales and drowning; (b) shoe size and reading age; (c) umbrella sales and rain.

12 Why can a strong curve give $r \approx 0$? Sketch an example with mean lines.

APPLICATION AND EXTENSION

13 A survey of 240 students at one school finds $r = -0.41$ between screen time and exam score. A headline says "Phones damage grades". Explain what the study supports, referring to strength, sample and confounding.

14 Coat sales and ice-cream sales are strongly negatively correlated over a year. Why does neither cause the other? Identify the driving variable.

15 **Extension.** Show that r is unchanged if every x is multiplied by positive k and every y by positive m. What does this say about units, and why does a negative constant change the sign?

MODELLING AND PREDICTION

9.5 Regression and prediction

A scatter graph shows a pattern; a line of best fit turns it into a model. A line by eye is reasonable but personal. The **least-squares regression line** removes the choice: it minimises the sum of squared vertical distances from the points. Vertical distances are used because the line predicts y from x.

For $y = a + bx$, the gradient b is the predicted change in y per unit increase in x; the intercept a is the predicted y at $x = 0$. The formulas use the same deviations as r:

$$b = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}, \qquad a = \bar{y} - b\bar{x}.$$

The line passes through $(\bar{x}, \bar{y})$. The gradient has units of y per unit of x; the intercept has units of y.

Prediction inside the observed range of x is **interpolation**. Prediction outside it is extrapolation, and risky: the pattern may bend or the mechanism may change. A model fitted to advertising spends from £100 to £800 has no authority at £10,000, however high its r^2.

The difference between observed and predicted values is a residual, $y - \hat{y}$. A residual plot scatters residuals against x. No pattern suggests a line is reasonable; a curve suggests non-linearity; a widening funnel suggests changing variability; a single large residual invites a check. Residuals sum to zero when the line has an intercept. The **coefficient of determination** r^2 is the proportion of variation in y accounted for by the linear model. If $r = 0.90$, then $r^2 = 0.81$: 81% of the variation is associated with the linear change in x. A high r^2 does not prove causation or make extrapolation safe.

DEFINITIONS

Least-squares line, residual, interpolation, extrapolation, r^2

The **least-squares regression line** of y on x is $y = a + bx$, with $b = \dfrac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$ and $a = \bar{y} - b\bar{x}$. The **predicted value** is $\hat{y} = a + bx$; the **residual** is $y - \hat{y}$. **Interpolation** predicts inside the observed x-range; **extrapolation** predicts outside it. The **coefficient of determination** r^2 is the proportion of variation in y explained by the linear regression.

EXAMPLE **9.9** FITTING, PREDICTING AND CHECKING

For the seven pairs of Example 9.8, find the least-squares line, predict y at $x = 4.5$ and $x = 9$, calculate the residuals, and find r^2.

SOLUTION

From Example 9.8, $\bar{x} = 4$, $\bar{y} = 5.571\ldots$, $\sum(x - \bar{x})^2 = 28$ and $\sum(x - \bar{x})(y - \bar{y}) = 31$. Hence

$$b = \frac{31}{28} = 1.1071\ldots \approx 1.11, \qquad a = 5.571\ldots - 1.1071\ldots(4) = 1.1429\ldots \approx 1.14.$$

The line is $y = 1.14 + 1.11x$. The gradient means that a one-unit increase in x is associated with about 1.11 more in y. The intercept is the prediction at $x = 0$, outside the data, so treat it as a positioning constant.

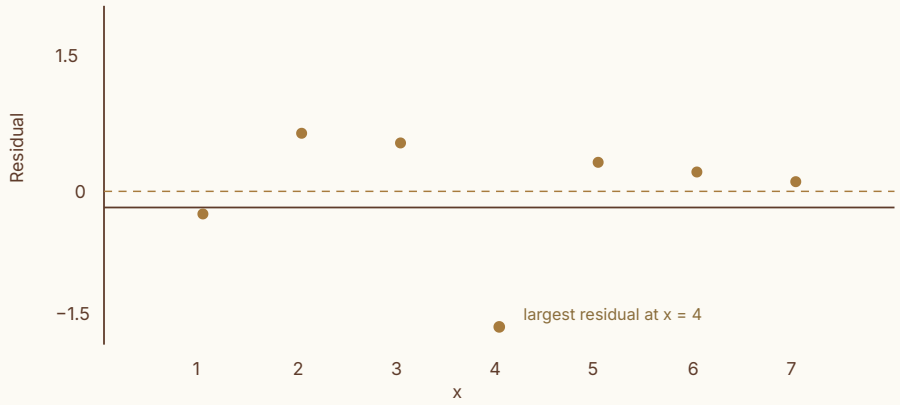
At $x = 4.5$: $\hat{y} = 6.125\ldots \approx 6.13$, an interpolation. At $x = 9$: $\hat{y} = 11.107\ldots \approx 11.1$, an extrapolation beyond the largest observed $x = 7$.

x	y	$\hat{y}$	RESIDUAL
1	2	2.25	-0.25
2	4	3.36	0.64
3	5	4.46	0.54
4	4	5.57	-1.57
5	7	6.68	0.32
6	8	7.79	0.21
7	9	8.89	0.11

The residuals sum to zero. Five are positive; the point at $x = 4$ lies 1.57 below the line. There is no smooth curve, but with seven points one observation dominates. Check whether $(4, 4)$ is an error or a genuine exception.

Since $r = 0.954$, $r^2 = 0.910$: about 91.0% of the variation in y is accounted for by the linear regression. The remaining 9.0% is not.

> A high r^2 describes fit inside the observed range; it does not license extrapolation.



Most residuals lie within about 0.6 of zero; the point at $x = 4$ is 1.57 below the line.

EXAMPLE **9.10** READING TECHNOLOGY OUTPUT

A spreadsheet fits the nine training pairs of Example 9.7 and reports intercept -0.0514, gradient 0.1696, $R^2 = 0.9502$, $r = 0.9748$, $n = 9$. Write the equation and interpret it.

SOLUTION

The equation is $\hat{y} = -0.0514 + 0.1696x$, or $\hat{y} = -0.0514 + 0.170x$ to 3 s.f. The gradient means each extra weekly hour is associated with an extra 0.170 seconds of improvement. The intercept is the prediction at 0 hours, outside the observed range (minimum 2), so it is not substantive.

At $x = 7$: $\hat{y} = 1.1358 \approx 1.14$ seconds, an interpolation. At $x = 12$: $\hat{y} = 1.9838 \approx 1.98$ seconds, an extrapolation beyond the maximum 8. $R^2 = 0.9502$ means about 95.0% of the variation in improvement is accounted for by the linear relationship in this sample; it does not mean a 95% chance the model is true, nor that training causes improvement.

> Report coefficients to 3 significant figures; keep unrounded values only for intermediate work.

MISCONCEPTION

"$r^2 = 0.95$ means a 95% chance the model is correct," or "the line proves x causes y." r^2 is a proportion of sample variation, not a probability, and regression describes co-variation. A model can have high r^2, be the wrong form, use a biased sample, or be applied far outside its data. Report n, the x-range and the data source with the equation.

GUIDED PRACTICE

1. For $y = 3 + 2x$, predict y at $x = 4$ and interpret the gradient. (11; y rises by 2 per unit x.)
2. Find the residual at $(4, 9)$ using $y = 3 + 2x$. (-2.)
3. Data range from $x = 10$ to 40. Is $x = 50$ interpolation or extrapolation? (Extrapolation.)
4. If $r = 0.9$, find and interpret r^2. (0.81; 81% of variation accounted for.)

EXERCISE 9E

REGRESSION LINES, RESIDUALS AND PREDICTION

FLUENCY

1 For $y = 2.4 + 0.8x$, predict y at $x = 5$ and 10, and interpret the gradient.

2 For $(1, 2), (2, 5), (3, 4), (4, 8), (5, 9)$, find $\bar{x}, \bar{y}, \sum(x - \bar{x})(y - \bar{y})$ and $\sum(x - \bar{x})^2$, then the least-squares line.

3 Using Question 2, calculate the five residuals and check that they sum to zero.

4 Interpret r^2: (a) 0.81; (b) 0.36; (c) 0.94.

EXERCISE 9E · CONTINUED REGRESSION LINES, RESIDUALS AND PREDICTION

5 A model uses $10 \leq x \leq 40$. Classify: (a) $x = 25$; (b) $x = 5$; (c) $x = 60$; (d) $x = 40$.

6 Output: intercept 12.3, gradient -0.45, $r^2 = 0.72$, $n = 20$. Write the equation, state r, and interpret the gradient.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student predicts at $x = 30$ from a model fitted for $2 \leq x \leq 8$ and gives 4 decimal places. Identify two problems and say what should be reported.

8 Why does least squares use squared vertical distances, not horizontal or unsquared distances?

9 An outlier is added at the right-hand end of a strong positive set. Describe the likely effect on gradient, intercept and r.

10 A residual plot shows a U-shape. What does this suggest, and what should the analyst do?

11 A model has $r^2 = 0.98$ but is used far beyond the data. Why is the prediction unreliable?

12 The gradient is 2.5 kg per cm. Why is r unitless while the gradient is not?

APPLICATION AND EXTENSION

13 Heating cost $C = 28.5 - 1.4T$ for $2 \leq T \leq 18$. Predict at $T = 10$ and $T = 0$, interpret the gradient, and comment on reliability.

14 Car value $V = 18.5 - 2.1a$ for $1 \leq a \leq 8$. Predict at $a = 5$ and $a = 12$. Why is the second unreliable before considering the arithmetic?

15 **Extension.** Show that $b = r\dfrac{s_y}{s_x}$, where s_x, s_y use denominator n. Interpret how gradient depends on correlation and spread.

PATTERN, EVIDENCE AND JUDGEMENT

9.6 Investigation: correlation, causation and reliability

A statistical claim rarely arrives with its data. It arrives as a headline and a recommendation. Mathematics can describe an association precisely, but it cannot decide whether the association is causal, whether the sample represents anyone beyond itself, or whether the funder had an interest in the conclusion.

Four questions expose weak claims. **Who was measured?** A volunteer sample or a single school is not a random sample. **How many?** Small samples give unstable estimates. **Compared with what?** A group using a product must be compared with a comparable group not using it. **Who paid?** Funding does not make a result false, but it shapes what was measured and published.

A confounding variable is associated with both explanatory and response variables, so its effect is tangled with the one being studied. Reverse causation is possible: low grades may lead to more phone use. Coincidence is possible: with enough pairs, some correlate by chance. **Regression to the mean** occurs when a group selected for an extreme measurement tends to move closer to the average on a second measurement. **Survivorship bias** occurs when only successful cases remain in the data.

INVESTIGATION 9 · THE TWELVE PER CENT CLAIM

"Our app raises results by 12%"

A company selling a numeracy app publishes: **"Schools using Daily Numeracy for twenty minutes a day achieve results 12% higher than schools that do not. The programme is proven to raise attainment."** The report compares 60 schools that volunteered to use the app with 60 that did not, was funded by the company, and excludes nine schools that stopped using the app.

1. Identify the explanatory and response variables, the population described and the sample measured. Is the sample likely to be representative? Justify.
2. List four possible confounders. For each, explain how it could produce a difference even if the app had no effect.
3. The adopting schools had lower results the previous year. Explain how regression to the mean could create an apparent improvement, and how a comparison group would help.
4. Evaluate the sample size. Is 60 versus 60 informative? What else would you need to judge the precision of the 12% figure, and how could a few very high-scoring schools distort the comparison?
5. Assess the provenance: funding, peer review, missing schools and the word "proven". Which most seriously weakens the claim?
6. Design a small randomised study of a 20-minute daily session. State participants, control condition, allocation, measurements and two ethical safeguards.
7. Rewrite the claim so that it matches the evidence. What evidence would justify the word "proven"?

Generalise. A causal claim needs more than a correlation: the cause must precede the effect, a mechanism must be plausible, alternatives must be controlled, results should replicate, and the effect must matter in context. Write a short set of questions a reader could apply to any statistical headline, and use them on one claim from an advertisement.

MODELLING · CRITERION D

Does advertising predict sales?

A shop records weekly advertising spend A, in hundreds of pounds, and weekly sales S, in thousands of pounds, for eight weeks.

WEEK	1	2	3	4	5	6	7	8
A	1	2	3	4	5	6	7	8
S	12	16	15	22	21	27	27	32

1. **Model.** Draw a scatter graph and describe direction, strength, form and unusual points. Calculate r.
2. **Fit.** Find $S = a + bA$ and interpret the gradient and intercept, with units.
3. **Predict.** Predict sales for advertising spends of £450 and £1200. Classify each and say which is more trustworthy.
4. **Evaluate.** Find r^2 and the residual for week 3. State two assumptions and two reasons not to treat the gradient as the return on advertising.

CUMULATIVE

9.7 Mixed practice

Attempt all questions without notes. Decide first whether the question asks you to describe, calculate or judge.

MIXED PRACTICE 9

DISPERSION, COMPARISON, CORRELATION AND REGRESSION

FLUENCY AND UNDERSTANDING

1 Find the median, quartiles and IQR of $14, 17, 19, 21, 23, 26, 29, 31, 35$.

2 Write the five-number summary of $11, 13, 14, 16, 18, 19, 21, 22, 45$ and test 45.

3 Find the mean and sd of $5, 6, 8, 11, 15$.

4 Estimate mean and sd: $10 < x \leq 20$: 4; $20 < x \leq 30$: 9; $30 < x \leq 40$: 12; $40 < x \leq 50$: 5.

5 Marks have mean 62 and sd 4.2. Interpret the sd.

6 Class A: mean 62, median 61, IQR 14, sd 9. Class B: 62, 66, 7, 4. Compare.

7 A rising scatter is tightly clustered around a line except for one point far below. Describe the association.

8 Interpret $r = -0.68$ between laptop age and resale value.

9 For $y = 3.2 + 0.8x$, predict y at $x = 10$.

10 Observed $y = 5$, predicted 5.6. Find the residual and the point's position relative to the line.

11 Model fitted for $5 \leq x \leq 25$. Classify: (a) 12; (b) 30; (c) 5; (d) 2.

12 Association or causal claim? (a) "Breakfast eaters score higher"; (b) "Breakfast improves concentration"; (c) "Ice-cream sales are linked to drowning".

REASONING AND APPLICATION

13 Box plots: A min 12, $Q_1 = 16$, median 20, $Q_3 = 24$, max 30; B min 18, $Q_1 = 20$, median 22, $Q_3 = 25$, max 27. Compare and state one conclusion the plots cannot support.

14 For $10, 12, 13, 14, 15, 16, 17, 18, 60$, find mean and median with and without 60. Which represents a typical value better?

15 Calculate r for $(1, 3), (2, 5), (3, 6), (4, 8), (5, 10)$.

16 For Question 15, find the least-squares line and predict y at $x = 6$.

17 Interpret $r^2 = 0.81$, and state one thing it does not tell you.

18 A website claims its revision service raises scores 15%, using schools that chose it. Write three questions you would ask.

19 **Extension.** Show that adding a constant k to every value leaves the sd unchanged. Explain using deviations.

20 **Extension.** Construct five values with mean 10 and sd 2. Verify both and say whether the answer is unique.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 20 marks

Measuring and interpreting spread

Instructions. Answer all questions. Show working and include units where they apply. Give non-integer answers to 3 significant figures.

1 The waiting times, in minutes, of nine customers are $4, 7, 7, 9, 11, 13, 16, 18, 40$. Find the five-number summary, the IQR and the range, and test 40 for being an outlier. [4]

2 Find the mean and the standard deviation of $6, 8, 9, 11, 16$. [5]

3 Estimate the mean and standard deviation of the grouped data: $0 < t \leq 10$: 6; $10 < t \leq 20$: 14; $20 < t \leq 30$: 8; $30 < t \leq 40$: 2. [5]

4 A journey has mean 24.6 minutes and standard deviation 3.1 minutes. Interpret both numbers in context. [3]

5 Every value in a set has mean 20 and standard deviation 4. State the new mean and standard deviation when (a) 5 is added to every value; (b) every value is doubled. [3]

Evidence: correct quartiles and fences; a complete deviation table; midpoint estimates; an interpretation in units; correct responses to both transformations.

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

One point, four statistics

Instructions. Work independently. Keep a full table of totals; do not round intermediate values.

A researcher records the number of hours of revision, x, and the score, y, of ten students.

x	1	2	3	4	5	6	7	8	9	10
y	3	5	4	6	7	5	8	9	7	18

1. Plot the ten points. Describe the association and identify the point that does not fit the pattern. [4]
2. Calculate the mean and the standard deviation of y for all ten values. [4]
3. Calculate the product-moment correlation coefficient r for all ten values. [5]
4. Remove the point $(10, 18)$. Recalculate the mean, the standard deviation and r for the remaining nine values. [6]
5. Compare the two sets of results. Explain how one point changed the mean, the standard deviation and r. [3]
6. The least-squares lines are $y = 1.27 + 1.08x$ for ten points and $y = 3.08 + 0.58x$ for nine. Predict y at $x = 10$ from each and comment. [2]

Evidence: a labelled scatter; correct totals and substitution; two means and two standard deviations; two values of r; a comparison that distinguishes the effect on centre, spread and linear fit; predictions interpreted as extrapolation.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

What the newsletter may claim

Instructions. Read the extract, then answer for a non-specialist audience. Use statistical terms only after explaining them. Write in continuous prose.

School newsletter: "Students who attended our new breakfast club at least three times a week scored 11 per cent higher in the summer examinations than those who did not. The club clearly raises attainment." The figures came from 28 volunteers among 37 club members; non-attenders were not surveyed.

1. Identify three weaknesses in the evidence and explain how each could distort the comparison. [6]
2. Explain, in plain language, what a standard deviation and a correlation coefficient each measure. Explain why neither can establish that breakfast caused the higher scores. [8]
3. Write a 300–350 word response for parents stating what the school can and cannot claim, and propose one change that would make a future study more trustworthy. [6]

Evidence: named weaknesses with mechanisms (self-selection, missing comparison group, sample size, response bias); accurate plain-language definitions; an explicit distinction between association and causation; a measured recommendation with a feasible design change.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

What is a used car worth?

Instructions. A dealer records the age, x years, and asking price, y thousand pounds, of eight cars of the same model.

x	1	2	3	4	5	6	7	8
y	18.4	15.2	15.1	11.4	12.3	9.0	7.9	5.9

1. Plot the data and describe the association, referring to direction, strength, form and any unusual point. [4]
2. Calculate r and interpret it in context. [4]
3. Find the equation of the least-squares regression line, giving both coefficients to 3 significant figures. [5]
4. Predict the price of a car aged 5.5 years and of one aged 12 years. Comment on the reliability of each prediction. [5]
5. Calculate the residual for the car aged 4 years and interpret its sign. State the value of r^2 and one thing it does not tell the dealer. [3]
6. Evaluate the model: state one assumption, one variable that has been omitted, and one reason the line should not be used to price a new car. [3]

Evidence: an accurate scatter with labelled axes; a correct r with a contextual interpretation; a correctly substituted regression line; predictions with interpolation and extrapolation identified; a residual with direction; a critical evaluation naming assumptions and omitted variables.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- Spread matters as much as centre. The range uses two values, the standard deviation uses all, and the IQR uses the middle half. Quartiles split the ordered data into four groups; the five-number summary and box plot display the result, and values beyond $1.5 \times \text{IQR}$ from the quartiles are flagged as outliers.
- Variance is the mean of the squared deviations; standard deviation is its square root and carries the units of the data. The divisor n gives the population form used here; $n - 1$ gives the sample estimate.
- Grouped data are summarised by class midpoints, so the estimated mean and standard deviation are approximations that ignore variation within each class.
- Comparing distributions requires centre, spread, shape and outliers, reported in context and in units. A single average, or a truncated axis, can mislead.
- Correlation coefficient r measures the strength and direction of a linear association, not its steepness and not a percentage. r^2 is the proportion of variation accounted for by the regression.
- The least-squares line $\hat{y} = a + bx$ minimises the sum of squared residuals, with $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$, where $S_{xy} = \sum(x - \bar{x})(y - \bar{y})$. Predictions inside the data range are interpolations; beyond it they are extrapolations and need justification.
- Association is not causation. Confounding variables, selection effects, sample size and provenance must be examined before a claim supports a policy.

$$\bar{x} = \frac{\sum x}{n}, \qquad s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}, \qquad \text{IQR} = Q_3 - Q_1$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}, \qquad b = \frac{S_{xy}}{S_{xx}}, \qquad a = \bar{y} - b\bar{x}, \qquad r^2 = 1 - \frac{\sum e_i^2}{\sum (y_i - \bar{y})^2}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can find the range, quartiles and IQR of a data set.
- I can write a five-number summary and draw a box plot to scale.
- I can test a value against the $1.5 \times \text{IQR}$ fences and describe its effect.
- I can read a cumulative frequency curve to estimate the median and quartiles.
- I can calculate a variance and standard deviation from a deviation table, and estimate both for grouped data using midpoints.
- I can explain what a standard deviation says about a distribution, in units.
- I can compare two distributions using centre, spread, shape and outliers.
- I can explain when the IQR is more useful than the standard deviation.
- I can describe a scatter graph and calculate r from summary totals.
- I can interpret r, r^2 and a residual without claiming causation.
- I can find a least-squares line and distinguish interpolation from extrapolation.
- I can identify confounding, selection bias and provenance problems in a published claim.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which measure of spread did you trust first, and what changed your mind about when to use it?
2. Describe one error you made in a calculation in this chapter and the check that caught it.
3. The statement of inquiry says that measuring spread and modelling relationships let us compare groups and judge claims. Use one example from this chapter to show how a summary changed a conclusion.
4. Why does a strong correlation never, on its own, justify a causal or policy claim? Give an example with a plausible confounding variable.
5. What question about data would you still like to investigate, and what would you need to measure?

EVIDENCE, RISK AND THE ARITHMETIC OF CONDITION

Chance and Evidence

10

LOGIC · RELATIONSHIPS · REPRESENTATION

Reading a test result, a table or a tree — and learning when a probability must be conditioned rather than merely counted.

STATEMENT OF INQUIRY

Probability is a measure on a set of outcomes; conditioning restricts that set, and the difference between $P(A \mid B)$ and $P(B \mid A)$ is often the difference between a sound decision and a misleading one.

KEY CONCEPT

Logic

RELATED CONCEPTS

Relationships · Representation · Patterns

GLOBAL CONTEXT

Fairness and development — risk, screening and fair decisions

FACTUAL

What does $P(A \mid B)$ measure, and which outcomes belong to the restricted sample space?

CONCEPTUAL

Why can a test be accurate and yet a positive result still be more likely wrong than right?

DEBATABLE

When a screening programme produces many false positives, is it fair to offer the test to everyone?

IN THIS CHAPTER

10.1 Sets, tables and probability
10.2 Conditional probability
10.3 Independence
10.4 Tree diagrams and total probability
10.5 Counting, permutations and combinations
10.6 Investigation: the birthday problem

OPENING PROBLEM · THE LETTER THAT SAYS POSITIVE

A test, a population and a difficult sentence

A public health programme screens 10,000 adults for a condition that affects 2% of them. The test correctly identifies 90% of people who have the condition, and it wrongly flags 5% of people who do not. A letter arrives: *your result is positive*.

Is the chance that this person has the condition 90%, 50%, or something much smaller — and how could a sentence in a letter change what they decide to do?

1. Before calculating, write down your intuitive estimate and your reasoning. Keep it.
2. Represent the 10,000 people in a two-way table. How many fall in each of the four cells?
3. Among all positive results, what fraction belong to people who have the condition? Give a fraction and a percentage.
4. Draft an accurate sentence and a misleading sentence for the letter. Explain the difference and state what support should accompany the result.

A probability is always a probability *given something*. Sometimes the given information is the whole population; sometimes it is a test result, a table row, or the first branch of a tree. This chapter makes that condition visible, then counts the outcomes on which it operates.

10.0 ORIENTATION Learning intentions

- Represent events with set notation, two-way tables and Venn diagrams, and apply the addition and complement rules.
- Compute conditional probabilities from a restricted sample space and use the multiplication law in both directions.
- Test independence, and distinguish it from mutual exclusivity.
- Use tree diagrams for two and three stages to find total and reverse conditional probabilities.
- Count arrangements and selections with permutations and combinations, including restrictions.
- Communicate a screening result with natural frequencies and evaluate the assumptions behind it.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

list a sample space and describe events as subsets; move between a two-way table, a Venn diagram and a probability statement; use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A') = 1 - P(A)$; recognise mutually exclusive events and explain why they are not independent; compute $P(A \mid B)$ by restricting the denominator; use $P(A \cap B) = P(A \mid B)P(B)$ and reverse the conditioning correctly; test independence from a table; build a tree whose branches sum to 1 at every node; multiply along and add across; recover a reverse conditional from a tree; solve "at least one" problems by complements; count with $n!$, ${}^{n}P_{r}$ and ${}^{n}C_{r}$; handle a repeated-item arrangement and a "particular person included" restriction; and explain a screening result without confusing the probability of a positive test with the probability of the condition.

ATL SKILLS · THINKING, TRANSFER AND COMMUNICATION

Critical thinking: when a conditional probability is quoted, identify the group in the denominator. Most errors here are not arithmetic but errors about *which group* is described. **Transfer:** the same structure appears in a table, a Venn diagram and a tree; choose the representation that makes the condition visible. **Communication:** natural frequencies — "of 1,000 people, 18 have the condition and test positive" — are clearer than a bare percentage because they show the base rate instead of hiding it.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 10.1.

1. A bag contains 5 red and 3 blue counters. What fraction are red?
2. Evaluate $\frac{5}{8} \times \frac{4}{7}$.
3. Find $0.02 \times 10{,}000$.
4. A jacket costs £80 and is reduced by 15%. Find the reduced price.
5. Write $\frac{18}{67}$ as a decimal, correct to three significant figures.
6. Expand and evaluate $n(n-1)(n-2)$ for $n = 5$.

REFRESHER · FRACTIONS, PERCENTAGES AND "OF"

A fraction of a quantity is found by multiplying: $\frac{5}{8}$ of 64 is 40. A percentage is a fraction with denominator 100, so 15% of 80 is $0.15 \times 80 = 12$. To convert a fraction to a decimal, divide: $\frac{18}{67} = 0.26865\ldots$, which is 0.269 to three significant figures. When two stages occur, "of" usually signals multiplication: 90% of the 2% who have a condition is $0.90 \times 0.02 = 0.018$ of the population. Keep exact fractions while calculating, round only at the end, and check that every probability lies between 0 and 1.

CONCEPT AND REPRESENTATION

10.1 Sets, tables and probability

A probability model begins with a sample space: the set of all outcomes of a described experiment. For a fair die it is $\{1, 2, 3, 4, 5, 6\}$. An event is a subset of the sample space, such as $A = \{2, 4, 6\}$, "the score is even". Probability assigns each event a number from 0 to 1, with disjoint events adding.

Set notation gives the algebra of events. The union $A \cup B$ means A or B or both; the intersection $A \cap B$ means both; the complement A' means not A. These correspond to the regions of a Venn diagram and to the cells of a two-way table.

DEFINITIONS

Sample space, event, union, intersection, complement

The **sample space** ξ is the set of all possible outcomes. An **event** is a subset of ξ. For events A and B: $A \cup B$ contains outcomes in at least one of A, B; $A \cap B$ contains outcomes in both; A' contains outcomes in ξ not in A.

Events are **mutually exclusive** if $A \cap B = \varnothing$, so $P(A \cap B) = 0$. **Addition rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Complement rule:

$$P(A') = 1 - P(A).$$

The subtraction corrects for double counting. For mutually exclusive events the intersection is empty, so the rule becomes $P(A \cup B) = P(A) + P(B)$. In a two-way table, the margins give totals and the cells give intersections; always ask which total belongs in the denominator.

Theoretical probability comes from a model, such as $\frac{1}{6}$ for a fair die. Experimental probability is a relative frequency: the proportion of trials in which the event occurred. The two approach one another as trials increase, but a short run can differ, and a difference is not proof of bias.

EXAMPLE **10.1** READING PROBABILITIES FROM A VENN DIAGRAM

In a group of 40 students, 18 study French, 14 study Spanish and 7 study both. One student is chosen at random. Find (a) $P(F)$, (b) $P(F \cap S)$, (c) $P(F \cup S)$, (d) $P(F')$, and (e) the number who study neither.

SOLUTION

French only: $18 - 7 = 11$. Spanish only: $14 - 7 = 7$. Both: 7. These regions contain 25 students, so 15 study neither.

(a) $P(F) = \frac{18}{40} = \frac{9}{20} = 0.45$. **(b)** $P(F \cap S) = \frac{7}{40} = 0.175$.

(c) $P(F \cup S) = \frac{25}{40} = \frac{5}{8} = 0.625$. Check: $\frac{18}{40} + \frac{14}{40} - \frac{7}{40} = \frac{25}{40}$.

(d) $P(F') = 1 - 0.45 = 0.55$. **(e)** 15 students, so $P(F' \cap S') = \frac{15}{40} = \frac{3}{8}$.

> The 7 students in the overlap belong to both events. Adding 18 and 14 counts them twice, which is why the addition rule subtracts 7.

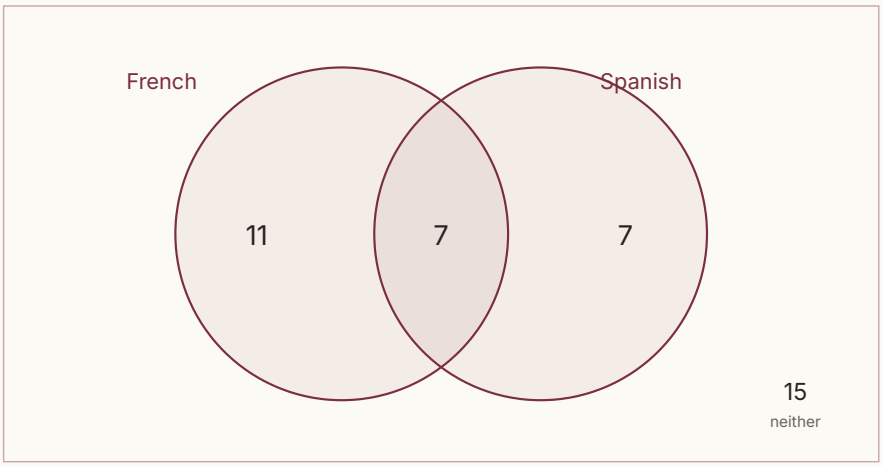


Figure 10.1 · The overlap contains the 7 students who study both languages; the 15 outside both circles study neither.

EXAMPLE 10.2 A TWO-WAY TABLE AND THE ADDITION RULE

A survey of 250 people records whether they recycle (R) and whether they are under 30 (U). There are 60 under-30 recyclers, 40 under-30 non-recyclers, 105 recyclers aged 30 or over and 45 non-recyclers aged 30 or over. Find $P(R)$, $P(U)$, $P(U \cap R)$, $P(U \cup R)$ and $P(R')$.

SOLUTION

	RECYCLES	DOES NOT	TOTAL
Under 30	60	40	100
30 or over	105	45	150
Total	165	85	250

$P(R) = \frac{165}{250} = 0.66$, $P(U) = \frac{100}{250} = 0.4$, $P(U \cap R) = \frac{60}{250} = 0.24$. By the addition rule, $P(U \cup R) = 0.4 + 0.66 - 0.24 = 0.82$. Direct counting gives $\frac{100 + 165 - 60}{250} = 0.82$. Also $P(R') = \frac{85}{250} = 0.34 = 1 - 0.66$.

> The 60 under-30 recyclers belong to both the "under 30" row total and the "recycles" column total.

MISCONCEPTION

"$P(A \cup B) = P(A) + P(B)$, always." This holds only when A and B are mutually exclusive. With an overlap, adding counts it twice and can even exceed 1. The correct statement subtracts $P(A \cap B)$. Nor does "mutually exclusive" mean "unrelated": events that cannot happen together are strongly related, because one rules out the other.

GUIDED PRACTICE

1. A fair die is rolled. For $A = \{2, 4, 6\}$ and $B = \{2, 3, 5\}$, find $P(A \cup B)$. (Answer: $\frac{5}{6}$.)
2. Of 30 students, 12 play chess, 15 play cards and 5 play both. How many play neither? (Answer: 8.)
3. Of 80 people, 30 drink tea, 25 drink coffee and 10 drink both. Find $P(\text{tea or coffee})$. (Answer: $\frac{45}{80} = \frac{9}{16}$.)
4. If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$, find $P(A \cap B)$ and decide whether the events are mutually exclusive. (Answer: 0.2; no.)

EXERCISE 10A SAMPLE SPACES, TABLES AND THE ADDITION RULE

FLUENCY

1 Write the sample space for (a) rolling a die, (b) tossing a coin twice, (c) choosing a letter from CHANCE.

2 A die is rolled. Let A be "even" and B be "prime". List $A \cup B$, $A \cap B$ and A', and find each probability.

3 In a class of 25, 14 study history, 9 study geography and 4 study both. Find the number who study neither.

4 In 100 people, 35 drink tea and coffee, 15 drink tea only, 25 drink coffee only and 25 drink neither. Find $P(T)$, $P(C)$, $P(T \cap C)$ and $P(T \cup C)$.

5 If $P(A) = 0.35$, $P(B) = 0.4$ and the events are mutually exclusive, find $P(A \cup B)$, $P(A')$ and $P(A' \cap B')$.

6 A coin lands heads 36 times in 80 tosses. Find the relative frequency and compare it with $\frac{1}{2}$. Does the result prove bias?

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes $P(A \cup B) = 0.6 + 0.5 = 1.1$. Identify the two errors and state what extra information is needed.

8 Explain, using the sample space, why $P(A') = 1 - P(A)$.

EXERCISE 10A · CONTINUED SAMPLE SPACES, TABLES AND THE ADDITION RULE

9 A game costs £1 and pays £3 with probability 0.2. Find the expected payout and net gain, and decide whether it is fair.

10 Are tea and coffee mutually exclusive in Question 4? Justify using the table, not only the probabilities.

11 A forecast gives a 30% chance of rain on each of Saturday and Sunday. Explain why the chance of rain over the weekend need not be 60%.

12 Explain the difference between theoretical and experimental probability, using 300 rolls of a die as an example.

APPLICATION AND EXTENSION

13 Of 200 students, 120 walk to school, 70 cycle and 40 do both. Draw a Venn diagram, find the number who use neither, and find $P(\text{walk or cycle})$.

14 Two fair dice are rolled. From the 36 outcomes, find $P(\text{total } 7)$ and $P(\text{total at least } 10)$. What payout for a £1 stake makes the "total 7" bet fair, ignoring costs?

15 **Extension.** Use a Venn diagram to explain why
$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.
Check the formula when the three events are pairwise disjoint.

CONCEPT AND METHOD

10.2 Conditional probability

Conditional probability answers: *given that B has happened, what is the chance of A?* The word "given" changes the sample space. We look only at outcomes in B, and among those count the outcomes also in A. The formula therefore divides by $P(B)$, rescaling the restricted space to total probability 1.

In a table, conditioning means fixing a row or column and reading across it; in a Venn diagram, it means ignoring everything outside the conditioning circle. Both make the same point: $P(A \mid B)$ and $P(B \mid A)$ describe different groups, because each has a different denominator. Reversing them without thought is the most common serious error in applied probability.

DEFINITION AND LAWS

Conditional probability and the multiplication law

For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Equivalently, the **multiplication law** is

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

The condition $P(B) > 0$ is necessary: conditioning on an impossible event is undefined.

EXAMPLE 10.3 CONDITIONING IN BOTH DIRECTIONS FROM A TABLE

Use the recycling table from Example 10.2. Find $P(R \mid U)$, $P(U \mid R)$ and $P(R \mid U')$. Explain why the first two differ.

SOLUTION

Given U, restrict to the 100 under-30s: $P(R \mid U) = \dfrac{60}{100} = 0.6$.

Given R, restrict to the 165 recyclers: $P(U \mid R) = \dfrac{60}{165} = \dfrac{4}{11} \approx 0.364$.

Given U', restrict to the 150 people aged 30 or over: $P(R \mid U') = \dfrac{105}{150} = 0.7$.

The denominators are different: 100 under-30s versus 165 recyclers. About 60% of young people recycle, but only about 36% of recyclers are young, because older people are the larger group.

> Check: $P(U \cap R) = 0.6 \times 0.4 = 0.24$ and also $\dfrac{4}{11} \times 0.66 = 0.24$.

EXAMPLE 10.4 CONDITIONING WITH A VENN DIAGRAM

Use the 40-student language group from Example 10.1. Find $P(F \mid S)$, $P(S \mid F)$ and $P(F' \mid S)$.

SOLUTION

Given S, there are 14 Spanish students, of whom 7 study French: $P(F \mid S) = \dfrac{7}{14} = \dfrac{1}{2}$.

Given F, there are 18 French students, of whom 7 study Spanish: $P(S \mid F) = \dfrac{7}{18} \approx 0.389$.

Within the Spanish group, 7 do not study French, so $P(F' \mid S) = \dfrac{7}{14} = \dfrac{1}{2} = 1 - P(F \mid S)$.

> Both conditional probabilities lie between 0 and 1. A value outside that range signals a misread denominator or intersection.

EXAMPLE 10.5 A SCREENING TEST IN NATURAL FREQUENCIES

In 10,000 people, 2% have a condition. A test detects 90% of cases and wrongly flags 5% of healthy people. Find $P(C \mid +)$ and $P(C' \mid +)$, and interpret the result.

SOLUTION

Of 10,000 people, 200 have the condition and 9,800 do not. Of the 200, 180 test positive and 20 negative. Of the 9,800, 490 test positive and 9,310 negative.

	POSITIVE	NEGATIVE	TOTAL
Condition C	180	20	200
No condition C'	490	9,310	9,800
Total	670	9,330	10,000

There are 670 positives, of which 180 are true:

$$P(C \mid +) = \frac{180}{670} = \frac{18}{67} \approx 0.269, \qquad P(C' \mid +) = \frac{490}{670} = \frac{49}{67} \approx 0.731.$$

"Of every 100 people with a positive result, about 27 have the condition and about 73 do not. A positive result needs confirmation; it is not a diagnosis."

> The sensitivity is $P(+ \mid C) = 0.90$, but a patient needs $P(C \mid +)$. The denominators are 200 people with the condition versus 670 people with a positive result.

THE OPENING PROBLEM, RESOLVED

The letter should not say "you are 90% likely to have the condition". The relevant probability is $P(C \mid +) = \dfrac{18}{67} \approx 27\%$, because 490 false positives outnumber 180 true positives. This is the **base-rate effect**: when a condition is rare, even a fairly specific test produces many more false positives than true positives. A fair letter would report natural frequencies, state that a positive result is not a diagnosis, and explain the confirmatory step, without language that treats the result as an identity or a certainty.

MISCONCEPTION

"$P(A \mid B)$ is the same as $P(B \mid A)$." The numerator is shared, but the denominator is not. In the screening example, $P(+ \mid C) = 0.90$ while $P(C \mid +) \approx 0.269$. Reversing a conditional is legitimate only when the base rates make the denominators equal, which is a special case, not a rule. Always ask which group the condition names.

GUIDED PRACTICE

1. Of 100 students, 40 play an instrument, 25 play a sport and 15 do both. Find $P(\text{instrument} \mid \text{sport})$. (Answer: 0.6.)
2. If $P(A) = 0.5$ and $P(A \cap B) = 0.2$, find $P(B \mid A)$. (Answer: 0.4.)
3. If $P(B \mid A) = 0.3$ and $P(A) = 0.6$, find $P(A \cap B)$. (Answer: 0.18.)
4. In 1,000 people, 2% have a condition; a test detects 90% of cases and falsely flags 10% of healthy people. Find $P(C \mid +)$. (Answer: $\dfrac{18}{116} = \dfrac{9}{58} \approx 0.155$.)

EXERCISE 10B

CONDITIONING, REVERSING AND NATURAL FREQUENCIES

FLUENCY

1 From the recycling table, find $P(R \mid U)$, $P(U \mid R)$ and $P(R \mid U')$.

2 Of 40 people, 20 own a cat, 12 own a dog and 5 own both. Find $P(\text{cat} \mid \text{dog})$ and $P(\text{dog} \mid \text{cat})$.

3 Given $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \mid B)$ and $P(B \mid A)$.

4 Given $P(A \mid B) = 0.3$ and $P(B) = 0.6$, find $P(A \cap B)$.

5 Given $P(A \cap B) = 0.12$ and $P(B) = 0.4$, find $P(A \mid B)$.

6 In a school, 60% own a phone, 45% own a laptop and 30% own both. Find $P(\text{phone} \mid \text{laptop})$ and $P(\text{laptop} \mid \text{phone})$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student writes $P(A \mid B) = P(B \mid A)$ when $P(A) = 0.5$, $P(B) = 0.2$ and $P(A \cap B) = 0.1$. Calculate both and explain the error.

8 In 1,000 people, 100 have a condition; a test detects 90 and wrongly flags 50 healthy people. Find $P(+ \mid C)$ and $P(C \mid +)$ and explain the difference.

9 Explain what "restricting the sample space" means, using a table of 100 people.

10 A table shows $P(A \mid B) = 0.4$ and $P(A \mid B') = 0.7$. Without calculating $P(A)$, are A and B independent? Explain.

11 Derive the multiplication law from the definition of $P(A \mid B)$.

12 Give events with $P(A \mid B) > P(A)$, and interpret the inequality in context.

APPLICATION AND EXTENSION

13 A programme tests 50,000 people. The condition affects 0.5%, the test detects 98% of cases and falsely flags 4% of healthy people. Find $P(C \mid +)$ and explain it to a patient in one sentence.

14 Of 400 parcels, 320 arrive on time, 40 are damaged and 16 are both. Find $P(\text{on time} \mid \text{damaged})$ and $P(\text{damaged} \mid \text{on time})$, and say which is more useful to the company.

15 **Extension.** Prove that $P(A \mid B) + P(A' \mid B) = 1$. Explain why $P(A \mid B) + P(A \mid B') = 1$ is false in general.

CONCEPT AND TEST

10.3 Independence

Two events are independent when knowing that one has occurred does not change the probability of the other. In symbols, $P(A \mid B) = P(A)$. Substituting into the multiplication law gives the usual test:

$$P(A \cap B) = P(A)P(B).$$

Independence is a property of the model, not a vague claim that events are "unrelated". It extends to repeated trials: if a coin is tossed five times independently, the probability of five heads is $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

The assumption can fail — drawing without replacement changes the bag, so the second draw depends on the first.

DEFINITION

Independent events

Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

Equivalently, if $P(B) > 0$, then $P(A \mid B) = P(A)$; and if $P(A) > 0$, then $P(B \mid A) = P(B)$. Independence is symmetric.

For three events, mutual independence requires every pair to satisfy the product rule and also $P(A \cap B \cap C) = P(A)P(B)P(C)$.

EXAMPLE 10.6 TESTING INDEPENDENCE FROM A TWO-WAY TABLE

Of 120 students, 24 play music and a sport, 36 play music only, 36 play sport only and 24 do neither. Are M and S independent?

SOLUTION

The margins are 60 music players and 60 sports players, so $P(M) = P(S) = 0.5$. The overlap is $P(M \cap S) = \frac{24}{120} = 0.2$.

If independent, the product would be $0.5 \times 0.5 = 0.25$. Since $0.2 \neq 0.25$, the events are **not independent**.

The conditional check agrees: $P(M \mid S) = \frac{24}{60} = 0.4$, whereas $P(M) = 0.5$. Knowing that a student plays a sport lowers the probability that they play music.

> The balanced margins make the dependence visible: the "both" cell is smaller than independence predicts, and the "neither" cell is larger.

EXAMPLE **10.7** INDEPENDENT REPEATED TRIALS

A fair die is rolled four times. Find the probability of (a) no six, (b) at least one six, (c) exactly one six.

SOLUTION

(a) $P(\text{no six}) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296} \approx 0.482.$

(b) $P(\text{at least one six}) = 1 - \frac{625}{1296} = \frac{671}{1296} \approx 0.518.$

(c) A six can occur in any of 4 positions, each with probability $\frac{1}{6}\left(\frac{5}{6}\right)^3 = \frac{125}{1296}$:

$$P(\text{exactly one six}) = 4 \times \frac{125}{1296} = \frac{500}{1296} = \frac{125}{324} \approx 0.386.$$

The three answers do not add to 1 because "exactly one six" is contained in "at least one six". The complement of "at least one" is "none".

MISCONCEPTION

"Independent events cannot happen together." That describes mutually exclusive events. Two independent events usually *can* occur together; independence says the chance of both is the product. On a die, $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$ are independent because $P(A \cap B) = \frac{1}{3} = \frac{2}{3} \times \frac{1}{2}$, yet they overlap. Conversely, mutually exclusive events with positive probability are dependent.

GUIDED PRACTICE

1. If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, are A and B independent? (Answer: yes.)
2. If $P(A) = 0.3$, $P(B) = 0.7$ and the events are mutually exclusive, are they independent? (Answer: no; $0 \neq 0.21$.)
3. A fair coin is tossed five times. Find $P(\text{five heads})$. (Answer: $\frac{1}{32}$.)
4. A fair die is rolled twice. Find $P(\text{at least one six})$. (Answer: $\frac{11}{36}$.)

EXERCISE 10C

INDEPENDENCE AND REPEATED TRIALS

FLUENCY

1 If $P(A) = 0.5$, $P(B) = 0.4$ and the events are independent, find $P(A \cap B)$ and $P(A \cup B)$.

2 Test independence when $P(A) = 0.3$, $P(B) = 0.6$ and $P(A \cap B) = 0.18$.

3 A coin is tossed three times. Find $P(\text{three heads})$, $P(\text{exactly two heads})$ and $P(\text{at least one head})$.

4 A die is rolled twice. Find $P(\text{six, then six})$, $P(\text{no six})$ and $P(\text{at least one six})$.

5 Of 200 shoppers, 100 have a loyalty card, 120 shopped online and 60 did both. Test independence.

6 If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0$, state whether the events are mutually exclusive and whether they are independent.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says "A and B are independent because they cannot happen together". Explain the confusion and give a die counterexample.

8 Of 300 people, 120 are left-handed, 90 are colour-blind and 45 are both. Test independence and describe the dependence.

9 Explain why two mutually exclusive events with non-zero probability must be dependent.

10 A die is rolled repeatedly. Find the smallest number of rolls for which the probability of at least one six exceeds 0.99.

EXERCISE 10C · CONTINUED INDEPENDENCE AND REPEATED TRIALS

11 If A and B are independent, show that A' and B are independent, starting from $P(A' \cap B) = P(B) - P(A \cap B)$.

12 Are two draws from a bag independent with replacement? Without replacement? Explain.

APPLICATION AND EXTENSION

13 A machine produces defectives independently with probability 0.02. Find the probability that the first two are defective and that at least one of 50 is defective.

14 A player scores a free throw with probability 0.8. Assuming independence, find $P(\text{three in a row})$ and $P(\text{at least one in four})$. Give one reason the assumption might fail.

15 **Extension.** If $P(A \mid B) = P(A)$, with $P(A) > 0$ and $P(B) > 0$, prove $P(B \mid A) = P(B)$. Explain why $P(A) > 0$ is needed.

REPRESENTATION AND METHOD

10.4 Tree diagrams and total probability

A tree diagram represents stages in sequence. Each branch carries the probability of the next event *given* the stage reached. Two rules govern the arithmetic: **multiply along a path** and **add across the paths** that produce the event you want. At every node the branches sum to 1. Trees are especially useful when the second stage depends on the first, as in drawing without replacement.

DEFINITION

A **tree diagram** shows stages, each branch carrying the probability of the next event given the stage reached. The total probability of an event is the sum of the probabilities of all paths ending in it.

KEY IDEA · READING A TREE

Multiply along: $P(A \cap B) = P(A)P(B \mid A)$. **Add across:** if $A_1, A_2, \ldots$ partition the first stage, $P(B) = \sum_i P(A_i)P(B \mid A_i)$. **Reverse:** $P(A_i \mid B) = \dfrac{P(A_i)P(B \mid A_i)}{P(B)}$. The denominator is the total probability of B; the numerator is the single path in question.

EXAMPLE 10.8 TWO STAGES WITHOUT REPLACEMENT

A bag contains 5 red and 3 blue counters. Two are drawn without replacement. Find $P(RR)$, $P(\text{exactly one red})$, $P(\text{at least one red})$ and $P(\text{red second})$.

SOLUTION

$$P(RR) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

Exactly one red occurs as RB or BR:

$$P(RB) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}, \qquad P(BR) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56},$$

so $P(\text{exactly one red}) = \dfrac{30}{56} = \dfrac{15}{28}$.

$P(BB) = \dfrac{3}{8} \times \dfrac{2}{7} = \dfrac{6}{56} = \dfrac{3}{28}$, so $P(\text{at least one red}) = 1 - \dfrac{3}{28} = \dfrac{25}{28}$.

$P(\text{red second}) = P(RR) + P(BR) = \dfrac{20}{56} + \dfrac{15}{56} = \dfrac{5}{8}$, the same as $P(\text{red first})$.

The four path probabilities sum to 1. Check each node: $\frac{5}{8} + \frac{3}{8} = 1$ and, after a red, $\frac{4}{7} + \frac{3}{7} = 1$.

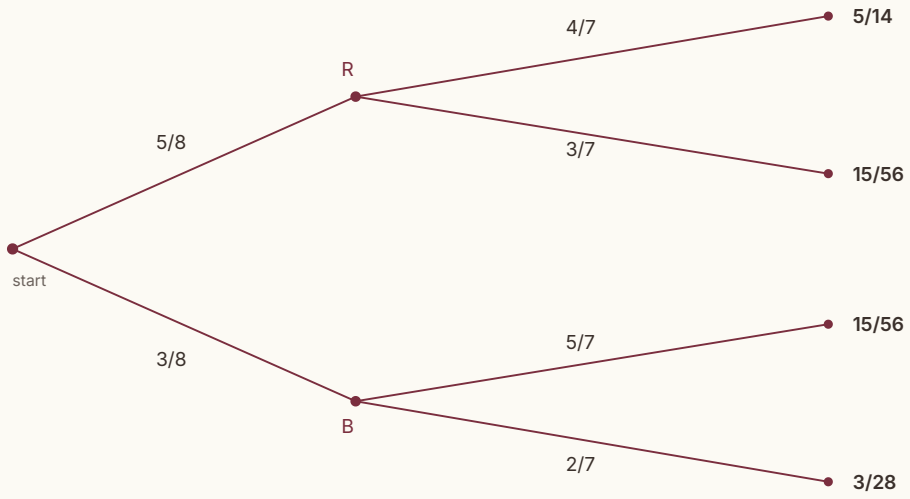


Figure 10.2 · With each red removed the bag changes: the second-stage probabilities $\frac{4}{7}$ and $\frac{3}{7}$ differ from $\frac{5}{8}$ and $\frac{3}{8}$.

EXAMPLE **10.9** THREE STAGES WITHOUT REPLACEMENT

A bag contains 4 red and 2 green counters. Three are drawn without replacement. Find $P(\text{all red})$, $P(\text{exactly two red})$ and $P(\text{at least one green})$.

SOLUTION

$$P(RRR) = \frac{4}{6} \times \frac{3}{5} \times \frac{2}{4} = \frac{24}{120} = \frac{1}{5}.$$

The three orders for two reds and one green each have probability $\frac{1}{5}$:

$$P(RRG) = \frac{4}{6} \times \frac{3}{5} \times \frac{2}{4}, \quad P(RGR) = \frac{4}{6} \times \frac{2}{5} \times \frac{3}{4}, \quad P(GRR) = \frac{2}{6} \times \frac{4}{5} \times \frac{3}{4}.$$

Each equals $\frac{1}{5}$, so $P(\text{exactly two red}) = \frac{3}{5}$.

"At least one green" is the complement of "all red": $1 - \frac{1}{5} = \frac{4}{5}$.

> After two reds, 2 red and 2 green remain, so the final branches are $\frac{1}{2}$ and $\frac{1}{2}$. After one of each, 3 red and 1 green remain.

EXAMPLE **10.10** REVERSING A CONDITIONAL FROM A TREE

Factory A makes 60% of components with 2% defective; Factory B makes 40% with 5% defective. A component is defective. Find the probability it came from A.

SOLUTION

The defective paths have probabilities $0.6 \times 0.02 = 0.012$ and $0.4 \times 0.05 = 0.020$. The other paths are $0.6 \times 0.98 = 0.588$ and $0.4 \times 0.95 = 0.380$. The four sum to 1.

$P(D) = 0.012 + 0.020 = 0.032$. Given D,

$$P(A \mid D) = \frac{0.012}{0.032} = \frac{12}{32} = \frac{3}{8} = 0.375.$$

So 37.5% of defectives come from A and 62.5% from B.

> A produces more components, but B has the higher defective rate, so B supplies most of the defectives. This is the same reversal structure as screening.

MISCONCEPTION

"On a tree, add along the branches and multiply across the paths." It is the reverse. Adding along one path is meaningless because the branches describe successive stages; their product is the probability of both. Adding is only for alternative paths leading to the same outcome. Nor are all final branches equally likely: 0.018 and 0.049 in the screening table are products, not counts.

GUIDED PRACTICE

1. A bag has 3 red and 2 blue counters. Two are drawn without replacement. Find $P(RB)$. (Answer: $\frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$.)

2. Repeat with replacement. (Answer: $\frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$.)

3. Factory A supplies 60% with 2% defective; B supplies 40% with 5% defective. Find $P(D)$. (Answer: 0.032.)

4. For the same factories, find $P(A \mid D)$. (Answer: 0.375.)

EXERCISE 10D

TREES, TOTAL PROBABILITY AND REVERSE CONDITIONING

FLUENCY

1 A bag has 3 red and 2 blue counters. Two are drawn without replacement. Find $P(RR)$, $P(BB)$ and $P(\text{exactly one red})$.

2 Repeat Question 1 with replacement.

3 A coin is tossed three times. Find $P(HHH)$, $P(\text{exactly two heads})$ and $P(\text{at least one head})$.

4 Seeds germinate independently with probability 0.9. Three are planted. Find $P(\text{all})$, $P(\text{none})$ and $P(\text{exactly two})$.

5 Factory A makes 70% with 3% defective; B makes 30% with 6% defective. Find $P(D)$ and $P(A \mid D)$.

6 A bag has 4 red and 4 blue. Two are drawn without replacement. Find $P(\text{same colour})$.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student computes $P(RR) = \frac{5}{8} + \frac{4}{7} = \frac{67}{56}$. Explain why this is impossible, identify the error, and give the correct value.

8 Explain why the branches leaving any node sum to 1, and what this implies for all final path probabilities.

9 A coin is tossed four times. Find $P(\text{at least one tail})$ and explain how the complement shortens the work.

10 Of 1,000 items, 700 come from A with 3% defective and 300 from B with 6% defective. Find $P(D)$ and $P(A \mid D)$.

11 Draw a three-stage tree for 2 red and 1 blue counter, drawn without replacement. Find $P(\text{blue last})$.

12 For 5 red and 3 blue, compare $P(RR)$ with and without replacement. Why are the answers close but not equal?

APPLICATION AND EXTENSION

13 A team wins 60% at home and 40% away, playing 15 home and 20 away matches. Find $P(\text{win})$ for a randomly chosen match.

14 A condition affects 4%. A test detects 85% of cases and falsely flags 10% of healthy people. Build a table for 10,000, find $P(C \mid +)$, and comment.

15 **Extension.** Derive

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A \mid B)P(B) + P(A \mid B')P(B')}$$

and explain in one sentence why the base rate cannot be omitted.

METHOD AND APPLICATION

10.5 Counting, permutations and combinations

When outcomes are equally likely,

$$P(E) = \frac{\text{number of outcomes in } E}{\text{number of outcomes in the sample space}}.$$

The difficulty is usually the counting. The **fundamental counting principle** says that if one choice can be made in m ways and a second in n ways, both can be made in mn ways; it extends to any number of stages.

A permutation is an arrangement in which order matters. A combination is a selection in which order does not. The same r objects can be arranged in $r!$ orders but selected in only one way, so ${}^{n}P_{r} = r!\,{}^{n}C_{r}$. Deciding which applies is the central skill of this section.

DEFINITIONS AND FORMULAE

Factorial, permutation, combination

For a positive integer n, the factorial is $n! = n(n-1)\cdots\times 2\times 1$, with $0! = 1$. The number of ordered arrangements of r objects from n distinct objects is

$${}^{n}P_{r} = \frac{n!}{(n-r)!}.$$

The number of unordered selections is

$${}^{n}C_{r} = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

These are related by ${}^{n}C_{r} = \dfrac{{}^{n}P_{r}}{r!}$, because each selection can be arranged in $r!$ orders. For repeated items, divide by the factorial of each repeat count: the letters of SUCCESS have $\dfrac{7!}{3!\,2!} = 420$ distinct arrangements.

EXAMPLE 10.11 ORDERED ARRANGEMENTS OF RUNNERS

Seven runners finish a race, with no ties. (a) How many finishing orders are possible? (b) How many have Ana first? (c) Find $P(\text{Ana in the first three})$ and the number of ordered first-three placings.

SOLUTION

(a) $7! = 5040$. **(b)** With Ana first, the other 6 can be arranged in $6! = 720$ ways.

(c) By symmetry, $P(\text{Ana in the top three}) = \dfrac{3}{7}$. Counting confirms it: $3\times 6! = 2160$ orders, and $\dfrac{2160}{5040} = \dfrac{3}{7}$.

The number of ordered first-three placings is ${}^{7}P_{3} = 7\times 6\times 5 = 210$.

> The event "Ana is in the top three" is about membership, while the sample space consists of ordered finishes; counting both consistently gives the same answer.

EXAMPLE **10.12** SELECTING A COMMITTEE

A committee of 4 is chosen from 6 women and 5 men. Find $P(\text{exactly 2 women})$, $P(\text{at least one woman})$ and $P(\text{Priya included})$, where Priya is one of the women.

SOLUTION

Total committees: ${}^{11}C_4 = 330$.

Exactly 2 women: ${}^{6}C_2 \times {}^{5}C_2 = 15 \times 10 = 150$, so $P = \dfrac{150}{330} = \dfrac{5}{11}$.

At least one woman: the complement is 4 men, ${}^{5}C_4 = 5$, so $P = 1 - \dfrac{5}{330} = \dfrac{65}{66}$.

Priya included: choose the other 3 from 10, ${}^{10}C_3 = 120$, so $P = \dfrac{120}{330} = \dfrac{4}{11}$.

> Fix the restricted person and choose the rest. Do not multiply by 4 for possible positions: a committee has no positions, so order does not matter.

MISCONCEPTION

"Use ${}^{n}P_r$ for every selection, because order can always be imposed afterwards." Order can be imposed, but that changes the question. If the sample space is a set of committees, the count is ${}^{n}C_r$; if it is a list of placings, it is ${}^{n}P_r$. Mixing them changes the probability by a factor of $r!$. Also, $0! = 1$, not 0.

GUIDED PRACTICE

1. Evaluate $5!$. (Answer: 120.)
2. Evaluate ${}^{7}P_3$. (Answer: 210.)
3. Evaluate ${}^{8}C_3$. (Answer: 56.)
4. A committee of 3 is chosen from 5 women and 4 men. Find $P(\text{exactly 2 women})$. (Answer: $\dfrac{40}{84} = \dfrac{10}{21}$.)

EXERCISE 10E

COUNTING ARRANGEMENTS AND SELECTIONS

FLUENCY

1 Evaluate (a) $6!$, (b) $\dfrac{9!}{7!}$, (c) ${}^{6}P_2$, (d) ${}^{6}C_2$.

2 A code uses 5 different letters from 10. How many codes are possible? How many with repetition allowed?

3 Eight people stand in a row. How many orders are possible? How many have a particular person at an end?

4 Evaluate (a) ${}^{9}C_4$, (b) ${}^{15}C_2$, (c) ${}^{9}P_4$, (d) ${}^{15}P_2$.

5 How many distinct arrangements are there of (a) BANANA, (b) LEVEL?

6 A team of 3 is chosen from 7. How many teams are possible? How many include a particular player?

UNDERSTANDING AND REASONING

7 **Error analysis.** A student calculates ${}^{8}C_3 = \dfrac{{}^{8}P_3}{3} = \dfrac{336}{3} = 112$. Identify the error and correct it.

8 Explain why ${}^{n}C_r = {}^{n}C_{n-r}$, by choosing which objects to leave out.

9 A committee of 4 is chosen from 7 women and 4 men. Find $P(\text{exactly 2 women})$ and $P(\text{at least one man})$.

10 Decide whether order matters, then calculate: (a) 3 toppings from 8; (b) 3 of 8 books arranged on a shelf.

11 Ten runners finish a race. Using counting, find $P(\text{a particular runner is in the top three})$.

12 Explain why $0! = 1$, using $n! = n(n-1)!$ and the case $n = 1$.

EXERCISE 10E · CONTINUED COUNTING ARRANGEMENTS AND SELECTIONS

APPLICATION AND EXTENSION

13 A password has 4 different letters followed by 2 different digits. How many passwords are possible? How many begin with a vowel?

14 A lottery chooses 6 numbers from 45. How many tickets are possible? Find $P(\text{a single ticket wins})$.

15 **Extension.** Prove ${}^{n}C_{r} = {}^{n-1}C_{r-1} + {}^{n-1}C_{r}$ by considering a committee of r from n, one of whom is distinguished. Use it to find ${}^{7}C_{3}$.

PATTERN AND GENERALISATION

10.6 Investigation: the birthday problem

How large must a group be before it is more likely than not that two people share a birthday? Most people guess far too large; the answer is a clean exercise in complements and products.

INVESTIGATION 10 · SHARED BIRTHDAYS

Counting the complement

Assume each birthday is equally likely among 365 days and that birthdays are independent. Ignore leap years.

1. Explain why $P(\text{at least two share})$ is hard to calculate directly, while $P(\text{all different})$ is straightforward.
2. For $n = 5$, write $P(\text{all different})$ as a product of five fractions. Evaluate it and find $P(\text{at least one shared})$.
3. Repeat for $n = 10$ and $n = 20$, recording n, $P(\text{all different})$ and $P(\text{shared})$ in a table.
4. Extend the table to $n = 23, 30, 40, 50$. Describe how the probability grows, and explain why the growth is not linear.
5. Find the smallest n for which $P(\text{shared})$ first exceeds 0.5. Test $n = 22$ and $n = 23$.
6. Compare the answer for $n = 23$ with the probability that someone shares *your* birthday in a group of 23. Why are they so different?
7. Discuss the assumptions: seasonal birth patterns, twins, and 366 days. Which matter, and which are harmless?

n	$P(\text{all different})$	$P(\text{at least one shared})$
5	0.9729	0.0271
10	0.8831	0.1169
20	0.5886	0.4114
23	0.4927	0.5073
30	0.2937	0.7063
40	0.1088	0.8912
50	0.0296	0.9704

Generalise. For $n \le 365$,

$$P(\text{at least one shared}) = 1 - \frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365 - n + 1}{365}.$$

Write the product as $1 - \dfrac{365!}{(365-n)!\,365^{n}}$ and explain each factor. What happens at $n = 366$? What changes with 366 equally likely days? Finally, explain why 23 people form 253 pairs, and why "253 small chances" is a better intuition than "23 people, each with 365 days".

MODELLING · CRITERION D

Writing the result letter

Return to the opening problem. The programme must send a result, and the wording will influence what each person does next.

1. **Model.** Choose a prevalence, sensitivity and false-positive rate. State the population and build a table in natural frequencies.
2. **Calculate.** Find $P(C \mid +)$, $P(C' \mid +)$ and $P(C \mid -)$. Say which the patient needs and which the manufacturer is likely to quote.
3. **Communicate.** Write a 120-word result letter for someone with a positive result. Use natural frequencies, avoid reversing the conditional, and state the next step.
4. **Evaluate.** State two assumptions and how each could fail. Discuss whether the programme is fair when many receive false positives, and recommend one change.

CUMULATIVE

10.7 Mixed practice

Attempt all questions without notes. Decide first whether the question asks for a probability, a conditional probability, or a count.

MIXED PRACTICE 10

SETS, CONDITIONALS, INDEPENDENCE, TREES AND COUNTING

FLUENCY AND UNDERSTANDING

1. If $P(A) = 0.45$, $P(B) = 0.3$ and $P(A \cap B) = 0.15$, find $P(A \cup B)$, $P(A \mid B)$ and test independence.
2. Of 150 adults, 45 read fiction and non-fiction, 30 read fiction only, 30 non-fiction only and 45 neither. Find $P(F)$, $P(N)$, $P(F \mid N)$ and test independence.
3. Of 50 people, 28 own a bicycle, 18 a helmet and 9 both. Find $P(\text{bicycle} \mid \text{helmet})$, $P(\text{helmet} \mid \text{bicycle})$ and $P(\text{neither})$.
4. A bag has 6 red and 4 blue. Two are drawn without replacement. Find $P(RR)$ and $P(\text{at least one red})$.
5. How many 5-digit codes can be formed from 1 to 9 with no repetition?
6. If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.8$, find $P(A \cap B)$ and test independence.
7. A die is rolled three times. Find $P(\text{at least one six})$.
8. In 200 people, 40 have a condition; a test detects 90% of cases and falsely flags 10% of healthy people. Find $P(C \mid +)$.
9. A committee of 5 is chosen from 5 women and 4 men. Find the number of committees and $P(\text{exactly 3 women})$.
10. A spinner lands on red 120 times in 500 spins. Find the relative frequency and compare it with $\frac{1}{5}$. Does this prove bias?
11. Explain the difference between mutually exclusive and independent events.
12. A game costs £1 and pays £5 with probability 0.25. Find the expected payout and decide whether it is fair.

REASONING AND APPLICATION

13. A programme tests 100,000 people. The condition affects 0.8%, the test detects 95% of cases and falsely flags 3% of healthy people. Find $P(C \mid +)$ and $P(C' \mid -)$, and interpret each.
14. A bag has 3 red and 3 blue. Three are drawn without replacement. Find $P(\text{all same colour})$ and $P(\text{exactly two reds})$.
15. If A and B are independent, show that A' and B are independent.
16. Six men and four women line up at random. Find $P(\text{the four women stand together})$.

MIXED PRACTICE 10 · CONTINUED

SETS, CONDITIONALS, INDEPENDENCE, TREES AND COUNTING

17 Find $P(\text{at least two share a birthday})$ in a group of 15, stating your assumptions.

18 **Error analysis.** A student writes $P(A \mid B) = \dfrac{P(A \cap B)}{P(A)}$. What does this compute, why is it wrong, and when would the two coincide?

19 A game uses two dice and pays on a total of 7. Find $P(\text{win})$ and the fair payout for a £1 stake.

20 Machine A makes 55% with 2% defective, B 30% with 4%, C 15% with 8%. Find $P(\text{defective from C} \mid \text{defective})$.

21 **Extension.** Prove $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$, and hence show that $P(A \mid B) > P(A)$ implies $P(B \mid A) > P(B)$.

22 **Extension.** A committee of 4 is chosen from 6 women and 5 men. Find $P(\text{more women than men})$ and $P(\text{at least one of each})$.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 20 marks

Sets, conditionals and counting

Instructions. Answer all questions. Show sufficient working. A calculator is permitted; give exact fractions where outcomes are equally likely.

In a group of 80 students, 40 own a bicycle (B) and 30 own a skateboard (S). There are 18 who own both.

1 Draw a Venn diagram and find $P(B)$, $P(S)$, $P(B \cap S)$ and $P(B \cup S)$. [4]

2 Find $P(S \mid B)$ and $P(B \mid S)$. Explain why they differ. [4]

3 Test whether B and S are independent, and interpret your answer. [4]

4 A bag has 4 red and 3 blue counters. Two are drawn without replacement. Find $P(RR)$, $P(\text{exactly one red})$ and $P(\text{at least one red})$. [5]

5 A four-digit code uses four different digits from 1 to 9. How many codes are possible? Find $P(\text{the code is even})$. [3]

Evidence: a correct Venn diagram and probabilities, two distinguished conditional probabilities, an independence test with interpretation, a complete tree or path calculation, and a counting argument for the even-code probability.

CRITERION B · INVESTIGATING PATTERNS — 40 minutes · 24 marks

The sum of two dice

Instructions. Work independently. Record results in a table, describe the pattern and justify your generalisation.

1 Copy and complete a 6×6 addition table for the totals of two dice. How many equally likely outcomes are there? [6]

2 Count the outcomes for each total from 2 to 12 and write the probability of each. [4]

3 Describe the pattern and conjecture a formula for $P(\text{total } s)$ in terms of s. [5]

4 Test the formula for $s = 4$ and $s = 10$. Explain the symmetry of the distribution. [4]

5 A game pays 5 credits for a 7 and 2 credits for a 6. Find the expected payout for a 1-credit stake and decide whom the game favours. [3]

6 For two fair n-sided dice numbered 1 to n, how many outcomes give $n + 1$? What is the probability? [2]

Evidence: an accurate table, correctly counted frequencies, a formula that fits all totals, a verified prediction and symmetry argument, an expected-value calculation, and a correct extension.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining a positive test result

Instructions. A friend says: "The test is 90% accurate, so if I test positive there is a 90% chance I have the condition." Write a 300–400 word explanation for someone who has not studied probability. Your explanation must: (i) distinguish $P(+ \mid C)$ from $P(C \mid +)$ in plain language; (ii) use a natural-frequency table for 10,000 people with prevalence 2%, sensitivity 90% and false-positive rate 5%; (iii) explain the base-rate effect without blaming or alarming the reader; (iv) suggest two sentences a clinic could use in a result letter.

Evidence: correct use of the two conditional probabilities and an accurate table [6] · a clear explanation of the base rate [6] · language suited to a non-specialist, with a fair tone [4] · a justified recommendation for the letter [4].

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

A screening decision

Instructions. A region is considering screening 10,000 adults for a condition that affects 1.5% of them. The test detects 90% of cases and wrongly flags 2% of people without the condition. A positive result is followed by a confirmatory test.

1. Build a two-way table in natural frequencies for the 10,000 adults. [5]
2. Find $P(+)$, $P(C \mid +)$ and $P(C' \mid -)$. Give each as a fraction and a percentage. [6]
3. Interpret $P(C \mid +)$ for a patient, and explain why it is much smaller than the sensitivity. [4]
4. State two assumptions and explain how each could alter the result. [5]
5. Evaluate the programme. Discuss the benefits and harms of false positives, and recommend whether it should be introduced with or without a confirmatory test. [4]

Evidence: an accurate table, correct probabilities with consistent denominators, a clear base-rate explanation, explicit assumptions, and a balanced evaluation distinguishing test accuracy from the meaning of a positive result.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 5 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- An event is a subset of the sample space. $A \cup B$ means at least one occurs, $A \cap B$ means both, and A' means A does not.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ corrects for double counting; it simplifies to $P(A) + P(B)$ only for mutually exclusive events.
- Conditioning restricts the sample space: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, and $P(A \cap B) = P(A \mid B)P(B)$.
- $P(A \mid B)$ and $P(B \mid A)$ generally differ. A screening result must be read with the base rate, preferably using natural frequencies.
- Events are independent when $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$. Independent events can overlap; mutually exclusive events with positive probability are dependent.
- On a tree, multiply along paths and add across alternatives; total probability is the sum of the paths producing an event, and reverse conditionals divide by that total.
- Counting uses $n!$, ${}^{n}P_{r} = \dfrac{n!}{(n-r)!}$ when order matters, and ${}^{n}C_{r} = \dfrac{n!}{r!(n-r)!}$ when it does not.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \qquad P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(A)P(B) \text{ if independent}$$

$$ {}^{n}P_{r} = \frac{n!}{(n-r)!}, \qquad {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} $$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works.

- I can list a sample space and describe events as subsets.
- I can use $A \cup B$, $A \cap B$ and A' correctly.
- I can read probabilities from a two-way table and a Venn diagram.
- I can apply the addition rule and identify mutually exclusive events.
- I can compute a conditional probability by restricting the denominator.
- I can use the multiplication law in both directions.
- I can explain why $P(A \mid B) \neq P(B \mid A)$ in general.
- I can test independence from a product or a conditional.
- I can distinguish independence from mutual exclusivity.
- I can draw a tree whose branches sum to 1 at each node.
- I can find total and reverse conditional probabilities from a tree.
- I can solve "at least one" problems using a complement.
- I can count with $n!$, ${}^{n}P_{r}$ and ${}^{n}C_{r}$, including a restriction.
- I can explain a screening result using natural frequencies and the base rate.

REFLECTION

Write short answers in your notebook. Use evidence from your work.

1. Which representation — table, Venn diagram or tree — made conditional probability clearest for you, and why?
2. Describe one occasion when you reversed a conditional by mistake. What check now prevents the error?
3. Use the screening example to explain what conditioning restricts and what is being measured.
4. Why is it misleading to describe a test as "90% accurate" without saying which conditional probability is meant?
5. Which assumption in the birthday problem surprised you most, and how would you test it with data from your class?

FROM A REAL QUESTION TO A DEFENDED ANSWER

The Modelling Project

11

Assumptions, data, a model, a recommendation — and an honest account of what the model cannot see.

RELATIONSHIPS · CHANGE · REPRESENTATION

STATEMENT OF INQUIRY

A mathematical model is an argument about the world: its value depends as much on the assumptions it makes and the evidence it ignores as on the accuracy of its arithmetic.

KEY CONCEPT
Relationships

RELATED CONCEPTS
Change · Representation · System

GLOBAL CONTEXT
Globalization and sustainability — decisions about shared resources, access and long-term cost

FACTUAL
What are the stages of the modelling cycle, and what evidence belongs at each stage?

CONCEPTUAL
How do assumptions shape a model's conclusion, even when the mathematics is correct?

DEBATABLE
When a model is uncertain, is it more responsible to act on it or to wait for better data?



IN THIS CHAPTER

11.1 The modelling cycle
11.2 Defining the problem
11.3 Building and testing a model
11.4 Optimising and evaluating
11.5 Presenting and defending

OPENING CASE · THE TIMETABLE THAT LOOKED FAIR

An average that hid the problem

A city transport department plans a bus timetable using the average journey time on each route. The averages come from ticket-tap data, which records when passengers enter and leave the network but not where they change vehicles. On paper, every route meets its target. After the timetable is introduced, complaints rise sharply on two routes that serve a hospital and a night shift.

The arithmetic is not wrong. The model is. **What did the average leave out, and what would a better model need to represent?**

1. List three quantities the department measured, and three quantities that matter to passengers but were not measured.
2. Explain how an average can meet a target while many individual journeys fail it.
3. Suggest one variable the department should record, and one assumption it should state publicly.
4. Would more data alone have prevented the problem? Explain what else is required.

This chapter is different from the others. It does not introduce a new algebraic technique. It teaches you to choose, build, test, evaluate and defend a model — the work that turns the mathematics of Chapters 1–10 into a reasoned answer to a real question.

CH 11

ORIENTATION

11.0 Learning intentions

- Describe the modelling cycle and explain why it is iterative rather than linear.
- Define a modelling question with stated scope, variables, units, assumptions and data plan.
- Select and fit an appropriate model, then test it against evidence not used in fitting.
- Optimise within constraints and evaluate a recommendation against alternatives and uncertainty.
- Present and defend a model in a written report and an oral response to questions.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

state a focused modelling aim and its boundaries; identify independent, dependent and controlled variables with units; distinguish assumptions from facts; choose a model family for a stated reason; fit parameters and report uncertainty; test a model on a validation set and describe residuals; optimise an objective under constraints; perform a simple sensitivity check; write a structured report with appropriate graphs and significant figures; acknowledge limitations without undermining the conclusion; answer a challenge about my assumptions with evidence.

ATL SKILLS · RESEARCH, THINKING AND COMMUNICATION

Research: record the source, date and sample size of every data set. A model is only as trustworthy as its data, and a number without provenance is an opinion. **Critical thinking:** separate three questions — Is the arithmetic right? Is the model appropriate? Is the conclusion justified? A yes to the first does not imply a yes to the others. **Communication:** write for a decision-maker who will not read your working. Lead with the recommendation and the uncertainty, then support it.

PREREQUISITE CHECK · 15 MINUTES

These skills are used throughout the project. If you cannot do three or more, revisit the chapter named beside each item.

1. Complete the square for $y = 2x^2 - 12x + 7$ and state the vertex. (Chapter 1)
2. Solve $1.06^t = 1.5$ for t. (Chapter 3)
3. Find the sum of the first 20 terms of $3, 7, 11, \ldots$. (Chapter 4)
4. Sketch the feasible region $x + y \leq 8$, $x \geq 0$, $y \geq 0$, and find its vertices. (Chapter 5)
5. Calculate the mean and standard deviation of $4, 7, 9, 10, 15$. (Chapter 9)
6. For a 2×2 matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, find the determinant. (Chapter 8)

REFRESHER · THE MATHEMATICS YOU WILL DRAW ON

A model is a simplified description, not a copy. Linear models describe constant rates; quadratic models describe constant acceleration or a single turning point; exponential models describe constant proportional change; trigonometric models describe periodic behaviour; geometric models describe shape and space; optimisation models compare feasible choices. The choice of family is an assumption, and it must be justified by the mechanism of the situation, not only by which curve fits the points.

CONCEPT

11.1 The modelling cycle

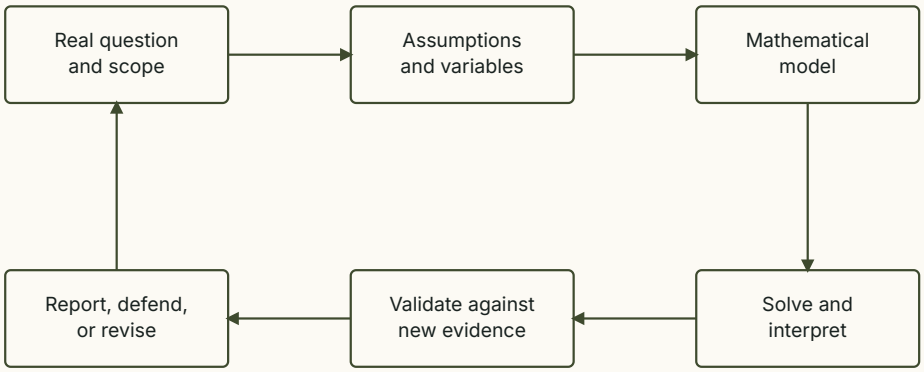
Modelling begins with a question about the world, not with an equation. The modeller simplifies the situation until it becomes mathematical, solves the mathematics, then translates the result back and asks whether it is credible. The process is a cycle because the answer usually reveals a weakness in the simplification, which sends the modeller back to the assumptions.

The stages are not a checklist to be completed once. A modeller may move from validation back to assumptions three times before a result is defensible. What matters is that each move is recorded: the reader must be able to see which assumption produced which consequence.

DEFINITIONS

Model, assumption, parameter, validation

A **mathematical model** is a simplified mathematical representation of a real situation, built to answer a question. An **assumption** is a statement accepted as true for the purposes of the model without being proved from the data. A **parameter** is a constant in the model whose value is estimated from data (for example the growth rate k in $P = Ae^{kt}$). **Validation** is the process of comparing the model's predictions with evidence that was not used to build it.



The cycle is entered at the top left and revisited whenever validation exposes a weakness. A model that is never revised is usually a model that was never tested.

EXAMPLE **11.1** READING A MODEL CRITICALLY

A school models its water consumption as $W(d) = 120 + 8d$ litres per day, where d is the number of days since the start of term. The model fits the first four weeks well. In week five, a sports tournament doubles attendance. Evaluate the model.

SOLUTION

The arithmetic is simple and the fit over the first four weeks is evidence that the linear form captures the routine use of the school. But the model assumes a constant daily increase of 8 litres and no dependence on attendance. The tournament changes a variable that the model treats as fixed. The prediction for week five is therefore an extrapolation beyond the conditions under which the model was validated.

A better model would include attendance a: for example $W = 120 + 8d + ka$, with k estimated from weeks with varying attendance. The original model is not wrong within its scope; it is being used outside it.

> The phrase “the model fits” is incomplete. A model fits a particular data set under particular conditions; state both.

MISCONCEPTION

“A model is either right or wrong.” Models are useful or misleading for a stated purpose, within a stated range. A model that predicts the height of a tide to within 10 cm may be excellent for deciding when to launch a boat and useless for calibrating a scientific instrument. Judgements of a model must name the purpose and the tolerance.

GUIDED PRACTICE

1. Name the stage of the cycle at which each activity belongs: collecting temperatures; writing $T = 20 + Ae^{-kt}$; checking a prediction for a new day. (Data/assumptions; model; validation.)

2. State one assumption behind a model that predicts a population by $P = 1000(1.03)^t$. (Constant 3% growth, unlimited resources.)

3. Why is a model that has never been tested not yet a model you can defend? (No evidence it predicts beyond the fitting data.)

4. Give an example of a question for which an average is an adequate summary and one for which it is not. (Average height of a class; waiting time at a hospital.)

EXERCISE 11A

THE CYCLE IN PRACTICE

FLUENCY

1 Order the stages: validate; define the question; interpret; build the model; state assumptions; report.

2 Classify each as a fact, an assumption or a parameter estimate.

a. The room is kept at 20°C.

b. The cooling constant is $k = 0.11$.

c. The cup does not gain heat from the table.

d. The population was 12 000 in 2020.

3 For each question, name a suitable model family and one variable.

a. How the height of a bouncing ball changes with each bounce

b. How the cost of a taxi depends on distance

c. How the number of bacteria changes with time

d. How the area of a circular ripple grows with radius

4 State the independent and dependent variable, with units, for "how the temperature of a drink changes over time".

5 A model predicts a value of 48; the measured value is 52. Calculate the percentage error.

6 Explain the difference between interpolation and extrapolation in one sentence each.

UNDERSTANDING AND REASONING

7 A model of a school's electricity use ignores weekends. Describe the effect on predictions for a holiday week and explain why the model may still be useful.

8 Two models fit the same data equally well: a quadratic and an exponential. Describe two pieces of evidence you would seek before choosing one.

9 Explain why adding more parameters to a model can make the fit better while making predictions worse.

10 A report says "the model is accurate to within 5%". What else must be stated before this claim can be assessed?

11 Give one situation in which a simple model is preferable to a complex one, and justify your answer.

12 Explain why the validation data must not be the same as the fitting data.

APPLICATION AND EXTENSION

13 A café records sales for four weeks and fits $S = 80 + 12d$. In week five it runs a promotion. Predict sales, state whether the prediction is interpolation or extrapolation, and explain the risk.

14 A model of a reservoir uses a linear decrease of 2% of the current volume per day. Explain why this is actually an exponential model, and write it in both forms.

15 **Extension.** A model is built on data from a sample of 40 volunteers, all aged 18–25. Discuss two ways the sample limits the model's scope, and state how you would report the limitation honestly.

METHOD

11.2 Defining the problem

A modelling project succeeds or fails at the definition stage. "How can the school save energy?" is too broad to model; "How much would switching the corridor lights to motion sensors reduce electricity use between 07:00 and 18:00 on school days?" can be modelled, measured and defended. A focused question names the decision, the population, the time frame and the quantity to be predicted.

Definition also fixes the variables, their units and the data plan. Units are not decoration: a rate in litres per minute and a time in hours cannot be combined without conversion, and a graph without axis units cannot be interpreted. The data plan must state what will be measured, how often, with what instrument, and what will be done about missing or implausible values.

KEY IDEA · A DEFENSIBLE PROJECT DEFINITION

State (1) the decision the model will inform; (2) the scope — population, place, time; (3) the dependent variable and its unit; (4) the independent variables and their units; (5) the variables held constant; (6) the assumptions; (7) the data source and sample size; (8) the accuracy required for the decision.

EXAMPLE **11.2** TURNING A VAGUE AIM INTO A MODEL QUESTION

A student wants to model "the effect of the canteen on student health". Rewrite this as a focused modelling question with variables and a data plan.

SOLUTION

Focused question: "Among Year 10 students who buy lunch at the canteen at least three days a week, how does the number of canteen purchases per week predict the amount of sugar in their lunch, measured in grams, over one term?"

Variables: dependent — sugar content (g) of the lunch purchased; independent — canteen purchases per week (count); controlled or recorded — year group, portion size, day of week, whether a drink is included. **Data plan:** collect canteen receipts and menu nutritional information for 30 volunteers over eight weeks; record missing days; check portion sizes against the menu; obtain consent and anonymise records.

Accuracy needed: the decision is whether to change the menu, so a difference of about 5 g per lunch matters; this sets the measurement precision required.

> The rewrite does not make the question small; it makes it answerable. "Health" has been replaced by a measurable quantity, with a stated population and time frame.

EXAMPLE **11.3** CHOOSING VARIABLES AND UNITS

A model will predict the time for a kettle to boil. List the variables, state which are controlled, and explain why each unit matters.

SOLUTION

Dependent: time to boil (s). Independent: volume of water (L), initial temperature (°C), power rating of the kettle (W). Controlled: starting temperature (if not varied), kettle model, lid position, altitude. Each unit matters: power in watts is energy per second, so the predicted time must use energy in joules; mixing litres and millilitres changes the mass and therefore the energy required.

A dimensional check is possible: energy to heat water is approximately $4.2\,\text{J g}^{-1}\,^\circ\text{C}^{-1}\times\text{mass}\times\text{temperature rise}$. Dividing by power in watts (J/s) gives seconds.

> A model with inconsistent units can look numerically reasonable and still be meaningless. Write the units into the equation as you build it.

MISCONCEPTION

"More variables make a model more scientific." A model with many variables may be harder to measure, harder to validate and easier to overfit. Each variable must earn its place by affecting the decision or by confounding a relationship you intend to measure. Parsimony — the simplest model that answers the question — is a virtue, not a shortcut.

GUIDED PRACTICE

1. Rewrite "How does homework affect learning?" as a focused question with one measurable outcome. (e.g. "How does weekly homework time predict the next unit test score for Year 10 students?")

2. State the units of the gradient of a distance–time graph. (m/s.)

3. Name one controlled variable in an experiment on plant growth and light. (Water volume, soil, temperature.)

4. Why must the accuracy required be stated before data collection? (It determines precision, sample size and method.)

EXERCISE 11B

AIMS, VARIABLES AND DATA

FLUENCY

1 Identify the dependent and independent variable, with units.
   a. Height of a plant measured weekly
   b. Cost of a journey against distance travelled
   c. Temperature of water against time after boiling
   d. Number of bacteria against temperature of incubation

2 State whether each is a model, an assumption, a parameter or a measurement: "the decay constant is 0.04 per minute"; "the room stays at 20°C"; "the mass is 250 g"; "the cup loses heat only to the air".

3 Write a focused modelling question for each broad aim.
   a. Reduce food waste in the canteen.
   b. Improve traffic flow at the school gate.
   c. Choose the best phone plan for a family.

4 Convert 45 minutes to hours and 2.5 litres to millilitres. Explain why each conversion matters in a model.

5 A table records distance in km and time in minutes. State the unit of the gradient and interpret it.

6 List three items that a data plan must specify.

UNDERSTANDING AND REASONING

7 Explain why "number of canteen purchases" may be a poor measure of "healthiness of lunch". Suggest a better dependent variable.

8 A student plans to model the height of a tide using data from one week. State two scope limitations and one way to reduce each.

9 Explain the difference between a controlled variable and an assumption, using an example.

10 Why can a sample of volunteers who respond to an online advert be biased? Describe the direction the bias might take in a sleep study.

11 A model uses time in days and a rate per week. Describe the error that results and how to prevent it.

12 "The model needs to be accurate enough to choose between two options." Explain how this statement determines the required precision.

APPLICATION AND EXTENSION

13 Design a data collection plan for comparing the waiting time at two canteen queues. Include sample size, timing, what to record and one ethical consideration.

14 A school wants to reduce paper use. Write a focused question, two variables with units, and one assumption. Explain how the answer could inform a decision.

15 **Extension.** A model is intended to compare two policies, not to predict an exact value. Explain why a biased model can still rank the policies correctly, and why this does not make the bias acceptable.

METHOD

11.3 Building and testing a model

Building a model means choosing a family and estimating its parameters. The choice should be justified by mechanism: a quantity that changes by a constant amount per unit suggests a linear model; one that changes by a constant proportion suggests an exponential; one with a single turning point suggests a quadratic. Fitting then uses the data — by least squares, by two points, or by a transformation such as taking logarithms.

Testing is separate from fitting. Reserve part of the data, or collect new data, and compare predictions with observations. Inspect residuals — the differences between observed and predicted values — for a pattern. A pattern in the residuals means the family is wrong even if the fit looked good. Report the size of the typical error, not only the correlation.

KEY IDEA · FIT, TEST, REPORT

Fit on one set of data; test on another. Report the typical size of the residuals in the units of the dependent variable. A model with a high r^2 but a systematic residual pattern is not yet a good model. Extrapolation beyond the data range is a separate, weaker claim.

EXAMPLE **11.4** FITTING AND VALIDATING AN EXPONENTIAL MODEL

Bacteria counts (in thousands) are measured every 30 minutes: 2.0, 2.8, 4.0, 5.6, 7.9, 11.1. Fit an exponential model, predict the next count, and evaluate the fit using the final observation as validation.

SOLUTION

Ratios: $2.8/2.0 = 1.40$, $4.0/2.8 = 1.43$, $5.6/4.0 = 1.40$, $7.9/5.6 = 1.41$. The ratio is roughly constant, so an exponential model is appropriate. Using the first and fourth points, $5.6 = 2.0b^3$, so $b^3 = 2.8$ and $b = 1.410$. With $a = 2.0$:

$$N(t) = 2.0(1.41)^t, \quad t \text{ in half-hours.}$$

Prediction for $t = 5$: $2.0(1.41)^5 = 11.15$, against the observed 11.1 — an error of about 0.5%. The fit is good over the observed range.

A prediction for $t = 10$ would be $2.0(1.41)^{10} = 62.2$ thousand, but this is extrapolation: the culture may run out of nutrients long before then. The model's validity is limited to the conditions and range observed.

> The ratios being constant is stronger evidence for the exponential family than a high r^2 alone, because it tests the mechanism, not just the curve.

EXAMPLE **11.5** COMPARING TWO MODELS

A ball is dropped and its height is measured each 0.1 s. Model A is linear; Model B is quadratic. Both are fitted to the first five points. On the remaining points, Model A has residuals $+4, -6, +8, -10$ cm and Model B has $+1, -1, +1, -1$ cm. Which model is better supported, and what else should be checked?

SOLUTION

Model B has smaller residuals and, more importantly, no systematic growth in their size. Model A's residuals alternate and increase, which suggests the linear family cannot capture the changing rate. Model B is better supported.

Still to check: the residuals should be plotted against time to confirm there is no curve; the measurement uncertainty (about ± 0.5 cm) should be compared with the residuals; and the model should not be used beyond the observed time, especially after the ball hits the floor.

> Small residuals do not prove the model; they fail to disprove it. Scientific language is calibrated: "supported by the evidence", not "proved".

CHECK	QUESTION TO ASK	EVIDENCE TO RECORD
Mechanism	Does the family match how the quantity changes?	Constant difference, ratio or second difference
Parameters	Were they estimated from data or assumed?	Method and data used
Residuals	Is there a pattern, and how large are they?	Plot or table, typical size in context units
Validation	Was the test data separate from the fitting data?	Prediction error on held-back points
Scope	Where is the model supported?	Range of inputs, conditions, sample

MISCONCEPTION

"The line with the highest r^2 is the best model." r^2 measures how much of the variation in the data is explained by a *linear* fit; it can be high for a model that is systematically wrong, and it says nothing about extrapolation. A model must also be checked for residual pattern, mechanism and scope.

GUIDED PRACTICE

1. Data double every step. Which family fits, and what is the parameter? (Exponential; base 2.)
2. Residuals are $2, -1, 0, 1, -2$ with measurement uncertainty ± 0.5. What do you conclude? (No clear pattern; residuals larger than uncertainty, so model imperfect but plausible.)
3. Why is r^2 not a validation statistic? (It is computed on fitting data and measures linear association, not predictive accuracy.)
4. A model is fitted to data from 5 to 60 minutes. Is a prediction at 120 minutes interpolation or extrapolation? (Extrapolation.)

EXERCISE 11C

FITTING, VALIDATING AND REPORTING

FLUENCY

1 State the family suggested by each table.
   a. Constant difference 4
   b. Constant ratio 1.2
   c. Constant second difference 6
   d. Values repeat every 12 units

2 For $y = ab^x$ passing through $(0, 5)$ and $(4, 80)$, find a and b.

3 A model predicts 84 and the observation is 78.

4 Residuals for five validation points are

EXERCISE 11C · CONTINUED FITTING, VALIDATING AND REPORTING

5 State whether each prediction is interpolation or extrapolation.
   a. Fitted 10–50, predicting 30
   b. Fitted 10–50, predicting 70
   c. Fitted 1–12, predicting 6
   d. Fitted 1–12, predicting 0.5

6 Explain why the units of a residual are the units of the dependent variable.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student reports "the model is excellent because $r^2 = 0.98$" after using all the data to fit and test. Identify two errors in the reasoning.

8 A residual plot shows a clear U-shape. What does this suggest about the chosen model family?

9 Explain why a model can be accurate on average but useless for an individual case. Give an example.

10 Two models have the same typical residual but different maximum residuals. Which might you prefer for a safety decision, and why?

11 Explain why a model fitted to data with a rounded input variable may overstate its precision.

12 A model's prediction is 100 with a typical error of 15. A decision threshold is 105. Explain the uncertainty in the decision.

APPLICATION AND EXTENSION

13 A shop's daily sales for 10 days are 120, 135, 128, 150, 160, 155, 170, 180, 175, 190. Fit a linear model by eye, predict day 12, and explain why the prediction may be unreliable.

14 A cooling model gives temperatures 72, 60, 51, 45, 41 at 0,2,4,6,8 minutes. Show that the drop is approximately exponential, estimate the half-life, and state one limitation.

15 **Extension.** A model with three parameters fits 8 data points exactly. Explain why this is not evidence of a good model, and describe a validation procedure that would expose the problem.

APPLICATION

11.4 Optimising and evaluating

Many modelling questions ask for a best choice: the largest volume, the lowest cost, the shortest time, the fairest allocation. Optimisation requires two things: an **objective** — the quantity to be made as large or small as possible — and **constraints** — the limits within which the choice must lie. Without constraints, an optimisation is usually trivial or impossible; with them, it becomes a mathematical problem.

The optimum is not the end of the work. A recommendation must be evaluated: Is the optimum attainable? How sensitive is it to the assumptions? What does it cost in a dimension the objective ignores? A model that minimises cost may maximise environmental harm; a model that maximises profit may reduce access. Evaluation makes those trade-offs explicit.

EXAMPLE 11.6 AN OPTIMISATION WITH A CONSTRAINT

A farmer has 60 m of fencing and wants to enclose a rectangular vegetable plot against an existing wall, using fencing on three sides. Find the dimensions that maximise the area, and evaluate the result.

SOLUTION

Let the two sides perpendicular to the wall be x m each and the side parallel to the wall be y m. The fencing constraint is $2x + y = 60$, so $y = 60 - 2x$. The area is

$$A(x) = x(60 - 2x) = 60x - 2x^2 = -2(x - 15)^2 + 450.$$

The maximum is 450 m² at $x = 15$ m, giving $y = 30$ m. The plot is twice as long as it is deep.

Evaluation: the model assumes the wall is straight and long enough, the fencing is used exactly, and the plot is a rectangle. In practice, a gate must be included (reducing the fenced length), the wall may need repair, and a plot 15 m deep may be inconvenient for access. The mathematical optimum is a starting point for a decision, not the decision.

> The domain $0 < x < 30$ is part of the model; at the endpoints the area is zero, which is mathematically consistent but not a useful plot.

EXAMPLE 11.7 SENSITIVITY OF A RECOMMENDATION

A model recommends 240 units of a product to maximise profit. The profit function is $P(n) = -2(n - 240)^2 + 18\,000$. The factory can actually produce only 220 units. What is the cost of the constraint, and how sensitive is the profit to a small error in the optimum?

SOLUTION

At $n = 240$, $P = 18\,000$. At $n = 220$, $P = -2(20)^2 + 18\,000 = 18\,000 - 800 = 17\,200$. The constraint costs 800 in profit, about 4.4%.

Sensitivity: at $n = 230$, $P = 18\,000 - 200 = 17\,800$, only 200 below the maximum. Near the vertex, a quadratic is flat, so small deviations cost little; far from it, the cost grows quadratically. This is useful information for a decision: a small shortfall in production is tolerable, a large one is not.

> Reporting only the optimal value hides how much the answer matters. Report the cost of departing from it.

MISCONCEPTION

"The mathematical optimum is the best real decision." The optimum is best only with respect to the stated objective and the stated constraints. Real decisions involve objectives the model omits — safety, equity, environmental impact, staff workload — and constraints that are not written down. Evaluation is the stage at which those omissions are acknowledged.

GUIDED PRACTICE

1. Identify the objective and one constraint: "A bakery wants to maximise daily profit using at most 40 kg of flour." (Profit; flour ≤ 40 kg.)
2. A quadratic profit model has vertex $(50, 900)$. What profit is lost by choosing 48? ($900 + 4a$; needs a; if $a = -5$, loss 20.)
3. Why must an optimum be checked against attainability? (Constraints may not be represented; whole-number restrictions.)
4. Name one objective a cost-minimising model might ignore. (Environmental impact, quality, access.)

EXERCISE 11D

OBJECTIVES, CONSTRAINTS AND EVALUATION

FLUENCY

1 State the objective and two constraints for each.
   a. Packing a container with boxes of two sizes
   b. Choosing a phone plan with a data limit
   c. Planning a school timetable with rooms and teachers

2 Complete the square and state the maximum or minimum.
   a. $y = -3x^2 + 18x - 5$
   b. $y = 2x^2 - 8x + 11$
   c. $y = -x^2 + 10x$

3 A profit model is $P = -4(n - 30)^2 + 2500$. Find the profit at $n = 30$, 28 and 35.

4 Explain why an optimum at a vertex may not be attainable in a whole-number problem.

5 State one trade-off that a fuel-minimising delivery model might neglect.

6 What does it mean for a recommendation to be "sensitive" to an assumption?

UNDERSTANDING AND REASONING

7 **Error analysis.** A student maximises $A = x(60 - 2x)$ and reports $x = 30$, $A = 0$. Explain the error in choosing the endpoint and the correct reasoning.

8 A model has two objectives that conflict. Describe how a decision-maker might choose between them, and why mathematics alone cannot settle the choice.

9 Explain how a sensitivity check can change a recommendation even when the optimal value does not change.

10 Why is it important to state the domain of an optimisation model? Give an example where the mathematical optimum lies outside it.

11 A model recommends closing a service to save money. List two quantities the objective omits and explain how they could be included.

12 Explain why "best" is incomplete without "for what purpose".

APPLICATION AND EXTENSION

13 A rectangular pen is to be built with 48 m of fencing on all four sides. Find the dimensions that maximise area, and evaluate whether a square is always the best shape in practice.

14 A cylindrical water tank is to hold 2 m^3. The material for the base costs twice as much per m^2 as the curved side. Set up an expression for cost in terms of radius and height, and describe how you would search for a minimum using a table (no calculus).

15 **Extension.** A linear programming model has its optimum at a vertex that is not integer. Explain why checking the nearest integer points is not always sufficient in larger problems, and describe a strategy that would be reliable for two variables.

COMMUNICATION

11.5 Presenting and defending

A model that cannot be understood cannot be used. The report is part of the modelling, not an afterthought. It must let a reader reconstruct what you did, judge whether it was appropriate, and see how confident the conclusion is. A decision-maker will read the summary first; if the recommendation and its uncertainty are buried on page nine, the report has failed.

Defence is not a performance of certainty. It is an account of the choices you made and the evidence for them. A good answer to "Why did you choose an exponential model?" names the constant ratio in the data and the mechanism it represents. A good answer to "What would change your conclusion?" names the assumption that matters most and the evidence that would overturn it.

KEY IDEA · REPORT STRUCTURE

1 Summary — question, model, recommendation, uncertainty. **2 Introduction** — decision, scope, stakeholders. **3 Data** — source, sample, cleaning, limitations. **4 Model** — assumptions, variables, units, equations, parameter estimates. **5 Results** — graphs, predictions, residuals, validation. **6 Evaluation** — sensitivity, trade-offs, what the model cannot see. **7 Conclusion** — recommendation with confidence. **8 References** — data and method sources.

EXAMPLE **11.8** WRITING A DEFENSIBLE CONCLUSION

Rewrite each weak conclusion as a defensible one. (a) "The model shows that the new timetable is better." (b) "The model is accurate." (c) "More data would solve the problem."

SOLUTION

(a) "Within the observed range of passenger flows, the new timetable reduces the modelled average wait from 7.2 to 6.1 minutes; this prediction is uncertain by about ± 0.4 minutes and does not include the effect of missed connections."

(b) "On the 12 validation journeys, the model predicted total time with a mean error of 1.8 minutes and a largest error of 4.5 minutes, which is adequate for comparing timetable options but not for individual arrival advice."

(c) "The largest uncertainty comes from unrecorded transfer times. Better data on transfers would test the assumption that they are constant; more data of the same kind would not address it."

> Each rewrite names the quantity, the evidence, the uncertainty and the scope. None claims more than the model supports.

EXAMPLE **11.9** ANSWERING A CHALLENGE

A reviewer asks: "Your model predicts a 20% saving, but it ignores staff time. Why should we believe the number?" Write a response of three or four sentences.

SAMPLE RESPONSE

The 20% figure is a saving in consumable cost, not in total cost; the model was built to compare consumable use between two options, and I should have stated that scope in the summary. Staff time is likely to differ because option B requires an extra preparation step, so the total saving is smaller than 20% and may be negative if staff time is valued at the current hourly rate. I can estimate the break-even value of staff time at which the options are equal, and I will add that sensitivity to the report. The conclusion should be read as conditional on staff time being below that threshold.

> A defence admits the limitation, quantifies its effect if possible, and states what would resolve it. It does not repeat the original claim more loudly.

MISCONCEPTION

"A long report with many graphs is a rigorous report." Length is not evidence. A rigorous report has a clear question, a justified model, a validation, an honest evaluation and a conclusion that follows. Extra graphs that do not support the argument make the report harder to assess, not more impressive.

GUIDED PRACTICE

1. Name the first thing a decision-maker should see. (The recommendation and its uncertainty.)
2. Rewrite "the data are good" as a defensible statement. (State source, sample size, missing values, range.)
3. What should a graph axis always include? (Quantity, unit, scale.)
4. Give one question a reviewer might ask about an exponential model. (Why exponential? Is the ratio constant? What limits growth?)

EXERCISE 11E

REPORTS, GRAPHS AND DEFENCE

FLUENCY

1 List the sections of a modelling report in order.

2 State the quantity and unit that should appear on each axis of a graph of temperature against time.

EXERCISE 11E · CONTINUED REPORTS, GRAPHS AND DEFENCE

3 Rewrite each as a defensible statement.
   a. "The model is very accurate."
   b. "The new system is cheaper."
   c. "Students prefer option A."

4 Give two questions a reviewer might ask about a linear model of long-term population.

5 Explain why a summary should state uncertainty as well as a value.

6 State one difference between a report for a specialist and a report for a decision-maker.

UNDERSTANDING AND REASONING

7 A graph's vertical axis starts at 90 instead of 0 and shows a "dramatic" rise. Explain the distortion and when such an axis is acceptable.

8 Explain why reporting only the mean error can hide a model that is sometimes badly wrong.

9 A conclusion says "the model proves that the policy works". Replace "proves" with a defensible verb and explain the change.

10 Describe how you would answer: "Why did you not include variable X?"

11 Why should the limitations section come before the conclusion rather than after it?

12 Explain the role of references in a modelling report.

APPLICATION AND EXTENSION

13 Write a 120-word summary for a model that predicts a 12% reduction in queue time with a typical error of 2 percentage points, based on four weeks of data at one site.

14 Design a one-page report template with headings and the evidence required under each.

15 **Extension.** A reviewer challenges your sample as biased. Write a four-sentence response that acknowledges the limitation, assesses its likely direction, and proposes a check.

PROJECT

11.6 The capstone project brief

Choose one of the three briefs below, or negotiate an equivalent question with your teacher. The project is assessed against Criteria B, C and D, with a short Criterion A check on the mathematics you use. It is not a research essay: the mathematics must do real work, and the evaluation must be honest about what it cannot do.

BRIEF	QUESTION	DATA YOU COULD COLLECT	MATHEMATICS
A · Water	How should the school size a rainwater tank to cover most of its garden watering?	Daily rainfall, garden area, watering volume, tank cost	Sequences, volumes, exponential smoothing, optimisation
B · Transport	Which route should the school bus take to minimise average journey time within a fixed fuel budget?	Route times at different times of day, distances, fuel use	Linear programming, averages, spread, sensitivity
C · Sport	How does training load predict performance, and what load is optimal?	Weekly training hours, test results, recovery days	Scatter, correlation, regression, quadratic optimisation

Milestones

Milestone 1 · A focused question, scope, variables with units, and a data plan (about 300 words).

Milestone 2 · Cleaned data table, provenance note, and a first graph with labelled axes.

Milestone 3 · Model with stated assumptions, fitted parameters, and a validation on held-back data.

Milestone 4 · Optimisation or comparison, sensitivity check, and a draft evaluation.

Milestone 5 · Full report (1 200–1 800 words plus appendix) and a five-minute defence.

What “good” looks like

A strong project does not need sophisticated mathematics. It needs a question that matters, a model whose assumptions are stated and tested, arithmetic that is checked, a graph that can be read without explanation, and an evaluation that names the strongest objection to the conclusion. A project that predicts a modest result honestly is stronger than one that claims a dramatic result it cannot support.

CAPSTONE · CRITERION B · INVESTIGATING PATTERNS — Project component · 24 marks

Pattern, model and generalisation

Evidence required. A data table with provenance; a described pattern supported by a table, graph or ratio; a model with parameters estimated and justified; a validation on data not used in fitting; a statement of the range over which the pattern holds. The generalisation must be more than “the graph goes up”: it must state what changes, by how much, and under what conditions.

CAPSTONE · CRITERION C · COMMUNICATING — Project component · 20 marks

Report and defence

Evidence required. A structured report with a summary that states the recommendation and its uncertainty; correct notation and units; graphs with labelled axes and sensible scales; a conclusion that follows from the evidence; coherent answers to two reviewer questions, including one about limitations. The report must be intelligible to a reader who has not seen the data.

CAPSTONE · CRITERION D · APPLYING MATHEMATICS — Project component · 24 marks

Decision and evaluation

Evidence required. A clear decision context and stakeholder; an objective and constraints; a recommendation that is attainable; a sensitivity check showing how the recommendation changes when an assumption changes; an evaluation of at least two limitations, including one the model cannot address with more data. The evaluation must distinguish uncertainty from ignorance.

Chapter summary

- Modelling is a cycle: define, assume, build, solve, validate, report, revise. It is iterative, not linear.
- A focused question names the decision, population, time frame and quantity; variables carry units; assumptions are stated, not hidden.
- The choice of model family should be justified by mechanism as well as fit. Constant difference suggests linear; constant ratio suggests exponential; constant second difference suggests quadratic.
- Fit on one data set and validate on another. Inspect residuals for pattern and size, and report errors in context units.
- Optimisation requires an objective and constraints; the mathematical optimum is not automatically the best real decision.
- Sensitivity analysis shows how much the recommendation depends on an assumption; it is part of the answer, not an optional extra.
- A defensible conclusion states the quantity, the evidence, the uncertainty and the scope, and does not claim more than the model supports.

$$\text{residual} = \text{observed} - \text{predicted}, \qquad \text{percentage error} = \frac{|\text{error}|}{|\text{observed}|} \times 100\%$$

Revision checklist

Tick each statement only when you can do it and explain why it matters.

- I can describe the modelling cycle and give an example of a revision.
- I can write a focused modelling question with a stated decision.
- I can identify variables, units and controlled quantities.
- I can distinguish an assumption from a measurement.
- I can choose a model family and justify the choice by mechanism.
- I can fit parameters and report the method.
- I can validate on held-back data and describe residuals.
- I can perform a sensitivity check.
- I can evaluate an optimum against real constraints.
- I can write a structured report with a defensible summary.
- I can answer a challenge about assumptions with evidence.
- I can state the scope and limitations of a model honestly.

REFLECTION

Write short answers in your notebook. Use evidence from your project or from the chapter.

1. Which stage of the cycle did you find hardest, and what made it hard: the mathematics, the data, or the judgement?
2. Describe one assumption in your project that you initially treated as a fact. How did you discover it was an assumption?
3. The statement of inquiry claims that a model's value depends on what it ignores. Use an example from this chapter to support and to complicate that claim.
4. What would you do differently if you had twice as much data? What would you do if you had none?
5. What question about modelling do you still want to answer?

CONSOLIDATION, CONNECTION AND NEXT STEPS

The Whole Picture

12

LOGIC · GENERALISATION · JUSTIFICATION

Cumulative practice across the course, unfamiliar problems that require judgement, and an honest account of the move from MYP to the Diploma Programme.

STATEMENT OF INQUIRY

Mathematical maturity is the ability to choose a method, justify it, connect it to other ideas, and recognise the limits of what has been shown.

KEY CONCEPT

Logic

RELATED CONCEPTS

Generalisation · Representation · Justification

GLOBAL CONTEXT

Identities and relationships — how mathematical habits shape the way we reason elsewhere

FACTUAL

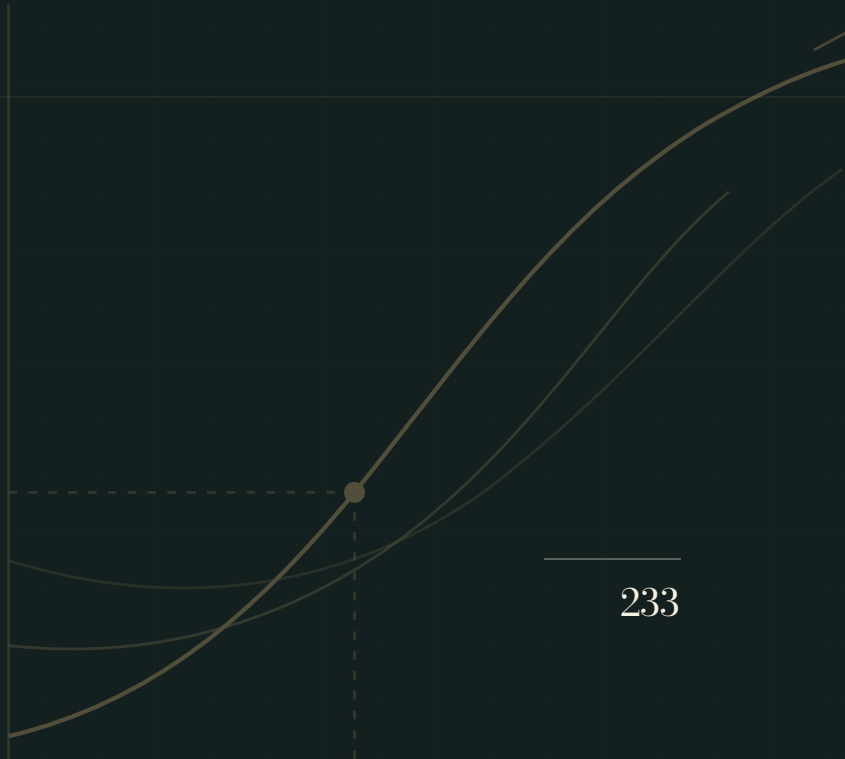
Which methods from this course apply across several topics, and which are specific to one?

CONCEPTUAL

What distinguishes a calculation from a justification, and why does the difference matter at the next level?

DEBATABLE

Can a textbook make a student "ready" for the Diploma Programme, or does readiness depend on habits the textbook cannot supply?



IN THIS CHAPTER

12.1 Cumulative practice: algebra and functions
12.2 Cumulative practice: geometry and trigonometry
12.3 Cumulative practice: statistics and probability
12.4 Unfamiliar problems and extended reasoning
12.5 From MYP to DP: expectations and readiness

OPENING PROBLEM · THE EXAMINATION QUESTION NOBODY HAD SEEN

Four marks, no familiar routine

A student is asked: "The function f satisfies $f(x+1) = 2f(x) + 3$ for all x, and $f(0) = 1$. Find $f(3)$ and explain whether f is linear." The student has practised linear functions, exponential functions and sequences, but none of the practised routines matches the question exactly.

Within two minutes the student writes $f(1) = 5$, $f(2) = 13$, $f(3) = 29$ and then answers "not linear, because the differences 4, 8, 16 are not constant". **What made the question solvable, and what would have happened if the student had searched only for a formula to memorise?**

1. Identify the three ideas the question combines: a recurrence, substitution, and a test for linearity.
2. Explain why "not linear" is justified by the differences, and whether "exponential" would be a better description.
3. Find the general solution $f(n) = 2^{n+2} - 3$ and verify it for $n = 0, 1, 2, 3$.
4. Name one habit from this course that made the unfamiliar question accessible.

This chapter does not teach new content. It asks you to retrieve, connect and apply what you have learned, under conditions that are less predictable than a single chapter's exercises. The final section describes what the Diploma Programme expects — not to promise that any one book is sufficient, but to make the expectations visible.

12.0 ORIENTATION How to use this chapter

Attempt each set without notes first, then mark it against the answer key. A question you cannot start is more useful than one you can: it identifies the exact idea to revisit. Record the chapter and section beside every error, and return to the worked example rather than re-reading the whole chapter.

The sets are cumulative, not arranged by difficulty. Within each set, the early questions test fluency; the later ones require selection, justification or transfer. The final cumulative tasks are criterion-mapped and should be attempted under timed conditions.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

retrieve the core methods of the course without prompts; select an appropriate representation for an unfamiliar problem; write a justification that names the reason for each step; connect algebraic, graphical and geometric representations of the same idea; evaluate a model and its limitations; describe the differences between MYP and DP expectations accurately; identify my own readiness gaps and act on them.

ATL SKILLS · SELF-MANAGEMENT AND REFLECTION

Self-management: use a error log with three columns — question, idea tested, action. Revisit the action after a week; if the same idea fails again, the action was too vague. **Reflection:** a "topic I find hard" is not a plan. "I cannot yet complete the square when the leading coefficient is not 1" is a plan, because it names the procedure and the condition.

RETRIEVAL

12.1 Cumulative practice: algebra and functions

This set draws on Chapters 1–5 and 8. Work without notes for 60 minutes, then mark. For each error, name the section you will revisit.

EXAMPLE 12.1 SELECTING A METHOD

Solve $x^3 - 2x^2 - 5x + 6 = 0$ given that $x = 1$ is a root.

SOLUTION

Use the factor theorem and division. Since $P(1) = 1 - 2 - 5 + 6 = 0$, $(x - 1)$ is a factor. Dividing,

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2).$$

The roots are $x = 1, 3, -2$.

> The given root is a gift: it converts a cubic into a quadratic. Check the quotient by expanding, not by re-dividing.

EXAMPLE 12.2 A FUNCTION AND ITS INVERSE

For $f(x) = \dfrac{3x - 2}{x + 1}$, find $f^{-1}(x)$ and state its domain.

SOLUTION

Let $y = \dfrac{3x - 2}{x + 1}$. Then $y(x + 1) = 3x - 2$, so $xy + y = 3x - 2$ and $x(y - 3) = -2 - y$. Hence

$$x = \frac{-2 - y}{y - 3} = \frac{y + 2}{3 - y}.$$

Therefore $f^{-1}(x) = \dfrac{x + 2}{3 - x}$, defined for $x \neq 3$. The original function has range excluding 3, which matches.

> Check with a point: $f(0) = -2$, and $f^{-1}(-2) = 0$. The check is faster than re-deriving.

REVISION SET 12A

ALGEBRA AND FUNCTIONS

FLUENCY

1 Expand $(2x - 3)(x^2 + x - 4)$.

2 Factorise completely: (a) $x^3 - 4x$; (b) $2x^3 + 3x^2 - 11x - 6$.

3 Divide $2x^3 - 5x^2 + x + 2$ by $x - 2$, stating quotient and remainder.

4 Simplify $\dfrac{x^2 - 9}{x^2 + x - 12}$, stating excluded values.

5 Solve $\dfrac{2}{x - 1} + \dfrac{3}{x + 2} = 1$.

6 Complete the square for $y = 2x^2 - 8x + 3$ and state the vertex.

7 Find the inverse of $f(x) = \dfrac{3x - 2}{x + 1}$ and state its domain.

8 For $f(x) = x^2 + 1$, $x \geq 0$, and $g(x) = \sqrt{x - 3}$, find $f(g(x))$ and its domain.

9 Describe the transformation from $y = 2^x$ to $y = 3 \cdot 2^{x-1} - 4$.

10 Solve $5^{2x-1} = 30$, giving x to 3 significant figures.

11 Solve $\log_3(x + 5) + \log_3(x - 1) = 2$.

12 Solve $e^{2x} - 3e^x - 10 = 0$.

UNDERSTANDING AND REASONING

13 The 7th term of an arithmetic sequence is 23 and the 12th is 38. Find the first term, the common difference and the sum of the first 20 terms.

14 A geometric series has first term 3 and common ratio $\frac{2}{3}$. Find S_4 and S_∞.

15 Evaluate $\displaystyle\sum_{k=1}^{15} (4k - 3)$.

16 Solve $x^2 - 3x - 10 \leq 0$ and illustrate the solution on a number line.

17 Maximise $P = 3x + 2y$ subject to $x + y \leq 6$, $x \leq 4$, $y \geq 1$, $x \geq 0$. List the vertices and the maximum.

18 A population of 5000 grows at 3.2% per year. Find the number of whole years until it exceeds 8000.

REVISION SET 12A · CONTINUED

ALGEBRA AND FUNCTIONS

19 For $y = \dfrac{2x-1}{x+3}$, state the equations of the asymptotes and the intercepts.

20 Show that $x-2$ is a factor of $x^3 - 3x^2 + 4$ and factorise completely.

21 Find k if $x+3$ is a factor of $2x^3 + kx^2 - 7x + 6$.

22 For $f(x) = 2x + 3$ and $g(x) = x^2$, solve $f(g(x)) = g(f(x))$.

APPLICATION AND EXTENSION

23 Sketch $y = (x-1)^2(x+3)$, stating the roots and their multiplicities, the y-intercept and the end behaviour.

24 A rectangular sheet is 20 cm by 14 cm. Squares of side x are cut from the corners and the sides folded up. Write the volume as a polynomial, state the domain, and find the value of x that gives the largest volume from a table of values.

25 **Extension.** Prove that $f(x) = x^3 + x$ is one-to-one, and explain why an inverse exists even though it cannot be written using square roots alone.

RETRIEVAL

12.2 Cumulative practice: geometry and trigonometry

This set draws on Chapters 6–8. Draw a diagram before calculating; label every known length and angle. A correct diagram is part of the solution, not an illustration of it.

EXAMPLE **12.3** CHOOSING BETWEEN THE RULES

In triangle ABC, $AB = 8$ cm, $AC = 5$ cm and angle $BAC = 120^\circ$. Find BC and angle ABC.

SOLUTION

Two sides and the included angle are known, so use the cosine rule:

$$BC^2 = 8^2 + 5^2 - 2(8)(5)\cos 120^\circ = 64 + 25 + 40 = 129,$$

since $\cos 120^\circ = -\tfrac{1}{2}$. Hence $BC = \sqrt{129} = 11.36$ cm (3 s.f.).

For angle ABC, use the sine rule: $\dfrac{\sin B}{5} = \dfrac{\sin 120^\circ}{\sqrt{129}}$, so $\sin B = \dfrac{5\sin 120^\circ}{\sqrt{129}} = 0.3812$, giving $B = 22.4^\circ$ (3 s.f.). The angle is acute, so no ambiguity.

> After the cosine rule, the sine rule is safe for an angle only when the angle is clearly acute or the triangle is otherwise determined. Check the angle sum: $C = 37.6^\circ$.

REVISION SET 12B

GEOMETRY AND TRIGONOMETRY

FLUENCY

1 State the size of the angle at the centre of a circle subtended by an arc, if the angle at the circumference on the same arc is 37°.

2 A cyclic quadrilateral has one angle of 112°. State the opposite angle.

3 A tangent touches a circle of radius 9 cm at T. O is the centre. State the size of angle OTP, where P is on the tangent.

4 Two chords intersect inside a circle. The segments of one chord are 4 cm and 6 cm; one segment of the other is 3 cm. Find the other segment.

5 In triangle ABC, $A = 42^\circ$, $B = 65^\circ$ and $a = 9$ cm. Find b.

6 Find the area of a triangle with sides 7 cm and 9 cm and included angle 50°.

UNDERSTANDING AND REASONING

7 In triangle ABC, $AB = 6$, $BC = 10$ and angle $ABC = 120^\circ$. Find AC.

8 A ship sails 8 km on a bearing of 060°, then 5 km on a bearing of 150°. Find the distance from the start.

9 A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of the space diagonal and the angle it makes with the base.

10 A chord of length 24 cm lies in a circle of radius 13 cm. Find the perpendicular distance from the centre to the chord.

REVISION SET 12B · CONTINUED

GEOMETRY AND TRIGONOMETRY

11 In a circle, P is an external point and PT is a tangent of length 12 cm. A secant from P meets the circle at A and B, with $PA = 8$ cm. Find AB.

12 Explain why the angle in a semicircle is a right angle, using the isosceles triangles formed by two radii.

APPLICATION AND EXTENSION

13 A vertical mast is supported by two wires, each 15 m long, attached to the ground 9 m from the base on opposite sides. Find the angle between the wires at the top.

14 A triangular field has sides 45 m, 60 m and 75 m. Find its area and the largest angle.

15 **Extension.** Prove the alternate segment theorem for a tangent and a chord, using the angle at the centre and the isosceles triangle formed by two radii.

RETRIEVAL

12.3 Cumulative practice: statistics and probability

This set draws on Chapters 9 and 10. State the sample or the sample space before calculating; a probability without a stated space is a guess.

EXAMPLE **12.4** CONDITIONING ON THE RIGHT EVENT

In a school, 60% of students study French, 30% study Spanish, and 15% study both. A student is chosen at random. (a) Find the probability the student studies French or Spanish. (b) Given that the student studies Spanish, find the probability they also study French.

SOLUTION

(a) $P(F \cup S) = P(F) + P(S) - P(F \cap S) = 0.60 + 0.30 - 0.15 = 0.75$.

(b) $P(F \mid S) = \dfrac{P(F \cap S)}{P(S)} = \dfrac{0.15}{0.30} = 0.5$. Given Spanish, the sample space is reduced to the 30% who study Spanish, and half of them also study French.

> The conditional probability is not 0.15: that is the probability of both, before any condition is applied.

REVISION SET 12C

STATISTICS AND PROBABILITY

FLUENCY

1 Find the mean, median and interquartile range of 4, 6, 7, 9, 11, 15, 18.

2 Calculate the standard deviation of 3, 5, 6, 8, 8 (use denominator 5).

3 A data set has lower quartile 12, upper quartile 20. Find the outlier boundaries using the 1.5×IQR rule.

4 A bag contains 4 red and 6 blue counters. Two are taken without replacement. Find the probability both are red.

5 How many ways can 3 students be chosen from 10?

6 A code consists of 4 different letters from 26. How many codes are possible if order matters?

UNDERSTANDING AND REASONING

7 In a class of 30 students, 18 play football, 12 play basketball and 6 play both. Find the probability that a randomly chosen student plays at least one of the two sports.

8 Two events A and B have $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$. Are they independent? Justify.

9 A test is positive for 95% of people with a condition and for 4% of people without it. The condition affects 1 in 500 people. Find the probability a positive result comes from someone with the condition.

10 A scatter graph of study time against test score has $r = 0.82$. Interpret the value and state one thing it does not show.

REVISION SET 12C · CONTINUED STATISTICS AND PROBABILITY

11 A regression line is $y = 42 + 0.8x$, where x is hours of study. Predict y for $x = 10$ and explain why a prediction for $x = 40$ is unreliable.

12 Explain why the mean is pulled toward a long tail while the median is not.

APPLICATION AND EXTENSION

13 The table shows 120 students by year and whether they walk to school.

	Walk	Do not walk	Total
Year 10	24	36	60
Year 11	18	42	60
Total	42	78	120

Find $P(\text{Walk} \mid \text{Year 10})$, $P(\text{Year 10} \mid \text{Walk})$ and state whether year group and walking are independent.

14 In a game, a fair die is rolled twice. You win if at least one roll is a 6. Find the probability of winning, and the fair price of a ticket if the prize is 10 and the game is fair.

15 **Extension.** Explain why a screening test with high sensitivity and specificity can still have a low positive predictive value when the condition is rare, using the numbers in Question 9.

TRANSFER

12.4 Unfamiliar problems and extended reasoning

An unfamiliar problem is not a problem outside the course; it is a problem whose route is not signposted. The strategy is always the same: restate the problem in your own words, identify what is given and what is required, choose a representation, try a small case, look for structure, and justify the step that turns the pattern into a general result.

STRATEGY · WHEN NO ROUTINE APPLIES

1 Restate. What exactly is being asked? **2 Specialise.** Try the smallest example, or a simple value. **3 Represent.** Draw it, tabulate it, or name it algebraically. **4 Conjecture.** State what seems true. **5 Justify.** Prove it, or find a counterexample. **6 Check.** Does the answer satisfy the original conditions and make sense in context?

EXAMPLE 12.5 A PROBLEM WITH TWO REPRESENTATIONS

A sequence is defined by $u_1 = 1$ and $u_{n+1} = 2u_n + 1$. Find a closed form for u_n, and prove that every term is odd.

SOLUTION

Specialise: $u_1 = 1$, $u_2 = 3$, $u_3 = 7$, $u_4 = 15$. The terms are one less than powers of 2: $u_n = 2^n - 1$.

Justification. Suppose $u_n = 2^n - 1$. Then

$$u_{n+1} = 2(2^n - 1) + 1 = 2^{n+1} - 2 + 1 = 2^{n+1} - 1,$$

so the formula passes from n to $n + 1$. Since it holds for $n = 1$, it holds for every positive integer n. This is a proof by induction, although the course has not named it; the logic is a chain of valid steps with a stated starting point.

Oddness. 2^n is even for $n \geq 1$, so $2^n - 1$ is odd. The recurrence also preserves oddness: twice an odd number is even, and adding 1 makes it odd.

> The closed form is a conjecture until the step from n to $n + 1$ is proved. Checking four terms is evidence, not proof.

REVISION SET 12D

UNFAMILIAR PROBLEMS

EXTENDED REASONING

1 A rectangle is inscribed in a circle of radius 5. Its sides are $2x$ and $2y$. Show that $x^2 + y^2 = 25$, and find the dimensions of the rectangle with the largest area.

2 The sum of the first n odd numbers is $1 + 3 + 5 + \cdots + (2n - 1)$. Calculate the sum for $n = 1, 2, 3, 4, 5$, conjecture a formula, and prove it using the pairing method.

3 A closed cylindrical can has surface area 600π cm^2. Its radius is r and height h. Show that $h = \dfrac{300 - r^2}{r}$, write the volume as a function of r, and use a table to estimate the radius that maximises the volume.

4 In a circle, two chords AB and CD intersect at P. Prove that $PA \cdot PB = PC \cdot PD$ by joining A to D and B to C and using similar triangles.

5 A triangle has sides a, b, c and area A. Starting from $A = \frac{1}{2}ab\sin C$, use the cosine rule to derive Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = \dfrac{a+b+c}{2}$. (This is a guided extension; the algebra is substantial.)

6 An investment scheme promises to double your money in 8 years. Find the implied annual rate, and explain why the answer depends on whether interest is compounded annually or continuously.

7 A game charges 2 to play. A fair coin is tossed up to three times; you win 5 if the first head occurs on the third toss, and nothing otherwise. Is the game fair? Justify with expected values.

8 Prove that the product of any three consecutive integers is divisible by 6.

ASSESSMENT

12.5 Cumulative examination

Attempt under timed conditions: 90 minutes. Calculators permitted. Show reasoning for every mark above 1. The tasks are criterion-mapped to the school's Year 5 descriptors; they are not past IB examination papers.

CRITERION A · KNOWING AND UNDERSTANDING

30 minutes · 20 marks

Core methods

1 Factorise $2x^3 + x^2 - 13x + 6$ completely, given that $x = 2$ is a root. [4]

2 Find the inverse of $f(x) = \dfrac{x+3}{2x-1}$ and state its domain. [4]

3 Solve $3^{x+2} = 7^x$, giving x to 3 significant figures. [3]

4 The 5th term of an arithmetic sequence is 17 and the 10th is 37. Find the sum of the first 25 terms. [3]

5 In triangle ABC, $a = 7$, $b = 9$ and $C = 52°$. Find c and the area. [4]

6 The table shows 80 students' choice of subject.

	Arts	Science	Total
Year 10	22	18	40
Year 11	14	26	40
Total	36	44	80

Find $P(\text{Science} \mid \text{Year 11})$ and state whether year and subject are independent. [2]

Evidence: correct factorisation, a valid inverse with restriction, a logarithmic solution, a correct sum, a cosine-rule calculation with area, and a justified independence decision.

CRITERION B · INVESTIGATING PATTERNS 20 minutes · 12 marks

A pattern in the differences

The table gives values of a quadratic function f.

x	0	1	2	3	4	5
$f(x)$	3	6	11	18	27	38

1. Calculate the first and second differences. [3]
2. Explain how the second differences determine the coefficient of x^2. [3]
3. Find the equation of f. [4]
4. Predict $f(10)$ and verify your formula. [2]

Evidence: a complete difference table, a justified coefficient, a correct equation and a verified prediction.

CRITERION C · COMMUNICATING 15 minutes · 8 marks

Explaining a decision

Instructions. A classmate claims that because $r = 0.75$ for hours of revision and test score, "revision causes better scores, and 75% of the score is caused by revision". Write a 200–250 word response that corrects both claims and explains what the correlation does and does not show. Use correct terminology.

Evidence: correct interpretation of r, the distinction between association and causation, the meaning of r^2 (if used) and a clear, coherent correction.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 25 minutes · 12 marks

A community water tank

A community tank holds 20 000 litres and drains at a constant rate. On Monday it is full; consumption over the week is recorded. The remaining volume after d days is modelled by $V(d) = 20\,000 - 1\,800d$.

1. State the daily consumption and the domain for which the model makes sense. [2]
2. Find the day on which the tank first falls below 5 000 litres. [3]
3. The community grows, and consumption rises by 15%. Write the new model and find the new day. [3]
4. State two assumptions of the model and evaluate whether a linear model is appropriate. [4]

Evidence: correct rate and domain, a solved inequality, an updated model, and an evaluation that names assumptions and their effect.

TRANSITION

12.6 From MYP to DP: expectations and readiness

The Diploma Programme is a different assessment system, not a harder version of the same one. MYP criteria describe the quality of thinking across a broad range of performances; DP examinations award marks for specific responses under time pressure. The mathematics also changes: calculus, vectors, complex numbers and statistical distributions become central. This section describes the shift honestly. It does not claim that completing this book guarantees success in any particular DP course; readiness depends on the course, the school and the student's habits.

DIMENSION	MYP YEAR 5	DIPLOMA PROGRAMME
Assessment	Criteria A–D, often with extended tasks and investigations; reasoning is valued alongside the answer.	Written papers with mark schemes; marks are allocated to specific steps; speed and accuracy matter.
Content	Functions, sequences, geometry, trigonometry, statistics and probability, with modelling.	Calculus, further functions, vectors, complex numbers, probability distributions and statistical inference.
Proof	Justification in words and structured arguments.	Formal proof, including induction and contradiction, in specific courses.
Technology	Calculators and graphing tools used to explore and check.	A permitted calculator is specified; algebraic fluency without it is assumed.
Independence	Scaffolded inquiry and reflection.	Independent study, self-marking and the ability to diagnose one's own errors.

COMMAND TERMS · WHAT A QUESTION IS ASKING

State — give a value or fact without explanation. **Describe** — give the features. **Explain** — give reasons. **Justify** — support with evidence or a mathematical argument. **Prove** — a deductive argument from stated assumptions. **Derive** — obtain a result from previous results. **Evaluate** — judge against criteria, including limitations. **Discuss** — a balanced account of more than one position.

Three habits transfer directly. First, write the method, not only the answer: a mark scheme rewards the step, and a reader cannot infer a correct method from a wrong number. Second, check units and reasonableness; a negative time or a probability above 1 signals an error before the marker sees it. Third, learn to work without a prompt: the DP will ask you to choose a method, not to recognise a labelled exercise.

Readiness audit

Rate each statement 1 (not yet), 2 (with support) or 3 (independently). Choose two 1s and write a specific action for each.

- I can expand, factorise and simplify algebraic expressions fluently without a calculator.
- I can solve linear, quadratic, rational, exponential and logarithmic equations.
- I can rearrange a formula to make any variable the subject.
- I can sketch and interpret quadratic, exponential and logarithmic graphs.
- I can work with functions, composites and inverses, including domains.
- I can use sequences and series, including sigma notation and convergence.
- I can solve inequalities and interpret a feasible region.
- I can use circle theorems and write a structured geometric proof.
- I can solve triangles with the sine and cosine rules and work in three dimensions.
- I can use vectors in two and three dimensions and matrices in two dimensions.
- I can calculate and interpret standard deviation, correlation and a regression line.
- I can use conditional probability, independence, trees and counting.
- I can choose a model, state its assumptions and evaluate it.
- I can explain my reasoning in writing with correct notation and units.
- I can diagnose my own errors and design a revision action.

EXAMPLE **12.6** A DP-STYLE TRANSITION TASK

The function f is defined by $f(x) = x^2 - 4x + 7$ for $x \geq 2$. (a) Write f in the form $(x - h)^2 + k$. (b) State the range. (c) Find f^{-1}. (d) Solve $f(x) = f^{-1}(x)$.

SOLUTION

(a) $x^2 - 4x + 7 = (x - 2)^2 + 3$. **(b)** Since $x \geq 2$, the minimum is 3, so the range is $[3, \infty)$.

(c) Let $y = (x - 2)^2 + 3$. Then $(x - 2)^2 = y - 3$ and, since $x \geq 2$, $x - 2 = \sqrt{y - 3}$. Hence $f^{-1}(x) = 2 + \sqrt{x - 3}$ for $x \geq 3$.

(d) Solve $(x - 2)^2 + 3 = x$. Then $x^2 - 4x + 7 = x$, so $x^2 - 5x + 7 = 0$, whose discriminant is $25 - 28 = -3 < 0$. There is no real solution: the graphs do not meet. This is consistent with $f(x) > x$ for all x in the domain.

> Part (d) rewards interpretation as well as algebra: "no real solution" is a result, and explaining why it is plausible strengthens the answer.

REVISION SET 12E

TRANSITION AND REFLECTION

REASONING

1 Distinguish between "explain" and "justify" with an example of each from this book.

2 Explain why a DP mark scheme may award a mark for a correct method even when the final answer is wrong, and why this does not make the answer unimportant.

3 Describe two differences between an MYP investigation and a timed DP examination question.

4 A student says "I understand it in class but cannot do it in the test". Suggest three specific changes to study habits that address this.

5 Why is calculus described as a change of content rather than simply an increase in difficulty?

6 Explain why the statement "this book guarantees DP readiness" would be misleading.

REFLECTION

7 Which chapter of this book changed the way you think about mathematics most? Give a specific example.

8 Name one mathematical habit you have developed and one you still need to develop.

9 What is the most useful question you asked in this course? What made it useful?

10 Write three sentences of advice to a student beginning Year 5 next year.

Chapter summary

- Cumulative practice reveals connections between topics; errors are most useful when traced to a named idea.
- Unfamiliar problems are solved by restating, specialising, representing, conjecturing, justifying and checking.
- A justification names the reason for each step; a calculation alone is not a proof.
- Correlation is not causation; a regression prediction is conditional on the model and the data range.
- The DP changes the assessment system, the content and the level of independence; it is not merely "more of the same".
- Readiness is a set of habits — fluency, method, checking, self-diagnosis — that can be practised, not a certificate conferred by a book.

$$\text{method} + \text{reason} = \text{justification}, \qquad \text{model} + \text{assumptions} + \text{evidence} = \text{defensible conclusion}$$

Revision checklist

Tick each statement only when you can do it without notes and explain the method.

- I can retrieve the core algebra methods without prompts.
- I can select a function model and justify it.
- I can solve equations from every chapter.
- I can draw and use a diagram in geometry and trigonometry.
- I can write a structured geometric proof.
- I can calculate and interpret statistical measures.
- I can condition a probability correctly.
- I can choose a representation for an unfamiliar problem.
- I can justify a generalisation.
- I can evaluate a model's assumptions and limitations.
- I can explain the difference between MYP and DP expectations.
- I can name my own readiness gaps and act on them.

FINAL REFLECTION

Write in your notebook. These answers are for you, not for assessment.

1. Look back at your first piece of work in this book. What can you now do that you could not do then?
2. Which error taught you the most? Describe the error and the correction.
3. The statement of inquiry defines mathematical maturity as choosing, justifying, connecting and recognising limits. Which of these four is your strongest, and which will you work on next?
4. What kind of mathematical question do you want to be able to answer in two years?
5. Write a short paragraph explaining to a younger student why mathematics is more than a collection of methods.

REFERENCE · ANSWERS, REASONING AND ASSESSMENT GUIDANCE

Answers *and guidance*

This section contains the answer to every numbered question in the coursebook, with the reasoning that makes each answer checkable. Steps are shown wherever a final value alone would hide the method: algebraic manipulation, function notation and domains, graph reading, proof structure, units and rounding.

Entries are grouped by chapter and follow the printed order. Question numbers match the printed numbers exactly, including questions that continue across a page. Lettered parts appear in the order in which they are asked.

What the entries contain

Most entries give the result and one or two decisive steps. A note in italics explains a common error, an alternative valid method, or a point of interpretation. For investigations, the entry states the result the pattern should reach. For the capstone and for open assessment tasks, the entry gives marking guidance and a sample response rather than a single "correct" answer.

Using the key responsibly

An answer is not a substitute for a solution. If your value differs, compare the method before changing your work: a sign error, an unrounded intermediate value, a lost domain restriction or a misread scale can produce a plausible but wrong result. Where a question asks you to explain, justify or evaluate, a correct number without reasoning does not meet the criterion.

> **Assessment tasks.** The Criterion A–D tasks are marked against the published MYP criteria, not against a single model answer. The guidance shows the evidence a response should contain and how the marks are usually distributed. A sample response illustrates one satisfactory route; other valid approaches can earn full marks.

Conventions

Unless a question states otherwise, give non-exact answers to three significant figures, use π and e from your calculator, and include units in context. Angles are in degrees unless radians are stated. Probability answers are exact fractions where outcomes are equally likely, and decimals or percentages where the context uses them. Logarithms are natural unless the base is written.

CHAPTER 1 · POLYNOMIALS, FACTOR THEOREM AND ALGEBRAIC FRACTIONS

Opening problem · the box that must hold its shape

1 Height x cm, length $(24-2x)$ cm, width $(18-2x)$ cm: a square of side x is removed from each end of every side.

2 $$V(x)=x(24-2x)(18-2x)=4x^3-84x^2+432x.$$
The degree is 3. The model is physical only while all three dimensions are positive, so $0<x<9$ (at $x=9$ the width has closed to zero).

3 $V(1)=352$, $V(2)=560$, $V(3)=648$, $V(4)=640$, $V(5)=560$, $V(6)=432$, $V(7)=280$, $V(8)=128$, all in cm^3. The largest tabulated value is $648\ \text{cm}^3$, at $x=3$ cm.

A finer table places the peak between $x=3$ and $x=4$, near $x=3.4$ cm, where $V\approx 655\ \text{cm}^3$; the integer table locates the maximum but slightly underestimates it.

4 The formula treats the card as a surface of zero thickness with perfect folds. It ignores the thickness of the card (which makes the internal dimensions smaller than $24-2x$ and $18-2x$), the overlap or glued tab needed to hold each corner together, and the absence of a lid. It also assumes the sides stay vertical and rigid, whereas a very shallow or very tall box sags; and it says nothing about strength or appearance, which a designer must also weigh.

Prerequisite check

1 $(x+4)(x-3)=x^2+x-12$ and $(2x-1)(x+5)=2x^2+9x-5$.

2 $x^2-7x+12=(x-3)(x-4)$ (since $3+4=7$ and $3\cdot 4=12$); $x^2-25=(x-5)(x+5)$, a difference of two squares.

3 $x^2-5x+6=(x-2)(x-3)=0$, so $x=2$ or $x=3$.

4 $\dfrac{6x^3}{2x}=3x^2$; $\dfrac{x^2-4}{x-2}=\dfrac{(x-2)(x+2)}{x-2}=x+2$, for $x\neq 2$.

5 $f(2)=2(4)-3(2)+1=3$ and $f(-1)=2(1)+3+1=6$.

6 $x^3\cdot x^4=x^7$ (add the indices) and $\dfrac{x^5}{x^2}=x^3$ (subtract the indices).

Exercise 1A

1 (a) Degree 4, leading coefficient 3. (b) Degree 3, leading coefficient -1. (c) Degree 5, leading coefficient -2 (writing the polynomial as $-2x^5+8$). (d) Degree $1+1+1=3$, leading coefficient 1 (the product of the three leading coefficients).

2 (a) $x^3-2x^2+3x+5x^2-10x+15=x^3+3x^2-7x+15$.

(b) $2x^3+2x^2-8x-x^2-x+4=2x^3+x^2-9x+4$.

(c) $(x^2-6x+9)(x+1)=x^3+x^2-6x^2-6x+9x+9=x^3-5x^2+3x+9$.

(d) $(x+2)(x-2)=x^2-4$, so $(x^2-4)(x^2+4)=x^4-16$.

3 (a) $x(x^2-4)=x(x-2)(x+2)$.

(b) $x(x^2-9x+20)=x(x-4)(x-5)$.

(c) $2x(x^2-4)=2x(x-2)(x+2)$.

(d) $x^4-16=(x^2-4)(x^2+4)=(x-2)(x+2)(x^2+4)$; the last factor has no real linear factor.

4 $P(0)=(-1)(2)(-4)=8$; $P(1)=(0)(3)(-3)=0$; $P(-2)=(-3)(0)(-6)=0$. The last two confirm that 1 and -2 are roots.

5 (a) Root 5 with multiplicity 2; root -1 with multiplicity 1.

(b) Root 0 with multiplicity 1; root 2 with multiplicity 3.

(c) Root -4 with multiplicity 1; root 1 with multiplicity 2; root 3 with multiplicity 1.

6 (a) Even degree 4, leading coefficient $3>0$: as $x\to\pm\infty$, $y\to+\infty$ (both ends up).

(b) Odd degree 3, leading coefficient $-2<0$: as $x\to+\infty$, $y\to-\infty$ and as $x\to-\infty$, $y\to+\infty$ (falls right, rises left).

(c) Odd degree 5, leading coefficient $1>0$: as $x\to+\infty$, $y\to+\infty$ and as $x\to-\infty$, $y\to-\infty$.

(d) Even degree 2, leading coefficient $-1<0$: as $x\to\pm\infty$, $y\to-\infty$ (both ends down).

7 The roots are -3 (multiplicity 2) and 2 (multiplicity 1). Error 1: the signs are reversed; $(x+3)^2=0$ gives $x=-3$, not 3, and $x-2=0$ gives $x=2$, not -2. A root is the value that makes each factor zero, so the factor $x+3$ points to -3. Error 2: the multiplicity of -3 is 2, not 1, because the factor occurs twice.

Check: the multiplicities must add to the degree. Here $2+1=3$, matching the cubic; the student's $1+1=2$ does not. The graph touches the axis at $x=-3$ and crosses it at $x=2$.

8 An odd-degree polynomial has opposite end behaviour at the two ends: as $x\to+\infty$ it tends to one of $\pm\infty$ and as $x\to-\infty$ it tends to the other. Since every polynomial is continuous, its graph cannot pass from large positive values to large negative values (or the reverse) without meeting the x-axis. That meeting point is a real root, so at least one real root exists. (Complex roots occur in conjugate pairs, which is the same obstruction seen algebraically.)

9 Write $y=a(x+1)^2(x-2)(x-5)$. At $x=0$: $a(1)^2(-2)(-5)=10a=20$, so $a=2$ and
$$y=2(x+1)^2(x-2)(x-5).$$
Check: expanding gives $2x^4-10x^3-6x^2+26x+20$, whose constant term is 20. ✓

10 Roots: -1 (multiplicity 2, so the graph touches and turns), 2 and 4 (multiplicity 1, so it crosses). Degree 4 with leading coefficient 1: both ends go up. y-intercept $y(0)=(1)^2(-2)(-4)=8$, so $(0,8)$ lies on the curve.

Sign by interval: for $x<-1$ the signs are $(+)(-)(-)$, positive; for $-1<x<2$, $(+)(-)(-)$, positive; for $2<x<4$, $(+)(+)(-)$, negative; for $x>4$, positive. So $y>0$ except on $2<x<4$, where $y<0$.

Shape: the curve comes down from $+\infty$, touches at $(-1,0)$, rises to a local maximum near $(0.8,12.4)$ (the y-intercept 8 is passed on the way up), crosses at $(2,0)$ going down, reaches a local minimum near $(3.2,-16.9)$, crosses at $(4,0)$ going up and then rises without limit.

11 Both graphs have the same x-intercepts, the same crossing or touching behaviour at each, and therefore the same factors up to a constant. If the leading coefficients are a_1 and a_2, then $P_2(x)=\dfrac{a_2}{a_1}P_1(x)$: every y-value is multiplied by $k=\dfrac{a_2}{a_1}$, so the y-intercept and all local maximum and minimum values are scaled by k and the graph is stretched or compressed vertically. The shapes are alike in their root structure but not in steepness; and if $k<0$ the second graph is the first reflected in the x-axis, so its end behaviour and its sign on every interval are reversed.

12 Near the origin x^3 is very small compared with x^2: at $x=0.1$, $x^3=0.001$ while $x^2=0.01$. The curve $y=x^3$ therefore approaches the axis almost horizontally — its tangent at the origin is the x-axis — so it looks flat there. Flatness is a statement about the gradient, not about multiplicity. Multiplicity 3 is odd, so the graph does cross the axis, but it crosses with a horizontal tangent (an inflection); multiplicity 2 would instead make it turn back without crossing.

13 $V=(x+1)(x+3)(x-2)=(x^2+4x+3)(x-2)=x^3+2x^2-5x-6\ \text{cm}^3$. All dimensions are positive when $x+1>0$, $x+3>0$ and $x-2>0$; the strongest condition is $x>2$.

14 Crossing at -2 gives the factor $(x+2)$; touching at 1 gives $(x-1)^2$. So $y=a(x+2)(x-1)^2$. At $x=0$: $a(2)(1)=2a=-4$, so $a=-2$ and
$$y=-2(x+2)(x-1)^2=-2x^3+6x-4.$$
Check: $y(3)=-2(5)(4)=-40$ and $-54+18-4=-40$. ✓

15 Write $P(x) = a(x-2)^2(px^2+qx+r)$. The degree is 4, so the remaining factor is a quadratic. If that quadratic had a real root, P would have a further real root; since there are none, px^2+qx+r keeps one sign for all x. Also $(x-2)^2 \geq 0$, with equality only at $x=2$. Hence $P(x)$ has the constant sign of $a \cdot$ (sign of the quadratic) everywhere except at $x=2$, where it is zero, so the graph cannot reach the axis anywhere else and only touches at 2.

If $a>0$ both ends rise and the graph touches $(2,0)$ from above, staying above the axis on either side, like a flattened W with one point of contact. If $a<0$ both ends fall and it touches $(2,0)$ from below, staying below the axis elsewhere. The two shapes are reflections of one another in the x-axis; in each case the turning point at $(2,0)$ is horizontal.

Exercise 1B

1 (a) $(x^2+5x+6) \div (x+2)$: quotient $x+3$, remainder 0, since $(x+2)(x+3) = x^2+5x+6$.

(b) $(x^3-8) \div (x-2)$: quotient x^2+2x+4, remainder 0 (difference of cubes).

(c) $(2x^3+x^2-8x+5) \div (x+3)$: quotient $2x^2-5x+7$, remainder -16. Check: $P(-3) = -54+9+24+5 = -16$. ✓

(d) $(x^4-3x^2+1) \div (x-1)$, written $x^4+0x^3-3x^2+0x+1$: quotient x^3+x^2-2x-2, remainder -1. Check: $P(1) = 1-3+1 = -1$. ✓

2 (a) Coefficients $1, -4, 1, 6$, root 2: bringing down and accumulating gives $1,\ -2,\ -3,\ 0$. Quotient x^2-2x-3, remainder 0, so $x^3-4x^2+x+6 = (x-2)(x-3)(x+1)$.

(b) Coefficients $2, 3, -11, -6$, root -3: $2,\ -3,\ -2,\ 0$. Quotient $2x^2-3x-2$, remainder 0, so the polynomial is $(x+3)(2x+1)(x-2)$.

(c) Coefficients $1, 0, 0, 0, -1$, root -2: $1,\ -2,\ 4,\ -8,\ 15$. Quotient x^3-2x^2+4x-8, remainder 15.

(d) Coefficients $3, 0, -7, 4$, root 1: $3,\ 3,\ -4,\ 0$. Quotient $3x^2+3x-4$, remainder 0.

Common error: for (b) the divisor $x+3$ is $x-(-3)$, so the number used is -3, not 3.

3 (a)
$$x^3+1 = (x+1)(x^2-x+1)+0.$$
(b) $x^3+1 = (x-1)(x^2+x+1)+2$; here $P(1)=2$, which is the remainder.

(c) Dividing by $2x+1$ means using the root $-\frac{1}{2}$:
$$6x^3-5x^2+4 = (2x+1)(3x^2-4x+2)+2.$$
Check: $(2x+1)(3x^2-4x+2) = 6x^3-5x^2+2$, plus 2 gives $6x^3-5x^2+4$. ✓

4 Substituting checks the remainder but not the quotient: $P(1) = 1-2+4 = 3$, so remainder 3 is correct. But $(x-1)(x^2-x-1)+3 = x^3-2x^2+4$, which is not P — an x^2 term has appeared. The student dropped the zero placeholder for x^2. Dividing x^3+0x^2-2x+4 by $x-1$ gives quotient x^2+x-1 and remainder 3, so
$$x^3-2x+4 = (x-1)(x^2+x-1)+3.$$
Check: $(x-1)(x^2+x-1)+3 = x^3-2x+1+3 = x^3-2x+4$. ✓

5 By the factor theorem, $x-4$ is a factor only if $P(4)=0$. But $P(4) = 64+2(16)-12+5 = 64+32-12+5 = 89 \neq 0$, so $x-4$ is not a factor; the remainder would be 89.

6 $P(x) = (x-2)(x^2+3x-1)+5 = x^3+3x^2-x-2x^2-6x+2+5 = x^3+x^2-7x+7$. Then $P(2) = 8+4-14+7 = 5$, which equals the remainder, as the remainder theorem requires.

7 In the division identity the remainder must satisfy $\deg R < \deg D$. A linear divisor has degree 1, so $\deg R < 1$, and the only polynomials of degree less than 1 are the constants. Hence the remainder is a single number, $P(a)$, possibly zero.

8 The factor theorem gives $P(3)=0$: $27+3k+6=0$, so $3k=-33$ and $k=-11$. Check: $x^3-11x+6$ at $x=3$ is $27-33+6=0$. ✓

9 $P(1) = (0)(1+4+c)+7 = 7$ for every c, so the first condition gives no information — it is automatically satisfied by the stated form. The second condition: $P(2) = (1)(4+8+c)+7 = 17$, so $12+c=10$ and $c=-2$. Check: $P(2) = 4+8-2+7 = 17$. ✓

10 Synthetic division by 2: coefficients $1,\ 4,\ -7,\ -22,\ 24$. Bring down 1; $1\times 2 = 2$, $4+2=6$; $6\times 2 = 12$, $-7+12=5$; $5\times 2 = 10$, $-22+10=-12$; $-12\times 2 = -24$, $24-24=0$. The quotient is $x^3+6x^2+5x-12$ with remainder 0, so $x-2$ is a factor.

Factor the quotient: $x=1$ gives $1+6+5-12=0$, so $x-1$ is a factor, and $x^3+6x^2+5x-12 = (x-1)(x^2+7x+12) = (x-1)(x+3)(x+4)$. Hence
$$P(x) = (x-2)(x-1)(x+3)(x+4),$$
with roots $2,\ 1,\ -3,\ -4$.

Check by expansion: $(x-2)(x-1)(x+3)(x+4) = x^4+4x^3-7x^2-22x+24$, and the four roots sum to -4, matching $-b/a$.

11 Substituting the three conditions: $1+a+b+c=0$, $8+4a+2b+c=6$ and $-1+a-b+c=0$. Subtracting the third from the first gives $2b=-2$, so $b=-1$; then $a+c=0$, so $c=-a$. The second condition becomes $8+4a-2-a=6$, that is $3a=0$, so $a=0$ and $c=0$. Hence $a=0$, $b=-1$, $c=0$ and $P(x) = x^3-x = x(x-1)(x+1)$. Check: $P(1)=0$, $P(2)=8-2=6$, $P(-1)=-1+1=0$. ✓

Exercise 1C

1 (a) Root 1: $P(1) = 1-2+5 = 4$.

(b) Root -2: $P(-2) = -16-2-7 = -25$.

(c) Root 2: $P(2) = 16-6+1 = 11$.

(d) The divisor $2x-1$ has root $\frac{1}{2}$: $P(\frac{1}{2}) = \frac{5}{8}-\frac{1}{2}+\frac{1}{2}-4 = \frac{5}{8}-4 = -\frac{27}{8}$, that is -3.375.

Common error: using 1 instead of -1 for the divisor $x+2 = x-(-2)$ in part (b).

2 (a) $P(3) = 27-18-15+6 = 0$, so $x-3$ is a factor (and $P = (x-3)(x^2+x-2)$).

(b) $P(-1) = -2+3+2-3 = 0$, so $x+1$ is a factor.

(c) $P(2) = 16-16 = 0$, so $x-2$ is a factor.

(d) $P(-3) = -27+27 = 0$, so $x+3$ is a factor.

3 (a) $P(2) = 8-4-28+24 = 0$, so $x-2$ is a factor. Dividing gives $x^2+x-12 = (x+4)(x-3)$, hence $x^3-x^2-14x+24 = (x-2)(x-3)(x+4)$.

(b) $P(2) = 8+8-10-6 = 0$. Dividing by $x-2$ gives $x^2+4x+3 = (x+1)(x+3)$, so the factorisation is $(x-2)(x+1)(x+3)$.

(c) $P(-1) = -2-5+4+3 = 0$. Dividing by $x+1$ gives $2x^2-7x+3 = (2x-1)(x-3)$, so the factorisation is $(x+1)(2x-1)(x-3)$. Check $P(\frac{1}{2}) = \frac{1}{4}-\frac{5}{4}-2+3 = 0$. ✓

(d) $P(1) = 1-6+11-6 = 0$. Dividing by $x-1$ gives $x^2-5x+6 = (x-2)(x-3)$, so the factorisation is $(x-1)(x-2)(x-3)$.

4 $P(1) = 1+1-1-2 = -1$, not 0. Since the remainder is -1, $x-1$ is not a factor, and the student's conclusion is false. The error is arithmetic: $1+1-1-2=-1$. The correction is to state that the remainder on division by $x-1$ is -1, so $x-1$ does not divide x^4+x^3-x-2. (Testing shows no small integer root: $P(-1)=-1$, $P(2)=20$, $P(-2)=8$.)

5 $P(2) = 8+4k-6+2 = 4k+4 = 0$, so $k=-1$. Check: x^3-x^2-3x+2 at $x=2$ is $8-4-6+2=0$. ✓

6 From the remainder at $x=1$: $1+a-4+b=2$, so $a+b=5$. From the remainder at $x=-2$: $-8+4a+8+b=-4$, so $4a+b=-4$. Subtracting: $3a=-9$, so $a=-3$ and $b=8$. Then $P(x) = x^3-3x^2-4x+8$. Check: $P(1) = 1-3-4+8 = 2$ and $P(-2) = -8-12+8+8 = -4$. ✓

7 From the division identity with $D(x) = x-a$, $P(x) = (x-a)Q(x)+P(a)$. Because $P(a)=0$ exactly when the remainder vanishes, the statements "$x-a$ is a factor of P", "a is a root of P" and "a is a solution of $P(x)=0$" are three descriptions of one condition. The factor theorem is the bridge: each linear factor of P names a solution of the equation, and each solution names a factor. That is why factorising solves polynomial equations completely.

8 $P(x) = 2(x-1)(x-2)(x+3)$. Now $(x-1)(x-2) = x^2-3x+2$, and $(x^2-3x+2)(x+3) = x^3+3x^2-3x^2-9x+2x+6 = x^3-7x+6$, so
$$P(x) = 2x^3-14x+12.$$
Verifying the factor theorem at $x=2$: $P(2) = 16-28+12 = 0$. ✓

9 For $P(x) = x^n-1$, $P(1) = 1^n-1 = 0$ for every positive integer n, so by the factor theorem $x-1$ is always a factor; in fact $x^n-1 = (x-1)(x^{n-1}+x^{n-2}+\cdots+x+1)$. For $x+1$, $P(-1) = (-1)^n-1$. This is 0 only when $(-1)^n = 1$, that is when n is even; when n is odd, $P(-1) = -2 \neq 0$ and $x+1$ is not a factor

10 Dividing $x^3+6x^2+11x+6$ by $x+1$ (root -1) gives quotient $x^2+5x+6=(x+2)(x+3)$, since $V(-1)=-1+6-11+6=0$. The two remaining dimensions differ by 1, so the length is $(x+3)$ cm and the width $(x+2)$ cm. Check: $(x+1)(x+2)(x+3)=x^3+6x^2+11x+6$. ✓

11 Because P is monic with roots 1 and -2, write $P(x)=(x-1)(x+2)(x-r)$. Then $P(0)=(-1)(2)(-r)=2r=-8$, so $r=-4$. Hence

$$P(x)=(x-1)(x+2)(x+4)=x^3+5x^2+2x-8,$$

and the third root is -4.

Check: $P(1)=0$, $P(-2)=0$, $P(0)=-8$. Without the word "monic" the data would only force $2ar=-8$, leaving a family of cubics; the leading coefficient is what makes the answer unique.

12 If $x-a$ divides both, then $P(a)=0$ and $Q(a)=0$, so $(P+Q)(a)=P(a)+Q(a)=0$; by the factor theorem $x-a$ divides $P+Q$. The converse fails: take $a=0$, $P(x)=x+1$ and $Q(x)=x-1$. Then $P+Q=2x$ is divisible by x, but $P(0)=1$ and $Q(0)=-1$, so x divides neither. Divisibility of a sum therefore does not pass back to the summands.

Exercise 1D

1 (a) $\dfrac{6x^2}{9x}=\dfrac{2x}{3}$, $x\neq 0$.

(b) $\dfrac{(x-4)(x+4)}{x+4}=x-4$, $x\neq -4$.

(c) $\dfrac{x(x+2)}{(x+2)(x+3)}=\dfrac{x}{x+3}$, $x\neq -2,\ -3$.

(d) $\dfrac{2(x-2)(x+2)}{x(x-2)}=\dfrac{2(x+2)}{x}$, $x\neq 0,\ 2$.

Common error: dropping $x=2$ in (d) because the factor cancelled — the restriction belongs to the original denominator.

2 (a) $\dfrac{(x+2)+x}{x(x+2)}=\dfrac{2x+2}{x(x+2)}=\dfrac{2(x+1)}{x(x+2)}$, $x\neq 0,\ -2$.

(b) $\dfrac{3x-(x-1)}{x(x-1)}=\dfrac{2x+1}{x(x-1)}$, $x\neq 0,\ 1$.

(c) $\dfrac{2(x-2)+5(x+3)}{(x+3)(x-2)}=\dfrac{7x+11}{(x+3)(x-2)}$, $x\neq -3,\ 2$.

(d) $\dfrac{x(x+1)-2(x-4)}{(x-4)(x+1)}=\dfrac{x^2-x+8}{(x-4)(x+1)}$, $x\neq 4,\ -1$.

3 (a) $\dfrac{(x-1)(x+1)(x-2)(x+2)}{(x+2)(x-1)}=(x+1)(x-2)=x^2-x-2$, $x\neq -2,\ 1$.

(b) $\dfrac{x(x+3)}{(x-3)(x+3)}\cdot\dfrac{x-3}{x}=1$, $x\neq 0,\ 3,\ -3$.

(c) $\dfrac{4x^2(x-1)}{(x-1)(x+1)(2x)}=\dfrac{2x}{x+1}$, $x\neq 0,\ 1,\ -1$.

(d) $\dfrac{(x-2)(x-3)}{(x-2)(x+2)}\cdot\dfrac{x+2}{x-3}=1$, $x\neq 2,\ -2,\ 3$.

4 The simplification is correct but incomplete: $\dfrac{x^2+5x}{x}=\dfrac{x(x+5)}{x}=x+5$ only where the original is defined, so $x\neq 0$ must be stated. At $x=0$ the original is $\frac{0}{0}$, which is undefined, while the simplified expression would give 5; the two expressions differ at that single point, where the graph has a hole at $(0,5)$. Saying "defined for all x" would wrongly claim a value the original never takes.

5 For $b,d\neq 0$, multiplying a fraction by $\dfrac{d}{d}=1$ (or by $\dfrac{b}{b}=1$) does not change its value, so $\dfrac{a}{b}=\dfrac{ad}{bd}$ and $\dfrac{c}{d}=\dfrac{bc}{bd}$. The two fractions now have the common denominator bd, and adding fractions with equal denominators adds numerators: $\dfrac{ad}{bd}+\dfrac{bc}{bd}=\dfrac{ad+bc}{bd}$. Each step is reversible, so the identity holds for every value at which $b\neq 0$ and $d\neq 0$.

6 $\dfrac{x^2-1}{x-1}=\dfrac{(x-1)(x+1)}{x-1}=x+1$ whenever $x-1\neq 0$, that is $x\neq 1$. At $x=1$ the original is $\frac{0}{0}$, which is undefined; the graph is the line $y=x+1$ with a hole at $(1,2)$. The simplified expression takes the value 2 there, but the original never does.

7 $\dfrac{1}{x-1}-\dfrac{1}{x+1}=\dfrac{(x+1)-(x-1)}{(x-1)(x+1)}=\dfrac{2}{x^2-1}$. At $x=0$ this is $\dfrac{2}{-1}=-2$, and the original gives $\dfrac{1}{-1}-\dfrac{1}{1}=-1-1=-2$. At $x=10$ it is $\dfrac{2}{99}\approx 0.0202$, and the original gives $\dfrac{1}{9}-\dfrac{1}{11}=\dfrac{11-9}{99}=\dfrac{2}{99}$. The values agree because the two expressions

8 Write $\dfrac{2}{(x-1)(x+1)}=\dfrac{A}{x-1}+\dfrac{B}{x+1}=\dfrac{A(x+1)+B(x-1)}{(x-1)(x+1)}$. Comparing numerators: $A(x+1)+B(x-1)=2$ for all x, so $A+B=0$ (coefficient of x) and $A-B=2$ (constant). Hence $A=1$, $B=-1$ and

$$\frac{2}{x^2-1}=\frac{1}{x-1}-\frac{1}{x+1},\qquad x\neq\pm 1.$$

Check at $x=0$: the right side gives $\dfrac{1}{-1}-\dfrac{1}{1}=-1-1=-2$, and the left side gives $\dfrac{2}{0-1}=-2$. ✓

9 Area $=\dfrac{x+3}{x-2}\cdot\dfrac{x-2}{x+1}=\dfrac{x+3}{x+1}$ square units, with $x\neq 2$ and $x\neq -1$. For a genuine rectangle with positive sides we also need $x>2$, which makes every side positive.

10 $\dfrac{1}{R}=\dfrac{1}{R_1}+\dfrac{1}{R_2}=\dfrac{R_2+R_1}{R_1R_2}$, so taking reciprocals gives $R=\dfrac{R_1R_2}{R_1+R_2}$ (with $R_1,R_2>0$, so $R_1+R_2\neq 0$). Check: for $R_1=R_2=4$, the formula gives $R=2$, and $\frac{1}{4}+\frac{1}{4}=\frac{1}{2}=\frac{1}{R}$. ✓

11 The common denominator is x^3: $\dfrac{1}{x}+\dfrac{2}{x^2}+\dfrac{3}{x^3}=\dfrac{x^2+2x+3}{x^3}$. The single fraction is undefined exactly when $x^3=0$, that is $x=0$, and for every other x both numerator and denominator are finite with a non-zero denominator, so the value exists. Each separate term also requires $x\neq 0$, so combining them has not removed the restriction. Note also that $x^2+2x+3=(x+1)^2+2>0$, so the numerator never vanishes and the expression has no zero.

Exercise 1E

1 (a) $3(x+2)=6x$, so $3x+6=6x$ and $x=2$. Check: $\dfrac{3}{2}=\dfrac{6}{4}=1.5$. ✓

(b) $5=2(x-1)$, so $x-1=2.5$ and $x=3.5$. Check: $\dfrac{5}{2.5}=2$. ✓

(c) $x=3(x+2)$, so $-2x=6$ and $x=-3$. Check: $\dfrac{-3}{-1}=3$. ✓

(d) $4(x+1)=6(x-3)$, so $4x+4=6x-18$, $2x=22$ and $x=11$. Check: $\dfrac{4}{8}=\dfrac{6}{12}=0.5$. ✓

2 (a) Multiply by $x(x+1)$: $(x+1)+x=x(x+1)$, so $2x+1=x^2+x$ and $x^2-x-1=0$. Hence $x=\dfrac{1\pm\sqrt{5}}{2}$, that is $x\approx 1.618$ or $x\approx -0.618$; neither is 0 or -1. Check for 1.618: $0.618+0.382=1$. ✓

(b) Multiply by $(x-1)(x+1)$: $2(x+1)-(x-1)=x^2-1$, so $x+3=x^2-1$ and $x^2-x-4=0$. Hence $x=\dfrac{1\pm\sqrt{17}}{2}\approx 2.562$ or -1.562, and neither is ± 1. Check for 2.562: $1.281-0.281=1$. ✓

(c) Multiply by $(x-2)(x+2)$: $x(x+2)-4=0$, so $x^2+2x-4=0$ and $x=-1\pm\sqrt{5}\approx 1.236$ or -3.236; neither is ± 2. Check for 1.236: $\dfrac{1.236}{-0.764}=-1.618$ and $\dfrac{4}{1.528-4}=-1.618$. ✓

(d) Multiply by $(x+2)(x-1)$: $3(x-1)+2(x+2)=2(x+2)(x-1)$, so $5x+1=2x^2+2x-4$, giving $2x^2-3x-5=0$, that is $(2x-5)(x+1)=0$. So $x=2.5$ or $x=-1$. Check $x=2.5$: $\dfrac{3}{4.5}+\dfrac{2}{1.5}=2$. ✓

3 (a) No solution. Multiplying by $x-3$ gives $x=3$, but $x=3$ makes both denominators zero, so the value is extraneous and the solution set is empty.

(b) One solution. The left side is $\dfrac{(x+2)+(x-2)}{(x-2)(x+2)}=\dfrac{2x}{x^2-4}$, so $\dfrac{2x}{x^2-4}=\dfrac{3}{x^2-4}$ gives $2x=3$, $x=\dfrac{3}{2}$. Check: $\dfrac{3}{2}\neq\pm 2$, so the denominator is non-zero and the solution stands.

(c) Two solutions. Multiply by $x(x-2)$: $3(x-2)+x=x(x-2)$, so $4x-6=x^2-2x$ and $x^2-6x+6=0$. Then $x=\dfrac{6\pm\sqrt{36-24}}{2}=3\pm\sqrt{3}$, giving $x=4.73$ or $x=1.27$ (2 d.p.). Neither is 0 or 2, so both are valid.

Common error: accepting the algebraic result without testing it against the excluded values. In (a) the only candidate is excluded; in (c) both survive.

4 The rejection is wrong. The restriction is $x\neq 1$, and the solution found is $x=1.5$, which is not 1, so nothing excludes it. Substituting: $\dfrac{1}{1.5-1}=\dfrac{1}{0.5}=2$. ✓ The correct solution is $x=\frac{3}{2}$; the student seems to have rejected the answer merely because a restriction had been written down.

5 If $x = 2$ is substituted into the multiplied equation, both sides were multiplied by $x - 2 = 0$, so both become 0 and the new equation is satisfied whatever the original fractions were; the step has produced a true statement from possibly false ones. Equivalently, multiplying by zero is not reversible, so the new equation can have solutions the old one lacked. For example $\frac{1}{x-2} = \frac{3}{x-2}$ becomes $1 = 3$, while $\frac{1}{x-2} = \frac{x-1}{x-2}$ becomes $1 = x - 1$, giving $x = 2$, which must be rejected because the original fractions are undefined there. This is why every candidate must be tested against the original denominators.

6 $\frac{1}{x+3} = \frac{1}{2}$, so $x + 3 = 2$ and $x = -1$. Check: $\frac{1}{-1+3} = \frac{1}{2}$. ✓

7 Let the integer be n, so the next integer is $n + 1$: $\frac{1}{n} + \frac{2}{n+1} = \frac{7}{6}$. Multiplying by $6n(n+1)$ gives $6(n+1) + 12n = 7n(n+1)$, so $18n + 6 = 7n^2 + 7n$ and $7n^2 - 11n - 6 = 0$. Then $(7n + 3)(n - 2) = 0$, giving $n = 2$ or $n = -\frac{3}{7}$. Only $n = 2$ is a positive integer. Check: $\frac{1}{2} + \frac{2}{3} = \frac{3}{6} + \frac{4}{6} = \frac{7}{6}$.

8 $\frac{1}{x-a} - \frac{1}{x+a} = \frac{(x+a)-(x-a)}{(x-a)(x+a)} = \frac{2a}{x^2-a^2}$. With $a = 3$ the left side of the equation equals $\frac{6}{x^2-9}$, so the equation reads $\frac{6}{x^2-9} = \frac{3}{x^2-9}$. For $x \neq \pm 3$ this requires $6 = 3$, which is impossible; so the equation has no solution (at $x = \pm 3$ it is undefined).

9 Let the speed in still water be v km/h; downstream the speed is $v + 3$ and upstream $v - 3$. Equal times give $\frac{30}{v+3} = \frac{18}{v-3}$, so $30(v - 3) = 18(v + 3)$, $30v - 90 = 18v + 54$, $12v = 144$ and $v = 12$. Check: downstream $\frac{30}{15} = 2$ h, upstream $\frac{18}{9} = 2$ h. ✓ The boat travels at 12 km/h in still water.

10 Let the faster worker take x hours alone; the slower takes $x + 6$. In one hour they complete $\frac{1}{x}$ and $\frac{1}{x+6}$ of the job, and together $\frac{1}{4}$: $\frac{1}{x} + \frac{1}{x+6} = \frac{1}{4}$. Multiplying by $4x(x + 6)$ gives $4(x + 6) + 4x = x(x + 6)$, so $8x + 24 = x^2 + 6x$ and $x^2 - 2x - 24 = 0$, that is $(x - 6)(x + 4) = 0$. Hence $x = 6$ (reject -4, since a time must be positive). One worker takes 6 h, the other 12 h. Check: $\frac{1}{6} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$. ✓

11 Let the width be w metres, so the length is $\frac{60}{w}$. After the change the dimensions are $\frac{60}{w} + 2$ and $w - 1$, and the area is unchanged:

$$\left(\frac{60}{w} + 2\right)(w - 1) = 60, \qquad w > 1.$$

Expanding: $60 - \frac{60}{w} + 2w - 2 = 60$, so $2w - 2 - \frac{60}{w} = 0$. Multiplying by $w > 0$: $2w^2 - 2w - 60 = 0$, that is $w^2 - w - 30 = 0$, so $(w - 6)(w + 5) = 0$ and $w = 6$ (reject -5). The garden is 10 m by 6 m. Check: $12 \times 5 = 60$ m^2. ✓

Investigation 1 · sign changes and multiplicity

1 Root $x = 1$ with multiplicity 2, root $x = -2$ with multiplicity 1. Expanding: $(x^2 - 2x + 1)(x + 2) = x^3 + 2x^2 - 2x^2 - 4x + x + 2 = x^3 - 3x + 2$, of degree $1 + 2 = 3$. ✓

2 At $x = -3$: $P(-3) = (-4)^2(-1) = -16$, negative. At $x = 0$: $P(0) = (1)^2(2) = 2$, positive. At $x = 2$: $P(2) = (1)^2(4) = 4$, positive. So $P < 0$ on $x < -2$, $P > 0$ on $-2 < x < 1$, and $P > 0$ on $x > 1$.

3 Cubic with leading coefficient 1: the graph rises from $-\infty$ on the left, crosses at $(-2, 0)$, reaches a local maximum at $(-1, 4)$, comes down to touch the axis at $(1, 0)$ (a local minimum), and then rises to $+\infty$. The y-intercept is $(0, 2)$. Check: $P(-1) = 4$ and $P(1) = 0$. ✓

4 Signs for Q: at $x = -3$, $(-4)^3(-1) = 64$, positive; at $x = 0$, $(-1)^3(2) = -2$, negative; at $x = 2$, $(1)^3(4) = 4$, positive. So $Q > 0$ for $x < -2$, $Q < 0$ for $-2 < x < 1$, $Q > 0$ for $x > 1$. Expanding gives $x^4 - x^3 - 3x^2 + 5x - 2$, degree 4 with positive leading coefficient, so both ends rise; the graph crosses at $(-2, 0)$ going down, has a y-intercept $(0, -2)$, and at $x = 1$ it now crosses with a horizontal tangent (an inflection) instead of touching. What changes at $x = 1$ is the sign: for P the graph stays positive on both sides, for Q it passes from negative to positive.

5 On $0.9 < x < 1.1$ the factor $x + 2 \approx 2.9$ is positive, so each sign is decided by the power of $x - 1$. For P, $(x - 1)^2$ is positive on both sides of 1 (and zero only at 1), so $P > 0$ throughout the interval and the graph stays above the axis. For Q, $(x - 1)^3 < 0$ when $x < 1$ and $(x - 1)^3 > 0$ when $x > 1$, so Q changes from negative to positive and the graph crosses. An even power erases the change of sign; an odd power preserves it.

6 For x just below r, $x - r < 0$, so $(x - r)^m$ has sign $(-1)^m$; just above r it is positive. Hence the factor changes sign at r exactly when m is odd, and keeps its sign when m is even. The other factors do not vanish near r, so they keep a fixed sign and cannot alter the conclusion. Therefore P changes sign at a root if and only if that root has odd multiplicity.

7 R has root -3 (multiplicity 1, crossing), root 2 (multiplicity 4, touching) and root 5 (multiplicity 2, touching). The degree is $1 + 4 + 2 = 7$, odd, and the leading coefficient is -1, so $R \to +\infty$ as $x \to -\infty$ and $R \to -\infty$ as $x \to +\infty$. Signs: for $x < -3$ all three factors except $(x + 3)$ are positive and $(x + 3) < 0$, so $R > 0$; for $-3 < x < 2$, $R < 0$; for $2 < x < 5$, $R < 0$; for $x > 5$, $R < 0$. Check at $x = -4$: $-(-1)(1296)(81) = 104\,976 > 0$; at $x = 0$: $-(3)(16)(25) = -1200 < 0$. ✓ The sign changes only at $x = -3$.

Generalise. The number of sign changes across the whole real line equals the number of distinct roots with odd multiplicity (counting each such root once). An even-multiplicity root contributes $(x - r)^m \geq 0$ on both sides, so the sign is unchanged there, while an odd-multiplicity root multiplies the sign by -1 as x passes it.

Modelling task · the open box, revisited

1 $V(x) = x(24 - 2x)(18 - 2x)$. Expanding $(24 - 2x)(18 - 2x) = 432 - 84x + 4x^2$, so

$$V(x) = 4x^3 - 84x^2 + 432x, \qquad 0 < x < 9.$$

The lower bound keeps the height positive and the upper bound keeps the width $18 - 2x$ positive.

2 $V(1) = 352.0$, $V(2) = 560.0$, $V(3) = 648.0$, $V(4) = 640.0$, $V(5) = 560.0$, $V(6) = 432.0$, $V(7) = 280.0$, $V(8) = 128.0$ (all cm^3 to 1 d.p.). The maximum in the table is 648.0 cm^3 at $x = 3.0$ cm.

3 The polynomial gives $V(0) = 0$ and $V(9) = 0$, because in each case one dimension has collapsed: at $x = 0$ nothing is cut and the card stays flat with no height, and at $x = 9$ the two cuts meet so the width $18 - 2x$ is zero. Both are boundary values of the model rather than boxes — there is no enclosed volume to hold anything, and the fold lines coincide.

4 Assumption 1: the card has negligible thickness, so the inside of the box measures $24 - 2x$ by $18 - 2x$ by x. In a workshop the card is perhaps 2 mm thick, so the internal dimensions are smaller, the real volume is slightly less than the formula predicts and the best x is marginally smaller. Assumption 2: the sides are cut and folded exactly, with no material used for gluing, and the box is open with vertical, rigid sides. If a glued tab or an overlapping corner is needed, or if the box must carry weight, extra material is consumed or the walls need to be thicker relative to the base; the usable height is then less than x, which again pushes the recommended size down. Neither assumption changes the shape of the model, but both shift its optimum.

Mixed practice 1

1 $(x - 2)(x^2 + 3x - 1) = x^3 + 3x^2 - x - 2x^2 - 6x + 2 = x^3 + x^2 - 7x + 2$, of degree 3 (predicted $1 + 2 = 3$).

2 (a) $x(x^2 - 7x + 12) = x(x - 3)(x - 4)$.

(b) $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, a sum of cubes.

(c) Group: $x^2(2x - 1) - 4(2x - 1) = (2x - 1)(x^2 - 4) = (2x - 1)(x - 2)(x + 2)$.

3 Root -2: $P(-2) = 16 + 16 - 2 - 5 = 25$. The remainder is 25.

4 $P(2) = 8 - 12 - 20 + 24 = 0$, so $x - 2$ is a factor. Synthetic division by 2 gives quotient $x^2 - x - 12 = (x - 4)(x + 3)$, so $P(x) = (x - 2)(x - 4)(x + 3)$. Check at $x = 4$: $64 - 48 - 40 + 24 = 0$. ✓

5 $\dfrac{(x-5)(x+3)}{(x-3)(x+3)} = \dfrac{x-5}{x-3}$, with $x \neq 3$ and $x \neq -3$.

6 $\dfrac{4(x+2)+3(x-2)}{(x-2)(x+2)} = \dfrac{4x+8+3x-6}{x^2-4} = \dfrac{7x+2}{x^2-4}$, $x \neq \pm 2$.

7 $3(x-2) = 2(x+1)$, so $3x - 6 = 2x + 2$ and $x = 8$. Check: $\dfrac{3}{9} = \dfrac{2}{6} = \dfrac{1}{3}$. ✓

8 Divide by $x - 3$: quotient $x^2 - x - 2 = (x - 2)(x + 1)$, so the roots are $3,\ 2,\ -1$. Check $x = -1$: $-1 - 4 - 1 + 6 = 0$. ✓

9 The graph crosses the axis at $x = -1$, because the factor $(x + 1)$ has odd multiplicity 1 so the sign changes there; it touches the axis at $x = 3$, because $(x - 3)^2$ is even and the sign does not change. The y-intercept is $y(0) = (1)(9) = 9$.

10 $P(-2) = -16 + 4k + 10 + 6 = 4k = 0$, so $k = 0$. Check: $2x^3 - 5x + 6$ at $x = -2$ is $-16 + 10 + 6 = 0$. ✓

11 $\dfrac{(x-1)(x+1)}{(x+1)^2} \cdot \dfrac{x+1}{x-1} = 1$ for $x \neq -1,\ 1$. Every factor cancels: two copies of $(x + 1)$ and one of $(x - 1)$ appear in both numerator and denominator.

12 A quotient is a number only when its denominator is non-zero. At $x = 0$ the denominator is 0, so the expression has no value there, whatever the numerator is; the graph has a vertical asymptote at $x = 0$ and no point on it. The numerator being always positive shows that the fraction never takes the value 0, but it says nothing about where the fraction is defined.

13 $y = a(x + 1)(x - 2)(x - 4)$. At $x = 0$: $a(1)(-2)(-4) = 8a = -16$, so $a = -2$. Now $(x - 2)(x - 4) = x^2 - 6x + 8$ and $(x^2 - 6x + 8)(x + 1) = x^3 - 5x^2 + 2x + 8$, so

$$y = -2x^3 + 10x^2 - 4x - 16.$$

Check: $y(0) = -16$ and $y(2) = -16 + 40 - 8 - 16 = 0$. ✓

14 $P(1) = 1 + a + b - 6 = 0$ gives $a + b = 5$. $P(-2) = -8 + 4a - 2b - 6 = 0$ gives $4a - 2b = 14$, that is $2a - b = 7$. Adding: $3a = 12$, so $a = 4$ and $b = 1$, and $P(x) = x^3 + 4x^2 + x - 6$. Dividing by $(x - 1)(x + 2) = x^2 + x - 2$ gives quotient $x + 3$, so the third factor is $x + 3$. Check: $(x - 1)(x + 2)(x + 3)$ has constant term -6. ✓

15 Since $x^2 + x - 2 = (x - 1)(x + 2)$, multiply by that: $2(x + 2) - 3(x - 1) = 1$, so $2x + 4 - 3x + 3 = 1$, $7 - x = 1$ and $x = 6$. Check: $\dfrac{2}{5} - \dfrac{3}{8} = 0.4 - 0.375 = 0.025$ and $\dfrac{1}{36 + 6 - 2} = \dfrac{1}{40} = 0.025$. ✓

16 Let the width be w; then the length is $14 - w$ (perimeter $2(l + w) = 28$). Area: $w(14 - w) = 45$, so $w^2 - 14w + 45 = 0$, that is $(w - 5)(w - 9) = 0$. Hence $w = 5$ and $l = 9$, or $w = 9$ and $l = 5$: the rectangle is 9 cm by 5 cm. Check: $9 + 5 = 14$ and $45 = 9 \times 5$. ✓

17 Divide by $x + 2$: quotient $x^2 + 7x + 12 = (x + 3)(x + 4)$. So the other two dimensions are $(x + 3)$ and $(x + 4)$. Check: $(x + 2)(x + 3)(x + 4) = x^3 + 9x^2 + 26x + 24$. ✓

18 Let the speed be v km/h: $\dfrac{240}{v} - \dfrac{240}{v + 20} = 1$. Multiplying by $v(v + 20)$: $240(v + 20) - 240v = v(v + 20)$, so $4800 = v^2 + 20v$ and $v^2 + 20v - 4800 = 0$, that is $(v - 60)(v + 80) = 0$. Hence $v = 60$ (reject -80). Check: 4 h at 60, 3 h at 80, a difference of 1 h. ✓

19 $\dfrac{1}{x-1} + \dfrac{1}{x+1} = \dfrac{(x+1)+(x-1)}{(x-1)(x+1)} = \dfrac{2x}{x^2-1}$. The identity holds for every x except $x = 1$ and $x = -1$, where both sides are undefined (the left side has a zero denominator and the right side has denominator $x^2 - 1 = 0$).

20 Crossing at -2 gives a single factor $(x + 2)$; touching at 1 gives a repeated factor $(x - 1)^2$. Multiplicities $1 + 2 = 3$ match the cubic, so

$$y = a(x + 2)(x - 1)^2.$$

At $x = 0$: $a(2)(1) = 2a = 18$, so $a = 9$. Expanding, $(x - 1)^2 = x^2 - 2x + 1$ and $(x + 2)(x^2 - 2x + 1) = x^3 - 3x + 2$, hence

$$y = 9x^3 - 27x + 18,$$

so $a = 9$, $b = 0$, $c = -27$, $d = 18$.

Checks: $y(-2) = -72 + 54 + 18 = 0$; $y(1) = 9 - 27 + 18 = 0$; $y(0) = 18$. Also $y' = 27x^2 - 27 = 27(x - 1)(x + 1)$, which is zero at $x = 1$, confirming that the graph turns there rather than crossing.

21 Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ have integer coefficients and let $\dfrac{p}{q}$ be a root in lowest terms, so p and q have no common factor and $q \neq 0$. Then $a_n \dfrac{p^n}{q^n} + a_{n-1}\dfrac{p^{n-1}}{q^{n-1}} + \cdots + a_1 \dfrac{p}{q} + a_0 = 0$. Multiplying by q^n: $a_n p^n + a_{n-1}p^{n-1}q + \cdots + a_1 pq^{n-1} + a_0 q^n = 0$. Rearranging, $a_0 q^n = -p\left(a_n p^{n-1} + a_{n-1}p^{n-2}q + \cdots + a_1 q^{n-1}\right)$, so p divides $a_0 q^n$. Since p and q are coprime, p divides a_0. (The same argument applied to the other end shows q divides a_n.)

22 Multiply by $(x - a)(x - b)$: $x(x - b) + x(x - a) = 2(x - a)(x - b)$, so $2x^2 - (a + b)x = 2x^2 - 2(a + b)x + 2ab$, giving $(a + b)x = 2ab$. Besides $x = 0$, the equation has the solution

$$x = \frac{2ab}{a+b} \qquad (a + b \neq 0).$$

Check with $a = 1$, $b = 3$: $x = \dfrac{6}{4} = 1.5$, and $\dfrac{1.5}{0.5} + \dfrac{1.5}{-1.5} = 3 - 1 = 2$. ✓

This root is extraneous when it makes an original denominator zero: $\dfrac{2ab}{a+b} = a$ means $2ab = a^2 + ab$, so $a(b - a) = 0$, that is $a = 0$ (since $a \neq b$); similarly it equals b when $b = 0$. So the second root is extraneous if $a = 0$ or $b = 0$, and if $a + b = 0$ the equation $(a + b)x = 2ab$ has no solution at all (it reads $0 = -2a^2$).

Criterion A · Knowing and understanding

1 Degree 4 (the highest power is x^4); leading coefficient 2 (the coefficient of x^4); constant term 5 (the term without x).

Marking guidance: [3] one mark for each correct statement; the degree must be read from the highest power, not from the number of terms.

2 $(2x - 1)(x^2 - 3x + 4) = 2x^3 - 6x^2 + 8x - x^2 + 3x - 4 = 2x^3 - 7x^2 + 11x - 4$. Check at $x = 1$: $(1)(2) = 2$ and $2 - 7 + 11 - 4 = 2$. ✓

Marking guidance: [3] one mark for the two correct outer products $2x^3$ and -4, one for the three collected middle terms, one for the final simplified cubic; a sign slip in one term should cost at most one mark.

3 $x^3 + 2x^2 - 7x - 10$ divided by $x - 2$: $x^3 \div x = x^2$, and $x^2(x - 2) = x^3 - 2x^2$, leaving $4x^2 - 7x - 10$. Then $4x^2 \div x = 4x$ and $4x(x - 2) = 4x^2 - 8x$, leaving $x - 10$. Then $x \div x = 1$ and $1(x - 2) = x - 2$, leaving -8. So the quotient is $x^2 + 4x + 1$ and the remainder is -8:

$$x^3 + 2x^2 - 7x - 10 = (x - 2)(x^2 + 4x + 1) - 8.$$

Check: $P(2) = 8 + 8 - 14 - 10 = -8$, which equals the remainder. ✓

Marking guidance: [4] one mark for each of three correct quotient terms, one for the remainder -8; the check $P(2) = -8$ may earn the remainder mark if the division is otherwise complete.

4 $P(-3) = -27 + 18 + 15 - 6 = 0$. By the factor theorem, $x + 3$ is a factor. Dividing (synthetic, root -3) gives quotient $x^2 - x - 2 = (x - 2)(x + 1)$, so

$$P(x) = (x + 3)(x - 2)(x + 1).$$

Check by expansion: $(x + 3)(x^2 - x - 2) = x^3 + 2x^2 - 5x - 6$. ✓

Marking guidance: [5] one mark for substituting -3 (not 3), one for the conclusion by the factor theorem, one for the correct quotient, one for factorising the quadratic, one for the complete product and a check.

5 $\dfrac{x^2 - 4x}{x^2 - 16} = \dfrac{x(x-4)}{(x-4)(x+4)} = \dfrac{x}{x+4}$, with $x \neq 4$ and $x \neq -4$ (the denominator is zero at both). The restriction $x \neq 4$ survives the cancellation.

Marking guidance: [3] one mark for each correct factorisation, one for the simplified fraction with both excluded values; omitting $x \neq 4$ loses the final mark.

6 $2(x + 1) = 5(x - 2)$, so $2x + 2 = 5x - 10$, $3x = 12$ and $x = 4$. The restrictions are $x \neq 2,\ -1$, and $x = 4$ satisfies them. Check: $\dfrac{2}{2} = 1$ and $\dfrac{5}{5} = 1$. ✓ No extraneous root arises.

Marking guidance: [2] one mark for the correct linear equation and solution $x = 4$, one for the check against the restrictions and by substitution; failure to mention the restrictions caps the question at one mark.

Criterion B · Investigating patterns

1 By the remainder theorem, $R_n = 2^n - 1$: $R_1 = 1$, $R_2 = 3$, $R_3 = 7$, $R_4 = 15$, $R_5 = 31$.

Marking guidance: [5] one mark per correct remainder; a candidate who divides each time rather than substituting still earns full marks if the values are right.

2 The sequence is $1,\ 3,\ 7,\ 15,\ 31$: each term is one less than a power of 2, the differences are $2,\ 4,\ 8,\ 16$ (doubling), and each term equals twice the previous term plus 1. It grows without bound, and every term is odd.

Marking guidance: [3] one mark for recognising powers of 2, one for the recursive rule (double and add 1) or the doubling differences, one for expressing the pattern in symbols.

3 The pattern predicts $R_6 = 63$ and $R_7 = 127$. Verification for $n = 6$: dividing $x^6 - 1$ by $x - 2$ (coefficients $1, 0, 0, 0, 0, 0, -1$, root 2) gives successive entries $1, 2, 4, 8, 16, 32$ and final value $64 - 1 = 63$, so the quotient is $x^5 + 2x^4 + 4x^3 + 8x^2 + 16x + 32$ and the remainder is 63. Check: $(x - 2)(x^5 + 2x^4 + 4x^3 + 8x^2 + 16x + 32) + 63 = x^6 - 64 + 63 = x^6 - 1$. ✓

Marking guidance: [4] one mark for each prediction, two for a complete and correctly aligned verification of one of them.

4 General formula: $R_n = 2^n - 1$. Justification: with $P(x) = x^n - 1$ and divisor $x - 2$, the remainder theorem gives $R_n = P(2) = 2^n - 1$. The value $2^n - 1$ is consistent with every entry of the table, and the division identity confirms it, since

$$x^n - 1 = (x - 2)(x^{n-1} + 2x^{n-2} + \cdots + 2^{n-2}x + 2^{n-1}) + (2^n - 1).$$

Marking guidance: [6] two marks for the formula, two for citing the remainder theorem with $a = 2$, two for showing the quotient pattern or otherwise verifying the identity for general n.

5 $x - 2$ is a factor exactly when $R_n = 0$, that is $2^n - 1 = 0$, so $2^n = 1$ and $n = 0$. For every positive integer n, $R_n \geq 1 > 0$, so $x - 2$ is never a factor of $x^n - 1$; the graph of $y = x^n - 1$ passes through $(2, 2^n - 1)$, always above the axis.

Marking guidance: [3] one mark for linking a factor to a zero remainder, one for solving $2^n = 1$, one for the conclusion "never, for positive n".

6 For division by $x + 1$ the remainder theorem gives $S_n = (-1)^n - 1$: $S_1 = -2$, $S_2 = 0$, $S_3 = -2$, $S_4 = 0$, $S_5 = -2$. This sequence is constant in magnitude and alternates: 0 when n is even, -2 when n is odd. So $x + 1$ is a factor exactly for even n, whereas $x - 2$ is never a factor. The two patterns differ because 2^n grows without bound while $(-1)^n$ only alternates; division by $x - 2$ samples the polynomial at a point outside $[-1, 1]$, division by $x + 1$ at a point where the power merely changes sign.

Marking guidance: [3] one mark for the alternating table, one for the general statement $(-1)^n - 1$, one for a comparison that explains the difference rather than merely restating both.

Criterion C · Communicating

1 **Model response.** When a polynomial $P(x)$ is divided by $x - a$, the division identity states that there are polynomials Q and R with

$$P(x) = (x - a)Q(x) + R, \qquad \deg R < 1.$$

Because the divisor is linear, the remainder has degree less than 1, so it is a constant. Now substitute $x = a$: the first term on the right becomes $0 \cdot Q(a) = 0$, leaving $P(a) = R$. The remainder is therefore not a by-product of long division at all — it is the value of the polynomial at the root of the divisor. This is the remainder theorem, and it means a remainder can always be found by substitution.

The special case $R = 0$ gives $P(x) = (x - a)Q(x)$: the polynomial is exactly the product of $x - a$ and a polynomial of degree one less, so $x - a$ divides P with nothing left over. Conversely, if $x - a$ divides P, the remainder is zero, so $P(a) = 0$. That two-way statement is the factor theorem: $x - a$ is a factor if and only if $P(a) = 0$.

Example: for $P(x) = x^3 - 4x^2 + x + 6$, $P(2) = 8 - 16 + 2 + 6 = 0$, so $x - 2$ is a factor, and division gives $P(x) = (x - 2)(x^2 - 2x - 3) = (x - 2)(x - 3)(x + 1)$. Non-example: for $P(x) = x^3 - 2x + 4$, $P(1) = 1 - 2 + 4 = 3 \neq 0$, so $x - 1$ is not a factor; the division is $P(x) = (x - 1)(x^2 + x - 1) + 3$, and the 3 is exactly $P(1)$.

In short, the factor theorem is the remainder theorem with the remainder set to zero: dividing tells you the remainder, and evaluating at the root of the divisor tells you the same number far more quickly.

Marking guidance: 20 marks in total — [3] states the division identity with $\deg R < 1$ and identifies each part; [4] substitutes $x = a$, explains why the first term vanishes and concludes $P(a) = R$; [3] treats the case $R = 0$ and states the factor theorem in both directions; [3] gives a correct worked example with the division shown; [3] gives a non-example that is clearly distinguished by a non-zero remainder, correctly computed; [4] for structure (introduction and conclusion), audience-appropriate language and correct notation. A response that quotes the theorem without the substitution argument should not exceed half marks; an unexplained non-example earns nothing for that element. Length 300–400 words is expected, but concision is not penalised if all four required elements are present and coherent.

Criterion D · Applying mathematics in real-life contexts

1 Height x cm, length $(30 - 2x)$ cm, width $(20 - 2x)$ cm, so

$$V(x) = x(30 - 2x)(20 - 2x) = 4x^3 - 100x^2 + 600x \text{ cm}^3, \qquad 0 < x$$

The upper bound follows from $20 - 2x > 0$, the stronger of the two conditions.

Marking guidance: [3] one mark for the three dimensions, one for the expanded cubic, one for the domain with a reason.

2 $V(1) = 504$, $V(2) = 832$, $V(3) = 1008$, $V(4) = 1056$, $V(5) = 1000$, $V(6) = 864$, $V(7) = 672$, $V(8) = 448$, $V(9) = 216$ (all cm^3). The maximum in the table is $1056\ \text{cm}^3$ at $x = 4$ cm.

Marking guidance: [4] two marks for a complete and correct table, one for identifying $x = 4$, one for the units and the value 1056 cm^3.

3 Solve $4x^3 - 100x^2 + 600x = 800$, that is $x^3 - 25x^2 + 150x - 200 = 0$. From the table, $V(1) = 504 < 800$ but $V(2) = 832 \geq 800$, so one boundary lies between 1 and 2; and $V(6) = 864 \geq 800$ but $V(7) = 672 < 800$, so the other lies between 6 and 7. Refining: $V(1.87) \approx 798$ and $V(1.88) \approx 801$; $V(6.36) \approx 800$ and $V(6.37) \approx 798$. Hence $V(x) \geq 800$ for approximately

$$1.88 \leq x \leq 6.36,$$

that is about 1.9 cm to 6.4 cm (the boundaries themselves are about 1.876 and 6.361). Check: $V(3) = 1008$ and $V(5) = 1000$ lie inside and satisfy the condition, while $V(1) = 504$ and $V(7) = 672$ lie outside and do not. ✓

Marking guidance: [6] two marks for setting up the equation or inequality, two for locating both boundaries from the table or a graph, one for refining to two decimal places or one decimal place with an interval statement, one for a check and for expressing the answer as an interval with units.

4 Assumption 1: the card has no thickness and each fold is exact, so the internal dimensions equal the external ones and no material is used at the corners. Real card 2–3 mm thick would reduce the internal base by roughly twice the thickness, cutting the internal volume by about 15% at this size and moving the optimum slightly below 4 cm. Assumption 2: the box is open, with vertical rigid sides and no glued tab, lid or overlap. A glued tab removes a strip from each end, effectively reducing the length available for the base; a lid would use part of the sheet and change the model completely, so the recommended cut would be smaller. A third assumption, that the sides can be folded without scoring and stay upright, affects whether a tall narrow box is practical at all.

Marking guidance: [5] one mark per assumption clearly stated, one mark for each explanation of its effect on the answer (direction of the shift, not just "it would change"), one mark for linking at least one assumption to the recommended size.

5 Recommend $x = 4\text{ cm}$. It gives the largest tabulated volume, 1056 cm^3, comfortably above the required 800 cm^3; the continuous maximum is near $x = 3.9\text{ cm}$, so 4 cm is within a millimetre of the best possible cut and is an easy dimension to set on a cutting machine. It is also robust: $V(3) = 1008$ and $V(5) = 1000$, so a cutting error of $\pm 1\text{ cm}$ still leaves the box above the specification, whereas $x = 6\text{ cm}$ (864 cm^3) is only just inside the acceptable band and a small error would fail. The box made at $x = 4$ has base 22 cm by 12 cm and height 4 cm, a stable open tray that is neither too shallow to hold its contents nor too tall for its base. If card thickness were allowed for, I would move to about $x = 3.8\text{ cm}$ to keep the internal volume at the maximum.

Marking guidance: [6] two marks for a specific value supported by the table, two for a justification that uses the 800 cm³ constraint and the maximum together, one for considering tolerance or robustness to error, one for evaluating the model (thickness, flaps) and adjusting the recommendation accordingly.

CHAPTER 2 · FUNCTIONS: TRANSFORMATIONS, COMPOSITES AND INVERSES

Opening problem · the shadow that shrinks

1 Side view: the lamp L stands on the floor 2 m from the wall, the ball has centre C and radius 0.5 m, and the line LC produced meets the wall at P, the centre of the shadow. The ball is x metres from the wall, so LC has horizontal run $(2-x)$ and the wall is 2 from the lamp. The triangle LCP and the triangle cut off by the ball's silhouette are similar, so the shadow is a copy of the ball enlarged in the ratio $2:(2-x)$. Its radius is that enlarged radius less the ball's own radius, since the shadow is measured from the rim of the silhouette:

$$Y=\frac{0.5\times 2}{2-x}=\frac{1}{2-x}\quad(\text{height of }P),\qquad r=Y-0.5=\frac{1}{2-x}-\frac{1}{2}$$

Check: $r(0)=0$ (ball touching the wall) and $r(1)=0.5\cdot 1/1=0.5$, equal to the ball's own radius, and $r(1.5)=0.75/0.5=1.5$, three times as large — the nearer the lamp, the bigger the shadow.

The model assumes a point lamp on the floor and a ball resting on the floor, and that x is the gap between the ball and the wall; with ball radius a the same similar triangles give $r=\dfrac{ax}{2-x}$.

2 The ball must lie between the wall and the lamp, so $0\le x<2$; $x=2$ is excluded because the denominator vanishes there, the ball would be at the lamp, and the shadow becomes unbounded. On this interval r increases from 0 without limit, so the range is $[0,\infty)$.

3 Write $r(x)=\dfrac{1}{2-x}-0.5$, since $\dfrac{1}{2-x}-\dfrac{1}{2}=\dfrac{2-(2-x)}{2(2-x)}=\dfrac{0.5x}{2-x}$. Starting from $y=\dfrac{1}{x}$: reflect in the y-axis to get $y=-\dfrac{1}{x}=\dfrac{1}{-x}$; translate 2 right to get $y=\dfrac{1}{2-x}$; translate 0.5 down to get $r(x)$. Key points and asymptotes track the moves: $(1,1)\mapsto(-1,1)\mapsto(1,1)\mapsto(1,0.5)$, while the asymptotes $x=0$, $y=0$ become $x=2$ and $r=-0.5$. Only the branch $0\le x<2$ is used: it rises steeply towards the asymptote $x=2$, and the horizontal asymptote $r=-0.5$ is never reached but explains why r stays above -0.5.

Common error: dropping the reflection and writing $r=\dfrac{1}{x-2}-0.5$. That branch is negative for $x<2$ (at $x=1$ it gives -1.5), so it cannot be a shadow radius; x appears as $2-x$, not $x-2$.

4 Yes, always, and uniquely. From $r=\dfrac{0.5x}{2-x}$, multiply up: $r(2-x)=0.5x$, so $2r-rx=0.5x$ and $2r=x(0.5+r)$, giving

$$x=\frac{2r}{0.5+r}=\frac{4r}{1+2r}\quad(r\ge 0).$$

Check: $r=0.5$ gives $x=\dfrac{2}{2}=1$; $r=1.5$ gives $x=\dfrac{3}{2}=1.5$. Since r is strictly increasing on $[0,2)$, no shadow radius arises twice — the horizontal line $r=r_0$ meets the graph once — so the rule is one-to-one and can be reversed. The inverse has domain $[0,\infty)$ and range $[0,2)$, matching the original domain and range swapped, and $x=\dfrac{4r}{1+2r}<2$ for every finite r, so a finite shadow always corresponds to a position short of the lamp.

Prerequisite check

1 $f(0)=3(0)-2=-2$; $f(4)=12-2=10$; $f(-1)=-3-2=-5$.

2 $y=x^2$ is a parabola opening upwards with its lowest point (vertex) at the origin, symmetric about the y-axis and passing through $(1,1)$ and $(-1,1)$. $y=x^3$ passes through the origin with a horizontal inflection there, rising from lower left $(-1,-1)$ to upper right $(1,1)$ and taking negative values for $x<0$. $y=\dfrac{1}{x}$ has two branches, in the first and third quadrants, with asymptotes $x=0$ and $y=0$, through $(1,1)$ and $(-1,-1)$; it is undefined at $x=0$.

3 $2x-5=11$ gives $2x=16$, so $x=8$. $x^2=16$ gives $x=\pm 4$, since both 4^2 and $(-4)^2$ equal 16.

4 $[-2,\infty)$. The square bracket includes -2; ∞ always takes a round bracket.

5 A square root needs a non-negative radicand: $x-3\ge 0$, so $x\ge 3$, that is $[3,\infty)$.

6 $y=2x+7$, so $2x=y-7$ and $x=\dfrac{y-7}{2}$. Check: $2\left(\dfrac{y-7}{2}\right)+7=y-7+7=y$.

Exercise 2A

1 $f(0)=-7$; $f(2)=8-7=1$; $f(-1)=-4-7=-11$; $f\left(\frac{1}{2}\right)=2-7=-5$.

2 $g(3)=9+6=15$; $g(-3)=9-6=3$; $g(0)=0$; $g(a)=a^2+2a$.

3 (a) Denominator zero at $x=-4$: domain $\mathbb{R}\setminus\{-4\}$, or $(-\infty,-4)\cup(-4,\infty)$. (b) $x+5\ge 0$: $[-5,\infty)$. (c) $x^2-9=(x-3)(x+3)\ne 0$: $\mathbb{R}\setminus\{-3,3\}$. (d) $7-2x\ge 0$, so $x\le 3.5$: $(-\infty,3.5]$. (e) $x^2+1\ge 1>0$ for every x, so the denominator never vanishes: domain $\mathbb{R}$.

4 (a) $x^2\ge 0$, so $y\ge 1$: range $[1,\infty)$. (b) $\sqrt{x}\ge 0$, so $3-\sqrt{x}\le 3$ and every smaller value occurs: $(-\infty,3]$. (c) $\dfrac{2}{x}\ne 0$ but takes every other value: $\mathbb{R}\setminus\{0\}$. (d) $(x-3)^2\ge 0$ with equality at $x=3$: $[0,\infty)$.

5 (a) $3x-12=0$, so $x=4$. (b) $x^2-5x+6=(x-2)(x-3)=0$, so $x=2$ or $x=3$. (c) A fraction is zero only when its numerator is zero, so $x-1=0$, $x=1$ (and $x\ne -2$, which is satisfied).

6 (a) $(-3,4]$. (b) $(-\infty,1)$. (c) $x\ne 5$ removes one point: $(-\infty,5)\cup(5,\infty)$. (d) $[2,7)$.

7 The domain is correct but the range is not: $\sqrt{x-1}$ is never negative and takes every value from 0 upward, so the range is $y\ge 0$, that is $[0,\infty)$. The student has shifted the range by 1, as if the rule were $\sqrt{x}-1$. In $\sqrt{x-1}$ the -1 acts on the input, moving the graph 1 right and the domain 1 right; only a term outside the root moves the range.

Common error: reading a horizontal shift as a vertical one. Test a point: $f(1)=0$ shows that 0 is in the range, which $y\ge 1$ forbids.

8 At $x=0$ the equation gives $y^2=9$, so $y=3$ or $y=-3$: the input 0 has two outputs. A function must assign exactly one output to each input, so the whole circle is not the graph of a function; the vertical line $x=0$ meets it twice. (The upper semicircle $y=\sqrt{9-x^2}$ is a function.)

9 Complete the square: $x^2-6x+5=(x-3)^2-4$. For $x\ge 3$, $(x-3)^2$ runs through every value in $[0,\infty)$, so the range is $[-4,\infty)$. Check: $f(3)=9-18+5=-4$ is the least value, and $f(4)=16-24+5=-3$ lies in the interval.

10 The inside $x-1$ shifts every input 1 right, so the domain moves from $[-2,4]$ to $[-1,5]$; the outside $+3$ shifts every output 3 up, so the range moves from $[1,9]$ to $[4,12]$. Endpoints confirm: $x=-1$ uses $f(-2)=1$, giving 4; $x=5$ uses $f(4)=9$, giving 12.

11 $f(x)=x^2$ has natural domain $\mathbb{R}$: every real number can be squared. Its range is $[0,\infty)$, not $\mathbb{R}$, because $x^2\ge 0$ for every real x, so negative outputs never occur; and every $y\ge 0$ does occur, as $f(\sqrt{y})=y$.

12 For $f(x)=x^2$, $f(2)=f(-2)=4$ while $2\ne -2$, so equal outputs need not come from equal inputs. The implication becomes true when f is one-to-one (injective), for example when its domain is restricted to $x\ge 0$ or, more generally, when f is strictly increasing or strictly decreasing on its domain.

13 $F(d)=3.50+1.20d$, with d in kilometres and F in pounds. A city journey of at most 30 km gives domain $[0,30]$ (a continuous model; the meter in practice jumps in small steps). The range is $[3.50,\ 3.50+1.20(30)]=[3.50,39.50]$. Check: $1.20\times 30=36$ and $36+3.50=39.50$.

14 The ball is in the air from $t=0$ until $h=0$: solving $5t^2-8t-1=0$, $t=\dfrac{8\pm\sqrt{64+20}}{10}=\dfrac{4\pm\sqrt{21}}{5}$, and the positive root is $\dfrac{4+\sqrt{21}}{5}\approx 1.72$. So the domain is $\left[0,\dfrac{4+\sqrt{21}}{5}\right]$. Completing the square:

$$h=1+8t-5t^2=4.2-5(t-0.8)^2,$$

so the height is greatest when $t=0.8$ s, and the maximum is 4.2 m. Check: $1+8(0.8)-5(0.64)=1+6.4-3.2=4.2$. Since h starts at 1, rises to 4.2 and falls to 0, the range is $[0,4.2]$.

15 Suppose $f(x) = \frac{a}{c}$. Then $c(ax+b) = a(cx+d)$, so $acx + bc = acx + ad$, that is $bc = ad$. If $bc \neq ad$ this is impossible, so no input produces $\frac{a}{c}$ and the range is $\mathbb{R} \setminus \left\{\frac{a}{c}\right\}$ — the horizontal asymptote is exactly the missing value. If $bc = ad$ then $ax + b = \frac{a}{c}(cx+d)$, so $f(x) = \frac{a}{c}$ for every $x \neq -\frac{d}{c}$: the function is constant, its range is the single value $\frac{a}{c}$, and the excluded value is the only value taken.

When $a = 0$ the rule is $f(x) = \frac{b}{cx+d}$, a scaled and shifted copy of $\frac{1}{x}$ with horizontal asymptote $y = 0$; the excluded value is $\frac{d}{c} = 0$, so for $b \neq 0$ the range is $\mathbb{R} \setminus \{0\}$. If also $b = 0$ then $f(x) = 0$ identically, the constant case above.

Exercise 2B

1 (a) Translate 7 down. (b) Translate 3 left. (c) Stretch vertically by factor 4. (d) Reflect in the y-axis. (e) Stretch horizontally by factor 3, since replacing x by $\frac{1}{3}x$ makes the graph reach each parent value three times as far out. (f) Reflect in the x-axis, then translate 2 up.

2 (a) $(6, -9)$. (b) $(3, -2)$. (c) $(6, -8)$. (d) $(-6, -2)$. (e) $(18, -2)$, because $f\left(\frac{18}{3}\right) = f(6) = -2$. (f) $(6, 4)$: the reflection sends -2 to 2, then 2 up gives 4.

3 (a) Parent $y = x^2$: translate 4 right and 1 up; vertex $(0,0) \mapsto (4,1)$, axis $x = 4$. (b) Parent $y = |x|$: translate 2 left and 5 down; vertex $(0,0) \mapsto (-2,-5)$, a minimum. (c) Parent $y = \frac{1}{x}$: stretch vertically by factor 3, so $(1,1) \mapsto (1,3)$; the asymptotes $x = 0$, $y = 0$ are unchanged and the branches stay in the first and third quadrants. (d) Parent $y = \sqrt{x}$: translate 1 right and 3 up; endpoint $(0,0) \mapsto (1,3)$ and the domain becomes $x \geq 1$. (e) Parent $y = x^3$: compress vertically by factor $\frac{1}{2}$ and reflect in the x-axis, so $(1,1) \mapsto (1,-0.5)$ and $(-1,-1) \mapsto (-1,0.5)$; the flat point at the origin is fixed.

4 (a) $(-6, -11)$, a minimum. (b) $(2, 7)$, a maximum because the coefficient -3 is negative. (c) $(-8, 0)$, the endpoint of the branch, with domain $x \geq -8$. (d) The asymptotes are $x = 5$ and $y = 2$, meeting at $(5, 2)$; the parent point $(1, 1)$ maps to $(6, 3)$, and $(0.5, 2) \mapsto (5.5, 4)$.

5 (a) $y = (x+3)^2 - 4$. (b) $y = -|x-2|$. (c) $y = 3\sqrt{x} + 1$. (d) $y = \frac{1}{2x}$: a horizontal compression by factor $\frac{1}{2}$ replaces x by $2x$, and the point $(1,1)$ moves to $(0.5, 1)$.

6 The claim is **false**. Test on $f(x) = x^2$, where $y = (2x-6)^2$ and $(1,1)$ lies on $y = f(x)$. Follow the student's order: compressing horizontally by factor $\frac{1}{2}$ sends $(1,1)$ to $(0.5,1)$; shifting 6 right then sends it to $(6.5, 1)$. But on the required graph, $x = 6.5$ gives $y = (2(6.5) - 6)^2 = 7^2 = 49 \neq 1$. So the image point is not on $y = f(2x-6)$, and the described order produces $y = f(2(x-6)) = f(2x-12)$, not $y = f(2x-6)$.

Factor the input before reading the shift: $2x - 6 = 2(x-3)$. The correct description is **compress horizontally by factor $\frac{1}{2}$, then translate 3 right**. That order sends $(1,1)$ to $(0.5,1)$ and then to $(3.5,1)$, and $(2(3.5)-6)^2 = 1^2 = 1$, which lies on the graph.

Equally valid, with the stages taken in the other order: translate 6 right first and then compress by $\frac{1}{2}$, because the compression then halves the shift as well. The single rule that always works is to factor b out of $bx + c$ and read the shift as $\frac{c}{b} = -3$; the shift is never c itself when $b \neq 1$.

7 Test with $f(x) = x^2$ and the point $(2, 4)$. Horizontally, shift 3 right then compress by factor $\frac{1}{2}$ replaces x by $2x$ only after the shift: $y = (2x-3)^2$, and $y = 4$ where $2x - 3 = 2$, that is $x = 2.5$. Compress by $\frac{1}{2}$ then shift 3 right: $y = (2(x-3))^2$, and $y = 4$ where $2(x-3) = 2$, that is $x = 4$. The two graphs differ, because the stretch is applied to the input and therefore rescales the translation as well: a shift of 3 parent units becomes a shift of 1.5 compressed units, so the printed "3" no longer describes the same displacement. Vertically, a stretch by 3 and a shift up by 2 both act on the output and never move an x-coordinate: $y = 3f(x) + 2$ and $y = 3(f(x) + 2) = 3f(x) + 6$ have the same shape factor 3 and differ only in the size of the constant, so the same graph is obtained in either order once the constant is adjusted (2 before the stretch corresponds to 6 after it). Horizontal order matters because the translation is measured along the axis the stretch changes; vertical order does not, because the translation and the stretch are measured on different scales that do not interact.

8 The maximum of f is at $(-3, 8)$. The inside $x + 1$ moves it 1 left: $x = -4$. The outside -2 reflects in the x-axis and doubles, so the output 8 becomes -16, and the -3 then lowers it to -19. The turning point is $(-4, -19)$, and because the graph has been reflected, it is a **minimum**: the arms that fell away either side of the maximum now rise without bound, so the transformed graph has no maximum at all.

Common error: answering $(-4, 19)$ or calling $(-4, -19)$ a maximum. A negative vertical factor reverses every maximum into a minimum.

9 First curve $g(x) = (x-2)^2 - 3$: vertex $(2, -3)$, a minimum; axis $x = 2$; y-intercept $g(0) = 1$; roots where $(x-2)^2 = 3$, that is $x = 2 \pm \sqrt{3} \approx 0.27$ and 3.73. Second curve $h(x) = -(x-2)^2 + 3$: vertex $(2, 3)$, a maximum; same axis; y-intercept -1; the same two roots. Since $h(x) = -g(x)$, the second graph is exactly the reflection of the first in the x-axis: every point (x, y) becomes $(x, -y)$, the minimum becomes the maximum, and the curves cross the axis at the same two points and nowhere else.

10 Symmetry about the y-axis means $f(-x) = f(x)$ for all x. (a) $f(-x) - 7 = f(x) - 7$: preserved. (b) $f(-x+3)$ is not $f(x+3)$ in general (at $x = 1$ it compares $f(2)$ with $f(4)$): not preserved — the axis of symmetry moves to $x = -3$. (c) $4f(-x) = 4f(x)$: preserved. (d) $f(-(-x)) = f(x)$: preserved. (e) $f\left(-\frac{x}{3}\right) = f\left(\frac{x}{3}\right)$: preserved. (f) $-f(-x) + 2 = -f(x) + 2$: preserved. So only the horizontal translation in (b) destroys the symmetry; horizontal stretches keep it because they leave $x = 0$ fixed.

11 Let $y(x) = f(x-2) + f(2-x)$. Reflecting an input in the line $x = 2$ replaces x by $4 - x$. Then

$$y(4-x) = f(4-x-2) + f(2-(4-x)) = f(2-x) + f(x-2) = y$$

The two terms simply exchange places, so every point (x, y) has a partner $(4-x, y)$ on the graph: the curve is unchanged by reflection in $x = 2$. (Equivalently, the graph is the sum of $f(x-2)$ and its own reflection in $x = 2$.)

12 Original profile $y = 6 - 0.5x^2$ on $-2\sqrt{3} \leq x \leq 2\sqrt{3}$: height 6 at the centre, zero at $x = \pm 2\sqrt{3}$, since $0.5(12) = 6$. Doubling the width is a horizontal stretch by factor 2, so replace x by $\frac{x}{2}$: $y = 6 - 0.5\left(\frac{x}{2}\right)^2 = 6 - 0.125x^2$. Raising the tunnel 2 m adds 2 to the output:

$$y = 8 - 0.125x^2, \qquad -4\sqrt{3} \leq x \leq 4\sqrt{3}.$$

Check: at $x = 4\sqrt{3}$, $0.125(48) = 6$ and $y = 2$; the apex is $(0, 8)$. The arch therefore springs from height 2, so 2 m vertical side walls complete the tunnel down to the floor.

13 Moving the endpoint of $y = \sqrt{x}$ to $(3, 5)$ gives $y = a\sqrt{x-3} + 5$ with domain $x \geq 3$, for any $a > 0$. Check: at $x = 3$ the root is zero, so $y = 5$. For example $a = 1$ gives $y = \sqrt{x-3} + 5$, which also passes through $(7, 7)$; $a = 2$ gives $y = 2\sqrt{x-3} + 5$, through $(4, 7)$. The equation is not unique: one point fixes the endpoint but not the steepness, so the vertical stretch factor stays free. Allowing a reflection gives a further family $y = a\sqrt{3-x} + 5$ with domain $x \leq 3$, whose endpoint is again $(3, 5)$.

14 Start from $y = f(x)$. Applying $y = f(x - h)$ and then replacing x by bx gives $y = f(bx - h) = f\left(b\left(x - \frac{h}{b}\right)\right)$, that is a compression by $\frac{1}{b}$ followed by a shift of $\frac{h}{b}$ right, with $c = -h$. Doing it the other way, $y = f(bx)$ first and then replacing x by $x - h$, gives $y = f(b(x - h)) = f(bx - bh)$, a shift of h right followed by a compression by $\frac{1}{b}$, with $c = -bh$. The two agree only when $h = 0$ or $b = 1$. Test with $f(x) = x^2$, $h = 3$, $b = 2$: the first order gives $(2x - 3)^2$ and the second $(2x - 6)^2$; at $x = 0$ these are 9 and 36. In the form $y = f(bx + c)$ the first is $c = -h$ and the second $c = -bh$.

Exercise 2C

1 $f(g(x)) = (2x - 1) + 3 = 2x + 2$. $g(f(x)) = 2(x + 3) - 1 = 2x + 5$. $f(g(2))$: $g(2) = 3$, so $f(3) = 6$; with the composite, $2(2) + 2 = 6$. $g(f(2))$: $f(2) = 5$, so $g(5) = 9$; with the composite, $2(2) + 5 = 9$. The two composites differ by the constant 3.

2 $f(g(x)) = (x - 4)^2 = x^2 - 8x + 16$. $g(f(x)) = x^2 - 4$. $f(g(-1))$: $g(-1) = -5$, so $f(-5) = 25$. $g(f(-1))$: $f(-1) = 1$, so $g(1) = -3$.

3 $f(g(x)) = \sqrt{x + 9}$. The inner function accepts every real x, but the outer needs a non-negative input: $x + 9 \geq 0$, so $x \geq -9$ and the domain is $[-9, \infty)$, with range $[0, \infty)$. Check: at $x = -9$, $f(g(-9)) = \sqrt{0} = 0$.

4 $f(g(x)) = \dfrac{1}{x - 3}$. The inner function is defined for all x, but the outer requires a non-zero input, so $x - 3 \neq 0$: domain $\mathbb{R} \setminus \{3\}$, that is $(-\infty, 3) \cup (3, \infty)$. The range is $\mathbb{R} \setminus \{0\}$.

5 (a) $f(2) = 4 - 1 = 3$, so $f(f(2)) = f(3) = 9 - 1 = 8$. (b) $f(2) = \dfrac{1}{3}$, so $f(f(2)) = f\left(\frac{1}{3}\right) = \dfrac{1}{\frac{1}{3} + 1} = \dfrac{1}{\frac{4}{3}} = \dfrac{3}{4}$.

6 (a) $g(x) = 4x + 1$, $f(u) = \sqrt{u}$. (b) $g(x) = x - 5$, $f(u) = u^2$. (c) $g(x) = 2x + 3$, $f(u) = \dfrac{1}{u}$. (d) $g(x) = 3x - 2$, $f(u) = |u|$. In each case the inner function is the expression handled first and the outer function is the operation applied to its result.

7 The student has squared x and then subtracted 1, which is $f(x) - 1$ (equivalently $g(f(x))$), not f applied to $g(x)$. The input to f is the whole of $x - 1$, so

$$f(g(x)) = (x - 1)^2 = x^2 - 2x + 1.$$

Test at $x = 2$: $g(2) = 1$ and $f(1) = 1$, whereas $x^2 - 1 = 3$.

Common error: treating $f(g(x))$ as $f(x)$ with a correction added on. The square acts on the whole of $x - 1$, not on x alone.

8 $f(g(x)) = x^2 + 1$ and $g(f(x)) = (x + 1)^2$. At $x = 2$: $f(g(2)) = f(4) = 5$, while $g(f(2)) = g(3) = 9$. Since $5 \neq 9$ at the same input, the composites are different functions, so composition is not commutative.

9 $f(g(x)) = \sqrt{\dfrac{1}{x} - 2}$. Two conditions: $x \neq 0$ from the inner function, and $\dfrac{1}{x} - 2 \geq 0$ from the outer. The inequality is $\dfrac{1}{x} \geq 2$; multiplying by $x^2 > 0$ preserves the direction and gives $x \geq 2x^2$, that is $2x^2 - x \leq 0$, so $x(2x - 1) \leq 0$ and $0 \leq x \leq \frac{1}{2}$. Removing $x = 0$ leaves the domain $\left(0, \frac{1}{2}\right]$. Check: at $x = \frac{1}{4}$, $\dfrac{1}{1/4} - 2 = 2 \geq 0$; at $x = 1$, $1 - 2 = -1 < 0$, rejected; at $x = -\frac{1}{2}$, $-2 - 2 = -4 < 0$, rejected.

Common error: solving $\dfrac{1}{x} \geq 2$ by multiplying by x without recording its sign, which wrongly admits negative x.

10 $f(f(x)) = 2(2x + 3) + 3 = 4x + 9$. Setting this to zero: $4x = -9$, so $x = -\frac{9}{4} = -2.25$. Check: $f(-2.25) = -4.5 + 3 = -1.5$ and $f(-1.5) = -3 + 3 = 0$.

11 For $f(g(x))$ to be defined at an input x, the first stage must produce a value, so x must already lie in the domain of g. The domain of $f \circ g$ is the set of such x for which $g(x)$ also lies in the domain of f — a subset of the domain of g, possibly smaller by the values that g maps outside the domain of f. Adding a condition can only remove inputs, never restore them, so the composite's domain can never be larger.

12 $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, so its domain is $[0, \infty)$, the domain of g. $g(f(x)) = \sqrt{x^2} = |x|$ for all real x, so its domain is $\mathbb{R}$. They differ because the inner function differs: in $f \circ g$ the square root is applied first and demands $x \geq 0$, while in $g \circ f$ the square is applied first and $x^2 \geq 0$ always. The simplification $(\sqrt{x})^2 = x$ is valid only on $[0, \infty)$; on $\mathbb{R}$ the correct simplification is $\sqrt{x^2} = |x|$, which is not the identity.

13 Let $d(p) = 0.8p$ (20% off) and $t(q) = 1.15q$ (15% added). The final price is

$$t(d(p)) = 1.15(0.8p) = 0.92p.$$

For a £240 item the discounted price is $0.8(240) = 192$ and the till takes $1.15(192) = 220.80$, so the customer pays £220.80; directly, $0.92(240) = 220.80$. The other order gives $d(t(240)) = 0.8(276) = 220.80$, the same: both stages are multiplications, and $1.15 \times 0.8 = 0.8 \times 1.15 = 0.92$. So the order does not matter here; it would matter if one stage were a fixed amount, say £20 off, since $0.8(p - 20) \neq 0.8p - 20$.

14 $$F(E(v)) = \frac{300}{E(v)}.$$

The inner function turns speed into efficiency in km per litre; the outer divides the 300 km of the journey by that efficiency, so the composite gives the fuel used, in litres, for the whole journey at speed v. The units confirm it: $\text{km} \div (\text{km per litre}) = \text{litres}$. It is a function of v alone: $v \mapsto$ litres consumed.

15 $f(f(x)) = a(ax + b) + b = a^2x + ab + b$. Matching $9x + 4$ gives $a^2 = 9$, so $a = 3$ or $a = -3$, and $b(a + 1) = 4$.

If $a = 3$: $4b = 4$, so $b = 1$ and $f(x) = 3x + 1$. Check: $3(3x + 1) + 1 = 9x + 4$. If $a = -3$: $-2b = 4$, so $b = -2$ and $f(x) = -3x - 2$. Check: $-3(-3x - 2) - 2 = 9x + 6 - 2 = 9x + 4$. Both satisfy the condition and both were needed, because the square $a^2 = 9$ has two roots.

Exercise 2D · Inverses and restricted domains

1 (a) $y = 3x + 2$ gives $x = \dfrac{y - 2}{3}$, so $f^{-1}(x) = \dfrac{x - 2}{3}$. (b) $y = 7 - x$ gives $x = 7 - y$, so $f^{-1}(x) = 7 - x$; the function is its own inverse, and reflecting the line in $y = x$ leaves it unchanged. (c) $y = \dfrac{x - 5}{2}$ gives $x = 2y + 5$, so $f^{-1}(x) = 2x + 5$. (d) $y = -\dfrac{2}{3}x + 4$ gives $\dfrac{2}{3}x = 4 - y$, so $x = \dfrac{3}{2}(4 - y) = 6 - \dfrac{3}{2}y$ and $f^{-1}(x) = 6 - \dfrac{3}{2}x$. Check (d): $f\left(6 - \dfrac{3}{2}x\right) = -\dfrac{2}{3}\left(6 - \dfrac{3}{2}x\right) + 4 = -4 + x + 4 = x$.

2 (a) $y = x^3$ gives $x = \sqrt[3]{y}$, so $f^{-1}(x) = \sqrt[3]{x}$ with domain $\mathbb{R}$.
(b) $y = \sqrt{x + 4}$ has domain $[-4, \infty)$ and range $[0, \infty)$. Then $y^2 = x + 4$, so $f^{-1}(x) = x^2 - 4$ with domain $x \geq 0$. Check: $f(f^{-1}(x)) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = x$ for $x \geq 0$.
(c) $y = \dfrac{1}{x - 1}$ gives $x - 1 = \dfrac{1}{y}$, so $f^{-1}(x) = 1 + \dfrac{1}{x} = \dfrac{x + 1}{x}$ with domain $x \neq 0$. Check: $f\left(1 + \dfrac{1}{x}\right) = \dfrac{1}{\left(1 + \frac{1}{x}\right) - 1} = \dfrac{1}{1/x} = x$.
(d) $y = 2 + \sqrt{x}$ has domain $x \geq 0$ and range $y \geq 2$. Then $y - 2 = \sqrt{x}$, so $f^{-1}(x) = (x - 2)^2$ with domain $x \geq 2$. Check: $f(f^{-1}(x)) = 2 + \sqrt{(x - 2)^2} = 2 + (x - 2) = x$ for $x \geq 2$.

3 (a) $(9, 2)$. (b) $(5, -3)$. (c) $(-1, 0)$. (d) (b, a), since $f(a) = b$ exactly when $f^{-1}(b) = a$.

4 For $x \neq 0$, $f(f(x)) = \dfrac{1}{1/x} = x$. Solving $y = \dfrac{1}{x}$ gives $x = \dfrac{1}{y}$, so $f^{-1}(x) = \dfrac{1}{x} = f(x)$ on $\mathbb{R} \setminus \{0\}$. The graph records it: reflection in $y = x$ maps each branch of the hyperbola onto the other, so the curve is its own image.

5 Solve $2x^2 + 1 = 3$: $x^2 = 1$, so $x = \pm 1$; the restriction $x \geq 0$ leaves $x = 1$. Hence $f^{-1}(3) = 1$. (The inverse function is $f^{-1}(x) = \sqrt{\dfrac{x - 1}{2}}$ for $x \geq 1$, which confirms it.)

6 On $\mathbb{R}$, $f(x) = (x - 1)^2$ is not one-to-one: $f(0) = f(2) = 1$, so the horizontal line $y = 1$ meets the graph twice and the reverse rule cannot decide between 0 and 2. The formula $1 + \sqrt{x}$ inverts only the restriction $x \geq 1$, where $x - 1 \geq 0$ and the positive root is the correct branch; on the restriction $x \leq 1$ the inverse is $1 - \sqrt{x}$. Without a stated restriction the function has no inverse at all, so the answer is incomplete twice over: it omits the restriction on the domain of f, and it does not state the domain $x \geq 0$ or the range $y \geq 1$ of the inverse it does give.

7 (a) One-to-one: $y = x^3$ is strictly increasing, so no horizontal line meets it twice. (b) Not one-to-one: $y = x^2 - 4$ is symmetric about the y-axis, and $y = 0$ meets it at $x = \pm 2$. (c) One-to-one: $y = \dfrac{1}{x}$ is strictly decreasing on each branch, and a horizontal line $y = c$ with $c \neq 0$ meets it only at $x = \dfrac{1}{c}$. (d) Not one-to-one: $y = |x - 2|$ takes the value a at $x = 2 \pm a$. (e) One-to-one: $y = \sqrt{4 - x}$ is strictly decreasing on $x \leq 4$.

8 A point (a, b) lies on the graph of f exactly when $f(a) = b$, which by definition means $f^{-1}(b) = a$, so (b, a) lies on f^{-1}. Reflection in $y = x$ is precisely the map $(a, b) \mapsto (b, a)$. The x-coordinates of the reflected graph are therefore the original y-coordinates, that is, the domain of f^{-1} is the range of f, and in the same way its range is the domain of f.

9 From $y = \dfrac{x + 3}{x - 1}$, $y(x - 1) = x + 3$, so $xy - y = x + 3$ and $x(y - 1) = y + 3$, giving $x = \dfrac{y + 3}{y - 1}$. Hence $f^{-1}(x) = \dfrac{x + 3}{x - 1}$ with domain $x \neq 1$: the function is its own inverse. Numerical check: $f(0) = -3$, and $f^{-1}(-3) = \dfrac{0}{-4} = 0$, recovering the original input.

10 The y-intercept $(0, 5)$ of f becomes the x-intercept $(5, 0)$ of f^{-1}, and the x-intercept $(4, 0)$ of f becomes the y-intercept $(0, 4)$. Intercepts swap roles because reflection in $y = x$ exchanges the two axes.

11 Assume $f(a) = f(b)$. Then $3a - 7 = 3b - 7$. Adding 7 to both sides gives $3a = 3b$, and dividing by 3 (which is not zero) gives $a = b$. Since equal outputs force equal inputs, f is one-to-one, and the steps are reversible because each operation used is invertible.

12 From $e = 0.92d$, $d = \dfrac{e}{0.92} = \dfrac{25e}{23} \approx 1.087e$. So $C^{-1}(e) = \dfrac{25e}{23}$: it converts euros back into dollars, the rate being the reciprocal of 0.92. The physical domain is $e \geq 0$, giving $d \geq 0$; the composition $C^{-1}(C(d)) = d$ checks the conversion both ways.

13 Squaring gives $v^2 = 9.8d$, so $d = \dfrac{v^2}{9.8}$. For $v = 7$, $d = \dfrac{49}{9.8} = 5$, so the depth is 5 m. Check: $\sqrt{9.8 \times 5} = \sqrt{49} = 7$.

14 Complete the square: $f(x) = (x - 3)^2 - 4$, so on $x \geq 3$ the function is increasing with range $y \geq -4$. Then $y + 4 = (x - 3)^2$ and, since $x - 3 \geq 0$, $x = 3 + \sqrt{y + 4}$. Hence $f^{-1}(x) = 3 + \sqrt{x + 4}$ with domain $x \geq -4$ and range $y \geq 3$. Check: $f(f^{-1}(x)) = \left(3 + \sqrt{x + 4} - 3\right)^2 - 4 = (x + 4) - 4 = x$. The restriction $x \leq 3$ uses the other branch, giving $f^{-1}(x) = 3 - \sqrt{x + 4}$ with range $y \leq 3$: the same curve reflected in $y = x$ but with the decreasing half selected, so the two restrictions produce genuinely different inverse functions.

Exercise 2E · Building and using transformed models

1 Use $y = a(x - h)^2 + k$ with the vertex as (h, k). (a) $8 = a(2 - 0)^2$, so $a = 2$ and $y = 2x^2$. (b) $5 = a(3 - 1)^2 - 3$, so $4a = 8$, $a = 2$ and $y = 2(x - 1)^2 - 3$. (c) $0 = a(0 + 2)^2 + 4$, so $4a = -4$, $a = -1$ and $y = -(x + 2)^2 + 4$.

2 The endpoint $(2, 1)$ fixes $h = 2$ and $k = 1$, so $y = a\sqrt{x - 2} + 1$. Substituting $(6, 5)$: $5 = a\sqrt{4} + 1 = 2a + 1$, so $a = 2$. Hence $a = 2$, $h = 2$, $k = 1$: $y = 2\sqrt{x - 2} + 1$ for $x \geq 2$, with range $y \geq 1$.

3 $y = \dfrac{1}{x - 3} - 1$. The vertical asymptote moves from $x = 0$ to $x = 3$ and the horizontal asymptote from $y = 0$ to $y = -1$; the branches lie either side of $x = 3$, approaching $y = -1$ at the far left and far right.

4 $y = -2(x - 1)^2 + 4$ has vertex $(1, 4)$, axis of symmetry $x = 1$ and maximum value 4, attained at $x = 1$. The coefficient -2 opens the parabola downward and makes it twice as steep as $y = x^2$.

5 A hanging cable is lowest at its centre and rises towards both towers, so its height has a minimum between the towers; a quadratic with $a > 0$ opens upward and has exactly that minimum at (h, k). A bridge arch is highest at the centre and falls to the road at both ends, so its height has a maximum and needs $a < 0$. Using $a > 0$ for the arch would place a dip where the bridge must be highest, and no choice of h and k can repair the wrong sign: the model would predict a lowest point at the crown and unbounded height at the ends, contradicting the measurements.

6 Extrapolation means predicting outside the interval on which the model was fitted. Here the parameters a, h and k were fixed by data from $x = 1$ to $x = 5$; between those values the fitted curve is pinned by nearby points, but at $x = 10$ nothing in the data constrains it. An error in the fitted stretch, or a change in the underlying process beyond the measured range, grows with distance from the fitted interval, and the assumed shape is untested there. Interpolation at, say, $x = 3$ is safer because the curve is held by measurements on both sides.

7 Since $\dfrac{0.5x}{2 - x} = \dfrac{-0.5(2 - x) + 1}{2 - x}$, $r(x) = -0.5 + \dfrac{1}{2 - x}$. Because $\dfrac{1}{2 - x} = -\dfrac{1}{x - 2}$, this is $r(x) = -0.5 - \dfrac{1}{x - 2}$: from $y = \dfrac{1}{x}$ the graph is reflected in the x-axis, translated 2 right and translated 0.5 down. The asymptotes are $x = 2$ and $r = -0.5$, and on $0 < x < 2$ the curve rises from $r = 0$ towards $+\infty$. The physical domain is $0 < x < 2$: the ball lies between wall and lamp and cannot touch either.

8 Domain $x \geq 2$ fixes the endpoint at $x = 2$, and range $y \leq 7$ fixes its highest value there, so the endpoint is $(2, 7)$ and the graph must fall away from it. One possible equation is $y = 7 - \sqrt{x - 2}$; more generally $y = 7 - a\sqrt{x - 2}$ with $a > 0$, or $y = 7 - \sqrt{b(x - 2)}$ with $b > 0$.

9 The inverse answers the design question rather than the prediction question: instead of "what output follows from this input?" it gives "what input produces this output?". An engineer who needs a period of 3 s uses $L = \left(\dfrac{T}{k}\right)^2$ directly instead of guessing and re-evaluating. Approximation enters only through k: the inverse of a monotonic model is itself a function, so a small error in k produces a small, predictable error in L, and the estimate can be corrected by one measurement. The inverse remains trustworthy as a starting point even when the model is only approximate, because it preserves the direction of the relationship.

10 $h(0) = 2 + 0.8(0) = 2$ cm; $h(4) = 2 + 0.8(2) = 3.6$ cm; $h(16) = 2 + 0.8(4) = 5.2$ cm. For 6 cm: $0.8\sqrt{t} = 4$, so $\sqrt{t} = 5$ and $t = 25$ weeks.

11 At the towers $x = 0$ and $x = 80$: $y = 0.02(40)^2 + 3 = 32 + 3 = 35$ m, the same at both ends as symmetry requires. At the lowest point $x = 40$: $y = 3$ m.

12 After replacing a by $2a$ and k by $2k$ the rule is

$$y = 2a\,f(b(x - h)) + 2k = 2\big(a\,f(b(x - h)) + k\big),$$

which is twice the old output at every x: on its own that change is *not* harmless, it is the original graph stretched vertically by factor 2 about the x-axis. Stretching the whole picture back by $\dfrac{1}{2}$ multiplies every output by $\dfrac{1}{2}$ and returns exactly the original y. Every point (x, y) leaves and comes back to the same place, so the final graph is identical.

The reason is that a and k both sit outside the function, so they scale with the output; they can be pulled out as a common factor, whereas b, h live inside and act on the input, where no vertical move can reach them. Only the *ratio* $a : k$ and the horizontal parameters b, h are fixed by the shape.

Implication for fitting. Parameters carry the vertical unit of the data, so they are meaningless until that unit is declared: recording the same experiment in centimetres instead of metres multiplies both a and k by 100 and leaves the physical model untouched. Two consequences follow. First, parameters from two fits may be compared only when the output scales agree, and a goodness-of-fit measured in raw units (a residual sum in metres versus the same in millimetres) is not comparable either — use a relative or standardised measure. Second, if the data source has a free gain, such as a sensor calibration factor, that gain is multiplied into both a and k, so output data alone can never separate the calibration from the model: the pair (a, k) is identifiable only up to a common factor, and fixing it requires one measurement on a known standard. What *is* safely recoverable from the data alone is the midline-to-amplitude structure and the horizontal parameters.

Investigation 2 · one-to-one and reversible

1 $y = x^2$ fails the test: the horizontal line $y = 4$ meets it at $(-2, 4)$ and $(2, 4)$. $y = x^3$ passes: it rises strictly, so each horizontal line meets it once. $y = |x|$ fails: $y = 1$ meets it at $(-1, 1)$ and $(1, 1)$. $y = \frac{1}{x}$ passes: for $c \neq 0$ the line $y = c$ meets it only at $\left(\frac{1}{c}, c\right)$, and $y = 0$ misses both branches.

2 $y = x^2$: the largest intervals containing 0 on which it is one-to-one are $[0, \infty)$ and $(-\infty, 0]$. On $x \geq 0$, $f(x) = x^2$ has inverse $f^{-1}(x) = \sqrt{x}$ with domain $x \geq 0$; on $x \leq 0$ the inverse is $f^{-1}(x) = -\sqrt{x}$, also for $x \geq 0$. $y = |x|$: on $x \geq 0$, $f^{-1}(x) = x$ with domain $x \geq 0$; on $x \leq 0$, $f^{-1}(x) = -x$. $y = x^3$ and $y = \frac{1}{x}$ already pass, so no restriction is needed: their inverses are $\sqrt[3]{x}$ and $\frac{1}{x}$, the second on $x \neq 0$.

3 ... every output value then comes from exactly one input, so the reverse rule assigns exactly one input to each output in the range and is therefore a function. Equivalently, the reflection of the graph in $y = x$ then meets each vertical line at most once and so passes the vertical line test.

4 Complete the square: $f(x) = x^2 - 4x = (x - 2)^2 - 4$, a parabola with vertex $(2, -4)$, decreasing on $(-\infty, 2]$ and increasing on $[2, \infty)$. The two natural restrictions are $x \geq 2$ and $x \leq 2$. On $x \geq 2$: $y + 4 = (x - 2)^2$ with $x - 2 \geq 0$, so $x = 2 + \sqrt{y + 4}$ and $f^{-1}(x) = 2 + \sqrt{x + 4}$ with domain $x \geq -4$ and range $y \geq 2$. On $x \leq 2$: $x = 2 - \sqrt{y + 4}$, so $f^{-1}(x) = 2 - \sqrt{x + 4}$ with domain $x \geq -4$ and range $y \leq 2$. Check the first: $f(f^{-1}(x)) = \left(2 + \sqrt{x + 4} - 2\right)^2 - 4 = x$.

5 With $f(x) = \frac{2x + 1}{x - 1}$, defined for $x \neq 1$,

$$f(f(x)) = \frac{2\left(\frac{2x+1}{x-1}\right) + 1}{\frac{2x+1}{x-1} - 1} = \frac{\frac{5x+1}{x-1}}{\frac{x+2}{x-1}} = \frac{5x + 1}{x + 2}.$$

The result is not x: iterating this function does not return the input, so f is not its own inverse, unlike $f(x) = \frac{x + 3}{x - 1}$ in Exercise 2D. Solving $y = \frac{2x + 1}{x - 1}$ gives $x(y - 2) = y + 1$, so $f^{-1}(x) = \frac{x + 1}{x - 2}$ with domain $x \neq 2$; it is a different formula from f, and the check $f^{-1}(f(x)) = x$ holds for every $x \neq 1$.

6 $f(x) = 2x + 1$ is a line of gradient 2 through $(0, 1)$; $f^{-1}(x) = \frac{x - 1}{2}$ is a line of gradient $\frac{1}{2}$ through $(1, 0)$, the reflection of the first in $y = x$. They meet where $2x + 1 = x$, so $x = -1$: the single intersection is $(-1, -1)$, on the line $y = x$. In general, if (a, b) lies on both graphs then $f(a) = b$ and $f(b) = a$. If $a = b$ the point is on $y = x$. If $a \neq b$, the same two equations show that (b, a) also lies on both graphs, so intersections away from $y = x$ must occur in pairs interchanged by the reflection. The self-inverse function $f(x) = 3 - x$ shows this: its graph coincides with its inverse, and the pairs $(1, 2)$ and $(2, 1)$ are both on it.

7 Yes: any strictly increasing function is invertible, and the same holds for strictly decreasing ones. Suppose f is strictly increasing and $f(a) = f(b)$. If $a < b$ then strict increase gives $f(a) < f(b)$, contradicting equality; if $b < a$ then $f(b) < f(a)$, again a contradiction. Hence $a = b$, so f is one-to-one and its inverse exists. For strictly decreasing f the inequalities reverse at each step and the conclusion is the same, so strict monotonicity guarantees invertibility.

Monotonicity is sufficient but not necessary: $f(x) = \frac{1}{x}$ is one-to-one on $\mathbb{R} \setminus \{0\}$ although its domain splits into two branches on which it is decreasing separately. Conversely, a function that increases on one interval and decreases on another can fail badly: on $[-1, 1]$, $f(x) = x^2$ decreases for $x < 0$ and increases for $x > 0$, and each branch is one-to-one on its own, yet $f(-1) = f(1) = 1$, so two different inputs share an output and no single inverse exists. Each branch has to be inverted on its own restricted domain, which is why the restriction must be stated as part of the answer.

Modelling task · the shrinking shadow

1 In the side view the ray from the lamp that grazes the ball meets the wall at the edge of the shadow. The ball's centre is x from the wall and $2 - x$ from the lamp, and the two similar right triangles give the proportion $\frac{r}{0.5} = \frac{x}{2 - x}$, so

$$r(x) = \frac{0.5x}{2 - x}.$$

Reading it: as the ball approaches the lamp, $2 - x \to 0$ and the shadow grows without bound. The domain is $0 < x < 2$, the ball lying strictly between wall and lamp; at $x = 2$ the model is undefined and the physics breaks down.

2 Dividing, $\frac{0.5x}{2 - x} = \frac{-0.5(2 - x) + 1}{2 - x}$, so

$$r(x) = -0.5 + \frac{1}{2 - x} = -0.5 - \frac{1}{x - 2}.$$

Thus $a = -0.5$ and $b = 1$. Starting from $y = \frac{1}{x}$: reflect in the x-axis, translate 2 right, then translate 0.5 down. The asymptotes move to $x = 2$ and $r = -0.5$, and the branch used is the one to the left of the vertical asymptote, where r rises from -0.5 towards $+\infty$; on $0 < x < 2$ it runs from 0 up to $+\infty$, giving range $r > 0$, as a shadow of positive size requires.

3 From $r(2 - x) = 0.5x$: $2r - rx = 0.5x$, so $x(0.5 + r) = 2r$ and

$$x = \frac{2r}{0.5 + r} = \frac{4r}{1 + 2r}.$$

The inverse is valid on the physical domain because r is strictly increasing there: $r'(x) = \frac{1}{(2 - x)^2} > 0$ on $0 < x < 2$, so each shadow size corresponds to exactly one position. Its domain is $r > 0$ and its range $0 < x < 2$. Check: $r = 0.5$ gives $x = \frac{2}{1} = 1$, and $r(1) = 0.5$. The inverse answers the practical question directly: measure the shadow and recover the ball's distance, up to the approximation of the model.

4 First assumption: the lamp is a point source. A real bulb has a filament of finite size, so the shadow has a penumbra — a blurred rim — and the measured radius would be larger and less sharply defined than the model predicts, more so as the ball nears the lamp. Second assumption: the ball is a flat disc of radius 0.5 facing the lamp, with no allowance for the sphere's curvature or its reflective surface. A real sphere presents a slightly larger silhouette, enlarging the shadow, and a glossy ball reflects light into the shadow, brightening it and making the edge harder to locate. A third assumption, that the wall is flat and perpendicular to the axis, fails on a curved or angled surface, where the shadow becomes elliptical rather than circular. Finally the model treats the ball as fixed; any motion blurs the shadow.

The reading is comparative: each assumption changes the size, sharpness or shape of the shadow, so the model is reliable for locating the ball approximately and unreliable for deciding a shadow's exact edge.

Mixed practice 2

1 Domain $x \neq 2$, since the denominator vanishes there; range $y \neq 0$, since a fraction with numerator 6 never equals 0. $f(5) = \frac{6}{3} = 2$ and $f(-1) = \frac{6}{-3} = -2$.

2 From $y = x^2$: translate 3 left, stretch vertically by factor 2, reflect in the x-axis, translate 4 up. The vertex moves from $(0, 0)$ to $(-3, 4)$ and the parabola opens downward, twice as steep as the parent.

3 The inside $3x$ must equal the parent input -2, so $x = -\frac{2}{3}$; the output becomes $6 + 1 = 7$. The image is $\left(-\frac{2}{3}, 7\right)$.

4 $f(g(x)) = 2(x^2 + 3) - 1 = 2x^2 + 5$. $g(f(x)) = (2x - 1)^2 + 3 = 4x^2 - 4x + 1 + 3 = 4x^2 - 4x + 4$. The two composites are different, so composition is not commutative here.

5 $5 - 2x \geq 0$, so $x \leq \frac{5}{2}$; the domain is $\left(-\infty, \frac{5}{2}\right]$.

6 $y = \frac{4x - 1}{2}$ gives $2y = 4x - 1$, so $x = \frac{2y + 1}{4}$ and $f^{-1}(x) = \frac{2x + 1}{4}$. Check: $f\left(\frac{2x+1}{4}\right) = \frac{4 \cdot \frac{2x+1}{4} - 1}{2} = \frac{2x}{2} = x$.

7 Complete the square: $y = (x-4)^2 - 9$, a parabola with vertex $(4, -9)$. It is not one-to-one, since $f(3) = f(5) = -8$: the horizontal line $y = -8$ meets it twice. Restricting to $x \geq 4$, where the graph increases, makes it invertible; the restriction $x \leq 4$, where it decreases, works equally well. On $x \geq 4$ the inverse is $f^{-1}(x) = 4 + \sqrt{x+9}$ with domain $x \geq -9$.

8 $y = x^2 + 2$ with $x \geq 0$ gives $x^2 = y - 2$ and $x = \sqrt{y-2}$, so $f^{-1}(x) = \sqrt{x-2}$ with domain $x \geq 2$. Check: $f(f^{-1}(x)) = \left(\sqrt{x-2}\right)^2 + 2 = x$.

9 $y = \dfrac{1}{x+2}$ is the hyperbola $y = \dfrac{1}{x}$ translated 2 left. The vertical asymptote is $x = -2$ and the horizontal asymptote is $y = 0$. The left branch lies below the x-axis for $x < -2$; the right branch is above it for $x > -2$, passing through $(0, 0.5)$ and $(-1, 1)$.

10 $f^{-1}(x) = 4$ means the input 4 produces the output x: $x = f(4) = 3(4) + 5 = 17$. Check: $f^{-1}(17) = 4$ since $f(4) = 17$.

11 Write $9 - 3x = 3(3 - x)$, so $y = \sqrt{3(3-x)} = \sqrt{3}\sqrt{3-x}$. From $y = \sqrt{x}$: reflect in the y-axis, translate 3 right, then stretch vertically by factor $\sqrt{3}$. The endpoint is $(3, 0)$, the graph passes through $(0, 3)$ and the domain is $x \leq 3$.

12 The inner function differs, so the conditions differ. For $f(g(x))$, x must lie in the domain of g and $g(x)$ in the domain of f; for $g(f(x))$ the roles are exchanged. With $f(x) = \sqrt{x}$ and $g(x) = x - 5$: $f(g(x)) = \sqrt{x-5}$ needs $x \geq 5$, while $g(f(x)) = \sqrt{x} - 5$ needs only $x \geq 0$. Neither domain contains the other, so the two composites are different functions with different domains.

13 With vertex $(-1, 2)$: $y = a(x+1)^2 + 2$. Substituting $(3, -6)$: $-6 = 16a + 2$, so $16a = -8$ and $a = -\dfrac{1}{2}$. Hence $y = -\dfrac{1}{2}(x+1)^2 + 2$, opening downward with maximum 2 at $x = -1$.

14 The inside $2x - 6 = 2(x - 3)$ equals 4 when $x = 5$, and the output is $-10 + 1 = -9$. Because the outer sign is negative, this point is a minimum: $y = -f(2x-6) + 1 \geq -9$ for all x. So the image of the maximum is $(5, -9)$, a minimum of the new graph, which has no maximum.

15 From $y = \dfrac{2x+5}{3-x}$: $y(3-x) = 2x + 5$, so $3y - xy = 2x + 5$ and $x(2+y) = 3y - 5$, giving $x = \dfrac{3y-5}{y+2}$. Hence $f^{-1}(x) = \dfrac{3x-5}{x+2}$ with domain $x \neq -2$. Check: $f(0) = \dfrac{5}{3}$ and $f^{-1}\left(\dfrac{5}{3}\right) = \dfrac{5-5}{5/3+2} = 0$.

16 $P(n) = -2(n^2 - 60n) = -2\left[(n-30)^2 - 900\right] = -2(n-30)^2 + 1800$. The maximum profit is £1800, at $n = 30$ units, which lies inside $0 \leq n \leq 60$.

17 From $16 = k(40)^2$: $k = \dfrac{16}{1600} = 0.01$. At 60 km/h: $d = 0.01(3600) = 36$ m. For 50 m: $v^2 = \dfrac{50}{0.01} = 5000$, so $v = \sqrt{5000} = 70.7$ km/h (3 s.f.).

18
$$f(f(x)) = \frac{\dfrac{x}{x-1}}{\dfrac{x}{x-1} - 1} = \frac{\dfrac{x}{x-1}}{\dfrac{1}{x-1}} = x.$$
Applying f twice returns the input, so f is its own inverse: $f^{-1}(x) = \dfrac{x}{x-1}$, with domain $x \neq 1$. The output cannot be 1, since $\dfrac{x}{x-1} = 1$ has no solution, which agrees with the domain restriction of the inverse.

19 Let (a, b) lie on the graph of f. Then $f(a) = b$, and by the definition of the inverse $f^{-1}(b) = a$, so (b, a) lies on f^{-1}. Reflection in $y = x$ is exactly the map $(a, b) \mapsto (b, a)$, so it carries every point of f to a point of f^{-1} and, reversing the argument, every point of f^{-1} back to f. The two graphs are therefore mirror images, and the line $y = x$ is the mirror.

20 If one side is x, the other is $20 - x$, so the area is $A(x) = x(20-x) = -(x-10)^2 + 100$ on $0 < x < 20$. The maximum area is 100 cm^2, at $x = 10$, that is for a square of side 10 cm; the domain excludes 0 and 20 because a degenerate rectangle has no area.

21 Suppose $(f \circ g)(a) = (f \circ g)(b)$, that is $f(g(a)) = f(g(b))$. Since f is one-to-one, $g(a) = g(b)$. Since g is one-to-one, $a = b$. Hence equal outputs of $f \circ g$ force equal inputs, so $f \circ g$ is one-to-one.

22 Complete the square: $f(x) = \left(x + \dfrac{k}{2}\right)^2 + 4 - \dfrac{k^2}{4}$. The graph is a parabola with axis $x = -\dfrac{k}{2}$, so $f\left(-\dfrac{k}{2} + t\right) = f\left(-\dfrac{k}{2} - t\right)$ for every $t \neq 0$: two distinct inputs always share an output. Hence f is not one-to-one on $\mathbb{R}$ for any real value of k, and the answer is all real k. It becomes invertible on either restriction $x \geq -\dfrac{k}{2}$ or $x \leq -\dfrac{k}{2}$, however large or small k is.

Criterion A · Functions and their properties

1 Complete the square: $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4$. The rule is a polynomial, so the domain is $\mathbb{R}$. Since $(x-3)^2 \geq 0$, the output is at least -4, and every value from -4 upward is attained, so the range is $[-4, \infty)$. The vertex, a minimum point, is $(3, -4)$ on the axis of symmetry $x = 3$.

Marking guidance: 4 marks — 1 for the completed square, 1 for the domain $\mathbb{R}$, 1 for the range $[-4, \infty)$, 1 for the vertex $(3, -4)$. A range written $y \geq -4$ is acceptable; a range of $\mathbb{R}$, or a vertex read with the wrong sign, loses that mark.

2 Start from the vertex $(0, 0)$ of $y = |x|$. The inside $x - 3$ translates the graph 3 right; the factor 2 stretches it vertically by 2; the minus sign reflects it in the x-axis; the $+1$ translates it 1 up. The vertex moves to $(3, 1)$, the arms open downward with gradients ± 2, and the range is $(-\infty, 1]$.

Marking guidance: 3 marks — 1 for the translations 3 right and 1 up, 1 for the vertical stretch by 2 together with the reflection in the x-axis, 1 for the image $(3, 1)$ of the vertex and the downward orientation. Reading the shift as 3 left costs the first mark.

3 $f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 1$. $g(f(x)) = (3x-2)^2 + 1 = 9x^2 - 12x + 4 + 1 = 9x^2 - 12x + 5$. The two results differ, which illustrates that composition is not commutative.

Marking guidance: 4 marks — 2 for $3x^2 + 1$, 2 for $9x^2 - 12x + 5$. An answer of $3x^2 + 3$ (forgetting the -2) or $9x^2 + 5$ (forgetting the cross term) scores 1 of the 2 for that composite.

4 Write $y = \dfrac{x+4}{2x-3}$ and solve for x. Then $y(2x-3) = x + 4$, so $2xy - 3y = x + 4$, and collecting the x terms gives $x(2y-1) = 3y + 4$. Hence $x = \dfrac{3y+4}{2y-1}$, and interchanging the names
$$f^{-1}(x) = \frac{3x+4}{2x-1}, \qquad x \neq \tfrac{1}{2}.$$
The original function excludes $x = \dfrac{3}{2}$, and its range excludes $y = \dfrac{1}{2}$, which is exactly the domain restriction of the inverse. Check: $f(0) = -\dfrac{4}{3}$ and $f^{-1}\left(-\dfrac{4}{3}\right) = \dfrac{-4+4}{-8/3-1} = 0$.

Marking guidance: 5 marks — 1 for multiplying out and collecting the x terms, 1 for solving for x, 1 for the correct formula, 1 for the domain $x \neq \frac{1}{2}$, 1 for a numerical or algebraic check. Omitting the domain caps the response at 4; writing $\frac{1}{f(x)}$ as the inverse scores 0.

5 On $\mathbb{R}$, $y = x^2 + 4$ is not one-to-one: $f(2) = f(-2) = 8$, so the horizontal line $y = 8$ meets the graph twice and a reverse rule could not choose between 2 and -2. Its reflection would fail the vertical line test, so no inverse function exists on $\mathbb{R}$. Restrict to $x \leq 0$, where the graph decreases from $+\infty$ to 4 and is therefore one-to-one. Then $y - 4 = x^2$ with $x \leq 0$, so $x = -\sqrt{y-4}$ and
$$f^{-1}(x) = -\sqrt{x-4}, \qquad x \geq 4.$$
The domain of the inverse is the range of the restricted f, namely $[4, \infty)$, and its range is $(-\infty, 0]$. Check: $f\left(-\sqrt{x-4}\right) = \left(-\sqrt{x-4}\right)^2 + 4 = x$.

Marking guidance: 4 marks — 1 for a valid reason that f is not one-to-one (a numerical pair or the horizontal line test), 1 for choosing $x \leq 0$ and taking the negative root, 1 for $f^{-1}(x) = -\sqrt{x-4}$, 1 for the domain $x \geq 4$. Giving $+\sqrt{x-4}$ shows the wrong branch and loses the root mark.

Criterion B · Iterating a function

1 With $f(x) = 2x + 1$: $f^{(2)}(x) = 2(2x + 1) + 1 = 4x + 3$; $f^{(3)}(x) = 2(4x + 3) + 1 = 8x + 7$; $f^{(4)}(x) = 2(8x + 7) + 1 = 16x + 15$. Checking one value: $f^{(2)}(1) = f(3) = 7$ and $4(1) + 3 = 7$.

Marking guidance: 6 marks — 2 for each iterate correctly formed by substituting the previous one. Adding 1 twice, or squaring instead of composing, loses the marks for that iterate; correct answers with no substitution working score at most 4.

2 For $n = 1, 2, 3, 4$ the coefficient of x is $2, 4, 8, 16$ and the constant is $1, 3, 7, 15$: the coefficient doubles each time and the constant is one less than the coefficient. Equivalently the constant doubles and 1 is added, so it also follows $2^n - 1$. The step from one iterate to the next is itself doubling, the signature of a geometric pattern rather than a linear one.

Marking guidance: 4 marks — 1 for the doubling coefficient, 1 for the coupling "constant = coefficient -1" (or $2^n - 1$), 1 for expressing the pattern in terms of n, 1 for evidence that it was checked on more than one iterate. "The numbers get bigger" earns nothing.

3 Prediction: $f^{(n)}(x) = 2^n x + (2^n - 1)$. It is plausible because each application doubles the coefficient and maps a constant c to $2c + 1$, so after n steps the constant is $1 + 2 + 4 + \ldots + 2^{n-1} = 2^n - 1$. Test $n = 5$: the formula gives $32x + 31$, while $f^{(5)}(x) = 2(16x + 15) + 1 = 32x + 31$, so the prediction holds. It also holds at $n = 1$, giving $2x + 1$.

Marking guidance: 5 marks — 2 for the formula, 2 for an independent test at $n = 5$ (substituting into $f^{(4)}$, not merely restating the formula), 1 for confirming $n = 1$ as the base case. A formula that fits $n = 2$ and $n = 3$ but is never tested earns 2.

4 The constant is $2^n - 1$, so the first time it exceeds 1000 is given by $2^n - 1 > 1000$, that is $2^n > 1001$. Then $n > \log_2 1001 = \dfrac{\ln 1001}{\ln 2} = \dfrac{6.9088}{0.6931} = 9.97$, so the least integer is $n = 10$. Check: $f^{(9)}$ has constant $2^9 - 1 = 511$ and $f^{(10)}$ has $1024 - 1 = 1023$, so $n = 10$ is the first.

Marking guidance: 4 marks — 1 for the inequality $2^n - 1 > 1000$, 1 for the logarithm step, 1 for rounding up to 10, 1 for checking $n = 9$ and $n = 10$. An answer of 9.97 or 9 shows the inequality was not interpreted for whole n.

5 For $g(x) = 3x$: $g^{(2)}(x) = 9x$, $g^{(3)}(x) = 27x$, and in general $g^{(n)}(x) = 3^n x$. There is no constant term at all, so no additive pattern appears. The reason is that $g(0) = 0$: the origin is fixed, and multiplying by 3 never creates an additive part. The function f has $f(0) = 1$, so each application doubles the previous constant and adds 1, producing the sum $1 + 2 + \ldots + 2^{n-1}$.

Marking guidance: 3 marks — 1 for $g^{(n)}(x) = 3^n x$ with the constant staying 0, 1 for the reason $g(0) = 0$ (the origin is fixed), 1 for the contrast with $f(0) = 1$, which generates the accumulated constant.

6 Iteration composes a function with itself, so $f^{(n)}$ applies f n times; its inverse applies f^{-1} n times, that is $(f^{-1})^{(n)}$. Solving $f^{(n)}(x) = c$ therefore means applying f^{-1} exactly n times to c. For $f(x) = 2x + 1$, $f^{-1}(x) = \dfrac{x - 1}{2}$, and undoing n steps from 1023 returns the starting value in n halvings and subtractions.

Marking guidance: 2 marks — 1 for stating that the inverse of the iterate is the iterate of the inverse, 1 for linking this to solving $f^{(n)}(x) = c$ by repeated application of f^{-1}.

Criterion C · Explaining the order of transformations

1 **Sample response.** I can see why the $\frac{1}{2}$ and the 6 look right, because both numbers really do appear in $2x - 6$. The trouble is that they are read at the wrong stage. Factor the inside first:

$$2x - 6 = 2(x - 3).$$

The multiplier 2 is the horizontal compression by factor $\frac{1}{2}$; the number that survives outside the multiplier is 3, and that is the shift. So the graph of $y = f(2x - 6)$ is the graph of $y = f(x)$ compressed horizontally by factor $\frac{1}{2}$, then translated **3** right — not 6.

Test it on $y = x^2$ with the parent point $(1, 1)$, whose image must satisfy $(2x - 6)^2 = 1$, that is $2x - 6 = \pm 1$, so $x = 3.5$ or $x = 2.5$. Take the branch matching the sign used below: the required image of $(1, 1)$ is $(3.5, 1)$. Now follow your recipe. Compressing by $\frac{1}{2}$ sends $(1, 1)$ to $(0.5, 1)$; shifting 6 right then sends it to $(6.5, 1)$. Check on the equation: $(2(6.5) - 6)^2 = 7^2 = 49 \neq 1$, so $(6.5, 1)$ is nowhere near the graph. Following the correct order instead, $(0.5, 1)$ shifted 3 right becomes $(3.5, 1)$, and $(2(3.5) - 6)^2 = 1^2 = 1$. ✓

Why does the size of the shift change? Because the compression acts on the coordinate, and a translation of 6 performed *after* it is measured on the compressed axis, where the same displacement of the parent graph is only 3. Your order builds $y = f\left(2\left(\frac{x}{2} - 6\right)\right)$, which simplifies to $y = f(x - 12)$ — a plain translation, with the compression cancelled by the doubled shift. The two operations do not commute: applying them in the other sequence, shift 6 right first and then compress every x-coordinate by $\frac{1}{2}$, sends $(1, 1)$ to $(7, 1)$ and then to $(3.5, 1)$, which *is* correct. The 6 is the shift measured before the compression; the 3 is the same movement measured after it.

The same reversal explains the signs. In $y = f(x - 3)$ the height shown at x is the parent height from 3 units to the left, so the picture moves right even though the sign is negative. In $y = f(2x)$ each parent height is reached at half the input, so the graph is squeezed by $\frac{1}{2}$ although the multiplier is 2. Outside the function there is no reversal: in $3f(x) + 1$ the output is trebled and then raised by 1, exactly as written.

General rule for $y = af(b(x - h)) + k$: factor the inside before reading anything from it. Then apply, in this order — horizontal compression by factor $\frac{1}{b}$ (reflection in the y-axis if $b < 0$), translation h right; vertical stretch by factor $|a|$ (reflection in the x-axis if $a < 0$), translation k up. For $y = f(2x - 6)$: $b = 2$, $h = 3$, $a = 1$, $k = 0$.

Marking guidance: 20 marks, typically 4 for factoring to $2(x - 3)$ and stating the correct order with $b = 2$, $h = 3$; 5 for the numerical demonstration, which must test the classmate's own recipe and show where it lands (the graph of $f(2x - 12)$, not $f(2x - 6)$); 5 for the causal explanation of why the shift is rescaled and why horizontal changes reverse their sign; 4 for a general rule that names all four parameters and factors the inside; 2 for notation, clarity and a diagram or table where used. A response that merely asserts "the shift is 3, not 6" with no numerical evidence cannot reach the top band, however correct it is.

Criterion D · Modelling a theme-park ride

1 $h(0) = 60 - 5(0)^2 = 60$, so the height at the instant of release is 60 m. On $0 \leq t \leq 2\sqrt{3}$ the term $5t^2$ only grows, so h only falls: the maximum height is 60 m and it occurs at $t = 0$ s, the moment of release. The vertex of the parabola $h = 60 - 5t^2$ is $(0, 60)$, which is why the model describes the fall alone and not the lift.

Marking guidance: 3 marks — 1 for $h(0) = 60$, 1 for maximum 60 m, 1 for the time $t = 0$. Award the third mark only if the student notices that the maximum is at the start rather than searching for a later turning point; a response that differentiates and "finds" a maximum at some positive time has misread the model.

2 Set $60 - 5t^2 = 40$: then $5t^2 = 20$, $t^2 = 4$ and $t = \pm 2$. Both are algebraic solutions of the equation, and only $t = 2$ lies in the model's domain $0 \le t \le 2\sqrt{3}$ (note $2\sqrt{3} \approx 3.46$). Interpreted: the carriage passes 40 m exactly once during the fall, 2 s after release. The rejected root $t = -2$ is the mirror image on the same parabola continued backwards in time — it is the instant at which a carriage *rising* along the mirrored branch would stand at 40 s — a reading the ride never performs, since before $t = 0$ the carriage was being lifted, not falling. The two roots are therefore evidence that a quadratic is many-to-one: a height does not determine a time until the physical branch is chosen.

Marking guidance: 5 marks — 2 for forming and solving $60 - 5t^2 = 40$, 1 for both roots including the negative one, 1 for rejecting $t = -2$ on domain grounds, 1 for interpreting what the extra root would mean rather than simply discarding it as "wrong". Discarding -2 only because "time cannot be negative", with no reference to the model's domain, scores 4.

3 Write $h = 60 - 5t^2$. Then $5t^2 = 60 - h$, so $t^2 = \dfrac{60 - h}{5}$ and

$$t = \sqrt{\frac{60 - h}{5}}$$

taking the positive root, since $t \ge 0$. The inverse converts a measured height into the elapsed time since release — the calculation an operator needs when a sensor reports a height and the control system must know where in the fall the carriage is. The domain of the inverse is the range of the original model, $0 \le h \le 60$: $h = 60$ at $t = 0$ and $h = 60 - 5(2\sqrt{3})^2 = 60 - 60 = 0$ at $t = 2\sqrt{3}$. A domain restriction must be stated because the unrestricted parabola $h = 60 - 5t^2$ on $\mathbb{R}$ fails the horizontal line test — t and $-t$ give the same height, so solving for t returns two values — and only becomes invertible once the branch $t \ge 0$ is kept. With the printed domain the model is strictly decreasing and therefore one-to-one, which is exactly why a single square root, not a $\pm$, is the answer.

Marking guidance: 6 marks — 2 for the algebra giving $t = \sqrt{(60 - h)/5}$, 1 for choosing the positive root with a reason, 1 for the domain $0 \le h \le 60$ obtained from the model's range, 1 for explaining the horizontal line test / the $\pm$ ambiguity, 1 for saying what the inverse calculates in context. A formula with no domain and no interpretation caps the response at 3.

4 **Assumption 1 — the carriage is released from rest and falls with constant acceleration, with no drag or friction.** The coefficient -5 is $-\frac{1}{2}g$ with $g = 10\ \text{m s}^{-2}$, so the model asserts free fall throughout. In reality air resistance and guide friction grow with speed, so the acceleration is less than $10\ \text{m s}^{-2}$ as the fall proceeds, the carriage takes longer to reach any given height, and the real graph lies above the model after the first second.

Assumption 2 — nothing decelerates the carriage and the ground is at $h = 0$. The model runs to $h = 0$ at $t = 2\sqrt{3} \approx 3.46$ s with speed $|dh/dt| = 10t = 34.6\ \text{m s}^{-1} \approx 125\ \text{km h}^{-1}$, i.e. it predicts an unrestrained arrival at ground level at full speed. A real drop tower brakes magnetically in its last metres, so the true graph flattens sharply near $h = 0$ and the carriage stops; times read from the model in the final second are too early.

A third, smaller assumption is that the carriage is a single point at height h; a real train has length, so front and rear rows pass a given height at different instants.

Marking guidance: 5 marks — 2 for two genuine assumptions (not restatements of the formula), 2 for the direction of each effect on the motion or on the graph, 1 for quantifying at least one effect (a speed, a time, or how the curve is displaced). An assumption named with no stated consequence scores 1 each.

5 A quadratic is the right family for the part of the ride the model claims to describe. Constant acceleration from rest gives $h = h_0 - \frac{1}{2}gt^2$, and $-\frac{1}{2}g = -5$ with $g = 10\ \text{m s}^{-2}$ is a defensible rounded value, so the shape of the curve is derived from physics rather than fitted: the graph starts at $(0, 60)$ with zero slope — the correct picture of a carriage just let go from the top — and falls faster and faster, which is what riders feel. The domain is also internally consistent: it stops precisely where the fall ends, at $h = 0$.

The model is unreasonable as a description of the ride as a whole. Its impact speed of about $34.6\ \text{m s}^{-1}$ ($\approx 125\ \text{km h}^{-1}$) after a 60 m unrestrained fall is far beyond what any passenger ride allows, so the absence of braking is not a small detail but the model's largest error; the last metre of the curve is the one part of the ride it gets most wrong. Real towers also release in stages, with the first fraction of a second gentler than free fall as the magnets release, so the measured curve starts below the model and ends well above it. And because the parabola is symmetric about $t = 0$, extending it even slightly off-domain predicts a rise before release, which is the lift, not the fall.

Verdict. A good model of the first 1–1.5 s of the drop and of the maximum speed the ride must arrest; not a model of the whole ride. For a design or safety calculation the quadratic should be replaced by a piecewise description — gentle release, free fall, magnetic deceleration — and used only for checking where braking must begin.

Marking guidance: 5 marks — 2 for justifying the quadratic by constant acceleration with the numerical link $\frac{1}{2}g = 5$, 2 for identifying where it fails and quantifying at least one failure (the impact speed, the missing brake, the release profile), 1 for a balanced verdict that states the range over which the model is usable rather than accepting or rejecting it outright. Praise or criticism with no mathematical support scores no more than 2.

CHAPTER 3 · EXPONENTIAL AND LOGARITHMIC FUNCTIONS

Opening problem · the cooling cup

1 $T(0) = 20 + 75(1) = 95^\circ\text{C}$; $T(5) = 20 + 75(0.88)^5 = 20 + 75(0.5277) = 59.6^\circ\text{C}$; $T(15) = 20 + 75(0.88)^{15} = 20 + 75(0.1471) = 31.0^\circ\text{C}$. The room temperature is the horizontal asymptote, 20°C, since $0.88^t \to 0$ as $t \to \infty$.

2 Because $0 < 0.88 < 1$, each minute the excess $T - 20$ is multiplied by 0.88, a 12% decrease, so the coffee cools. Since $75(0.88)^t > 0$ for every real t, $T = 20 + 75(0.88)^t > 20$ always: the graph approaches $T = 20$ from above but never reaches or crosses it.

3 From a table, $T(4) = 65.0^\circ\text{C}$ and $T(5) = 59.6^\circ\text{C}$, so the coffee first falls below 60°C between 4 and 5 minutes; solving $75(0.88)^t = 40$ gives $t = 4.92$ min. Also $T(15) = 31.0^\circ\text{C}$ and $T(16) = 29.7^\circ\text{C}$, so it falls below 30°C between 15 and 16 minutes; solving $75(0.88)^t = 10$ gives $t = 15.8$ min. The pleasant window is roughly 4.9 to 15.8 minutes, about 11 minutes long.

4 The model assumes the room stays at exactly 20°C, that the coffee is unstirred and uniform in temperature, that no milk or sugar is added, and that the cup loses heat at a rate proportional to the excess (a constant factor 0.88 each minute). A warming room, a lid, a cold cup or a draught would all change the factor. The model is still useful: it reproduces the readings closely and locates the window to within a minute, and its purpose is prediction over a limited range, not exact description.

The factor describes the temperature difference, not the temperature: $T - 20$, not T, is what is multiplied by 0.88 each minute.

Prerequisite check

1 $2^5 = 32$, $3^4 = 81$, $5^0 = 1$, $2^{-3} = \dfrac{1}{2^3} = \dfrac{1}{8}$.

2 $x^3 \cdot x^4 = x^{3+4} = x^7$; $\dfrac{x^7}{x^3} = x^{7-3} = x^4$; $(x^2)^5 = x^{2\times 5} = x^{10}$.

3 $32 = 2^5$, so $2^x = 2^5$ and $x = 5$.

4 $200 \times 1.07 = 214$, then $214 \times 0.93 = 199.02$. The result is below 200 because the 7% decrease is taken from the larger amount: the combined multiplier is $1.07 \times 0.93 = 0.9951$, a 0.49% net decrease.

5 $0.85 = 1 - 0.15$, a decrease of 15%.

6 Divide by 3: $2^x = \dfrac{y}{3}$, so x is the power to which 2 must be raised: $x = \log_2\left(\dfrac{y}{3}\right)$, defined for $y > 0$. (Logarithms are developed in Section 3.3.)

Exercise 3A · Growth factors, graphs and models

1 (a) Initial value 15, factor 1.3, growth of 30% per unit. (b) Initial value 200, factor 0.82, decay of 18%. (c) Initial value 4, factor 2.5, growth of 150%. (d) Initial value 0.5, factor 0.95, decay of 5%. (e) $7(4)^{-x} = 7\left(\frac{1}{4}\right)^x$: initial value 7, factor 0.25, decay of 75%.

2 (a) $3(2)^4 = 3(16) = 48$. (b) $100(0.9)^3 = 100(0.729) = 72.9$. (c) $5(1.5)^{-2} = \dfrac{5}{2.25} = \dfrac{20}{9} \approx 2.22$. (d) $2(0.5)^5 = 2\left(\dfrac{1}{32}\right) = \dfrac{1}{16} = 0.0625$.

3 (a) $1 + 0.08 = 1.08$. (b) $1 - 0.12 = 0.88$. (c) $1 + 0.005 = 1.005$. (d) $1 - 0.33 = 0.67$.

4 $P(t) = 12\,000(1.04)^t$. $P(6) = 12\,000(1.04)^6 = 12\,000(1.26532) = 15\,184$ people (nearest whole number).

5 24 days is $24 \div 8 = 3$ half-lives, so the fraction is $\left(\frac{1}{2}\right)^3 = \dfrac{1}{8}$. 40 days is 5 half-lives, so the fraction is $\left(\frac{1}{2}\right)^5 = \dfrac{1}{32}$.

6 Both graphs pass through $(0, 1)$ and have the x-axis as asymptote, with range $(0, \infty)$. $y = 2^x$ rises through $(1, 2)$ and approaches $y = 0$ as $x \to -\infty$; $y = 2^{-x} = \left(\frac{1}{2}\right)^x$ falls through $(-1, 2)$ and approaches $y = 0$ as $x \to +\infty$. Each is the reflection of the other in the y-axis, so the asymptote $(y = 0)$ and the y-intercept $((0, 1))$ are the same for both.

7 The student has read the base as the rate. The multiplier is $0.7 = 1 - 0.3$, so the decrease is 30% per unit, not 7%. A 7% decrease would have factor 0.93.

8 $0.5^{-x} = \left(\frac{1}{2}\right)^{-x} = 2^x$ by the index law $\left(\frac{1}{b}\right)^{-x} = b^x$. The two rules are therefore identical for every x, so the graphs coincide.

9 Each value is double the one before, so the ratio of successive values is constant: $10/5 = 20/10 = 40/20 = 80/40 = 2$. A constant ratio, not a constant difference, is the signature of exponential growth, with factor $b = 2$. Taking $x = 0$ at the first value gives $a = 5$, so $y = 5(2)^x$.

10 (a) The whole graph is stretched vertically by factor 2: every output and the y-intercept double, the asymptote $y = 0$ is unchanged. (b) $(b^2)^x = b^{2x}$, so each unit of x now multiplies the output by b twice; equivalently the graph of $y = ab^x$ is compressed horizontally by factor $\frac{1}{2}$. For $b > 1$ the doubling time is halved. (c) Adding 4 translates the graph 4 units up: the asymptote becomes $y = 4$, the y-intercept becomes $a + 4$, and the range becomes $(4, \infty)$.

11 An exponential with $b > 1$ has a constant ratio b between values one step apart, so doubling the input squares the output; a quadratic has constant second difference, and the ratio of consecutive values tends to 1. Comparing 2^x with $100x^2$: at $x = 10$, $2^{10} = 1024$ but $100(10)^2 = 10\,000$; at $x = 20$, $2^{20} = 1\,048\,576$ but $100(20)^2 = 40\,000$; at $x = 30$, $2^{30} = 1\,073\,741\,824$ but $100(30)^2 = 90\,000$. Once the exponential overtakes the quadratic it stays ahead, because $\dfrac{2^x}{100x^2} \to \infty$; no choice of leading coefficient prevents this.

12 Reflecting in the y-axis replaces x by $-x$: $y = 4(0.8)^{-x} = 4\left(\frac{1}{0.8}\right)^x = 4(1.25)^x$. The result is exponential growth of 25% per unit, passing through $(0, 4)$ and rising, with asymptote $y = 0$ on the left.

13 Value $= 32\,000(0.82)(0.88)^3 = 32\,000(0.82)(0.681472) = 17\,881.83$, so £17 882 (nearest pound). The first year uses the 18% factor and only the remaining three years use 12%.

14 After 18 h, $500\left(\frac{1}{2}\right)^{18/5} = 500(0.08247) = 41.2$ mg, still above 40 mg. After 19 h, $500\left(\frac{1}{2}\right)^{19/5} = 500(0.07177) = 35.9$ mg, below 40 mg. The exact crossing is $t = 5\dfrac{\ln 0.08}{\ln 0.5} = 18.2$ h, so the first whole hour at which less than 40 mg remains is 19 hours.

15 Let the annual factor be b. Five years give $b^5 = 1.2$, so $b = 1.2^{1/5} = e^{(\ln 1.2)/5} = e^{0.036464} = 1.03714$. The annual rate is 3.71% (2 d.p.). Check: $1.03714^5 = 1.2000$.

The rate is not $20\% \div 5 = 4\%$: the 20% itself compounds over the five years.

Exercise 3B · The number e and continuous models

1 (a) $e^2 = 7.389$. (b) $e^{-1} = 0.3679$. (c) $e^{0.5} = 1.649$. (d) $e^0 = 1.000$.

2 (a) $y = 200e^{0.03t}$. (b) $y = 50e^{-0.04t}$. (c) $y = 8e^{0.015t}$.

3 Doubling time $= \dfrac{\ln 2}{k}$. (a) $\dfrac{0.6931}{0.02} = 34.7$. (b) $\dfrac{0.6931}{0.07} = 9.90$. (c) $\dfrac{0.6931}{0.12} = 5.78$.

4 Half-life $= \dfrac{\ln 2}{-k}$. (a) $\dfrac{0.6931}{0.05} = 13.9$. (b) $\dfrac{0.6931}{0.2} = 3.47$. (c) $\dfrac{0.6931}{0.011} = 63.0$.

5 $1500e^{0.04\times 10} = 1500e^{0.4} = 1500(1.49182) = 2237.74$.

6 Monthly rate $\dfrac{0.035}{12} = 0.0029167$, with $12 \times 6 = 72$ periods: $10\,000(1.0029167)^{72} = 10\,000(1.23330) = 12\,333.01$.

7 The student has replaced e by 2.7 before exponentiating, which changes the base: $2.7^{0.4} = 1.48780$ while $e^{0.4} = 1.49182$. The correct value is $2000e^{0.4} = 2983.65$; the substitution gives 2975.61 an error of about 8 because the rounding is d by the ex

8 Doubling means $P_0e^{kt} = 2P_0$. Since $P_0 > 0$, divide by it: $e^{kt} = 2$, so $kt = \ln 2$ and $t = \dfrac{\ln 2}{k}$. The starting value has cancelled, so the doubling time depends only on k.

9 Not exactly. The annual multiplier is $e^{0.08} = 1.08329$, so the population grows by 8.33% per year, not 8%. The 8% is the continuous (instantaneous) rate k; applying 8% once a year gives 1.08, which is smaller than $e^{0.08}$.

10 Compounding n times a year at annual rate r gives $\left(1+\frac{r}{n}\right)^{nt}$. As $n\to\infty$ this expression tends to e^{rt}, which is the same interest calculated in the limit of ever-shorter periods. Continuous compounding is therefore the limiting case of ordinary compound interest, not a separate financial product; the interest rate is unchanged, only the frequency of application tends to infinity.

11 $0.7=e^{12k}$, so $k=\dfrac{\ln 0.7}{12}=\dfrac{-0.35667}{12}=-0.0297$, a continuous decay rate of 2.97% per year. Half-life $=\dfrac{\ln 2}{0.0297}=23.3$ years.

12 Write $2^{x/7}$ with base e: $2^{x/7}=e^{(x/7)\ln 2}=e^{0.09902x}$, whereas $e^{0.1x}$ has exponent $0.1x$. Since $0.1>0.09902$, $e^{0.1x}$ grows faster for large x. Their ratio is $e^{(0.1-0.09902)x}=e^{0.00098x}$, which increases without bound, though the difference is small for moderate x.

13 $P(3)=400e^{0.54}=400(1.71601)=686.4$, so 686 fish. For 1000: $e^{0.18t}=2.5$, so $t=\dfrac{\ln 2.5}{0.18}=\dfrac{0.91629}{0.18}=5.09$ years.

14 $A=80e^{-0.18t}$. After 4 h, $A=80e^{-0.72}=80(0.48675)=38.9$ mg. For 10 mg: $e^{-0.18t}=0.125$, so $t=\dfrac{\ln 8}{0.18}=\dfrac{2.07944}{0.18}=11.6$ hours.

15 Doubling in time T means $Ae^{kT}=2A$, so $e^{kT}=2$ and $kT=\ln 2$, giving $k=\dfrac{\ln 2}{T}$. Then

$$e^{kt}=e^{(\ln 2)t/T}=\left(e^{\ln 2}\right)^{t/T}=2^{t/T},$$

so $y=Ae^{kt}=A\,2^{t/T}$. Check with $A=800$, $T=3$: $k=\ln 2/3=0.2310$, and $800e^{0.2310t}=800\cdot 2^{t/3}$.

Exercise 3C · Logarithms, laws and change of base

1 (a) $\log_2 64=6$. (b) $\log_{10}0.01=-2$. (c) $\log_9 3=\frac{1}{2}$. (d) $\ln 1=0$.

2 (a) $3^3=27$. (b) $10^3=1000$. (c) $2^{-3}=\frac{1}{8}$. (d) $e^0=1$.

3 (a) $4^2=16$, so $\log_4 16=2$. (b) $5^3=125$, so $\log_5 125=3$. (c) $\frac{1}{16}=2^{-4}$, so $\log_2\left(\frac{1}{16}\right)=-4$. (d) $27^{1/3}=3$, so $\log_{27}3=\frac{1}{3}$. (e) $8^0=1$, so $\log_8 1=0$.

4 (a) $\log_2 5+\log_2 7=\log_2 35$. (b) $\log_3 54-\log_3 2=\log_3 27=3$. (c) $3\log_2 2=3(1)=3$. (d) $\log 4+\log 25=\log 100=2$.

5 (a) $\log_3 10=\dfrac{\ln 10}{\ln 3}=\dfrac{2.30259}{1.09861}=2.096$. (b) $\log_6 200=\dfrac{5.29832}{1.79176}=2.957$. (c) $\log_{12}5=\dfrac{1.60944}{2.48491}=0.6477$.

6 (a) $x=2^5=32$. (b) $x=5^0=1$. (c) $x^2=49$, so $x=7$ (the base must be positive and not 1, so $x=-7$ is rejected).

7 The first line is correct: $8=2\times 4$, so the product law gives $\log_2 8=\log_2 2+\log_2 4=1+2=3$. The second line is wrong because $10\neq 5\times 5$; the split $10=5+5$ is a sum, and there is no law for the logarithm of a sum. Correctly, $\log_2 10=3.32$, whereas $2\log_2 5=\log_2 25=4.64$.

8 $\log_b 0$ would be the power y with $b^y=0$. No power of a positive base is zero, since $b^y>0$ for all real y, so there is no such y. Consistently, $\log_b x\to-\infty$ as $x\to 0^+$, so the graph has a vertical asymptote at $x=0$.

9 $\log_b\left(\dfrac{1}{x}\right)=\log_b(x^{-1})=-1\cdot\log_b x=-\log_b x$, using the power law with $n=-1$.

10 $y=\log_2 x$ means $2^y=x$. As $x\to 0^+$, the power y must tend to $-\infty$, so the graph falls without bound and $x=0$ is a vertical asymptote. Also $2^0=1$ gives the point $(1,0)$ and $2^1=2$ gives $(2,1)$; the graph is the reflection of $y=2^x$ in $y=x$, so its domain is $x>0$ and its range is $\mathbb{R}$.

11 Let $y=\log_b x$, so $b^y=x$. Take $\log_c$ of both sides: $\log_c(b^y)=\log_c x$, so $y\log_c b=\log_c x$ by the power law. Since $b\neq 1$, $\log_c b\neq 0$, and dividing gives $y=\dfrac{\log_c x}{\log_c b}$, that is $\log_b x=\dfrac{\log_c x}{\log_c b}$.

12 Compare powers. $7^5=16\,807>2^{14}=16\,384$, so $5\log_2 7>14$ and $\log_2 7>2.8$. Also $20^5=3\,200\,000<3^{14}=4\,782\,969$, so $5\log_3 20<14$ and $\log_3 20<2.8$. Hence $\log_2 7>2.8>\log_3 20$, so $\log_2 7$ is larger (2.807 against 2.727).

13 $L=10\log_{10}\left(\dfrac{10^{-5}}{10^{-12}}\right)=10\log_{10}(10^7)=10(7)=70$ dB.

14 $\text{pH}=-\log_{10}(2.5\times 10^{-4})=-(-3.602)=3.60$. For pH 8.3, $\log_{10}[\text{H}^+]=-8.3$, so $[\text{H}^+]=10^{-8.3}=5.0\times 10^{-9}$ mol/L (2 s.f.).

15 Change all three logarithms to natural logarithms: $\log_a b=\dfrac{\ln b}{\ln a}$, $\log_b c=\dfrac{\ln c}{\ln b}$, $\log_c a=\dfrac{\ln a}{\ln c}$. The product is $\dfrac{\ln b}{\ln a}\cdot\dfrac{\ln c}{\ln b}\cdot\dfrac{\ln a}{\ln c}=1$, since every factor cancels. The result is valid because $a,b,c>0$ and none equals 1, so no denominator is zero.

Exercise 3D · Exponential and logarithmic equations

1 (a) $2^x=2^6$, so $x=6$. (b) $3^{x+1}=3^4$, so $x+1=4$ and $x=3$. (c) $5^{2x}=5^3$, so $2x=3$ and $x=\frac{3}{2}$. (d) $9^x=3^{2x}$ and $27=3^3$, so $2x=3$ and $x=\frac{3}{2}$.

2 (a) $x=\dfrac{\ln 30}{\ln 2}=\dfrac{3.4012}{0.6931}=4.91$. (b) $x=\dfrac{\ln 100}{\ln 5}=\dfrac{4.6052}{1.6094}=2.86$. (c) $x-2=\dfrac{\ln 50}{\ln 7}=2.0104$, so $x=4.01$. (d) $x=\dfrac{\ln 2}{\ln 1.08}=\dfrac{0.6931}{0.07696}=9.01$.

3 (a) $x=\ln 12=2.48$. (b) $2x=\ln 5$, so $x=0.805$. (c) $-x=\ln 0.4$, so $x=-\ln 0.4=0.916$. (d) $x+1=\log_{10}250=2.3979$, so $x=1.40$.

4 (a) $x=2^4=16$. (b) $x+2=3^2=9$, so $x=7$. (c) $x=e^{1.5}=4.48$. (d) $2x-1=5^2=25$, so $x=13$.

5 Take natural logs: $(x+1)\ln 3=x\ln 4$, so $x\ln 3+\ln 3=x\ln 4$ and $x(\ln 4-\ln 3)=\ln 3$. Hence

$$x=\frac{\ln 3}{\ln(4/3)}=\frac{1.09861}{0.28768}=3.82\quad(3\text{ s.f.}).$$

Check: $3^{4.819}=199.2$ and $4^{3.819}=199.2$.

6 Let $u=2^x$, so $2^{2x}=u^2$: $u^2-6u+8=0$, $(u-2)(u-4)=0$, so $u=2$ or $u=4$. Both are positive, so $2^x=2$ gives $x=1$ and $2^x=4$ gives $x=2$.

7 The algebra gives $x(x-2)=2^3=8$, so $x^2-2x-8=0$ and $(x-4)(x+2)=0$, that is $x=4$ or $x=-2$. The domain requires $x>0$ and $x-2>0$, so $x>2$; $x=-2$ makes both arguments negative and must be rejected. Only $x=4$ is valid: $\log_2 4+\log_2 2=2+1=3$.

8 $\log_b$ is strictly increasing on $x>0$, so it is one-to-one: equal inputs give equal outputs and equal outputs force equal inputs. Applying it to both sides of a true equality therefore preserves the solution set, and exponentiating reverses it. Squaring is not one-to-one, since $(-3)^2=3^2$, so it can create solutions of the squared equation that do not satisfy the original.

9 With $u=e^x$: $u^2-7u+10=0$, $(u-2)(u-5)=0$, so $u=2$ or $u=5$. Both roots are positive, and e^x takes every positive value exactly once, so both are admissible: $x=\ln 2=0.693$ and $x=\ln 5=1.609$.

10 Domain: $x+1>0$ and $x-1>0$, so $x>1$. Then $\log\left(\dfrac{x+1}{x-1}\right)=1$, so $\dfrac{x+1}{x-1}=10$. Thus $x+1=10x-10$, $11=9x$ and $x=\dfrac{11}{9}=1.222$, which satisfies $x>1$. Check: $\log\dfrac{20/9}{2/9}=\log 10=1$.

11 $2^x=3^{x-1}$ gives $x\ln 2=(x-1)\ln 3=x\ln 3-\ln 3$. So $x(\ln 3-\ln 2)=\ln 3$ and

$$x=\frac{\ln 3}{\ln(3/2)}=\frac{1.09861}{0.40547}=2.71.$$

Check: $2^{2.7095}=6.545$ and $3^{1.7095}=6.545$.

12 For $a>1$, a^x is strictly increasing; for $0<a<1$ it is strictly decreasing. Either way it is one-to-one, so it cannot take the same value b twice, and its range is $(0,\infty)$, so every $b>0$ is attained. Hence there is exactly one solution, $x=\log_a b$.

13 Solve $5000(1.045)^n>8000$, that is $1.045^n>1.6$. Taking logs, $n>\dfrac{\ln 1.6}{\ln 1.045}=\dfrac{0.47000}{0.04402}=10.68$. Check: $5000(1.045)^{10}=7764.85<8000$ and $5000(1.045)^{11}=8114.27>8000$. So 11 whole years are required.

14 For 2000: $e^{0.028t}=\dfrac{2000}{1200}=\dfrac{5}{3}$, so $t=\dfrac{\ln(5/3)}{0.028}=\dfrac{0.51083}{0.028}=18.2$ years. Doubling time $=\dfrac{\ln 2}{0.028}=\dfrac{0.69315}{0.028}=24.8$ years.

15 Write $4^x=(2^x)^2$ and $3\cdot 2^{x+1}=6\cdot 2^x$. With $u=2^x$: $u^2-6u+8=0$, $(u-2)(u-4)=0$, so $u=2$ or $u=4$. Hence $x=1$ or $x=2$. Check $x=2$: $4^2-3(2)^3+8=16-24+8=0$.

Exercise 3E · Logarithmic scales and exponential data

1 (a) $10\log_{10}\left(\dfrac{10^{-9}}{10^{-12}}\right)=10\log_{10}(10^3)=30$ dB. (b) $10\log_{10}(10^8)=80$ dB. (c) $10\log_{10}(2\times 10^6)=10(6.3010)=63.0$ dB.

3 The magnitudes differ by 3, so the amplitude ratio is $10^3 = 1000$: the magnitude 8 earthquake has 1000 times the amplitude.

4 (a) Linear: the differences are $13-10=16-13=19-16=3$, constant, so $y = 10 + 3x$. (b) Exponential: the ratios are $15/10 = 22.5/15 = 33.75/22.5 = 1.5$, constant, so $y = 10(1.5)^x$.

5 $\log y = \log 5 + x\log 1.25 = 0.6990 + 0.09691x$. The gradient is $\log 1.25 = 0.0969$ and the intercept is $\log 5 = 0.699$, so the graph of $\log y$ against x is a straight line.

6 At $x=0$, $y = a = 8$. At $x=3$: $8b^3 = 27$, so $b^3 = \dfrac{27}{8}$ and $b = \dfrac{3}{2} = 1.5$. Hence $y = 8(1.5)^x$.

7 The difference is $100 - 50 = 50$ dB, so the intensity ratio is $10^{50/10} = 10^5 = 100\,000$: 100 dB is 100 000 times as intense as 50 dB, not twice. Each increase of 10 dB multiplies intensity by 10; a doubling of intensity adds only $10\log_{10} 2 \approx 3$ dB.

8 A logarithmic scale plots $\log x$, and $\log 0$ is undefined because no power of the base equals 0. As $x \to 0^+$ the logarithm tends to $-\infty$, so a finite axis could never reach zero; the scale can only start at a small positive value.

9 $\log_{10} y = c - 0.3x$, so $y = 10^c\left(10^{-0.3}\right)^x = A(0.5012)^x$. The relationship is exponential decay with factor 0.5012 per unit, a decrease of about 49.9% for each unit increase in x.

10 A straight line on semilog paper means $\log y$ is linear in x, so $y = ab^x$: the data are exponential, with constant ratio b per step. A curve would show that the ratio is not constant, so the relationship is not exponential — it might be quadratic, a power law, or exponential only in parts.

11 Hydrogen ion concentrations are usually much less than 1 mol/L, so their logarithms are negative. The minus sign makes pH positive and increasing in the direction of lower acidity, so a small pH means a large concentration. It also keeps the familiar order: pH 3 is ten times more acidic than pH 4.

12 The 3 measures the logarithm of the ratio: $1000 = 10^3$, so the quantity has increased by three factors of 10, that is by three orders of magnitude, a multiplicative factor of 1000. It does not mean the quantity increased by 3 units or by 3%, and it says nothing about the time taken.

13 $320 = 50b^6$, so $b^6 = 6.4$ and $b = 6.4^{1/6} = 1.36258$, an annual factor of 1.363 (a 36.3% increase per year). The continuous rate is $k = \dfrac{\ln 6.4}{6} = \dfrac{1.85630}{6} = 0.30938$, that is 30.9% per year.

14 The ratios are $566/800 = 0.7075$, $400/566 = 0.7067$, $283/400 = 0.7075$ and $200/283 = 0.7067$: approximately constant at $0.707 \approx \frac{1}{\sqrt{2}}$, so the data are exponential. Two steps of 2 hours multiply the activity by about $\frac{1}{2}$, so the half-life is 4 hours (800 to 400 in 4 h, 400 to 200 in 4 h). Model: $A = 800\left(\frac{1}{2}\right)^{t/4} = 800e^{-0.1733t}$. Check at $t=2$: $800(0.7071) = 565.7$ Bq.

15 Take $\log_{10}$ of $y = ab^x$: $\log_{10} y = \log_{10} a + x\log_{10} b$, which is linear in x with gradient $\log_{10} b$ and intercept $\log_{10} a$. For a quadratic $y = cx^2$, $\log_{10} y = \log_{10} c + 2\log_{10} x$, which is not linear in x. So a straight semilog graph identifies exponential growth, while a curve rules it out; the gradient also gives $b = 10^{\text{gradient}}$.

Investigation 3 · the first digits of powers

1 A $(k+1)$-digit number lies from 10^k (1 followed by k zeros) up to but not including 10^{k+1}. Taking $\log_{10}$, which is increasing, gives $k \le \log_{10} N < k+1$. So k is the greatest integer not exceeding $\log_{10} N$, that is $k = \lfloor \log_{10} N \rfloor$, and the number of digits is $k + 1 = \lfloor \log_{10} N \rfloor + 1$.

2 $\log_{10} 2^{10} = 10(0.30103) = 3.0103$, so the number of digits is $\lfloor 3.0103 \rfloor + 1 = 4$, and $2^{10} = 1024$ indeed has 4 digits. $\log_{10} 2^{50} = 15.0515$, giving 16 digits; $2^{50} = 1\,125\,899\,906\,842\,624$, which has 16 digits. $\log_{10} 2^{100} = 30.103$, giving 31 digits.

3 $\log_{10} 3^{200} = 200(0.47712) = 95.424$, so 3^{200} has 96 digits. $\log_{10} 7^{100} = 100(0.84510) = 84.510$, so 7^{100} has 85 digits.

4 The leading digits for $n = 1$ to 20 are 2, 4, 8, 1, 3, 6, 1, 2, 5, 1, 2, 4, 8, 1, 3, 6, 1, 2, 5, 1. Each is read from 10^f, where f is the fractional part of $n\log_{10} 2$; for example $n = 10$ gives $f = 0.0103$, $10^{0.0103} = 1.024$, leading digit 1.

5 For $n = 1$ to 20 the counts are: digit 1 six times; digit 2 four times; digits 3, 4, 5, 6 and 8 twice each; digits 7 and 9 never. The distribution is not uniform — 1 is three times as common as 3 and some digits do not appear at all. Extending to $n = 1$ to 60 gives counts 18, 12, 6, 6, 6, 4, 2, 5, 1 for the digits 1 to 9, again with 1 the most frequent.

6 Write $2^n = 10^{k+f}$ with $k = \lfloor n\log_{10} 2 \rfloor$ and f the fractional part. Then $2^n = 10^f \times 10^k$, and 10^f lies in $[1, 10)$, so it carries the leading digit. The leading digit is 1 exactly when $1 \le 10^f < 2$, that is $0 \le f < \log_{10} 2 = 0.30103$. The fractional parts are spread approximately evenly over $[0, 1)$, so about 30.1% of powers of 2 begin with 1.

7 Taking $\log_{10}$ turns the product a^n into the sum $n\log_{10} a$, so questions about size become questions about position between consecutive powers of 10. The integer part counts the completed powers of 10 (the digits), and the fractional part fixes the leading digits. In general a^n has $\lfloor n\log_{10} a \rfloor + 1$ digits. Doubling the exponent nearly doubles this count, but not exactly: if 2^n has d digits then 2^{2n} has either $2d-1$ or $2d$, because $\lfloor 2a \rfloor$ is $2\lfloor a \rfloor$ when the fractional part of $a = n\log_{10} 2$ is below 0.5 and $2\lfloor a \rfloor + 1$ otherwise. For example, 2^{100} has 31 digits but 2^{200} has 61, not 62.

Modelling task · cooling, but how fast?

1 $T = 60$: $75(0.88)^t = 40$, so $0.88^t = \dfrac{40}{75} = 0.53333$ and $t = \dfrac{\ln 0.53333}{\ln 0.88} = \dfrac{-0.62861}{-0.12783} = 4.92$ min. $T = 30$: $0.88^t = \dfrac{10}{75} = 0.13333$, so $t = \dfrac{-2.01490}{-0.12783} = 15.76$ min. These agree with the table estimates of between 4 and 5 minutes, and between 15 and 16 minutes.

2 $T - 20$ is the temperature excess over the room. Newton's law of cooling states that the rate of loss of heat is proportional to that excess, so it is the excess, not the temperature itself, that is multiplied by the same factor each minute: $T - 20 = 75(0.88)^t$. The 75 is the initial excess and 0.88 is the fraction of the excess remaining after one minute.

3 $75(0.88)^t = 37.5$ gives $0.88^t = 0.5$, so $t = \dfrac{\ln 0.5}{\ln 0.88} = \dfrac{-0.69315}{-0.12783} = 5.42$ minutes. The half-life does not depend on the starting temperature, because it is fixed by the factor 0.88 alone; any starting excess halves in the same 5.42 minutes.

4 Two assumptions: the room remains at 20°C, and the coffee cools as a single uniform body with a constant factor 0.88 (no milk, no lid, no draught, no heat absorbed by the cup). If the room warmed, the excess would fall more slowly and the drinkable window would start later and last longer; if the cup were insulated or lidded, 0.88 would be closer to 1, the coffee would stay in the window longer, and the model would underestimate the window. Neither assumption makes the model useless, but both limit it to a kitchen whose conditions match the data used to fit it.

The model predicts temperatures, not preferences: the 60 °C and 30 °C limits are choices, and changing them changes the window without changing the model.

Mixed practice 3

1 $y = 60(1.045)^x$: factor 1.045, increase of 4.5% per unit. $y = 90(0.93)^x$: factor 0.93, decrease of 7% per unit.

2 $4(3)^2 = 36$; $2^{-4} = \dfrac{1}{16} = 0.0625$; $e^{0.2} = 1.221$; $\ln 5 = 1.609$.

3 $5^3 = 125$ becomes $\log_5 125 = 3$; $\log_4 64 = 3$ becomes $4^3 = 64$.

4 $\log_2 12 + \log_2\left(\dfrac{4}{3}\right) = \log_2\left(12 \times \dfrac{4}{3}\right) = \log_2 16 = 4$.

5 $32 = 2^5$, so $2^{3x-1} = 2^5$, $3x - 1 = 5$ and $x = 2$.

6 $x = \dfrac{\ln 250}{\ln 7} = \dfrac{5.52146}{1.94591} = 2.84$.

7 $2x - 1 = e^2 = 7.38906$, so $x = \dfrac{8.38906}{2} = 4.19$. The domain $x > 0.5$ is satisfied.

8 Let $u = e^x$: $u^2 - 3u - 4 = 0$, $(u-4)(u+1) = 0$, so $u = 4$ or $u = -1$. Since $e^x > 0$, reject $u = -1$; $e^x = 4$ gives $x = \ln 4 = 1.39$.

9 $Q = Q_0\left(\frac{1}{2}\right)^{t/12} = Q_0 e^{kt}$ with $k = \dfrac{\ln 0.5}{12} = \dfrac{-0.69315}{12} = -0.0578$, a continuous decay rate of 5.78% per year.

10 $\text{pH} = -\log_{10}(4.0 \times 10^{-6}) = -(-5.39794) = 5.40$.

11 $\log_{10} 6^{80} = 80\log_{10} 6 = 80(0.77815) = 62.252$, so 6^{80} has $\lfloor 62.252 \rfloor + 1 = 63$ digits.

12 $2^x > 0$ for every real x, so the graph never reaches $y = 0$; the x-axis is a horizontal asymptote. An x-intercept would require $2^x = 0$, which has no solution.

13 $(1 + r)^9 = 2$, so $1 + r = 2^{1/9} = e^{0.69315/9} = e^{0.077016} = 1.08006$. The annual rate is 8.01%.

14 Value $= 28\,000(0.80)(0.90)^4 = 22\,400(0.6561) = 14\,696.64$, so £14 697.

15 Domain: $x + 4 > 0$ and $x - 2 > 0$, so $x > 2$. Then $\log_3[(x + 4)(x - 2)] = 2$, so $(x + 4)(x - 2) = 9$ and $x^2 + 2x - 17 = 0$. Hence $x = \dfrac{-2 \pm \sqrt{4 + 68}}{2} = -1 \pm 3\sqrt{2}$, giving $x = 3.243$ or $x = -5.243$. Only $x = -1 + 3\sqrt{2} = 3.24$ satisfies $x > 2$. Check: $7.243 \times 1.243 = 9.00$, and $\log_3 9 = 2$.

16 $\Delta L = 10\log_{10}\left(\dfrac{2I}{I_0}\right) - 10\log_{10}\left(\dfrac{I}{I_0}\right) = 10\log_{10} 2 = 3.01$. The level rises by about 3.0 dB, not by a factor of 2: decibels add where intensities multiply.

17 Over 10 years the factor is $\dfrac{52\,000}{45\,000} = 1.15556$, so the annual factor is $1.15556^{1/10} = e^{0.14459/10} = 1.01456$, a rate of 1.46% per year. For 2035, ten more years multiply by the same factor: $52\,000(1.15556) = 60\,089$, about 60 100 people.

18 $\left(\frac{1}{2}\right)^{t/30} = 0.2$, so $t = 30\dfrac{\ln 0.2}{\ln 0.5} = 30\left(\dfrac{-1.60944}{-0.69315}\right) = 30(2.32193) =$ 69.7 years.

19 Change base to 2: $\log_3 8 = \dfrac{\log_2 8}{\log_2 3} = \dfrac{3}{\log_2 3}$. Then $\log_2 3 \cdot \log_3 8 = \log_2 3 \cdot \dfrac{3}{\log_2 3} = 3$.

20 Let $u = 3^x$: $u^2 - 10u + 9 = 0$, $(u - 1)(u - 9) = 0$, so $u = 1$ or $u = 9$. Hence $3^x = 1$ gives $x = 0$, and $3^x = 9$ gives $x = 2$.

21 Put $x = a$, which is allowed since $a > 0$. Then $\log_a a = 1$, so the hypothesis gives $\log_b a = 1$, which means $b^1 = a$, that is $a = b$. (The argument needs $a, b > 0$ and $a, b \neq 1$ so that both logarithms are defined.)

22 $N = 2N_0$ at $t = 5$ gives $e^{5k} = 2$, so $k = \dfrac{\ln 2}{5}$. For $N = 3N_0$: $e^{kt} = 3$, so $t = \dfrac{\ln 3}{k} = \dfrac{5\ln 3}{\ln 2} = 7.92$. Exactly, $k = \dfrac{\ln 2}{5}$ and $t = \dfrac{5\ln 3}{\ln 2}$.

Criterion A · Exponentials and logarithms

1 Initial value $a = 250$; decay factor $b = 0.86$; percentage decrease $= (1 - 0.86) \times 100\% = 14\%$ per unit.

Marking guidance: one mark each for the initial value, the factor and the 14% decrease; 0.86 written as "86% decrease" earns no marks.

2 $y = 2^x$: horizontal asymptote $y = 0$, passing through $(0, 1)$ and $(1, 2)$, rising. $y = \log_2 x$: vertical asymptote $x = 0$, passing through $(1, 0)$ and $(2, 1)$, rising slowly. The second graph is the reflection of the first in the line $y = x$; each point (p, q) on one curve corresponds to (q, p) on the other. The exponential has range $y > 0$ and the logarithm domain $x > 0$, which is exactly why the asymptotes swap from horizontal to vertical.

Marking guidance: 4 marks for two correct curves with correct asymptotes and two labelled points each; a missing or misplaced asymptote costs one mark.

3 $3^4 = 81$, so $\log_3 81 = 4$. $\dfrac{1}{125} = 5^{-3}$, so $\log_5\left(\dfrac{1}{125}\right) = -3$. $7^0 = 1$, so $\log_7 1 = 0$.

Marking guidance: one mark per correct value; a positive 3 in the second part indicates the reciprocal has been mishandled.

4 Take natural logs: $(2x + 1)\ln 5 = \ln 100$, so $2x + 1 = \dfrac{\ln 100}{\ln 5} = \dfrac{4.60517}{1.60944} = 2.86135$. Hence $2x = 1.86135$ and $x = 0.931$ (3 s.f.). Check: $5^{2(0.93068)+1} = 5^{2.86136} = 100.0$.

Marking guidance: 3 marks for taking logs correctly, isolating $2x + 1$ and giving 0.931; a final answer of 1.43 comes from forgetting to subtract 1 before halving.

5 Domain: $x - 3 > 0$ and $x > 0$, so $x > 3$. Combine: $\log_2[x(x - 3)] = 2$, so $x(x - 3) = 4$ and $x^2 - 3x - 4 = 0$, giving $(x - 4)(x + 1) = 0$. So $x = 4$ or $x = -1$; $x = -1$ is rejected because it fails $x > 3$. Check $x = 4$: $\log_2 1 + \log_2 4 = 0 + 2 = 2$.

Marking guidance: 4 marks for the domain, the quadratic, both roots and the rejection with a check; failing to reject -1 caps the mark at 2.

6 Initial population $P(0) = 800$. Doubling time $= \dfrac{\ln 2}{0.035} = \dfrac{0.69315}{0.035} = 19.8$ years. At $t = 12$: $P = 800e^{0.42} = 800(1.52196) =$ 1218 (nearest whole number).

Marking guidance: one mark each for 800, 19.8 years and 1218; $800e^{0.42}$ left unevaluated earns at most 2 of 3.

Criterion B · Powers, digits and leading digits

1 Adding 0.30103 each step gives $\log_{10} 2^n$, and the digit count is $\lfloor n\log_{10} 2 \rfloor + 1$. For $n = 1$ to 12 the entries $(\log_{10} 2^n,\ \text{digits},\ \text{leading digit})$ are: $(0.3010, 1, 2)$, $(0.6021, 1, 4)$, $(0.9031, 1, 8)$, $(1.2041, 2, 1)$, $(1.5051, 2, 3)$, $(1.8062, 2, 6)$, $(2.1072, 3, 1)$, $(2.4082, 3, 2)$, $(2.7093, 3, 5)$, $(3.0103, 4, 1)$, $(3.3113, 4, 2)$, $(3.6124, 4, 4)$. These agree with $2^n =$ $2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096$.

Marking guidance: 6 marks for a complete, consistent table; one arithmetic slip costs one mark, a systematically wrong digit rule costs more.

2 Increasing n by 1 adds $\log_{10} 2 = 0.30103$ to the logarithm, so the number of digits increases by 0 (when the fractional part does not pass 1) or by 1 (when it does). It stays the same whenever 2^n and 2^{n+1} lie between the same consecutive powers of 10, as with 4 and 8 (both one digit) or 16, 32 and 64 (all two digits). On average a new digit appears every $1/0.30103 \approx 3.3$ steps.

Marking guidance: 4 marks for the additive change, the two cases and an example; "it sometimes stays the same because the number is still below the next power of 10" is the key idea.

3 The number of digits of 2^n is $\lfloor n\log_{10} 2 \rfloor + 1$. For $n = 200$: $200(0.30103) = 60.206$, so the count is $60 + 1 = 61$. The rule is justified because $10^{60} \leq 2^{200} < 10^{61}$, so 2^{200} lies between a 61-digit lower bound and a 62-digit upper bound.

Marking guidance: 5 marks for the formula, the substitution, 60.206, the floor step and the answer 61; 60 or 62 signals an off-by-one in the floor rule.

4 For $n = 1$ to 40 the counts for digits 1 to 9 are 12, 8, 4, 4, 4, 4, 0, 4, 0. Digit 1 occurs most often, 12 times out of 40; digits 7 and 9 do not occur at all. The pattern is not uniform. Extending to 60 powers gives 18, 12, 6, 6, 6, 4, 2, 5, 1, still led by 1, and the counts fall roughly as the Benford proportions $30.1\%,\ 17.6\%,\ 12.5\%, \ldots$ as n grows.

Marking guidance: 5 marks for systematic recording, correct counts, the identification of 1 as most frequent and an explicit statement that the distribution is not uniform; claiming uniformity contradicts the data.

5 Write $2^n = 10^{k+f}$ with f the fractional part of $n\log_{10} 2$. The leading digit is 1 when $10^f \in [1, 2)$, that is $0 \leq f < \log_{10} 2$, an interval of width 0.301; it is 9 when $10^f \in [9, 10)$, that is $\log_{10} 9 \leq f < 1$, a width of only 0.046. Since the fractional parts fall roughly evenly across $[0, 1)$, each digit's share is its interval width, so 1 is about $0.301/0.046 = 6.6$ times as likely as 9. This is why 1 leads far more often than 9, and why the counts above are not uniform.

Marking guidance: 4 marks for using the fractional part, the interval for 1, the interval for 9 and the comparison of widths; a correct answer that only says "1 occurs more often" without the interval argument earns 2.

Criterion C · Why logarithms solve exponential equations

1 **Sample response.** A logarithm answers the question "to what power?". Just as subtraction undoes addition, the logarithm base 2 undoes raising 2 to a power: $\log_2 x$ is the exponent y for which $2^y = x$. Because a positive base raised to any power is positive, the logarithm is defined only for $x > 0$, a restriction that every solution must respect. The word "logarithm" is therefore a name for an exponent, not a new operation to memorise. That is why $2^x = 32$ is easy: $32 = 2^5$, so $x = 5$, without any new tool.

When the number is not a power of the base, as in $2^x = 30$, we apply the inverse operation to both sides: $\log_2(2^x) = \log_2 30$. This is legitimate because $\log_2$ is one-to-one, so it can neither create nor destroy solutions. By the definition the left side is x, so $x = \log_2 30$, which is exact. A calculator may not have a base 2 button, so use change of base: $x = \dfrac{\ln 30}{\ln 2} = \dfrac{3.4012}{0.6931} = 4.91$. Checking in the original equation, $2^{4.91} = 30.0$, so the value is correct to two decimal places.

The same steps work for any base $a > 0$, $a \neq 1$: taking $\log_a$ of $a^x = b$ gives $x = \log_a b = \dfrac{\ln b}{\ln a}$, because $\log_a$ is the inverse of a^x and a^x is one-to-one. For example, $5^{2x+1} = 100$ gives $(2x + 1)\ln 5 = \ln 100$, so $2x + 1 = 2.861$ and $x = 0.931$. Trial and improvement would locate these exponents too, but only approximately and only after several substitutions; the logarithm gives them in one line, and the same line works whatever the size of the numbers.

A common error is to write $\dfrac{\ln 30}{\ln 2}$ as $\ln 15$, applying the quotient law to a quotient of logarithms rather than to a difference. A second error is to write $\log(2^x) = (\log 2)^x$; the correct power law is $\log(2^x) = x \log 2$, which is exactly the step that brings the unknown down. In short, the logarithm does not solve a new kind of equation; it names the exponent that was already there.

Marking guidance: 20 marks, typically 4 for the inverse definition, 5 for the complete solution of $2^x = 30$ with a check, 4 for the general-base justification, 4 for the worked example with correct notation, and 3 for a clearly diagnosed error. A response that solves the equation but never explains why the inverse operation is legitimate cannot reach the top band for communicating, however accurate the arithmetic.

Criterion D · A cooling drink and a warming room

1 The room temperature is the asymptote, 21°C. The initial temperature is $T(0) = 21 + 74 = 95$°C, which matches the stated serving temperature and confirms the model's initial value.

Marking guidance: 2 marks, one for each temperature; "21 °C" described as the starting temperature earns nothing.

2 For 55°C: $74(0.91)^t = 34$, so $0.91^t = \dfrac{34}{74} = 0.45946$ and $t = \dfrac{\ln 0.45946}{\ln 0.91} = \dfrac{-0.77782}{-0.09431} = 8.25$ min. For 35°C: $0.91^t = \dfrac{14}{74} = 0.18919$, so $t = \dfrac{-1.66501}{-0.09431} = 17.65$ min. The drink is in the preferred range from about 8.2 to 17.7 minutes, a window of about 9.4 minutes.

Marking guidance: 6 marks — 1 for each correctly formed equation, 1 for each solved time, 1 for correct rounding and units, 1 for the window; times given without the subtraction of 21 (for example solving $74(0.91)^t = 55$) score at most 2.

3 The temperature difference is $T - 21 = 74(0.91)^t$. Half of it remains when $0.91^t = 0.5$, so $t = \dfrac{\ln 0.5}{\ln 0.91} = \dfrac{-0.69315}{-0.09431} = 7.35$ minutes. In context, the drink closes half of the gap between its current temperature and room temperature every 7.35 minutes: from 95 °C it reaches 58 °C after 7.35 min, then 39.5 °C after another 7.35 min.

Marking guidance: 5 marks — 1 for identifying $T - 21$, 1 for $0.91^t = 0.5$, 2 for the value 7.35 min, 1 for the interpretation; halving the temperature itself (solving $74(0.91)^t = 47.5$, giving 4.7 min) is not the half-life of the excess.

4 Assumption 1: the room stays at 21°C. In a warming room the excess $T - 21$ falls more slowly, so both times increase and the window starts later and lasts longer; a cooling room shortens it. Assumption 2: the drink cools as one uniform body with a constant factor 0.91, with no milk, lid, draught or heat absorbed by the cup. Milk added at serving lowers the starting temperature and moves the whole window earlier without changing its length, provided the cooling factor stays at 0.91; a lid makes the factor closer to 1, lengthening the window. The model therefore describes an idealised still room and an uncovered cup.

Marking guidance: 5 marks — 2 for two valid assumptions, 2 for the direction of each effect, 1 for linking the effect to the predicted window; assertions without a direction of change score 1 per assumption.

5 The most practical change is to slow the cooling, for example by serving the drink in a preheated cup or with a lid, so that the factor rises from 0.91 towards 1. The window length is $\dfrac{\ln(34/14)}{\ln(1/0.91)} = 9.41$ min; with factor 0.94 it becomes $\dfrac{\ln(34/14)}{\ln(1/0.94)} = 14.34$ min, an increase of 4.9 min, so a 6% loss per minute would achieve the target. Serving the drink hotter does not help: the window is $\dfrac{\ln(34/14)}{\ln(1/0.91)}$ whatever the initial temperature, so a hotter drink merely shifts the window later. The model supports the recommendation in principle, but it cannot predict the new factor — that must be measured — so the café should test the change and refit 0.91 from data before promising five extra minutes.

Marking guidance: 6 marks for a practical change, a quantitative test using the model, the correct conclusion about whether five minutes is achievable, and an evaluation of the model's limits. The strongest responses notice that changing the serving temperature shifts the window without lengthening it.

CHAPTER 4 · SEQUENCES AND SERIES

Opening problem · the thirty-day contract

1 Option B is geometric with first term 1 and ratio 2: the daily pay is $u_n = 2^{n-1}$. The first ten terms are $1, 2, 4, 8, 16, 32, 64, 128, 256, 512$, and adding by hand gives $1023 = 2^{10} - 1$.

2 A: $u_n = 1000$ (constant, $d = 0$); B: $u_1 = 1,\ u_{n+1} = 2u_n$, so $u_n = 2^{n-1}$; C: $u_n = 10n$ (arithmetic, $a = 10$, $d = 10$). Thirty-day totals: A $= 30 \times 1000 = 30\,000$; C $= \frac{30}{2}(10 + 300) = 15 \times 310 = 4650$.

3 Daily pay: $2^{n-1} > 1000$ first holds at $n = 11$, since $2^{10} = 1024$ while $2^9 = 512$. Running total: $2^n - 1 > 30\,000$ needs $2^n > 30\,001$; $2^{14} = 16\,384$ (total 16 383) is short and $2^{15} = 32\,768$ (total 32 767) passes, so day 15.

4 Sign Option B: over thirty days it pays $2^{30} - 1 = 1\,073\,741\,823$ against 30 000 for A and 4650 for C. The decision rests on the sum of all thirty terms, not on one term: day 1 B pays 1 against A's 1000, yet by day 11 B's single day already exceeds A's whole day's pay, and its cumulative total passes A's on day 15. Judging a sequence by its first term compares a term with a sum and ignores how the rule accumulates.

A term answers "how much next?"; a sum answers "how much altogether?". The two questions have different answers here.

Prerequisite check

1 21, 25, 29. The rule is "add 4 to the previous term", i.e. $u_1 = 5,\ u_{n+1} = u_n + 4$, or in closed form $u_n = 4n + 1$.

2 $3n - 7 = 41 \Rightarrow 3n = 48 \Rightarrow n = 16$. $2^n = 64 = 2^6 \Rightarrow n = 6$.

3 $2^5 = 32$; $\left(\frac{1}{2}\right)^3 = \frac{1}{8} = 0.125$; $1.05^2 = 1.1025$; $5^0 = 1$.

4 $x^2 + 3x - 40 = (x + 8)(x - 5)$, so $x = -8$ or $x = 5$.

5 $0.75 = \frac{3}{4}$ (lowest terms); $\frac{3}{8} = 0.375$.

6 Multiply by $\frac{2}{n}$: $\frac{2S}{n} = a + l$, so

$$l = \frac{2S}{n} - a.$$

Exercise 4A · Arithmetic sequences

1 (a) $d = 7$; next terms 25, 32. (b) $d = -4$; next 18, 14. (c) $d = 1.5$; next 7.0, 8.5. (d) $d = 3$; next 1, 4.

2 (a) $a = 3$, $d = 4$: $u_n = 3 + 4(n - 1) = 4n - 1$. (b) $a = 20$, $d = -3$: $u_n = 20 - 3(n - 1) = 23 - 3n$. (c) $a = -4$, $d = 5$: $u_n = -4 + 5(n - 1) = 5n - 9$. (d) $a = 9$, $d = 0$: $u_n = 9$ for every n.

3 $u_1 = 1$, $u_{10} = 55$, $u_{50} = 295$. Solve $6n - 5 = 61$: $6n = 66$, $n = 11$, so the 11th term is 61.

4 $a = 100$, $d = -7$, so $u_n = 100 - 7(n - 1) = 107 - 7n$. Then $u_{15} = 2$. The first negative term needs $107 - 7n < 0$, i.e. $n > 15.28$, so $n = 16$: $u_{16} = -5$ (and $u_{15} = 2 > 0$).

5 (a) $d = \frac{35 - 7}{8 - 1} = 4$, $a = 7$, $u_n = 4n + 3$. (b) $d = \frac{31 - 11}{9 - 4} = 4$, $a = 11 - 3(4) = -1$, $u_n = 4n - 5$ (check $u_4 = 11$, $u_9 = 31$). (c) $d = \frac{-19 - 2}{10 - 3} = -3$, $a = 2 - 2(-3) = 8$, $u_n = 11 - 3n$ (check $u_3 = 2$, $u_{10} = -19$).

6 Use $n = \frac{l - a}{d} + 1$. (a) $\frac{85 - 5}{4} + 1 = 21$ terms. (b) $\frac{96 - 12}{7} + 1 = 13$ terms. (c) $\frac{5 - 60}{-5} + 1 = 12$ terms.

7 At $n = 1$ the student's rule gives $3 + 4(1) = 7$, but the first term is 3; the rule is one step ahead. Correct: $u_n = 3 + 4(n - 1) = 4n - 1$, which gives 3, 7, 11. ✓

Common error: writing $a + dn$ instead of $a + (n - 1)d$; there are only $n - 1$ steps from the first term.

8 Here $a = 8$, $d = 7$, so $u_n = 7n + 1$. Solving $7n + 1 = 175$ gives $7n = 174$, $n = \frac{174}{7} = 24.7\ldots$, not a positive integer, so 175 is not a term. Taking the next lower integer index, $u_{24} = 169$ (and $u_{25} = 176 > 175$), so the largest term below 175 is 169.

9 $u_n = 3n - 2$ has constant difference $d = 3$ (e.g. $u_2 - u_1 = 4 - 1 = 3$). For $v_n = 2^n$: $v_2 - v_1 = 4 - 2 = 2$ but $v_3 - v_2 = 8 - 4 = 4$. The differences are not equal, so no single d exists and $\{v_n\}$ is not arithmetic (it is geometric with ratio 2).

10 $u_{11} = u_3 + 8d$ gives $46 = 14 + 8d$, so $d = 4$; then $a = 14 - 2(4) = 6$ and $u_n = 4n + 2$. Hence $u_{20} = 82$. Check: $u_3 = 14$, $u_{11} = 46$. ✓

11 If $u_m = u_n$ with $m \neq n$, then $a + (m - 1)d = a + (n - 1)d$, so $(m - n)d = 0$; since $m - n \neq 0$, this forces $d = 0$. With $d \neq 0$ every term is different, so no value can repeat. When $d = 0$ the sequence is constant $(a, a, a, \ldots)$ and every value repeats infinitely often.

12 $d = -2 < 0$, so the sequence decreases $(3, 1, -1, \ldots)$. $u_n > 0$ means $5 - 2n > 0$, i.e. $2n < 5$, i.e. $n < 2.5$ (dividing by the positive 2; the inequality is rearranged by subtracting $2n$, so no reversal is needed here). The largest integer index is $n = 2$, with $u_2 = 1$; $u_3 = -1$.

13 $a = 18$, $d = 3$, so $u_n = 3n + 15$. Last row: $u_{20} = 75$ seats. $3n + 15 = 54$ gives $n = 13$, so row 13 has 54 seats. $3n + 15 = 100$ gives $n = \frac{85}{3} = 28.3\ldots$, not an integer, so no row has 100 seats (every row count is a multiple of 3), and the stand has only 20 rows in any case.

14 Counting from the bottom, $a = 20$, $d = -1$, so $u_n = 21 - n$. The twelfth row holds $u_{12} = 9$ pipes. $21 - n = 8$ gives $n = 13$, so the thirteenth row from the bottom has 8 pipes. Requiring $21 - n \geq 4$ gives $n \leq 17$: the greatest possible number of rows is 17, the top row holding 4 pipes (row 18 would hold 3, which the rule forbids). That stack holds

$$S_{17} = \frac{17}{2}(20 + 4) = 17 \times 12 = 204 \text{ pipes}.$$

Common error: counting the bottom row as $u_1 = 20$ but then writing $u_n = 20 - n$, which loses the one-row offset and gives 8 pipes in row 12.

15 $u_9 = 3 + 8d$ and $u_4 = 3 + 3d$, so $3 + 8d = 3(3 + 3d) = 9 + 9d$, giving $-d = 6$ and $d = -6$. Check: $u_9 = -45 = 3u_4 = 3(-15)$. ✓ Proof: $u_m + u_n = 2a + (m + n - 2)d$ and $u_p + u_q = 2a + (p + q - 2)d$; when $m + n = p + q$ the two right-hand sides are identical, so $u_m + u_n = u_p + u_q$. This is exactly why Gauss pairing works.

Exercise 4B · Arithmetic series and sigma notation

1 (a) $n = \frac{59 - 2}{3} + 1 = 20$, $S = \frac{20}{2}(2 + 59) = 610$. (b) $n = \frac{5 - 100}{-5} + 1 = 20$, $S = 10(105) = 1050$. (c) $S = \frac{100}{2}(1 + 100) = 50 \times 101 = 5050$. (d) $n = \frac{99 - 12}{3} + 1 = 30$, $S = 15(111) = 1665$.

2 (a) 10 terms from 5 to 23: $S = 5(28) = 140$. (b) 12 terms from 4 to 59: $S = 6(63) = 378$. (c) $20 - 4 + 1 = 17$ terms from 12 to 60: $S = \frac{17}{2}(72) = 612$. (d) 8 terms from 37 to 16: $S = 4(53) = 212$.

3 (a) $\sum_{k=1}^{20} 3k$ (since $3 \times 20 = 60$). (b) $\sum_{k=1}^{10}(4k + 1)$ (since $4 \times 10 + 1 = 41$). (c) $\sum_{k=1}^{34}(3k - 2)$ (since $3 \times 34 - 2 = 100$).

4 (a) The sum of the first n odd numbers is $S_n = \frac{n}{2}(1 + 2n - 1) = n^2 = 900$, so $n = 30$ (positive root). Check: $30^2 = 900$. (b) $S_n = \frac{n}{2}(4 + 3n + 1) = \frac{n(3n + 5)}{2} = 175$, so $3n^2 + 5n - 350 = 0$, $n = \frac{-5 + \sqrt{4225}}{6} = \frac{-5 + 65}{6} = 10$; the negative root is rejected. Check: $10(35)/2 = 175$. ✓

5 $u_1 = 1$, $u_{15} = 57$, and $S_{15} = \frac{15}{2}(1 + 57) = 15 \times 29 = 435$.

6 $S_{18} = \frac{18}{2}(2(6) + 17(5)) = 9(12 + 85) = 9 \times 97 = 873$.

7 Two errors: the value 47 is the last term l, not the number of terms, and the number of terms was therefore wrong (there is also no reason to halve 47 rather than n). Correct count: $n = \frac{47 - 5}{3} + 1 = 15$. Correct sum: $S_{15} = \frac{15}{2}(5 + 47) = 15 \times 26 = 390$.

Common error: substituting the last term into the place of n in $\frac{n}{2}(a + l)$.

8 First term $4(1) + 1 = 5$, nth term $4n + 1$, n terms, so $S_n = \frac{n}{2}(5 + 4n + 1) = \frac{n}{2}(4n + 6) = n(2n + 3)$. ✓ Setting $n(2n + 3) = 275$: $2n^2 + 3n - 275 = 0$, $n = \frac{-3 + \sqrt{9 + 2200}}{4} = \frac{-3 + 47}{4} = 11$ (check $11 \times 25 = 275$); the negative root is rejected.

9 Reading from both ends, the terms one step up from the start and one step down from the end always add to the same value: $u_k + u_{n+1-k} = [a + (k-1)d] + [a + (n-k)d] = 2a + (n-1)d = a + l$. So the n terms form $\frac{n}{2}$ pairs each worth $a + l$, giving $S_n = \frac{n}{2}(a + l)$. For $1 + 2 + \cdots + 60$: 30 pairs of 61, so $S = 30 \times 61 = 1830$.

10 $u_1 = S_1 = 1 + 3 = 4$. $u_2 = S_2 - S_1 = 10 - 4 = 6$, so $d = 2$. In general $u_n = S_n - S_{n-1} = (n^2 + 3n) - \big((n-1)^2 + 3(n-1)\big) = 2n + 2$. Re-summing: $S_n = \frac{n}{2}(4 + (2n+2)) = \frac{n}{2}(2n+6) = n(n+3) = n^2 + 3n$ ✓, the given formula.

11 $u_1 = S_1 = 4$; $S_2 = 10 - 4 = 6$ so $u_2 = 6 - 4 = 2$; $S_3 = 15 - 9 = 6$ so $u_3 = 6 - 6 = 0$. The sequence is $4, 2, 0, -2, \ldots$ with $d = -2$. $S_3 = S_2$ because $u_3 = 0$: adding the third term changes nothing. (The sums later fall again once the terms become negative.)

12 $a + l = 2a + (n-1)d$. With n odd, $n - 1$ is even, so $(n-1)d$ is even and $a + l$ has the same parity as $2a$, i.e. $a + l$ is even; equivalently a and l have the same parity. Then $\frac{n}{2}(a + l) = n \times \frac{a+l}{2}$ is an integer times an integer, so the sum of integers comes out an integer with no fraction left over.

13 $a = 18$, $d = 2$, $n = 26$: last row $u_{26} = 18 + 2(25) = 68$. Total $S_{26} = \frac{26}{2}(18 + 68) = 13 \times 86 = 1118$ seats.

14 $a = 80$, $d = 15$, $n = 36$: final deposit $u_{36} = 80 + 15(35) = 605$. Total $S_{36} = \frac{36}{2}(80 + 605) = 18 \times 685 = 12\,330$.

15 $u_{10} = S_{10} - S_9 = (3(100) + 20) - (3(81) + 18) = 320 - 261 = 59$. First term $u_1 = S_1 = 5$; $u_2 = S_2 - S_1 = 16 - 5 = 11$, so $d = 6$ and $u_n = 5 + 6(n-1) = 6n - 1$ (check $6(10) - 1 = 59$). Verification: $S_n = \frac{n}{2}(5 + (6n-1)) = \frac{n}{2}(6n+4) = n(3n+2) = 3n^2 + 2n$ ✓.

Exercise 4C · Geometric sequences

1 (a) $r = 5$; next 250, 1250. (b) $r = \frac{1}{3}$; next 3, 1. (c) $r = -2$; next -40, 80. (d) $r = 2$; next 3.2, 6.4.

2 (a) $u_n = 4 \cdot 3^{n-1}$. (b) $u_n = 64\left(\frac{1}{2}\right)^{n-1} = 2^{7-n}$. (c) $u_n = -3(-2)^{n-1}$. (d) $u_n = 0.5(0.1)^{n-1} = 5 \times 10^{-n}$.

3 $u_1 = 5$, $u_6 = 5 \cdot 2^5 = 160$, $u_{10} = 5 \cdot 2^9 = 2560$. Exceeding 5000 needs $2^{n-1} > 1000$; $2^{10} = 1024$, so $n - 1 = 10$, $n = 11$: $u_{11} = 5120$ (and $u_{10} = 2560 < 5000$).

4 (a) $r^3 = \frac{24}{3} = 8 \Rightarrow r = 2$, $a = \frac{10}{2} = 5$, $u_n = 5 \cdot 2^{n-1}$. (b) $r^3 = \frac{80}{10} = 8 \Rightarrow r = 2$, $a = \frac{12}{r^2} = 3$, $u_n = 3(-2)^{n-1}$ (check $u_6 = 3(-32) = -96$). (c) $r^3 = \frac{-96}{12} = -8 \Rightarrow r = -2$, $a = \frac{12}{r^2} = 3$, $u_n = 3(-2)^{n-1}$ (check $u_6 = 3(-32) = -96$).

5 $r^2 = \frac{32}{8} = 4 \Rightarrow r = \pm 2$ (even index gap, so both signs survive); $a = \frac{8}{r^2} = 2$ either way. The sequences are $2, 4, 8, 16, 32, \ldots$ with $u_n = 2 \cdot 2^{n-1}$, and $2, -4, 8, -16, 32, \ldots$ with $u_n = 2(-2)^{n-1}$.

6 (a) 1.07 (b) 0.88 (c) 3 (d) 0.5 (e) 1.005. A percentage change of p per cent gives multiplier $1 + \frac{p}{100}$, so a fall keeps it below 1 but positive.

7 The index is wrong: $u_n = ar^{n-1}$, so the tenth term uses r^9, not r^{10}. Correct: $u_{10} = 3 \cdot 2^9 = 1536$. (The value $3072 = 3 \cdot 2^{10}$ is the eleventh term.)

8 Model $V_n = 2000(1.045)^n$. Then $V_6 = 2000(1.045)^6 = 2604.52$. Exceeding 3000 needs $1.045^n > 1.5$, so $n > \frac{\log 1.5}{\log 1.045} = 9.21$, giving $n = 10$: $V_{10} = 3105.94$ while $V_9 = 2972.19 < 3000$.

9 Multiplying by a negative ratio reverses the sign at every step, so the signs alternate; multiplying by a number of magnitude below 1 shrinks $|u_n|$ towards 0, so the terms get closer to zero. At $r = 1$ the sequence is constant, $a, a, a, \ldots$ (also arithmetic with $d = 0$); at $r = -1$ it alternates $a, -a, a, -a, \ldots$ with constant magnitude and no limit.

10 Write the four terms as $5, 5r, 5r^2, 5r^3$. Then $5r^3 = 40$, $r^3 = 8$, $r = 2$ (the only real root, and positive terms are required). The numbers are 10 and 20: 5, 10, 20, 40. ✓

11 Four hours multiply by 2, so one hour multiplies by $2^{1/4} = \sqrt[4]{2} = 1.1892$ (a growth of about 18.9%). Over 24 hours the factor is $2^{24/4} = 2^6 = 64$. The hourly factor is not 2, because four doublings per day would give $2^4 = 16$ over four hours, not 2; factors compose by multiplication, not by division of the time.

12 Require $3(1.2)^{n-1} > 100$, so $1.2^{n-1} > \frac{100}{3}$ and $n - 1 > \frac{\log(100/3)}{\log 1.2} = 19.23$. Since $n - 1$ is an integer, $n - 1 = 20$, i.e. $n = 21$: $u_{21} = 3(1.2)^{20} = 115.01$, while $u_{20} = 95.84 < 100$. The 21st term is the first to exceed 100.

13 $V_n = 45\,000(0.82)^n$. Then $V_5 = 45\,000(0.82)^5 = 16\,683.29$. Below 10 000 needs $0.82^n < \frac{10\,000}{45\,000}$, so $n > \frac{\log(1/4.5)}{\log 0.82} = 7.58$, i.e. $n = 8$: $V_8 = 9198.63$ while $V_7 = 11\,217.85$. The machine is first worth less than 10 000 in the 8th year.

14 $P_n = 12\,000(1.025)^n$, so $P_{10} = 12\,000(1.025)^{10} = 15\,361$ people (nearest whole person). Doubling needs $1.025^n > 2$, so $n > \frac{\log 2}{\log 1.025} = 28.07$, i.e. $n = 29$: $P_{29} = 24\,557$ while $P_{28} = 23\,958 < 24\,000$. It doubles during the 29th year.

15 $u_1 + u_2 = a(1 + r) = 12$ and $u_3 + u_4 = ar^2(1 + r) = 27$. Dividing gives $r^2 = \frac{27}{12} = \frac{9}{4}$, so $r = \pm\frac{3}{2}$. If $r = \frac{3}{2}$, $a = \frac{12}{1 + 3/2} = \frac{24}{5} = 4.8$; if $r = -\frac{3}{2}$, $a = \frac{12}{1 - 3/2} = -24$. Two answers because the two pieces of information are two positions apart, so only r^2 is determined and the sign of r stays free; each pair checks out ($-24 + 36 = 12$, 27).

Exercise 4D · Geometric series and convergence

1 (a) $a = 3$, $r = 2$: $S_8 = \frac{3(2^8 - 1)}{2 - 1} = 3(255) = 765$. (b) $a = 2$, $r = 3$: $S_6 = \frac{2(3^6 - 1)}{2} = 728$. (c) $a = 100$, $r = 0.5$: $S_8 = \frac{100(1 - 0.5^8)}{0.5} = 200\left(1 - \frac{1}{256}\right) = 199.21875$. (d) $a = 1$, $r = -3$: $S_7 = \frac{1 - (-3)^7}{1 - (-3)} = \frac{1 + 2187}{4} = 547$.

2 (a) $r = \frac{1}{2}$, $S_\infty = \frac{12}{1 - \frac{1}{2}} = 24$. (b) $r = \frac{1}{3}$, $S_\infty = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$. (c) $r = -\frac{1}{2}$, $S_\infty = \frac{8}{1 + \frac{1}{2}} = \frac{16}{3} = 5.33$. (d) $r = \frac{1}{4}$, $S_\infty = \frac{5}{1 - \frac{1}{4}} = \frac{20}{3} = 6.67$.

3 (a) $\frac{0.7}{0.9} = \frac{7}{9}$. (b) $\frac{0.36}{0.99} = \frac{36}{99} = \frac{4}{11}$. (c) $0.1 + \frac{0.06}{0.9} = \frac{1}{10} + \frac{1}{15} = \frac{1}{6}$. (d) $0.41 + \frac{0.006}{0.9} = \frac{41}{100} + \frac{1}{150} = \frac{123 + 2}{300} = \frac{5}{12}$.

For a mixed decimal such as (c) and (d) only the recurring tail is summed, so the non-recurring part is added separately.

4 (a) $|1.03| > 1$: diverges, no sum. (b) Converges, $S_\infty = \frac{10}{1.6} = 6.25$. (c) $|5| > 1$: diverges. (d) Converges, $S_\infty = \frac{9}{1/3} = 27$. (e) $|-1| = 1$: diverges, since the partial sums alternate $a, 0, a, 0, \ldots$ and never settle. (f) Converges (barely, $|0.999| < 1$), $S_\infty = \frac{2}{0.001} = 2000$.

5 $\frac{20}{1 - r} = 50 \Rightarrow 1 - r = 0.4 \Rightarrow r = 0.6$. With $S_\infty = 36$, $r = 0.25$: $a = 36(1 - 0.25) = 27$.

6 $S_{10} = \frac{4(1.5^{10} - 1)}{1.5 - 1} = 8(57.6650390625 - 1) = 453.3203125$, i.e. 453.32 (exactly $\frac{58\,025}{128}$). Since $r > 1$ the sums grow without bound, so there is no S_∞.

7 Here $a = 1$, $r = 2$ and $|r| > 1$, so the condition for $S_\infty = \frac{a}{1 - r}$ fails; the division is algebra outside its own domain. The partial sums are $S_1 = 1$, $S_2 = 3$, $S_3 = 7$ (in general $S_n = 2^n - 1$), all positive and increasing without bound, so the total cannot be -1.

8 $S_n = a + ar + \cdots + ar^{n-1}$ and $rS_n = ar + ar^2 + \cdots + ar^n$. Subtracting, every middle term cancels: $(1 - r)S_n = a - ar^n = a(1 - r^n)$. Dividing by $1 - r$ is legal only when $r \neq 1$; when $r = 1$ that step gives $0 \times S_n = 0$ and yields nothing, while the series is n copies of a, so $S_n = na$.

9 Nobody adds infinitely many terms. S_∞ is defined as $\lim_{n\to\infty} S_n$, the value approached by the finite partial sums: for any required accuracy, some finite n makes $\left|S_n - \frac{a}{1-r}\right| = \left|\frac{ar^n}{1-r}\right|$ smaller than that accuracy. It is a statement about the sequence of totals, not a completed infinite addition.

10 $\frac{4}{1-r} = 16 \Rightarrow 1-r = \frac{1}{4} \Rightarrow r = \frac{3}{4}$. Then $u_3 = 4\left(\frac{3}{4}\right)^2 = 4 \cdot \frac{9}{16} = \frac{9}{4} = 2.25$.

11 $S_4 = 8 - 4 + 2 - 1 = 5$ (formula: $\frac{8(1-(-0.5)^4)}{1.5} = 5$). $S_\infty = \frac{8}{1.5} = \frac{16}{3} = 5.33$. The partial sums $8,\ 4,\ 6,\ 5,\ 5.5,\ 5.25, \ldots$ alternate above and below $\frac{16}{3}$, each oscillation half the previous one, so they close in on the limit without approaching it from one side only.

12 $S_\infty - S_n = \frac{a}{1-r} - \frac{a(1-r^n)}{1-r} = \frac{ar^n}{1-r}$. With $a > 0$ and $0 < r < 1$ both $r^n > 0$ and $1 - r > 0$, so the difference is positive: $S_n < S_\infty$ for every n. The sums increase because $S_{n+1} - S_n = u_{n+1} = ar^n > 0$: each term added is positive. So the partial sums rise monotonically towards the limit and never reach it.

13 (a) Day 30: $u_{30} = 2^{29} = 536\,870\,912$. Total: $S_{30} = \frac{2^{30}-1}{2-1} = 1\,073\,741\,823$. (b) A: $30 \times 1000 = 30\,000$; C: $\sum_{n=1}^{30} 10n = 10 \cdot \frac{30}{2}(1+30) = 10 \times 465 = 4650$. (c) Daily pay: $2^{n-1} > 1000$ first at $n = 11$ ($1024 > 1000$; day 10 gives 512). Running total: $2^n - 1 > 30\,000$ first at $n = 15$ (32 767; day 14 gives 16 383). (d) Option B: its thirty-day total is about 1.07 billion, roughly 36 000 times A's 30 000 and about 231 000 times C's 4650. The comparison must be made between sums S_{30}, not between the single terms $u_1 = 1,\ 1000,\ 10$, which rank the offers in the opposite order.

14 Down 2 m; then every rebound height is travelled twice (up and down): heights $1.4,\ 0.98,\ 0.686, \ldots$ form a geometric series with $a = 1.4$, $r = 0.7$. Total

$$2 + 2\left(\frac{1.4}{1-0.7}\right) = 2 + \frac{2.8}{0.3} = \frac{34}{3} = 11.3\text{ m} \quad (11.33\text{ m to 2 d.p.}).$$

15 Here $a = 1$ and $r = 2x$, so the sum to infinity exists only when $|2x| < 1$, i.e. $-\frac{1}{2} < x < \frac{1}{2}$, and then $\frac{1}{1-2x} = 3 \Rightarrow 1 - 2x = \frac{1}{3} \Rightarrow x = \frac{1}{3}$. Check: $|\frac{2}{3}| < 1$ and $\frac{1}{1-\frac{2}{3}} = 3$ ✓. The only value is $x = \frac{1}{3}$.

Exercise 4E · Saving, borrowing and comparing

1 $B_5 = 2500(1.035)^5 = 2500(1.187686) = 2969.22$.

2 $V_4 = 12\,000(0.8)^4 = 12\,000(0.4096) = 4915.20$.

3 $B_8 = \frac{150(1.03^8-1)}{0.03} = 150(8.892336) = 1333.85$. Deposits alone total 1200, so 133.85 is interest.

4 $p = \frac{6000(0.07)}{1-1.07^{-4}} = \frac{420}{1-0.762895} = \frac{420}{0.237105} = 1771.37$. Total repaid $4p = 7085.47$, so total interest $= 1085.47$.

5 Recurrence $B_{n+1} = 1.04B_n + 100$, $B_0 = 1000$: $B_1 = 1140.00$, $B_2 = 1285.60$, $B_3 = 1437.02$. Check: $1000(1.04)^3 + \frac{100(1.04^3-1)}{0.04} = 1124.86 + 312.16 = 1437.02$ ✓ — the initial sum grows by the power rule and the deposits form the annuity.

6 $\left(1 + \frac{0.048}{12}\right)^{12} - 1 = 1.004^{12} - 1 = 0.049070\ldots$, i.e. an effective annual rate of 4.91% (2 d.p.).

7 Only money already in the account earns interest, and the deposits arrive one month at a time, so almost none of the 1200 is present for the whole year; multiplying the whole 1200 by 1.05 gives every deposit a full year of interest and so overstates the return. The first deposit (earliest) earns the most; the last, made at the end of the year, earns nothing.

8 Written from the last deposit backwards, the values at the end are $200,\ 200(1.04),\ 200(1.04)^2, \ldots$: a geometric series with first term $a = 200$, ratio $r = 1.04$ and $n = 10$ terms, so $B_{10} = \frac{200(1.04^{10}-1)}{1.04-1} = \frac{200(1.04^{10}-1)}{0.04} = 2401.22$. The denominator is $R - 1$ because $R > 1$.

9 With $p = 1899.17$: year 1 interest $= 8000(0.06) = 480.00$ (capital repaid 1419.17, balance 6580.83); year 5 interest $= 1791.68(0.06) = 107.50$. Interest falls (480.00, 394.85, 304.59, 208.92, 107.50) as the balance shrinks, while the capital part of each instalment grows by the factor 1.06 each year; the total 1495.85 is unchanged.

10 Effective annual rates: annually, 3.9%. Monthly, $\left(1 + \frac{0.038}{12}\right)^{12} - 1 = 0.038669$, i.e. 3.87%. Since $3.87\% < 3.9\%$, the 3.8% compounded monthly option is cheaper (and it is the smaller debt growth for a borrower), even though $3.8 < 3.9$ already suggests it; here the nominal comparison and the effective comparison agree.

11 A start-of-year deposit earns one extra year, so every term of the series is multiplied by $1 + i$: $B_n^{(\text{start})} = (1+i)\frac{d((1+i)^n-1)}{i}$. For 100 a year, five years, 5%: end-of-year $= \frac{100(1.05^5-1)}{0.05} = 552.56$; start-of-year $= 552.56 \times 1.05 = 580.19$; the difference is 27.63.

12 $p = \frac{Li}{1-(1+i)^{-n}}$: a larger n makes $(1+i)^{-n}$ smaller, so the denominator grows and p falls. But the total repaid is np, and interest accrues on the outstanding balance for longer, so $np - L$ rises. Example ($L = 8000$, $i = 0.06$): three years gives $p = 2992.88$, total 8978.64, interest 978.64; five years gives $p = 1899.17$, total 9495.86, interest 1495.86 — smaller payments, larger bill.

13 Solve $15\,000 = \frac{d(1.045^6-1)}{0.045}$, so $d = \frac{15\,000(0.045)}{1.045^6-1} = \frac{675}{0.302260} = 2233.18$ (round up to be sure the target is reached). The deposits total $6d = 13\,399.05$, so the interest earned is 1600.95.

14 At 6.0%: $p = \frac{12\,000(0.06)}{1-1.06^{-3}} = 4489.32$, total 13 467.95, interest 1467.95. At 5.4%: $p = \frac{12\,000(0.054)}{1-1.054^{-3}} = 4439.57$, total 13 318.71, interest 1318.71. The lower rate saves 49.75 a year and 149.24 over the term.

15 $B_{n+1} = 1.015B_n - 150$, $B_0 = 1200$ gives $B_1 = 1068.00$, $B_2 = 934.02$, $B_3 = 798.03$. Unrolling: $B_n = 1200(1.015)^n - \frac{150\,(1.015^n-1)}{0.015} = 10\,000 - 8800(1.015)^n$. Setting $B_n < 0$: $1.015^n > \frac{10\,000}{8800}$, $n > \frac{\log(25/22)}{\log 1.015} = 8.59$, so 9 months. After eight payments $B_8 = 86.87$, so the ninth payment need only cover $86.87 \times 1.015 = 88.17$: the standing 150 would overpay, since a payment is set to clear the whole remaining balance plus that month's interest, not to repeat the original schedule.

Investigation 4 · what does an infinite sum mean?

1 $S_3 = \frac{7}{8} = 0.875$, $S_4 = \frac{15}{16} = 0.9375$, $S_5 = \frac{31}{32} = 0.96875$, $S_6 = \frac{63}{64} = 0.984375$. In general $S_n = \frac{2^n-1}{2^n}$, so $S_{10} = \frac{1023}{1024} = 0.9990234375$ and $S_\infty = 1$. The gap to 1 is $\frac{1}{2^n}$: it is halved at every step, so it can be made smaller than any stated positive number, yet it is never zero.

2 After n steps the shaded area is $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$ and the unshaded remainder is exactly $\frac{1}{2^n}$. Since $\frac{1}{2^n} \to 0$ the shaded area tends to 1, the whole square. For every finite n, however, $\frac{1}{2^n} > 0$, so a strip of area $\frac{1}{2^n}$ is always left white: the limit is reached by the sequence of areas, not by any single drawing.

3 $0.999\ldots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \cdots$ with $a = 0.9$, $r = 0.1$, so $S_\infty = \frac{0.9}{0.9} = 1$. Separately $\frac{1}{3} = 0.333\ldots$ gives $1 = 3 \times \frac{1}{3} = 3 \times 0.333\ldots = 0.999\ldots$. The two arguments agree. The notation $0.999\ldots$ names the limit of $0.9,\ 0.99,\ 0.999, \ldots$, so it is not "just less than 1": there is no number strictly between $0.999\ldots$ and 1, and two decimal names can denote one value.

4 Distances: $100 + 50 + 25 + \cdots$, $a = 100$, $r = \frac{1}{2}$, so the total is $\frac{100}{1 - \frac{1}{2}} = 200$ m. If the first stage takes time t and each stage takes half the previous one, the times form $t + \frac{t}{2} + \frac{t}{4} + \cdots = \frac{t}{1 - \frac{1}{2}} = 2t$ — a finite time. The paradox is only in the description: an infinite number of stages is completed in a finite distance and a finite time, because infinitely many tasks whose durations shrink geometrically have a finite total.

5 $S_{10} \approx 2.93$, $S_{100} \approx 5.19$, $S_{1000} \approx 7.49$: still climbing. Grouping gives $\left(\frac{1}{3} + \frac{1}{4}\right) > 2 \cdot \frac{1}{4} = \frac{1}{2}$, $\left(\frac{1}{5} + \cdots + \frac{1}{8}\right) > 4 \cdot \frac{1}{8} = \frac{1}{2}$, $\left(\frac{1}{9} + \cdots + \frac{1}{16}\right) > 8 \cdot \frac{1}{16} = \frac{1}{2}$, and so on, so $S_{2^k} > 1 + \frac{k}{2}$, which exceeds any bound as k grows. The harmonic series therefore diverges even though its terms tend to zero.

6 With $a = 1$, $r = 0.5$: $S_5 = 1.9375$, $S_{10} = 1.998046875$, approaching $\frac{1}{0.5} = 2$ quickly. With $r = 0.9$: $S_5 = 4.0951$, $S_{10} = 6.5132$, creeping towards $\frac{1}{0.1} = 10$ slowly (the limit is still 10, but 0.9^n shrinks slowly). With $r = 1.1$: $S_5 = 6.1051$, $S_{10} = 15.9374$ and rising, with no limit, since 1.1^n grows. So $|r| < 1$ approaches a limit and $r > 1$ does not.

7 "An infinite geometric series converges exactly when $-1 < r < 1$, i.e. $|r| < 1$, and then its sum is $\frac{a}{1 - r}$ $(a \neq 0)$."
Justification from $S_n = \frac{a(1 - r^n)}{1 - r}$: if $|r| < 1$ then $r^n \to 0$ and $S_n \to \frac{a}{1 - r}$; if $r > 1$ then $r^n \to \infty$ and the sums grow without bound; if $r < -1$ the powers grow in magnitude and alternate in sign, so the sums swing ever wider; if $r = 1$ then $S_n = na$, unbounded for $a \neq 0$; and if $r = -1$ the powers alternate $-1, 1$, so S_n alternates between a and 0 and never settles. In the generalising step, the harmonic series is the caution: $u_n \to 0$ is necessary but not sufficient for convergence, since $1 + \frac{1}{2} + \frac{1}{3} + \cdots$ has terms tending to zero yet partial sums exceeding every bound.

Modelling task · designing a repayment plan

1 $B_{n+1} = 1.006B_n - p$ with $B_0 = 15\,000$. Unrolling, $B_n = 15\,000(1.006)^n - \frac{p\,(1.006^n - 1)}{0.006}$. Setting $B_{36} = 0$:
$$p = \frac{15\,000(0.006)}{1 - 1.006^{-36}} = \frac{90}{0.193744} = 464.53.$$
Total repaid $36p = 16\,723.06$; total interest $= 1723.06$.

2 Over 48 months, $p = \frac{90}{1 - 1.006^{-48}} = 360.59$, total 17 308.18, interest 2308.18. The three-year plan asks $464.53 - 360.59 = 103.94$ more each month, but the balance is cleared twelve months earlier, so less interest accrues: it costs $2308.18 - 1723.06 = 585.12$ less overall. Paying more per month buys a shorter time in debt, and interest is charged on time, not on the original sum.

3 With $p = 464.53 + 50 = 514.53$, solve $B_n = 0$: $n = -\frac{\log\left(1 - \frac{15\,000(0.006)}{514.53}\right)}{\log 1.006} = \frac{0.192277}{0.0059821} = 32.14$. So 32 full instalments plus a final part-payment clear it — about 4 months early. $B_{32} = 72.42$, so the last payment is $72.42 \times 1.006 = 72.86$; the total paid is $32(514.53) + 72.86 = 16\,537.82$, interest 1537.82, a saving of about 185.

4 Assumption 1: the rate stays 0.6% per month (nominal 7.2% a year, effective annual 7.44%) for the whole term. A rise to 0.8% per month lifts the instalment to 481.20 and the interest to 2323.05, so the plan must be affordable at the higher rate, or a cap on the rate is needed. Assumption 2: payments are made on time, in equal amounts, at month ends, with interest on the outstanding balance only. A missed month adds a month of interest to the whole balance and pushes the term out; a bonus payment made early saves more, because it removes principal before it can compound. Assumption 3: no fees, insurance or tax, no inflation, and the equipment earns the cash the repayments need from day one. Arrangement fees or insurance would add to the cost and could reverse the ranking against a rival lender, while inflation makes the fixed instalment easier in real terms but does not reduce the nominal cash the enterprise must find each month.

Mixed practice 4

1 $a = 6$, $d = 4$, so $u_n = 6 + 4(n - 1) = 4n + 2$; $u_{25} = 102$; $4n + 2 = 142$ gives $n = 35$, the 35th term.

2 $r = \frac{54}{81} = \frac{2}{3}$ (check $\frac{36}{54} = \frac{2}{3}$), so $u_n = 81\left(\frac{2}{3}\right)^{n-1}$. Then $u_6 = 81\left(\frac{2}{3}\right)^5 = 81 \cdot \frac{32}{243} = \frac{32}{3} = 10.67$.

3 12 terms from $5(1) - 2 = 3$ to $5(12) - 2 = 58$: $S = \frac{12}{2}(3 + 58) = 6 \times 61 = 366$.

4 $r = \frac{12}{18} = \frac{2}{3}$, $|r| < 1$, so $S_\infty = \frac{18}{1 - \frac{2}{3}} = \frac{18}{\frac{1}{3}} = 54$.

5 $\frac{0.72}{0.99} = \frac{72}{99} = \frac{8}{11}$. Check: $8 \div 11 = 0.7272\ldots$ ✓

6 Five terms $2, 2r, 2r^2, 2r^3, 2r^4 = 162$ give $r^4 = 81$, so $r = 3$ (positive terms). The inserted numbers are 6, 18, 54: 2, 6, 18, 54, 162. ✓

7 $d = \frac{41 - 17}{10 - 4} = 4$, $a = 17 - 3(4) = 5$, so $u_n = 4n + 1$ and $u_{20} = 81$. Check $u_4 = 17$, $u_{10} = 41$. ✓

8 $r^3 = \frac{u_5}{u_2} = \frac{160}{20} = 8$, so $r = 2$ (odd gap gives a unique ratio) and $a = \frac{20}{2} = 10$; $u_n = 10 \cdot 2^{n-1}$.

9 First term 2, nth term $3n - 1$: $S_n = \frac{n}{2}(2 + 3n - 1) = \frac{n(3n + 1)}{2} = 155$, so $3n^2 + n - 310 = 0$, $n = \frac{-1 + \sqrt{3721}}{6} = \frac{-1 + 61}{6} = 10$ (check $10 \times \frac{31}{2} = 155$); the negative root is rejected.

10 $B_{24} = \frac{120\left(1.004^{24} - 1\right)}{0.004} = 120(25.13705) = 3016.45$. Deposits total 2880, so interest is 136.45.

11 (a) Converges: $r = 0.2$, $|r| < 1$, $S_\infty = \frac{5}{0.8} = 6.25$. (b) Diverges: $r = -1$, $|r| = 1$, partial sums alternate 3, 0, 3, 0. (c) Diverges: $r = 1$, $S_n = 7n$ grows. (d) Diverges: $r = 1.5$, $|r| > 1$, terms grow.

12 The reasoning mistakes the last term for the number of terms, and so uses $n = 94$. Correct: $n = \frac{94 - 4}{5} + 1 = 19$, and $S_{19} = \frac{19}{2}(4 + 94) = 19 \times 49 = 931$.

13 Down 5 m; rebound heights 4, 3.2, 2.56, … ($a = 4$, $r = 0.8$) are each travelled twice: total $= 5 + 2\left(\frac{4}{1 - 0.8}\right) = 5 + 40 = 45$ m.

14 $u_1 + u_n = a + [a + (n - 1)d] = 2a + (n - 1)d$, and $u_2 + u_{n-1} = (a + d) + [a + (n - 2)d] = 2a + (n - 1)d$. The two are identical, so $u_1 + u_n = u_2 + u_{n-1}$. The same working with u_k and u_{n+1-k} shows every paired sum equals $a + u_n = a + l$; with $\frac{n}{2}$ such pairs, $S_n = \frac{n}{2}(a + l)$, which is exactly what Gauss pairing computes.

15 First year 32 000, so year n pays $32\,000(1.03)^{n-1}$. Tenth year: $32\,000(1.03)^9 = 41\,752.74$. Ten-year total: $S_{10} = \frac{32\,000\left(1.03^{10} - 1\right)}{0.03} = 366\,844.14$. (If the first rise is taken as already applied, both figures are multiplied by 1.03.)

16 $p = \frac{5000(0.008)}{1 - 1.008^{-12}} = \frac{40}{0.091189} = 438.65$ a month. Total $12p = 5263.80$, so interest $= 263.80$.

17 $a = 24(1 - 0.25) = 18$; $u_3 = 18(0.25)^2 = 18 \times \frac{1}{16} = \frac{9}{8} = 1.125$.

18 A is geometric, $a_n = 100(0.9)^{n-1}$; B is arithmetic, $b_n = 100 - 5(n-1)$. Tenth terms: $a_{10} = 100(0.9)^9 = 38.74$ and $b_{10} = 55$, so B is larger at first — the geometric sequence falls faster early. $100(0.9)^{n-1} < 1$ gives $0.9^{n-1} < 0.01$, so $n - 1 > \dfrac{\log 0.01}{\log 0.9} = 43.71$, i.e. $n = 45$: $a_{45} = 0.97$ (while $a_{44} = 1.08$).

19 $\dfrac{a}{1-r} = 9$ gives $a = 9(1-r)$, and $a + ar = a(1+r) = 5$ gives $9(1-r)(1+r) = 5$, so $1 - r^2 = \dfrac{5}{9}$, $r^2 = \dfrac{4}{9}$, $r = \pm\dfrac{2}{3}$ — both admissible since $|r| < 1$. If $r = \dfrac{2}{3}$, $a = 3$; if $r = -\dfrac{2}{3}$, $a = 15$. Checks: $\dfrac{3}{1/3} = 9$, $3 + 2 = 5$; $\dfrac{15}{5/3} = 9$, $15 - 10 = 5$. ✓

20 First eight terms: $S_8 = \dfrac{8}{2}(2(5) + 7d) = 40 + 28d$. Next four (terms 9 to 12): $\dfrac{4}{2}[(5 + 8d) + (5 + 11d)] = 20 + 38d$. Equating: $40 + 28d = 20 + 38d \Rightarrow d = 2$. Verification by direct addition: $5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 96$ and $21 + 23 + 25 + 27 = 96$. ✓

21 New hearers are geometric: 3, 6, 12, … with $u_n = 3 \cdot 2^{n-1}$. (a) Day 6: $3 \cdot 2^5 = 96$ new people. (b) Total by the end of day 6: $S_6 = 3(2^6 - 1) = 3 \times 63 = 189$. (c) Require $3(2^n - 1) > 5000$, so $2^n > 1667.7$; $2^{10} = 1024$ (total 3069) is short and $2^{11} = 2048$ (total 6141) passes, so the town is exceeded on day 11.

Criterion A · Sequences and series

1 (a) $d = 15 - 11 = 4$. (b) $u_n = 11 + 4(n-1) = 4n + 7$. (c) $u_{30} = 4(30) + 7 = 127$. (d) Solve $4n + 7 = 203$: $4n = 196$, $n = 49$, a positive integer, so 203 is a term — it is the 49th.

Marking guidance: [5] one mark each for d, for $u_n = 4n + 7$, for $u_{30} = 127$, and two for part (d) (one for the equation and $n = 49$, one for the stated reason that an integer index makes it a term). The form $11 + 4n$ in (b) costs the method mark even if later parts are consistent.

2 (a) $r = \dfrac{18}{6} = 3$, $u_n = 6 \cdot 3^{n-1}$. (b) $u_6 = 6 \cdot 3^5 = 6(243) = 1458$. (c) Require $6 \cdot 3^{n-1} > 5000$, so $3^{n-1} > 833.3$ and $n - 1 > \dfrac{\log(5000/6)}{\log 3} = 6.12$; hence $n - 1 = 7$, $n = 8$. Check: $u_7 = 4374 < 5000$ but $u_8 = 6 \cdot 3^7 = 13\,122 > 5000$, so the eighth term is the first to exceed 5000.

Marking guidance: [5] one mark for $r = 3$, one for the closed form $6 \cdot 3^{n-1}$, one for $u_6 = 1458$, and two for (c): one for the logarithm or an equivalent trial of powers of 3, one for $n = 8$ justified by checking $u_7 = 4374 < 5000$. Answering $n = 7$ by rounding 6.12 down loses both marks in (c).

3 (a) 15 terms from $4(1) - 3 = 1$ to $4(15) - 3 = 57$: $S = \dfrac{15}{2}(1 + 57) = 15 \times 29 = 435$. (b) $a = 24$, $r = \frac{1}{2}$, and $|r| < 1$, so $S_\infty = \dfrac{24}{1 - \frac{1}{2}} = 48$.

Marking guidance: [4] two marks each: one for identifying the count of terms (or the ratio and testing $|r| < 1$) and one for the value 435 / 48.

4 Iterating $B_{n+1} = 1.04B_n + 300$ from $B_0 = 2000$: $B_1 = 2380.00$, $B_2 = 2775.20$, $B_3 = 3186.21$. Check by formula: $2000(1.04)^3 + \dfrac{300(1.04^3 - 1)}{0.04} = 2249.73 + 936.48 = 3186.21$.

Marking guidance: [3] one mark for growing the initial investment to 2249.73, one for the annuity value 936.48 (the final 300 earns no interest), one for the correct total to two decimal places.

5 (a) $r = 3$ and $|r| > 1$, so the powers 3^n grow without bound; the partial sums $S_1 = 1$, $S_2 = 4$, $S_3 = 13$ (in general $S_n = \dfrac{3^n - 1}{2}$) keep rising and pass any number, so the series diverges and has no finite sum. (b) $a = 1$, $r = -\dfrac{1}{3}$, $|r| < 1$:

$$S_\infty = \frac{1}{1 - \left(-\frac{1}{3}\right)} = \frac{1}{\frac{4}{3}} = \frac{3}{4}.$$

The partial sums $1, \frac{2}{3}, \frac{7}{9}, \frac{20}{27}, \ldots$ oscillate towards $\frac{3}{4}$.

Marking guidance: [3] one mark for three correct partial sums, one for linking divergence to $|r| > 1$ (not to "many terms"), one for $\frac{3}{4}$ with the condition $|r| < 1$ noticed. Using $\dfrac{a}{1-r}$ on the divergent series to obtain $-\frac{1}{2}$ scores no mark in (a).

Criterion B · A tower of cans

1 Cans in row n: 2, 5, 8, 11, 14, 17. Totals T_n: 2, 7, 15, 26, 40, 57. Each new row adds 3 cans, and the totals add the row: $2,\ 2 + 5 = 7,\ 7 + 8 = 15,\ 15 + 11 = 26,\ 26 + 14 = 40,\ 40 + 17 = 57$.

Marking guidance: [6] one mark per completed correct pair of entries in the last four columns; the row entry for n and the total for n together count as one. A table generated by repeatedly adding 3 is acceptable; a gap in the totals costs the mark for that column only.

2 Words: the number of cans in a row increases by a constant 3 , so the rows form an arithmetic sequence; the totals increase by ever-larger amounts, and their first differences (5, 8, 11, 14, 17) are exactly the row counts, themselves increasing by 3. Formulas: row n has $u_n = 2 + 3(n-1) = 3n - 1$ cans, and

$$T_n = \frac{n}{2}(2 + (3n-1)) = \frac{n(3n+1)}{2}.$$

Test at $n = 6$: $u_6 = 17$ ✓ and $T_6 = \dfrac{6 \times 19}{2} = 57$ ✓.

Marking guidance: [6] two marks for the two verbal descriptions (constant row difference; differences of T_n equal the rows), two for the row formula, two for the total formula, provided both are tested with $n = 6$. An untested formula loses the verification half-mark.

3 Write the sum of rows forwards and backwards:

$T_n = 2 + 5 + 8 + \cdots + (3n-1)$, $\quad T_n = (3n-1) + \cdots + 8 + 5 +$

Column by column the entries add to $2 + (3n-1) = 3n + 1$, because moving one row down in the first line is offset by moving one row up in the second; with n columns, $2T_n = n(3n+1)$, so $T_n = \dfrac{n(3n+1)}{2}$. ✓ Difference check:

$$T_n - T_{n-1} = \frac{3n^2 + n - \big(3(n-1)^2 + (n-1)\big)}{2} = \frac{6n - 2}{2} = 3n - 1 =$$

so the first differences reproduce the row totals exactly.

Marking guidance: [5] three marks for a pairing argument that states why every pair is equal and ends with the formula, two for the difference check. A proof that only substitutes n into the formula (verification, not proof) reaches at most the middle band.

4 Solve $\dfrac{n(3n+1)}{2} = 100$: $3n^2 + n - 200 = 0$, so $n = \dfrac{-1 \pm \sqrt{1 + 2400}}{6} = \dfrac{-1 \pm 49}{6}$, giving $n = 8$ or $n = -\dfrac{25}{3}$. The negative, non-integer root is rejected because n counts rows, and T_n increases strictly for $n > 0$, so at most one positive integer can give 100. Hence $n = 8$ rows; the bottom row holds $u_8 = 3(8) - 1 = 23$ cans. Check: $\dfrac{8}{2}(2 + 23) = 4 \times 25 = 100$ ✓.

Marking guidance: [4] one mark for the quadratic, one for both roots, one for $n = 8$ with the rejection justified, one for 23 cans. "The negative root is impossible" without mentioning that n must also be an integer still earns full marks here, since the other root is not an integer either.

5 New design: row n has $v_n = 3 + 4(n-1) = 4n - 1$ cans, and $T'_n = \dfrac{n}{2}(3 + (4n-1)) = n(2n+1) = 2n^2 + n$. Solving $2n^2 + n = 105$: $2n^2 + n - 105 = 0$, $n = \dfrac{-1 + \sqrt{1 + 840}}{4} = \dfrac{-1 + 29}{4} = 7$. Check: $2(49) + 7 = 105$ ✓, so the tower has 7 rows (bottom row 25 cans).

Marking guidance: [3] one mark for the row formula, one for the total formula, one for $n = 7$ with the rejected root or a check.

Criterion C · Explaining an infinite sum

1 **Sample response.** A series has no "answer" until we build one from what we can actually do. Adding the first n terms gives a *partial sum* $S_n = u_1 + u_2 + \cdots + u_n$; this is a finite addition, always a definite number. As n runs through $1, 2, 3, \ldots$ these sums form a new sequence. We say the series **converges** with sum S if that sequence of partial sums closes in on one fixed number: however small a tolerance you name, from some point on every partial sum stays within it. The sum of an infinite series is therefore a limit, not a finished addition, and a series that has no such limit has no sum at all.

Marking guidance: [3] one mark for defining S_n as the sum of the first n terms, one for convergence as "the partial sums approach a fixed value", one for insisting the infinite sum is that limit. Language such as "the answer you get if you keep going forever" without reference to partial sums is not sufficient.

2 For $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots$ the partial sums are $\frac{1}{2}=0.5$, $\frac{3}{4}=0.75$, $\frac{7}{8}=0.875$, $\frac{15}{16}=0.9375$, $\frac{31}{32}=0.96875$, $\frac{63}{64}=0.984375$. Each is $S_n=1-\frac{1}{2^n}$, so the shortfall to 1 is halved every row of the table: after ten terms it is only $\frac{1}{1024}$. The sums never exceed 1 and never stop moving towards it, so the series converges to $S_\infty=1$. A square of side 1 shaded half, then half the remainder, shows the same thing geometrically.

Marking guidance: [4] two marks for a correct table of at least four partial sums, one for the pattern $S_n=1-\frac{1}{2^n}$ (or "the gap halves"), one for stating the limit 1 as the behaviour of the table rather than as a guess.

3 Let $S_n=a+ar+ar^2+\cdots+ar^{n-1}$. Multiplying by r shifts every term one place, so $rS_n=ar+ar^2+\cdots+ar^n$. Subtracting cancels all the middle terms and leaves $(1-r)S_n=a-ar^n$, hence for $r\neq 1$

$$S_n=\frac{a(1-r^n)}{1-r}=\frac{a}{1-r}-\frac{ar^n}{1-r}.$$

The second fraction is the distance from the partial sum to $\frac{a}{1-r}$. If $|r|<1$, then r^n shrinks towards zero, so that distance becomes as small as we like and $S_n\to\frac{a}{1-r}$; that limit is what we call S_∞. For the decimal $0.999\ldots$ the series is $0.9+0.09+\cdots$ with $a=0.9$, $r=0.1$, giving $S_\infty=\frac{0.9}{0.9}=1$.

Marking guidance: [5] two marks for the subtraction producing $(1-r)S_n=a-ar^n$, one for dividing with the condition $r\neq 1$ stated, two for splitting $\frac{a}{1-r}-\frac{ar^n}{1-r}$ and explaining that $r^n\to 0$ only when $|r|<1$.

4 The same formula cannot be used when $|r|\geq 1$, because the step $r^n\to 0$ is false there: the term $\frac{ar^n}{1-r}$ does not disappear, so nothing approaches $\frac{a}{1-r}$. Counterexample: for $1+2+4+8+\cdots$ the formula $\frac{a}{1-r}$ gives $\frac{1}{1-2}=-1$, yet the partial sums are 1, 3, 7, 15, all positive and rising past any bound. A number produced by a formula outside its conditions is arithmetic with no meaning; the series diverges. (At $r=1$ the sum is na, which grows without limit unless $a=0$; at $r=-1$ the partial sums alternate and never settle.)

Marking guidance: [4] one mark for identifying the failed step $r^n\to 0$, two for a counterexample with correct partial sums, one for saying explicitly that -1 is not the sum. Claiming the series "equals -1 because the formula says so" scores zero.

5 Writing $0.999\ldots=0.9+0.09+0.009+\cdots$ with $a=0.9$ and $r=0.1$ gives $S_\infty=\frac{0.9}{1-0.1}=1$. The nth partial sum is $1-\frac{1}{10^n}$, so the gap to 1 is $\frac{1}{10^n}$, smaller than any positive number for large enough n; that gap has no room to contain a number, so there is no value "just below 1" for $0.999\ldots$ to be. Since $1=3\times\frac{1}{3}=3\times 0.333\ldots=0.999\ldots$, the two names 1 and $0.999\ldots$ denote one and the same number.

Marking guidance: [2] one mark for the geometric series and its sum, one for the conclusion that $0.999\ldots$ is exactly 1, not approximately.

6 Notation used throughout: u_n for a term, S_n for the sum of the first n terms, $S_\infty=\lim_{n\to\infty}S_n$ for the limit, $\sum$ for "add", and the condition written as $S_\infty=\frac{a}{1-r}$ only with $|r|<1$ attached. Every symbol is defined when it first appears, quantities that are not pure numbers carry units, and the two tables (partial sums of $\frac{1}{2}+\frac{1}{4}+\cdots$, and the shaded-square diagram with the remainder $\frac{1}{2^n}$ marked) do the work that a paragraph of words would only describe.

Marking guidance: [2] one mark for consistent, defined notation (including $|r|<1$ stated beside the formula), one for a table or diagram that is correctly labelled and actually used to support an argument.

Criterion D · Save now or borrow now?

1 Monthly rate $i=\frac{3.6}{12}\%=0.3\%=0.003$, $n=36$. Balance recurrence $B_{n+1}=1.003B_n+d$ with $B_0=0$ unrolls to the annuity $B_{36}=\frac{d(1.003^{36}-1)}{0.003}$. Setting it to 9500:

$$d=\frac{9500(0.003)}{1.003^{36}-1}=\frac{28.5}{0.113868}=250.29\quad\text{(per month)}.$$

Total deposited $36d=9010.46$; interest earned $9500-9010.46=489.54$. (A deposit of 250.30 guarantees the target is reached.)

Marking guidance: [5] one mark for the monthly rate 0.003, one for choosing the annuity formula, one for the substitution, one for 250.29, one for both totals (deposited and interest) consistent with it.

2 Monthly rate $i=\frac{7.2}{12}\%=0.6\%=0.006$, $L=9500$, $n=36$:

$$p=\frac{Li}{1-(1+i)^{-n}}=\frac{9500(0.006)}{1-1.006^{-36}}=\frac{57}{0.193744}=294.20.$$

Total repaid $36p=10\,591.27$; interest paid $=10\,591.27-9500=1091.27$.

Marking guidance: [5] one mark for the monthly rate, one for the loan formula (or the equivalent condition $B_{36}=0$), one for the instalment 294.20, one for the total 10 591.27, one for the interest 1091.27. Dividing 9500 by 36 and then adding 7.2% of 9500 once is a method error and caps the response at the lower bands.

3 Monthly cost: saving needs 250.29, borrowing needs 294.20, so borrowing costs 43.91 more each month. Total cost: the saving route turns 9010.46 of contributions into the 9500 equipment, the loan route costs 10 591.27 for the same 9500, a difference of 1580.81 (made up of 489.54 interest earned instead of 1091.27 interest paid). Timing: saving only delivers the equipment at month 36; borrowing delivers it immediately, so the group has the use of the equipment for the whole three years while it is still paying for it. Advantage of S: it is cheaper by about 1580.81 over the term and builds a reserve. Advantage of B: immediate access, which matters if the equipment is needed to earn income now.

Marking guidance: [5] two marks for a comparison on one common basis (monthly and total figures quoted for both options), one for the timing difference, two for one clearly stated advantage of each option, each tied to a number or a dated consequence.

4 Assumption 1: the deposit rate stays 3.6% a year. If it falls to 2% a year the required deposit rises to 256.27 and the interest earned drops from 489.54 to 274.24, so saving becomes less attractive while a fixed-rate loan is unaffected, and the gap of 1580.81 in favour of saving shrinks to about 1366. Assumption 2: the loan rate stays 7.2% a year with no fees or insurance. A variable rate raises the instalment above 294.20; an arrangement fee of 2% alone (borrowing 9690) gives $p=300.09$, which already breaks a budget of 300 a month — so the ranking on monthly cost can change before the ranking on total cost does. Assumption 3: the equipment price stays 9500 , and no money is withdrawn early. If the price rises with inflation at 3% a year, the three-year plan must target about 10 381, pushing the deposit to 273.50, and an emergency withdrawal stops the compounding; borrowing fixes the price on day one, which is the strongest argument for Option B.

Marking guidance: [5] three marks for three distinct, relevant assumptions (rate, charges, price or liquidity) and two for showing a directional change in the comparison for each; an assumption that could not change the recommendation earns no mark.

5 Both routes fit a budget of 300 per month: 250.29 leaves about 50 of headroom, 294.20 leaves only 5.80, so affordability alone does not decide. Recommendation: if the equipment is not needed before month 36, choose Option S — it costs 1580.81 less, the monthly payment is easier, and the habit of saving builds a buffer that a loan does not. If the group needs the equipment now and the equipment generates income or saves cost greater than about 30 a month, choose Option B: paying 294.20 for an asset in use beats waiting three years, and the difference of 43.91 a month is inside the budget. Limitations: the recommendation rests on both rates holding, on no fees, on a constant price of 9500, and on the assumption that the group will not draw on the savings early; it also does not value the income the equipment produces, which is the only thing that could justify borrowing at 7.2%, and a single early repair bill could exhaust the 50 headroom that makes Option S comfortable.

Marking guidance: [4] one mark for a recommendation that follows from the figures in parts 1–3, one for the condition under which the other option is better, two for evaluating limitations honestly (stating what would overturn the advice) rather than restating the models.

CHAPTER 5 · QUADRATIC INEQUALITIES AND LINEAR PROGRAMMING

Opening problem · two products, three limits

1 With x desks and y shelves the five inequalities are $3x + y \leq 54$ (assembly hours), $x + 2y \leq 40$ (finishing hours), $x + y \leq 22$ (storage), $x \geq 0$ and $y \geq 0$; all boundaries are solid. Test point $(1, 1)$ satisfies every one ($4 \leq 54$, $5 \leq 40$, $2 \leq 22$), so the allowed region is on the origin side of each line: the pentagon with vertices $(0, 0)$, $(18, 0)$, $(16, 6)$, $(4, 18)$, $(0, 20)$.

2 At $(10, 8)$: assembly $30 + 8 = 38 \leq 54$ (16 hours spare), finishing $10 + 16 = 26 \leq 40$ (14 spare), storage $10 + 8 = 18 \leq 22$ (4 spare). Every limit is met, so the plan is feasible, and its profit is $50(10) + 40(8) = 500 + 320 = £820$.

Feasible is not best: no constraint is tight, so output can still be increased.

3 Yes: $(16, 6)$ is feasible (54, 28, 22) and gives $50(16) + 40(6) = 800 + 240 = £1040 > £900$. Because the region is bounded, the corner-point principle settles the question: $P = 50x + 40y$ is 0 at $(0, 0)$, 900 at $(18, 0)$, 1040 at $(16, 6)$, 920 at $(4, 18)$ and 800 at $(0, 20)$. No two vertices tie, so there is a single best plan — 16 desks and 6 shelves.

4 Four extra storage units ($x + y \leq 26$): the optimum moves to $3x + y = 54$ with $x + y = 26$, giving $2x = 28$, $x = 14$, $y = 12$; finishing $14 + 24 = 38 \leq 40$ keeps it feasible and profit is $700 + 480 = £1180$, a gain of £140, i.e. £35 per unit. Four extra assembly hours ($3x + y \leq 58$): with $x + y = 22$ this gives $2x = 36$, $x = 18$, $y = 4$, finishing $18 + 8 = 26 \leq 40$, profit $900 + 160 = £1060$, a gain of only £20, i.e. £5 per hour. Storage is worth more, on the assumption that the extra items can be stored and sold.

Prerequisite check

1 $x^2 - 5x + 6 = (x - 2)(x - 3)$ (2 and 3 multiply to 6, sum to 5); $x^2 - 9 = (x - 3)(x + 3)$, a difference of two squares.

2 $x^2 - 5x + 6 = 0$ gives $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$. $2x^2 + x - 3 = (2x + 3)(x - 1) = 0$ gives $x = 1$ or $x = -\frac{3}{2}$.

3 $y = x^2 - 4x + 3 = (x - 1)(x - 3)$: x-intercepts $(1, 0)$ and $(3, 0)$, y-intercept $(0, 3)$, vertex $(2, -1)$ since $y = 4 - 8 + 3 = -1$; the parabola opens upward.

4 $3x - 7 \leq 8 \Rightarrow 3x \leq 15 \Rightarrow x \leq 5$. $-2x < 6 \Rightarrow x > -3$ (dividing by -2 reverses the sign).

5 $[-1, 4)$: square bracket at -1 because it is included, round bracket at 4 because it is excluded.

6 Gradient $\frac{3-0}{2-4} = -\frac{3}{2}$, so $y - 0 = -\frac{3}{2}(x - 4)$, i.e. $y = -\frac{3}{2}x + 6$, or $3x + 2y = 12$. Check $(2, 3)$: $6 + 6 = 12$.

Exercise 5A · Quadratic inequalities in one variable

1 (a) Critical values 2, 6; the product of two factors is positive outside them (test $x = 0$: $(-2)(-6) = 12 > 0$): $x < 2$ or $x > 6$. (b) Critical values -5, -1; negative between them (test $x = -3$: $(2)(-2) = -4 \leq 0$): $-5 \leq x \leq -1$. (c) Critical values $-3, 0$: $x \leq -3$ or $x \geq 0$ (test $x = 1$: $1(4) = 4 \geq 0$).

2 (a) $(x - 4)(x + 4) < 0$: $-4 < x < 4$. (b) $(x - 4)(x + 3) \geq 0$: $x \leq -3$ or $x \geq 4$. (c) $2x^2 + 7x + 3 = (2x + 1)(x + 3) \leq 0$: $-3 \leq x \leq -\frac{1}{2}$ (test $x = 0$: $3 > 0$, so the outside intervals fail).

3 (a) $x^2 - 5x < 0$, $x(x - 5) < 0$: $0 < x < 5$. (b) $x^2 + 3x - 10 \geq 0$, $(x + 5)(x - 2) \geq 0$: $x \leq -5$ or $x \geq 2$. (c) $4x^2 - 9 > 0$, $(2x - 3)(2x + 3) > 0$: $x < -\frac{3}{2}$ or $x > \frac{3}{2}$. Never divide by x: $x = 0$ is a value to be tested, not cancelled.

4 (a) $(-\infty, 2) \cup (6, \infty)$. (b) $[-5, -1]$. (c) $(-\infty, -3] \cup [0, \infty)$. The union sign is essential: no number lies in both parts of (a) or (c).

5 (a) A square is never negative, so $(x - 3)^2 \leq 0$ forces equality: $x = 3$ only. (b) $(x + 2)^2 > 0$ for every x except the one that makes it zero: $x \neq -2$, i.e. $\mathbb{R} \setminus \{-2\}$. (c) $(x - 1)^2 < 0$ has no solution, $\varnothing$.

6 (a) $x^2 - 2x + 5 = (x - 1)^2 + 4 \geq 4$: always positive ($\Delta = 4 - 20 = -16 < 0$, $a > 0$). (b) $-2x^2 + 4x - 5 = -2(x - 1)^2 - 3 \leq -3$: always negative ($\Delta = 16 - 40 < 0$, $a < 0$). (c) $9x^2 - 6x + 1 = (3x - 1)^2 \geq 0$, zero at $x = \frac{1}{3}$: never negative, but not always positive.

7 $x = -4$ satisfies $x^2 > 9$ since $16 > 9$, but $-4 < 3$, so it is not in the student's answer. Correctly: $x^2 - 9 > 0 \Rightarrow (x - 3)(x + 3) > 0 \Rightarrow x < -3$ or $x > 3$. Taking a square root is reversible only for known non-negative quantities.

8 $x^2 + 6x + 10 = (x + 3)^2 + 1$. Since $(x + 3)^2 \geq 0$ for every real x, the expression is at least $1 > 0$; its minimum value 1 occurs at $x = -3$, and $\Delta = 36 - 40 = -4 < 0$ confirms the graph never reaches the axis.

9 The parabola opens upward, so it is below the axis between the intercepts, $-1 < x < 5$, and on or above it outside them, $x \leq -1$ or $x \geq 5$. Test $x = 0$: $y = -5 < 0$, confirming the middle interval is the below-axis part.

10 A repeated root needs $\Delta = k^2 - 4(1)(25) = k^2 - 100 = 0$, so $k = 10$ or $k = -10$. For $k = 10$, $x^2 + 10x + 25 = (x + 5)^2 \geq 0$ for all x: solution $\mathbb{R}$ (equality at $x = -5$). For $k = -10$, $(x - 5)^2 \geq 0$: solution $\mathbb{R}$ (equality at $x = 5$).

11 $x^2 - 2x + 3 = (x - 1)^2 + 2$. As $(x - 1)^2 \geq 0$ for all real x, the expression is $\geq 2 > 0$; therefore it is strictly positive for every real x. (Equivalently $\Delta = 4 - 12 = -8 < 0$ and $a = 1 > 0$.)

12 Take $x = -3$, $y = 1$: $x^2 = 9 > 1 = y^2$ holds while $x = -3 < 1 = y$, so the implication is false. Squaring compares magnitudes, not order: $x^2 > y^2$ is the same as $|x| > |y|$, and a large negative x satisfies it while failing $x > y$.

13 $20t - 5t^2 > 15 \Rightarrow -5t^2 + 20t - 15 > 0 \Rightarrow t^2 - 4t + 3 < 0$ (dividing by -5 reverses the sign) $\Rightarrow (t - 1)(t - 3) < 0 \Rightarrow 1 < t < 3$. The ball is higher than 15 m between 1 s and 3 s, i.e. for 2 s: it passes 15 m on the way up at $t = 1$ and on the way down at $t = 3$.

14 $P > 0 \Rightarrow -2x^2 + 14x - 20 > 0 \Rightarrow x^2 - 7x + 10 < 0 \Rightarrow (x - 2)(x - 5) < 0 \Rightarrow 2 < x < 5$: profit is positive for production strictly between 200 and 500 units (break-even at 200 and 500). The maximum is at $x = -\frac{14}{2(-2)} = 3.5$, where $P = -2(12.25) + 49 - 20 = 4.5$ hundred pounds $= £450$ from 350 units.

15 Since $a = 1 > 0$, the parabola is strictly above the axis exactly when it has no real roots: $\Delta = m^2 - 16 < 0 \Rightarrow -4 < m < 4$. At $m = 4$, $x^2 + 4x + 4 = (x + 2)^2$, which is 0 at $x = -2$, so the strict inequality fails there; at $m = -4$ it fails at $x = 2$. Thus $m = \pm 4$ are the boundary cases where the graph touches the axis, giving ≥ 0 for all x but not > 0.

Exercise 5B · Inequalities in two variables

1 $(0, 0)$: $0 \leq 18$, strictly inside. $(5, 2)$: $15 + 4 = 19 > 18$, outside. $(2, 6)$: $6 + 12 = 18$, on the boundary (and included). $(6, 0)$: $18 + 0 = 18$, on the boundary (and included).

2 (a) Boundary $x + y = 5$, solid, through $(5, 0)$ and $(0, 5)$; test $(0, 0)$ gives $0 \leq 5$, true, so shade the origin side (below). (b) Boundary $y = 2x - 1$, solid, through $(0.5, 0)$ and $(0, -1)$; test $(0, 0)$ gives $0 \geq -1$, true, so shade the origin side (above the line). (c) Boundary $x = 4$, solid vertical line; test $(0, 0)$ gives $0 \leq 4$, true, so shade to the left. (d) Boundary $2x - y = 3$, dashed (strict), through $(1.5, 0)$ and $(0, -3)$; test $(0, 0)$ gives $0 > 3$, false, so shade the side away from the origin, i.e. below the line where $y < 2x - 3$.

3 $x \geq 3$ ("at least"), $y \geq 0$ ("non-negative"), $x + y \leq 20$ ("total at most"). All three boundaries are solid; test point $(5, 5)$ satisfies them all.

4 $x = 3 \geq 0$ yes; $y = 4 \geq 0$ yes; $x + y = 3 + 4 = 7 \leq 8$ yes; $y \leq 2x$ is $4 \leq 6$ yes. Every inequality holds, so $(3, 4)$ satisfies the system.

5 The line is $3x + 2y = 12$ (intercepts $(4, 0)$ and $(0, 6)$). Testing the origin: $0 \leq 12$ is true, so the half-plane containing the origin is $3x + 2y \leq 12$, with a solid boundary.

6 No: the inequality is strict, so the boundary $y = 2x + 1$ is excluded and drawn dashed. $(0, 0)$ gives $0 < 1$, which is true, so the origin satisfies it and lies in the region below the line.

7 Substituting $(0, 0)$ into $2x + y \leq 10$ gives $0 \leq 10$, true, so the origin's side — below the line — is the solution; the student has shaded $2x + y \geq 10$, the side containing points such as $(6, 6)$ where $18 > 10$. The rule "$\leq$ means shade below" fails whenever the inequality is written with x and y on one side, so a test point is required.

8 $x \geq 0$ and $x \leq 0$ together force $x = 0$, while y is unrestricted. The set $\{(0, y) : y \in \mathbb{R}\}$ is the y-axis: it has no width, hence no area, and no point of it is interior. A two-dimensional region needs some variable to range over an interval.

9 $x \geq 1$, $x \leq 4$, $y \geq 2$, $y \leq 6$; all four boundaries are solid and the region is their overlap. Test point $(2,4)$: $2 \geq 1$, $2 \leq 4$, $4 \geq 2$, $4 \leq 6$, all true.

10 $(2,2)$: $x + y = 4 \geq 6$ is false, so it does not belong. $(10,0)$: $10 \geq 6$, $0 \leq 8$, $10 \geq 0$, all true, so it belongs. The region is unbounded because for every $s \geq 6$ the point $(s,0)$ satisfies all three inequalities, so it contains an indefinitely long ray along the positive x-axis; its only vertices are $(0,6)$ and $(0,8)$, where $x = 0$ meets $x + y = 6$ and $y = 8$.

11 Twice as many adults as children is $a \geq 2c$ (multiply the children by 2 and compare with the adults). $a \geq 0$ and $c \geq 0$ are stated separately because $a \geq 2c$ alone admits negative values — $(a,c) = (-2,-3)$ satisfies $-2 \geq -6$ — and a negative number of people is meaningless, so non-negativity is a modelling condition, not a consequence.

12 $x \geq 0$, $y \geq 0$, $3x + 2y \leq 18$, $x + 2y \leq 10$. The edges $3x + 2y = 18$ and $x + 2y = 10$ meet where subtracting gives $2x = 8$: $(4,3)$; $3x + 2y = 18$ cuts the x-axis at $(6,0)$ and $x + 2y = 10$ cuts the y-axis at $(0,5)$. Test point $(2,1)$ inside: $8 \leq 18$ and $4 \leq 10$, true, so both signs are correct.

13 $2x + 3y \leq 24$ (garage spaces), $x + y \leq 10$ (drivers), $y \geq 2$ (minibuses), $x \geq 0$. At $(4,4)$: $8 + 12 = 20 \leq 24$, $4 + 4 = 8 \leq 10$, $4 \geq 2$ — all satisfied, so the fleet is feasible.

14 $x + y \geq 5$ (weight), $y \leq 3$ (B limit), $4x + 6y \leq 48$ i.e. $2x + 3y \leq 24$ (cost), with $x \geq 0$, $y \geq 0$. Test plan $(4,1)$: $5 \geq 5$, $1 \leq 3$, $16 + 6 = 22 \leq 48$, all satisfied.

15 The same quantity $x + y$ cannot be at most 4 and at least 6 at once, so the two half-planes do not overlap: their boundary lines are parallel with the whole strip $4 < x + y < 6$ between them. In context the feasible region is empty, which means the requirements are mutually impossible — no plan exists, so one figure, or one of the words "at most"/"at least", must be wrong.

Exercise 5C · Feasible regions and vertices

1 (a) $(0,0)$, $(6,0)$ (where $x + y = 6$ meets $y = 0$), $(0,6)$. (b) $(0,0)$, $(4,0)$ (from $2x + y = 8$), $(0,4)$ (from $x + 2y = 8$), and the crossing of the two lines: subtracting gives $x = y$, so $3x = 8$, $\left(\frac{8}{3}, \frac{8}{3}\right)$, which satisfies both bounds exactly. Test point $(1,1)$ lies inside, so each line's origin side is taken.

2 The region's vertices are $(0,0)$, $(6,0)$, $(2,4)$ (where $x + y = 6$ meets $y = 4$) and $(0,4)$. Of the listed points, $(0,0)$ and $(6,0)$ are vertices. $(4,2)$ is feasible ($6 \leq 6$, $2 \leq 4$) but lies on the edge $x + y = 6$ between $(6,0)$ and $(2,4)$, so it is a boundary point, not a corner; $(2,5)$ is outside the region since $2 + 5 = 7 > 6$ and $5 > 4$.

3 (a) Bounded: the triangle $(1,1)$, $(7,1)$, $(1,7)$, inside a circle of radius about 8. (b) Unbounded: $(s,0)$ is feasible for every $s \geq 4$, so the region runs off along the x-axis. (c) Bounded: the rectangle $(2,1)$, $(6,1)$, $(6,5)$, $(2,5)$.

4 From $x - y = 1$, $x = y + 1$; substituting: $2(y + 1) + 3y = 12 \Rightarrow 5y = 10 \Rightarrow y = 2$, $x = 3$. The lines meet at $(3,2)$; check $2(3) + 3(2) = 12$.

5 $(2,0)$ (from $x = 2$ and $y = 0$), $(9,0)$ (from $x + y = 9$ and $y = 0$) and $(2,7)$ (from $x = 2$ and $x + y = 9$). The region is the triangle they bound, with solid boundaries.

6 Count row by row in y: the largest x is $8 - 2y$, so the rows hold $9, 7, 5, 3, 1$ points for $y = 0, 1, 2, 3, 4$. Total $9 + 7 + 5 + 3 + 1 = 25$ lattice points.

7 It fails the storage constraint: $13.6 + 13.2 = 26.8 > 22$. A point outside the feasible region cannot be a vertex, so it must be discarded and the corners on the storage line used instead: $(16,6)$ and $(4,18)$. (Note that the true crossing of $3x + y = 54$ with $x + 2y = 40$ is $(13.6, 13.2)$, with sum 26.8, which fails the same constraint.)

8 Because $x = 0$ is a boundary line like any other: $(0,20)$ is where $x + 2y = 40$ meets the y-axis. It satisfies every constraint — $3(0) + 20 = 20 \leq 54$, $0 + 2(20) = 40 \leq 40$, $0 + 20 = 20 \leq 22$ — and the boundary changes direction there, so it is a vertex.

9 Vertices $(0,4)$ (where $x = 0$ meets $x + y = 4$) and $(3,1)$ (where $y = 1$ meets $x + y = 4$); $(0,1)$ is not feasible since $0 + 1 < 4$. The boundary is the ray up the y-axis from $(0,4)$, the segment to $(3,1)$, and the ray along $y = 1$ to the right. There is no top-right corner because all three inequalities are lower bounds, which stop the region only from below and from the left; x and y may both grow without limit.

10 Rows $y = 0, 1, 2, 3$ hold $7, 5, 3, 1$ points: 16 in total. The triangle $(0,0)$, $(6,0)$, $(0,3)$ has area $\frac{1}{2} \times 6 \times 3 = 9$ square units. Counting is not measuring: each lattice point stands for a whole unit while the boundary cuts partial units away, and the boundary points are counted in full. By Pick's rule, $A = I + \frac{B}{2} - 1$ with $B = 12$ boundary points gives $I = 9 - 6 + 1 = 4$, and $4 + 12 = 16$ as counted.

11 $x \geq 1$, $y \geq 1$, $x + y \leq 7$. The first two give the edges through $(1,1)$, and the line through $(6,1)$ and $(1,6)$ is $x + y = 7$; the test point $(2,2)$ gives $4 \leq 7$, so the origin side is taken.

12 Without $x + y \leq 22$ the system $3x + y \leq 54$, $x + 2y \leq 40$, $x, y \geq 0$ has vertices $(0,0)$, $(18,0)$, $(0,20)$ and the crossing $(13.6, 13.2)$ of the two resource lines. Adding $x + y \leq 22$ removes that crossing, since $13.6 + 13.2 = 26.8 > 22$. The new line then cuts the two old ones: with $3x + y = 54$ it gives $2x = 32$, $(16,6)$ (finishing $16 + 12 = 28 \leq 40$, feasible), and with $x + 2y = 40$ it gives $y = 18$, $(4,18)$ (assembly $12 + 18 = 30 \leq 54$, feasible) — so $(16,6)$ is created where the storage line meets the assembly line.

13 Vertices $(0,0)$, $(9,0)$ (from $2x + y = 18$ with $y = 0$; $9 + 0 = 9 \leq 12$), $(6,6)$ (subtracting the two lines gives $x = 6$, then $y = 6$) and $(0,12)$ (from $x + y = 12$ with $x = 0$; $0 + 12 = 12 \leq 18$). Sketch: solid lines $x = 0$, $y = 0$, $x + y = 12$ and $2x + y = 18$, shaded overlap; test point $(1,1)$ satisfies both $2 \leq 12$ and $3 \leq 18$.

14 Vertices $(2,0)$, $(8,0)$ (from $2x + y = 16$ with $y = 0$; $8 + 0 = 8 \leq 10$), $(6,4)$ (subtracting gives $x = 6$, then $y = 4$) and $(2,8)$ (from $x = 2$ with $x + y = 10$; $4 + 8 = 12 \leq 16$). The region is a bounded convex quadrilateral with a right angle at $(2,0)$ (the edges there lie on $y = 0$ and $x = 2$); its other edges have gradients -2 and -1, so no two sides are parallel and it is not a trapezium.

15 The feasible set is an intersection of half-planes, so it is a convex polygon whose sides lie on boundary lines; a polygon needs at least three sides, and two lines cut the plane into a wedge, strip or angle, all of which escape to infinity — hence at least three vertices. An unbounded region's boundary contains rays that never meet, so it can close only part of the way round: $x \geq 0$, $y \geq 0$, $x + y \geq 4$ has just the two vertices $(4,0)$ and $(0,4)$, the rest of its boundary being the ray along $y = 0$ for $x \geq 4$ and the ray up the y-axis for $y \geq 4$.

Exercise 5D · Linear programming

1 $P = 2x + 3y$: $(0,0) \to 0$, $(4,0) \to 8$, $(3,2) \to 12$, $(0,5) \to 15$. Minimum 0 at $(0,0)$; maximum 15 at $(0,5)$.

2 $P = 5x + 4y$: $(0,0) \to 0$, $(6,0) \to 30$, $(4,4) \to 20 + 16 = 36$, $(0,3) \to 12$. The region is bounded, so the corner-point principle applies: maximum $P = 36$ at $(4,4)$.

3 $P = x + y$: $(0,0) \to 0$, $(6,0) \to 6$, $(4,4) \to 8$, $(0,3) \to 3$. Minimum 0 at $(0,0)$, which is a vertex, as the principle predicts.

4 Vertices: $(0,0)$, $(8,0)$ (from $x + y = 8$ with $y = 0$; $8 + 0 = 8 \leq 18$), $(3,5)$ (subtracting the two lines gives $2y = 10$), and $(0,6)$ (from $x + 3y = 18$ with $x = 0$; $0 + 6 = 6 \leq 8$). Then $P = 3x + 2y$ is 0, 24, $9 + 10 = 19$, 12: maximum 24 at $(8,0)$.

5 The level line $4x + 3y = 12$ meets the axes at $(3,0)$ and $(0,4)$. At the origin $P = 0 < 12$, and at $(4,0)$, $P = 16 > 12$, so P increases on the side away from the origin — in the direction of the normal $(4,3)$, perpendicular to the line, whose gradient is $-\frac{4}{3}$.

6 $P = x + 3y$: $(0,0) \to 0$, $(10,0) \to 10$, $(6,4) \to 6 + 12 = 18$, $(0,6) \to 18$. Maximum 18, attained at both $(6,4)$ and $(0,6)$. What is unusual is the tie: the edge joining them has gradient $\frac{6-4}{0-6} = -\frac{1}{3}$, the same as the level lines of P, so every point of that edge is optimal.

7 The corner-point principle requires testing every vertex; two of the four were skipped. $(0,6)$ already gives $P = 18$ and $(6,4)$ gives 18, both above 10, and no vertex beats them. The objective weights y three times, so the high-y corners — exactly the ones left out — carry the maximum.

8 Along an edge the objective changes steadily, so its values at the ends bound its value in between. If the level line $P = c$ is parallel to the edge, then P is constant on that whole edge, so the two end vertices take the same value; neither can be beaten, because a larger value would need a level line beyond it, which would miss the region entirely. That is why the parallel level line last touches the region along the whole edge, not at one point.

9 The principle is a statement about bounded regions. Here $x \geq 0$, $y \geq 1$, $x + y \geq 4$ has vertices $(0, 4)$ and $(3, 1)$ only, and its boundary includes the ray $y = 1$, $x \geq 3$. On that ray $P = 2x + y = 2x + 1$, which grows past any value as $x \to \infty$, so no maximum exists; the vertices give the minimum ($P = 4$ at $(0, 4)$) but say nothing about the largest value.

10 Values: $(0, 0) \to 0$, $(8, 0) \to 8$, $(6, 4) \to 6 + 4k$, $(0, 10) \to 10k$. Equal at $(6, 4)$ and $(0, 10)$: $6 + 4k = 10k \Rightarrow k = 1$ (both then give 10, and no vertex exceeds 10). For $(0, 10)$ to be the unique maximum we need $10k > 6 + 4k \Rightarrow k > 1$ (which also gives $10k > 8 \geq P$ at $(8, 0)$ and $P \geq 0$ at $(0, 0)$). So $k > 1$; at $k = 1$ the whole edge ties instead.

11 The continuous vertices are $(0, 0) \to 0$, $\left(\frac{10}{3}, 0\right) \to \frac{50}{3} \approx 16.7$, $(2.2, 3.4) \to 11 + 6.8 = 17.8$ and $(0, 4.5) \to 9$. Testing the lattice points on the boundary near that vertex: $(3, 1) \to 17$ (feasible, $9 + 1 = 10$ and $3 + 2 = 5 \leq 9$), $(2, 3) \to 16$, $(3, 0) \to 15$, $(2, 2) \to 14$, $(1, 4) \to 13$. The integer maximum is 17 at 3 and 1 — below 17.8, and not a rounding of $(2.2, 3.4)$, which rounds to $(2, 3) \to 16$.

12 At $(16, 6)$: assembly $48 + 6 = 54$, slack 0; finishing $16 + 12 = 28$, slack $40 - 28 = 12$; storage 22, slack 0. So assembly and storage are fully used, finishing is not. At $(4, 18)$: assembly $12 + 18 = 30$, slack 24; finishing $4 + 36 = 40$, slack 0; storage 22 , slack 0. There finishing and storage are fully used, assembly is not.

13 Vertices: $(0, 0)$, $(22, 0)$ (from $2x + y = 44$), $(16, 12)$ ($2x + y = 44$ with $x + y = 28$; $16 + 24 = 40 \leq 46$), $(10, 18)$ ($x + y = 28$ with $x + 2y = 46$; $20 + 18 = 38 \leq 44$) and $(0, 23)$ (from $x + 2y = 46$; $23 \leq 28$). The crossing $(14, 16)$ of the first two lines fails $x + y \leq 28$ ($30 > 28$), so it is not a vertex. Profit $P = 5x + 6y$: 110, 152, 158, 138 — maximum £158 from 10 booklets and 18 posters.

14 Vertices: $(0, 0)$, $(20, 0)$ (where $x + y = 20$ and $3x + 2y = 60$ meet), $(10, 10)$ ($x + y = 20$ with $x + 2y = 30$; $30 + 20 = 50 \leq 60$) and $(0, 15)$ (from $x + 2y = 30$; $15 \leq 20$). The crossing $(15, 7.5)$ of $3x + 2y = 60$ with $x + 2y = 30$ fails $x + y \leq 20$ ($22.5 > 20$). Returns $P = 20x + 30y$: 400, 500, 450 — best planting is 10 herb beds and 10 salad beds, return £500.

15 Vertices: $(0, 6)$ (from $x + 2y = 12$ with $x = 0$; $6 \leq 10$), $(8, 2)$ (where $x + 2y = 12$ and $x + y = 10$ meet, giving $y = 2$, $x = 8$) and $(0, 10)$. Cost $C = 40x + 90y$: 540, 500, 900 — minimum £500 with 8 vans and 2 lorries, ten vehicles in all. The fewest vehicles is 6, the fleet $(0, 6)$, but it costs £540: a lorry supplies 2 units of capacity for £90 (£45 per unit) while a van supplies 1 unit for £40, so the cheaper fleet buys capacity from vans and accepts more vehicles.

Exercise 5E · Modelling and optimisation

1 Objective $P = 2.2x + 1.8y$ (profit in pounds). Constraints $x \leq 120$ (coffees), $y \leq 80$ (cakes), $x + y \leq 160$ (total items), $x \geq 0$, $y \geq 0$; all in items, with profit in £.

2 Vertices: $(0, 0)$, $(120, 0)$, $(120, 40)$ (where $x = 120$ meets $x + y = 160$), $(80, 80)$ (where $y = 80$ meets $x + y = 160$) and $(0, 80)$; $(120, 80)$ fails the total limit. P: 0, 264, 336, 320, 144. Maximum profit £336 from 120 coffees and 40 cakes (coffee limit and total both tight; 40 cake slots spare).

3 $P = 3x + 4y$: $(0, 0) \to 0$, $(9, 0) \to 27$, $(6, 3) \to 18 + 12 = 30$, $(0, 6) \to 24$. Maximum 30 at $(6, 3)$.

4 At $(6, 4)$: $x + y = 10$ against 12, slack 2; $2x + y = 16$ against 20, slack 4; $x \geq 0$ slack 6; $y \geq 0$ slack 4. It is not a vertex — every constraint is slack, so the point is strictly inside the region, whose vertex in that corner is $(8, 4)$, where $x + y = 12$ and $2x + y = 20$ meet.

5 $P = 35x + 80y$, with P in pounds and x, y in items (chairs and tables).

6 (i) Every loaf sells at the same price, so demand does not limit output and no markdowns are needed. (ii) Ingredient, labour and energy costs per loaf do not change with the batch size (no bulk discount, no overtime premium, no economies of scale), and no loaves are wasted.

7 The Dovetail optimum $(16, 6)$ earns £1040 while using all 54 assembly hours and all 22 storage places but only $16 + 12 = 28$ of the 40 finishing hours — 12 hours are left idle. If using every resource were necessary, some plan with finishing full, such as $(4, 18)$, would beat it, but $(4, 18)$ earns only £920. Only the binding limits constrain the optimum; the rest is slack.

8 Slack at the optimum means the marginal (shadow) price of that resource is zero: the best plan is already feasible using less of it, so the binding limits still cut the region at the same vertex and giving more of an unused resource changes nothing — however large the increase, as Question 9 shows for finishing $40 \to 50$. Reducing it is a different matter. Each unit taken back shrinks the region only in a direction the optimum does not use, so the plan survives until the reduction is large enough that the limit passes through the old optimum; beyond that point it becomes binding and the optimum must move. For a limit with slack s at the optimum, reductions up to s change nothing and reductions greater than s do.

A limit that is fully used behaves the opposite way: storage at 22 is worth £35 per extra unit until about 26.8, after which assembly and finishing bind instead and further storage is worth nothing.

9 Relaxing the finishing constraint to $x + 2y \leq 50$ leaves $(16, 6)$ feasible and its finishing use is $28 \leq 50$. The region's vertices become $(0, 0)$, $(18, 0)$, $(16, 6)$ and $(0, 22)$, giving $P = 50x + 40y$ as 0, 900, 1040, 880: still 1040 at $(16, 6)$. The optimum is set by assembly with storage, which are untouched, and the new assembly-with-finishing crossing $\left(\frac{58}{5}, \frac{96}{5}\right)$ fails storage ($30.8 > 22$), so no better corner appears.

10 No. At the optimum $(16, 24)$ the ink use is $2(16) + 24 = 56$ of 60, slack 4, while press hours are $16 + 24 = 40$ of 40 and the banner limit is exactly reached. Ink therefore has shadow price zero, so extra ink cannot lift profit; buying press hours or a larger banner market could, because those are where the region is actually cut.

11 To sell a larger run the studio may have to discount posters, so the £12 contribution per poster falls as x grows. Then the objective is no longer linear, its level lines are no longer parallel straight lines, and the corner-point principle cannot be used: the best point may move inside an edge or off the vertices altogether, so the recommendation $(16, 24)$ is no longer justified.

12 (a) Search the lattice points around the vertex — the four corners of the unit cell, $(10, 6)$, $(10, 7)$, $(11, 6)$, $(11, 7)$, plus the next points along each binding boundary such as $(9, 7)$ or $(12, 5)$ — discard the infeasible ones and take the largest P. (b) Round to a feasible point, then step along the binding constraint in the improving direction while P keeps rising. Neither is guaranteed: the rounded point may be infeasible, and in Example 5.10 the feasible rounding $(2, 3)$ gives 23 while the unrounded neighbour $(1, 4)$ gives 24.

13 Vertices: $(6, 0)$, $(16, 0)$ (from $3x + y = 48$ with $y = 0$), $(10, 18)$ (where $3x + y = 48$ meets $y = 18$) and $(6, 18)$ (where $x = 6$ meets $y = 18$). The crossing $(9, 21)$ of $3x + y = 48$ with $x + y = 30$ fails $y \leq 18$, and $(12, 18)$ fails $3x + y \leq 48$ ($54 > 48$). Profit $P = 15x + 22y$: 90, 240, $150 + 396 = 546$, $90 + 396 = 486$ — maximum £546 from 10 posters and 18 banners; the total limit is slack by $30 - 28 = 2$.

14 Extra banner places ($y \leq 22$): vertices become $(6, 0)$, $(16, 0)$, $(9, 21)$, $(8, 22)$, $(6, 22)$ with $P = 15x + 22y$ equal to 240, 597, $120 + 484 = 604$, 574; the best is now 8 posters and 22 banners at £604, a gain of £58. Extra total capacity ($x + y \leq 32$): its boundary meets $3x + y = 48$ at $(8, 24)$ and $y = 18$ at $(14, 18)$, both infeasible, so the constraint stays redundant, the vertices are unchanged and the maximum remains £546, a gain of £0. Recommend the four extra banner places.

15 With $y \le 18$ removed the vertices are $(6,0)$, $(16,0)$, $(9,21)$ and $(6,24)$, giving $P = 15x + 22y$ as 90, 240, 597, $90 + 528 = 618$: maximum £618 from 6 posters and 24 banners. A maximum still exists because the region stays bounded — $3x + y \le 48$ with $x, y \ge 0$ forces $x \le 16$ and $y \le 48$, so the corner-point principle still applies. The gain is only £72 for six more banner places (about £12 each), and it depends entirely on 24 banners being saleable and storable; the deleted limit expressed exactly that restriction, so it should be replaced by a realistic larger one rather than removed.

Investigation 5 · hunting the best lattice point

1 Vertices: $(0,0)$, $\left(\frac{10}{3}, 0\right)$ from $3x + y = 10$, $\left(0, \frac{9}{2}\right)$ from $x + 2y = 9$, and the crossing $3x + y = 10$ with $x + 2y = 9$: substituting $y = 10 - 3x$ gives $x + 20 - 6x = 9$, so $x = 2.2$, $y = 3.4$. Then $P = 4x + 5y$ is 0, $\frac{40}{3} \approx 13.3$, 22.5 and $8.8 + 17 = 25.8$: continuous maximum 25.8 at $(2.2, 3.4)$.

2 For a fixed x the binding bound is the smaller of $10 - 3x$ and $\frac{9-x}{2}$. $x = 0$: $y = 0, 1, 2, 3, 4$, $P = 0, 5, 10, 15, 20$. $x = 1$: $y = 0, \ldots, 4$, $P = 4, 9, 14, 19, 24$. $x = 2$: $y = 0, \ldots, 3$, $P = 8, 13, 18, 23$. $x = 3$: $y = 0, 1$, $P = 12, 17$. That is 16 lattice points, and the best is $P = 24$ at $(1,4)$.

3 Over the same 16 points, $P = 5x + 2y$ is largest at $(3,1)$ with $15 + 2 = 17$ (next best $(2,3) = 16$), and $P = 2x + 3y$ is largest at $(1,4)$ with $2 + 12 = 14$ (next $(2,3) = 13$). The continuous maxima at $(2.2, 3.4)$ are $4(2.2) + 5(3.4) = 25.8$, $11 + 6.8 = 17.8$ and $4.4 + 10.2 = 14.6$, giving gaps 1.8, 0.8 and 0.6 exactly as printed.

4 With $3x + y \le 11$: $y = 11 - 3x$ in $x + 2y = 9$ gives $x + 22 - 6x = 9$, so $x = 2.6$, $y = 3.2$, and the continuous maximum of $4x + 5y$ is $10.4 + 16 = 26.4$. The best lattice point is still $(1,4)$ with 24 ($3 + 4 = 7 \le 11$ and $1 + 8 = 9 \le 9$); $(2,4)$ fails $x + 2y \le 9$ and $(3,2)$ gives only 22. The gap becomes $26.4 - 24 = 2.4$, so it grows from 1.8.

5 $(1,4)$ wins for $4x + 5y$ and for $2x + 3y$; $(3,1)$ wins for $5x + 2y$. Relative to the fractional vertex $(2.2, 3.4)$ both winners sit on one of the two binding boundary lines, one step inside: $(1,4)$ lies on $x + 2y = 9$ to the upper left, $(3,1)$ lies on $3x + y = 10$ to the lower right. The rounding $(2,3)$ is strictly inside both lines ($9 < 10$, $8 < 9$) and loses in all three cases, so the winner is on the boundary, not in the middle.

6 Rounding $(2.2, 3.4)$ gives $(2,3)$, which is feasible but scores 23, 16 and 13 against the true best 24, 17 and 14. Rounding the vertex of Question 4, $(2.6, 3.2)$, gives $(3,3)$, and $3(3) + 3 = 12 > 11$, so it is not even feasible. The rule gives the best lattice point in 0 of the four cases and a feasible point in only 3 of 4.

7 Vertices $(0,0)$, $(5,0)$, $(0,5)$ and the crossing of $2x + y = 10$ with $x + 2y = 10$: subtracting gives $x = y$, so $x = y = \frac{10}{3}$. Continuous maximum $P = 7x + 4y = \frac{70}{3} + \frac{40}{3} = \frac{110}{3} \approx 36.7$ at $\left(\frac{10}{3}, \frac{10}{3}\right)$, which is not integral. Checking lattice points on the boundary: $(4,2) \to 36$, $(5,0) \to 35$, $(3,3) \to 33$, $(2,4) \to 30$, so the integer maximum is 36 with gap $\frac{2}{3}$; the rounded point $(3,3)$ is feasible but not best.

Generalise. When the optimal vertex is integral the two optima coincide there, as in the bike workshop below, whose vertex $(27, 12)$ is already whole. When it is not, the candidates worth checking are the lattice points on the two binding lines nearest the vertex: take the integers just below and just above the vertex's x, read the largest admissible y off each binding line, and add one further step along each edge — four to six points. On R that list is $(1,4)$, $(2,3)$, $(3,1)$, $(3,0)$, and it holds the winner for all three objectives; the bike workshop needs no search at all, since its vertex is integral.

Modelling task · The bike workshop

1 Objective $P = 80x + 150y$ (weekly contribution in £), subject to $x + y \le 40$ (frames), $2x + 3y \le 90$ (assembly hours), $y \le 12$ (motors), $x \ge 0$, $y \ge 0$. Both $x + y \le 40$ and $y \le 12$ are needed because they count different resources: frames cap every machine built, motors cap only e-bikes, and neither statement contains the other — $(45, 0)$ obeys $y \le 12$ yet breaks the frame limit, while $(0, 40)$ obeys $x + y \le 40$ yet breaks the motor limit.

2 Vertices: $(0,0)$, $(40,0)$, $(30,10)$ (from $x + y = 40$ with $2x + 3y = 90$), $(27,12)$ (from $2x + 3y = 90$ with $y = 12$) and $(0,12)$. The candidate $(28,12)$ fails the hours: $56 + 36 = 92 > 90$. Then $P = 80x + 150y$ is 0, 3200, $2400 + 1500 = 3900$, $2160 + 1800 = 3960$, 1800, so the best weekly plan is 27 bicycles and 12 e-bikes for £3960. Slack: frames $40 - 39 = 1$, assembly hours $90 - 90 = 0$, motors $12 - 12 = 0$.

3 Three more motors ($y \le 15$): the continuous optimum moves to $(22.5, 15)$ worth £4050, which is not whole. Along the binding hour limit $P = 80x + 150y$ with $x \le \frac{90-3y}{2}$ gives $P \le 3600 + 30y$, so more e-bikes help: at $y = 15$ the largest whole x is 22, £4010; at $y = 14$, $x = 24$, £4020; at $y = 13$, $x = 25$, £3950. Best whole plan 24 bicycles and 14 e-bikes, £4020 — a gain of £60. Six more hours ($2x + 3y \le 96$): vertices $(0,0)$, $(40,0)$, $(28,12)$, $(0,12)$ (the crossing of $2x + 3y = 96$ with $y = 12$, $(30,12)$, breaks the frame limit), so £4040 at 28 bicycles and 12 e-bikes, already whole — a gain of £80 with 4 hours spare. Take the six extra assembly hours.

4 Assumptions: every machine assembled is sold at the stated contribution, and the 90 hours, 40 frames and 12 motors are exact, fixed and usable by either model; also that hours and frames split in whole machines with no minimum batch and that the £80 and £150 figures already net out all costs. The recommendation could fail because the six extra hours may only be bought as overtime, which raises the hourly cost and may cut the contribution of the extra bicycles below the £80 gain, or because demand for e-bikes is really 10 rather than 12: then the plan keeps 2 motors and idle capacity, and hours — the binding resource — should be switched to bicycles instead.

Mixed practice 5 · Inequalities, regions and optimisation

1 $x^2 - 5x - 14 = (x - 7)(x + 2) \le 0$, critical values -2 and 7, negative between them (test $x = 0$: $-14 \le 0$): $-2 \le x \le 7$, i.e. $[-2, 7]$.

2 $3x^2 - 5x - 2 = (3x + 1)(x - 2) > 0$, critical values $-\frac{1}{3}$ and 2, positive outside (test $x = 0$: $-2 < 0$): $x < -\frac{1}{3}$ or $x > 2$.

3 $x^2 + 6x + 10 = (x + 3)^2 + 1 \ge 1 > 0$ for every real x; $\Delta = 36 - 40 = -4 < 0$ and $a > 0$, so the solution is $\mathbb{R}$.

4 A square is never negative and is 0 at $x = 5$, so $(x - 5)^2 < 0$ has no solution: $\varnothing$.

5 $(-3, 2]$: round bracket at -3 (excluded), square bracket at 2 (included).

6 Boundary $3x + 2y = 12$, solid (non-strict), through $(4,0)$ and $(0,6)$; test $(0,0)$ gives $0 \ge 12$, false, so shade the side away from the origin, above the line. $(2,3)$: $6 + 6 = 12$, so it is on the boundary and is included.

7 $(0,0)$, $(8,0)$ (from $2x + y = 16$ with $y = 0$; $8 \le 10$), $(6,4)$ (subtracting gives $x = 6$, then $y = 4$) and $(0,10)$ (from $x + y = 10$ with $x = 0$; $10 \le 16$).

8 $P = 4x + 3y$: $(0,0) \to 0$, $(8,0) \to 32$, $(6,4) \to 24 + 12 = 36$, $(0,10) \to 30$. Maximum 36 at $(6,4)$, where both resource lines are tight.

9 At $(6,4)$: $x + y = 10$ against 10, slack 0; $2x + y = 12 + 4 = 16$ against 16, slack 0; $x \ge 0$ slack 6; $y \ge 0$ slack 4. Both resource limits are fully used, which is why the plan is a vertex.

10 Because size of y is not the objective: $(0,10)$ gives $P = 30$ while $(6,4)$ gives 36. The objective weights x at 4 and y at 3, so its level lines have gradient $-\frac{4}{3}$, flatter than the edge from $(6,4)$ to $(0,10)$ whose gradient is -1; sliding the level line outward therefore leaves the region last at $(6,4)$, not at the highest vertex.

11 Continuous vertices $(0,0)$, $(4.5,0)$, $(3,3)$ (subtracting the two lines gives $x = y = 3$) and $(0,4.5)$; $P = 3x + 7y$ is 0, 13.5, 30, 31.5 , so the continuous maximum is 31.5 at $(0,4.5)$. Among lattice points: $(1,4) \to 3 + 28 = 31$ (feasible, $2 + 4 = 6 \le 9$ and $1 + 8 = 9 \le 9$), $(3,3) \to 30$, $(0,4) \to 28$, $(2,3) \to 27$, while $(0,5)$ and $(2,4)$ break $x + 2y \le 9$. Maximum 31 at 1 and 4.

12 Vertices: $(0,0)$, $(18,0)$ ($18 + 0 = 18 \le 20$), $(17,3)$ (from $3x + y = 54$ with $x + y = 20$: $2x = 34$) and $(0,20)$ (from $x + 2y = 40$ with $x = 0$, which also gives $x + y = 20$). The crossings $(4,18)$ and $(13.6, 13.2)$ both fail $x + y \le 20$. Profit: 900 at $(18,0)$, $850 + 120 = 970$ at $(17,3)$, 800 at $(0,20)$ — new maximum £970, £70 below the old £1040, and the old optimum $(16,6)$ is now infeasible ($22 > 20$).

13 Vertices: $(0,0)$, $(15,0)$, $(10,10)$ ($2x+y=30$ with $x+y=20$; $10+20=30\le 36$), $(4,16)$ ($x+y=20$ with $x+2y=36$; $8+16=24\le 30$) and $(0,18)$; the crossing $(8,14)$ of the first two lines fails $x+y\le 20$ ($22>20$). Profit $P=4x+5y$: 60, 90, $16+80=96$, 90 — best plan 4 chairs and 16 tables, £96. Raising the total limit to 22 admits $(8,14)$, where all three limits are exactly met, and $P=32+70=102$: the best plan becomes 8 chairs and 14 tables for £102, a gain of £6.

14 Vertices: $(0,0)$, $(45,0)$ (from $2x+y=90$), $(35,20)$ ($2x+y=90$ with $x+y=55$; $35+40=75\le 80$), $(30,25)$ ($x+y=55$ with $x+2y=80$; $60+25=85\le 90$) and $(0,40)$ (from $x+2y=80$); the crossing $\left(\frac{100}{3},\frac{70}{3}\right)$ of the first two lines fails $x+y\le 55$ ($\frac{170}{3}\approx 56.7>55$). Profit $P=1.2x+2y$: 54, $42+40=82$, $36+50=86$, 80 — maximum £86 from 30 loaves and 25 cakes. The binding limits are $x+y\le 55$ and $x+2y\le 80$; $2x+y\le 90$ has slack $90-85=5$.

15 Values of $P=x+ky$: $(0,0)\to 0$, $(10,0)\to 10$, $(4,6)\to 4+6k$, $(0,8)\to 8k$. With $k=0.5$ they are 10, 7, 4, so $(10,0)$ wins; with $k=1.5$ they are 10, 13, 12, so $(4,6)$ wins; with $k=3$ they are 10 , 22, 24, so $(0,8)$ wins. Raising k walks the optimum along the boundary from the x-axis corner to the y-axis corner, because a bigger k makes y relatively more valuable; the change-overs are $k=1$ ($10=4+6k$, tie at $(10,0)$ and $(4,6)$) and $k=2$ ($4+6k=8k$, tie at $(4,6)$ and $(0,8)$).

16 Check $(7,2)$, $(7,3)$, $(8,2)$, $(8,3)$ — the corners of the unit cell that contains $(7.6,2.8)$, obtained by taking each coordinate as its floor or its ceiling. The list is not arbitrary: the objective grows as one moves toward the vertex, so a lattice point farther out in either coordinate can never beat the closest ones, and only these four are within one unit of it. Two cautions: discard any that break a constraint, and extend one step along each binding boundary (for instance $(6,3)$ or $(9,2)$), since Example 5.10 shows a feasible cell corner $(2,3)$ can be beaten by the off-cell point $(1,4)$.

17 Vertices: $(2,0)$, $(8,0)$ (from $3x+2y=24$ with $y=0$; $8\ge 2$), $(4,6)$ (subtracting the two resource lines gives $2x=8$) and $(2,7)$ (from $x=2$ with $x+2y=16$; $6+14=20\le 24$); the candidate $(2,9)$ from $x=2$ with $3x+2y=24$ fails $x+2y\le 16$ ($20>16$). $P=5x+4y$: 10, 40, $20+24=44$, $10+28=38$ — maximum 44 at $(4,6)$, where both resource lines are tight.

18 (a) The objective may not capture A's full value: A might pull customers or orders toward B, complete a product range or hold a contract, so dropping A could cost more than the model shows. (b) The coefficients are estimates: if A's contribution is understated, or demand for B is optimistic, the corner with no A is a fragile all-or-nothing plan, so a manager runs a small batch of A as a hedge and to keep skills, tooling and supplier relationships live.

19 Let $P=ax+by$ with $P(A)=P(B)=c$. Any point of the segment is $(1-t)A+tB$ for $0\le t\le 1$, and P is linear, so $P\big((1-t)A+tB\big)=(1-t)P(A)+tP(B)=(1-t)c+tc=c$. Every point of AB gives the same value, and the segment lies in the feasible region because an intersection of half-planes is convex. Hence when two vertices tie the whole edge between them is optimal, as in Exercise 5D Question 6.

20 The claim is true. Adding a constraint replaces the feasible region R by $R\cap S\subseteq R$, and the largest value of the objective over a subset cannot exceed the largest value over the whole set: if P has maximum M at $v\in R$, then every new feasible point is old-feasible and so scores at most M. Example: adding $x+y\le 20$ to the Dovetail system cut the maximum from £1040 to £970 (Question 12); if the new constraint is already satisfied at the old optimum the maximum is unchanged, as when Dovetail's finishing limit rose. The one subtlety is an unbounded programme with no maximum: there a new constraint creates a maximum where none existed, which is not an increase.

Criterion A · Inequalities and feasible regions

1 (a) $x^2-7x+10=(x-2)(x-5)>0$: $x<2$ or $x>5$. (b) $2x^2+x-6=(2x-3)(x+2)\le 0$: $-2\le x\le\frac{3}{2}$. (c) $x^2+2x+5=(x+1)^2+4\ge 4>0$ always ($\Delta=4-20=-16<0$), so $x^2+2x+5<0$ has no solution, $\varnothing$.

Marking guidance: 6 marks, 2 per part — 1 for correct critical values or completed square, 1 for the correct set. (a) loses a mark if $x<2$ is omitted; (c) scores full marks only if "no solution" is stated, not "no real roots".

2 (a) $(x-4)^2\ge 0$ for every real x: solution $\mathbb{R}$ (equality at $x=4$). (b) $x^2-4x+4=(x-2)^2>0$ for every x except $x=2$: $\mathbb{R}\setminus\{2\}$. (c) $\Delta=(-3)^2-4(1)(7)=9-28=-19<0$, so there are no real roots and the sign never changes; since $a=1>0$ the parabola lies wholly above the axis, giving $x^2-3x+7>0$ for all real x.

Marking guidance: 4 marks — 1 for (a), 1 for (b), 2 for (c), one for the discriminant value and one for linking a negative discriminant with $a>0$ to a constant positive sign. Answering (c) by testing $x=0$ only earns 1.

3 Boundary $2x-y=4$, i.e. $y=2x-4$, solid because the inequality is non-strict, through $(2,0)$ and $(0,-4)$. Test $(0,0)$: $0\le 4$ is true, so shade the side containing the origin (above the line). $(3,2)$: $6-2=4$, on the boundary and included. $(0,-1)$: $0+1=1\le 4$, inside. $(5,1)$: $10-1=9>4$, outside.

Marking guidance: 4 marks — 1 for the correct boundary with a solid line and shading on the origin side, and 1 for each of the three tests. Dashed boundaries or shading below the line cap the first mark at 0.

4 $(0,0)$; $(7,0)$ where $2x+y=14$ meets $y=0$ (and $7+0=7\le 9$); $(5,4)$ where $x+y=9$ meets $2x+y=14$ (subtracting gives $x=5$, then $y=4$); $(0,9)$ where $x+y=9$ meets $x=0$ (and $2(0)+9=9\le 14$). All four satisfy every inequality, and the crossing $(9,0)$ of $x+y=9$ with $y=0$ fails $2x+y\le 14$, so it is not a vertex.

Marking guidance: 3 marks for all four correct vertices, 2 marks for three, and no mark for a point that breaks a constraint; the axes corners $(7,0)$ and $(0,9)$ count as much as $(5,4)$.

5 $x+y$ is a single number, and no number is simultaneously at most 3 and at least 5; the boundary lines $x+y=3$ and $x+y=5$ are parallel and the two half-planes face away from each other, so their intersection is empty. In a production model this means the requirements contradict one another — the plant is ordered to make at least 5 while its capacity allows at most 3 — so there is no plan at all, not even "make nothing", which still breaks the minimum. The figures or the direction of one of the two inequalities must be corrected before the model can be used.

Marking guidance: 3 marks — 1 for the arithmetic contradiction, 1 for naming the empty (infeasible) region, 1 for the contextual meaning with a stated remedy. "The graph does not meet" without interpretation earns 1.

Criterion B · How the objective chooses a corner

1 $P=x+ky$: $k=0.5$ gives $A=0$, $B=10$, $C=4+3=7$, $D=4$ — optimum B. $k=1$ gives 0, 10, 10, 8 — B and C tie. $k=1.5$ gives 0, 10, 13, 12 — C. $k=2$ gives 0, 10, 16, 16 — C and D tie. $k=3$ gives 0, 10, 22, 24 — D.

Marking guidance: 5 marks — one per row of the table; a row is correct only if the values at all four vertices and the named optimum (or tie) are right. Only the maximum is asked for, but recording the minimum, always 0 at A, supports the pattern in Question 2.

2 As k rises the winning vertex moves steadily round the boundary from $B(10,0)$ to $C(4,6)$ and then to $D(0,8)$, passing through a two-vertex tie at each change-over. Larger k weights y more, so plans with more of y become better; the ties occur exactly when k makes a level line parallel to the joining edge, and A is always the minimum.

Marking guidance: 4 marks — 1 for the order $B\to C\to D$, 1 for the reason (the y-coefficient grows), 1 for noticing the ties at the change-overs, 1 for stating the pattern as a rule about k rather than about the five cases listed.

3 B ties C when $10 = 4 + 6k \Rightarrow k = 1$. C ties D when $4 + 6k = 8k \Rightarrow 4 = 2k \Rightarrow k = 2$. At $k = 1$ the tied vertices are $B(10, 0)$ and $C(4, 6)$ with $P = 10$; at $k = 2$ they are $C(4, 6)$ and $D(0, 8)$ with $P = 16$. The apparent change-over B against D, $10 = 8k \Rightarrow k = 1.25$, is not a winner change, because at $k = 1.25$ the value at C is $4 + 7.5 = 11.5$, which beats both.

Marking guidance: 5 marks — 2 for each transition value found by equating two vertex values, 1 for identifying the correct pairs of tying vertices; stating $k = 1.25$ as a third transition without testing C loses the last mark.

4 The level lines $x + ky = c$ have gradient $-\frac{1}{k}$. Edge BC has gradient $\frac{6-0}{4-10} = -1$, which matches $-\frac{1}{k}$ exactly when $k = 1$; edge CD has gradient $\frac{8-6}{0-4} = -\frac{1}{2}$, matching when $k = 2$. When the level line is parallel to an edge it last touches the region along that whole edge, so its two ends tie; for $1 < k < 2$ the level line is steeper than CD and flatter than BC, so it leaves the region at C alone. Increasing k flattens the level lines, which is why the contact point climbs from B to C to D.

Marking guidance: 4 marks — 1 for the level-line gradient, 1 for each of the two edge gradients, 1 for the parallel-when-ties argument; correct arithmetic with no mention of gradients earns at most 2.

5 $k = 10 > 2$, past the second transition, so the optimum is $D(0, 8)$ with $P = 80$; C gives only $4 + 60 = 64$. $k = 0.1 < 1$, below the first transition, so the optimum is $B(10, 0)$ with $P = 10$; C gives $4 + 0.6 = 4.6$ and D gives 0.8. Each prediction follows from the rule in Questions 3 and 4 rather than from a new table.

Marking guidance: 3 marks — 1 for each correct vertex with its value, 1 for justifying both by the transition values $k = 1$ and $k = 2$.

6 At a transition the level line $P = c$ coincides with the whole edge. For $k = 1$ a point of BC is $(10 - 6t, 6t)$ with $0 \le t \le 1$, and $P = x + y = 10 - 6t + 6t = 10$ for every t; the edge is feasible because a feasible region is convex. No point off the edge scores more, since the vertices bound the objective on a bounded region and the two ends already attain 10. So the endpoints are only two members of a whole set of optimal plans — the reason a tie must be reported as an edge, not as two separate answers.

Marking guidance: 3 marks — 1 for the parametric substitution showing P is constant, 1 for feasibility of the edge, 1 for the conclusion that no other point can beat it; asserting "the line touches along the edge" without working earns 1.

Criterion C · Why the best point is a corner

1 **Sample response.** A **constraint** is a linear inequality that limits the plan, such as $3x + y \le 54$ for Dovetail's assembly hours; the **feasible region** is the set of points satisfying every constraint at once, the overlap of their half-planes; the **objective** is the separate expression to be maximised, here $P = 50x + 40y$. Your rule — take the point with the largest coordinates — fails on the workshop numbers: $(18, 0)$ has the largest x and earns £900, $(0, 20)$ the largest y and earns £800, yet the best plan $(16, 6)$, smaller in both coordinates, earns £1040. Why a corner: the level lines $50x + 40y = c$ are parallel, of gradient $-\frac{5}{4}$, and a bigger c slides one away from the origin. The last line that still meets the region cannot meet it at an interior point, since from any interior point one may step further in the direction of increase and stay feasible; and along an edge P changes steadily, so its best point there is an end. Hence the optimum is a vertex, and testing the five settles it: 0, 900, 1040, 920, 800. On an unbounded region this fails: for $x \ge 0$, $y \ge 1$, $x + y \ge 4$ the objective $P = 2x + y$ has minimum 4 at $(0, 4)$ but no maximum, because $(t, 1)$ is feasible for every $t \ge 3$ and $2t + 1$ grows without limit — the vertices then prove only the minimum. The model assumes both products sell at the stated contribution and that the 54 hours exist; if demand for desks is only 12, then $(16, 6)$ over-produces desks and the best plan lies on the edge $x + y = 22$.

Marking guidance: 20 marks — [4] for the three definitions, [5] for a Dovetail counterexample to the classmate's rule with values, [5] for a causal level-line argument for boundary then vertex, [3] for unbounded regions, with an example that has no maximum, [3] for one assumption and one limitation in context. A response that only restates the corner-point principle, with no counterexample, reaches at most the middle band.

Criterion D · The school canteen

1 Let x be vegetarian meals and y chicken meals per day. Objective $C = 1.5x + 2y$ (contribution, in £ per day). Constraints: $x + y \le 28$ (meals, kitchen capacity per day), $x + 2y \le 44$ (oven tray-units), $y \le 18$ (chicken meals, fridge capacity), $x \ge 0$, $y \ge 0$ (meals). Counts are in meals, oven space in tray-units, money in £.

Marking guidance: 4 marks — 1 for a correct objective, 3 for the three resource constraints with variables and units identified; $x \ge 0, y \ge 0$ expected but counted once at most.

2 All five boundaries are solid. Test point $(1, 1)$ satisfies $2 \le 28$, $3 \le 44$, $1 \le 18$, so the region is on the origin side of each line. Vertices: $(0, 0)$, binding at $x = 0$ and $y = 0$; $(28, 0)$ where $x + y = 28$ meets $y = 0$ (oven use $28 \le 44$); $(12, 16)$ where $x + y = 28$ meets $x + 2y = 44$ (subtracting gives $y = 16$, $x = 12$, and $16 \le 18$); $(8, 18)$ where $x + 2y = 44$ meets $y = 18$ (kitchen $26 \le 28$); $(0, 18)$ where $x = 0$ meets $y = 18$ (oven $36 \le 44$). The crossing $(10, 18)$ of $x + y = 28$ with $y = 18$ fails the oven limit ($46 > 44$), so it is not a vertex. Binding pairs: kitchen only at $(28, 0)$; kitchen and oven at $(12, 16)$; oven and fridge at $(8, 18)$; fridge and axis at $(0, 18)$.

Marking guidance: 6 marks — 2 for an accurate graph with the correct overlap, 3 for all five vertices, 1 for naming the binding constraints at each; $(10, 18)$ included as a vertex caps the vertex marks at 2.

3 $C = 1.5x + 2y$: $(0, 0) \to 0$, $(28, 0) \to 42$, $(12, 16) \to 18 + 32 = 50$, $(8, 18) \to 12 + 36 = 48$, $(0, 18) \to 36$. The region is bounded, so the corner-point principle applies: the optimal daily plan is 12 vegetarian and 16 chicken meals with contribution £50. Kitchen capacity and oven space are exactly full; the fridge has $18 - 16 = 2$ meals of slack.

Marking guidance: 4 marks — 1 for evaluating all five vertices, 2 for the correct plan and value, 1 for stating the slack; a plan taken from the graph without the table earns at most 2.

4 Four more meals of kitchen capacity ($x + y \le 32$): vertices become $(0, 0)$, $(32, 0) \to 48$, $(20, 12)$ (where $x + y = 32$ meets $x + 2y = 44$) $\to 30 + 24 = 54$, $(8, 18) \to 48$, $(0, 18) \to 36$; the best plan is 20 vegetarian and 12 chicken meals, £54, a gain of £4. Six more tray-units ($x + 2y \le 50$): now the kitchen and the fridge bind first, giving vertices $(0, 0)$, $(28, 0) \to 42$, $(10, 18) \to 15 + 36 = 51$ (oven use $10 + 36 = 46 \le 50$, so the oven is no longer tight), $(0, 18) \to 36$; the best plan is 10 vegetarian and 18 chicken, £51, a gain of £1. Recommend the four extra meal places: on identical assumptions they are worth four times as much, because the 28-meal kitchen limit is the constraint that binds at the current optimum together with the oven, while the extra oven space is largely absorbed by the unchanged fridge ceiling of 18.

Marking guidance: 6 marks — 2 for each option's new optimum with its value, 1 for the numerical comparison, 1 for a recommendation justified by which limits bind; a recommendation with no re-solved vertices earns at most 2.

5 Assumption 1: every meal cooked is served and sold at the stated contribution, so $x + y \le 28$ is a capacity limit rather than a demand limit and nothing is wasted. Assumption 2: contributions stay at £1.50 and £2.00 whatever the numbers made, tray-units scale exactly (one for vegetarian, two for chicken), and any extra capacity is available every day. Evaluation: the recommendation is worth only £4 a day, so if four more meal places cost more than £4 a day it fails. Its size is sensitive to the mix — cutting the chicken contribution to £1.75 changes the two optima to £46 and £51, so the places are then worth £5 — but the direction survives, and a fall in chicken demand is tolerable because the recommended plan uses only 12 chicken meals; what would break it is unsold vegetarian meals, since 20 of the 32 places are vegetarian.

Marking guidance: 4 marks — 2 for two assumptions specific to this model, 1 for testing the recommendation against one of them, 1 for a quantified or clearly directional conclusion about whether it survives; generic statements such as "the data may be wrong" earn at most 1.

CHAPTER 6 · CIRCLE GEOMETRY AND GEOMETRIC PROOF

Opening problem · A circular gateway and a delivery

1 The arc through the two ground points $A(-4,0)$, $B(4,0)$ and the crown $(0,2.4)$ has its centre O on the perpendicular bisector of chord AB, i.e. on the vertical centre line; the chord AB is the 8 m opening and the highest point is 2.4 m above it.

2 With half-chord 4 and rise 2.4, the centre is $r-2.4$ below the crown, so $r^2 = 4^2 + (r-2.4)^2 = 16 + r^2 - 4.8r + 5.76$, giving $4.8r = 21.76$, $r = \dfrac{68}{15} \approx 4.53$ m; O is $r - 2.4 = \dfrac{32}{15} \approx 2.13$ m below ground level on the centre line.

3 Height at $x = 1.1$: $y = \sqrt{4.5333^2 - 1.1^2} - 2.1333 = \sqrt{19.3411} - 2.1333 \approx 4.3979 - 2.1333 = 2.26$ m, which is under the van's 2.3 m.

4 The claim is false, although the fact the architect quotes is true: the crown at 2.4 m does stand above the 2.3 m van. At the van's top corners, ± 1.1 m from the centre, the arch is only 2.26 m high, a shortfall of about 0.04 m (roughly 3.5 cm) at each corner, so the van does not pass. The deciding measurement is the arch height at 1.1 m from the centre, not the crown height. Assumption: the van is a rectangular box driven exactly on the centre line; mirrors or a roof rack (extra height) or driving off-centre (less clearance) would change the answer, while a lower roofline could make it pass.

Common error: comparing only the crown height with the van; the binding constraint is the height at the van's top corners, because the arch falls away on either side of the centre.

Prerequisite check

1 $\angle C = \angle B = 48^\circ$ (base angles), so $\angle A = 180^\circ - 96^\circ = 84^\circ$.

2 $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

3 $180^\circ - 118^\circ = 62^\circ$ (angles on a straight line).

4 $3x = 168$, so $x = 56$.

5 RHS (right angle–hypotenuse–side).

6 A circle is the set of points at a fixed distance (the radius) from a fixed point (the centre); the segment joining two points on the circle is a chord.

Exercise 6A · Angles at the centre and circumference

1 $\angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2}(140^\circ) = 70^\circ$ (angle at the centre is twice the angle at the circumference).

2 $\angle AOB = 2 \times 38^\circ = 76^\circ$; reflex $\angle AOB = 360^\circ - 76^\circ = 284^\circ$.

3 $\angle ACB = 90^\circ$ (angle in a semicircle), so $\angle BAC = 180^\circ - 90^\circ - 52^\circ = 38^\circ$.

4 $\angle ADB = 47^\circ$ (angles in the same segment, chord AB); central angle on minor arc $AB = 2 \times 47^\circ = 94^\circ$ (angle at the centre).

5 C is on the minor arc, so the relevant central angle is reflex: reflex $\angle AOB = 2 \times 125^\circ = 250^\circ$; smaller $\angle AOB = 360^\circ - 250^\circ = 110^\circ$.

6 $OA = OB$ (radii), so $\angle OBA = \angle OAB = 35^\circ$ and $\angle AOB = 180^\circ - 70^\circ = 110^\circ$; then $\angle ACB = \frac{1}{2}(110^\circ) = 55^\circ$ (angle at the centre).

7 With $\angle ACB = 130^\circ > 90^\circ$, C lies on the minor arc, so the central angle to compare with is the reflex $\angle AOB = 360^\circ - 100^\circ = 260^\circ$. Indeed $260^\circ = 2 \times 130^\circ$: the theorem holds. The error is doubling the smaller $\angle AOB$ instead of the reflex one.

8 $OA = OC$ is justified by "radii of the same circle are equal". $\angle AOD = 2\alpha$ is justified by "an exterior angle of a triangle equals the sum of the two interior opposite angles" applied to isosceles $\triangle OAC$, whose base angles are both α.

9 C and D both stand on arc AB with the central angle $\angle AOB$ on the arc not containing them. By Theorem 6.1, $\angle AOB = 2\angle ACB$ and $\angle AOB = 2\angle ADB$; hence $2\angle ACB = 2\angle ADB$, so $\angle ACB = \angle ADB$.

10 $AB = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$ cm. Since $\angle ACB = 90^\circ$, the converse of the semicircle theorem (6.3) makes AB a diameter, so the radius is 7.5 cm.

11 Same segment: $3x + 10 = 5x - 20$, so $2x = 30$, $x = 15$; both angles are 55°.

12 A non-diameter chord cuts a minor arc with central angle $c < 180^\circ$ and a major arc with central angle $360^\circ - c > 180^\circ$. An angle in the major segment is $\frac{1}{2}c < 90^\circ$ (acute); an angle in the minor segment is $\frac{1}{2}(360^\circ - c) > 90^\circ$ (obtuse), both by the

13 $\angle ACB = 90^\circ$ (angle in a semicircle, AB a diameter) and $\angle BAC = 180^\circ - 90^\circ - 34^\circ = 56^\circ$. The brace is right-angled because AB is a diameter of the (circular table's) circle through A, B, C.

14 Central angle $\angle AOB = 2 \times 30^\circ = 60^\circ$. $OA = OB$ (radii) with vertex angle 60° gives base angles $(180^\circ - 60^\circ) \div 2 = 60^\circ$ each, so $\triangle AOB$ is equilateral and radius $= OA = AB = 40$ m.

15 Draw diagonal AC. Since $AB \parallel CD$, $\angle BAC = \angle ACD$ (alternate angles). $\angle BAC$ stands on arc BC and $\angle ACD$ on arc AD, so equal angles at the circumference give equal arcs, hence equal chords $BC = AD$. One pair of sides parallel and the legs equal: $ABCD$ is an isosceles trapezium (the equal-chords case gives a rectangle, which the statement still admits).

Exercise 6B · Cyclic quadrilaterals and their converses

1 $\angle C = 180^\circ - 85^\circ = 95^\circ$ (opposite angles of a cyclic quadrilateral).

2 $\angle D = 180^\circ - 112^\circ = 68^\circ$ (opposite angles of a cyclic quadrilateral).

3 $\angle ABC = \angle CDE = 75^\circ$ (exterior angle of a cyclic quadrilateral equals the interior opposite angle).

4 $2x + 3x + 4x + 3x = 360$, so $12x = 360$, $x = 30$; the angles are $60^\circ, 90^\circ, 120^\circ, 90^\circ$. Opposite pairs: $60^\circ + 120^\circ = 180^\circ$ and $90^\circ + 90^\circ = 180^\circ$, so yes, it is cyclic (converse of Theorem 6.4).

5 $(x + 20) + (2x - 5) = 180$, so $3x = 165$, $x = 55$; $\angle A = 75^\circ$, $\angle C = 105^\circ$.

6 Base angles equal: $\angle B = \angle A = 68^\circ$; co-interior angles on AD: $\angle D = 180^\circ - 68^\circ = 112^\circ = \angle C$. Opposite pairs $\angle A + \angle C = \angle B + \angle D = 180^\circ$, so by the converse of Theorem 6.4 it is cyclic.

7 Counterexample: a (non-square) kite with angles $60^\circ, 80^\circ, 140^\circ, 80^\circ$ has the equal pair $(80^\circ, 80^\circ)$ opposite, but $60^\circ + 140^\circ = 200^\circ \neq 180^\circ$, so it is not cyclic. Correct test: a pair of **opposite** angles must be supplementary (converse of Theorem 6.4).

8 Arcs BCD and BAD partition the whole circle, so their central angles make one full turn and add to 360°. Theorem 6.1 (angle at the centre is twice the angle at the circumference) writes them as 2α and 2γ.

9 $\angle ABC + \angle ADC = 180^\circ$ (opposite angles of cyclic $ABCD$) and $\angle ABC + \angle CBE = 180^\circ$ (angles on straight line ABE). Both equal $180^\circ - \angle ABC$, so $\angle CBE = \angle ADC$.

10 (a) Rectangle: yes — opposite angles $90^\circ + 90^\circ = 180^\circ$, cyclic by the converse of 6.4. (b) Non-square rhombus: no — opposite angles are equal, and equal supplements force 90° each, which would make it a square. (c) Kite with a pair of opposite right angles: yes — that pair sums to 180°, so cyclic (the diagonal joining the other two vertices is a diameter). (d) Non-rectangular parallelogram: no — equal opposite angles must be supplementary, forcing rectangles.

11 Three non-collinear points A, B, C determine a unique circle. If D were not on it, line AD would meet the circle again at some $D' \neq D$; then $\angle ABC + \angle AD'C = 180^\circ$ (cyclic $ABCD'$) while $\angle ABC + \angle ADC = 180^\circ$ (given), giving $\angle AD'C = \angle ADC$. But for distinct points on line AD one of these is an exterior angle of $\triangle DCD'$, so they cannot be equal (unless $C \in AD$, impossible here). Hence $D' = D$: D lies on the circle through A, B, C.

12 In the circle through A, B, C, $\angle ACB = 90^\circ$ makes AB a diameter (converse of 6.3); likewise in the circle through A, B, D. A diameter fixes a unique circle, so the two circles coincide: A, B, C, D are concyclic with AB a diameter. Radius $= 5$ cm; half of CD is 3 cm, so distance from centre to $CD = \sqrt{5^2 - 3^2} = 4$ cm.

13 $\angle ABC + \angle CDA = 96^\circ + 84^\circ = 180^\circ$, a supplementary pair of opposite angles, so by the converse of Theorem 6.4 the posts are concyclic. If $\angle CDA = 80^\circ$ the sum is $176^\circ \neq 180^\circ$; concyclic points in order A, B, C, D must have $\angle B + \angle D = 180^\circ$, so with the other data unchanged they could **not** be concyclic.

14 Cyclic order $95^\circ, 100^\circ, 85^\circ, 80^\circ$: opposite pairs $95^\circ + 85^\circ = 180^\circ$ and $100^\circ + 80^\circ = 180^\circ$. Order matters because the

15 ($\Rightarrow$) In a parallelogram opposite angles are equal; if cyclic they are also supplementary, so each is 90°: a rectangle. ($\Leftarrow$) A rectangle has opposite angles $90^\circ + 90^\circ = 180^\circ$, so it is cyclic by the converse of Theorem 6.4.

Exercise 6C · Tangents, radii and incircles

1 $\angle OTP = 90^\circ$ (tangent $\perp$ radius), so $PT = \sqrt{15^2 - 9^2} = \sqrt{144} = 12$.

2 $OP = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$.

3 Quadrilateral $OTPQ$: $\angle TOQ = 360^\circ - 90^\circ - 90^\circ - 70^\circ = 110^\circ$; OP bisects $\angle TPQ$ (equal tangents), so $\angle TPO = 35^\circ$.

4 In right $\triangle OTP$: $\angle TPO = 180^\circ - 90^\circ - 40^\circ = 50^\circ$; OP bisects both angles, so $\angle TPQ = 2 \times 50^\circ = 100^\circ$ and $\angle TOQ = 2 \times 40^\circ = 80^\circ$.

5 $\angle OTP = 90^\circ$ (tangent perpendicular to radius at the point of contact); $OP = \sqrt{6^2 + 8^2} = 10$.

6 Area $= \frac{1}{2} \cdot 5 \cdot 12 = 30$; semiperimeter $s = \frac{5+12+13}{2} = 15$; $r = \dfrac{\text{area}}{s} = \dfrac{30}{15} = 2$.

7 Wrong line: it is the radius to the point of contact, $OT \perp PT$, not $OP \perp PT$. OP bisects $\angle TPQ$ (and $\angle TOQ$); it is generally oblique to the tangents.

8 In right triangle OTN the hypotenuse is OT, so the leg $ON < OT = r$; a point closer to O than r lies inside the circle. But a line through an interior point meets the circle twice, contradicting the hypothesis that the line is a tangent (one point of contact).

9 $\angle OTP = \angle OQP = 90^\circ$ (tangent $\perp$ radius), hypotenuse OP common, $OT = OQ$ (radii). So $\triangle OTP \cong \triangle OQP$ by RHS, giving $PT = PQ$ (matching parts).

10 From $\triangle OTP \cong \triangle OQP$ (RHS, as in Q9) the matching angles $\angle TOP = \angle QOP$, so OP bisects $\angle TOQ$; the same congruence gives $\angle TPO = \angle QPO$ for $\angle TPQ$.

11 Height 0.1 km, $R = 6400$. Tangent length $d = \sqrt{(6400.1)^2 - 6400^2} = \sqrt{2(6400)(0.1) + 0.1^2} = \sqrt{1280.01} \approx 35.8$, i.e. about 36 km (nearest kilometre).

12 $s = \frac{13+14+15}{2} = 21$, so $r = \dfrac{84}{21} = 4$ cm. Tangent lengths from the vertices are $s - a$ opposite each side: $21 - 13 = 8$, $21 - 14 = 7$, $21 - 15 = 6$ cm. Check: $8 + 7 = 15$, $8 + 6 = 14$, $7 + 6 = 13$.

13 $OP = \sqrt{OT^2 + PT^2} = \sqrt{6^2 + 8^2} = 10$ m (tangent $\perp$ radius).

14 Right triangle: area $= 30$ m^2, $s = 15$ m, so $r = 30 \div 15 = 2$ m; water surface area $= \pi r^2 = 4\pi$ m^2.

15 $\angle OTP = \angle OQP = 90^\circ$ and quadrilateral $OTPQ$ has angle sum 360°, so $\angle TPQ + \angle TOQ = 360^\circ - 180^\circ = 180^\circ$. Also $OT = OQ$ and $PT = PQ$, so both O and P are equidistant from T and Q; two points equidistant from the endpoints of a segment determine its perpendicular bisector, hence $OP \perp$ bisects TQ.

Exercise 6D · Chords, intersecting chords and the alternate segment

1 The perpendicular from the centre bisects the chord (Theorem 6.8), so $d = \sqrt{13^2 - 12^2} = 5$ cm.

2 Half-chord $= 8$, so $d = \sqrt{10^2 - 8^2} = 6$ cm.

3 Half-chord $= 9$, so each distance $= \sqrt{15^2 - 9^2} = 12$ cm (equal chords are equidistant, as expected).

4 Chords equidistant from the centre are equal (converse of Theorem 6.9), so the other chord is 20 cm.

5 Intersecting chords: $PA \cdot PB = PC \cdot PD$, so $3 \cdot 8 = 4 \cdot PD$ and $PD = 6$.

6 Tangent–secant: $PT^2 = PA \cdot PB = 4 \cdot 9 = 36$, so $PT = 6$.

7 False: a general line from the centre to a chord does not bisect it. Correct statement: the **perpendicular** from the centre bisects a chord (Theorem 6.8), equivalently the line through the centre and the chord's midpoint. Special case: if the chord is a diameter, its midpoint is the centre itself.

8 $OA = OB$ (radii) — these are the hypotenuses because they join the centre to the endpoints of the chord; OM is a common leg; $\angle OMA = \angle OMB = 90^\circ$. Congruence test: RHS, giving $AM = MB$.

9 $OM \perp AB$ and $ON \perp CD$, so $AM = \frac{1}{2}AB$, $CN = \frac{1}{2}CD$ (6.8); with $AB = CD$, $AM = CN$. Right triangles OMA, ONC have $OA = OC$ (radii), so by RHS (or Pythagoras) they are congruent; hence $OM = ON$.

10 Alternate-segment angle $= 58^\circ$; central angle on the chord $= 2 \times 58^\circ = 116^\circ$; angle between chord and radius at contact $= 90^\circ - 58^\circ = 32^\circ$.

11 $PB = PA + AB = 4 + 5 = 9$; $PT^2 = PA \cdot PB = 4 \cdot 9 = 36$, so $PT = 6$.

12 24 cm chord: $\sqrt{13^2 - 12^2} = 5$ cm. 10 cm chord: $\sqrt{13^2 - 5^2} = 12$ cm. The longer chord is closer to the centre: $d = \sqrt{r^2 - (\text{half-chord})^2}$ decreases as the chord lengthens.

13 1.6 m strip: $d = \sqrt{1^2 - 0.8^2} = 0.6$ m. 1.2 m strip: $d = \sqrt{1^2 - 0.6^2} = 0.8$ m. The 1.6 m strip is closer to the centre.

14 $r = \dfrac{4^2 + 2.4^2}{2(2.4)} = \dfrac{21.76}{4.8} \approx 4.53$ m; the centre lies $r - 2.4 \approx 2.13$ m below the springing line. Height at 1.1 m from the centre: $\sqrt{4.5333^2 - 1.21} - 2.1333 = \sqrt{19.3411} - 2.1333 \approx 2.26$ m < 2.3 m, so the corners (each 1.1 m off-centre) do not clear; vertical shortfall $\approx 2.3 - 2.2645 \approx 0.04$ m at each corner.

The crown at 2.4 m does clear the van, which is exactly the trap: clearance at the centre proves nothing, because the arch falls away towards the sides.

15 In triangles PTA and PBT: $\angle PTA = \angle TBA = \angle TBP$ (alternate segment, chord TA) and $\angle TPA = \angle BPT$ (the same angle at P). So $\triangle PTA \sim \triangle PBT$ (AA), giving $\dfrac{PT}{PB} = \dfrac{PA}{PT}$, hence $PT^2 = PA \cdot PB$.

Exercise 6E · Proof, reasons and converses

1 (a) Hypothesis: the chord is a diameter; conclusion: it is the longest chord. (b) Hypothesis: the quadrilateral is a rectangle; conclusion: it is cyclic.

2 (a) Converse: "if a chord is the longest chord, it is a diameter" — true (longer chords lie closer to the centre; the longest has distance 0, so it passes through the centre). (b) Converse: "if a quadrilateral is cyclic, it is a rectangle" — false; e.g. a quadrilateral with angles $60^\circ, 90^\circ, 120^\circ, 90^\circ$ in order has opposite pairs supplementary, so it is cyclic, yet it is not a rectangle.

3 (a) Radii of the same circle. (b) Angle at the centre is twice the angle at the circumference (Theorem 6.1). (c) Angles on a straight line sum to 180°.

4 Statement 1: $\angle AOB = 180^\circ$ — Reason: AB is a diameter, so AOB is a straight angle at the centre. Statement 2: $\angle AOB = 2\angle ACB$ — Reason: angle at the centre (Theorem 6.1), C on the circle. Statement 3: $\angle ACB = 90^\circ$ — Reason: substitution, $\angle ACB = \frac{1}{2}(180^\circ)$.

5 (a) vertically opposite angles; (b) radii of the same circle; (c) tangent perpendicular to radius at the point of contact; (d) opposite angles of a cyclic quadrilateral; (e) angles in the same segment.

6 1. $\angle OTP = \angle OQP = 90^\circ$ — tangent $\perp$ radius. 2. $OT = OQ$ — radii of the same circle. 3. $OP = OP$ — common side. 4. $\triangle OTP \cong \triangle OQP$ — RHS. 5. $PT = PQ$ — matching parts of congruent triangles.

7 Error 1: in isosceles $\triangle OAC$ (with $OA = OC$) the equal base angles are $\angle OAC = \angle OCA$; $\angle AOC$ is the vertex angle and is not equal to either base angle, so "$\angle AOC = 2\angle OCA$" has no justification. Error 2: the doubling step belongs to the **exterior** angle: $\angle AOD = \angle OAC + \angle OCA = 2\angle OCA$ (exterior angle equals sum of interior opposite angles). Correct opening: "Since $OA = OC$, $\angle OAC = \angle OCA = \alpha$; joining CO produced to D, the exterior angle $\angle AOD = 2\alpha$."

8 Bisecting diagonals is a parallelogram property; a rectangle is both a parallelogram and cyclic, so the two happen to coincide there. Counterexample: an isosceles trapezium with angles $60^\circ, 90^\circ, 120^\circ, 90^\circ$ is cyclic but its diagonals do not bisect each other (the parallel sides are unequal).

9 A kite with angles $60^\circ, 80^\circ, 140^\circ, 80^\circ$ has perpendicular diagonals but opposite pairs 200° and 160°, so it is not cyclic. The converse "cyclic $\Rightarrow$ perpendicular diagonals" is also false: a non-square rectangle is cyclic but its diagonals are not perpendicular.

10 1. Draw $OM \perp AB$, $ON \perp CD$ — construction. 2. $AM = \frac{1}{2}AB$, $CN = \frac{1}{2}CD$ — perpendicular from centre bisects a chord. 3. $AB = CD$, so $AM = CN$ — given. 4. $OA = OC$ — radii. 5. $\triangle OMA \cong \triangle ONC$ — RHS. 6. $OM = ON$ — matching parts. ■

11 1. $OM \perp AB$, $ON \perp CD$ with $OM = ON$ — given distances. 2. $OA = OC$ — radii. 3. $\angle OMA = \angle ONC = 90^\circ$ — construction. 4. $\triangle OMA \cong \triangle ONC$ — RHS. 5. $AM = CN$ — matching parts. 6. $AB = 2AM = 2CN = CD$ — perpendicular from centre bisects each chord. ■

12 A theorem is "if P then Q"; its converse swaps them, "if Q then P", and is a new claim needing separate proof. Example: "if a quadrilateral is a rectangle then it is cyclic" is true, but its converse "if a quadrilateral is cyclic then it is a rectangle" is false (a $60^\circ, 90^\circ, 120^\circ, 90^\circ$ trapezium is cyclic yet not a rectangle).

13 Let A, B, C lie on a circle with centre O and C on the major arc AB. Construction: join C to O and produce CO to meet the circle again at D. Since $OA = OC$ (radii), $\triangle OAC$ is isosceles, so $\angle OAC = \angle OCA = \alpha$; the exterior angle $\angle AOD = \alpha + \alpha = 2\alpha$. Similarly, with $\angle OCB = \beta$, $\angle BOD = 2\beta$. Adding, $\angle AOB = \angle AOD + \angle DOB = 2\alpha + 2\beta = 2\angle ACB$. Reflex case: if C is on the minor arc the same argument gives reflex $\angle AOB = 2\angle ACB$. ■

14 Let the tangent at T be parallel to chord AB; join TA, TB. The angle between the tangent and chord TA equals $\angle TBA$ (alternate segment). Since the tangent $\parallel AB$ with transversal TA, that same angle equals $\angle TAB$ (alternate angles). Hence $\angle TAB = \angle TBA$, so $\triangle TAB$ is isosceles with $TA = TB$. ■

15 Let $ABCD$ be a cyclic parallelogram. Parallelogram property used: $\angle A = \angle C$ (opposite angles equal). Cyclic property used: $\angle A + \angle C = 180^\circ$ (opposite angles supplementary). Hence $2\angle A = 180^\circ$, $\angle A = 90^\circ = \angle C$; since adjacent angles of a parallelogram are supplementary, $\angle B = \angle D = 90^\circ$ too, so $ABCD$ is a rectangle. ■

Investigation 6 · Moving, reflecting and rotating a circle theorem

1 $\angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2}(100^\circ) = 50^\circ$ (angle at the centre).

2 At all five positions $\angle ACB = 50^\circ$ unchanged, while CA and CB vary (equal only when C is the midpoint of the major arc, where $CA = CB$). Invariant: the angle; changing: the two chord lengths.

3 On the minor arc the new angle is 130°, the supplement of 50°. It is explained by the reflex central angle: reflex $\angle AOB = 360^\circ - 100^\circ = 260^\circ = 2 \times 130^\circ$ (angle at the centre on the major arc).

4 $\angle ADB = 50^\circ$, equal to every value in step 2. Conjecture: all angles subtended by a fixed chord at points on the same arc are equal (angles in the same segment), each being half the central angle on the opposite arc.

5 C maps to the opposite side of AB, on the reflected circle; the angle at the image is still 50° (reflection is an isometry and A, B are fixed). The major and minor arcs swap sides of AB. Against the original circle, a point taken to the other side of AB sits where the angle is the supplement, 130°.

6 After a 40° rotation about O every measurement is identical: $\angle AOB = 100^\circ$, $\angle ACB = 50^\circ$, chord AB unchanged. Invariant under rotation: all angles, all chord lengths, and the $2:1$ central-to-inscribed ratio (the circle maps to itself and rotation preserves distances and angles).

7 Join C to O and produce it to D on the circle. Equality of radii is used first: $OA = OC$ makes $\triangle OAC$ isosceles with $\angle OAC = \angle OCA = \alpha$, and $OB = OC$ gives $\angle OBC = \angle OCB = \beta$. The exterior-angle property is used next: $\angle AOD = 2\alpha$, $\angle BOD = 2\beta$. Adding, $\angle AOB = 2(\alpha + \beta) = 2\angle ACB$. For any other point C' on the same arc the identical argument gives the same $\angle AOB$, so $\angle AC'B = \angle ACB$: the conjecture of step 4. The $2:1$ ratio is forced by the circle itself, not by one drawing.

Modelling task · The arch, revisited

1 The perpendicular from O to the half-span chord bisects it, so $r^2 = a^2 + (r-h)^2$, i.e. $2rh = a^2 + h^2$ and $r = \dfrac{a^2 + h^2}{2h}$. With $a = 4$, $h = 2.4$: $r = \dfrac{16 + 5.76}{4.8} = \dfrac{21.76}{4.8} = \dfrac{68}{15} \approx 4.53$ m.

2 Height at 1.1 m from the centre: $\sqrt{r^2 - 1.1^2} - (r - h) = \sqrt{19.3411} - 2.1333 \approx 4.3979 - 2.1333 = 2.2645$ m. Shortfall for the 2.3 m van: $2.3 - 2.2645 \approx 0.04$ m (about 3.5 cm) at each top corner, so the van does not clear the arch.

3 Require $\sqrt{r^2 - 1.21} - (r - h) \geq 2.4$ with $r = \dfrac{16 + h^2}{2h}$. Test: $h = 2.50$ m gives $r = 4.45$ and height ≈ 2.36 m (fails); $h = 2.55$ m gives $r \approx 4.412$ and height ≈ 2.41 m (passes). Smallest rise to the nearest 0.05 m: 2.55 m.

4 Assumption 1: the van is a rectangular $2.2\text{ m} \times 2.3\text{ m}$ box driven exactly on the centre line — mirrors or a roof box add height, and any offset from the centre moves the corners into a lower part of the arch, raising the required rise. Assumption 2: the arch is an exact circular arc of the given rise with level ground and no structural thickness — a flattened or thicker arch reduces clearance, so the modelled rise under-estimates what is needed.

Mixed practice 6 · Circle geometry and geometric proof

1 $\angle ACB = \frac{1}{2}(76^\circ) = 38^\circ$ (angle at the centre); reflex $\angle AOB = 360^\circ - 76^\circ = 284^\circ$.

2 $(3x + 8) + (5x + 12) = 180$ (opposite angles of cyclic quadrilateral), $8x = 160$, $x = 20$; $\angle A = 68^\circ$, $\angle C = 112^\circ$.

3 $PT = \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ (tangent $\perp$ radius, then Pythagoras).

4 Half-chord $= 5$, so $d = \sqrt{13^2 - 5^2} = 12$ cm (perpendicular from centre bisects a chord).

5 Intersecting chords: $6 \cdot 4 = 3 \cdot PD$, so $PD = 8$.

6 Alternate-segment angle $= 47^\circ$; central angle on the chord $= 2 \times 47^\circ = 94^\circ$.

7 $\angle TOQ = 360^\circ - 90^\circ - 90^\circ - 58^\circ = 122^\circ$; $\angle TPO = 58^\circ \div 2 = 29^\circ$ (OP bisects the angle between the tangents).

8 $\angle ACB = 90^\circ$ (angle in a semicircle), so $\angle ABC = 180^\circ - 90^\circ - 28^\circ = 62^\circ$.

9 $\angle C = 118^\circ$ (exterior angle of a cyclic quadrilateral equals the interior opposite angle).

10 Tangent–secant: $8^2 = 4 \cdot PB$, so $PB = 16$ and $AB = PB - PA = 16 - 4 = 12$.

11 Cyclic order: $95^\circ, 100^\circ, 85^\circ, 80^\circ$ — opposites $95^\circ + 85^\circ = 180^\circ$, $100^\circ + 80^\circ = 180^\circ$ (converse of Theorem 6.4). Non-cyclic order: $95^\circ, 85^\circ, 100^\circ, 80^\circ$ — opposites $95^\circ + 100^\circ = 195^\circ \neq 180^\circ$, so it fails the test.

12 Construction: join C to O, produce to D. $OA = OC$ and $OB = OC$ (radii), so $\triangle OAC$ and $\triangle OBC$ are isosceles: $\angle OAC = \angle OCA = \alpha$, $\angle OBC = \angle OCB = \beta$. Exterior angles: $\angle AOD = 2\alpha$, $\angle BOD = 2\beta$. Since OD lies inside $\angle AOB$, $\angle AOB = 2\alpha + 2\beta = 2\angle ACB$. ■

13 Invalid: the same-segment condition needs C, D on the **same** side of AB. On opposite sides, if the four points were concyclic then in cyclic quadrilateral $ACBD$ the opposite angles at C and D would sum to 180°, not be equal; $40^\circ + 40^\circ = 80^\circ \neq 180^\circ$. Extra condition that would make it valid: $\angle ACB + \angle ADB = 180^\circ$ (converse of Theorem 6.4), or C, D on the same side with $\angle ACB = \angle ADB$.

14 Second chord's half-length $= \sqrt{13^2 - 12^2} = 5$ cm, so the chord is 10 cm long.

15 $\angle A + \angle C = 180^\circ$: $2k + 4k = 180$, $k = 30$. $\angle A = 60^\circ$, $\angle B = 90^\circ$, $\angle C = 120^\circ$; $\angle D = 180^\circ - \angle B = 90^\circ$. Check: angles sum to 360°.

16 $\angle ASB = 90^\circ$ makes AB a diameter (converse of the semicircle theorem), so radius $= 6$ km. In the right triangle, $SB = AB \sin\angle SAB = 12 \sin 35^\circ = 6.88$ km (3 s.f.).

17 $r = \dfrac{5^2 + 2.5^2}{2(2.5)} = \dfrac{31.25}{5} = 6.25$ m. Height 2 m from the centre: $\sqrt{6.25^2 - 2^2} - (6.25 - 2.5) = \sqrt{35.0625} - 3.75 \approx 5.921 - 3.75 = 2.17$ m.

18 Join AC and DB. $\angle APC = \angle DPB$ (vertically opposite); $\angle CAP = \angle BDP$ (angles in the same segment on arc BC). So $\triangle PAC \sim \triangle PDB$ (AA), giving $\dfrac{PA}{PD} = \dfrac{PC}{PB}$, hence $PA \cdot PB = PC \cdot PD$. ■

19 Area $= 30$, $s = 15$, so $r = 30 \div 15 = 2$. Tangent lengths from the vertices: $s - a$ for each opposite side: $15 - 5 = 10$, $15 - 12 = 3$, $15 - 13 = 2$. Check: $10 + 3 = 13$, $10 + 2 = 12$, $3 + 2 = 5$.

20 Drop perpendiculars $OM \perp AB$, $ON \perp CD$; then $AM = \frac{1}{2}AB = \frac{1}{2}CD = CN$ (6.8) and $OA = OC$ (radii), so right triangles OMA, ONC are congruent (RHS), giving $\angle AOM = \angle CON$; doubling, $\angle AOB = \angle COD$. Equal central angles cut off equal arcs, so the minor arcs of equal chords are equal. ■

Criterion A · Knowing and understanding · Angles, tangents and

1 $\angle ACB = \frac{1}{2}(118^\circ) = 59^\circ$ (angle at the centre); reflex $\angle AOB = 360^\circ - 118^\circ = 242^\circ$; $\angle ADB = \frac{1}{2}(242^\circ) = 121^\circ$ (D on the minor arc subtends the major arc).

Marking guidance: [4] — 1 mark each for 59°, 242°, 121° and 1 mark for correct theorem use (halving the correct, including reflex, central angle). A top-band response identifies which arc each point subtends before doubling or halving.

2 $(4x - 10) + (2x + 40) = 180$ (opposite angles of cyclic quadrilateral), $6x + 30 = 180$, $x = 25$; $\angle A = 90^\circ$, $\angle C = 90^\circ$. Exterior angle at B equals the interior opposite angle, so $\angle D = 115^\circ$.

Marking guidance: [4] — 1 mark for the equation, 1 for $x = 25$, 1 for $\angle A, \angle C$, 1 for $\angle D = 115^\circ$ with the exterior-angle theorem named. Top band: reasons stated, not just answers.

3 $\angle OTP = 90^\circ$ (tangent $\perp$ radius), so $PT = \sqrt{13^2 - 5^2} = 12$ cm. $\sin\angle TPO = \dfrac{5}{13}$, so $\angle TPO = \sin^{-1}\left(\frac{5}{13}\right) = 22.6^\circ$ (1 d.p.).

Marking guidance: [4] — 1 for the right angle at T, 1 for $PT = 12$, 1 for the correct trig ratio, 1 for 22.6° to the required accuracy. Top band: correct line chosen (radius over hypotenuse for the sine).

4 The perpendicular from the centre bisects the chord: $d = \sqrt{10^2 - 8^2} = 6$ cm. Intersecting chords: $PA \cdot PB = PC \cdot PD$, $6 \cdot 8 = 4 \cdot PD$, $PD = 12$ cm.

Marking guidance: [5] — 1 for bisecting the chord, 2 for Pythagoras and $d = 6$, 1 for the intersecting-chords product, 1 for $PD = 12$. Top band: theorems named and units given.

5 If AB is a diameter, the central angle $\angle AOB = 180^\circ$ (straight angle); by Theorem 6.1, $\angle ACB = \frac{1}{2}\angle AOB = 90^\circ$. Converse: if $\angle ACB = 90^\circ$, Theorem 6.1 forces the central angle on the same arc to be 180°, so A, O, B are collinear and AB passes through the centre — a diameter.

Marking guidance: [3] — 1 mark each for the central angle 180°, the application of Theorem 6.1 giving 90°, and a valid converse argument via collinearity with O. Top band: the converse is argued, not asserted.

Criterion B · Investigating patterns · The arc of constant angle

1 Construct the arc (Parts 2–3 locate it) and mark five points C on it with $\angle ACB = 60^\circ$. Sample table (cm): arc midpoint C_1: distance to $AB = 3\sqrt{3} \approx 5.20$, $CA = CB = 6$ (equilateral); C_2: 3.46, 3.46, 6.93; C_3: 4.7, 4.2, 6.4; C_4, C_5 mirror pairs. Record: the perpendicular distance and CA, CB vary (distance from just above 0 to 5.20), while $\angle ACB$ stays 60°.

Marking guidance: [6] — construction of five accurate points (2–3), systematic table with all three quantities (2), statement of what changes and what is invariant (1). Top band: constructions within tolerance, table ordered and complete, values consistent with the arc.

2 The locus is an arc, not a line: the major arc of a circle through A and B (the arc capable of 60° angles), with A, B themselves excluded. There are two symmetric loci, one on each side of AB, because reflecting any valid point in line AB preserves CA, CB and hence the angle.

Marking guidance: [4] — arc correctly identified (1–2), excluded endpoints noted (1), explanation of the two mirror loci (1). Top band: explains why points off the arc give a different angle.

3 Central angle $= 2 \times 60^\circ = 120^\circ$, so radius $R = \dfrac{AB}{2\sin 60^\circ} = \dfrac{6}{\sqrt{3}} = 2\sqrt{3} \approx 3.46$ cm; the centre is on the perpendicular bisector of AB, on the same side as C, at distance $\sqrt{R^2 - 3^2} = \sqrt{3} \approx 1.73$ cm from AB. Verify: each measured point satisfies $OC \approx 3.46$ cm.

Marking guidance: [5] — centre located by construction (2), radius value (2), verification against the table (1). Top band: verification within measurement tolerance.

4 For $\angle ACB = 90^\circ$ the centre lies on AB itself (midpoint of AB) and $R = \dfrac{6}{2\sin 90^\circ} = 3$ cm: the two arc loci combine into the full circle with diameter AB (semicircle theorem and its converse).

Marking guidance: [3] — centre at midpoint (1), radius 3 (1), the two-semicircle observation (1). Top band: links to the semicircle theorem.

5 Prediction: $R = \dfrac{6}{2\sin 30^\circ} = 6$ cm (central angle 60° makes $\triangle AOB$ equilateral, so $R = AB$). Test by construction: draw the circle of radius 6 cm through A, B with centre on the perpendicular bisector at $3\sqrt{3} \approx 5.20$ cm from AB; measured angles at points of its major arc are 30°.

Marking guidance: [3] — prediction (1), construction (1), measured confirmation (1). Top band: prediction justified from the pattern of the earlier radii.

6 $R = \dfrac{AB}{2\sin\theta}$. Proof: $OA = OB = R$, so $\triangle OAB$ is isosceles; the perpendicular from O bisects AB and the central angle 2θ, so in the right triangle $\sin\theta = \dfrac{AB/2}{R}$, hence $R = \dfrac{AB}{2\sin\theta}$. (Sine is increasing on $0^\circ < \theta < 90^\circ$, so smaller θ gives larger R, matching the 30° prediction.)

Marking guidance: [3] — formula stated (1), proof from the isosceles triangle (2). Top band: formula checked against all earlier cases ($\theta = 30, 60, 90^\circ$) and validity range noted.

Criterion C · Communicating · Explaining why the centre angle doubles

1 **Sample response.** *Angle at the centre theorem.* In a circle with centre O, if A and B are points on the circle, the angle $\angle AOB$ at the centre standing on arc AB is twice the angle $\angle ACB$ at any point C on the remaining circumference. The arc that matters is the one **not** containing C. *Construction.* Draw the radius CO and extend it to meet the circle again at D; a diagram should show O inside the triangle region and D between the rays CA and CB produced. *Two isosceles triangles.* Since $OA = OC$ (radii), $\triangle OAC$ is isosceles, so $\angle OAC = \angle OCA$; call it α. Likewise $OB = OC$, so $\triangle OBC$ is isosceles with base angle $\beta = \angle OCB = \angle OBC$. Note $\angle ACB = \alpha + \beta$. *Exterior-angle step.* $\angle AOD$ is an exterior angle of $\triangle OAC$, so it equals the two interior opposite angles together: $\angle AOD = \alpha + \alpha = 2\alpha$. Similarly $\angle BOD = 2\beta$. Adding: $\angle AOB = 2\alpha + 2\beta = 2(\alpha + \beta) = 2\angle ACB$. *Reflex case.* If C lies on the minor arc AB, exactly the same argument gives reflex $\angle AOB = 2\angle ACB$; the ordinary angle $\angle AOB$ is then $360^\circ - 2\angle ACB$. *Conclusion.* Every step used only equal radii, isosceles base angles and the exterior-angle property — all things that must hold in **every** drawing, not features of one picture. That is why the $2:1$ ratio is a theorem and not a measurement: measuring one diagram can suggest the result, but only this chain of reasons guarantees it.

Marking guidance: 20 marks on the Criterion C band descriptors (0–8). Top band requires all five labelled moves (statement with correct arc, construction of D, two isosceles triangles, justified exterior-angle step, separate reflex case), an accurate labelled diagram, audience-appropriate language, and a closing statement distinguishing proof from measurement; coherent organisation throughout.

Criterion D · Applying mathematics in real-life contexts · A circular stained-glass panel

1 In the proposed order $80^\circ, 100^\circ, 95^\circ, 85^\circ$ the opposite pairs are $80^\circ + 95^\circ = 175^\circ$ and $100^\circ + 85^\circ = 185^\circ$; neither is 180°, so the pane cannot be cyclic (Theorem 6.4). A workable order is $80^\circ, 95^\circ, 100^\circ, 85^\circ$: opposites $80^\circ + 100^\circ = 180^\circ$, $95^\circ + 85^\circ = 180^\circ$.

Marking guidance: [4] — identifying the opposite pairs in the given order and both failing sums (2), a valid reordering with both sums 180° verified (2). Top band: names the theorem and its converse.

2 The four corners lie on the rim, i.e. on one circle (the window's circumference), so the quadrilateral is cyclic by definition and the window circle is its circumcircle — any four concyclic points joined in order form a cyclic quadrilateral. Order matters because the supplementary condition $\angle A + \angle C = 180^\circ$, $\angle B + \angle D = 180^\circ$ applies to **opposite** angles, and which angles are opposite is decided by the cyclic order of the corners.

Marking guidance: [4] — inscription justified from "vertices on one circle" (2), role of order explained with reference to opposite pairs (2). Top band: links back to the theorem rather than asserting.

3 Chord 2.4 m: half is 1.2 m, distance $= \sqrt{1.5^2 - 1.2^2} = \sqrt{0.81} = 0.9$ m. Chord 1.8 m: half 0.9, distance $= \sqrt{1.5^2 - 0.9^2} = 1.2$ m. The 2.4 m chord is closer to the centre (longer chords lie closer).

Marking guidance: [4] — bisection of each chord (1), both Pythagoras calculations (2), correct comparison (1). Top band: units and a stated reason for the comparison.

4 Assumption 1: the lead lines have negligible width. Real lead of width w covers a strip along each side, so the glass area is the pane area minus roughly (perimeter $\times w$); thicker lead reduces every pane and demands the corners be recut. Assumption 2: the angles are achieved exactly. A one-degree error at a corner slides that vertex along the rim and distorts both the adjacent chord lengths and the total glass area; on a $1.5\ \text{m}$ radius window the edge displacement is about $1.5\times$ (error in radians) $\approx 2.6\ \text{cm}$ per degree, which then forces gaps or overlaps at the lead.

Marking guidance: [6] — two clearly stated assumptions (2), each with a quantified or well-reasoned effect on glass area (2 per assumption). Top band: assumptions are realistic for glazing and their effects are directional (increase/decrease), not vague.

5 Proposal 2, $70^\circ, 110^\circ, 70^\circ, 110^\circ$ in order, pairs opposite angles 70° with 70° and 110° with 110°: sums 140° and 220°, so **as written** it cannot be cyclic (the equal-opposite layout is the parallelogram arrangement, cyclic only for rectangles). It is, however, highly symmetric and its angles admit the cyclic ordering $70^\circ, 70^\circ, 110^\circ, 110^\circ$ — an isosceles trapezium, symmetric about a diameter, with opposite sums 180°. Recommendation: build the reordered Proposal 2 ($70^\circ, 70^\circ, 110^\circ, 110^\circ$): it is cyclic, and its mirror symmetry is easier to lay out and more attractive on the rim than Proposal 1's scalene cyclic order $80^\circ, 95^\circ, 100^\circ, 85^\circ$, which works but has no symmetry.

Marking guidance: [6] — testing Proposal 2 against the 180° test (2), testing Proposal 1 (1), recognising a cyclic reordering and the symmetry of both (2), a recommendation justified by both cyclic validity and symmetry (1). Top band: distinguishes "these angles" from "this order" throughout.

CHAPTER 7 · SINE AND COSINE RULES; TRIGONOMETRY IN THREE DIMENSIONS

Opening problem · The fire between the towers

1 Yes. The bearing of B from A is $090°$, so the angle at A is $090° - 040° = 50°$; the bearing of A from B is $270°$, so the angle at B is $330° - 270° = 60°$; hence $\angle AFB = 180° - 50° - 60° = 70°$.

2 The pair (AB, F) is complete, so $\dfrac{AF}{\sin B} = \dfrac{AB}{\sin F}$: $AF = \dfrac{8\sin 60°}{\sin 70°} = 7.37$ km and $BF = \dfrac{8\sin 50°}{\sin 70°} = 6.52$ km (3 s.f.). Perpendicular distance to the firebreak $= AF\sin 50° = 5.65$ km (check: $BF\sin 60° = 5.65$ km).

3 area $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2}(8)(5.648) = 22.6$ km^2 (check: $\frac{1}{2}(7.373)(6.516)\sin 70° = 22.6$ km^2).

4 Tower B is nearer: 6.52 km against 7.37 km, a difference of 0.85 km. With bearings $042°, 328°$ the angles are $48°, 58°, 74°$, so $AF = 7.06$ km, $BF = 6.18$ km and the height is 5.25 km; the position moves 0.40 km toward the firebreak. Two degrees per bearing is worth nearly half a kilometre, but B stays nearer.

Prerequisite check

1 $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$ cm.

2 opposite $= 10\sin 35° = 5.74$ cm (3 s.f.).

3 $\sin 30° = \frac{1}{2}$, $\cos 60° = \frac{1}{2}$, $\tan 45° = 1$.

4 $x = \sin^{-1} 0.6 = 36.9°$ (1 d.p.); the supplement $143.1°$ is outside the given range.

5 $N30°E$ is $030°$; the back bearing of $120°$ is $120° + 180° = 300°$.

6 $\dfrac{8\sin 50°}{\sin 70°} = \dfrac{6.128}{0.9397} = 6.52$ (3 s.f.).

Exercise 7A · The sine rule, including the ambiguous case

1 $C = 180° - 47° - 63° = 70°$; $b = \dfrac{11.2\sin 63°}{\sin 47°} = 13.6$ cm, $c = \dfrac{11.2\sin 70°}{\sin 47°} = 14.4$ cm (3 s.f.). Check: the largest angle $70°$ faces the longest side.

2 (a) $q = \dfrac{6.4\sin 80°}{\sin 35°} = 11.0$ cm. (b) $\sin Y = \dfrac{14.0\sin 28°}{9.0} = 0.7303$, so $Y = 46.9°$ or $133.1°$: two triangles, since X is acute, $x < y$ and $28° + 133.1° = 161.1° < 180°$. (c) $d = \dfrac{8.5\sin 112°}{\sin 34°} = 14.1$ cm ($D > E$, so $d > e$, as required).

3 $C = 180° - 60° - 45° = 75°$. $b = \dfrac{8\sqrt{3}\sin 45°}{\sin 60°} = 8\sqrt{2}$ cm; $c = \dfrac{8\sqrt{3}\sin 75°}{\sin 60°} = 16\sin 75° = 4(\sqrt{6} + \sqrt{2})$ cm ≈ 15.5 cm.

4 $\sin B = \dfrac{10.2\sin 39°}{7.4} = 0.8674$; two triangles exist (given angle acute, $a < b$, $h = b\sin A = 6.41$ cm $< a$): **first** $B = 60.2°$, $C = 80.8°$, $c = \dfrac{7.4\sin 80.8°}{\sin 39°} = 11.6$ cm; **second** $B = 119.8°$, $C = 21.2°$, $c = 4.25$ cm.

5 (a) Yes—ASA: $C = 180° - 40° - 65° = 75°$ gives the pair (c, C). (b) No—SAS, no angle faces a known side; use the cosine rule. (c) Yes—SSA: A faces a. (d) No—SSS; use the cosine rule. (e) Yes—AAS: the pair (a, A) is complete.

6 The third angle is $180° - 58° - 44° = 78°$. The shortest side faces $44°$ and the longest faces $78°$: $c = \dfrac{9.6\sin 78°}{\sin 44°} = 13.5$ cm.

7 $\sin B = \dfrac{8\sin 35°}{7} = 0.6555$ is correct, but $\sin^{-1}$ returns only the acute angle and $\sin(180° - B) = \sin B$. Since A is acute and $a < b$, the second value $B = 180° - 41.0° = 139.0°$ also fits ($35° + 139.0° = 174.0° < 180°$). So $B = 41.0°$ or $139.0°$: two triangles, and the calculator's single value loses the second one.

8 In right triangle ABE, $BE = c\sin A$; in right triangle CBE, $BE = a\sin C$. Equating: $c\sin A = a\sin C$, so $\dfrac{a}{\sin A} = \dfrac{c}{\sin C}$; a second altitude (from A to BC, or from C to AB) gives $\dfrac{b}{\sin B} = \dfrac{c}{\sin C}$, and hence the full rule.

As printed, the altitude from B yields $\dfrac{a}{\sin A} = \dfrac{c}{\sin C}$, not $\dfrac{b}{\sin B} = \dfrac{c}{\sin C}$; the stated target needs the altitude from A or C.

9 AB_1C (wider, $c = AB_1 = 12.0$ cm) has $B = 56.4°$, $C = 93.6°$; AB_2C (short base $c = 5.34$ cm) has the obtuse $B_2 = 123.6°$, $C = 26.4°$. Both keep $A = 30°$, $b = 10$ cm, $a = 6$ cm: B_1, B_2 are the two points where the arc of radius 6 cm about C cuts the base line, and $\sin 56.4° = \sin 123.6°$.

10 One triangle only. $\sin B = \dfrac{6\sin 40°}{8} = 0.4821$ gives $B = 28.8°$ or $151.2°$, and $40° + 151.2° = 191.2° > 180°$ rejects the supplement ($a > b$ also forces $A > B$). Hence $C = 111.2°$, $c = \dfrac{8\sin 111.2°}{\sin 40°} = 11.6$ cm.

11 $\sin B = \dfrac{9\sin 30°}{5} = 0.9$, so $B = 64.2°$ or $115.8°$. Both fit: $30° + 115.8° < 180°$, A is acute and $h = b\sin A = 4.5$ cm $< a = 5$ cm $< b$. Two triangles exist.

12 $\sin\theta = \sin(180° - \theta)$, so the sine rule gives two roots in $0° < \theta < 180°$ and cannot separate them. Test each: the supplement must leave a positive third angle, and the larger side must face the larger angle, so an angle facing the shorter known side is the smaller candidate. Equivalently the sign of $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ settles acute or obtuse uniquely.

13 Angles inside the triangle: at A, $090° - 052° = 38°$; at B, $338° - 270° = 68°$; so $F = 180° - 38° - 68° = 74°$. $AF = \dfrac{6\sin 68°}{\sin 74°} = 5.79$ km, $BF = \dfrac{6\sin 38°}{\sin 74°} = 3.84$ km, perpendicular distance to $AB = AF\sin 38° = 3.56$ km.

14 The larger elevation marks the nearer point, so B is 20 m from A toward the kite. In triangle ABK: $\angle A = 38°$, the exterior angle at B is $52°$ so $\angle ABK = 128°$ and $\angle AKB = 14°$. Then $AK = \dfrac{20\sin 128°}{\sin 14°} = 65.1$ m and the height is $AK\sin 38° = 40.1$ m (check: $\frac{h}{\tan 38°} - \frac{h}{\tan 52°} = 51.3 - 31.3 = 20$ m).

With A and B read on opposite sides of the kite's foot, the same data give $\angle AKB = 90°$, $AK = 20\sin 52° = 15.8$ m, $h = 9.70$ m.

15 By the sine rule $a = \dfrac{b\sin A}{\sin B} = \dfrac{b\sin 2B}{\sin B} = \dfrac{b(2\sin B\cos B)}{\sin B} = 2b\cos B$. □ (Check $B = 30°$, $A = 60°$, $C = 90°$, $b = 1$: then $a = 2\cos 30° = \sqrt{3}$, the side opposite $60°$ in a $30°$–$60°$–$90°$ triangle.)

Exercise 7B · The cosine rule and the choice of method

1 SAS: $a^2 = 8.4^2 + 6.2^2 - 2(8.4)(6.2)\cos 54° = 47.78$, $a = 6.91$ cm (3 s.f.).

2 The largest angle faces the 11 cm side: $\cos A = \dfrac{9^2 + 6^2 - 11^2}{2(9)(6)} = \dfrac{-4}{108} = -0.03704$, so $A = 92.1°$ (1 d.p.); the negative cosine confirms it is obtuse.

3 SSS, largest angle first: $\cos^{-1}\dfrac{7^2 + 8^2 - 10^2}{2(7)(8)} = \cos^{-1}\dfrac{13}{112} = 83.3°$ opposite 10 cm; $\cos^{-1}\dfrac{7^2 + 10^2 - 8^2}{2(7)(10)} = 52.6°$ opposite 8 cm; third angle $44.0°$ opposite 7 cm. Sum $180°$, and side order $10 > 8 > 7$ matches angle order.

4 $c^2 = 5^2 + 12^2 - 2(5)(12)\cos 90° = 169$, $c = 13$ cm: the $-2ab\cos C$ term vanished because $\cos 90° = 0$, so the cosine rule reproduces Pythagoras and $5, 12, 13$ is a Pythagorean triple.

5 (a) Cosine rule—SAS, the angle lies between the two given sides. (b) Cosine rule—SSS, no angle is given. (c) Sine rule—ASA: $C = 180° - 52° - 61° = 67°$ supplies the pair (c, C). (d) Sine rule—SSA, the pair (a, A) exists (then test the ambiguous case). (e) Right-triangle ratios or Pythagoras; neither general rule is needed.

6 Opposite 7 cm: $\cos C = \dfrac{5^2 + 6^2 - 7^2}{2(5)(6)} = \dfrac{12}{60} = 0.2$, $C = 78.5°$. Opposite 6 cm: $\cos B = \dfrac{5^2 + 7^2 - 6^2}{2(5)(7)} = \dfrac{38}{70}$, $B = 57.1°$. Opposite 5 cm: $\cos A = \dfrac{6^2 + 7^2 - 5^2}{2(6)(7)} = \frac{5}{7}$, $A = 44.4°$. Total $180°$; side order matches angle order.

7 With $A = 120^\circ$, $\cos A = -0.5$, so the correct term $-2bc\cos A = -2(6)(7)(-0.5) = +42$ is *added*: $a^2 = 36 + 49 + 42 = 127$, $a = 11.3$ cm. The student's $a^2 = b^2 + c^2 + 2bc\cos A$ gives 43, $a = 6.56$ cm, impossible for an obtuse angle since a^2 must exceed $b^2 + c^2 = 85$.

At $A = 90^\circ$ the sign error is invisible, so it often goes unnoticed.

8 Put $A(0,0)$, $B(c,0)$, $C(b\cos A, b\sin A)$. Then $a^2 = (b\cos A - c)^2 + (b\sin A)^2 = b^2(\cos^2 A + \sin^2 A) + c^2 - 2bc\cos A = b^2 + c^2 - 2bc\cos A$. $\square$

9 At $A = 90^\circ$, $\cos A = 0$ and $a^2 = b^2 + c^2$, Pythagoras. If A is obtuse then $\cos A < 0$, so $-2bc\cos A > 0$ and $a^2 > b^2 + c^2$: the side opposite an obtuse angle is longer than Pythagoras predicts.

10 Included 30°: $x^2 = 7^2 + 9^2 - 2(7)(9)\cos 30^\circ = 130 - 109.12 = 20.88$, $x = 4.57$ cm. Included 150°: $x^2 = 130 + 109.12 = 239.12$, $x = 15.5$ cm. Since $\cos 150^\circ = -\cos 30^\circ$ the correction term only changes sign, and the squares differ by $2(7)(9)\cos 30^\circ = 109.1$. Both satisfy the triangle inequality ($4.57 + 7 > 9$; $15.5 < 7 + 9$).

11 The longest side is 8 cm and $8^2 = 64 > 4^2 + 5^2 = 41$, so the triangle is obtuse: $\cos C = \dfrac{4^2 + 5^2 - 8^2}{2(4)(5)} = -0.575$, giving largest angle $C = 125.1^\circ$ (1 d.p.).

12 A complete matching pair (a side with the angle opposite it) means the sine rule: ASA, AAS and also SSA, where the two-triangle check is still needed. SAS (the angle *between* two sides) or SSS give no pair, so the cosine rule is required. With a right angle, Pythagoras and the basic ratios are quickest. For area, $\frac{1}{2}ab\sin C$ needs the included angle; with SSS find an angle first or use Heron's formula.

13 Angle at the port $= 145^\circ - 060^\circ = 85^\circ$, so $d^2 = 12^2 + 18^2 - 2(12)(18)\cos 85^\circ = 430.35$, $d = 20.7$ km. At the first ship $\sin P = \dfrac{18\sin 85^\circ}{20.74} = 0.8644$, $P = 59.8^\circ$; the port bears $060^\circ + 180^\circ = 240^\circ$ from it and the second ship lies anticlockwise from the port, so the bearing is $240^\circ - 59.8^\circ = 180.2^\circ \approx 180^\circ$ (due south).

14 $45^2 + 60^2 = 5625 = 75^2$, so by the converse of Pythagoras the angle opposite the 75 m side is 90° (cosine-rule check: $\cos C = 0$); it is the largest angle. Area $= \frac{1}{2}(45)(60) = 1350$ m^2.

15 By the cosine rule $c^2 = a^2 + b^2 - 2ab\cos C$, so $a^2 + b^2 < c^2 \iff -2ab\cos C > 0 \iff \cos C < 0$ (as $2ab > 0$) $\iff C$ obtuse; every step reverses, giving the "if and only if". Applied to the longest side this classifies any triangle—$c^2 = a^2 + b^2$ right, $c^2 < a^2 + b^2$ acute, $c^2 > a^2 + b^2$ obtuse—without computing an angle.

Exercise 7C · Area, parallelograms and working backwards

1 C lies between a and b, so $K = \frac{1}{2}(9.2)(7.5)\sin 48^\circ = 34.5 \times 0.7431 = 25.6$ cm^2 (3 s.f.).

2 $K = \frac{1}{2}(12)(10)\sin 115^\circ = 60 \times 0.9063 = 54.4$ m^2 (check: the 90° area would be 60 m^2 and $\sin 115^\circ < 1$).

3 $\sin C = \dfrac{2K}{ab} = \dfrac{48}{8 \times 7} = \dfrac{6}{7} = 0.8571$, so $C = 59.0^\circ$ or 121.0°. Both are admissible, giving third sides 7.44 cm or 13.1 cm.

4 Parallelogram area $= ab\sin\theta = (7.4)(5.6)\sin 62^\circ = 36.6$ cm^2, twice the 18.3 cm^2 triangle cut off by a diagonal.

5 (a) $\frac{1}{2}(14)(9) = 63$ cm^2. (b) $\frac{1}{2}(7)(12)\sin 30^\circ = 21$ cm^2; the methods agree because $\frac{1}{2} \times$ base $\times$ height and $\frac{1}{2}ab\sin C$ are the same formula with height $= b\sin C$.

6 $\sin C = \dfrac{2(36)}{(9)(10)} = 0.8$, so the included angle is 53.1° or 126.9°.

7 $\frac{1}{2}ab\sin C$ needs the angle *between* the two sides; here 40° faces the 6 cm side, so 17.4 cm^2 is the area of a different triangle. The formula becomes valid once the angle between the 6 cm and 9 cm sides is given. From the data supplied, $\sin\theta = \dfrac{9\sin 40^\circ}{6} = 0.9642$, so the angle facing 9 cm is 74.6° or 105.4°, the included angle is 65.4° or 34.6°, and the area is 24.5 cm^2 or 15.3 cm^2.

8 Take $AC = b$ as base and drop $BF \perp AC$. In right triangle BFC the hypotenuse is $BC = a$, so $h = BF = a\sin C$ and $K = \frac{1}{2}b(a\sin C) = \frac{1}{2}ab\sin C$. (For obtuse C, F falls outside the segment but $h = a\sin(180^\circ - C) = a\sin C$ still.)

9 Each expression is half a base times its height, so all equal K; algebraically the sine rule gives $a\sin C = c\sin A$ and $b\sin A = a\sin B$, hence $\frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$. With $K = \frac{1}{2}ah_a = \frac{1}{2}bh_b = \frac{1}{2}ch_c$ the altitudes satisfy $h_a : h_b : h_c = \dfrac{1}{a} : \dfrac{1}{b} : \dfrac{1}{c}$:

10 Parallel lines are a constant distance apart, so both triangles have base b, height h and area $\frac{1}{2}bh$. In the $\frac{1}{2}ab\sin C$ form the sloping sides and included angle change as the vertex slides, but (side) $\times$ sin(included angle) $= h$ always, so K is unchanged.

11 $\sin C = \dfrac{2(20)}{(6)(9)} = \dfrac{20}{27} = 0.7407$, so $C = 47.8^\circ$ or 132.2°. Two triangles, because two sides and the area fix only the sine of the included angle and $\sin C = \sin(180^\circ - C)$. The third side is 6.67 cm or 13.8 cm.

12 $\cos C = \dfrac{13^2 + 14^2 - 15^2}{2(13)(14)} = \dfrac{140}{364} = 0.3846$, $C = 67.4^\circ$, so $K = \frac{1}{2}(13)(14)\sin 67.4^\circ = 84$ m^2. Heron: $s = \dfrac{13 + 14 + 15}{2} = 21$, $K = \sqrt{21(8)(7)(6)} = \sqrt{7056} = 84$ m^2 — agreement. Heron is more direct here: three sides are given, so no angle and no rounding are needed.

13 $K = \frac{1}{2}(18)(24)\sin 72^\circ = 205$ m^2 (3 s.f.), so the cost is $205.43 \times 8.50 = £1746.14$, about $£1750$ (3 s.f.).

14 Area $= (65)(42)\sin 58^\circ = 2315$ m^2 (2.32×10^3 m^2 to 3 s.f.). Panels: $\dfrac{2315.17}{5} = 463.03$, so **463** complete 5 m^2 panels fit—the leftover strip of 0.17 m^2 is too small for another.

15 $\sin C = \dfrac{2K}{ab}$. Whenever $0 < \dfrac{2K}{ab} < 1$ there are exactly two solutions in $0^\circ < C < 180^\circ$, namely C and $180^\circ - C$, supplementary and equal in area. Exactly one occurs when $\dfrac{2K}{ab} = 1$, i.e. $K = \frac{1}{2}ab$ and $C = 90^\circ$, the maximum area for the two sides; if $K > \frac{1}{2}ab$ no triangle exists.

Exercise 7D · Bearings, back bearings and navigation

1 $N35^\circ E = 035^\circ$; $S20^\circ W = 180^\circ + 20^\circ = 200^\circ$; north-west $= 315^\circ$; due east $= 090^\circ$; $S40^\circ W = 180^\circ + 40^\circ = 220^\circ$.

2 Add or subtract 180°: $047^\circ \to 227^\circ$, $128^\circ \to 308^\circ$, $300^\circ \to 120^\circ$, $355^\circ \to 175^\circ$.

3 At the turning point the back bearing is $030^\circ + 180^\circ = 210^\circ$, so the interior angle is $210^\circ - 120^\circ = 90^\circ$ and the distance $= \sqrt{5^2 + 4^2} = \sqrt{41} = 6.40$ km. Angle at the start $\tan^{-1}\frac{4}{5} = 38.7^\circ$, taken clockwise from the first leg, so the bearing is $030^\circ + 38.7^\circ = 068.7^\circ \approx 069^\circ$.

4 Interior angle at the turn: $075^\circ + 180^\circ - 165^\circ = 90^\circ$, so the distance is $\sqrt{20^2 + 15^2} = 25$ km. Angle at the start $= \tan^{-1}\frac{15}{20} = 36.9^\circ$ clockwise from 075°, giving bearing $111.9^\circ \approx 112^\circ$ (components: east (23.2), north (-9.31)).

5 The harbour bears $230^\circ - 180^\circ = 050^\circ$ from P, at the same distance 12 km.

6 Distance $= 240 \times \frac{45}{60} = 180$ km. East $= 180\sin 110^\circ = 169$ km, north $= 180\cos 110^\circ = -61.6$ km, i.e. 61.6 km south (check: the track is 20° south of due east, and $180\cos 20^\circ = 169$, $180\sin 20^\circ = 61.6$).

7 A bearing is measured clockwise from north, not anticlockwise from east, so an angle 70° above the horizontal is the bearing $090^\circ - 70^\circ = 020^\circ$. The correct diagram has a north arrow at the point with the ray 70° clockwise from north, only 20° above the east–west line.

8 North is the same direction everywhere on a map (meridians drawn parallel; their convergence is negligible over a small area). The north line at B is parallel to that at A, so the angle facts for parallel lines (co-interior angles summing to 180°) give the bearing of A from B as the bearing of B from A plus 180°, which is why a bearing transfers from vertex to vertex.

9 The east–west baseline bears 090° from A and 270° from B. Angle at $A = 090^\circ - 040^\circ = 50^\circ$; angle at $B = 330^\circ - 270^\circ = 60^\circ$; angle at $F = 180^\circ - 50^\circ - 60^\circ = 70^\circ$.

10 Reverse both bearings to get directions out of A: B bears $050^\circ + 180^\circ = 230^\circ$ and C bears $320^\circ - 180^\circ = 140^\circ$, hence $\angle BAC = 230^\circ - 140^\circ = 90^\circ$. Only the two directions matter, not the distances AB and AC, which are needed for the sides.

11 The legs are the perpendicular sides of a right triangle whose hypotenuse is the displacement: east 10, north 10, so the angle with north is $\tan^{-1}\frac{10}{10} = 45^\circ$, bearing 045° (displacement $10\sqrt{2} = 14.1$ km). Not 090°, because a bearing describes the straight line from start to finish, not the final leg.

12 On an east–west line the interior angles are $90^\circ - 40^\circ = 50^\circ$ at the station reporting 040° and $310^\circ - 270^\circ = 40^\circ$ at the other, so the flare subtends 90° and the rays meet cleanly. If the stations are not east–west the baseline no longer bears $090^\circ/270^\circ$, so both interior angles are wrong by the tilt and the rays meet at a point that is not the flare (equal bearings would be parallel and never meet). A third bearing from a third station gives an independent fix: the three rays should meet in one point, and the small triangle they form measures the error.

Strictly, two non-parallel rays from known points always intersect; what a wrong baseline assumption destroys is the position of that intersection and the calculated distances.

13 Angle at the port $= 140^\circ - 055^\circ = 85^\circ$, so $PQ^2 = 24^2 + 30^2 - 2(24)(30)\cos 85^\circ = 1350.5$ and $PQ = 36.7$ km. At P, $\sin P = \dfrac{30 \sin 85^\circ}{36.75} = 0.8132$, $P = 54.4^\circ$; the port bears $055^\circ + 180^\circ = 235^\circ$ from P and Q lies anticlockwise from it, so the bearing of Q from P is $235^\circ - 54.4^\circ = 180.6^\circ \approx 181^\circ$.

14 The legs are $8 \times 3 = 24$ km on 215° and $8 \times 5 = 40$ km on 125°. The back bearing at the turn is $215^\circ - 180^\circ = 035^\circ$, so the interior angle is $125^\circ - 35^\circ = 90^\circ$ and the distance $= \sqrt{24^2 + 40^2} = \sqrt{2176} = 46.6$ km. Angle at the start $\tan^{-1}\frac{40}{24} = 59.0^\circ$, and the second leg turns anticlockwise from the first, so the bearing is $215^\circ - 59.0^\circ = 156.0^\circ$ (components: east (19.0), north (-42.6)). Returning at 10 km/h takes $\dfrac{46.65}{10} = 4.66$ h ≈ 4 h 40 min.

15 $\angle BAC = 145^\circ - 060^\circ = 85^\circ$, so $BC^2 = 40^2 + 55^2 - 2(40)(55)\cos 85^\circ = 4241.5$ and $BC = 65.1$ km. At B, $\sin B = \dfrac{55 \sin 85^\circ}{65.13} = 0.8414$, $B = 57.3^\circ$; A bears $060^\circ + 180^\circ = 240^\circ$ from B and C lies anticlockwise from A, so C bears $240^\circ - 57.3^\circ = 182.7^\circ \approx 183^\circ$ from B. Area $= \frac{1}{2}(40)(55)\sin 85^\circ = 1095.8$ km$^2 \approx 1100$ km^2.

Exercise 7E · Lines, planes and three-dimensional triangles

1 Base diagonal $\sqrt{8^2 + 5^2} = \sqrt{89} = 9.43$ m (right triangle in the base); space diagonal $\sqrt{89 + 4^2} = \sqrt{105} = 10.2$ m. In the vertical right triangle whose legs are the base diagonal and the height, $\tan\theta = \dfrac{4}{\sqrt{89}}$, $\theta = 23.0^\circ$.

2 Base diagonal $= \sqrt{10^2 + 10^2} = 10\sqrt{2} = 14.1$ m; space diagonal $= \sqrt{200 + 6^2} = \sqrt{236} = 2\sqrt{59} = 15.4$ m; $\tan\theta = \dfrac{6}{10\sqrt{2}} = 0.4243$, so $\theta = 23.0^\circ$.

3 Half-diagonal $= \dfrac{12\sqrt{2}}{2} = 6\sqrt{2} = 8.49$ m, so in the right triangle of height, half-diagonal and sloping edge, $l = \sqrt{8^2 + 72} = \sqrt{136} = 2\sqrt{34} = 11.7$ m and $\tan\alpha = \dfrac{8}{6\sqrt{2}}$, $\alpha = 43.3^\circ$. The dihedral angle comes from the cross-section through the apex and the midpoint of a base side: apothem 6 m, slant height $\sqrt{8^2 + 6^2} = 10$ m, so $\tan\beta = \dfrac{8}{6}$, $\beta = 53.1^\circ$.

4 Floor diagonal $= \sqrt{6^2 + 4^2} = \sqrt{52} = 7.21$ m; diagonal $= \sqrt{52 + 2.8^2} = \sqrt{59.84} = 7.74$ m; $\tan\theta = \dfrac{2.8}{\sqrt{52}} = 0.3883$, $\theta = 21.2^\circ$ above the floor.

5 The 4 m by 3 m face is perpendicular to the 5 m edge, so the diagonal projects on that face as its diagonal $\sqrt{4^2 + 3^2} = 5$ m and the perpendicular to the face is the 5 m edge. Thus $\tan\phi = \dfrac{5}{5} = 1$, $\phi = 45.0^\circ$.

6 The sloping face stands on the hypotenuse of the cross-section, $\sqrt{3^2 + 4^2} = 5$ m, so it is a 5 m by 6 m rectangle; the longest line joining two of its vertices is its diagonal $\sqrt{5^2 + 6^2} = \sqrt{61} = 7.81$ m.

7 Height over space diagonal is the sine of the angle, not its tangent: the diagonal is the hypotenuse of the right triangle, not the side next to the angle. The denominator must be the projection of the diagonal on the base, i.e. the base diagonal $\sqrt{l^2 + w^2}$: $\tan\theta = \dfrac{h}{\sqrt{l^2 + w^2}}$ and $\sin\theta = \dfrac{h}{d}$.

8 Let PQ meet the plane at Q, drop the perpendicular PR to the plane and join R to Q. Then QR is the projection of PQ, $\angle PQR$ is the line–plane angle and $\sin\angle PQR = \dfrac{PR}{PQ}$ in right triangle PQR. For any other plane line QS making angle ϕ with QR, $\cos\angle PQS = \cos\angle PQR\cos\phi \le \cos\angle PQR$, so $\angle PQS \ge \angle PQR$: the projection gives the smallest angle the line makes with anything in the plane.

9 Triangle AFG, with AF the base diagonal: $AF = \sqrt{6^2 + 4^2} = \sqrt{52} = 7.21$ m, $FG = 3$ m, $AG = \sqrt{52 + 9} = \sqrt{61} = 7.81$ m. Angle with the base $ABCD$: $\tan\theta = \dfrac{3}{\sqrt{52}}$, $\theta = 22.6^\circ$. Face $BCGF$ is the 4 m by 3 m end; the projection of AG on it is $GF = \sqrt{4^2 + 3^2} = 5$ m and the perpendicular is the 6 m edge AB, so $\tan\phi = \dfrac{6}{5}$, $\phi = 50.2^\circ$.

10 The two roof planes meet along the ridge. In each plane take the line perpendicular to the ridge at one point (a rafter running straight down the pitch); the angle between those two lines, measured in the single plane containing both, is the dihedral angle. A cut perpendicular to the ridge shows both lines at once, and the apex angle of that triangular section is the dihedral angle, so it can be measured or calculated directly.

11 Right triangle: ladder 5 m (hypotenuse), foot 1.8 m from the wall, height $\sqrt{25 - 1.8^2} = \sqrt{21.76} = 4.66$ m. With the ground $\cos\theta = \dfrac{1.8}{5} = 0.36$, $\theta = 68.9^\circ$; with the wall the complementary angle is $90^\circ - 68.9^\circ = 21.1^\circ$ (check: $\sin = \frac{1.8}{5}$).

12 Since $w > 0$ and $h > 0$, $d^2 = l^2 + w^2 + h^2 > l^2$, so $d > l$; cycling the letters gives $d > w$ and $d > h$. Also $(l + w + h)^2 = l^2 + w^2 + h^2 + 2(lw + wh + hl) > d^2$ because the cross terms are positive, and square roots preserve order for positive numbers, so $d = \sqrt{l^2 + w^2 + h^2} < l + w + h$.

13 The ridge is $7 - 4 = 3$ m above the eaves on the centre line of an 8 m width, so each face runs 4 m horizontally. In the cross-sectional right triangle $\tan\theta = \dfrac{3}{4}$, $\theta = 36.9^\circ$, and a rafter's sloping length is $\sqrt{4^2 + 3^2} = 5$ m. Each face is 12 m $\times 5$ m $= 60$ m^2, so the two faces total 120 m^2.

14 Half-diagonal $= \dfrac{230\sqrt{2}}{2} = 115\sqrt{2} = 162.6$ m, so $l = \sqrt{147^2 + (115\sqrt{2})^2} = \sqrt{48059} = 219$ m (3 s.f.). Edge to base: $\tan\alpha = \dfrac{147}{115\sqrt{2}} = 0.9038$, $\alpha = 42.1^\circ$. The dihedral angle uses the apothem 115 m: $\tan\beta = \dfrac{147}{115} = 1.2783$, $\beta = 52.0^\circ$ (slant height $\sqrt{147^2 + 115^2} = 187$ m).

15 Take the edges from the vertex as l, w, h with $d = \sqrt{l^2 + w^2 + h^2}$. The diagonal's projection on the edge l is l itself, so $\cos\alpha = \dfrac{l}{d}$, and similarly $\cos\beta = \dfrac{w}{d}$, $\cos\gamma = \dfrac{h}{d}$; hence $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \dfrac{l^2 + w^2 + h^2}{d^2} = 1$. (Unit cube: each equals $\frac{1}{\sqrt{3}}$, three equal angles.)

Investigation 7 · How far is it, really?

1 Read the elevation θ from the hanging string, measure the horizontal distance d to the wall and the eye height e, then $h = d\tan\theta + e$. With $\theta = 32^\circ$, $d = 18$ m, $e = 1.5$ m: $h = 18\tan 32^\circ + 1.5 = 12.75 \approx 12.8$ m. The straw must lie along the sight line and the string hang freely; an unzeroed protractor reads 90° minus the elevation.

2 From 20 m farther back, $d = 38$ m and the same wall should give $\theta = \tan^{-1}\dfrac{11.25}{38} = 16.5^\circ$, so $h = 38\tan 16.5^\circ + 1.5 = 12.8$ m, agreeing with the first reading. Two reasons for any difference: the shallow angle is harder to read (a 1° slip changes h by 0.7 m at 16.5° but 0.4 m at 32°), and the distance runs over sloping ground, so the horizontal distance and e are not what the formula assumes.

3 With the 40 m east–west baseline AB, suppose T bears 035° from A and 325° from B: interior angles $090^\circ - 035^\circ = 55^\circ$ at A and $325^\circ - 270^\circ = 55^\circ$ at B, so $\angle ATB = 70^\circ$. Sine rule: $AT = BT = \dfrac{40\sin 55^\circ}{\sin 70^\circ} = 34.9$ m, and the river width is $AT\sin 55^\circ = 28.6$ m. A scale drawing at 1 cm to 5 m should agree within its 2–3% tolerance.

4 Similar triangles give $\dfrac{s}{L} = \dfrac{p}{D}$ (p eye separation, L distance to the screen, s apparent shift, D finger distance), so $D = \dfrac{pL}{s}$: a larger shift means a nearer finger. With $p = 6.4$ cm, $L = 1.0$ m, $s = 16$ cm: $D = \dfrac{6.4 \times 100}{16} = 40$ cm, a typical arm's length.

5 Readings $31^\circ, 32^\circ, 32^\circ, 33^\circ, 32^\circ$ have mean 32.0° and range 2°. With $d = 18$ m, $e = 1.5$ m: $18\tan 31^\circ + 1.5 = 12.3$ m up to $18\tan 33^\circ + 1.5 = 13.2$ m, a range of 0.9 m (± 0.45 m), while a 0.5 m error in distance alone shifts h by $0.5\tan 32^\circ = 0.31$ m. The angle dominates, more so the shallower the angle or the longer the distance, since $h = d\tan\theta$ multiplies the angular error by $d\sec^2\theta$.

6 Falling ground: the tape measures a slope distance larger than the horizontal one, so h is over-estimated. A wall leaning away puts the top beyond $d\tan\theta$, so h is under-estimated, and a gutter inflates d. A straw out of the vertical plane reads a smaller elevation, under-estimating h. Eye height ignored: h too small by exactly e. A curved bank means the baseline is not straight and the tree is off the opposite line, so the sine-rule triangle is the wrong one; magnetic instead of true north biases both angles alike, which partly cancels but moves the fix sideways.

7 With the 8 km baseline and bearings $040^\circ, 330^\circ$ the angles are $50^\circ, 60^\circ, 70^\circ$, so $AF = \dfrac{8\sin 60^\circ}{\sin 70^\circ} = 7.37$ km, $BF = 6.52$ km, height 5.65 km. Bearings $042^\circ, 328^\circ$ give $48^\circ, 58^\circ, 74^\circ$, so $AF = 7.06$ km, $BF = 6.18$ km, height 5.25 km: a shift of 0.40 km (about 5%) toward the firebreak, and tower B is still nearer, by 0.88 km rather than 0.85 km, so the same crew is dispatched.

Generalise: from $c = \dfrac{a\sin C}{\sin A}$, a change ΔA gives relative error $\dfrac{\Delta c}{c} \approx |\cot A|\,\Delta A$ (radians), so sensitivity grows as the third angle nears 0° or 180°: $\cot 70^\circ = 0.36$ against $\cot 20^\circ = 2.75$. Choose a baseline that makes the third angle near 90°, keep it long relative to the distance, and confirm with a third station.

Modelling task · The fire crew's decision

1 The baseline bears 090° from A and 270° from B, so triangle ABF has angles $50^\circ, 60^\circ, 70^\circ$: $AF = \dfrac{8\sin 60^\circ}{\sin 70^\circ} = 7.37$ km, $BF = \dfrac{8\sin 50^\circ}{\sin 70^\circ} = 6.52$ km, and the perpendicular distance to the firebreak is $BF\sin 60^\circ = AF\sin 50^\circ = 5.65$ km. Tower B is closer, by 0.85 km.

2 Crew at B: $\dfrac{6.52}{45}$ h $= 8.7$ minutes; helicopter at A: $\dfrac{7.37}{160}$ h $= 2.8$ minutes, so the helicopter is first by about 6 minutes—a margin the errors in item 4 do not overturn. Assumptions: straight-line travel at constant speed, no road detour or river crossing, and no time for scrambling, boarding or loading; a road detour could erase the ground crew's shorter distance.

3 Area $= \frac{1}{2}(8)(5.648) = 22.6$ km^2. A retardant strip 0.8 km wide along BF covers $0.8 \times 6.52 = 5.2$ km^2 as a rectangle, about 23% of the triangle; only the part lying inside the worked area may be claimed and the ends overhang, so the treated area is a little under 5.2 km^2.

4 Repeating with $042^\circ, 328^\circ$ gives $AF = 7.06$ km, $BF = 6.18$ km, height 5.25 km, area 21.0 km^2, treated strip 4.9 km^2; the position moves 0.40 km toward the firebreak and the times fall to 8.2 and 2.7 minutes, so the response choice and treated area are stable. Assumptions: the smoke column marks the fire at the instant of both sightings (the plume drifts downwind) and the two north references agree. Recommendation: launch the helicopter on the calculated bearing, treat the 0.4 km uncertainty as a search circle, and get a third bearing before committing ground crews.

Mixed practice 7

1 $C = 180^\circ - 36^\circ - 81^\circ = 63^\circ$; $b = \dfrac{14.5\sin 81^\circ}{\sin 36^\circ} = 24.4$ cm.

2 $\sin B = \dfrac{9.0\sin 32^\circ}{6.5} = 0.7337$, so $B = 47.2^\circ$ or 132.8°; two triangles, since A is acute, $a < b$ and $32^\circ + 132.8^\circ = 164.8^\circ < 180^\circ$.

3 SAS: $a^2 = 9.5^2 + 7.2^2 - 2(9.5)(7.2)\cos 63^\circ = 79.98$, $a = 8.94$ cm.

4 The largest angle faces 9 cm: $\cos A = \dfrac{6^2 + 7^2 - 9^2}{2(6)(7)} = \dfrac{4}{84} = 0.0476$, $A = 87.3^\circ$.

5 $K = \frac{1}{2}(11.4)(8.6)\sin 47^\circ = 49.02 \times 0.7314 = 35.9$ cm^2.

6 $118^\circ + 180^\circ = 298^\circ$ (the distance stays 14 km); within 0° to 360° adding 180° is the only possibility, so there is no second value.

7 Base diagonal $\sqrt{9^2 + 6^2} = \sqrt{117} = 10.8$ m; diagonal $\sqrt{117 + 4^2} = \sqrt{133} = 11.5$ m; $\tan\theta = \dfrac{4}{\sqrt{117}}$, $\theta = 20.3^\circ$.

8 (a) Sine rule—ASA: $C = 80^\circ$ gives the pair (c, C). (b) Cosine rule—SAS, C lies between a and b. (c) Cosine rule—SSS. (d) Sine rule—SSA, the pair (a, A) exists, then test the ambiguous case: $\sin B = \frac{9\sin 25^\circ}{7} = 0.5432$ gives $B = 32.9^\circ$ or 147.1°, and $25^\circ + 147.1^\circ < 180^\circ$, so both triangles are possible.

9 $b = \dfrac{10\sqrt{2}\sin 60^\circ}{\sin 45^\circ} = 10\sqrt{3}$ cm ≈ 17.3 cm.

10 The angle in $\frac{1}{2}ab\sin C$ must be the one *between* the two sides; 35° faces the 8 cm side, so $\frac{1}{2}(8)(11)\sin 35^\circ = 25.2$ cm^2 is the area of a different triangle. Find the included angle instead: the angle facing 11 cm satisfies $\sin A = \dfrac{11\sin 35^\circ}{8} = 0.7887$, so $A = 52.1^\circ$ or 127.9° (both valid, as $35^\circ + 127.9^\circ < 180^\circ$); the included angle is 92.9° or 17.1°, giving areas 43.9 cm^2 or 12.9 cm^2.

11 $\sin B = \dfrac{7.2\sin 48^\circ}{9.5} = 0.5632$, so $B = 34.3^\circ$ or 145.7°, but only one triangle is possible: $145.7^\circ + 48^\circ = 193.7^\circ > 180^\circ$, and $a > b$ forces $A > B$, so the angle facing the shorter given side is acute. Then $C = 180^\circ - 48^\circ - 34.3^\circ = 97.7^\circ$ and $c = \dfrac{9.5\sin 97.7^\circ}{\sin 48^\circ} = 12.7$ cm.

12 $\sin C = \dfrac{2(45)}{(8.5)(12)} = \dfrac{90}{102} = 0.8824$, so the included angle is 61.9° or 118.1°—supplementary angles with equal sines and equal areas.

13 At the turn the back bearing is $055^\circ + 180^\circ = 235^\circ$, so the interior angle is $235^\circ - 125^\circ = 110^\circ$: $d^2 = 26^2 + 18^2 - 2(26)(18)\cos 110^\circ = 1320.1$, $d = 36.3$ km. Angle at the start: $\sin\theta = \dfrac{18\sin 110^\circ}{36.33} = 0.4655$, $\theta = 27.7^\circ$ clockwise from 055°, so the bearing is $082.7^\circ \approx 083^\circ$.

14 Each face runs 5 m horizontally (half the 10 m width) and rises $6.5 - 3.5 = 3$ m: $\tan\theta = \dfrac{3}{5} = 0.6$, $\theta = 31.0^\circ$, sloping length $\sqrt{5^2 + 3^2} = \sqrt{34} = 5.83$ m. Each face is $15 \times 5.831 = 87.5$ m^2, so the two total 175 m^2 (3 s.f.).

15 Half-diagonal $= 8\sqrt{2} = 11.31$ m, so $l = \sqrt{\sqrt{\quad}}$ (illegible) $10^2 + (8\sqrt{2})^2$ $\sqrt{\quad}$ $228\sqrt{\quad}$ 57 15.1 m and $\tan\alpha = \dfrac{10}{8\sqrt{2}} = 0.8839$, $\alpha = 41.5^\circ$. The dihedral angle uses the apothem 8 m and slant height $\sqrt{10^2 + 8^2} = \sqrt{164} = 12.8$ m: $\tan\beta = \dfrac{10}{8} = 1.25$, $\beta = 51.3^\circ$.

16 Drop $CD \perp AB$ with $CD = h$, $AD = x$, so in right triangle ACD, $h^2 = b^2 - x^2$. If A is acute, D lies on the ray AB with $DB = c - x$, so $a^2 = h^2 + (c - x)^2 = b^2 + c^2 - 2cx = b^2 + c^2 - 2bc\cos A$, since $x = b\cos A$. If A is obtuse, D falls beyond A, $DB = c + x$ with $x = -b\cos A > 0$, so $a^2 = b^2 + c^2 + 2cx = b^2 + c^2 - 2bc\cos A$ again—the same formula, because $\cos A$ is then negative. (Check $A = 120^\circ, b = 2, c = 7$: $a^2 = 67$, $a = 8.19$ cm $< 2 + 7$.)

17 The baseline bears $090^\circ/270^\circ$, so the angles are $090^\circ - 048^\circ = 42^\circ$ at A, $322^\circ - 270^\circ = 52^\circ$ at B, 86° at F. $AF = \dfrac{10\sin 52^\circ}{\sin 86^\circ} = 7.90$ km, $BF = \dfrac{10\sin 42^\circ}{\sin 86^\circ} = 6.71$ km, height above the line $AF\sin 42^\circ = 5.29$ km, so area $= \frac{1}{2}(10)(5.286) = 26.4$ km^2.

18 Re-running the extreme combinations of $\pm 1.5^\circ$ on the two bearings (angle at A: 48.5° to 51.5°; at B: 58.5° to 61.5°) moves the position 0.22 km when both errors run the same way and up to 0.33 km when they oppose. Over that range AF spans 7.13 to 7.64 km and BF 6.27 to 6.80 km, so $BF < AF$ always and the nearer tower does not change. The intersection is sensitive along the baseline because the angle at the fire is only about 70°.

19 The diagonal joining the ends of the two sides closes the angle $180^\circ - \theta$ at the opposite vertex, so $d_1^2 = a^2 + b^2 - 2ab\cos(180^\circ - \theta) = a^2 + b^2 + 2ab\cos\theta$; the other diagonal uses the included angle θ: $d_2^2 = a^2 + b^2 - 2ab\cos\theta$. For acute θ, $\cos\theta > 0$, so $d_1 > d_2$: the diagonal across the obtuse corner is longer, and $d_1^2 + d_2^2 = 2(a^2 + b^2)$. Check $a = 5, b = 3, \theta = 60^\circ$: $2ab\cos\theta = 15$, giving $\sqrt{49} = 7$ cm and $\sqrt{19} = 4.36$ cm, whose squares sum to $68 = 2(25 + 9)$.

20 Take unit edges, so the space diagonal is $\sqrt{3}$. Its projection on any edge through that vertex is the edge itself, so each edge meets the diagonal at $\cos^{-1}\frac{1}{\sqrt{3}} = 54.7^\circ$; its projection on a face is that face's diagonal $\sqrt{2}$, so the angle with the face is $\cos^{-1}\dfrac{\sqrt{2}}{\sqrt{3}} = \sin^{-1}\dfrac{1}{\sqrt{3}} = 35.3^\circ$. (Check: the edge outside that face is its normal, so the two must be complementary—$54.7^\circ + 35.3^\circ = 90^\circ$.)

Criterion A · Solving triangles

1 $C = 180^\circ - 41^\circ - 68^\circ = 71^\circ$, giving the pair (c, C). $a = \dfrac{12.4\sin 41^\circ}{\sin 71^\circ} = 8.60$ cm, $b = \dfrac{12.4\sin 68^\circ}{\sin 71^\circ} = 12.2$ cm. Check: $C > B > A$ matches $c > b > a$.

Marking guidance: 4 marks—C 1, substituted sine rule 1, a and b 1 each. Top band names the pair (c, C) before substituting and checks the side–angle ordering.

2 SAS: $c^2 = 8.5^2 + 6.2^2 - 2(8.5)(6.2)\cos 52^\circ = 45.80$, $c = 6.77$ cm. Then $\sin A = \dfrac{8.5\sin 52^\circ}{6.7675} = 0.9898$, so $A = 81.8^\circ$; the supplement 98.2° is excluded because SAS fixes one triangle ($\cos A = \dfrac{6.2^2 + 6.77^2 - 8.5^2}{2(6.2)(6.77)} = +0.143 > 0$). $B = 180^\circ - 52^\circ - 81.8^\circ = 46.2^\circ$.

Marking guidance: 4 marks—substitution 1, c 1, A and B 1 each. Top band finds c before any angle and rejects the ambiguous sine value with a reason.

3 $K = \frac{1}{2}(9.6)(12.4)\sin 47^\circ = 43.5$ cm^2. Since $\sin\theta = \sin(180^\circ - \theta)$, the obtuse angle $180^\circ - 47^\circ = 133.0^\circ$ between the same two sides gives the same area; it is the only obtuse possibility.

Marking guidance: 3 marks—formula and substitution 1, 43.5 cm^2 with units 1, 133° justified 1. 119 cm^2 (parallelogram value, factor $\frac{1}{2}$ omitted) scores nothing.

4 Base diagonal $\sqrt{10^2 + 6^2} = \sqrt{136} = 11.7$ m (right triangle in the base); space diagonal $\sqrt{136 + 5^2} = \sqrt{161} = 12.7$ m (Pythagoras again on the base diagonal and the height). The projection on the base is the base diagonal, so $\tan\theta = \dfrac{5}{\sqrt{136}} = 0.4289$, $\theta = 23.2^\circ$.

Marking guidance: 3 marks—base diagonal 1, 12.7 m 1, 23.2° 1. $\tan\theta = \frac{5}{12.7}$, giving 21.5°, loses the last mark.

5 At the turn the back bearing is $065^\circ + 180^\circ = 245^\circ$, so the interior angle is $245^\circ - 155^\circ = 90^\circ$. Distance $= \sqrt{18^2 + 12^2} = \sqrt{468} = 21.6$ km. Angle at the start $\tan^{-1}\frac{12}{18} = 33.7^\circ$, clockwise from the first leg, so the bearing is $065^\circ + 33.7^\circ = 098.7^\circ \approx 099^\circ$.

Marking guidance: 4 marks—interior 90° 1, 21.6 km 1, angle at the start 1, bearing 099° 1. Top band justifies the interior angle with the back bearing.

6 The altitude from C to AB is $h = b\sin A$; with A acute and $h < a < b$ the arc of radius a about C cuts the base line twice, so $\sin B = \dfrac{b\sin A}{a}$ admits both B and $180^\circ - B$, each leaving a positive third angle. Sketch: base line with A, $AC = b$ at angle A, circle centre C radius a meeting the base at B_1, B_2. Condition for two triangles: A acute and $b\sin A < a < b$.

Marking guidance: 2 marks—two-intersection reason with a labelled sketch 1, stated condition 1. "The calculator gives one angle" earns at most 1.

Criterion B · The ambiguous triangle

1 With $A = 30^\circ$, $b = 10$ cm, $\sin B = \dfrac{5}{a}$. Entries a, $\sin B$, possible B, number of triangles: 4, 1.25, none, 0. 5, 1.000, 90.0°, 1. 6, 0.833, 56.4°/123.6°, 2. 7, 0.714, 45.6°/134.4°, 2. 8, 0.625, 38.7°/141.3°, 2. 9, 0.556, 33.7°/146.3°, 2. 10, 0.500, 30.0° only, since 150.0° is degenerate, 1. 12, 0.417, 24.6° only, since $155.4^\circ + 30^\circ > 180^\circ$, 1. 15, 0.333, 19.5° only, 1.

Marking guidance: 6 marks—one per row of three consistent entries. Top band tests $30^\circ + B_2 < 180^\circ$ row by row instead of inferring the counts from the pattern.

2 The count runs $0, 1, 2, 2, 2, 2, 1, 1, 1$ as a increases: none while $a < 5$, one at $a = 5$, two for $5 < a < 10$, one for $a \geq 10$. The thresholds are $a = b\sin A = 5$ (where $\sin B = 1$) and $a = b = 10$ (where the obtuse option closes up to $C = 0^\circ$).

Marking guidance: 4 marks—sequence 2, each threshold recognised as $b\sin A$ and b 1.

3 If $a < 5$ then $\sin B = \dfrac{5}{a} > 1$, which no angle satisfies; geometrically the shortest distance from C to the base line is the altitude $h = b\sin A = 5$ cm, so the circle of radius a never reaches it. If $a = 5$ the circle is tangent: $\sin B = 1$, $B = 90^\circ$ is the single solution (its supplement coincides with it), giving one right-angled triangle with $C = 60^\circ$ and $c = \dfrac{5\sin 60^\circ}{\sin 30^\circ} = 8.66$ cm.

Marking guidance: 5 marks—$\sin B > 1$ 2, naming the altitude 1, tangency at $a = 5$ 1, the resulting triangle 1. Top band links algebra to construction.

4 For acute A, fixed b and $h = b\sin A$: zero triangles if $a < h$; one if $a = h$ (right-angled) or $a \geq b$; two if $h < a < b$. (If A is right or obtuse, one triangle exists exactly when $a > b$, since the obtuse sine option cannot be added to an already non-acute A.)

Marking guidance: 4 marks—1 per stated case, capped at 3 if the acuteness of A is not restricted. Justify from $\sin B = \dfrac{b\sin A}{a}$ and the angle sum.

5 At $a = 9.9$: $\sin B = \dfrac{5}{9.9} = 0.5051$, $B = 30.3^\circ$ or 149.7°, and $30^\circ + 149.7^\circ = 179.7^\circ < 180^\circ$, so two triangles—one thin, with $C = 0.3^\circ$, $c = 0.104$ cm. At $a = 10$ the candidates are 30.0° and 150.0°; the second gives $C = 0^\circ$, a collapsed segment with $c = 0$, so one triangle remains. The condition $h < a < b$ is strict and the count drops exactly at $a = b$.

Marking guidance: 3 marks—$a = 9.9$ test 1, degenerate solution at $a = 10$ 1, reason the supplement reaches the angle-sum limit 1.

6 Draw the ray from A at 30° to the base line, mark C on it with $AC = b = 10$ cm, then swing an arc of radius a about C onto the base line. It misses while $a < 5$ (0 triangles), touches once at $a = 5$ (1), cuts twice while $5 < a < 10$ (the ambiguous pair B_1, B_2), and for $a \geq 10$ the second intersection falls behind A, off the ray, leaving 1. The thresholds match because the perpendicular from C to the base line is exactly $b\sin A$.

Marking guidance: 2 marks—arc of radius a about C 1, matching 0/1/2 intersections to $a < 5$, $a = 5$, $5 < a < 10$ 1.

Criterion C · Why two rules?

1 Sample response. The sine rule needs a side paired with the angle opposite it; SAS and SSS give no such pair, so a second relation is required. Place A at the origin, B at $(c, 0)$, C at $(b\cos A, b\sin A)$; then a is the distance from B to C:

$$a^2 = (b\cos A - c)^2 + (b\sin A)^2 = b^2 + c^2 - 2bc\cos A.$$

The term $-2bc\cos A$ is Pythagoras' correction for the angle not being 90°: it vanishes at $A = 90^\circ$, and for $A > 90^\circ$ it is positive, so $a^2 > b^2 + c^2$—the longest side faces the largest angle. Use the sine rule for ASA, AAS, SSA, the cosine rule for SAS, SSS, ratios for a right angle. The sine rule cannot pick between acute and obtuse, since $\sin B = \sin(180^\circ - B)$: with $A = 30^\circ$, $a = 6$ cm, $b = 10$ cm, $\sin B = 0.8333$ gives $B = 56.4^\circ$ or 123.6°. The cosine rule settles it as a quadratic in c: $36 = 100 + c^2 - 20c\cos 30^\circ$, so $c^2 - 17.32c + 64 = 0$ and $c = 12.0$ cm or 5.34 cm—two genuine triangles. Errors to avoid: using an angle not between the two sides, or writing $a^2 = b^2 + c^2 + 2bc\cos A$; the sign is minus.

Marking guidance: 20 marks—coordinate derivation 5, 90° limit 3, comparing required information 4, ambiguous example resolved by the cosine rule 5, identified error 3, with communication through all strands. Top band: consistent 3 s.f., correct notation, both sine-rule angles shown as legitimate.

Criterion D · A coastal survey

1 Diagram: A west, B east with $AB = 4.2$ km, a north arrow at each headland, L north between them, C south-east of A. The baseline bears 090° from A, 270° from B, so $\angle LAB = 090^\circ - 062^\circ = 28^\circ$, $\angle LBA = 318^\circ - 270^\circ = 48^\circ$, $\angle ALB = 180^\circ - 28^\circ - 48^\circ = 104^\circ$.

Marking guidance: 4 marks—diagram with both north arrows and C on the correct side 1, then 1 per angle; bearings subtracted in the wrong order cost that mark.

2 The pair $(AB, \angle ALB)$ is complete: $AL = \dfrac{4.2\sin 48^\circ}{\sin 104^\circ} = 3.22$ km, $BL = \dfrac{4.2\sin 28^\circ}{\sin 104^\circ} = 2.03$ km. Check: 104° faces the longest side $AB = 4.2$ km, and $3.22 + 2.03 > 4.2$.

Marking guidance: 5 marks—matching pair stated 1, AL 2, BL 2, all to 3 s.f. Top band quotes the sine rule as a proportion first.

3 $\angle CAL = 145^\circ - 062^\circ = 83^\circ$ with $AC = 2.8$ km, $AL = 3.217$ km (SAS): $LC^2 = 3.217^2 + 2.8^2 - 2(3.217)(2.8)\cos 83^\circ = 15.99$, $LC = 4.00$ km. In ALC, $\cos\angle ALC = \dfrac{3.217^2 + 4.00^2 - 2.8^2}{2(3.217)(4.00)} = 0.719$, so $\angle ALC = 44.0^\circ$. The bearing of A from L is 242° and C lies anticlockwise from A, so C bears $242^\circ - 44.0^\circ = 198.0^\circ \approx 198^\circ$ (with A as origin, $L = (2.84, 1.51)$, $C = (1.61, -2.29)$).

Marking guidance: 6 marks—$\angle CAL = 83^\circ$ 1, cosine rule with $LC = 4.00$ km 2, angle at L and bearing orientation 2, three-figure bearing 1. Giving 44.0° as the bearing loses the last two.

4 Distance $AL + LC = 3.217 + 3.999 = 7.22$ km; time $= \dfrac{7.216}{25}$ h $= 0.2886$ h $= 17.3$ minutes, so 17 minutes to the nearest minute.

Marking guidance: 4 marks—total distance, $t = \dfrac{d}{s}$, the minutes, rounding as asked. Top band keeps unrounded values until the end.

5 Worst case $063.5^\circ, 316.5^\circ$: angles $26.5^\circ, 46.5^\circ, 107^\circ$ and $AL = \dfrac{4.2\sin 46.5^\circ}{\sin 107^\circ} = 3.19$ km, so L moves from $(2.84, 1.51)$ to $(2.85, 1.42)$— a shift of 0.09 km, about 90 m; opposite-signed errors shift it up to about 0.11 km, so the position is reliable to ±0.12 km. Then LC = 3.92 km and the journey is 7.10 km in 17.0 minutes, so neither distance nor plan changes materially. Assumptions: the headland positions and the east–west alignment of AB are exact, and both bearings are simultaneous true-north readings. Recommendation: add a third bearing from a second point, repeat and average sightings, and check LC directly.

Marking guidance: 5 marks—quantified error from re-solving 2, two assumptions with their effect 1, independent check 1, recommendation tied to the error size 1. Top band evaluates the model, not just the recomputation.

CHAPTER 8 · VECTORS AND MATRICES

Opening problem · the logo that collapsed

1 The corners are $\vec{a} = \begin{pmatrix}0\\0\end{pmatrix}$, $\vec{b} = \begin{pmatrix}3\\0\end{pmatrix}$, $\vec{c} = \begin{pmatrix}4\\2\end{pmatrix}$, $\vec{d} = \begin{pmatrix}1\\2\end{pmatrix}$. Since $L\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}x+0.5y\\y\end{pmatrix}$: $A(0,0) \to A'(0,0)$, $B(3,0) \to B'(3,0)$, $C(4,2) \to C'(5,2)$, $D(1,2) \to D'(2,2)$. On the grid both shapes are parallelograms with base 3 and height 2 (area 6 square units); the lean slides the top edge one unit right and fixes the bottom edge on the x-axis.

2 $LF = \begin{pmatrix}1 & 0.5\\0 & 1\end{pmatrix}\begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}$: row 1 of L times column 1 of F is $1(-1) + 0.5(0) = -1$; completing all four entries, $LF = \begin{pmatrix}-1 & 0.5\\0 & 1\end{pmatrix}$. Similarly $FL = \begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}1 & 0.5\\0 & 1\end{pmatrix} = \begin{pmatrix}-1 & -0.5\\0 & 1\end{pmatrix}$. Then $LF\begin{pmatrix}4\\2\end{pmatrix} = \begin{pmatrix}-4+1\\2\end{pmatrix} = \begin{pmatrix}-3\\2\end{pmatrix}$ and $FL\begin{pmatrix}4\\2\end{pmatrix} = \begin{pmatrix}-4-1\\2\end{pmatrix} = \begin{pmatrix}-5\\2\end{pmatrix}$. In LF the flip F acts first (rightmost is innermost), so $\boldsymbol{LF}$ matches the storyboard; in FL the lean acts first.

3 $\det L = 1\cdot 1 - 0.5\cdot 0 = 1 \neq 0$, so $L^{-1} = \frac{1}{1}\begin{pmatrix}1 & -0.5\\0 & 1\end{pmatrix} = \begin{pmatrix}1 & -0.5\\0 & 1\end{pmatrix}$. Check: $LL^{-1} = \begin{pmatrix}1\cdot 1 + (-0.5)\cdot 0 & 1(-0.5) + 0.5\cdot 1\\0 & 1\end{pmatrix} = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = I$. Recovering the corner: $L^{-1}\begin{pmatrix}5\\2\end{pmatrix} = \begin{pmatrix}5-1\\2\end{pmatrix} = \begin{pmatrix}4\\2\end{pmatrix}$, which is the original C.

4 $E\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}x+y\\x+y\end{pmatrix}$: the two coordinates of the image are equal, so every point lands on the line $y = x$ and the logo flattens to a segment. Since $\det E = 1\cdot 1 - 1\cdot 1 = 0$, E is singular: e.g. $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ both map to $\begin{pmatrix}1\\1\end{pmatrix}$, so no matrix can undo it. L and F have determinants 1 and -1, both nonzero, so they invert and their edits are reversible.

Prerequisite check

1 Adding the equations eliminates y: $3x = 12$, so $x = 4$ and $y = 11 - 2(4) = 3$. Check: $4 - 3 = 1$.

2 $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

3 $\left(\frac{-2+6}{2}, \frac{3+(-5)}{2}\right) = (2, -1)$.

4 $4y = 12 - 3x$, so $y = 3 - \dfrac{3x}{4}$.

5 In the x-axis: $(3, 2)$. In $y = x$ (coordinates swap): $(-2, 3)$.

6 $ad - bc = 3(4) - 1(2) = 12 - 2 = 10$.

Exercise 8A

1 (a) $\begin{pmatrix}4\\3\end{pmatrix}$ (b) $\begin{pmatrix}-2\\-5\end{pmatrix}$ (c) $\begin{pmatrix}6\\0\end{pmatrix}$ (d) $\begin{pmatrix}0\\-7\end{pmatrix}$.

2 (a) $\vec{a} + \vec{b} = \begin{pmatrix}2-3\\5+1\end{pmatrix} = \begin{pmatrix}-1\\6\end{pmatrix}$. (b) $\vec{b} + \vec{c} = \begin{pmatrix}1\\-1\end{pmatrix}$. (c) $\vec{a} - \vec{b} = \begin{pmatrix}2-(-3)\\5-1\end{pmatrix} = \begin{pmatrix}5\\4\end{pmatrix}$. (d) $2\vec{a} + 3\vec{b} = \begin{pmatrix}4\\10\end{pmatrix} + \begin{pmatrix}-9\\3\end{pmatrix} = \begin{pmatrix}-5\\13\end{pmatrix}$.

3 $\sqrt{9+16} = 5$; $\sqrt{25+144} = 13$; $\sqrt{64+36} = 10$; $\sqrt{0+16} = 4$.

4 Divide each vector by its magnitude from Question 3: $\frac{1}{5}\begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}3/5\\4/5\end{pmatrix}$; $\begin{pmatrix}5/13\\-12/13\end{pmatrix}$; $\begin{pmatrix}-4/5\\3/5\end{pmatrix}$.

5 $\begin{pmatrix}7\\-2\end{pmatrix} = 7\vec{i} - 2\vec{j}$; $\begin{pmatrix}-3\\0\end{pmatrix} = -3\vec{i}$; $\begin{pmatrix}0\\5\end{pmatrix} = 5\vec{j}$.

6 Use $\vec{p} = \begin{pmatrix}3\\-4\end{pmatrix}$, $\vec{q} = \begin{pmatrix}-1\\2\end{pmatrix}$. (a) $\vec{p} + \vec{q} = \begin{pmatrix}2\\-2\end{pmatrix} = 2\vec{i} - 2\vec{j}$. (b) $\vec{p} - \vec{q} = \begin{pmatrix}4\\-6\end{pmatrix} = 4\vec{i} - 6\vec{j}$. (c) $2\vec{p} - 3\vec{q} = \begin{pmatrix}6\\-8\end{pmatrix} - \begin{pmatrix}-3\\6\end{pmatrix} = \begin{pmatrix}9\\-14\end{pmatrix} = 9\vec{i} - 14\vec{j}$. (d) $|\vec{p} + \vec{q}| = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \approx 2.83$.

7 Error 1: the components were added instead of applying Pythagoras: $|(6,8)| = \sqrt{36+64} = 10$, not 14. Error 2: the magnitude of a sum is not the sum of magnitudes; here $(6,8) + (-6,-8) = (0,0)$, so $|(6,8) + (-6,-8)| = 0$. Correct values: 10 and 0.

8 Let $A(x_1, y_1)$, $B(x_2, y_2)$. Then $\overrightarrow{AB} = \begin{pmatrix}x_2 - x_1\\y_2 - y_1\end{pmatrix}$ and $\overrightarrow{BA} = \begin{pmatrix}x_1 - x_2\\y_1 - y_2\end{pmatrix} = -\overrightarrow{AB}$, since each component is negated: the same movement with the direction reversed.

9 $|\vec{a}| = \sqrt{16+9} = 5$, so double it: $2\vec{a} = \begin{pmatrix}8\\-6\end{pmatrix}$ (same direction, length 10) and $-2\vec{a} = \begin{pmatrix}-8\\6\end{pmatrix}$ (opposite direction, length 10).

10 $x^2 + 6^2 = 10^2$, so $x^2 = 64$ and $x = 8$ or $x = -8$.

11 Magnitude is $\sqrt{x^2 + y^2}$: the squares remove any signs of the components and a square root is defined to be non-negative, so a magnitude (a length) can never be negative. E.g. $\left|\begin{pmatrix}-3\\-4\end{pmatrix}\right| = 5$.

12 $\vec{u} = \begin{pmatrix}1\\7\end{pmatrix} - \begin{pmatrix}3\\1\end{pmatrix} = \begin{pmatrix}-2\\6\end{pmatrix}$; $|\vec{u}| = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10} \approx 6.32$ m if components are metres.

13 $\begin{pmatrix}5-3+1\\-2-4+6\end{pmatrix} = \begin{pmatrix}3\\0\end{pmatrix}$, so the final displacement is 3 m east; distance from the start is $\left|\begin{pmatrix}3\\0\end{pmatrix}\right| = 3$ m.

14 Resultant $= \begin{pmatrix}8-2\\3+5\end{pmatrix} = \begin{pmatrix}6\\8\end{pmatrix}$ N, magnitude $\sqrt{36+64} = 10$ N. Equilibrium needs the balancing force $\begin{pmatrix}-6\\-8\end{pmatrix}$ N (same 10 N magnitude, opposite direction).

15 Let $\vec{a} = \begin{pmatrix}x\\y\end{pmatrix}$. Then $|k\vec{a}| = \sqrt{(kx)^2 + (ky)^2} = \sqrt{k^2(x^2+y^2)} = \sqrt{k^2}\sqrt{x^2+y^2} = |k|\,|\vec{a}|$, since $\sqrt{k^2} = |k|$ (for $k < 0$ the root is still positive). A unit vector parallel to $\vec{a} \neq \vec{0}$ must be $k\vec{a}$ with $|k|\,|\vec{a}| = 1$, i.e. $k = \pm\frac{1}{|\vec{a}|}$: exactly two choices, $\frac{\vec{a}}{|\vec{a}|}$ and $-\frac{\vec{a}}{|\vec{a}|}$.

Exercise 8B

1 $\overrightarrow{AB} = \begin{pmatrix}-1-2\\3-5\end{pmatrix} = \begin{pmatrix}-3\\-2\end{pmatrix}$; $\overrightarrow{BC} = \begin{pmatrix}5\\-5\end{pmatrix}$; $\overrightarrow{CA} = \begin{pmatrix}-2\\7\end{pmatrix}$; $|\overrightarrow{AB}| = \sqrt{9+4} = \sqrt{13} \approx 3.61$.

2 (a) $\left(\frac{1+7}{2}, \frac{3+9}{2}\right) = (4, 6)$. (b) $\left(\frac{-4+2}{2}, \frac{2-8}{2}\right) = (-1, -3)$.

3 With $m : n = 2 : 1$ the section formula gives $\vec{p} = \frac{1\cdot\vec{a} + 2\vec{b}}{3} = \frac{1}{3}\left[\begin{pmatrix}1\\-2\end{pmatrix} + 2\begin{pmatrix}7\\4\end{pmatrix}\right] = \frac{1}{3}\begin{pmatrix}15\\6\end{pmatrix} = \begin{pmatrix}5\\2\end{pmatrix}$, i.e. $P(5, 2)$.

4 $\overrightarrow{AB} = \begin{pmatrix}3\\1\end{pmatrix}$ and $\overrightarrow{DC} = \vec{c} - \vec{d} = \begin{pmatrix}5-2\\5-4\end{pmatrix} = \begin{pmatrix}3\\1\end{pmatrix}$, so $\overrightarrow{AB} = \overrightarrow{DC}$: one pair of opposite sides is equal and parallel, hence a parallelogram. Moreover $|AB| = \sqrt{10} = |AD|$ (as $\overrightarrow{AD} = \begin{pmatrix}1\\3\end{pmatrix}$), so all four sides are equal: it is in fact a rhombus (not a square, since $\overrightarrow{AB} \cdot \overrightarrow{AD} = 3 + 3 \neq 0$).

5 (a) Parallel: $\begin{pmatrix}2\\4\end{pmatrix} = 2\begin{pmatrix}1\\2\end{pmatrix}$. (b) Parallel: $\begin{pmatrix}-6\\2\end{pmatrix} = -2\begin{pmatrix}3\\-1\end{pmatrix}$ (antiparallel, still parallel by definition).

6 $\overrightarrow{AB} = \begin{pmatrix}4\\3\end{pmatrix}$, distance $= 5$. $\overrightarrow{CD} = \begin{pmatrix}5\\-12\end{pmatrix}$, distance $= 13$.

7 $\overrightarrow{AB} = 2\overrightarrow{CD}$ says only that lines AB and CD are parallel and that AB is twice as long. A parallelogram needs $\overrightarrow{AB} = \overrightarrow{DC}$, equal vectors; a factor 2 (and the fact the arrows may lie on different lines with unrelated ends) breaks this. What the equation does tell you: $AB \parallel CD$ and $AB = 2\,CD$.

8 $\overrightarrow{AB} = \begin{pmatrix}3\\4\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}9\\12\end{pmatrix} = 3\overrightarrow{AB}$: parallel and sharing A, so the points are collinear. $\overrightarrow{BC} = \begin{pmatrix}6\\8\end{pmatrix} = 2\overrightarrow{AB}$, so $AB : BC = 1 : 2$.

9 Parallel requires equal component ratios: $\frac{k}{4} = \frac{3}{k-1}$, i.e. $k(k-1) = 12$, so $k^2 - k - 12 = 0$, $(k-4)(k+3) = 0$: $k = 4$ or $k = -3$. Check: $k = 4$ gives identical vectors $\begin{pmatrix}4\\3\end{pmatrix}$; $k = -3$ gives $\begin{pmatrix}-3\\3\end{pmatrix}$ and $\begin{pmatrix}4\\-4\end{pmatrix} = -\frac{4}{3}\begin{pmatrix}-3\\3\end{pmatrix}$.

10 In $OACB$ the diagonal rule gives $\overrightarrow{OC} = \vec{a} + \vec{b}$. Then M, the midpoint of AC, has position vector $\vec{a} + \frac{1}{2}\vec{b}$, and N, the midpoint of BC, has $\vec{b} + \frac{1}{2}\vec{a}$. Hence $\overrightarrow{MN} = \left(\vec{b} + \frac{1}{2}\vec{a}\right) - \left(\vec{a} + \frac{1}{2}\vec{b}\right) = \frac{1}{2}(\vec{b} - \vec{a}) = \frac{1}{2}\overrightarrow{AB}$. So $\overrightarrow{MN}$ is parallel to $\overrightarrow{AB}$ and half its length.

11 If the diagonals bisect each other, their midpoints coincide: $\frac{1}{2}(\vec{a} + \vec{c}) = \frac{1}{2}(\vec{b} + \vec{d})$, so $\vec{a} + \vec{c} = \vec{b} + \vec{d}$. Rearranging, $\vec{b} - \vec{a} = \vec{c} - \vec{d}$, i.e. $\overrightarrow{AB} = \overrightarrow{DC}$: one pair of opposite sides is equal and parallel, so $ABCD$ is a parallelogram.

12 $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} 9 \\ 6 \end{pmatrix} = 3\overrightarrow{AB}$: parallel with a shared point A, so yes, C lies on line AB. $\overrightarrow{BC} = \begin{pmatrix} 6 \\ 4 \end{pmatrix} = 2\overrightarrow{AB}$, so $AB : BC = 1 : 2$.

13 $\overrightarrow{AB} = \begin{pmatrix} 60 \\ 0 \end{pmatrix} = \overrightarrow{DC}$ (as $\vec{c} - \vec{d} = \begin{pmatrix} 80 - 20 \\ 40 - 40 \end{pmatrix} = \begin{pmatrix} 60 \\ 0 \end{pmatrix}$), so the field is a parallelogram. $\overrightarrow{AC} = \begin{pmatrix} 80 \\ 40 \end{pmatrix}$, so $AC = \sqrt{6400 + 1600} = \sqrt{8000} = 40\sqrt{5} \approx 89.4$ m.

14 Resultant $= \begin{pmatrix} 15 \\ 14 \end{pmatrix}$ kN, magnitude $\sqrt{225 + 196} = \sqrt{421} \approx 20.5$ kN. The third tug must supply $\begin{pmatrix} -15 \\ -14 \end{pmatrix}$ kN (a force of 20.5 kN in the opposite direction) for equilibrium.

15 $\overrightarrow{AB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, $\overrightarrow{CD} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$, $\overrightarrow{DA} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$, each of length $\sqrt{17}$, so all four sides are equal. Midpoint of $AC = \left(\frac{3}{2}, \frac{5}{2}\right)$ = midpoint of $BD = \left(\frac{4-1}{2}, \frac{1+4}{2}\right)$: the diagonals bisect each other and meet at $(1.5, 2.5)$. Equal sides alone do not fix the figure: the lengths say nothing about the order in which the vertices are joined, so a crossed (non-simple) equilateral quadrilateral is possible; proving the diagonals bisect shows opposite sides are parallel, making it a parallelogram with equal sides, i.e. a rhombus.

Exercise 8C

1 (a) $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$. (b) $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix}$. (c) $\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \\ -3 \end{pmatrix}$.

2 $\sqrt{4 + 9 + 36} = \sqrt{49} = 7$; $\sqrt{1 + 4 + 4} = 3$; $\sqrt{16 + 16 + 49} = \sqrt{81} = 9$; $\sqrt{0 + 9 + 16} = 5$.

3 (a) $\left(\frac{7}{2}, 4, 4\right) = (3.5, 4, 4)$. (b) $\left(\frac{1}{2}, -\frac{3}{2}, 2\right) = (0.5, -1.5, 2)$. (c) $(1.5, 1.5, 1.5)$.

4 Divide by the magnitudes of Question 2: $\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} = \begin{pmatrix} 2/7 \\ 3/7 \\ 6/7 \end{pmatrix}$ and $\frac{1}{3}\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/3 \\ -2/3 \\ 2/3 \end{pmatrix}$.

5 (a) $\vec{a} + \vec{b} = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$. (b) $\vec{a} - \vec{b} = \begin{pmatrix} 3 \\ -5 \\ 3 \end{pmatrix}$. (c) $3\vec{a} - 2\vec{b} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix} - \begin{pmatrix} -2 \\ 8 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ -11 \\ 9 \end{pmatrix}$. (d) $|\vec{a} + \vec{b}| = \sqrt{1 + 9 + 9} = \sqrt{19} \approx 4.36$.

6 $(3, 0, 2)$: $y = 0$, the xz-plane. $(0, 4, 0)$: the y-axis. $(5, 0, 0)$: the x-axis. $(1, 1, 0)$: $z = 0$, the xy-plane.

7 The student added the components, which measures a staircase path along the edges, not the straight-line length. Correctly, $\left|\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}\right| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.

8 $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} = 2.5\overrightarrow{AB}$: parallel and sharing A, so the points are collinear. $\overrightarrow{BC} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = 1.5\overrightarrow{AB}$, so $AB : BC = 2 : 3$.

9 $\overrightarrow{PQ} = \begin{pmatrix} 3 \\ 3 \\ z - 4 \end{pmatrix}$, so $PQ^2 = 9 + 9 + (z - 4)^2 = 27$, giving $(z - 4)^2 = 9$, $z - 4 = \pm 3$: $z = 7$ or $z = 1$.

10 Diagonal $= \sqrt{2^2 + 3^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$: the base diagonal is $\sqrt{4 + 9} = \sqrt{13}$ and Pythagoras is applied again with the height 6. It is not $2 + 3 + 6 = 11$ because that sums three perpendicular edges, a path along the surface; the diagonal is the straight line inside the cuboid.

11 $\overrightarrow{VA} = \vec{a} - \vec{v} = \begin{pmatrix} -2 \\ -2 \\ -6 \end{pmatrix}$, so $VA = \sqrt{4 + 4 + 36} = \sqrt{44} = 2\sqrt{11} \approx 6.63$ m. The centre of the base is $M(2, 2, 0)$, so the height $VM = 6$ m (a vertical segment). Midpoint of $AC = \left(\frac{0+4}{2}, \frac{0+4}{2}, 0\right) = (2, 2, 0) = M$, as required.

12 The midpoint only averages the coordinates (equivalently $\vec{m} = \frac{1}{2}(\vec{a} + \vec{b})$ componentwise), and each axis is treated independently, so the third coordinate simply adds a third average: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$.

13 Position $= \begin{pmatrix} 10 \\ 20 \\ 30 \end{pmatrix} + 5\begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 35 \\ 25 \end{pmatrix}$ m. Distance from the origin $= \sqrt{0 + 1225 + 625} = \sqrt{1850} = 5\sqrt{74} \approx 43.0$ m.

14 The top of the mast is $(3, 4, 12)$. Wire length $= \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$ m.

15 $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$, $AB = 3$; $\overrightarrow{BC} = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix}$, $BC = \sqrt{32} = 4\sqrt{2}$; $\overrightarrow{AC} = \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix}$, $AC = \sqrt{41}$. Since $AB^2 + BC^2 = 9 + 32 = 41 = AC^2$, the converse of Pythagoras gives a right angle at B (the vertex where sides AB and BC meet, opposite the longest side AC).

Exercise 8D

1 (a) Order 2×3; $a_{13} = 3$ (row 1, column 3). (b) Order 2×1; $a_{21} = -2$.

2 $A + B = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}$; $A - B = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}$; $3A = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$; $2A - B = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 4 & 9 \end{pmatrix}$.

3 Row 1 of A times column 1 of B: $1(0) + 2(2) = 4$; the other entries likewise give $AB = \begin{pmatrix} 4 & -1 \\ 8 & -1 \end{pmatrix}$. For BA: $0(1) + 1(3) = 3$ for the first entry, giving $BA = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$. The products are not equal, so matrix multiplication is not commutative.

4 (a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. (b) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. (c) $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = 4I$.

5 (a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$. (b) $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$. (c) $\begin{pmatrix} 8 \\ -20 \end{pmatrix}$.

6 Rotation: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 3 - 1 \cdot 1 \\ 1 \cdot 3 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Shear factor 2 parallel to the x-axis: $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 + 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$.

7 The student multiplied matching entries (as for addition), which is not matrix multiplication; entries of a product come from rows times columns. Correctly: row 1×column 1 is $1(5) + 2(7) = 19$, and continuing, $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$.

8 For AB the inner numbers must match: a 2×3 against another 2×3 gives $3 \neq 2$, so neither AB nor BA is defined. Against a 3×1 matrix the inner numbers match ($3 = 3$), so the product exists and has order 2×1 (outer numbers).

9 Its columns are the images of the basis vectors: $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix} \to \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, a rotation of $90°$ clockwise about O (equivalently $270°$ anticlockwise). It is the inverse of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (their product is I) and its transpose; the two matrices are the same rotation in opposite senses.

10 The columns are the images of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$: $M = \begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}$. Then $M\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 - 1 \\ 2 + 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$.

11 Reflection in the x-axis first, then the y-axis, is the product $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$; entry by entry this is $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, exactly the matrix of a $180°$ rotation about O. Both also send $(x, y) \to (-x, -y)$.

12 $M\begin{pmatrix}0\\0\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$ for every matrix, so the origin maps to itself. A matrix transformation is linear and cannot translate, so any other point with position vector $\vec{v}$ generally moves to a different point $M\vec{v}$; a whole shape away from the origin has all its vertices moved, changing its position (and usually its orientation) while the origin stays put.

13 $\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+y\\y\end{pmatrix}$, giving $(0,0)\to(0,0)$, $(2,0)\to(2,0)$, $(3,1)\to(4,1)$, $(1,1)\to(2,1)$. The bottom edge on the x-axis is fixed, the top edge slides 1 unit right: the parallelogram is slanted further but keeps base 2, height 1 and area 2, as $\det=1$.

14 Total displacement $=\begin{pmatrix}12-4\\-5+9\end{pmatrix}=\begin{pmatrix}8\\4\end{pmatrix}$ cm; length $=\sqrt{64+16}=\sqrt{80}=4\sqrt{5}\approx 8.94$ cm.

15 $RS=\begin{pmatrix}0&-1\\1&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}$: first entry $0(1)+(-1)(0)=0$; the product is $RS=\begin{pmatrix}0&1\\1&0\end{pmatrix}$, the reflection in $y=x$. $SR=\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&-1\\1&0\end{pmatrix}=\begin{pmatrix}0&-1\\-1&0\end{pmatrix}$, the reflection in $y=-x$. Since $\begin{pmatrix}0&1\\1&0\end{pmatrix}\neq\begin{pmatrix}0&-1\\-1&0\end{pmatrix}$, $RS\neq SR$.

Exercise 8E

1 (a) $3(4)-2(1)=10$. (b) $5(1)-(-2)(3)=5+6=11$. (c) $2(3)-6(1)=0$. (d) $(-1)(-5)-3(2)=5-6=-1$.

2 (a) $A^{-1}=\frac{1}{10}\begin{pmatrix}4&-2\\-1&3\end{pmatrix}$. (b) $B^{-1}=\frac{1}{11}\begin{pmatrix}1&2\\-3&5\end{pmatrix}$. (c) $\det=0$: singular, no inverse. (d) $\det=-1$: $D^{-1}=\frac{1}{-1}\begin{pmatrix}-5&-3\\-2&-1\end{pmatrix}=\begin{pmatrix}5&3\\2&1\end{pmatrix}$. Check (d): $\begin{pmatrix}-1&3\\2&-5\end{pmatrix}\begin{pmatrix}5&3\\2&1\end{pmatrix}=\begin{pmatrix}-5+6&-3+3\\10-10&6-5\end{pmatrix}=I$.

3 $\det A=4-3=1$, so $A^{-1}=\begin{pmatrix}1&-1\\-3&4\end{pmatrix}$. Then $A^{-1}A=\begin{pmatrix}1&-1\\-3&4\end{pmatrix}\begin{pmatrix}4&1\\3&1\end{pmatrix}$: row 1×column 1 is $1(4)+(-1)(3)=1$; all four entries give $\begin{pmatrix}1&0\\0&1\end{pmatrix}=I$, as required.

4 (a) $A=\begin{pmatrix}4&1\\3&1\end{pmatrix}$, $\det A=1$, $A^{-1}=\begin{pmatrix}1&-1\\-3&4\end{pmatrix}$, so $\begin{pmatrix}x\\y\end{pmatrix}=A^{-1}\begin{pmatrix}9\\7\end{pmatrix}=\begin{pmatrix}9-7\\-27+28\end{pmatrix}=\begin{pmatrix}2\\1\end{pmatrix}$: $x=2$, $y=1$ (checks: $8+1=9$, $6+1=7$). (b) $A=\begin{pmatrix}3&2\\1&4\end{pmatrix}$, $\det A=10$, $A^{-1}=\frac{1}{10}\begin{pmatrix}4&-2\\-1&3\end{pmatrix}$, so $\begin{pmatrix}x\\y\end{pmatrix}=\frac{1}{10}\begin{pmatrix}32+4\\-8-6\end{pmatrix}=\begin{pmatrix}18/5\\-7/5\end{pmatrix}$: $x=3.6$, $y=-1.4$. Checks: $3(3.6)+2(-1.4)=8$ and $3.6+4(-1.4)=-2$.

5 Singular means $\det=k(k+1)-4(3)=k^2+k-12=(k-3)(k+4)=0$, so $k=3$ or $k=-4$. (Check $k=3$: $\begin{pmatrix}3&4\\3&4\end{pmatrix}$, identical rows, $\det=0$.)

6 (a) $\det=2(3)-1(1)=5\neq 0$: invertible. (b) $\det=3(2)-6(1)=0$: singular, no inverse. (c) $\det=1(4)-2(2)=0$: singular, no inverse.

7 Error 1: the entries of the adjugate were not formed $\frac{1}{7}\begin{pmatrix}2&3\\1&5\end{pmatrix}$ keeps A itself; the diagonal entries must be interchanged. Error 2: the off-diagonal signs were not changed. The determinant 7 is correct. The inverse is $A^{-1}=\frac{1}{7}\begin{pmatrix}5&-3\\-1&2\end{pmatrix}$. Check: $\frac{1}{7}\begin{pmatrix}5&-3\\-1&2\end{pmatrix}\begin{pmatrix}2&3\\1&5\end{pmatrix}=\frac{1}{7}\begin{pmatrix}7&0\\0&7\end{pmatrix}=I$.

8 A zero determinant collapses the plane onto a line (or a point): two different input vectors, e.g. the multiples of the parallel columns' difference, give the same output. The transformation therefore loses information, and no matrix can reconstruct which pre-image was used, so no inverse exists; algebraically, A^{-1} would require division by $\det A=0$.

9 Areas are multiplied by $|-3|=3$ (tripled), and the negative sign means orientation is reversed: a clockwise vertex order in the image corresponds to anticlockwise in the original (a reflection is involved).

10 $\det A=2-1=1$; $\det B=3-1=2$; $AB=\begin{pmatrix}2(3)+1(1)&2(1)+1(1)\\1(3)+1(1)&1(1)+1(1)\end{pmatrix}=\begin{pmatrix}7&3\\4&2\end{pmatrix}$; $\det(AB)=14-12=2$. Comment: $\det(AB)=\det A\cdot\det B$ ($2=1\times 2$): area scale

11 The coefficient matrix is $\begin{pmatrix}2&3\\4&6\end{pmatrix}$ with $\det=12-12=0$, so there is no unique solution. The second equation is exactly twice the first, so both lines coincide: infinitely many solutions, the whole line $2x+3y=5$, i.e. $\left(t,\frac{5-2t}{3}\right)$ for any real t.

12 AB applies B first, then A; to undo it you must undo A first, then B, i.e. $B^{-1}A^{-1}$. Check with Question 10: $(AB)^{-1}=\frac{1}{2}\begin{pmatrix}2&-3\\-4&7\end{pmatrix}$ (as $\det AB=2$); $A^{-1}=\begin{pmatrix}1&-1\\-1&2\end{pmatrix}$, $B^{-1}=\frac{1}{2}\begin{pmatrix}1&-1\\-1&3\end{pmatrix}$, and $B^{-1}A^{-1}=\frac{1}{2}\begin{pmatrix}1&-1\\-1&3\end{pmatrix}\begin{pmatrix}1&-1\\-1&2\end{pmatrix}=\frac{1}{2}\begin{pmatrix}2&-3\\-4&7\end{pmatrix}$. The two agree.

13 $\det M=2-1=1$, so $M^{-1}=\begin{pmatrix}1&-1\\-1&2\end{pmatrix}$. Decoding: $M^{-1}\begin{pmatrix}7\\4\end{pmatrix}=\begin{pmatrix}7-4\\-7+8\end{pmatrix}=\begin{pmatrix}3\\1\end{pmatrix}$, i.e. the original pair was $(3,1)$. Check: $M\begin{pmatrix}3\\1\end{pmatrix}=\begin{pmatrix}6+1\\3+1\end{pmatrix}=\begin{pmatrix}7\\4\end{pmatrix}$.

Common error: applying M instead of M^{-1}, which encodes the received pair again rather than undoing the encoding.

14 The area scale factor is $60\div 24=2.5$, so $|\det|=|k\times 2.5-0|=2.5|k|=2.5$, giving $k=\pm 1$; taking $k=1$ (the enlargement suggested by the context) gives $\det=2.5$. Check: $\begin{pmatrix}1&0\\0&2.5\end{pmatrix}$ stretches vertically by 2.5, multiplying all areas by 2.5.

15 From $AA^{-1}=I$ and the product rule, $\det A\cdot\det(A^{-1})=\det I=1$, so when $\det A\neq 0$, $\det(A^{-1})=\dfrac{1}{\det A}$. For $\begin{pmatrix}3&1\\2&4\end{pmatrix}$, $\det=10$, so the inverse has determinant $\dfrac{1}{10}=0.1$: the undoing transformation shrinks area by 10 because the original enlarged it by 10.

Investigation 8

1 Row 2: image of (x,y) is $(2x+y,\ x+2y)$; vertices $(0,0),(2,1),(3,3),(1,2)$ form a parallelogram of area 3; area factor 3, $|\det A|=3$. Row 3: $(0,0),(1,1),(0,2),(-1,1)$ is a square of side $\sqrt{2}$, area 2; factor 2, $|\det A|=2$. Row 4: $(0,0),(3,0),(3,-1),(0,-1)$ is a 3×1 rectangle, area 3; factor 3, $|\det A|=|-3|=3$. Row 5: $(0,0),(1,2),(3,6),(2,4)$ all lie on the line $y=2x$: the square collapses to a segment, area 0; factor 0, $|\det A|=0$.

2 For every row, image area ÷ original area $=|\det A|$: the area scale factor of a matrix is the absolute value of its determinant.

3 Conjecture: any 2×2 matrix multiplies every area by $|\det A|$. A negative determinant does not give negative area: it scales area by $|\det A|$ and reverses orientation instead.

4 The triangle fits in a 3×2 rectangle of area 6; the three right-angled triangles cut off have legs $(3,1)$, $(2,1)$, $(1,2)$ and areas $1.5,1,1$, so the triangle's area is $6-1.5-1-1=2.5$. Under $A=\begin{pmatrix}2&1\\1&2\end{pmatrix}$: $(0,0)\to(0,0)$, $(3,1)\to(7,5)$, $(1,2)\to(4,5)$. Shoelace: $\frac{1}{2}|0\cdot 5+7\cdot 5+4\cdot 0-(0\cdot 7+5\cdot 4+5\cdot 0)|=\frac{1}{2}|35-20|=7.5=2.5\times 3$, confirming the factor $|\det A|=3$.

5 Under $C=\begin{pmatrix}1&-1\\1&1\end{pmatrix}$: $(3,1)\to(2,4)$ and $(1,2)\to(-1,3)$. Shoelace on $(0,0),(2,4),(-1,3)$: $\frac{1}{2}|0+6+0-(0-4+0)|=\frac{1}{2}|10|=5=2.5\times 2$. The ratio is $2=\det C$ itself (positive, so $|\det C|=\det C$): orientation is preserved, as the triangle keeps its anticlockwise order.

6 For $\begin{pmatrix}1&2\\2&4\end{pmatrix}$ the columns are $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}=2\begin{pmatrix}1\\2\end{pmatrix}$, so every image $x\begin{pmatrix}1\\2\end{pmatrix}+y\begin{pmatrix}2\\4\end{pmatrix}=(x+2y)\begin{pmatrix}1\\2\end{pmatrix}$ is a multiple of one vector: all image points lie on the line $y=2x$. The plane collapses onto a line, area becomes 0 and information is destroyed — exactly the singular matrices of Section 8.5, which have no inverse.

Modelling task

1 $L\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+0.5y\\y\end{pmatrix}$ gives $A'(0,0)$, $B'(3,0)$, $C'(5,2)$, $D'(2,2)$. Both logo and image are parallelograms with base 3 and height 2, so each has area 6 square units. This agrees with $\det L=1$: the area scale factor is $|\det L|=1$, so a shear preserves area.

2 Flip first, lean second: the composite is $LF = \begin{pmatrix}1 & 0.5\\0 & 1\end{pmatrix}\begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix} = \begin{pmatrix}-1 & 0.5\\0 & 1\end{pmatrix}$. Then $LF\begin{pmatrix}4\\2\end{pmatrix} = \begin{pmatrix}-4+1\\2\end{pmatrix} = \begin{pmatrix}-3\\2\end{pmatrix}$: the final position of C is $(-3, 2)$.

3 $\det L = 1$, so $L^{-1} = \begin{pmatrix}1 & -0.5\\0 & 1\end{pmatrix}$ (check: LL^{-1} gives row 1 × column 2: $1(-0.5) + 0.5(1) = 0$, and the other entries give I). Applying it: $L^{-1}\begin{pmatrix}5\\2\end{pmatrix} = \begin{pmatrix}5-1\\2\end{pmatrix} = \begin{pmatrix}4\\2\end{pmatrix}$, which is exactly the original corner $C(4, 2)$.

4 $\det E = 1\cdot 1 - 1\cdot 1 = 0$, so E is singular and no inverse formula can be applied (it would divide by zero).

Geometrically $E\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}x+y\\x+y\end{pmatrix}$ lands on the line $y = x$: e.g. $(1, 0)$ and $(0, 1)$ share the image $(1, 1)$, so the original point cannot be recovered. Assumption: screen pixels form a square Cartesian grid with the logo corner A as origin. On a non-square pixel grid the true distances and areas are stretched differently in x and y, so the matrix no longer matches the on-screen geometry — e.g. the shear would not visibly preserve area and "collapse onto $y = x$" would appear as a slanted segment.

Mixed practice 8

1 $\vec{a}+\vec{b} = \begin{pmatrix}-1\\-1\end{pmatrix}$; $\vec{a}-\vec{b} = \begin{pmatrix}7\\-3\end{pmatrix}$; $2\vec{a}+3\vec{b} = \begin{pmatrix}6\\-4\end{pmatrix} + \begin{pmatrix}-12\\3\end{pmatrix} = \begin{pmatrix}-6\\-1\end{pmatrix}$; $|\vec{a}| = \sqrt{9+4} = \sqrt{13}$.

2 Length $\sqrt{9+16} = 5$, so the unit vectors are $\pm\frac{1}{5}\begin{pmatrix}-3\\4\end{pmatrix} = \begin{pmatrix}-3/5\\4/5\end{pmatrix}$ or $\begin{pmatrix}3/5\\-4/5\end{pmatrix}$.

3 $\overrightarrow{AB} = \begin{pmatrix}4\\3\end{pmatrix}$, $|\overrightarrow{AB}| = 5$, midpoint $\left(\frac{1+5}{2}, \frac{3+6}{2}\right) = (3, 4.5)$.

4 $\vec{p} = \frac{2\vec{a}+3\vec{b}}{5} = \frac{1}{5}\left[\begin{pmatrix}-2\\8\end{pmatrix} + \begin{pmatrix}27\\-3\end{pmatrix}\right] = \frac{1}{5}\begin{pmatrix}25\\5\end{pmatrix} = \begin{pmatrix}5\\1\end{pmatrix}$, so $P(5, 1)$. Check: $\overrightarrow{AP} = \begin{pmatrix}6\\-3\end{pmatrix} = 1.5\overrightarrow{PB}$, so $AP : PB = 3 : 2$ as required.

5 $\overrightarrow{AB} = \begin{pmatrix}4\\2\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}10\\5\end{pmatrix} = 2.5\overrightarrow{AB}$: parallel and sharing A, so collinear. $\overrightarrow{BC} = \begin{pmatrix}6\\3\end{pmatrix} = 1.5\overrightarrow{AB}$, so $AB : BC = 4 : 6 = 2 : 3$.

6 $\overrightarrow{AB} = \begin{pmatrix}3\\4\\12\end{pmatrix}$, $|\overrightarrow{AB}| = \sqrt{9+16+144} = \sqrt{169} = 13$.

7 $\left(\frac{1+7}{2}, \frac{-2+6}{2}, \frac{4-2}{2}\right) = (4, 2, 1)$.

8 Order 2×3 (2 rows, 3 columns); $a_{21} = 0$ (row 2, column 1).

9 Row 1 of A times column 1 of B: $1(0) + 2(2) = 4$, and similarly $AB = \begin{pmatrix}4 & -1\\8 & -1\end{pmatrix}$. For the other order, $BA = \begin{pmatrix}3 & 2\\-1 & 0\end{pmatrix}$. They are not equal.

10 Reflection in $y = x$: $\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}2\\-5\end{pmatrix} = \begin{pmatrix}-5\\2\end{pmatrix}$. Rotation 90° anticlockwise: $\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}2\\-5\end{pmatrix} = \begin{pmatrix}5\\2\end{pmatrix}$. Enlargement factor 3: $3I\begin{pmatrix}2\\-5\end{pmatrix} = \begin{pmatrix}6\\-15\end{pmatrix}$.

11 $\det = 4(5) - 3(2) = 20 - 6 = 14 \neq 0$, so $A^{-1} = \frac{1}{14}\begin{pmatrix}5 & -3\\-2 & 4\end{pmatrix}$. Check: $\frac{1}{14}\begin{pmatrix}5 & -3\\-2 & 4\end{pmatrix}\begin{pmatrix}4 & 3\\2 & 5\end{pmatrix} = \frac{1}{14}\begin{pmatrix}20-6 & 15-15\\-8+8 & -6+20\end{pmatrix} = I$.

12 $\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{14}\begin{pmatrix}5 & -3\\-2 & 4\end{pmatrix}\begin{pmatrix}18\\16\end{pmatrix} = \frac{1}{14}\begin{pmatrix}90-48\\-36+64\end{pmatrix} = \frac{1}{14}\begin{pmatrix}42\\28\end{pmatrix} = \begin{pmatrix}3\\2\end{pmatrix}$: $x = 3$, $y = 2$. Checks: $12 + 6 = 18$, $6 + 10 = 16$.

13 $\det = k(k+1) - 12 = k^2 + k - 12 = (k-3)(k+4) = 0$, so $k = 3$ or $k = -4$.

14 $PQ = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix} = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$: first entry $0(1) + (-1)(0) = 0$; this is the reflection in $y = x$. $QP = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix} = \begin{pmatrix}0 & -1\\-1 & 0\end{pmatrix}$: the reflection in $y = -x$. The two composites differ, so the order of multiplication matters.

15 Let the vertices have position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ with $\overrightarrow{AB} = \overrightarrow{DC}$, i.e. $\vec{b} - \vec{a} = \vec{c} - \vec{d}$. Then $\overrightarrow{AX} = \frac{1}{2}(\vec{b} - \vec{a})$ and $\overrightarrow{DY} = \frac{1}{2}(\vec{c} - \vec{d}) = \frac{1}{2}(\vec{b} - \vec{a})$. So $\overrightarrow{AX} = \overrightarrow{DY}$: one pair of opposite sides of $AXYD$ is equal and parallel, hence $AXYD$ is a parallelogram.

16 Sides: $\overrightarrow{AB} = \begin{pmatrix}5\\0\end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix}3\\4\end{pmatrix}$, $\overrightarrow{CD} = \begin{pmatrix}-5\\0\end{pmatrix}$, $\overrightarrow{DA} = \begin{pmatrix}-3\\-4\end{pmatrix}$, each of length 5; opposite sides are parallel ($\overrightarrow{AB} = -\overrightarrow{CD}$ etc.), so it is a parallelogram with equal sides: a rhombus. Diagonals: $\begin{pmatrix}8\\4\end{pmatrix}$ has length $\sqrt{80} = 4\sqrt{5} \approx 8.94$; $\overrightarrow{(5,0)\to(3,4)} = \begin{pmatrix}-2\\4\end{pmatrix}$ has length $\sqrt{20} = 2\sqrt{5} \approx 4.47$.

17 The columns are the images of the basis vectors: $M = \begin{pmatrix}3 & -1\\1 & 2\end{pmatrix}$. Then $M\begin{pmatrix}2\\3\end{pmatrix} = \begin{pmatrix}6-3\\2+6\end{pmatrix} = \begin{pmatrix}3\\8\end{pmatrix}$.

18 $\det\begin{pmatrix}2 & 0\\0 & 3\end{pmatrix} = 6$, so the image area is $8 \times 6 = 48$ cm^2: the matrix stretches x by 2 and y by 3, and every area is multiplied by the product $2 \times 3 = 6$, the determinant.

19 $AB = \begin{pmatrix}ae+bg & af+bh\\ce+dg & cf+dh\end{pmatrix}$, so $\det(AB) = (ae+bg)(cf+dh) - (af+bh)(ce+dg)$. Expanding: $aecf + aedh + bgcf + bghd - afce - afdg - bhce - bhdg$; the terms $aecf = afce$ and $bghd = bhdg$ cancel, leaving $aedh + bgcf - afdg - bhce$. On the other side, $\det A \cdot \det B = (ad-bc)(eh-fg) = adeh - adfg - bceh + bcfg$, the same four products. Hence $\det(AB) = \det A \cdot \det B$.

20 $\det = 2 - 1 = 1$, so $A^{-1} = \frac{1}{1}\begin{pmatrix}1 & -1\\-1 & 2\end{pmatrix} = \begin{pmatrix}1 & -1\\-1 & 2\end{pmatrix}$ (check: $\begin{pmatrix}2 & 1\\1 & 1\end{pmatrix}\begin{pmatrix}1 & -1\\-1 & 2\end{pmatrix} = \begin{pmatrix}2-1 & -2+2\\1-1 & -1+2\end{pmatrix} = I$). The adjugate only swaps and sign-changes integers, so it stays integral; dividing by $\det = 1$ cannot introduce fractions. Another example: $\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$ has integer entries, $\det = 1$, and inverse $\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix}$.

Criterion A · Vectors, matrix products and inverses

1 $2\vec{a} - \vec{b} = \begin{pmatrix}8\\-6\end{pmatrix} - \begin{pmatrix}-2\\5\end{pmatrix} = \begin{pmatrix}10\\-11\end{pmatrix}$. $|\vec{a}| = \sqrt{16+9} = 5$. Unit vector $= \frac{1}{5}\begin{pmatrix}4\\-3\end{pmatrix} = \begin{pmatrix}4/5\\-3/5\end{pmatrix}$; check: $(0.8)^2 + (-0.6)^2 = 1$.

Marking guidance: [4 marks] ≈ 1 for the linear combination, 1 for $|\vec{a}| = 5$, 2 for the unit vector (correct divisor and both components). A 7–8 response also checks the unit vector has length 1 and keeps signs exact.

2 $\overrightarrow{AB} = \begin{pmatrix}3\\4\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}9\\12\end{pmatrix} = 3\overrightarrow{AB}$: the vectors are parallel and share the point A, so A, B, C are collinear. $\overrightarrow{BC} = \begin{pmatrix}6\\8\end{pmatrix} = 2\overrightarrow{AB}$, so $AB : BC = 1 : 2$.

Marking guidance: [4 marks] ≈ 1 each for $\overrightarrow{AB}$ and $\overrightarrow{AC}$, 1 for stating the scalar multiple with the shared-point reason (parallel vectors alone are not enough), 1 for the ratio 1 : 2. Top band: full justification of why a common point is needed.

3 Row 1 of A times column 1 of B: $2(0) + (-1)(1) = -1$; the other entries likewise: $AB = \begin{pmatrix}-1 & 12\\4 & 7\end{pmatrix}$. Reversing: row 1 of B times column 1 of A: $0(2) + 5(3) = 15$, so $BA = \begin{pmatrix}15 & 20\\-4 & -9\end{pmatrix}$. The products differ.

Marking guidance: [4 marks] ≈ 2 per product (full-mark scheme: 1 method of row-by-column, 1 accuracy per matrix). A 7–8 answer shows at least one entry computed and does not multiply entry by entry.

4 $\det A = 5(1) - 2(3) = -1 \neq 0$. $A^{-1} = \frac{1}{-1}\begin{pmatrix}1 & -2\\-3 & 5\end{pmatrix} = \begin{pmatrix}-1 & 2\\3 & -5\end{pmatrix}$. Verify: $AA^{-1} = \begin{pmatrix}5 & 2\\3 & 1\end{pmatrix}\begin{pmatrix}-1 & 2\\3 & -5\end{pmatrix}$: row 1 × column 1 is $5(-1) + 2(3) = -5 + 6 = 1$; all entries give $\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = I$. Solve: $\begin{pmatrix}x\\y\end{pmatrix} = A^{-1}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}-1\\3\end{pmatrix}$, so $x = -1$, $y = 3$. Checks: $5(-1) + 2(3) = 1$ and $3(-1) + 3 = 0$.

Marking guidance: [8 marks] ≈ 1 determinant, 2 inverse (adjugate and division by −1), 2 the explicit $AA^{-1} = I$ verification, 3 setting up $\vec{x} = A^{-1}\vec{b}$, the result and a substitution check. Top band: verification carried out entry by entry and the solution checked in both equations.

Criterion B · Repeating a shear

1 $S^2 = \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$: row 1 × column 2 is $1(1)+1(1)=2$, so $S^2 = \begin{pmatrix}1 & 2\\0 & 1\end{pmatrix}$. $S^3 = S^2S = \begin{pmatrix}1 & 3\\0 & 1\end{pmatrix}$, $S^4 = S^3S = \begin{pmatrix}1 & 4\\0 & 1\end{pmatrix}$. Table of top rows: $(1,1)$, $(1,2)$, $(1,3)$, $(1,4)$ for S, S^2, S^3, S^4.

Marking guidance: [6 marks] ≈ 2 per correct power (or table rows); a 7–8 response records all powers in one systematic, labelled table.

2 The diagonal stays 1, the bottom row stays $(0,1)$, and the top-right entry equals the exponent. Predict $S^n = \begin{pmatrix}1 & n\\0 & 1\end{pmatrix}$.

Marking guidance: [4 marks] pattern described (2) and general prediction stated (2). Top band: the pattern is described in terms of every entry, not just noticed informally.

3 $n=6$: the formula predicts $\begin{pmatrix}1 & 6\\0 & 1\end{pmatrix}$; from the powers, $S^6 = S^4S^2 = \begin{pmatrix}1 & 4\\0 & 1\end{pmatrix}\begin{pmatrix}1 & 2\\0 & 1\end{pmatrix} = \begin{pmatrix}1 & 4+2\\0 & 1\end{pmatrix}$, agreeing. Induction: $S^1 = \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$ is true; assume $S^k = \begin{pmatrix}1 & k\\0 & 1\end{pmatrix}$; then $S^{k+1} = S^kS$, whose top-right entry is $1\cdot 1 + k\cdot 1 = k+1$, so $S^{k+1} = \begin{pmatrix}1 & k+1\\0 & 1\end{pmatrix}$: true for all positive integers n. $S^{-1} = \begin{pmatrix}1 & -1\\0 & 1\end{pmatrix}$, since $\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix} = I$; it fits the pattern as the formula with $n=-1$ (a shear back by 1).

Marking guidance: [8 marks] ≈ 2 verification at $n=6$, 4 a complete induction (base, assumption, $k \to k+1$ step), 2 for S^{-1} checked and linked to $n=-1$. Top band: the inductive step shows the entry computation and the shear is interpreted geometrically.

4 $R^2 = \begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}$ (rotation 180°), $R^3 = R^2R = \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}$ (270° anticlockwise), $R^4 = R^3R = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = I$. Geometrically each power is another 90° anticlockwise turn, so the images cycle with period 4: every fourth power returns to the identity.

Marking guidance: [4 marks] ≈ 1 per matrix and 1 for describing the cycle. Top band: predicts $R^5 = R$, $R^6 = R^2$, etc. from the period.

5 $\det(S^n) = (\det S)^n = 1^n = 1$ and $\det(R^n) = (\det R)^n = 1^n = 1$ (both base matrices have determinant 1): by the product rule, applying a transformation n times multiplies the determinant n times. Shears and rotations preserve area and orientation exactly.

Marking guidance: [2 marks] rule stated (1) and justified via $\det(AB) = \det A \det B$ (1). Total for the criterion: $6+4+8+4+2=24$.

Criterion C · Explaining why order matters

1 **Sample response.** The product AB means "do B first, then A": $AB\vec{v} = A(B\vec{v})$, so the right-hand matrix acts on $\vec{v}$ first, like the inner bracket of a composite function. This also fixes the meaning of each column: the columns of M are the images of $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$. Counterexample with the rotation $R = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}$ (90° anticlockwise) and the reflection $S = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$ (in the x-axis): $RS = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$ but $SR = \begin{pmatrix}0 & -1\\-1 & 0\end{pmatrix}$, and these are different matrices. Following the basis vector $\begin{pmatrix}1\\0\end{pmatrix}$: RS sends it to $\begin{pmatrix}0\\1\end{pmatrix}$, while SR sends it to $\begin{pmatrix}0\\-1\end{pmatrix}$. Rotating first turns the axis onto the y-axis before the reflection, so the reflection acts on a different vector; the two composites are reflections in $y=x$ and $y=-x$. So $AB\vec{v} = BA\vec{v}$ fails even for nice transformations. The lesson for solving $A\vec{x} = \vec{b}$: multiply by A^{-1} on the left only: $A^{-1}A\vec{x} = A^{-1}\vec{b}$, giving $\vec{x} = A^{-1}\vec{b}$, because order cannot be rearranged freely; $(AB)^{-1} = B^{-1}A^{-1}$ for the same reason.

Marking guidance: 20 marks across Criterion C strands (language and symbols; moving between representations; organising; explaining with degree of accuracy). A 7–8 response must: state that the rightmost matrix acts first; include the computed counterexample $RS \neq SR$; trace the images of both basis vectors; draw the inverse-method conclusion ($\vec{x} = A^{-1}\vec{b}$, left multiplication); and use consistent notation throughout, with a diagram or table of the two image pairs.

Criterion D · Mapping a survey by drone

1 M is a 2×2 matrix. $\det M = 1.5(1.0) - 0.5(0.5) = 1.5 - 0.25 = 1.25 \neq 0$, so the mapping can be reversed: M^{-1} exists.

Marking guidance: [3 marks] order (1), determinant (1), decision with reason (1).

2 $M\begin{pmatrix}150\\80\end{pmatrix} = \begin{pmatrix}1.5(150)+0.5(80)\\0.5(150)+1.0(80)\end{pmatrix} = \begin{pmatrix}225+40\\75+80\end{pmatrix} = \begin{pmatrix}265\\155\end{pmatrix}$: map coordinates $(265, 155)$ m.

Marking guidance: [4 marks] setup (1), both entries computed (2), answer with units (1).

3 $M^{-1} = \frac{1}{1.25}\begin{pmatrix}1.0 & -0.5\\-0.5 & 1.5\end{pmatrix} = \begin{pmatrix}0.8 & -0.4\\-0.4 & 1.2\end{pmatrix}$. Then $M^{-1}\begin{pmatrix}300\\200\end{pmatrix} = \begin{pmatrix}0.8(300)-0.4(200)\\-0.4(300)+1.2(200)\end{pmatrix} = \begin{pmatrix}240-80\\-120+240\end{pmatrix} = \begin{pmatrix}160\\120\end{pmatrix}$: camera coordinates $(160, 120)$ m. Verify: $M\begin{pmatrix}160\\120\end{pmatrix} = \begin{pmatrix}240+60\\80+120\end{pmatrix} = \begin{pmatrix}300\\200\end{pmatrix}$, as required.

Marking guidance: [5 marks] inverse (2), recovered point (2), verification (1).

4 Camera area $= 200\times 150 = 30\,000$ m^2. The map area is $|\det M|$ times this: $1.25\times 30\,000 = 37\,500$ m^2. The determinant is the area scale factor of the transformation, so it converts camera-scale areas to map-scale areas without needing the image vertices.

Marking guidance: [4 marks] camera area (1), scale-factor use (2), result (1).

5 Ground velocity $= \begin{pmatrix}4\\-3\end{pmatrix} + \begin{pmatrix}-1\\-1\end{pmatrix} = \begin{pmatrix}3\\-4\end{pmatrix}$ m/s, speed $\sqrt{9+16} = 5$ m/s. Time for 500 m over the ground: $500 \div 5 = 100$ s.

Marking guidance: [3 marks] vector sum (1), speed (1), time (1).

6 Assumptions (any two): the ground is flat, so planar coordinates suffice and height is ignored; M is exact, constant over the whole survey, and maps metres to metres linearly; the wind is constant, so the ground velocity is constant; the camera and map axes are aligned. Evaluation: over hilly ground a fixed linear map ignores elevation and camera tilt (parallax displaces features), so the recovered coordinates carry a terrain-dependent error and the shear-like distortion of M no longer matches reality. With a changing wind the velocity varies along the path, so speed 5 m/s and time 100 s are averages only; the model remains acceptable for short, low-relief flights at constant altitude but must be re-fitted section by section for mountains or gusts.

Marking guidance: [5 marks] two stated assumptions (2), critique of flat-ground assumption (1), critique of constant-wind assumption (1), overall judgement of validity (1). Total for the criterion: $3+4+5+4+3+5=24$.

CHAPTER 9 · STATISTICS: DISPERSION, CORRELATION AND REGRESSION

Opening problem · same average, different story

1 Both sets total 145.0 s, so each has mean $145.0 \div 10 = 14.5$ s. With $n = 10$ the median is the mean of the 5th and 6th ordered values, $\frac{14.5+14.5}{2} = 14.5$ s for each. The two centres are identical, which hides that programme B contains both the fastest time (12.4 s) and the slowest (16.6 s) while programme A clusters near the middle.

2 Quartiles use halves of five values (medians of each half). A: range $15.9 - 13.1 = 2.8$ s, $Q_1 = 14.0$ s, $Q_3 = 15.0$ s, IQR $= 1.0$ s. B: range $16.6 - 12.4 = 4.2$ s, $Q_1 = 13.9$ s, $Q_3 = 15.1$ s, IQR $= 1.2$ s. Programme A is the more consistent (smaller range and IQR).

3 Same mean (14.5 s) and same median (14.5 s), but A has range 2.8 s, IQR 1.0 s and $s = 0.782$ s against B's range 4.2 s, IQR 1.2 s and $s = 1.192$ s. Since all three spread measures are smaller for A, choose **A** when a predictable team time matters; choose B only to gamble on one exceptional run.

4 Evidence that would strengthen a causal claim: random assignment of runners to programmes, and comparable baseline times (a matched control, ruling out confounders such as age or prior training). Evidence that would undermine it: runners choosing their own programme (self-selection) or the school that ran the trial also funding and reporting it.

Prerequisite check

1 Sum $= 40$, so mean $= 40 \div 5 = 8$. Ordered $4, 7, 7, 9, 13$: median $= 7$, mode $= 7$.

2 $n = 6$, so the median is the mean of the 3rd and 4th values, $\frac{18+21}{2} = 19.5$.

3 $(3-5)^2 + (5-5)^2 + (7-5)^2 + (9-5)^2 = 4 + 0 + 4 + 16 = 24$.

4 About 80% of scores are at or below 72; only about 20% are higher. It is a position, not a quality judgement.

5 The points rise from left to right (despite one dip at $(3, 4)$), so the trend is positive: larger x goes with larger y.

6 Multiply by 4: $k = 0.35 \times 4 = 1.4$.

Exercise 9A · Range, quartiles, box plots and outliers

1 (a) $n = 7$: range $18 - 4 = 14$, median 10, $Q_1 = 7$, $Q_3 = 15$, IQR $= 8$. (b) $n = 8$: range $41 - 21 = 20$, median $\frac{27+30}{2} = 28.5$, $Q_1 = 24$, $Q_3 = 34$, IQR $= 10$. (All in the units of the data.)

2 $n = 9$, median $= 10$. $Q_1 = \frac{6+8}{2} = 7$, $Q_3 = \frac{13+15}{2} = 14$. Five-number summary: $5,\ 7,\ 10,\ 14,\ 22$.

3 $n = 9$, median 8, $Q_1 = 5.5$, $Q_3 = 10.5$, IQR $= 5$. Upper fence $= 10.5 + 1.5(5) = 18$. Since $32 > 18$, 32 is a potential outlier.

4 Cumulative frequencies: $6,\ 20,\ 38,\ 48,\ 50$. The median is the $n/2 = 25$th value, which lies in the $20 < t \leq 30$ class (cumulative $20 \to 38$). Interpolating: $20 + \dfrac{25 - 20}{18} \times 10 = 22.8$ min.

5 Her time is at the 80th percentile, so about $0.80 \times 150 = 120$ times are at or below hers. As smaller is better, those 120 runners were as quick or quicker, so she is in the **slowest fifth** (only about 30 were slower).

6 About 25% of patients waited 12 minutes or less, and about 75% waited longer than 12 minutes. A quarter are seen within 12 min.

7 The student placed Q_1 a quarter of the way along the range; quartiles are instead positions in the ordered data (medians of the halves). The lower half is $12, 14, 15, 17, 18, 19$, so $Q_1 = \dfrac{15 + 17}{2} = 16$, not 15.

8 The mean and the median each increase by 5 (they move with the data). The range (max − min) and the IQR ($Q_3 - Q_1$) are unchanged, because the same $+5$ is added to both endpoints of each difference.

9 IQR 2: $0, 10, 10, 11, 12, 12, 20$ ($Q_1 = 10$, $Q_3 = 12$, range 20). IQR 15: $0, 2, 5, 10, 12, 17, 20$ ($Q_1 = 2$, $Q_3 = 17$, range 20). Range uses only the extremes, so it cannot reveal how the middle half is spread.

10 It may be (i) a genuine extreme value in a skewed population, or (ii) a real value from a different sub-population. Before removing it, check the recording and the units, and ask whether it is a data error or a true observation.

11 Positively (right-) skewed: the long right tail holds a few large values that pull the mean above the median, while the median sits in the bulk of the data.

12 The class intervals lose the individual values inside them; the median is read off the cumulative curve by assuming values are spread evenly across the median class. If the true within-class distribution is uneven, that estimate moves, so it is only an estimate.

13 P: $22,\ 25.5,\ 30,\ 34,\ 48$ (IQR $= 8.5$, range 26). Q: $24,\ 27.5,\ 30,\ 32.5,\ 34$ (IQR $= 5$, range 10). Both medians 30 min, but Q is far more predictable. P's 48 exceeds its upper fence $34 + 1.5(8.5) = 46.75$, so it is an outlier suggesting an occasional breakdown, making P less reliable (times in minutes).

14 A few very long waits (a long right tail) raise the mean to 31 min above the median 18 min, while the middle half, whose IQR is 7 min, is unaffected. Advise a patient to expect roughly 18 min, with the middle half inside a 7-minute band, but to be ready for occasional much longer waits.

15 New median $2m + 3$; new $Q_1 = 2q_1 + 3$; new $Q_3 = 2q_3 + 3$; new IQR $= (2q_3 + 3) - (2q_1 + 3) = 2(q_3 - q_1)$, i.e. doubled. Adding 3 raises every value equally, so it cancels in the difference $Q_3 - Q_1$: a shift changes centre but not spread, while the factor 2 rescales the spread.

Exercise 9B · Variance, standard deviation and grouped estimates

1 $\bar{x} = 42 \div 6 = 7$. Squared deviations $16, 4, 1, 1, 4, 16$ total 42. Variance $= 42 \div 6 = 7$, $s = \sqrt{7} = 2.65$ (2 d.p.), in the units of the data.

2 $\bar{x} = 104 \div 8 = 13$. Squared deviations $49, 25, 9, 1, 1, 9, 25, 49$ total 168. Variance $= 168 \div 8 = 21$, $s = \sqrt{21} = 4.58$ (2 d.p.).

3 Midpoints $5, 15, 25, 35, 45$; $\sum f = 50$, $\sum fx = 1230$, $\sum fx^2 = 36\,450$. Mean $= 1230 \div 50 = 24.6$ years; $s^2 = 36450/50 - 24.6^2 = 729 - 605.16 = 123.84$, so $s = 11.1$ years.

4 $\bar{x} = 44 \div 5 = 8.8$. Deviations $-4.8, -2.8, 0.2, 2.2, 5.2$ (sum 0); squares $23.04, 7.84, 0.04, 4.84, 27.04$ total 62.8. $s^2 = 62.8 \div 5 = 12.56$, $s = 3.54$.

5 4.2 cm $= 42$ mm $= 0.042$ m. The standard deviation carries the data's units, so it rescales with them.

6 Midpoints $15, 25, 35, 45$; $\sum f = 40$, $\sum fx = 1180$. Mean $\approx 1180 \div 40 = 29.5$ g.

7 Signed deviations always total zero, so their average is 0 no matter how spread the data are. The standard deviation uses the *squared* deviations: $s = \sqrt{\dfrac{\sum (x - \bar{x})^2}{n}}$. Squaring removes the sign, so the total is positive unless every value equals the mean.

8 (a) 6 — adding a constant shifts all values equally, so spread is unchanged. (b) $3 \times 6 = 18$ — multiplying scales the standard deviation by the factor. (c) $6 \div 2 = 3$ — dividing scales it by $\frac{1}{2}$. The standard deviation has the data's units, so it tracks any multiplicative scaling.

9 σx divides by n (population/descriptive form) and sx divides by $n - 1$ (sample estimate). This section uses σx with denominator n. Use sx when the data are a random sample meant to estimate a larger population's standard deviation.

10 Both classes centre on 60, but with $s = 3$ class 1's marks sit close to 60 (typical deviation about 3), while with $s = 12$ class 2 is far more spread out (typical deviation about 12). Class 1 is the more homogeneous group; class 2 contains much higher and much lower scores.

11 Within-class placement errors in the mean partly cancel because low and high classes offset, so the mean stays close. The standard deviation depends on squared distances from the mean, and the midpoint assumption places every value at the class centre, erasing real within-class spread, so it is more easily distorted.

12 The two standard deviations are in different units, so a raw comparison of magnitudes is meaningless. To compare variability across different measures or units one must use a scale-free statistic such as the coefficient of variation, $s/\bar{x}$.

13 B is more consistent ($s = 1.5$ g against A's 4 g). A risks underfilling, because its mean is exactly the 500 g target, so about half its bags fall below, and its larger spread puts some well under. Recommend **B**: the trade-off is that B uses about 2 g more product per bag, so it costs slightly more to guarantee a reliable fill.

14 Choose Route 2. Though its mean is 4 min longer, its spread is small ($s = 1.4$ min), so lateness is predictable; Route 1 ($s = 5.1$ min) sometimes runs much longer than 22 min and would risk the fixed deadline. If there were slack in the timing, Route 1 would be faster on average.

15 Mean $\frac{a+b}{2}$. Deviations are $a - \frac{a+b}{2} = \frac{a-b}{2}$ and $b - \frac{a+b}{2} = \frac{b-a}{2}$, each squaring to $\frac{(a-b)^2}{4}$. So $s^2 = \dfrac{2 \cdot (a-b)^2/4}{2} = \dfrac{(a-b)^2}{4}$ and $s = \dfrac{|a-b|}{2}$. With only two values the range is the whole spread, and the mean sits midway, so the standard deviation is exactly half the range.

Exercise 9C · Comparing centres, spreads and shapes

1 X: median 11, $Q_1 = 8.5$, $Q_3 = 14.5$, IQR $= 6$; upper fence $14.5 + 1.5(6) = 23.5$, so 25 is an outlier. Y: median 11, $Q_1 = 9.5$, $Q_3 = 12$, IQR $= 2.5$; fences 5.75 and 15.75, no outliers. Same centre but Y is far more consistent (smaller IQR and range); X is dragged by one extreme value. (In the units of the data, minutes.)

2 P: median 33, range $51 - 21 = 30$. Q: median 38, range $56 - 22 = 34$. Q has both the higher centre and the wider span.

3 Most consistent: 9Q (smallest IQR 7 and smallest $s = 4$). Highest typical score: 9Q (highest mean 64 and highest median 65).

4 Both are zero only when every value is identical. range $= 0 \Rightarrow$ max $=$ min $\Rightarrow$ all equal $\Rightarrow s = 0$, and conversely. So no — one cannot be zero while the other is not.

5 Mean 45, median 50: here mean $<$ median, the signature of negative skew (a longer left tail).

6 (a) middle half of salaries $\rightarrow$ IQR. (b) full span of temperatures $\rightarrow$ range. (c) uses every value, sensitive to extremes $\rightarrow$ standard deviation.

7 The single highest score is one extreme value and says nothing about typical performance or consistency, which a match is decided by. Fairer summaries: compare the medians (or means) for centre and the IQR (or standard deviation) for spread.

8 $4, 5, 6$ and $1, 5, 9$: both mean 5. First: squared deviations $1, 0, 1$ total 2, $s = \sqrt{2/3} = 0.82$. Second: squared deviations $16, 0, 16$ total 32, $s = \sqrt{32/3} = 3.27$. Same mean, different spread, as required.

9 Adding 50 pulls the mean up (well above 20); the median rises only slightly; the range increases, since 50 becomes the new maximum ($50 -$ min > 30); the IQR changes little because it describes the middle half and resists the new extreme.

10 For skewed data the mean and standard deviation are dragged toward the tail. In the salary example one £90 000 outlier raised the mean to £29 100 and the SD to £21 600, while the median £22 000 and IQR £5 000 still described the typical worker, so median and IQR are fairer.

11 With the vertical axis starting at 55, a 2-mark gap occupies a large fraction of the drawn height, so a small difference looks big. Fair-graph principle: run the measurement axis from zero (or use a clearly labelled, proportional scale) so the visual difference matches the real one.

12 They are the same: 0.02 m $= 2$ cm. The spreads are equal; only the units differ, so neither group is more variable.

13 Reason 1: a selective school admits higher prior attainment, so the mean reflects intake, not teaching. Reason 2: the mean hides spread and a small sample's mean is unstable. Better: compare progress/value-added from prior attainment, and report median with IQR for schools of similar intake.

14 "Half of patients wait about 18 min or less, and the middle half of waits lie within a 7-minute band around that; however a few waits are extremely long (one was 190 min), which is why the average is 31 min." The mean alone is unfair because the 190-min outlier inflates it above the typical patient's experience.

15 Range depends on just the smallest and largest value, so one faulty item changes it completely; SD uses every value, so it tracks overall day-to-day variation and is more informative when all output is good. But a genuine extreme still inflates the SD, whereas the IQR, based only on the middle half, ignores that single value — so IQR is preferable once real extremes occur.

Exercise 9D · Scatter graphs, association and correlation

1 (a) hours studied explanatory, test score response. (b) temperature explanatory, heating cost response. (c) age explanatory, height response. (d) advertising explanatory, sales response.

2 (a) 0.92: strong positive. (b) -0.45: moderate negative. (c) 0.03: essentially no linear association. (d) -0.88: strong negative.

3 $\bar{x} = 4.5$, $\bar{y} = 7.5$. $S_{xy} = 20.5$, $S_{xx} = 17.5$, $S_{yy} = 25.5$. $r = \dfrac{20.5}{\sqrt{17.5 \times 25.5}} = \dfrac{20.5}{21.12} = 0.970$, a strong positive linear association.

4 strong positive $= 0.95$; strong negative $= -0.86$; almost none $= 0.12$; weak negative $= -0.34$.

5 There is a moderate positive linear association: people who do more sport tend to report greater happiness. Limitation: r only measures association, so it cannot show that sport causes happiness (more happiness might lead to sport, or a third factor such as health could drive both).

6 The band rises, so the association is positive and (ignoring the point) fairly strong and linear. The single point far below adds a large negative cross-product and pulls the trend down, reducing r; removing it would move r closer to $+1$.

7 Error 1: $|r|$ can never exceed 1, so $r = 1.15$ is impossible and signals a calculation mistake. Error 2: describing an out-of-range value as "extremely strong" — r is not a percentage, so no reported strength can be trusted when the number is invalid.

8 An $r = 0.90$ association does not establish causation: the direction may be reversed or a third variable may cause both. Two confounders: parental income and parental education (or the reading time parents spend), each of which raises both the number of books and the reading score without the books themselves causing it.

9 Removing a point that lies far from a strong positive pattern increases r, moving it closer to $+1$, because that point was the main departure from the straight line and was reducing the ratio.

10 Restriction of range. Selecting only the high scorers shrinks the spread of the response, so the linear association is compressed toward 0 and the correlation looks smaller than over the full year.

11 (a) ice-cream and drowning: no direct causation; confounder is temperature/season. (b) shoe size and reading age: no direct causation; confounder is the child's age. (c) umbrella sales and rain: causation is plausible — rain causes the sales, so no separate confounding variable is needed.

12 For a symmetric curve such as $y = (x - \bar{x})^2$, points to the left of $\bar{x}$ and to the right at equal height above $\bar{y}$ give cross-products $(x - \bar{x})(y - \bar{y})$ of opposite sign, which cancel, so the numerator $\sum(x - \bar{x})(y - \bar{y}) \approx 0$ and $r \approx 0$. Pearson's r measures only straight-line association, so a strong curved relationship is invisible to it.

Common error: concluding "$r \approx 0$ means unrelated" — the curve is a genuine relationship, just not a linear one.

13 The study supports only a weak-to-moderate negative linear association ($r = -0.41$) among these 240 students. Because $|r|$ is modest, the sample is a single (non-random) school, and there was no random assignment, confounders (sleep, prior attainment, motivation) and reverse causation remain open, so the causal headline "damage" is not supported.

14 Neither causes the other: sales of both are driven by a third variable, the season/temperature. Cold months raise coat sales and lower ice-cream sales while warm months do the opposite, producing a strong negative correlation without any direct link.

15 With $x \to kx$ and $y \to my$, the deviations scale to $k(x-\bar{x})$ and $m(y-\bar{y})$. The numerator becomes $km\,S_{xy}$ and the denominator $\sqrt{k^2S_{xx}\cdot m^2S_{yy}} = km\sqrt{S_{xx}S_{yy}}$ for $k, m > 0$, so r is unchanged. Hence r is a pure number, independent of the units used. If k or m is negative, the sign of one set of deviations flips, reversing the numerator's sign but not the denominator's, so r changes sign while $|r|$ stays the same.

Exercise 9E · Regression lines, residuals and prediction

1 At $x = 5$: $\hat{y} = 2.4 + 0.8(5) = 6.4$. At $x = 10$: $\hat{y} = 2.4 + 0.8(10) = 10.4$. The gradient 0.8 means each increase of 1 in x raises the predicted y by 0.8 units.

2 $\bar{x} = 3$, $\bar{y} = 5.6$. $S_{xy} = \sum(x-\bar{x})(y-\bar{y}) = 17$, $S_{xx} = \sum(x-\bar{x})^2 = 10$. $b = 17 \div 10 = 1.7$, $a = 5.6 - 1.7(3) = 0.5$. Line: $y = 0.5 + 1.7x$ (passes through $(\bar{x}, \bar{y}) = (3, 5.6)$).

3 Predicted values $2.2, 3.9, 5.6, 7.3, 9.0$; residuals $y - \hat{y}$ are $-0.2,\ 1.1,\ -1.6,\ 0.7,\ 0.0$, which sum to 0. ✓

4 (a) 0.81: 81% of the variation in y is accounted for by the linear regression on x. (b) 0.36: 36% accounted for (much is unexplained). (c) 0.94: 94% accounted for (a close linear fit).

5 (a) $x = 25$: interpolation. (b) $x = 5$: extrapolation (below the range). (c) $x = 60$: extrapolation (above). (d) $x = 40$: interpolation (on the upper boundary, inside the range).

6 Equation $y = 12.3 - 0.45x$. Since the gradient is negative, $r = -\sqrt{0.72} = -0.849$. Gradient: each extra unit of x is associated with a decrease of 0.45 in y.

7 Problem 1: $x = 30$ is far outside the fitted range $2 \le x \le 8$, so it is extrapolation and the prediction is unreliable. Problem 2: four decimal places is false precision. Report the fitted x-range, flag the prediction as extrapolation, round to the data's precision, and include n, r^2 and the units.

8 Vertical distances are used because the line predicts y from a given x, so the error of interest is in y. They are squared because signed deviations cancel (an unsquared total could be 0), squaring penalises large misses and yields a unique smooth minimum; horizontal distances would wrongly attribute error to the explanatory variable.

9 A far-right point has high leverage. If it sits above the trend it steepens the gradient and lowers the intercept (if below, the reverse); because it is an exception to the main cluster it adds scatter, so r falls (the association looks less tight) unless the point happens to lie exactly on the extension of the line.

10 A U-shaped residual plot shows systematic curvature: the straight line is the wrong model. The analyst should fit a curved model (for example a quadratic) or transform a variable, and only then use it to predict.

11 $r^2 = 0.98$ only describes the fit inside the observed x-range. Far beyond the data the relationship may bend, level off or change mechanism, and the line assumes the straight trend continues, so the prediction is unreliable however high r^2 is.

12 The gradient is a rate of change, in units of y per unit of x, so it is 2.5 kg per cm. But $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, and the units in the numerator (kg·cm) cancel with those under the root in the denominator, leaving a pure number between -1 and 1.

13 At $T = 10$: $C = 28.5 - 1.4(10) = 14.5$, an interpolation. At $T = 0$: $C = 28.5 - 0 = 28.5$, but $T = 0$ lies outside the fitted range $2 \le T \le 18$, so it is extrapolation and unreliable (the intercept is only a positioning constant). Gradient -1.4: each extra 1°C is associated with 1.4 less heating cost.

14 At $a = 5$: $V = 18.5 - 2.1(5) = 8.0$ thousand £, an interpolation. At $a = 12$: $V = 18.5 - 2.1(12) = -6.7$, but $a = 12$ is beyond the fitted range $1 \le a \le 8$, so it is extrapolation and unreliable before the arithmetic is even examined — a straight-line depreciation cannot continue, since a car's value levels off near a salvage value and never goes negative.

15 With $s_x = \sqrt{S_{xx}/n}$ and $s_y = \sqrt{S_{yy}/n}$, $r\dfrac{s_y}{s_x} = \dfrac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \cdot \dfrac{\sqrt{S_{yy}/n}}{\sqrt{S_{xx}/n}} = \dfrac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \cdot \sqrt{\dfrac{S_{yy}}{S_{xx}}} = \dfrac{S_{xy}}{S_{xx}} = b$ (the n cancels). So the gradient equals the correlation scaled by the ratio of the two spreads: for a fixed r the line is steeper when y varies more than x, and the sign of b follows the sign of r.

Investigation 9 · the twelve per cent claim

1 Explanatory variable: whether the school uses the Daily Numeracy app. Response: attainment (exam results). The population described is "schools" in general, but the sample measured is the 60 volunteering schools plus 60 non-users. The sample is not representative: the app schools volunteered (self-selection), so they are likely more motivated or better resourced, and the comparison schools were not matched or randomly chosen.

2 (i) School motivation/leadership — improvement-minded schools adopt apps and also raise results anyway. (ii) Prior attainment/entry level — the groups may start from different places. (iii) Resources and teacher effort — adopting schools may get more IT or dedicated staff. (iv) Pupil intake/home support. Each could produce a higher mean for the app group even if the app itself did nothing.

3 Because the adopting schools had been measured at an unusually low result the year before, part of that dip was random, so their next-year scores tend to drift back up toward their usual level regardless of the app (regression to the mean) — making the rebound look like an app gain. A comparison group also selected after a low year, but not using the app, would show the same natural bounce, and the difference-in-differences would then isolate the app's real effect.

4 Sixty schools per group is a moderate count, but the precision of the 12% depends on the spread of results within each group and on clustering of pupils inside schools, so a standard error and confidence interval are needed, not just the point figure. Because the mean is not resistant, a handful of very high-scoring adopting schools could account for most of the gap and inflate the 12%.

5 Funding by the company is a conflict of interest; there is no sign of peer review; nine schools that stopped using the app were excluded (survivorship bias, which inflates the measured effect); and "proven" asserts causation from an observational comparison. The most serious weaknesses are the self-selected, non-random design together with the dropped-out schools, since these bias the estimate itself; funding then undermines confidence in what was measured and published.

6 Participants: matched pupils (or schools) of similar prior attainment. Allocation: randomly assign to app or control. Control condition: usual lessons, or an equal-time non-numeracy activity so time is held constant. Measurements: a baseline numeracy test and an identical, blind-marked end-of-period test. Two ethical safeguards: informed consent (with parental permission) and the right to withdraw, and offering the app to the control group afterwards so no benefit is permanently withheld.

7 Rewritten claim: "In this study, schools that chose the app scored about 12% higher than schools that did not; this is an observed association and does not on its own show that the app raised attainment." The word "proven" would be justified only by a randomised, preregistered, peer-reviewed trial with a matched control group, no dropout bias, replication in independent settings, a plausible mechanism, and a statistically and practically significant effect.

Generalise. Questions to apply to any statistical headline: Who was measured, and were they volunteers or a random sample? How many, and how spread out (what interval surrounds the figure)? Compared against what control? Who funded it and was it peer-reviewed? Did the proposed cause precede the effect? Were confounders ruled out, and is the effect large enough to matter? Applied to an advertisement claiming "our app raises results by 12%", these questions show that an uncontrolled, self-selected, company-funded comparison with excluded dropouts cannot support a causal claim, only an association.

Modelling task · does advertising predict sales?

1 With $\bar{A} = 4.5$ and $\bar{S} = 21.5$: $S_{AA} = 42$, $S_{SS} = 334$, $S_{AS} = 115$. The cloud rises in a tight, straight band, so direction positive, strength strong, form linear; the week-3 sales of 15 sits a little below the band but not far enough to be an outlier. $r = \dfrac{115}{\sqrt{42 \times 334}} = 0.971$: a very strong positive linear association.

2 $b = \dfrac{115}{42} = 2.74$, $a = 21.5 - 2.7381 \times 4.5 = 9.18$. Line: $S = 9.18 + 2.74A$. Gradient: each extra hundred pounds of weekly advertising is associated with about 2.74 thousand pounds more sales (£2740 per £100). Intercept: at $A = 0$ the model gives 9.18 thousand pounds — but $A = 0$ is outside the observed range, so treat it as the sales baseline before advertising, not a measured value.

3 £450 is $A = 4.5$: $\hat{S} = 9.18 + 2.74(4.5) = 21.5$ thousand pounds, i.e. £21 500. This lies inside the observed range $1 \le A \le 8$, so it is an **interpolation**. £1200 is $A = 12$: $\hat{S} = 9.18 + 2.74(12) = 42.0$ thousand pounds, i.e. £42 000, which lies well beyond the largest spend recorded, so it is an **extrapolation**. The £450 prediction is the more trustworthy: the line is supported by data on both sides of $A = 4.5$ — which is also $\bar{A}$, so the estimate is the mean sales £21 500 — whereas $A = 12$ assumes the same straight-line response continues where no observation exists, and returns to advertising usually fall away at high spend. Note also that £450 is the only one of the two whose answer is pinned down by the fitted points; at $A = 12$ a slightly different slope, or a curve, would move the figure by several thousand pounds.

4 $r^2 = 0.971^2 = 0.943$: about 94.3% of the variation in weekly sales is accounted for by the linear model. Week-3 residual: observed 15 minus predicted $9.179 + 2.738(3) = 17.4$ gives -2.39 thousand pounds (that week sold about £2400 below the line). Two assumptions: each week is a comparable data point (same products and prices), and the linear relationship holds across the spend range. Two reasons the gradient is not the return on advertising: advertising was not randomly assigned (other factors — season, promotions, quality — vary with it), and the data are only an eight-week correlation, so no causal link is established and extrapolation is unsafe.

Mixed practice 9 · dispersion, comparison, correlation and regression

1 $n = 9$: median 23. $Q_1 = \frac{17+19}{2} = 18$, $Q_3 = \frac{29+31}{2} = 30$, IQR $= 12$ (units of the data).

2 Median 18, $Q_1 = 13.5$, $Q_3 = 21.5$, IQR $= 8$. Five-number summary $11,\ 13.5,\ 18,\ 21.5,\ 45$. Upper fence $21.5 + 1.5(8) = 33.5$; $45 > 33.5$, so 45 is a potential outlier.

3 $\bar{x} = 45 \div 5 = 9$. Squared deviations $16, 9, 1, 4, 36$ total 66. $s^2 = 66 \div 5 = 13.2$, $s = 3.63$ (units of the data).

4 Midpoints $15, 25, 35, 45$; $\sum f = 30$, $\sum fx = 930$, $\sum fx^2 = 31\,350$. Mean $= 930 \div 30 = 31$; $s^2 = 31350/30 - 31^2 = 1045 - 961 = 84$, $s = 9.17$ (units of the data).

5 On average a mark differs from the mean of 62 by about 4.2 marks, so most marks fall within a few marks of 62 (roughly 62 ± 4.2 for a typical student). The unit is marks, not a percentage.

6 Same mean (62) but B is far more consistent (IQR 7 vs 14, SD 4 vs 9). B's higher median (66 vs 61) shows most of its marks sit above A's, while A's wider spread puts more pupils in a low tail. Overall B performs more reliably at a higher level.

7 A strong positive linear association — points cluster tightly on a rising line — with one outlier sitting well below the band. That single low point pulls the trend down and reduces r; removing it would strengthen the association.

8 $r = -0.68$ is a moderately strong negative linear association: older laptops tend to have lower resale values. The value is a correlation, not a percentage, and does not itself prove causation.

9 $\hat{y} = 3.2 + 0.8(10) = 3.2 + 8 = 11.2$.

10 Residual $= y - \hat{y} = 5 - 5.6 = -0.6$. The negative sign means the point lies 0.6 below the line (the model over-predicted).

11 (a) $x = 12$: interpolation. (b) $x = 30$: extrapolation (above). (c) $x = 5$: interpolation (on the lower boundary). (d) $x = 2$: extrapolation (below).

12 (a) an association (correlation, not proof breakfast helps). (b) a causal claim ("improves"). (c) an association, and almost certainly a spurious one — both are driven by a confounder (summer/temperature), not by each other.

13 A: median 20, IQR 8, range 18. B: median 22, IQR 7, range 9. B has the higher centre and is more consistent (shorter box and whiskers). One conclusion the plots cannot support is any statement about the mean or about statistical significance — a box plot shows positions only, not the mean, not the sample size, and not whether the difference is real.

14 With 60: mean $175 \div 9 = 19.4$, median 15. Without 60: mean $115 \div 8 = 14.4$, median 14.5. The outlier lifts the mean by about 5 but barely moves the median, so the median better represents a typical value.

15 $\bar{x} = 3$, $\bar{y} = 6.4$. $S_{xy} = 17$, $S_{xx} = 10$, $S_{yy} = 29.2$. $r = \dfrac{17}{\sqrt{10 \times 29.2}} = \dfrac{17}{17.09} = 0.995$: a very strong positive linear association.

16 $b = 17/10 = 1.7$, $a = 6.4 - 1.7(3) = 1.3$. Line $y = 1.3 + 1.7x$. At $x = 6$: $\hat{y} = 1.3 + 1.7(6) = 11.5$, an extrapolation (largest observed x is 5).

17 $r^2 = 0.81$: about 81% of the variation in y is accounted for by the linear regression on x. One thing it does not tell you is whether x causes y (nor whether the line predicts safely outside the data range).

18 (i) How were the schools chosen — volunteers or a random sample — and were the two groups comparable at the start? (ii) How many schools and pupils were involved, and what spread or confidence interval surrounds the 15%? (iii) Was there a proper control group, who funded the study, was it peer-reviewed, and were any dropouts excluded?

19 Let the original mean be $\bar{x}$. The shifted values $y_i = x_i + k$ have mean $\bar{y} = \bar{x} + k$, so each deviation is unchanged: $y_i - \bar{y} = (x_i + k) - (\bar{x} + k) = x_i - \bar{x}$. Therefore $\sum(y_i - \bar{y})^2 = \sum(x_i - \bar{x})^2$, and the variance $\frac{1}{n}\sum(y - \bar{y})^2$ and SD equal the originals. Adding k slides the whole distribution along the axis without changing how far each value sits from the centre, so spread is preserved while the mean, median and mode each move by k.

20 One valid set is $7,\ 9,\ 10,\ 11,\ 13$: mean $(7 + 9 + 10 + 11 + 13) \div 5 = 50 \div 5 = 10$. Deviations from 10 are $-3,\ -1,\ 0,\ 1,\ 3$; their squares are $9,\ 1,\ 0,\ 1,\ 9$ and sum to 20, so population variance $s^2 = 20 \div 5 = 4$ and SD $s = \sqrt{4} = 2$, as required. The answer is not unique. Another valid set is $9,\ 9,\ 9,\ 9,\ 14$ (deviations $-1,\ -1,\ -1,\ -1,\ 4$; squares sum to $1 + 1 + 1 + 1 + 16 = 20$, same SD). In general any five numbers summing to 50 with $\sum(x_i - 10)^2 = 20$ works, so there are infinitely many such lists (even up to permutation and allowing non-integers).

Criterion A · Knowing and understanding

1 Ordered: $4, 7, 7, 9, 11, 13, 16, 18, 40$. Five-number summary: min 4, $Q_1 = 7$ (median of $4, 7, 7, 9$), median 11, $Q_3 = 17$ (median of $13, 16, 18, 40$), max 40. IQR $= 17 - 7 = 10$ min. Range $= 40 - 4 = 36$ min. Upper fence $= 17 + 1.5(10) = 32$ min; since $40 > 32$ min, 40 min is a potential outlier.

Marking guidance: [4] one for the five-number summary, one for $\text{IQR} = 10$ *min, one for range* $= 36$ *min, one for the fence at* 32 *min and the outlier conclusion. Top-band answers use the halves convention stated in 9.1, include the units, and note that* 40 *min is only a flag, not proof of an error.*

2 $\sum x = 50$, $\bar{x} = 10$ min. Deviations $-4, -2, -1, 1, 6$; squared deviations $16, 4, 1, 1, 36$ sum 58. $s^2 = 58 \div 5 = 11.6$, $s = \sqrt{11.6} = 3.41$ min (3 s.f.).

Marking guidance: [5] one for $\bar{x} = 10$*, one for the deviation column summing to* 0*, one for the squared-deviation column summing to* 58*, one for* $s^2 = 11.6$*, one for* $s = 3.41$ *min. Top-band answers state the divisor convention and use minutes.*

3 Midpoints $5, 15, 25, 35$. $\sum f = 30$, $\sum fx = 510$, $\sum fx^2 = 10\,750$. $\bar{x} = 510 \div 30 = 17$ min. $s^2 = 10750 \div 30 - 17^2 = 358.33 - 289 = 69.3$, so $s = 8.33$ min (3 s.f.).

Marking guidance: [5] one for the midpoint column, one for $\sum fx = 510$ *and mean* 17 *min, one for* $\sum fx^2 = 10\,750$*, one for* $s^2 \approx 69.3$*, one for* $s \approx 8.33$ *min. Top-band answers recognise both are estimates that place every value at its midpoint.*

4 Mean 24.6 min is the typical length of a journey averaged over all journeys; standard deviation 3.1 min is the typical distance of an individual journey time from 24.6 min. Most journeys fall in a band around 24.6 min, roughly 21.5 to 27.7 min within one standard deviation, so a journey is unlikely to differ from average by, say, an hour.

Marking guidance: [3] one for the mean, one for the SD, one for the units and a contextual comparison. Top-band answers give a specific range and avoid claiming every value lies within one SD.

5 (a) Adding 5 to every value shifts the centre only: new mean 25, new standard deviation still 4. (b) Doubling every value scales both centre and spread: new mean 40, new standard deviation 8.

Marking guidance: [3] one for (a), one for (b), one for the explanation in terms of deviations ($(x + 5) - (\bar{x} + 5) = x - \bar{x}$*;* $2x - 2\bar{x} = 2(x - \bar{x})$*). Top-band answers justify by the deviation formula, not by re-calculation.*

Criterion B · Investigating patterns

1 Plot the ten points on labelled axes (x hours of revision, y score). The association is positive and roughly linear apart from one point: the cloud rises, but $(10, 18)$ sits well above every other point and above the general line. Direction positive, form linear, strength moderate once the outlier is considered, and the point $(10, 18)$ is the one that does not fit.

Marking guidance: [4] one for a correctly scaled labelled scatter, one for plotting every point accurately, one for naming the direction and form, one for identifying $(10, 18)$ *as the off-pattern point. Top-band answers describe all four features (direction, strength, form, outliers).*

2 $\sum y = 72$, so $\bar{y} = 7.2$. Deviations $y - \bar{y}$: $-4.2, -2.2, -3.2, -1.2, -0.2, -2.2, 0.8, 1.8, -0.2, 10.8$. Squares: $17.64, 4.84, 10.24, 1.44, 0.04, 4.84, 0.64, 3.24, 0.04, 116.64$, total 159.60. $s^2 = 159.6 \div 10 = 15.96$, $s = 4.00$ (to 3 s.f.; using $n - 1$, $s = \sqrt{17.73} = 4.21$).

Marking guidance: [4] one for the mean, one for the deviation column, one for the squared-deviation total, one for the SD. Top-band answers state the denominator they used and keep the unrounded values for later parts.

3 $\bar{x} = 5.5$. $S_{xx} = \sum(x - \bar{x})^2 = 82.5$, $S_{yy} = \sum(y - \bar{y})^2 = 159.6$, $S_{xy} = \sum(x - \bar{x})(y - \bar{y}) = 89$. $r = \dfrac{89}{\sqrt{82.5 \times 159.6}} = \dfrac{89}{\sqrt{13\,167}} = 0.776$.

Marking guidance: [5] one for $\bar{x} = 5.5$ *and* $\bar{y} = 7.2$*, one for* $S_{xx} = 82.5$*, one for* $S_{yy} = 159.6$*, one for* $S_{xy} = 89$*, one for* $r = 0.776$*. Top-band answers check that* $|r| \leq 1$ *and give it as a pure number.*

4 For the nine remaining points, $\bar{x} = 5$, $\bar{y} = 6$. $S_{xx} = \sum(x - 5)^2 = 60$, $S_{yy} = \sum(y - 6)^2 = 30$, $S_{xy} = \sum(x - 5)(y - 6) = 35$. Then $s_y^2 = 30 \div 9 = 3.333$, $s_y = 1.83$; and $r = \dfrac{35}{\sqrt{60 \times 30}} = \dfrac{35}{\sqrt{1800}} = 0.825$.

Marking guidance: [6] one for the revised totals, one for $\bar{x}, \bar{y}$*, one for* S_{xx}, S_{yy}, S_{xy} *individually, one for the correct SD, one for* $r = 0.825$*, one for consistent use of the same convention as in parts 2–3. Top-band answers keep the arithmetic unrounded and verify* $|r| \leq 1$*.*

5 Removing $(10, 18)$ lowered the mean from 7.2 to 6.0 marks, lowered the standard deviation from 4.00 to 1.83 marks, and raised r from 0.776 to 0.825. The single point sat at a high x and a much higher y than the rest, so it inflated the centre, spread the values more widely, and — because it did not follow the same line as the other nine — weakened the linear association; the remaining nine fit a tight upward pattern that the outlier obscured.

Marking guidance: [3] one mark per statistic discussed (mean, SD, r) with the correct direction of change and a mechanism. Top-band answers distinguish effects on centre, spread and linear fit.

6 From the ten-point line $y = 1.27 + 1.08x$: $\hat{y}(10) = 1.27 + 10.8 = 12.07$. From the nine-point line $y = 3.08 + 0.58x$: $\hat{y}(10) = 3.08 + 5.8 = 8.88$. The observed y at $x = 10$ is 18, which lies well above both predictions. Both are interpolations for the ten-point fit ($x = 10$ is inside its range) but extrapolations for the nine-point fit, whose data stop at $x = 9$. Neither line can honestly predict a score of 18 — that value is exactly the outlier the ten-point line was fitted through.

Marking guidance: [2] one for both predictions correct and to sensible precision, one for the reliability comment that recognises the different roles of interpolation and extrapolation. Top-band answers note that the outlier drives its own prediction.

Criterion C · Communicating

1 **Model answer.** Weakness 1 — self-selection: the 28 attenders were volunteers, so they were probably more motivated or better supported than average and would have scored higher even without the club. Weakness 2 — no real comparison group: the non-attenders were never surveyed, so the "11 per cent higher" has no measured baseline to compare with. Weakness 3 — sample size and response: only 28 of 37 members, and attendance may be mis-remembered, so a small or unrepresentative group could show an 11% gap by chance. Each of these makes the attenders look better than the club alone can explain.

Marking guidance: [6] two marks per weakness — one for naming it, one for explaining how it could distort the comparison; top-band answers give three distinct weaknesses (self-selection, missing comparison group, small sample or response bias) in language a non-specialist can follow.

2 **Model answer.** A standard deviation is the typical distance between the individual values and their average — a small SD means the scores cluster closely around the mean, a large SD means they are spread out; it is measured in the same units as the data (marks here), not a percentage. A correlation coefficient, r, is a number between -1 and $+1$ that says how strongly and in which direction two quantities move together in a straight line. Neither statistic can show that breakfast caused the higher scores: both only describe a pattern that was observed. Because the attenders chose to attend, some third factor — family routine, motivation or prior attainment — could raise both attendance and scores, so the association need not be a cause. Correlation, however strong, is not causation.

Marking guidance: [8] about four marks for accurate plain-language definitions of standard deviation and of r (with units/range stated and no claim that r is a percentage), and about four for a clear association-versus-causation argument tied to this study's design; top-band answers suit the non-specialist audience and avoid unexplained jargon.

3 **Sample response.** Our newsletter reported that breakfast-club attenders scored 11% higher. We must be careful about what that proves. The 11% is a genuine difference we saw in the 28 pupils who chose to come, and we can fairly say that these pupils tended to do better. We cannot honestly say the club made them do better. The pupils who attended were not chosen at random — they volunteered — so they may already have been more motivated or better supported at home, and those qualities by themselves can raise exam scores. We also never measured the children who did not attend, so there is no proper comparison group, and 28 pupils from one club is a small number from which to draw a school-wide conclusion. What the school can claim is an association worth investigating; what it cannot claim is a cause. To make a future study trustworthy, we should compare the club members with a matched group of non-members chosen at random from the same year, of similar prior attainment, and record everyone's scores. Better still, we could offer the club to a randomly chosen half of the year and keep the other half as a control, so the only systematic difference is the club itself. That design would let us talk about cause rather than coincidence.

Marking guidance: [6] for a coherent 300–350-word parent-facing response that states clearly what the school can and cannot claim, uses statistical ideas in plain language, and proposes one concrete design improvement (a matched or randomised comparison group); top-band answers are measured, well organised and free of unsupported causal language.

Criterion D · Applying mathematics in real-life contexts

1 Plot x (age, years) against y (price, thousands of pounds). The association is negative, strong and approximately linear: prices fall as age rises, with the eight points lying close to a downward band. The point $(3, 15.1)$ sits just above the line and $(5, 12.3)$ slightly above, but neither is a serious outlier; the trend runs from £18.4k at age 1 down to £5.9k at age 8.

Marking guidance: [4] one for a correctly labelled scatter with all points, one for direction, one for strength and form, one for commenting on any unusual point; top-band answers name all four features in context (thousands of pounds per year of age).

2 $\bar{x} = 4.5$, $\bar{y} = 11.9$, $S_{xx} = 42$, $S_{yy} = 124.20$, $S_{xy} = -70.7$. $r = \dfrac{-70.7}{\sqrt{42 \times 124.20}} = \dfrac{-70.7}{72.20} = -0.979$. The strong negative value means older cars of this model are reliably cheaper, and the linear pattern fits the observed ages 1 to 8 very closely.

Marking guidance: [4] one for each of S_{xx}, S_{yy}, S_{xy} and one for $r = -0.979$ (accept -0.98); the sign must be negative and $|r| \leq 1$; top-band answers check this and interpret direction and strength in context.

3 $b = \dfrac{S_{xy}}{S_{xx}} = \dfrac{-70.7}{42} = -1.6833$, $a = \bar{y} - b\bar{x} = 11.9 - (-1.6833)(4.5) = 19.475$. To 3 s.f.: $y = 19.5 - 1.68x$. Each extra year of age is associated with about £1.68k less price; the intercept £19.5k would be the model's (extrapolated) value at age 0.

Marking guidance: [5] one for the substitution of $b = S_{xy}/S_{xx}$, one for the correct b, one for $a = \bar{y} - b\bar{x}$, one for the correct a, one for the finished equation to 3 s.f.; the line must pass through $(\bar{x}, \bar{y}) = (4.5, 11.9)$.

4 At $x = 5.5$: $y = 19.475 - 1.6833(5.5) = 10.2$, i.e. about £10.2k — an interpolation inside $1 \leq x \leq 8$, so it is reasonably reliable. At $x = 12$: $y = 19.475 - 1.6833(12) = -0.73$, a negative price — an extrapolation well beyond age 8, and clearly unreliable, since depreciation cannot stay linear and a car's value never falls below zero.

Marking guidance: [5] two for the 5.5-year prediction with interpolation identified, two for the 12-year result with extrapolation identified, one for noting the negative value is impossible; top-band answers judge reliability from the range, not just the arithmetic.

5 At $x = 4$: $\hat{y} = 19.475 - 1.6833(4) = 12.74$, so the residual is $11.4 - 12.74 = -1.34$ (thousand pounds). The negative residual means this car is priced about £1.34k below the line — cheaper than the model expects for its age. $r^2 = 0.979^2 = 0.958$: about 95.8% of the variation in price is explained by age. It does not tell the dealer anything about other factors (mileage, condition, brand demand) or whether the relationship is causal.

Marking guidance: [3] one for the residual -1.34 with correct sign, one for interpreting that sign, one for $r^2 = 0.958$ plus a valid limitation; top-band answers note that r^2 measures fit, not cause or omitted variables.

6 Assumption: that price falls linearly with age alone, at a constant rate, in the range 1 to 8 years. Omitted variable: mileage (or condition/service history or market demand), which strongly affects price. Reason not to use the line for a new car: a new car has $x = 0$, outside the fitted range, so the prediction is extrapolation and the intercept £19.5k ignores the large first-year depreciation and dealer margins that a brand-new price includes — the model gives no authority there.

Marking guidance: [3] one for a stated assumption, one for a named omitted variable, one for a reason the line should not price a new car (extrapolation beyond the data); top-band answers link each point to the real pricing context.

CHAPTER 10 · PROBABILITY: CONDITIONAL PROBABILITY AND COUNTING

Opening problem · a test, a population and a difficult sentence

1 Estimates vary; what matters is that the 5% false-positive rate applies to about 9,800 people while the 90% detection rate applies to only 200, so false positives (490) outnumber true positives (180) by roughly 3 to 1. An answer near 90% confuses $P(+ \mid C)$ with $P(C \mid +)$; about a quarter is the right order of magnitude.

2 Of the 200 with the condition $0.90 \times 200 = 180$ test positive and 20 negative; of the 9,800 without, $0.05 \times 9800 = 490$ test positive and 9,310 negative. Cells 180, 20, 490, 9,310; margins 670 positive, 9,330 negative, and $670 + 9330 = 200 + 9800 = 10\,000$.

3 On the restricted space of the 670 positives, $P(C \mid +) = \dfrac{180}{670} = \dfrac{18}{67} \approx 26.9\%$ — far below 90%, and not 50% either.
Complement: $P(C' \mid +) = \dfrac{49}{67} \approx 73.1\%$, and the two sum to 1.

4 *Accurate:* "Of every 100 people with a result like yours, about 27 have the condition and 73 do not; this test does not diagnose, so we are arranging a confirmatory test."
Misleading: "Your test is 90% accurate, so you have a serious condition." The second reverses the conditional: it names the 200 people who have the condition instead of the 670 who test positive. The letter should also state the next step.

Prerequisite check

1 $\dfrac{5}{5+3} = \dfrac{5}{8} = 0.625$ of the counters are red.

2 $\dfrac{5}{8} \times \dfrac{4}{7} = \dfrac{20}{56} = \dfrac{5}{14} = 0.35714\ldots$

3 $0.02 \times 10\,000 = 200$.

4 The multiplier is $1 - 0.15 = 0.85$, so the reduced price is $80 \times 0.85 = £68$.

5 $\dfrac{18}{67} = 0.268656\ldots$, which is 0.269 to three significant figures.

6 $n(n-1)(n-2) = 5 \times 4 \times 3 = 60$, the product of three decreasing factors, which is ${}^5P_3$.

Exercise 10A · Sample spaces, tables and the addition rule

1 (a) $\xi = \{1, 2, 3, 4, 5, 6\}$, six equally likely outcomes. (b) $\xi = \{HH, HT, TH, TT\}$, order separating HT from TH. (c) $\xi = \{C_1, H, A, N, C_2, E\}$: six equally likely tiles, five distinct letters, so $P(C) = \dfrac{2}{6} = \dfrac{1}{3}$ and each other letter has probability $\dfrac{1}{6}$.

2 A=\{2,4,6} and $B = \{2, 3, 5\}$, so $A \cup B = \{2, 3, 4, 5, 6\}$ with $P = \dfrac{5}{6}$, $A \cap B = \{2\}$ with $P = \dfrac{1}{6}$, and $A' = \{1, 3, 5\}$ with $P = \dfrac{1}{2}$. Note $\dfrac{1}{2} + \dfrac{1}{6} \neq \dfrac{5}{6}$, because A and B overlap at 2.

3 $n(H \cup G) = 14 + 9 - 4 = 19$, so $25 - 19 = 6$ students study neither.

4 The cells sum $35 + 15 + 25 + 25 = 100$. $P(T) = \dfrac{50}{100} = \dfrac{1}{2}$, $P(C) = \dfrac{60}{100} = \dfrac{3}{5}$, $P(T \cap C) = \dfrac{35}{100} = \dfrac{7}{20}$, $P(T \cup C) = \dfrac{75}{100} = \dfrac{3}{4}$; the addition rule returns $\dfrac{1}{2} + \dfrac{3}{5} - \dfrac{7}{20} = \dfrac{3}{4}$.

5 Mutually exclusive, so $P(A \cup B) = 0.35 + 0.4 = 0.75$, $P(A') = 1 - 0.35 = 0.65$, and $A' \cap B'$ ("neither") is the complement of $A \cup B$, so $P(A' \cap B') = 1 - 0.75 = 0.25$.

6 Relative frequency $\dfrac{36}{80} = 0.45$, which is 0.05 below the theoretical $\dfrac{1}{2}$. No: for 80 tosses the standard deviation is $\sqrt{80 \times 0.25} \approx 4.5$, so 36 heads against an expectation of 40 is under one deviation — sampling variation, not proof of bias.

7 Error 1: the probabilities are added without subtracting $P(A \cap B)$, assuming an unstated mutual exclusivity. Error 2: a probability cannot exceed 1, so 1.1 is impossible. Needed: $P(A \cap B)$ (or a statement that A and B are mutually exclusive); then $P(A \cup B) = 1.1 - P(A \cap B)$ forces $P(A \cap B) \geq 0.1$.

8 $A \cap A' = \varnothing$ and $A \cup A' = \xi$: the two events split the sample space into two pieces, so adding them gives $P(A) + P(A') = P(\xi) = 1$, that is $P(A') = 1 - P(A)$.

9 Expected payout $= 0.2 \times £3 = £0.60$ against a £1 stake, so the expected net gain is $-£0.40$ a game: not fair, and it favours the operator. A fair game needs a prize of $\dfrac{1}{0.2} = £5$.

10 No: the "tea and coffee" cell holds 35 people, so $T \cap C \neq \varnothing$ and $P(T \cap C) = 0.35 \neq 0$. Mutual exclusivity needs an empty intersection cell, not merely a small probability.

11 It can rain on both days, so the events are not mutually exclusive and $P(\text{Sat} \cup \text{Sun}) = 0.3 + 0.3 - P(\text{both})$, which is 60% only if $P(\text{both}) = 0$. Consecutive days are positively correlated, so $P(\text{both}) > 0$ and the weekend chance is below 60%; on independent days it is $0.6 - 0.09 = 0.51$, that is 51%.

12 Theoretical probability comes from the model: for a fair die $P(\text{six}) = \dfrac{1}{6}$, so 300 rolls predict 50 sixes. Experimental probability is the observed relative frequency, e.g. $\dfrac{58}{300} \approx 0.193$. It settles toward $\dfrac{1}{6}$ as rolls increase but varies in a finite run (standard deviation $\sqrt{300 \times \dfrac{1}{6} \times \dfrac{5}{6}} \approx 6.5$), so 58 sixes is not evidence of bias.

13 Walk only $120 - 40 = 80$, both 40, cycle only $70 - 40 = 30$, so 150 use one or both and 50 neither: $P(\text{walk or cycle}) = \dfrac{150}{200} = \dfrac{3}{4} = 0.75$. The Venn regions 80, 40, 30, 50 sum to 200; check $\dfrac{120}{200} + \dfrac{70}{200} - \dfrac{40}{200} = \dfrac{150}{200}$.

14 ξ is the 36 ordered pairs. A total of 7 arises 6 ways and totals of at least 10 arise $3 + 2 + 1 = 6$ ways, so both probabilities are $\dfrac{6}{36} = \dfrac{1}{6}$. A fair £1 bet must return £1 in expectation, so the payout is $\dfrac{1}{1/6} = £6$.

15 Adding $P(A) + P(B) + P(C)$ counts each pairwise overlap twice, so those are subtracted; that removes the triple overlap three times although it was added three times, so it returns once. With pairwise disjoint events every intersection term is 0 and the rule reduces to $P(A \cup B \cup C) = P(A) + P(B) + P(C)$.

Exercise 10B · Conditioning, reversing and natural frequencies

1 Conditioning restricts to one row or column of the recycling table (60 under-30 recyclers, 40 under-30 non-recyclers, 105 older recyclers, 45 older non-recyclers; margins 100 under 30, 150 older, 165 recyclers, 250 in all). $P(R \mid U) = \dfrac{60}{100} = \dfrac{3}{5} = 0.6$, $P(U \mid R) = \dfrac{60}{165} = \dfrac{4}{11} = 0.3636\ldots$, $P(R \mid U') = \dfrac{105}{150} = 0.7$. The first two share a numerator but not a denominator, so they describe different groups.

2 Given a dog owner, 5 of the 12 own a cat: $P(\text{cat} \mid \text{dog}) = \dfrac{5}{12} = 0.4166\ldots$. Given a cat owner, 5 of the 20 own a dog: $P(\text{dog} \mid \text{cat}) = \dfrac{5}{20} = \dfrac{1}{4} = 0.25$.

3 $P(A \mid B) = \dfrac{0.2}{0.5} = 0.4$ and $P(B \mid A) = \dfrac{0.2}{0.4} = 0.5$. Since $0.4 \times 0.5 = 0.2 = P(A \cap B)$ the events happen to be independent, yet the two conditionals still differ.

4 By the multiplication law $P(A \cap B) = P(A \mid B)P(B) = 0.3 \times 0.6 = 0.18$.

5 $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.12}{0.4} = 0.3$.

6 Let P be phone and L laptop. $P(P \mid L) = \dfrac{0.30}{0.45} = \dfrac{2}{3} = 0.666\ldots$ and $P(L \mid P) = \dfrac{0.30}{0.60} = \dfrac{1}{2}$.

7 $P(A \mid B) = \dfrac{0.1}{0.2} = 0.5$ and $P(B \mid A) = \dfrac{0.1}{0.5} = 0.2$: not equal. The error is reversing the condition — one intersection rescaled by two denominators. Here $P(A \mid B) = P(A) = 0.5$, so A and B are independent, and the student has mistaken "the condition does not change the probability" for "the two conditionals are equal".

8 $P(+ \mid C) = \frac{90}{100} = 0.9$, the sensitivity; positives total $90 + 50 = 140$, so $P(C \mid +) = \frac{90}{140} = \frac{9}{14} \approx 64\%$. The two answer different questions: 90% of the 100 people with the condition test positive, but only 64% of the 140 positives are correct, since the extra 50 come from the 900 healthy people.

9 Given B, outcomes outside B are discarded and the sample space becomes that row or column: of its 40 people 15 are in both, so $P(A \mid B) = \frac{15}{40}$ rather than $\frac{15}{100}$. Inside the row the parts $A \cap B$ and $A' \cap B$ are $\frac{15}{40}$ and $\frac{25}{40}$, still summing to 1.

10 Not independent: independence would need $P(A \mid B) = P(A \mid B')$, and $0.4 \neq 0.7$, so knowing B changes the chance of A. Also, for $0 < P(B) < 1$, $P(A) = 0.4P(B) + 0.7P(B')$ lies strictly between the two values, so it equals neither conditional.

11 From $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, defined when $P(B) > 0$, multiply by $P(B)$: $P(A \cap B) = P(A \mid B)P(B)$. Interchanging A and B in the same definition gives $P(A \cap B) = P(B \mid A)P(A)$, so the two products are equal.

12 A bag of 3 red and 2 blue counters, two drawn without replacement; let A be "second is red" and B be "first is blue". Then $P(A) = \frac{3}{5}$ (the second draw has the same marginal distribution as the first) but $P(A \mid B) = \frac{3}{4}$, since removing a blue leaves all 3 reds among 4 counters. Thus $P(A \mid B) > P(A)$: a positive association between A and B', not independence.

13 Prevalence 0.5% of $50\,000 = 250$ affected and 49,750 not; $0.98 \times 250 = 245$ test positive (5 negative) and $0.04 \times 49\,750 = 1990$ test positive (47,760 negative), giving margins $245 + 1990 = 2235$ positive and $5 + 47\,760 = 47\,765$ negative, totalling 50,000. Then $P(C \mid +) = \frac{245}{2235} = \frac{49}{447} \approx 11.0\%$. To a patient: "of every 100 positive first tests about 11 have the condition, so yours needs a confirmatory test."

14 Cells: both 16, on time only 304, damaged only 24, neither 56, summing to 400. $P(\text{on time} \mid \text{damaged}) = \frac{16}{40} = \frac{2}{5} = 0.4$ against $P(\text{damaged} \mid \text{on time}) = \frac{16}{320} = \frac{1}{20} = 0.05$. The second is more useful to the company, since 5% is the rate at which apparently successful deliveries are damaged and handling changes can lower it; the first matters to a customer reporting damage. The denominators differ: the 40 damaged parcels against the 320 on-time ones.

15 Within B the events A and A' are disjoint and fill B, so $P(A \cap B) + P(A' \cap B) = P(B)$; dividing by $P(B) > 0$ gives $P(A \mid B) + P(A' \mid B) = 1$. But $P(A \mid B) + P(A \mid B') = 1$ is false in general, because the terms have different denominators and both count A; the correct form is $P(A) = P(A \mid B)P(B) + P(A \mid B')P(B')$. Counterexample from the recycling table: $0.6 + 0.7 = 1.3 \neq 1$.

Exercise 10C · Independence and repeated trials

1 Independent, so $P(A \cap B) = P(A)P(B) = 0.5 \times 0.4 = 0.2$, and $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

2 $P(A)P(B) = 0.3 \times 0.6 = 0.18 = P(A \cap B)$, so the product rule holds and A and B are **independent**; equivalently $P(A \mid B) = \frac{0.18}{0.6} = 0.3 = P(A)$.

3 Independent tosses: $P(\text{three heads}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$; exactly two heads on HHT, HTH, THH, so $3 \times \frac{1}{8} = \frac{3}{8}$; at least one head $= 1 - P(TTT) = \frac{7}{8}$. Check: $\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$.

4 Rolls independent: $P(\text{six, then six}) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$ and $P(\text{no six}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$, so $P(\text{at least one six}) = \frac{11}{36} = 0.3055\ldots$, matching the count $6 + 6 - 1 = 11$ of ordered pairs containing a six.

5 $P(L) = \frac{100}{200} = 0.5$, $P(O) = \frac{120}{200} = 0.6$, $P(L \cap O) = \frac{60}{200} = 0.3$. Since $0.5 \times 0.6 = 0.3 = P(L \cap O)$, loyalty-card ownership and online shopping are **independent** in this sample; $P(L \mid O) = \frac{60}{120} = 0.5 = P(L)$ agrees.

6 Mutually exclusive — yes, since $P(A \cap B) = 0$ with both probabilities positive. Independent — no: that would need $P(A \cap B) = 0.4 \times 0.5 = 0.2 \neq 0$; indeed $P(A \mid B) = 0 \neq P(A)$, so the events are strongly dependent.

7 The student has confused mutual exclusivity with independence. Exclusive events cannot happen together, so $P(A \mid B) = 0 \neq P(A)$ — dependent whenever both probabilities are positive — while independence requires overlap with $P(A \cap B) = P(A)P(B)$. Die example: $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$ are independent ($P(A \cap B) = \frac{1}{3} = \frac{2}{3} \times \frac{1}{2}$) yet overlap; $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$ are the exclusive pair, with $0 \neq \frac{1}{4}$.

8 $P(\text{left-handed}) = \frac{120}{300} = 0.4$, $P(\text{colour-blind}) = \frac{90}{300} = 0.3$, $P(\text{both}) = \frac{45}{300} = 0.15$. Since $0.4 \times 0.3 = 0.12 \neq 0.15$ the events are **not independent**: the overlap holds 45 people where 36 are expected, and $P(\text{left-handed} \mid \text{colour-blind}) = \frac{45}{90} = 0.5$ against 0.4 overall.

9 Mutually exclusive gives $P(A \cap B) = 0$; independence gives $P(A \cap B) = P(A)P(B)$, which is positive when both probabilities are. So for $P(A) > 0$ and $P(B) > 0$ the two cannot both hold; equivalently $P(A \mid B) = 0 \neq P(A) - B$ removes A altogether, the strongest possible dependence.

10 $P(\text{at least one six}) = 1 - \left(\frac{5}{6}\right)^n > 0.99$ needs $\left(\frac{5}{6}\right)^n < 0.01$, so $n > \frac{\ln 0.01}{\ln(5/6)} = \frac{-4.60517}{-0.182322} = 25.26$: smallest whole number $n = 26$. Check: $1 - (5/6)^{25} = 0.9895 < 0.99$ and $1 - (5/6)^{26} = 0.9913 > 0.99$.

11 Since $B = (A \cap B) \cup (A' \cap B)$ disjointly, $P(A' \cap B) = P(B) - P(A \cap B) = P(B) - P(A)P(B) = P(B)(1 - P(A)) = P(B)P(A')$ — exactly the product rule for A' and B, so they are independent.

12 With replacement the bag is restored, so for 3 red and 2 blue $P(R_2 \mid R_1) = \frac{3}{5} = P(R_2)$ and the draws are independent. Without replacement $P(R_2 \mid R_1) = \frac{2}{4} = \frac{1}{2}$ and $P(R_2 \mid B_1) = \frac{3}{4}$, neither equal to $P(R_2) = \frac{3}{5}$, so they are dependent. Note that $P(R_2) = \frac{3}{5} = P(R_1)$ still holds, so equal single-draw probabilities do not prove independence.

13 Independence gives $P(\text{first two defective}) = 0.02 \times 0.02 = 0.0004 = \frac{1}{2500}$. The complement gives $P(\text{at least one of 50}) = 1 - 0.98^{50} = 1 - 0.36417 \approx 0.636$, so a batch of 50 is more likely than not to contain a fault.

14 $P(\text{three in a row}) = 0.8^3 = 0.512$; $P(\text{at least one in four}) = 1 - 0.2^4 = 0.9984$. The assumption can fail because attempts are not independent: confidence, fatigue or a crowd reacting to earlier attempts change the next shot, so successes cluster.

15 $P(A \mid B) = P(A)$ with $P(B) > 0$ gives $P(A \cap B) = P(A \mid B)P(B) = P(A)P(B)$, and since $P(A) > 0$ division is legal, $P(B \mid A) = \frac{P(A)P(B)}{P(A)} = P(B)$ — independence is symmetric. The condition is needed because $P(B \mid A)$ divides by $P(A)$: if $P(A) = 0$ the ratio is undefined.

Exercise 10D · Trees, total probability and reverse conditioning

1 Drawn without replacement from 3 red and 2 blue, so the second stage uses 4 counters: $P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$, $P(BB) = \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$, and exactly one red is RB or BR, $2 \times \frac{3}{5} \times \frac{2}{4} = \frac{3}{5}$. Check: $\frac{3}{10} + \frac{1}{10} + \frac{3}{5} = 1$.

2 With replacement both stages use $\frac{3}{5}$ and $\frac{2}{5}$ and the branches are independent: $P(RR) = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$, $P(BB) = \frac{4}{25}$, $P(\text{exactly one red}) = 2 \times \frac{3}{5} \times \frac{2}{5} = \frac{12}{25}$. Check: $\frac{9 + 4 + 12}{25} = 1$.

3 $P(HHH) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$. Exactly two heads: 3 paths $\times \frac{1}{8} = \frac{3}{8}$. At least one head: $1 - P(TTT) = 1 - \frac{1}{8} = \frac{7}{8}$.

4 Independent germination with $p = 0.9$: $P(\text{all three}) = 0.9^3 = 0.729$, $P(\text{none}) = 0.1^3 = 0.001$, $P(\text{exactly two}) = 3 \times 0.9^2 \times 0.1 = 0.243$, the factor 3 counting which seed fails. Check with exactly one: $0.729 + 0.243 + 0.027 + 0.001 = 1$.

5 Total probability: $P(D) = 0.7 \times 0.03 + 0.3 \times 0.06 = 0.021 + 0.018 = 0.039$. Reversing: $P(A \mid D) = \dfrac{0.021}{0.039} = \dfrac{21}{39} = \dfrac{7}{13} \approx 53.8\%$. A supplies most components and most defectives, yet $P(D \mid A) = 0.03$ is still below $P(D \mid B) = 0.06$.

6 "Same colour" is RR or BB, drawn without replacement from 8 counters: $2 \times \dfrac{4}{8} \times \dfrac{3}{7} = \dfrac{24}{56} = \dfrac{3}{7} = 0.42857\ldots$, below $\dfrac{1}{2}$ because removing one counter of a colour makes the other colour likelier next time.

7 $\dfrac{67}{56} > 1$, and no probability can exceed 1, so the value is impossible — the tell-tale of the error. The branches $\dfrac{5}{8}$ and $\dfrac{4}{7}$ are successive stages of one path, so they must be multiplied: $P(RR) = \dfrac{5}{8} \times \dfrac{4}{7} = \dfrac{20}{56} = \dfrac{5}{14} = 0.35714\ldots$

8 The branches leaving a node are the exhaustive, mutually exclusive ways the next stage can happen *given* that node, so their conditional probabilities sum to 1; multiplying along paths and adding the disjoint path sets, the final path probabilities sum to 1 too — for 5 red and 3 blue, $\dfrac{20 + 15 + 15 + 6}{56} = 1$. This is the tree form of $P(\xi) = 1$, and a cheap check on a diagram.

9 $P(\text{at least one tail}) = 1 - P(HHHH) = 1 - \left(\dfrac{1}{2}\right)^4 = 1 - \dfrac{1}{16} = \dfrac{15}{16} = 0.9375$. The complement turns a sum of 15 paths into one, because "at least one" is everything except the single outcome $HHHH$.

10 Natural frequencies: A supplies 700 items of which $0.03 \times 700 = 21$ are defective and B supplies 300 of which $0.06 \times 300 = 18$ are defective, so $P(D) = \dfrac{39}{1000} = 0.039$. Given D the sample space is those 39, of which 21 come from A, so $P(A \mid D) = \dfrac{21}{39} = \dfrac{7}{13} \approx 0.538$ — the same values as the tree in Question 5.

11 Three stages, drawn without replacement: branches $\dfrac{2}{3}, \dfrac{1}{3}$ at the root, then $\dfrac{1}{2}$ or 1, then 1. Blue last is the path RRB, $P(RRB) = \dfrac{2}{3} \times \dfrac{1}{2} \times 1 = \dfrac{1}{3}$; the orders RBR and BRR also give $\dfrac{1}{3}$ each, and the three sum to 1. Hence $P(\text{blue last}) = \dfrac{1}{3}$, since the single blue counter is equally likely in any of the three positions.

12 Without replacement $P(RR) = \dfrac{5}{8} \times \dfrac{4}{7} = \dfrac{5}{14} \approx 0.357$; with replacement $P(RR) = \left(\dfrac{5}{8}\right)^2 = \dfrac{25}{64} \approx 0.391$; the difference is $\dfrac{25}{64} - \dfrac{5}{14} = \dfrac{15}{448} \approx 0.034$. Close because only one of eight counters leaves the bag, but not equal: after a red is taken the proportion falls from $\dfrac{5}{8} = 0.625$ to $\dfrac{4}{7} \approx 0.571$, so the first red lowers the chance of the second — dependence.

13 A randomly chosen match is home with probability $\dfrac{15}{35}$ and away with $\dfrac{20}{35}$, so by total probability $P(\text{win}) = \dfrac{15}{35} \times 0.6 + \dfrac{20}{35} \times 0.4 = \dfrac{9 + 8}{35} = \dfrac{17}{35} = 0.48571\ldots \approx 48.6\%$. Check by expected wins: $15 \times 0.6 + 20 \times 0.4 = 9 + 8 = 17$ wins from 35 matches.

14 Of 10,000 people $0.04 \times 10\,000 = 400$ have the condition and 9,600 do not; $0.85 \times 400 = 340$ and $0.10 \times 9600 = 960$ test positive, so the margins are $340 + 960 = 1300$ positive and $60 + 8640 = 8700$ negative, totalling 10,000. Then $P(C \mid +) = \dfrac{340}{1300} = \dfrac{17}{65} \approx 26.2\%$. Even at a 4% base rate with 85% detection, 960 of the positives are false because 9,600 healthy people are tested, so a positive result must be confirmed, not announced.

15 By definition $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$, and the multiplication law gives $P(A \cap B) = P(A \mid B)P(B)$. Since B and B' partition the sample space, $A = (A \cap B) \cup (A \cap B')$ disjointly, so $P(A) = P(A \mid B)P(B) + P(A \mid B')P(B')$, which is total probability; substituting both expressions gives the stated formula. The base rate cannot be omitted: the numerator is only the true-positive stream, while the denominator adds the false-positive stream, which dominates whenever B is rare.

Exercise 10E · Counting arrangements and selections

1 (a) $6! = 720$. (b) $\dfrac{9!}{7!} = 9 \times 8 = 72$. (c) ${}^6P_2 = \dfrac{6!}{4!} = 6 \times 5 = 30$, ordered. (d) ${}^6C_2 = \dfrac{6 \times 5}{2!} = 15$, unordered; note ${}^6P_2 = 2!\,{}^6C_2$.

2 A code is an ordered list, so ${}^{10}P_5 = 10 \times 9 \times 8 \times 7 \times 6 = 30\,240$ codes with five different letters. With repetition allowed each of the 5 positions has 10 independent choices, so $10^5 = 100\,000$.

3 Order matters in a row, so $8! = 40\,320$ orders. With a particular person at an end: 2 choices of end and $7! = 5040$ orders for the rest, giving $2 \times 5040 = 10\,080$.

4 (a) ${}^9C_4 = \dfrac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$. (b) ${}^{15}C_2 = \dfrac{15 \times 14}{2} = 105$. (c) ${}^9P_4 = 3024$. (d) ${}^{15}P_2 = 210$. Check (a) against (c): $126 \times 4! = 3024$.

5 (a) BANANA has 6 letters with A three times and N twice, so $\dfrac{6!}{3!\,2!} = \dfrac{720}{12} = 60$. (b) LEVEL has 5 letters with L twice and E twice, so $\dfrac{5!}{2!\,2!} = 30$. Dividing by the repeated factorials removes orders that are indistinguishable.

6 A team is a selection, not an ordering, so ${}^7C_3 = 35$ teams. Teams including a particular player: fix that player and choose the other 2 from the remaining 6, ${}^6C_2 = 15$, which is $\dfrac{15}{35} = \dfrac{3}{7}$ of all teams.

7 The value ${}^8P_3 = 8 \times 7 \times 6 = 336$ is right, but each selection of 3 corresponds to $3! = 6$ orders, so one must divide by 6, not by 3: ${}^8C_3 = \dfrac{336}{6} = 56$. The error uses the number of objects chosen instead of the number of their arrangements.

8 Choosing r objects to take decides, at the same moment, which $n - r$ are left out, and every choice of $n - r$ determines exactly one choice of r: a one-to-one correspondence, so ${}^nC_r = {}^nC_{n-r}$. Algebraically both equal $\dfrac{n!}{r!\,(n-r)!}$, symmetric in r and $n - r$.

9 Total committees ${}^{11}C_4 = 330$, unordered because a committee has no positions. Exactly 2 women: ${}^7C_2 \times {}^4C_2 = 21 \times 6 = 126$, so $P = \dfrac{126}{330} = \dfrac{21}{55} \approx 0.382$. At least one man: the complement is all-women, ${}^7C_4 = 35$, so $P = 1 - \dfrac{35}{330} = \dfrac{59}{66} \approx 0.894$. Split check: $35 + 140 + 126 + 28 + 1 = 330$.

10 (a) Toppings are a selection and order is irrelevant: ${}^8C_3 = 56$ pizzas. (b) Books on a shelf are an arrangement and order matters: ${}^8P_3 = 336$ displays, which is $3! = 6$ times as many as the selections.

11 Sample space of full finishing orders: 10!. Favourable: the runner takes one of 3 places and the other 9 runners fill the rest in 9! ways, so $P = \dfrac{3 \times 9!}{10!} = \dfrac{3}{10} = 0.3$. The same ratio from the ordered top three: ${}^{10}P_3 = 720$ lists, of which $3 \times {}^9P_2 = 216$ contain the runner.

12 The recursion $n! = n(n-1)!$ with $n = 1$ reads $1! = 1 \times 0!$, and $1! = 1$, so $0! = 1$. Consistency elsewhere demands it: ${}^nP_n = \dfrac{n!}{(n-n)!}$ needs $0! = 1$ to give $n!$, and combinatorially there is exactly one way to arrange nothing, as one way to choose nothing (${}^nC_0 = 1$).

13 The 4 letters are distinct and ordered: ${}^{26}P_4 = 26 \times 25 \times 24 \times 23 = 358\,800$. The 2 digits are distinct and ordered: ${}^{10}P_2 = 90$. By the counting principle, $358\,800 \times 90 = 32\,292\,000$ passwords. Beginning with a vowel (taking the vowels to be A, E, I, O, U): $5 \times {}^{25}P_3 = 69\,000$ letter parts, so $69\,000 \times 90 = 6\,210\,000$, which is $\dfrac{5}{26} \approx 0.192$ of the total.

14 A ticket is an unordered set of 6 numbers, so ${}^{45}C_6 = \dfrac{45\times44\times43\times42\times41\times40}{6!} = 8\,145\,060$ tickets. With one winning combination, $P(\text{a single ticket wins}) = \dfrac{1}{8\,145\,060} \approx 1.23\times10^{-7}$, about 1 chance in 8.1 million.

15 Fix one distinguished person among the n. A committee of r either contains them, with the other $r-1$ members from the remaining $n-1$ (${}^{n-1}C_{r-1}$ ways), or excludes them, with all r from those $n-1$ (${}^{n-1}C_r$ ways); the cases are disjoint and exhaustive, so ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$. Then ${}^7C_3 = {}^6C_2 + {}^6C_3 = 15 + 20 = 35$, as direct calculation also gives.

Investigation 10 · shared birthdays

1 "At least two share" is the union of many overlapping cases (one shared pair, two disjoint shared pairs, a triple, and so on); they are not disjoint and their number grows quickly, so the addition rule would need a long inclusion-exclusion chain. The complement "all different" is a single ordered chain of choices with no overlap, so one product gives it and $P(\text{shared}) = 1 - P(\text{all different})$.

2 Person 1 may have any birthday and each later person must avoid all earlier days: $P(\text{all different}) = \dfrac{365}{365}\times\dfrac{364}{365}\times\dfrac{363}{365}\times\dfrac{362}{365}\times\dfrac{361}{365} = 0.9729$, so $P(\text{at least one shared}) = 0.0271$, matching the printed table.

3 For $n=10$ the product runs to $\dfrac{356}{365}$, giving 0.8831 and 0.1169; for $n=20$ it runs to $\dfrac{346}{365}$, giving 0.5886 and 0.4114. Both agree with the printed table, and the pattern is always n factors, the k-th being $\dfrac{365-k+1}{365}$.

4 Extending the table: $n=23$ gives 0.4927 and 0.5073; $n=30$ gives $0.2937, 0.7063$; $n=40$ gives $0.1088, 0.8912$; $n=50$ gives $0.0296, 0.9704$. The rise is slow at first, steepest through the twenties, then flattens toward 1, so the curve is S-shaped. It cannot be linear because each new person multiplies $P(\text{all different})$ by the shrinking factor $\dfrac{365-n+1}{365}$, so the losses compound — in logarithms $-\ln P(\text{all different}) \approx \dfrac{n(n-1)}{730}$, growing like n^2 — and a probability cannot exceed 1.

5 $n=22$: $P(\text{all different}) = \prod_{k=0}^{21}\dfrac{365-k}{365} = 0.5243$, so $P(\text{shared}) = 0.4757 < 0.5$. $n=23$: $P(\text{all different}) = 0.4927$, so $P(\text{shared}) = 0.5073 > 0.5$. The smallest such group is therefore **23** people, because the product crosses $\frac{1}{2}$ between 22 and 23.

6 With a fixed date (yours) each of the other $n-1$ people must miss it: $P(\text{someone matches you}, n=23) = 1 - \left(\dfrac{364}{365}\right)^{22} = 0.0586\ldots$, about 5.9%, far below the 50.7% for any shared pair. Matching you uses 22 pairings all against one specified day, whereas a shared birthday uses all $\binom{23}{2} = 253$ pairs, each free to match on any of 365 days.

7 Seasonal birth patterns make some days likelier, and any departure from uniformity *raises* the chance of a match, so the true probability at $n=23$ is slightly above 0.5073, about 0.51 to 0.52 in real populations; the conclusion "more likely than not" is safe. Twins violate independence but are a small fraction of births and harmless at this precision, and using 365 days instead of 366 changes the probability by about 0.001; the assumption that carries the argument is independence between people. **Generalise.** The product is $\dfrac{{}^{365}P_n}{365^n} = \dfrac{365!}{(365-n)!\,365^n}$, where the numerator counts ordered assignments of n distinct days (order matters because the people are distinguishable) and the denominator counts all 365^n assignments. At $n=366$ the factor $\dfrac{365-366+1}{365} = 0$ appears, so $P(\text{shared}) = 1$: the pigeonhole principle. With 366 equally likely days the threshold stays at $n=23$, where $P(\text{shared}) = 0.5063$. The pair intuition is right: $\binom{23}{2} = 253$ chances, each $\dfrac{1}{365}$, expecting $\dfrac{253}{365} = 0.693$ matches, so $1 - e^{-0.693} \approx 0.500$.

Modelling task · writing the result letter

1 Chosen model: 20 000 adults screened, prevalence 1% (200 with the condition, 19,800 without), sensitivity 95%, false-positive rate 8%. Cells: true positives $0.95\times200 = 190$, false negatives 10, false positives $0.08\times19\,800 = 1584$, true negatives $19\,800 - 1584 = 18\,216$. Margins: 1,774 positive, 18,226 negative, totalling 20,000, so the table closes.

2 From the table, $P(C\mid+) = \dfrac{190}{1774} = \dfrac{95}{887} \approx 10.7\%$ and $P(C'\mid+) = \dfrac{1584}{1774} = \dfrac{792}{887} \approx 89.3\%$, while $P(C\mid-) = \dfrac{10}{18\,226} = \dfrac{5}{9113} \approx 0.055\%$ and $P(C'\mid-) \approx 99.95\%$ (complements $95+792=887$ and $5+9113=9113$). The patient needs $P(C\mid+)$ and $P(C\mid-)$, which are measured on the result they hold; the manufacturer quotes $P(+\mid C) = 95\%$ and $P(-\mid C') = 92\%$, which describe the device, not the person.

3 **Sample letter (about 120 words).** "Your screening result is positive. Please read this carefully: a positive screening result is not a diagnosis. In this programme 20,000 people were tested. Among them, 1,774 results were positive, but only about 190 of those people actually have the condition; roughly 11 in every 100 positive results are correct, and the other 89 come from healthy people whose test reacted anyway. Your own chance of having the condition is therefore about 11%, not 95%. Nothing is decided. We have arranged a confirmatory test, which is far more reliable, and a specialist nurse will go through that result with you and answer your questions. Do not start, change or stop any treatment because of this letter. If you would like to speak to someone before your appointment, please call the number below."

The letter must name the 1,774 positives as the group, never the 200 people with the condition, and must state the next step.

4 Assumption 1: the quoted prevalence of 1% applies to this screened population. Volunteers differ from average, so at 0.5% prevalence $P(C\mid+)$ falls to about 5.6%, while a higher-risk group raises it. Assumption 2: the sensitivity and false-positive rate hold in this setting; if the false-positive rate is really 12%, false positives rise to $0.12\times19\,800 = 2376$, positives total $190+2376 = 2566$, and only $\dfrac{190}{2566} \approx 7.4\%$ are correct. Fairness: anxiety, follow-up procedures and cost fall on about 1,584 wrongly flagged people while the benefit is an early diagnosis for about 190. Recommended change: screen by estimated risk rather than everyone, and confirm every positive before releasing the result.

Mixed practice 10

1 $P(A\cup B) = 0.45 + 0.3 - 0.15 = 0.6$ and $P(A\mid B) = \dfrac{0.15}{0.3} = 0.5$. Independence would need $P(A)P(B) = 0.45\times0.3 = 0.135$, but $P(A\cap B) = 0.15 \neq 0.135$, so the events are **not independent** (equivalently $P(A\mid B) = 0.5 \neq 0.45 = P(A)$).

2 Cells: both 45, fiction only 30, non-fiction only 30, neither 45, total 150. $P(F) = P(N) = \dfrac{75}{150} = \dfrac{1}{2}$, $P(F\cap N) = \dfrac{45}{150} = 0.3$, $P(F\mid N) = \dfrac{45}{75} = \dfrac{3}{5} = 0.6$. Since $P(F)P(N) = 0.25 \neq 0.3 = P(F\cap N)$, the events are **not independent**: readers of non-fiction are more likely to read fiction (60%) than the population (50%).

3 $P(\text{bicycle}\mid\text{helmet}) = \dfrac{9}{18} = \dfrac{1}{2}$ and $P(\text{helmet}\mid\text{bicycle}) = \dfrac{9}{28} = 0.3214\ldots$, different because the denominators are the 18 helmet owners and the 28 bicycle owners. The union holds $28+18-9 = 37$, so $P(\text{neither}) = \dfrac{50-37}{50} = \dfrac{13}{50} = 0.26$.

4 Drawn without replacement, from 6 red and 4 blue: $P(RR) = \dfrac{6}{10}\times\dfrac{5}{9} = \dfrac{30}{90} = \dfrac{1}{3}$, and $P(\text{at least one red}) = 1 - P(BB) = 1 - \dfrac{4}{10}\times\dfrac{3}{9} = 1 - \dfrac{2}{15} = \dfrac{13}{15} = 0.8666\ldots$

5 Positions are ordered and digits distinct, so ${}^9P_5 = 9\times8\times7\times6\times5 = 15\,120$ codes.

6 $P(A\cap B) = P(A) + P(B) - P(A\cup B) = 0.6 + 0.5 - 0.8 = 0.3$. Since $P(A)P(B) = 0.6\times0.5 = 0.3 = P(A\cap B)$, the events are **independent**; equivalently $P(A\mid B) = \dfrac{0.3}{0.5} = 0.6 = P(A)$.

7 The rolls are independent, so $P(\text{no six}) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$ and $P(\text{at least one six}) = 1 - \frac{125}{216} = \frac{91}{216} = 0.4213\ldots$, slightly better than even odds on three rolls.

8 Of the 40 with the condition $0.90 \times 40 = 36$ test positive; of the 160 without, $0.10 \times 160 = 16$, so positives total 52 and $P(C \mid +) = \frac{36}{52} = \frac{9}{13} \approx 69\%$. Margin check: negatives $4 + 144 = 148$ and $52 + 148 = 200$.

9 A committee is an unordered selection, so ${}^9C_5 = 126$ committees. Exactly 3 women: ${}^5C_3 \times {}^4C_2 = 10 \times 6 = 60$, so $P = \frac{60}{126} = \frac{10}{21} = 0.4762\ldots$

10 Relative frequency $= \frac{120}{500} = 0.24$, above the theoretical $\frac{1}{5} = 0.20$. No: 500 spins of a fair spinner give an expected 100 with standard deviation $\sqrt{500 \times 0.2 \times 0.8} \approx 8.9$, so 120 reds is about 2.2 deviations away — unusual, and evidence to investigate, but not proof of bias.

11 Mutually exclusive means $A \cap B = \varnothing$ and $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$; such events are necessarily *dependent* when both probabilities are positive, since $P(A \mid B) = 0 \neq P(A)$. Independent events may occur together, with $P(A \cap B) = P(A)P(B) > 0$, and knowing one does not change the probability of the other.

12 Expected payout $= 0.25 \times £5 = £1.25$ against a £1 cost, so the expected net gain is $+0.25$ pounds a game: the game is not fair and favours the player. A fair game would pay $\frac{1}{0.25} = £4$.

13 Prevalence 0.8% of $100\,000 = 800$ have the condition and 99,200 do not; $0.95 \times 800 = 760$ test positive (40 negative) and $0.03 \times 99\,200 = 2976$ test positive (96,224 negative), so the margins are 3,736 positive and 96,264 negative, totalling 100,000. Then $P(C \mid +) = \frac{760}{3736} = \frac{95}{467} \approx 20.3\%$ and $P(C' \mid -) = \frac{96\,224}{96\,264} \approx 99.96\%$. About one person in five who tests positive has the condition, so a positive needs confirming; of every 10,000 negative results only about 4 are wrong.

14 Three draws without replacement from 3 red and 3 blue; the sample space is the ${}^6C_3 = 20$ equally likely sets, since the order of drawing does not affect these events. All one colour: $1 + 1 = 2$ sets, so $P = \frac{2}{20} = \frac{1}{10}$. Exactly two reds: ${}^3C_2 \times {}^3C_1 = 9$ sets, so $P = \frac{9}{20} = 0.45$. Check over 3, 2, 1 and 0 reds: $\frac{1 + 9 + 9 + 1}{20} = 1$.

15 Since $B = (A \cap B) \cup (A' \cap B)$ disjointly, $P(A' \cap B) = P(B) - P(A \cap B) = P(B) - P(A)P(B) = P(B)(1 - P(A)) = P(B)P(A')$. The product rule holds for A' and B, so A' and B are independent.

16 The sample space is the $10!$ orders of ten distinct people. Treat the four women as one block: the block with the six men is 7 objects, arrangeable in $7!$ ways, with the women inside the block in $4!$ ways, so $P = \frac{7! \times 4!}{10!} = \frac{24}{720} = \frac{1}{30} = 0.0333\ldots$, about 1 chance in 30. The same ratio comes from the women's positions: of the ${}^{10}C_4 = 210$ sets of four places, 7 are consecutive.

17 Assume 365 equally likely days and independence between people, ignoring leap years and twins. Then $P(\text{all different}) = \prod_{k=0}^{14} \frac{365 - k}{365} = 0.7471$, so $P(\text{at least one shared}) = 1 - 0.7471 = 0.2529$, about 25.3% — roughly one group in four, far above most estimates.

18 $\frac{P(A \cap B)}{P(A)}$ computes $P(B \mid A)$, the probability of B *given* A — the conditioning has been reversed. It is wrong whenever $P(A) \neq P(B)$, because the same intersection must be rescaled by the denominator of the group actually given, whereas $P(A \mid B)$ requires $P(B)$. The two coincide exactly when $P(A) = P(B)$ with both positive, and trivially when $P(A \cap B) = 0$, when both conditionals are 0.

19 From the 36 equally likely ordered pairs a total of 7 occurs in 6 ways, so $P(\text{win}) = \frac{6}{36} = \frac{1}{6}$. A fair game returns the stake in expectation, so $\frac{1}{6} \times \text{payout} = £1$: a payout of £6 per £1 stake.

20 By total probability $P(D) = 0.55 \times 0.02 + 0.30 \times 0.04 + 0.15 \times 0.08 = 0.011 + 0.012 + 0.012 = 0.035$, and the defective stream from C has probability 0.012, so $P(C \mid D) = \frac{0.012}{0.035} = \frac{12}{35} \approx 34.3\%$. C makes only 15% of components yet supplies over a third of the defects, so it is the best target for quality work.

21 By definition $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, and the multiplication law gives $P(A \cap B) = P(B \mid A)P(A)$, so $P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$, valid when $P(A) > 0$ and $P(B) > 0$. If $P(A \mid B) > P(A)$, dividing $\frac{P(A \cap B)}{P(B)} > P(A)$ by $P(B) > 0$ gives $P(A \cap B) > P(A)P(B)$, and dividing that by $P(A) > 0$ gives $P(B \mid A) > P(B)$. Dependence is therefore symmetric.

22 Total committees ${}^{11}C_4 = 330$. More women than men means 3 women and 1 man, or 4 women: ${}^6C_3{}^5C_1 + {}^6C_4{}^5C_0 = 20 \times 5 + 15 \times 1 = 115$, so $P = \frac{115}{330} = \frac{23}{66} \approx 0.348$. At least one of each: subtract the all-women (${}^6C_4 = 15$) and all-men (${}^5C_4 = 5$) committees, giving $330 - 20 = 310$ and $P = \frac{310}{330} = \frac{31}{33} \approx 0.939$.

Criterion A · Sets, conditionals and counting

1 Of 80 students: bicycle only $40 - 18 = 22$, both 18, skateboard only $30 - 18 = 12$, so $22 + 18 + 12 = 52$ own at least one and 28 own neither. The Venn diagram has $\xi = 80$ and $n(B \cap S) = 18$ in the overlap. Then $P(B) = \frac{40}{80} = \frac{1}{2}$, $P(S) = \frac{30}{80} = \frac{3}{8}$, $P(B \cap S) = \frac{18}{80} = \frac{9}{40}$, $P(B \cup S) = \frac{52}{80} = \frac{13}{20} = 0.65$. Check: $\frac{1}{2} + \frac{3}{8} - \frac{9}{40} = \frac{26}{40} = \frac{13}{20}$, and the complement $\frac{28}{80} = 0.35$ sums to 1 with it.

Marking guidance: 4 marks — 1 for the Venn diagram (18 in the overlap, 22 and 12 in the outer parts, 28 outside), 1 for $P(B)$ and $P(S)$, 1 for $P(B \cap S)$, 1 for $P(B \cup S)$ found by subtracting the overlap; the top band needs all four probabilities exact and the diagram consistent with them.

2 Given a bicycle owner, 18 of the 40 also skate: $P(S \mid B) = \frac{18}{40} = \frac{9}{20} = 0.45$. Given a skateboard owner, 18 of the 30 cycle: $P(B \mid S) = \frac{18}{30} = \frac{3}{5} = 0.6$. The numerators are the same intersection of 18 people, but the denominators are different groups — 40 cyclists against 30 skaters — so they measure different things: the chance a cyclist skates, and the chance a skater cycles.

Marking guidance: 4 marks — 1 for each correct fraction, 1 for the decimals or percentages, 1 for naming the two different restricted sample spaces; a response that calls the two values equal, or computes without explaining, cannot reach the top band.

3 $P(B)P(S) = \frac{1}{2} \times \frac{3}{8} = \frac{3}{16} = 0.1875$, while $P(B \cap S) = \frac{18}{80} = 0.225$. Since $0.225 \neq 0.1875$, the events are **not independent**: the overlap holds 18 people where independence predicts $80 \times 0.1875 = 15$, so bicycle owners skate more often than the group as a whole suggests ($P(S \mid B) = 0.45 > P(S) = 0.375$).

Marking guidance: 4 marks — 1 each for $P(B)P(S)$ and $P(B \cap S)$, 1 for the conclusion, 1 for an interpretation in context; "not independent because the numbers differ" without context caps the response at 3.

4 Two counters drawn from 4 red and 3 blue *without replacement*, so the second stage uses the 6 counters left. $P(RR) = \frac{4}{7} \times \frac{3}{6} = \frac{12}{42} = \frac{2}{7}$, $P(BB) = \frac{3}{7} \times \frac{2}{6} = \frac{1}{7}$, and $P(\text{exactly one red}) = P(RB) + P(BR) = \frac{12}{42} + \frac{12}{42} = \frac{4}{7}$. These sum to $\frac{2 + 1 + 4}{7} = 1$. Then $P(\text{at least one red}) = 1 - P(BB) = 1 - \frac{1}{7} = \frac{6}{7}$.

Marking guidance: 5 marks — 1 for a tree or path listing with second-stage denominators of 6, 2 for $P(RR)$ and $P(\text{exactly one red})$ (both paths required), 1 for the complement step, 1 for the paths summing to 1; $\frac{4}{7} \times \frac{4}{7}$ (with replacement) loses accuracy throughout.

5 The four positions are ordered and the digits distinct, so ${}^{9}P_{4} = 9 \times 8 \times 7 \times 6 = 3024$ codes. A code is even exactly when its last digit is one of $2, 4, 6, 8$: choose it in 4 ways, then fill the first three positions from the remaining 8 digits in ${}^{8}P_{3} = 336$ ways, giving $4 \times 336 = 1344$ even codes, so $P(\text{even}) = \dfrac{1344}{3024} = \dfrac{4}{9} = 0.4444\ldots$

Marking guidance: 3 marks — 1 for 3024, 1 for 1344 counted by fixing the units digit first, 1 for $\frac{4}{9}$; symmetry gives the same result, since four of the nine digits are even and each is equally likely in the units place.

Criterion B · The sum of two dice

1 The 6×6 table has the first die's score along one axis, the second along the other, and their total in each cell, the rows reading $2, 3, 4, 5, 6, 7$ through $7, 8, 9, 10, 11, 12$ down the last column. Every cell is one outcome of rolling two distinguishable dice, so there are $6 \times 6 = 36$ equally likely outcomes.

Marking guidance: 6 marks — 4 for a complete, correctly filled table, 1 for stating 36, 1 for stating that the outcomes are equally likely, which is what licenses counting cells as probabilities.

2 Counting diagonals of the table gives the frequencies 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 for the totals 2 to 12, which sum to 36. Dividing each by 36: $P(2) = \dfrac{1}{36}$, $P(3) = \dfrac{2}{36}$, $P(4) = \dfrac{3}{36}$, $P(5) = \dfrac{4}{36}$, $P(6) = \dfrac{5}{36}$, $P(7) = \dfrac{6}{36}$, $P(8) = \dfrac{5}{36}$, $P(9) = \dfrac{4}{36}$, $P(10) = \dfrac{3}{36}$, $P(11) = \dfrac{2}{36}$, $P(12) = \dfrac{1}{36}$, so the eleven probabilities total 1.

Marking guidance: 4 marks — 3 for all eleven probabilities correct with denominator 36 or in lowest terms, 1 for noting the total 36; allow 2 marks for one or two slips, and credit only frequencies traceable to the table in part 1.

3 The frequency rises by 1 at each step up to 7 and falls by 1 afterwards, symmetrically about 7, the most likely total. The conjectured rule is $P(\text{total } s) = \dfrac{s-1}{36}$ for $2 \le s \le 7$ and $\dfrac{13-s}{36}$ for $7 \le s \le 12$, both captured by $P(\text{total } s) = \dfrac{6 - |s-7|}{36}$. Justification: for a total s the first die may show any a with $1 \le a \le 6$ and $1 \le s - a \le 6$, so a runs from $\max(1, s-6)$ to $\min(6, s-1)$ — that is $s-1$ values while $s \le 7$ and $13-s$ values after it.

Marking guidance: 5 marks — 1 for describing the rise and fall, 2 for a rule correct on both sides of 7, 2 for the justification from the range of first scores; $\frac{s-1}{36}$ offered for all s is a conjecture, not a rule, and reaches at most 3.

4 For $s = 4$ the rising branch gives $\dfrac{4-1}{36} = \dfrac{1}{12}$, matching $(1,3), (2,2), (3,1)$; for $s = 10$ the falling branch gives $\dfrac{13-10}{36} = \dfrac{1}{12}$, matching $(4,6), (5,5), (6,4)$. The symmetry comes from the map $(a, b) \mapsto (7-a, 7-b)$, which sends a total s to a total $14 - s$ and pairs outcomes one-to-one, so $P(s) = P(14-s)$, with the peak at $s = 7 = \dfrac{14}{2}$.

Marking guidance: 4 marks — 2 for the two verified predictions, 2 for an explanation identifying a genuine one-to-one pairing; a bare "the table looks symmetric" earns 1.

5 $P(7) = \dfrac{6}{36} = \dfrac{1}{6}$ and $P(6) = \dfrac{5}{36}$, so the expected payout per play is $5 \times \dfrac{6}{36} + 2 \times \dfrac{5}{36} = \dfrac{40}{36} = \dfrac{10}{9} = 1.111\ldots$ credits against a 1-credit stake: an expected net gain of $+\dfrac{1}{9}$ credit, about 11.1%. The game favours the **player**; a fair 1-credit stake returns exactly 1 credit in expectation.

Marking guidance: 3 marks — 1 for the expectation, 1 for comparing it with the stake, 1 for naming the favoured party with the reason; averaging the payouts 5 and 2 without their probabilities scores 1.

6 For two n-sided dice the total $n+1$ needs $a + b = n + 1$ with $1 \le a, b \le n$, i.e. $a = 1, 2, \ldots, n$: exactly n outcomes of the n^2, so $P(\text{total } n+1) = \dfrac{n}{n^2} = \dfrac{1}{n}$. With $n = 6$ this returns $\dfrac{6}{36} = \dfrac{1}{6}$, and with $n = 2$ it gives $\dfrac{2}{4} = \dfrac{1}{2}$, since both ordered pairs $(1,2)$ and $(2,1)$ total 3.

Marking guidance: 2 marks — 1 for the count n, 1 for the simplified probability $\frac{1}{n}$; quoting the $n = 6$ value without the general count earns 1.

Criterion C · Explaining a positive test result

1 **Sample response.** "Your friend's sentence sounds reasonable but it swaps two questions. The 90% figure means: of 100 people who *have* the condition, 90 test positive. It says nothing directly about the people who test positive and want to know whether they have it. Those are different groups, so their answers can differ a lot. Here is what happens if 10,000 people are screened, when 2% have the condition, the test finds 90% of them and wrongly flags 5% of healthy people: 200 people have the condition, and 180 of them test positive. The other 9,800 do not have it, and 5% of them, which is 490 people, also test positive. So 670 people get a positive letter, and only 180 of them are right. That is $\dfrac{180}{670} = \dfrac{18}{67}$, about 27% — roughly 1 person in 4, not 9 in 10. Why is it so low? Because the condition is uncommon. There are about 49 times as many healthy people as people with the condition, so even a small error rate among the healthy produces far more false alarms than the test produces true ones. The test is not bad; it is being applied to a large healthy group, and this is nobody's fault. It is arithmetic. Nothing is decided yet, and a positive screening result is never a diagnosis. Two sentences a clinic could use: *'Your screening result is positive, which means the test reacted; it does not mean you have the condition. Among people with a result like yours, about 1 in 4 turns out to have it.'* and *'The next step is a confirmatory test, which is much more accurate, and a nurse will go through that result with you before anything is decided.'*"

Marking guidance: 20 marks in the bands printed in the question's evidence line — 6 for correct use of the two conditional probabilities with an accurate table (180, 20, 490, 9,310, margins 670 and 9,330 totalling 10,000, and $\frac{18}{67} \approx 27\%$); 6 for explaining the base-rate effect and naming which group each probability describes; 4 for language suited to a non-specialist in a fair, non-alarming tone; 4 for two justified sentences for the letter. Top-band responses never write $P(+ \mid C) = P(C \mid +)$ and give the reverse probability as an exact fraction before rounding.

Criterion D · A screening decision

1 Prevalence 1.5% of $10\,000 = 150$ people have the condition and 9,850 do not. Of the 150, $0.90 \times 150 = 135$ test positive and 15 negative; of the 9,850, $0.02 \times 9850 = 197$ test positive and $9850 - 197 = 9653$ negative. The cells are 135, 15, 197 and 9,653, and the margins are 150 with the condition, 9,850 without, $135 + 197 = 332$ positive, $15 + 9653 = 9668$ negative, with $332 + 9668 = 10\,000$, so the table closes.

Marking guidance: 5 marks — 2 for the row totals from the prevalence, 2 for the four cells, 1 for margins summing to 10,000; applying the 2% to all 10,000 rather than to the 9,850 healthy people loses the cell marks.

2 With positives 332 and negatives 9,668: $P(+) = \dfrac{332}{10\,000} = \dfrac{83}{2500} = 3.32\%$; $P(C \mid +) = \dfrac{135}{332} \approx 40.7\%$ (already in lowest terms, as $332 = 4 \times 83$); $P(C' \mid -) = \dfrac{9653}{9668} \approx 99.8\%$. Complement check: $P(C \mid -) = \dfrac{15}{9668} \approx 0.16\%$, which with 99.84% sums to 100%, and $P(C' \mid +) = \dfrac{197}{332} \approx 59.3\%$ sums to 100% with 40.7%.

Marking guidance: 6 marks — 2 for $P(+)$, 2 for $P(C \mid +)$ as a fraction and a percentage, 2 for $P(C' \mid -)$; the denominators must be the restricted groups 10,000, 332 and 9,668 in turn, and reporting 90% for the second loses that part.

3 For a patient, $P(C \mid +) = \frac{135}{332} \approx 40.7\%$ means that of every 100 people who receive a positive letter about 41 have the condition and 59 do not, so the chance is about 2 in 5 rather than "almost certain", and a confirmatory test is needed. It is much smaller than the sensitivity $P(+ \mid C) = 90\%$ because the denominators differ: the sensitivity is measured on the 150 people with the condition, $P(C \mid +)$ on the 332 positives, 197 of whom come from the 9,850 healthy people. A 2% error rate is enough to produce that many false alarms when the healthy group is 65 times larger, so the size of the healthy group, not the quality of the test, drives the difference.

Marking guidance: 4 marks — 1 for a patient-facing interpretation in natural frequencies, 1 for naming the two denominators, 2 for the base-rate explanation; calling the test "inaccurate" loses the interpretation mark, since the arithmetic shows it behaving as specified.

4 Assumption 1: the condition affects 1.5% of *this* screened population. The 1.5% is a regional average, so at 0.5% prevalence there are 50 affected, 45 true positives and $0.02 \times 9950 = 199$ false ones, giving $P(C \mid +) = \frac{45}{244} \approx 18.4\%$, while at 5% prevalence it rises to $\frac{450}{640} = \frac{45}{64} \approx 70.3\%$. Assumption 2: the 90% and 2% rates transfer to this setting, though they come from the manufacturer's samples; at a 4% false-positive rate $P(C \mid +)$ falls from 40.7% to $\frac{135}{135 + 394} = \frac{135}{529} \approx 25.5\%$. Both are checkable: record the actual positives, negatives and confirmation outcomes in a pilot round and rebuild the table from the observed figures.

Marking guidance: 5 marks — 2 for two mathematical assumptions, 2 for the direction and size of each effect on $P(C \mid +)$, 1 for a check that would detect failure; remarks that do not change the table earn at most 1.

5 Benefits: 135 affected people are found early, when treatment is usually cheaper and more effective, and only 15 of the 9,668 negative results are wrong. Harms: 197 healthy people, 59% of all positives, receive an alarming letter and face anxiety, time off work and unnecessary procedures. Since a positive is right only about 2 times in 5, screening alone cannot justify treatment. Recommendation: introduce it **with** a mandatory confirmatory test before any result is communicated as a diagnosis, since that replaces a 40.7% chance with much stronger evidence and removes almost all 197 false alarms; add natural-frequency wording and consider restricting screening to a higher-risk group, as a higher base rate raises $P(C \mid +)$.

Marking guidance: 4 marks — 1 for quantified benefits, 1 for quantified harms, 1 for a clear recommendation, 1 for linking it to parts 2 and 3; a one-sided verdict cannot reach the top band.

CHAPTER 11 · CAPSTONE: A MATHEMATICAL MODELLING PROJECT

Opening case · the timetable that looked fair

1 **Model response. Measured:** tap-in and tap-out timestamps, the mean journey time per route derived from them, in minutes, taps per route per hour. **Not measured:** the interchange wait and walk between stands (minutes); the spread of journey times, such as the 90th percentile; the deadline a passenger works to, a shift start or a hospital appointment.

Any three of each are acceptable if the unmeasured ones are invisible in tap data rather than merely unused.

2 A target on the mean only requires the mean to pass: eight journeys of 25 minutes and two of 70 give $(8 \times 25 + 2 \times 70) \div 10 = 34$ minutes, and ten journeys of 33 minutes give 33, so both routes pass a 35-minute target although 2 in 10 on the first exceed an hour. Short direct trips hold the average down while the changed, night-shift and hospital journeys, which taps cannot separate, sit unnoticed in the tail.

3 **Model response. Record** the interchange time, minutes between alighting and re-boarding, from matched tap sequences or timed observation. **State publicly** that journey time is measured tap-in to tap-out, so a bus change is an unobserved gap inside one journey, and that the target is set on the mean, not on the proportion of journeys completed within a stated time.

Any recordable quantity with a unit plus a statement naming the measured quantity is acceptable.

4 **Model response.** No: more taps estimate the same quantity better and still miss the transfers. Also needed are a summary that sees the tail, a percentile or a proportion under a stated time; a structure in which a journey is leg plus wait plus leg; and the question of whose journeys the target protects. The case says the arithmetic was right, so the fault is in the quantity modelled, not the sample size.

A yes answer is defensible only if it argues that the unseen tail, not the missing leg, decided the outcome.

Prerequisite check

1 $y = 2(x^2 - 6x) + 7 = 2(x-3)^2 - 18 + 7 = 2(x-3)^2 - 11$. Vertex $(3, -11)$, a minimum since $2 > 0$. Check at $x = 3$: $18 - 36 + 7 = -11$.

2 $t = \frac{\ln 1.5}{\ln 1.06} = 0.40547 \div 0.058269 = 6.96$ (2 d.p.), about 7 years if the 6% period is a year. Check: $1.06^{6.9585} = 1.50$.

3 $a = 3$, $d = 4$, 20th term $3 + 19(4) = 79$, so $S_{20} = \frac{20}{2}(3 + 79) = 10(82) = 820$.

4 The triangle bounded by the axes and $x + y = 8$, on the origin side since $0 + 0 \le 8$; vertices $(0,0)$, $(8,0)$, $(0,8)$. It is closed, so a linear objective attains its optimum at a vertex.

5 $\bar{x} = 45 \div 5 = 9$; squared deviations $25, 4, 0, 1, 36$, sum 66; variance $66 \div 5 = 13.2$; standard deviation $\sqrt{13.2} = 3.63$ by the Chapter 9 convention (with $n - 1$, $\sqrt{16.5} = 4.06$).

6 The array printed is a 2×2 matrix although the sentence calls it 3×3. For the matrix shown the determinant is $2 \times 3 - 1 \times 1 = 5$, so it is invertible.

Printed defect: the stated size and the array disagree; the intended item is the 2×2 determinant, answered above.

11.1 Guided practice · The modelling cycle

1 Collecting temperatures sits at assumptions and variables, since it fixes what will be represented; $T = 20 + Ae^{-kt}$ is the mathematical-model stage, with A and k still unestimated; checking a prediction for a new day is validation.

2 That the factor stays exactly 1.03 for every unit of t, which needs unlimited resources, no migration and a consistently defined population. Any one may be named; it is an assumption, not a fact.

3 A model tested only on the data it was fitted to has shown it can reproduce them, which any flexible rule does. Without held-out evidence there is no estimate of prediction error, so the tolerance and range the claim can carry are unknown.

4 Adequate: mean height of a class when ordering 30 identical chairs, where one number sets the specification. Not adequate: emergency-department waits, where the decision concerns the longest waits, so the proportion seen within four hours matters.

Exercise 11A · The cycle in practice

1 Define the question, state assumptions, build the model, interpret, validate, report: the six boxes of the cycle diagram, whose last arrow returns to the first, so the order is one turn of a cycle, not a single pass.

2 (a) An **assumption**, unless a logger recorded it. (b) A **parameter estimate**: $k = 0.11$ comes from data. (c) An **assumption**, excluding a heat path without proof. (d) A **fact**, a recorded measurement used as the model's initial value.

3 (a) Exponential, $h_n = h_0 r^n$, variable bounce number n. (b) Linear, $C = F + pd$, variable distance d in km. (c) Exponential, $N = N_0 b^t$, variable time t in hours. (d) Quadratic, $A = \pi r^2$, variable radius r in cm.

4 Independent: time since the drink was made, in minutes. Dependent: temperature of the drink, in °C. Controls: room temperature (°C), volume (mL), cup and lid, stirring.

5 Residual = observed − predicted $= 52 - 48 = +4$, and the chapter formula bases the percentage on the observed value: $\frac{4}{52} \times 100 = 7.7\%$ (exactly 7.69%). Dividing by 48 would give 8.3%, so the base must be named.

6 Interpolation predicts at an input inside the range used for fitting, with evidence on both sides. Extrapolation predicts outside it, assuming a relationship continues where it was never tested, and is the weaker claim.

7 It prices a holiday week at school-day rates: where a weekday uses 100 kWh and the base load is 15 kWh it predicts $5 \times 100 = 500$ kWh against $5 \times 15 = 75$ kWh, some 6.7 times too high. It stays useful for the weeks it was fitted to, comparing term weeks or spotting faults; the error is use outside its scope, not the model.

8 First the diagnostic pattern, and with it the mechanism: whether successive ratios are roughly constant (exponential) or second differences are (quadratic), which a plot of $\ln y$ against x settles. Then behaviour beyond the data: a quadratic turns and can go negative, an exponential never turns, so which can hold where the decision is made, with held-out residuals, decides.

9 Each parameter spends a degree of freedom matching the fitting set's noise, leaving $n - p$ for the relationship. A degree-7 polynomial through 8 points fits exactly, wiggles between them and misbehaves outside, so a better fit comes with worse prediction.

10 The input range it was checked over; whether the figure is from fitting or validation data; which error measure, mean, 90th percentile or maximum, and against which base; sample size; measurement uncertainty. Accuracy of 5% means little where the instrument reads to 3%.

11 Cooling a drink over the 20 minutes a student has: a one-parameter exponential beats a fluid model, because few readings exist, so extra parameters would be fitted to noise and could not be validated, and the decision needs a degree or two. Parsimony buys testability and an answer the user can check.

12 A model is chosen to minimise error on the fitting data, so re-testing there measures flexibility, not skill. Fresh data give predictions the model never accommodated, which is the error the user meets and the only test separating a real relationship from an overfitted one.

13 Taking d as days from the first day recorded, four weeks cover $d = 0$ to 27, so week five is $d = 28$ to 34: $S(28) = 416$, $S(31) = 452$, $S(34) = 488$, about $7 \times 452 = 3164$ for the week. All are **extrapolation**, beyond the fitted range. A promotion is a change of regime, so the steady $+12$ a day will miss the uplift and, after it ends, the trend may overshoot.

Printed defect: no units or origin for d. With d starting at 1 the week is $d = 29$ to 35, giving 428 to 500 and 3248; state the convention used.

14 The amount removed depends on the current volume, not the day: $V_{t+1} = V_t - 0.02V_t = 0.98V_t$, a constant ratio, hence exponential. Forms: $V_t = V_0(0.98)^t$, or $V(t) = V_0e^{-kt}$ with $k = -\ln 0.98 = 0.0202$ per day; a linear decrease would remove 2% of the *initial* volume daily. Check at 30 days: $0.98^{30} = 0.545$, so 54.5% left where the linear reading leaves 40% and empties on day 50; half-volume time $\ln 0.5 \div \ln 0.98 = 34.3$ days.

15 Age: all are 18 to 25, so the parameters describe young adults and carry no authority over children or over-40s. Selection and size: 40 volunteers are likelier than the population to be motivated or already concerned, and 40 gives a wide interval. Report it by naming that population in the aim and quoting band and sample size with every figure.

11.2 Guided practice · Defining the problem

1 **Model response.** "For the Year 10 cohort at this school, how does weekly homework time, in minutes, predict the score on the next unit test, in per cent, over the four units of one term?" It names the population, both variables with units, the time frame and the decision it informs; "learning" has become a measurable outcome.

A strong aim is answerable with numbers from one defined group and period; cohort, subject and metric may all differ.

2 Metres per second: a gradient carries the vertical unit divided by the horizontal one, so km against h gives km/h. Read without its unit, a walking gradient of 1.4 is mistaken for 1.4 km/h when it is 84.

3 Water volume per plant per week (mL), or soil type and depth, temperature (°C), hours of light per day, any one named as held fixed so its effect cannot be confused with the light being varied.

4 Because it fixes instrument, repetitions and sample size before any data exist: if the decision turns on a 5 g difference, readings precise to 10 g cannot answer it however many are taken.

Exercise 11B · Aims, variables and data

1 (a) Dependent: height, cm; independent: time, weeks. (b) Dependent: cost, £; independent: distance, km. (c) Dependent: temperature, °C; independent: time since boiling, minutes. (d) Dependent: number of bacteria, count; independent: incubation temperature, °C.

Printed defect in (a): only one variable is stated, so time is implied; "height of a plant, in cm, measured weekly" would name both.

2 "The decay constant is 0.04 per minute" is a **parameter**, in $\text{min}^{(-1)}$; "the room stays at 20°C" is an **assumption** unless logged; "the mass is 250 g" is a **measurement**; "the cup loses heat only to the air" is an **assumption**.

3 **Model responses.** (a) "For the Year 10 menu, how many kilograms of food are discarded at each serving station during the 11:30 to 13:30 service, and which tray size keeps it lowest?" (b) "In the 08:10 to 08:30 arrival window on school days, how does the number of vehicles queuing depend on the bays open, and how many bays keep the queue under four cars?" (c) "For a family of four using 24 GB a month, which plan costs least over 12 months in £ per month, including excess-data charges?" Each names a decision, a population, a period and a quantity with units.

Reject aims that only "investigate" or "explore", or that lack a number with a unit and a named group.

4 45 min = 0.75 h and 2.5 L = 2500 mL. A rate in L per hour multiplied by 45 rather than 0.75 overstates the volume 60 times; a volume read in mL but treated as L is wrong by 1000.

5 Kilometres per minute, the gradient being change of distance over change of time: it is the average speed, so 0.2 km/min means 200 m a minute, $0.2 \times 60 = 12$ km/h.

6 Any three of: what is measured, with what instrument and precision; how often and over what period; source, sample size and provenance; the rule for missing or implausible values; consent and anonymity; how entries are recorded so they can be checked.

7 A count measures how often a student buys, not what they buy: daily fruit scores high as though unhealthy and a weekly soft drink low, while price and queue length confound frequency. Better dependent variables are sugar in grams per lunch, energy in kcal, or portions of fruit and vegetables per week.

8 One week contains no spring-to-neap cycle, which takes about 14 days, so the main source of variation in range is invisible; record four weeks across two cycles or validate against a published gauge for another month. That week's weather is also credited to the tide, so log wind and pressure and state the scope as this site under these conditions.

9 A controlled variable is held the same by the design so its effect cannot be confused with the variable tested; an assumption is taken true inside the model without verification. Watering every pot with 200 mL weekly is control; believing the seeds are one variety, unchecked, is an assumption.

10 Responders differ from the whole group: they are more interested in the topic and likelier to have an opinion about it. In a sleep study volunteers often have sleep concerns, so poor sleep is over-represented, modelled duration is biased downward and reported problems upward, making any recommendation more alarming than the reality.

11 A per-week rate applied to a day count runs seven times too fast: 12 L per week treated as 12 per day predicts 84 L daily, and the arithmetic looks consistent because only the units are wrong. Write units into every variable, convert first ($12 \div 7 = 1.71$ L per day) and check one prediction against a known observation.

12 The precision is set by the gap between the options, not by the instrument: if A and B differ by 3 units on the decision measure, the typical error must be well under 3, by convention below half of it, which fixes resolution, repetitions and sample size, since the error of a mean shrinks like $1/\sqrt{n}$.

13 **Model response. Measure:** waiting time from joining the queue to receiving a tray, in seconds, timed identically at both queues. **Design:** one observer per queue with a stopwatch, ten journeys per queue in each of four 15-minute slots between 11:30 and 13:00, on 5 school days, 200 journeys paired in the same slots. **Also record:** date, slot, staff on point, queue length on entry, interruptions such as a failed card reader. **Cleaning:** discard any journey where the observer had to join a queue. **Ethics:** no filming or naming of students, anonymous records, staff permission, no delay to service.

Criterion B i. What is marked is a protocol another student could follow and a rule for every value that will be discarded.

14 **Model response. Question:** "For the three networked printers, how many reams a week would making double-sided printing the default save over one term, and is it worth the slower turnaround?" **Variables:** dependent, sheets per week (1 ream = 500 sheets); independent, duplex setting, with jobs per week recorded; controlled, department and class size. **Assumption:** pages per job are unchanged by duplexing, so halving sheets does not halve demand. The figure decides whether to change the default, so it must separate a few reams a week from nothing.

Aim, variables with units and one stated assumption are all required; any decision-relevant school quantity is acceptable.

15 A bias lifting both policies alike cancels in the difference: if demand is overstated by 10% whatever is done, both options rise about 10% and the cheaper usually stays cheaper. Still not acceptable, because the budget test is on absolute cost, not order; the bias need not be common-mode, since a policy can change the very quantity mis-measured; and with an unquantified bias a 1% margin proves nothing.

11.3 Guided practice · Building and testing a model

1 Exponential: doubling each step is a constant ratio of 2, so $y = a(2)^x$, base $b = 2$ the parameter and a the starting value; equivalently $y = ae^{kx}$ with $k = \ln 2 = 0.693$ per step.

2 They alternate in sign with no trend and no growth in size, so there is no evidence the family is wrong; but they reach four times the quoted reading uncertainty of ± 0.5, so reading error alone does not explain them. The model is plausible yet imperfect; report the typical size, about 1 unit.

3 r^2 is computed on the fitting data the model was chosen to match, so it records linear association there, not accuracy on new points, and it is blind to curved residual patterns and to error in context units.

4 Extrapolation: 120 minutes lies outside the fitted range of 5 to 60, twice the largest input used, so the prediction assumes an untested continuation and its uncertainty should be reported as larger.

Exercise 11C · Fitting, validating and reporting

1 (a) Linear: constant first difference 4, a rate of 4 per step. (b) Exponential: constant ratio 1.2, growth of 20% per step, $y = a(1.2)^x$. (c) Quadratic: constant second difference gives $2a = 6$, so $a = 3$ for unit steps. (d) Trigonometric: period 12, e.g. $y = A\cos\left(\frac{2\pi x}{12}\right) + c$.

2 $(0, 5)$ gives $a = 5$; then $5b^4 = 80$, so $b^4 = 16$ and $b = 16^{1/4} = 2$, the positive root only, a negative base being undefined for all real x. Model $y = 5(2)^x$; check $5(2)^4 = 80$.

3 Residual = observed − predicted $= 78 - 84 = -6$, in the dependent variable's units, the sign recording an over-prediction. Percentage error $= \frac{6}{78} \times 100 = 7.7\%$ (exactly 7.69%), the observed value being the base.

4 The residuals sum to 0.3, so the mean residual is $0.3 \div 5 = 0.06$ in the units of the dependent variable. That near-zero mean only shows cancellation: the mean absolute residual is $0.9 \div 5 = 0.18$ and the rms value 0.195, which is the figure describing typical error.

5 (a) Interpolation, 30 lies inside 10 to 50. (b) Extrapolation, 70 is above the fitted range. (c) Interpolation, 6 lies inside 1 to 12. (d) Extrapolation, 0.5 is below the smallest input used.

6 A residual is observed minus predicted, a difference of two values of the dependent variable, so it carries that variable's unit, °C rather than minutes. Only the percentage error, a ratio of residual to observed value, is unitless.

7 First, there was no validation: the same data were used to fit and to test, so $r^2 = 0.98$ measures how flexibly the line matched those points, not how it predicts anything new. Second, r^2 measures *linear* association on the fitting set: it ignores a curved residual pattern and the error in context units, and 0.98 can still be too imprecise for the decision.

8 The U-shape makes error depend systematically on the input, under-predicting in the middle and over-predicting at both ends, the signature of a too-simple family for curved data, so a quadratic term or a model with a minimum is needed.

9 Opposite errors cancel in an average: waits predicted as 30 minutes when they are 0 and 60 have mean error 0 yet both patients are badly served, and a model can be right for two schools together and 20% wrong for each. For a single case report the spread and the worst case.

10 Prefer the smaller maximum residual: a safety limit is set by the worst case, a load or a dose, not by typical behaviour, and two models identical on average can differ in how far they ever stray. Check the maximum and 95th percentile on held-out data and design the limit against it.

11 Rounding discards information the equation then pretends to have: times logged to the nearest 10 minutes give a gradient printed to five figures, implying a precision the input never carried, and rounded values can fall exactly on a line, flattering the fit.

12 The threshold 105 is 5 above the prediction while the typical error is 15, so it lies inside the model's own band of roughly 85 to 115; if 15 is a standard deviation the true value exceeds 105 with probability $P(Z > 5/15) = 0.37$, close to a coin toss. Say whether 15 is a standard deviation, a maximum or a tolerance, then reduce the error or state the decision as conditional.

13 A line through the endpoints has slope $(190 - 120) \div 9 = 7.8$, giving $y = 120 + 7.8(t - 1)$; the least-squares line on all ten points is $y = 115.5 + 7.42t$, from mean $t = 5.5$ and mean $y = 156.3$. Both give about 205 items on day 12: $115.5 + 7.42(12) = 204.6$. Day 12 is beyond the data, so this is **extrapolation**; the fitted residuals run from -9.7 to $+7.4$ items, with residual standard error $\sqrt{314.8 \div 8} = 6.3$ items, so 205 with an error of about 6 to 10 items is the honest statement. Ten days, no held-back check and no day-of-week effect further limit reliability.

14 The drops over successive 2-minute intervals are 12, 9, 6, 4, summing to $31 = 72 - 41$ as they must, with ratios 0.75, 0.667, 0.667, roughly constant near 0.69, so the drop over a fixed interval shrinks by a constant factor, the signature of exponential cooling. From 0.694 per 2 minutes the half-life is $2\ln 2 \div \ln(1 \div 0.694) = 3.8$ minutes. Limitation: the room temperature is not given and it is the excess above it that halves, so at 20°C the excesses $52, 40, 31, 25, 21$ give $k = 0.113$ per minute and a half-life of 6.1 minutes.

15 Three parameters on eight points consume degrees of freedom, not evidence: the exact fit is interpolation, and a zero residual sum of squares, hence $r^2 = 1$, follows from the count of parameters. To expose it, fit on five points, test the three held back against a one- or two-parameter model on the same points, then collect data outside the fitted range; if the simpler model predicts as well, the extra parameters were fitting noise.

11.4 Guided practice · Optimising and evaluating

1 **Model response. Objective:** maximise daily profit, in £. **Constraint:** flour used is at most 40 kg, $F \leq 40$ kg; a strong answer adds whole numbers of each product and oven or labour time, since flour alone does not cap output. Without constraints the objective is unbounded.

An objective is a quantity to maximise or minimise with a unit; a constraint is a limit the decision cannot cross. Extra constraints are acceptable if genuinely binding.

2 Write $P(n) = a(n - 50)^2 + 900$ with $a < 0$. Then $P(48) = 900 + 4a$, so the loss is $-4a = 4|a|$: the vertex alone does not determine it, the coefficient is needed, and with $a = -5$ the loss is 20, since $P(48) = 880$.

Printed defect: the hint "$900 - 4a$" carries a sign slip; with $a = -5$ it gives 920, a gain rather than the loss of 20 its own note reports. Corrected hint: $900 + 4a$.

3 Because the optimum is found inside a model that may omit limits outside it: capacity not written as a constraint, whole numbers or batch sizes, lead time, budget, approval. An optimum of 240 units is worth nothing if 220 can be made.

4 Environmental impact, such as packaging waste or carbon, since only purchase price is in the objective. Also acceptable: quality or shelf life, access for users affected, staff workload, supply resilience.

Exercise 11D · Objectives, constraints and evaluation

1 (a) Objective: maximise boxes fitted, or volume used; constraints: the container's internal dimensions and weight limit, whole numbers of each box size. (b) Objective: minimise 12-month cost, in £; constraints: the allowance must cover the family's usage, and the monthly budget is capped. (c) Objective: maximise curriculum coverage, or minimise clashes; constraints: each teacher and room in one place per period, and a fixed number of periods per class per week.

2 (a) $y = -3(x^2 - 6x) - 5 = -3(x - 3)^2 + 22$: a **maximum** of 22 at $x = 3$, as $-3 < 0$; check $-27 + 54 - 5 = 22$. (b) $y = 2(x^2 - 4x) + 11 = 2(x - 2)^2 + 3$: a **minimum** of 3 at $x = 2$, as $2 > 0$; check $8 - 16 + 11 = 3$. (c) $y = -(x^2 - 10x) = -(x - 5)^2 + 25$: a **maximum** of 25 at $x = 5$; check $-25 + 50 = 25$.

3 $P(30) = 2500$, the vertex and a maximum since $-4 < 0$; $P(28) = -4(2)^2 + 2500 = 2484$; $P(35) = -4(5)^2 + 2500 = 2400$. Two units below costs 16, five above costs 100: the penalty grows with the *square* of the departure, which is the sensitivity figure a decision-maker needs.

4 Because a continuous vertex can fall between whole numbers: for $P = -4(n - 30.4)^2 + 2500$ only $n = 30$ and $n = 31$ are possible and both must be evaluated, since the parabola is steeper on one side; other constraints may exclude the nearer integer altogether.

5 Driver time and wage: the shortest route can be the slowest, so fuel is saved while cost rises and delivery windows are missed. Also acceptable: time-window penalties, vehicle wear, congestion charges, road safety.

6 A small change in the assumption produces a large change in the answer: the recommendation may switch from A to B if an assumed wage rises 5%. A sensitive recommendation is reported with the range of the assumption over which it holds.

7 At $x = 30$ the constraint gives $y = 60 - 2(30) = 0$, so $A = 30(0) = 0$: the student took an endpoint of the domain and read it as the answer. On $0 \le x \le 30$ both endpoints give zero area, so they are the **minima**; the maximum is the vertex $x = -60 \div (2(-2)) = 15$, with $A = 15(30) = 450$ m^2, and $A(10) = A(20) = 400$ confirms it lies between. The error is confusing a root with a turning point.

8 The decision-maker can value each objective and combine them in a weighted score, fix one as a constraint and optimise the other, or be shown the frontier of trade-offs and choose a point on it. Mathematics supplies the frontier and the exchange rate between objectives; the weights and the constraint level express priorities no equation supplies.

9 Each model's own optimum, its cheapest schedule, can survive the change while the comparison between options flips: A costs £12 000 and B £12 400, and if A is labour-heavy a 10% wage rise takes A to £13 200 while B moves only to £12 600, so the recommended option changes from A to B although neither model's internal optimum moved. Sensitivity therefore tests the choice, not only the numbers.

10 The domain says where the model may be used, so an optimum outside it cannot be attained. With $P(n) = -2(n - 240)^2 + 18\,000$ and a ceiling of 200 units the vertex is infeasible, the best attainable choice is $n = 200$ with $P = -2(40)^2 + 18\,000 = 14\,800$, 3200 below the unconstrained optimum; quoting 18 000 would recommend an impossible plan.

11 The saving counts operating cost only, omitting travel time and cost for users reaching the next provider and outcomes for those affected, such as later diagnosis or pupils with nowhere to eat. Include them by pricing the user cost per journey into the objective, or by constraining access, say no user travels beyond 30 minutes.

12 An optimum is optimal only for the objective, constraints and range stated, so "best" alone names no property: the same data make one option cheapest, another fastest, a third safest. "Least cost per delivery within 40 km on weekdays" can be checked; "best route" cannot, since the reader cannot tell what was traded away.

13 Four sides fenced: $2x + 2y = 48$, so $A = x(24 - x) = -(x - 12)^2 + 144$, a **maximum** of 144 m^2 at $x = y = 12$ m, the square, since the coefficient of x^2 is negative. In practice a square is not always best: with one side already a wall, 48 m on three sides gives $x = 12$ m and $y = 48 - 2(12) = 24$ m, an area of 288 m^2, twice as much, while gates, a trough, drainage and animals needing a long run can all favour a rectangle.

14 Let r and h be in metres and c the cost per m^2 of side material, the base costing $2c$ per m^2. With the open top the cost is $2c\pi r^2 + 2c\pi rh$, subject to $\pi r^2 h = 2$ m^3, so $h = \frac{2}{\pi r^2}$ and

$$C(r) = c\left(2\pi r^2 + \frac{4}{r}\right).$$

Search by table, taking h from the volume each time: $r = 0.50$, $h = 2.55$, $C/c = 9.57$; 0.60, 1.77, 8.93; 0.70, 1.30, 8.79; 0.80, 0.99, 9.02. The values fall then rise, so the least is a **minimum**, near $r = 0.68$ m with $h = 2 \div (\pi(0.68)^2) = 1.38$ m and $C/c = 8.79$; check $\pi(0.68)^2(1.38) = 2.00$ m^3. The optimum has $h = 2r$, height equal to the diameter. A lid at side cost would give $3\pi r^2 + 4 \div r$, minimised near $r = 0.60$ m with $C/c = 10.06$, so whether the tank is covered must be stated.

Printed defect: only the base and the curved side are priced, so a lid is neither stated nor costed. The open-top reading is answered first; corrected wording would add "with an open top" or give the lid's price per m².

15 The LP optimum might be $(4.7, 6.2)$, the four nearest integer points infeasible or beaten by $(2, 8)$ two steps away, because the feasible lattice is not a disc round the vertex. For two variables, bound then enumerate: take z^* from the LP and a feasible integer value z_{best}, then for each integer x between the extreme coordinates use the constraints to get the admissible interval of y and test only the integers in it whose objective lies between z_{best} and z^*. Nothing better than z^* exists, so the finite check cannot miss the integer optimum.

11.5 Guided practice · Presenting and defending

1 The recommendation and how uncertain it is, in the first sentences: what to do, by how much, over what range. A decision-maker reads no further if the answer and its confidence are not visible there.

2 **Model response.** "Rainfall from the school weather station for the 13 weeks from 1 March to 31 May, totalling 97 mm, with no missing days; two weeks flagged as sensor faults were removed and recorded. Weekly totals agreed with the public gauge 6 km away to within 4 mm, so the data support a term-time estimate for this site only."

Four things must appear: source, period, sample size and what was cleaned; a different but consistent cleaning rule is acceptable.

3 The quantity, its unit and a scale with values that can be read, for instance "time after boiling (minutes)" and "temperature (°C)", with the fitted curve distinguished from the held-out points so the reader sees the validation and not only the fit.

4 "Why an exponential rather than a quadratic, and is the ratio of successive values really constant?" and "what limits the growth, and at what value does the model stop being believable?" Each is answered by the observed ratio, the mechanism and the validated range.

Exercise 11E · Reports, graphs and defence

1 Summary; Introduction (decision, scope, stakeholders); Data (source, sample, cleaning, limitations); Model (assumptions, variables, units, equations, parameter estimates); Results (graphs, predictions, residuals, validation); Evaluation (sensitivity, trade-offs, what the model cannot see); Conclusion (recommendation with confidence); References.

2 Horizontal axis: time, in the unit used, e.g. "time after boiling (minutes)". Vertical axis: temperature, in °C. Both need numbered scales, with a title and a key separating the fitted curve from the held-out readings.

3 **Model responses.** (a) "On the 12 held-out days predictions were within 1.2°C on 10 days and 2.4°C low on the two coldest, so the model suits scheduling, not control at 0.5°C." (b) "Over 10 months option B cost £1 240 against £1 480 for A, £240 or 16% less, but £180 of that is a promotional rate ending in month 4, so the ongoing saving is £60, 4%." (c) "28 of 41 chose A when it was offered first and 24 when it was offered second, so preference leans to A but question order affects the count, in one year group at one school."

Each restatement needs the quantity, its unit, the error or margin, and the range it holds over; figures may differ if the three clauses are all present.

4 "Why should a constant annual increase hold for decades, and what would change it?" and "Over what range was the line validated, and what does it predict for a year no census covers?" A third: "is growth proportional, or does it slow as housing and services fill?"

5 Because the value alone invites a decision the evidence cannot carry: a 12% reduction with a typical error of 2 percentage points may or may not clear the threshold the decision turns on. The uncertainty says whether the effect is worth acting on and which measurement would settle it.

6 A decision-maker's report leads with the recommendation, its cost and uncertainty, keeps derivations in an appendix and avoids jargon; a specialist's report foregrounds method, parameter estimation and residual diagnostics. A specialist reads the model itself for information, a decision-maker only its consequence.

7 Truncation stretches a small change over most of the visible height: 92 to 98 differ by 6 units, a 6.5% rise, but on an axis from 90 to 100 the segment looks like a doubling. It is acceptable where the quantity legitimately never reaches zero, a temperature or an index, provided the baseline is labelled; not where the chart claims a change is large in proportion.

8 The mean cancels opposite mistakes: residuals of $+30, -30, +30, -30$ have mean 0 while every case is wrong by 30, so a model can look excellent in aggregate and fail the single patient being decided on. Report mean absolute or rms error, the maximum or 95th percentile, and errors by group.

9 Use "is consistent with", or "supports". Proof belongs to mathematics, where assumptions are stipulated; a model rests on estimates from a sample and on unverified assumptions, so evidence can confirm its predictions without establishing them. "Proves" closes the question, when the honest message is that the evidence favours the policy by this margin and within this scope.

10 Say X was considered; give the inclusion criterion and whether X changes the decision; report the evidence, a measurement or a sensitivity run showing the recommendation holds while X moves over a plausible range; and state the limitation with the check that would settle it: "transfer time was not modelled because a 5-minute shift changed the ranking in none of 40 runs; beyond 12 minutes it reverses."

11 So that the conclusion is read qualified: a reader who meets the limitation afterwards has already formed a judgement and tends to discount it. Putting evaluation first also forces the writer to state the uncertainty beside the number, the order a decision-maker needs.

12 References give provenance, the evidence that numbers were not invented: source, date and sample size of each data set, and the method behind any borrowed figure. They let a reader re-derive or challenge a parameter, separate measurement from assumption, and repeat the validation; without them the model's claims are opinion.

13 **Model response (about 120 words).** The school wants to know whether a second till at the main canteen queue would cut waiting time. From four weeks of timed journeys at one site, 214 journeys between 11:50 and 12:40, a linear model of wait against arrivals per minute predicts that adding one till reduces mean queue time by 12%, from 18.0 to 15.8 minutes, with a typical error of 2 percentage points, so between 10% and 14%. The recommendation is to add one till for the upper-school service only. The estimate holds for this site and term, assumes arrivals are unchanged by the change, and excludes assembly and event days; measure a second site before claiming the saving more widely.

Check the arithmetic: $18.0 \times 0.12 = 2.16$, *so* $18.0 - 2.16 = 15.84 \approx 15.8$ *minutes. All eight clauses must appear inside about 120 words.*

14 **Model response. Title and question:** decision, population, period in one sentence. **Summary:** recommendation, effect with units, uncertainty as a range, scope. **Data:** source, dates, instrument, sample size, cleaning rule, controls, one labelled graph of the raw data. **Model:** numbered assumptions; variables with units; the equation with each parameter's value, unit and estimation method. **Results:** fitted curve with hold-out points plotted; validation errors tabulated; typical and maximum error in context units. **Evaluation:** sensitivity to the two assumptions that matter, with the point where the recommendation flips; two limitations, one beyond the reach of more data. **Conclusion:** recommendation, conditions, next measurement. **References:** sources with dates.

Criterion C. The checklist must be itemised against the brief; wording may differ as long as every section carries its stated evidence.

15 **Model response.** The sample came from volunteers replying to one advert, so it is not a random sample of the group the decision concerns, and the report labels it accordingly. Volunteer samples usually over-represent people who already have the characteristic measured, which would inflate the effect, so the 12% reduction is likelier an upper than a lower estimate. The bias affects size rather than direction, since both queues were timed in the same weeks by the same observers and the ranking held in every week separately. The check proposed is one week at a second site recruited without an advert; a reduction under 5% there would withdraw the recommendation.

Criterion D v. Name the direction of the bias, its effect on the quoted figure, and a test that could overturn the recommendation; a different diagnosis is acceptable if argued.

Milestones · Capstone project brief (model project, Brief A · Water)

1 **Model response. Question:** what rainwater tank capacity, in litres, covers at least 80% of the garden's watering demand on watering days from 1 March to 31 May, and does the water saved pay for the tank? **Decision and stakeholders:** the site manager buys, so the bursar and grounds team are stakeholders; **scope:** one roof, one garden, one term. **Variables:** dependent, weekly water supplied (L) and coverage (%); independent, tank capacity (L), weekly rainfall (mm), watering days per week; held constant, roof 320 m^2, garden 150 m^2, 5 L per m^2 per watering day. **Assumptions:** (1) each mm on 320 m^2 yields $320 \times 0.85 = 272$ L, runoff coefficient 0.85; (2) water above capacity overflows and is lost; (3) the tank is drawn down weekly to the watering target; (4) demand is $150 \times 5 = 750$ L on each of 3 days, 2250 L a week, 29 250 L a term; (5) mains water costs £1.60 per m^3. **Data plan:** daily rainfall from the school station totalled weekly and cross-checked against the public gauge 6 km away, draw from the irrigation-line meter, the gauge's three earlier years for validation, missing days listed not interpolated, staff permission recorded and no personal data.

Criterion D i. A strong plan names the decision, the population, the period and every unit, and the demand arithmetic is checkable; roof area, site and tariff may differ.

2 **Model response.** Weekly rainfall in mm: 6, 9, 4, 11, 7, 5, 10, 8, 3, 9, 6, 12, 7, totalling 97 mm. School station read daily at 09:00, checked against the public gauge with a largest weekly disagreement of 4 mm in week 9, where the school gauge under-read after a blocked funnel so the gauge figure was used; no weeks missing. Harvest at 272 L per mm gives 1632, 2448, 1088, 2992, 1904, 1360, 2720, 2176, 816, 2448, 1632, 3264, 1904 L, a total of 26 384 L, or $26\,384 \div 29\,250 = 90.2\%$ of demand before storage loss. The first graph plots weekly harvest against week with the 2250 L demand as a horizontal line, labelled axes, weeks above tank capacity marked.

Every harvest figure must equal $272\times$ *the printed rainfall and the total 26 384 L; coverage changes with the record used, so state which weeks are in scope.*

3 **Model response.** With capacity C and weekly harvest $H_w = 272r_w$, the weekly supply is $U_w = \min(H_w, C)$, rain arriving at a full tank being lost, so term coverage is $\sum U_w \div 29\,250$. The single fitted parameter is the runoff coefficient, estimated as 0.85 by regressing metered draw on rainfall over the earlier three years. For $C = 2400$ L the weekly supplies total 24 512 L, $24\,512 \div 29\,250 = 83.8\%$. Validation on the last four weeks, unused in fitting: predicted $2400 + 1632 + 2400 + 1904 = 8336$ L against 8100 L metered, an error of 236 L, $236 \div 8100 = 2.9\%$, with no trend in the residuals.

Criterion B iii and iv. Validation data must be data not used in fitting and the error must be quoted in litres as well as per cent.

4 **Model response. Comparison** (objective: 80% coverage at least installed cost): 1200 L at £600 supplies 15 104 L, 51.6%, and fails; 2400 L at £900 supplies 24 512 L, 83.8%, the cheapest meeting the target; 3500 L at £1150 supplies 26 384 L, 90.2%, at £0.0436 per litre supplied against £0.0367. **Sensitivity**: at coefficient 0.75 the harvest is 240 L per mm, so the 2400 L tank covers 77.1% and even the 3500 L tank only 79.6%, so no size reaches 80% and the answer becomes reduce demand or accept less; at 0.95 coverage is 88.7%, and 4 watering days rather than 3 a week cuts the 2400 L tank to 62.9%. **Draft evaluation:** at £1.60 per m^3 the tank saves $24.512 \times 1.60 = 39.22$, that is £39.22 a term, so £900 needs about 23 terms to repay, past its working life, and a 10-year payback needs a tariff above roughly £3.67 per m^3. The model cannot see the value of stored water during a supply restriction, nor its use for teaching.

Criterion D iv and v. Name the flip point, here a coefficient near 0.79, and give the saving in money and in per cent coverage, not one or the other.

5 **Model response.** The report recommends the 2400 L tank as a resilience and teaching measure, not a saving: it covers 83.8% of this term's demand, 24 512 L modelled with a 2.9% error on four held-out weeks, but returns £39.22 a term at the current tariff, a payback near 23 terms, and it falls below the 80% target if the runoff coefficient is nearer 0.75 than 0.85. The limitation more data cannot fix is that watering happens only when mains pressure is adequate and the grounds team chooses, so demand is a decision, not a physical constant. The defence opens with the recommendation and those two numbers, then names the measurement that would resolve the challenge on the coefficient: metered draw against rainfall over a second term.

Criterion C vi and D v. State the recommendation, its uncertainty and one limitation no amount of data removes; a different but consistent tank choice is acceptable.

Capstone · Criterion B · Investigating patterns

1 **Criterion B, Investigating patterns, 24 marks.** Allocation implied by the evidence required: data table with provenance, source, dates and sample size [4]; the pattern described from a table, graph or ratio, saying what changes and by how much [4]; the model with every parameter's value, unit and estimation method [6]; validation on data not used in fitting, errors in context units [6]; a scope statement of input range, conditions and population [4]. Deduct where fitting and testing share data, a parameter is asserted rather than estimated, or the generalisation is a picture with no numbers or no range.

Capstone · Criterion C · Communicating

1 **Criterion C, Communicating, 20 marks.** Allocation: summary stating recommendation and uncertainty in the first paragraph [4]; the eight sections present and in order [3]; notation and units correct, including every axis [3]; graphs readable unaided, labelled axes, sensible scales, fitted points separated from held-out ones [4]; a conclusion claiming nothing beyond the evidence [3]; coherent oral answers to two reviewer questions, one on limitations, each naming the assumption and the check that would settle it [3]. Length is not evidence, so graphs carrying no argument earn nothing.

Capstone · Criterion D · Applying mathematics

1 **Criterion D, Applying mathematics, 24 marks.** Allocation: decision context with named stakeholder and the accuracy the decision needs [3]; an objective and at least two explicit constraints forming the optimisation or comparison [4]; an attainable recommendation with any binding constraint's cost quantified [5]; a sensitivity check varying a stated assumption and reporting where the recommendation changes [6]; evaluation of two limitations, one beyond the reach of more data, distinguishing measurable uncertainty from unmeasured ignorance [6]. The short Criterion A check tests technique, so sound methods with a debatable recommendation can still reach its top band.

The printed weightings, 24 for B, 20 for C and 24 for D, exceed the MYP maximum of 8 per criterion and are usable only with a declared scaling, for example three times the bands for B and D and two and a half for C. Criterion A is mentioned in the brief but carries no marks.

Reflection

1 **Model response.** Validation was hardest, and not for the arithmetic: subtracting a prediction from a held-out reading is easy, but choosing which weeks to hold back and deciding what error is acceptable is judgement, since the tolerance comes from the decision, not from the data.

Any stage honestly argued is acceptable; the credit is in naming a judgement rather than a calculation.

2 **Model response.** I took the runoff coefficient 0.85 as a fact because a handbook gave it. It showed itself as an assumption when I estimated it from my own metered draws and got a different figure, and when coverage moved from 83.8% to 77.1% for a change of that one number: something I had to look up, whose value swings the answer, is an assumption.

A strong answer identifies the move by its signature, a quantity not measured here whose change alters the recommendation.

3 **Model response.** The bus timetable supports the claim: the averages were right and what the model ignored, interchange waiting, decided the outcome for hospital and night-shift passengers. Example 11.1 complicates it, since the water model also ignores attendance yet remains the right model for the four weeks it was built for. What matters is not how much is left out but whether the omission changes the answer asked for.

Both directions of argument are acceptable; the mark is for using the two cases as evidence rather than restating the slogan.

4 **Model response.** With twice as much data I would not refit but split it, fitting one half and validating on the other, tightening the interval on the coefficient and testing the weeks I had assumed away. With none, I would keep the structure, run it on published figures from a neighbouring school, report a range rather than a number, and name the first measurement that would settle it.

Legitimate answers differ; each should say what more data can fix and what it cannot, such as the decision-shaped demand.

5 **Model response.** Where is the line between an assumption that simplifies and one that distorts? I would answer it by varying one assumption at a time in a model I know well and comparing how far the recommendation moves with the error the decision can tolerate, since the opening case suggests the question is never whether a model is simple but whether its simplicity survives the people who depend on it.

An open question, not a fact to recall: credit goes to a test the writer could actually run.

CHAPTER 12 · CUMULATIVE REVISION AND THE TRANSITION TO DP MATHEMATICS

Opening problem · the examination question nobody had seen

1 The question combines a recurrence relation (each value built from the previous one), substitution (feeding $x = 0, 1, 2$ through $f(x+1) = 2f(x) + 3$), and a test for linearity (constant first differences).

2 The values are $1, 5, 13, 29$ with first differences $4, 8, 16$; a linear function has constant first differences, so f is not linear. "Exponential" alone is not right either: the ratios $5, 13/5, 29/13$ are not constant. The exact form $f(n) = 2^{n+2} - 3$ shows $f(x) + 3$ is exponential (it doubles at each step), so the honest description is an exponential function with a constant shift.

3 Put $g(x) = f(x) + 3$; then $g(x+1) = 2g(x)$ and $g(0) = 4$, so $g(n) = 4 \cdot 2^n = 2^{n+2}$ and $f(n) = 2^{n+2} - 3$. Verify: $f(0) = 1$, $f(1) = 5$, $f(2) = 13$, $f(3) = 29$, and $2f(n) + 3 = 2(2^{n+2} - 3) + 3 = 2^{n+3} - 3 = f(n+1)$.

4 **Model response.** Computing the first few values before giving up: the sequence $1, 5, 13, 29$ exposed the doubling differences and made both parts answerable. Any named habit works if it is specific and tied to the question.

A strong answer names one transferable habit (specialise first, check with small cases, restate in your own words) and says how it unlocked this question, not just that it "helped".

Revision set 12A · Algebra and functions

1 $(2x-3)(x^2+x-4) = 2x^3 + 2x^2 - 8x - 3x^2 - 3x + 12 = 2x^3 - x^2 - 11x + 12$. Check at $x = 0$: $(-3)(-4) = 12$.

2 (a) $x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2)$.

(b) $x = 2$ is a root since $16 + 12 - 22 - 6 = 0$; dividing gives $2x^3 + 3x^2 - 11x - 6 = (x-2)(2x^2 + 7x + 3) = (x-2)(2x+1)(x+3)$.

3 Synthetic division by 2 gives quotient $2x^2 - x - 1$ and remainder 0. Check: $(x-2)(2x^2 - x - 1) = 2x^3 - 5x^2 + x + 2$; the quotient factors further as $(2x+1)(x-1)$.

4 $\dfrac{x^2-9}{x^2+x-12} = \dfrac{(x-3)(x+3)}{(x-3)(x+4)} = \dfrac{x+3}{x+4}$, with excluded values $x \neq 3$ and $x \neq -4$ (the original denominator and the cancelled factor).

5 Multiply by $(x-1)(x+2)$: $2(x+2) + 3(x-1) = (x-1)(x+2)$, so $5x + 1 = x^2 + x - 2$, i.e. $x^2 - 4x - 3 = 0$, giving $x = 2 \pm \sqrt{7}$. Neither equals 1 or -2, and substituting $x = 2 + \sqrt{7}$ gives $0.549 + 0.451 = 1$.

6 $y = 2(x^2 - 4x) + 3 = 2\big((x-2)^2 - 4\big) + 3 = 2(x-2)^2 - 5$. Vertex $(2, -5)$.

7 As in Example 12.2: $f^{-1}(x) = \dfrac{x+2}{3-x}$, domain $x \neq 3$ (matching the range of f). Check: $f(0) = -2$ and $f^{-1}(-2) = 0$.

8 $f(g(x)) = \big(\sqrt{x-3}\big)^2 + 1 = x - 2$. g requires $x \geq 3$, and $g(x) \geq 0$ always satisfies f's restriction, so the domain is $x \geq 3$.

9 Translate 1 unit right, then stretch vertically by scale factor 3 (parallel to the y-axis), then translate 4 units down: $2^x \to 2^{x-1} \to 3 \cdot 2^{x-1} \to 3 \cdot 2^{x-1} - 4$. The asymptote moves from $y = 0$ to $y = -4$; the y-intercept is $3 \cdot 2^{-1} - 4 = -2.5$. (Equivalent: stretch by factor $\frac{3}{2}$ then translate down 4, since $3 \cdot 2^{x-1} = \frac{3}{2} \cdot 2^x$.)

10 $(2x-1)\ln 5 = \ln 30$, so $x = \dfrac{1}{2}\left(\dfrac{\ln 30}{\ln 5} + 1\right) = \dfrac{1}{2}(2.1133 + 1) = 1.56$ (3 s.f.). Check: $5^{2(1.5566)-1} = 5^{2.1133} = 30$.

11 $\log_3(x+5) + \log_3(x-1) = 2$ gives $(x+5)(x-1) = 3^2 = 9$, so $x^2 + 4x - 14 = 0$ and $x = -2 \pm \sqrt{18} = -2 \pm 3\sqrt{2}$. Domain requires $x > 1$, so $x = -2 + 3\sqrt{2} \approx 2.24$; $-2 - 3\sqrt{2}$ is rejected.

12 Let $u = e^x$: $u^2 - 3u - 10 = (u-5)(u+2) = 0$. Since $e^x > 0$, $u = 5$ and $x = \ln 5 \approx 1.609$.

13 $a + 6d = 23$, $a + 11d = 38$ give $5d = 15$, so $d = 3$ and $a = 23 - 18 = 5$. $S_{20} = \dfrac{20}{2}(2(5) + 19(3)) = 10(67) = 670$.

14 $S_4 = \dfrac{3\left(1 - (2/3)^4\right)}{1 - 2/3} = 9\left(1 - \dfrac{16}{81}\right) = \dfrac{65}{9} \approx 7.22$. $S_\infty = \dfrac{3}{1 - 2/3} = 9$ since $\left|\frac{2}{3}\right| < 1$.

15 Arithmetic: $1, 5, 9, \ldots, 57$ (15 terms), so $\displaystyle\sum_{k=1}^{15}(4k-3) = 4 \cdot \frac{15 \cdot 16}{2} - 3 \cdot 15 = 480 - 45 = 435$. Check: $\dfrac{15}{2}(1 + 57) = 435$.

16 $(x-5)(x+2) \leq 0$. The parabola opens upward, so $x^2 - 3x - 10 \leq 0$ between the roots: $-2 \leq x \leq 5$. On the number line: closed (filled) dots at -2 and 5 with the segment between them shaded.

17 Vertices: $(0,1)$, $(4,1)$, $(4,2)$ (intersection of $x = 4$ and $x + y = 6$), $(0,6)$. Values of $P = 3x + 2y$: $2, 14, 16, 12$. Maximum $P = 16$ at $(4,2)$.

18 $5000(1.032)^n > 8000$ gives $n > \dfrac{\ln 1.6}{\ln 1.032} = \dfrac{0.47000}{0.03150} = 14.9\ldots$, so 15 whole years. Check: $5000(1.032)^{14} = 7769$ but $5000(1.032)^{15} = 8017 > 8000$.

19 Vertical asymptote $x = -3$ (denominator zero), horizontal asymptote $y = 2$ (ratio of leading coefficients). Intercepts: $\left(0, -\dfrac{1}{3}\right)$ and $\left(\dfrac{1}{2}, 0\right)$.

20 $P(2) = 8 - 12 + 4 = 0$, so by the factor theorem $x - 2$ is a factor. Then $x^3 - 3x^2 + 4 = (x-2)(x^2 - x - 2) = (x-2)^2(x+1)$. Check by expanding: $(x^2 - 4x + 4)(x+1) = x^3 - 3x^2 + 4$.

21 $x + 3$ a factor means $P(-3) = 0$: $2(-27) + k(9) - 7(-3) + 6 = -54 + 9k + 21 + 6 = 9k - 27 = 0$, so $k = 3$. Check: $P(-3) = -54 + 27 + 21 + 6 = 0$.

22 $f(g(x)) = 2x^2 + 3$ and $g(f(x)) = (2x+3)^2 = 4x^2 + 12x + 9$. Equating: $2x^2 + 12x + 6 = 0$, i.e. $x^2 + 6x + 3 = 0$, so $x = -3 \pm \sqrt{6}$. Check at $x = -3 + \sqrt{6}$: both sides equal $33 - 12\sqrt{6}$.

23 Roots: $x = 1$ with multiplicity 2 (graph touches the axis and turns) and $x = -3$ with multiplicity 1 (graph crosses). y-intercept: $(0-1)^2(0+3) = 3$, so $(0, 3)$. End behaviour: cubic with positive leading coefficient, $y \to -\infty$ as $x \to -\infty$ and $y \to +\infty$ as $x \to +\infty$. Stationary points from the expanded form $x^3 + x^2 - 5x + 3$: local maximum at $x = -\frac{5}{3}$ with value $\frac{256}{27} \approx 9.5$, local minimum at $(1, 0)$. Sketch: down from the left, crosses at -3, peaks near $(-1.67, 9.5)$, descends to touch the axis at $(1, 0)$, then rises.

24 Base $(20 - 2x)$ by $(14 - 2x)$, height x: $V(x) = x(20-2x)(14-2x) = 4x^3 - 68x^2 + 280x$. All dimensions positive require $0 < x < 7$. Table: $V(1) = 216$, $V(2) = 320$, $V(3) = 336$, $V(4) = 288$, $V(5) = 200$ cm^3, so from the table $x = 3$ cm gives the largest volume, 336 cm^3.

A finer table (step 0.1) places the true maximum near $x \approx 2.7$ cm with $V \approx 339$ cm^3; the integer table locates but slightly underestimates it.

25 Take $a < b$. Then $a^3 < b^3$ and $a < b$, so $f(a) = a^3 + a < b^3 + b = f(b)$: f is strictly increasing, hence one-to-one, so an inverse function exists on its range. The inverse cannot be obtained by rearranging with square roots, but the printed claim that it cannot be written using square and cube roots is not correct: solving $y = x^3 + x$ by Cardano's method (not taught in this course) gives $f^{-1}(y) = \sqrt[3]{\frac{y}{2} + \sqrt{\frac{y^2}{4} + \frac{1}{27}}} + \sqrt[3]{\frac{y}{2} - \sqrt{\frac{y^2}{4} + \frac{1}{27}}}$, which uses only square and cube roots.

Defect in the printed item: the intended point is presumably that the inverse has no simple rearrangeable form (e.g. read "cannot be written using square roots only"). The one-to-one proof above is the assessed part.

Revision set 12B · Geometry and trigonometry

1 $2 \times 37^\circ = 74^\circ$: the angle at the centre is twice the angle at the circumference on the same arc.

2 Opposite angles of a cyclic quadrilateral sum to 180°, so the opposite angle is $180^\circ - 112^\circ = 68^\circ$.

3 $\angle OTP = 90^\circ$: a tangent is perpendicular to the radius at the point of contact.

4 Intersecting chords: $4 \times 6 = 3x$, so $x = 8$ cm. Check: both products equal 24.

5 Sine rule: $\dfrac{b}{\sin 65^\circ} = \dfrac{9}{\sin 42^\circ}$, so $b = \dfrac{9 \sin 65^\circ}{\sin 42^\circ} = \dfrac{9(0.90631)}{0.66913} = 12.2$ cm (3 s.f.).

6 Area $= \dfrac{1}{2} \cdot 7 \cdot 9 \sin 50^\circ = 31.5(0.76604) = 24.1$ cm^2 (3 s.f.).

7 Cosine rule: $AC^2 = 6^2 + 10^2 - 2(6)(10)\cos 120^\circ = 36 + 100 + 60 = 196$ (since $\cos 120^\circ = -\frac{1}{2}$), so $AC = 14$ exactly.

8 At the turning point the back-bearing of the first leg is $60^\circ + 180^\circ = 240^\circ$, so the angle between the legs is $240^\circ - 150^\circ = 90^\circ$. Distance $= \sqrt{8^2 + 5^2} = \sqrt{89} = 9.43$ km (3 s.f.). Check by components: east $8\sin 60^\circ + 5\sin 150^\circ = 9.43$, north $8\cos 60^\circ + 5\cos 150^\circ = -0.33$; $\sqrt{9.43^2 + 0.33^2} = 9.43$.

9 Diagonal $= \sqrt{3^2 + 4^2 + 12^2} = \sqrt{169} = 13$ cm. The base diagonal is $\sqrt{3^2 + 4^2} = 5$, and $\tan\theta = \dfrac{12}{5}$ gives $\theta = 67.4^\circ$ (3 s.f.) with the base.

10 The perpendicular from the centre bisects the chord, so distance $= \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

11 Tangent–secant: $PT^2 = PA \cdot PB$, so $144 = 8 \cdot PB$, $PB = 18$ cm, hence $AB = PB - PA = 10$ cm.

12 Let AB be a diameter with centre O, and P any other point on the circle. Then $OP = OA = OB$ (radii), so triangle OAP is isosceles with $\angle OAP = \angle OPA = \alpha$ and triangle OBP is isosceles with $\angle OBP = \angle OPB = \beta$. Angle sum in triangle APB: $\alpha + \beta + (\alpha + \beta) = 180^\circ$, so $\angle APB = \alpha + \beta = 90^\circ$.

13 Mast height $= \sqrt{15^2 - 9^2} = \sqrt{144} = 12$ m. The triangle formed by the two wires has sides 15, 15 and base 18, so $\cos\theta = \dfrac{15^2 + 15^2 - 18^2}{2(15)(15)} = \dfrac{126}{450} = 0.28$, giving $\theta = 73.7^\circ$ (3 s.f.). Check: $\sin(\theta/2) = 9/15 = 0.6$ gives $\theta = 2(36.87^\circ) = 73.7^\circ$.

14 Since $45^2 + 60^2 = 2025 + 3600 = 5625 = 75^2$, the field is a right triangle (converse of Pythagoras) with legs 45 m and 60 m. Area $= \dfrac{1}{2}(45)(60) = 1350$ m^2; the largest angle is 90°, opposite the longest side 75 m.

15 Let XY be a chord, the tangent at X meeting the chord on the side of the alternate segment, and Z a point on the arc cut off by XY on the far side. Join OX and OY. Then $OX = OY$, so triangle OXY is isosceles: $\angle OXY = \angle OYX = 90^\circ - \frac{1}{2}\angle XOY$. The tangent is perpendicular to OX, so the angle between tangent and chord XY is $90^\circ - \angle OXY = \frac{1}{2}\angle XOY$. By the centre–circumference theorem, $\angle XZY = \frac{1}{2}\angle XOY$ (same arc XY). Hence the angle between tangent and chord equals the angle in the alternate segment.

A strong proof labels the diagram, states each theorem as the reason for each step, and treats the tangent–radius fact and the isosceles triangle explicitly.

Revision set 12C · Statistics and probability

1 Mean $= \dfrac{4 + 6 + 7 + 9 + 11 + 15 + 18}{7} = \dfrac{70}{7} = 10$; median $= 9$ (4th value). Convention: median excluded from the halves, so $Q1 = 6$ (of 4, 6, 7) and $Q3 = 15$ (of 11, 15, 18); IQR $= 15 - 6 = 9$. (The position formulae $\frac{n+1}{4}$ and $\frac{3(n+1)}{4}$ give the same quartiles here.)

2 Mean $= \dfrac{30}{5} = 6$. Squared deviations: $9 + 1 + 0 + 4 + 4 = 18$; variance $= \dfrac{18}{5} = 3.6$; standard deviation $= \sqrt{3.6} = 1.90$ (3 s.f.), population convention (denominator 5) as instructed.

3 IQR $= 20 - 12 = 8$, $1.5 \times 8 = 12$. Boundaries: $12 - 12 = 0$ and $20 + 12 = 32$; values below 0 or above 32 are outliers.

4 Without replacement: $P(\text{RR}) = \dfrac{4}{10} \cdot \dfrac{3}{9} = \dfrac{12}{90} = \dfrac{2}{15} \approx 0.133$.

5 $\binom{10}{3} = \dfrac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$ ways (order does not matter).

6 Permutations: $26 \times 25 \times 24 \times 23 = 358\,800$ codes.

7 $|F \cup B| = 18 + 12 - 6 = 24$ students play at least one sport, so the probability is $\dfrac{24}{30} = 0.8$. The 6 who play both must be subtracted once, since they are counted in both 18 and 12.

Common error: adding $18 + 12 = 30$ and reporting a probability of 1; the two groups overlap and the union is not the sum.

8 Test: $P(A)P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$. The condition holds, so A and B are independent.

9 Bayes/tree: $P(\text{cond} \mid \text{pos}) = \dfrac{0.95 \times 0.002}{0.95 \times 0.002 + 0.04 \times 0.998} = \dfrac{0.0019}{0.0019 + 0.03992} = \dfrac{190}{4182} = \dfrac{95}{2091} \approx 0.045$, about 4.5%. Most positives come from the healthy 99.8%, even at a 4% false-positive rate.

10 $r = 0.82$ indicates a strong positive linear correlation: more study time tends to go with higher scores. It does not show causation (nor does it measure curved relationships, or the effect of hidden variables such as prior ability), and it says nothing about whether extra hours cause the gain.

11 $\hat{y} = 42 + 0.8(10) = 50$. For $x = 40$ the prediction $42 + 32 = 74$ is unreliable: it extrapolates far beyond the range of the data, the linear form cannot continue (scores have a ceiling), and 40 hours of study is itself outside plausible values.

12 The mean uses every value as a distance from the centre ($\bar{x} = \frac{\sum x}{n}$), so extreme values in a long tail pull the balance point toward them; the median only needs the middle position, so moving tail values further out does not move it.

13 $P(\text{Walk} \mid \text{Year 10}) = \dfrac{24}{60} = 0.4$. $P(\text{Year 10} \mid \text{Walk}) = \dfrac{24}{42} = \dfrac{4}{7} \approx 0.571$. Not independent: $P(\text{Walk}) = \dfrac{42}{120} = 0.35 \neq 0.4 = P(\text{Walk} \mid \text{Year 10})$ (equivalently $\dfrac{24}{120} = 0.2 \neq 0.35 \times 0.5 = 0.175$).

14 $P(\text{win}) = 1 - \left(\dfrac{5}{6}\right)^2 = 1 - \dfrac{25}{36} = \dfrac{11}{36} \approx 0.306$. A fair ticket costs the expectation: $\dfrac{11}{36} \times 10 = \dfrac{110}{36} \approx 3.06$ (same units as the prize). Since 3.06 exceeds the stated charge of 2, this game actually favours the player.

15 **Model response.** In Question 9, out of 100 000 people about 200 have the condition and 95\,\% of them (190) test positive, but about 99 800 do not and 4\,\% of them (3992) test positive anyway. Of 4182 positives only 190 are true, so a positive result means the condition in under 5\,\% of cases. High sensitivity and specificity fix the conditional probabilities the wrong way round; when the condition is rare the huge healthy group overwhelms the few true positives, so the positive predictive value is low.

A strong answer quotes the numbers of Question 9, names the base rate, and distinguishes $P(\text{pos} \mid \text{cond})$ from $P(\text{cond} \mid \text{pos})$.

Revision set 12D · Unfamiliar problems

1 The diagonal of the rectangle is a diameter of the circle, so $(2x)^2 + (2y)^2 = 10^2$, i.e. $4x^2 + 4y^2 = 100$ and $x^2 + y^2 = 25$. Area $= 4xy$. Since $(x - y)^2 \geq 0$, $xy \leq \dfrac{x^2 + y^2}{2} = 12.5$, with equality when $x = y$. So the greatest area is $4(12.5) = 50$, attained by the square with side $2x = 5\sqrt{2}$ (about 7.07) each way.

2 The sums are $1, 4, 9, 16, 25$, suggesting $S_n = n^2$. Pairing: write $S_n = 1 + 3 + \cdots + (2n - 1)$ and, below it, the same sum reversed; each column adds to $1 + (2n - 1) = 2n$ and there are n columns, so $2S_n = n(2n)$ and $S_n = n^2$. Check at $n = 5$: 25 terms sum $= 5^2$.

3 $2\pi r^2 + 2\pi rh = 600\pi$ gives $r^2 + rh = 300$, hence $h = \dfrac{300 - r^2}{r}$. Volume $V = \pi r^2 h = \pi r(300 - r^2) = \pi(300r - r^3)$, valid for $0 < r < \sqrt{300} \approx 17.3$ (so that $h > 0$). Table: $r = 8, 9, 10, 11, 12$ give $V = 1888\pi, 1971\pi, 2000\pi, 1969\pi, 1872\pi$, so the table estimates $r \approx 10$ cm, $V = 2000\pi \approx 6280$ cm^3 (and then $h = 20$ cm).

4 Join AD and BC. In triangles APD and CPB: $\angle DAP = \angle BCP$ (angles in the same segment, on arc BD) and $\angle APD = \angle CPB$ (vertically opposite), so the triangles are similar (AA). Hence $\dfrac{PA}{PC} = \dfrac{PD}{PB}$, i.e. $PA \cdot PB = PC \cdot PD$. Check against 12B Q4: $4 \times 6 = 3 \times 8$.

5 From the cosine rule, $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$. Then $16A^2 = 4a^2b^2\sin^2 C = 4a^2b^2\left(1 - \cos^2 C\right) = 4a^2b^2 - (a^2 + b^2 - c^2)^2$. Difference of two squares: $= \left(c^2 - (a - b)^2\right)\left((a + b)^2 - c^2\right) = (c - a + b)(c + a - b)(a + b - c)(a + b + c)$. With $s = \frac{a+b+c}{2}$ these four factors are $2(s - a)$, $2(s - b)$, $2(s - c)$, $2s$, so $16A^2 = 16s(s - a)(s - b)(s - c)$ and $A = \sqrt{s(s - a)(s - b)(s - c)}$. Check on 3, 4, 5: $s = 6$, $A = \sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6 = \frac{1}{2} \cdot 3 \cdot 4$.

6 Compounded annually: $(1 + r)^8 = 2$ gives $r = 2^{1/8} - 1 = 0.0905$, about 9.05\,\% per year. Compounded continuously: $e^{8r} = 2$ gives $r = \frac{\ln 2}{8} = 0.0866$, about 8.66\,\%. The quoted rate differs because interest credited more often earns interest on itself sooner, so a smaller nominal rate suffices under continuous compounding; "annual rate" must state which convention it means.

7 "First head on the third toss" means the sequence TTH, of probability $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. Expected win $= 5 \cdot \frac{1}{8} = 0.625$, but the charge is 2, so expected gain $= 0.625 - 2 = -1.375$: the game is not fair — it favours the organiser. A fair charge would be 0.625 (or a fair prize of 16 for a charge of 2).

8 Let the integers be n, $n+1$, $n+2$. Among any two consecutive integers one is even, so the product is even (divisible by 2). Of any three consecutive integers, the residues modulo 3 are $0, 1, 2$ in some order, so exactly one is a multiple of 3 and the product is divisible by 3. Since 2 and 3 are coprime, the product is divisible by $2 \times 3 = 6$. Check: $4 \cdot 5 \cdot 6 = 120 = 6 \times 20$.

Criterion A · Knowing and understanding

1 $P(2) = 16 + 4 - 26 + 6 = 0$, so $x - 2$ is a factor; dividing gives $2x^3 + x^2 - 13x + 6 = (x-2)(2x^2 + 5x - 3) = (x-2)(2x-1)(x+3)$. Check: the constant term $(-2)(-1)(3) = 6$.

Marking guidance: [4] — factor theorem check 1, correct quotient $2x^2 + 5x - 3$ 1, factorising the quotient 1, fully factorised statement 1.

2 $y = \dfrac{x+3}{2x-1}$ gives $x(2y-1) = y + 3$, so $f^{-1}(x) = \dfrac{x+3}{2x-1}$ with domain $x \neq \dfrac{1}{2}$: f is its own inverse. Check: $f(1) = 4$ and $f^{-1}(4) = \frac{7}{7} = 1$.

Marking guidance: [4] — valid rearrangement 2, correct inverse stated 1, restriction $x \neq \frac{1}{2}$ 1.

3 $(x+2)\ln 3 = x \ln 7$, so $x(\ln 7 - \ln 3) = 2 \ln 3$ and $x = \dfrac{2\ln 3}{\ln 7 - \ln 3} = \dfrac{2.19720}{0.84730} = 2.59$ (3 s.f.).

Marking guidance: [3] — taking logs 1, correct isolation of x 1, value to 3 s.f. 1.

4 $a + 4d = 17$, $a + 9d = 37$ give $5d = 20$, $d = 4$, $a = 1$. $S_{25} = \dfrac{25}{2}(2(1) + 24(4)) = \dfrac{25}{2}(98) = 1225$.

Marking guidance: [3] — common difference 1, first term 1, sum 1.

5 $c^2 = 7^2 + 9^2 - 2(7)(9)\cos 52^\circ = 130 - 126(0.61566) = 52.43$, so $c = 7.24$ (3 s.f.). Area $= \dfrac{1}{2}(7)(9)\sin 52^\circ = 31.5(0.78801) = 24.8$ (3 s.f.).

Marking guidance: [4] — cosine rule substituted 1, c 1, area formula with $\sin C$ 1, area value 1.

6 $P(\text{Science} \mid \text{Year 11}) = \dfrac{26}{40} = 0.65$. Not independent: $P(\text{Science} \mid \text{Year 10}) = \dfrac{18}{40} = 0.45 \neq 0.65$ (equivalently $\frac{26}{80} = 0.325$ but $0.55 \times 0.5 = 0.275$).

Marking guidance: [2] — conditional probability 1, justified independence decision (comparison of two stated probabilities) 1.

Criterion B · Investigating patterns

1 First differences: $3, 5, 7, 9, 11$. Second differences: $2, 2, 2, 2$.

Marking guidance: [3] — first differences all correct 2, second differences all correct 1.

2 For $f(x) = ax^2 + bx + c$ the first differences are $a(2x+1) + b$, which increase by $2a$ as x increases by 1, so the constant second difference is $2a$. Here $2a = 2$, hence $a = 1$.

Marking guidance: [3] — general second difference of a quadratic is constant 1, it equals $2a$ 1, conclusion $a = 1$ 1.

3 With $f(x) = x^2 + bx + c$: $f(0) = 3$ gives $c = 3$; $f(1) = 1 + b + 3 = 6$ gives $b = 2$. So $f(x) = x^2 + 2x + 3$. Check: $f(3) = 9 + 6 + 3 = 18$ and $f(5) = 25 + 10 + 3 = 38$.

Marking guidance: [4] — form stated 1, c found 1, b found 1, verified against two further table values 1.

4 $f(10) = 100 + 20 + 3 = 123$. Verify by extending the difference table: adding $13, 15, 17, 19, 21$ to 38 gives $51, 66, 83, 102, 123$, matching the formula.

Marking guidance: [2] — prediction 1, verification by a second route 1.

Criterion C · Communicating

1 **Model response.** Both claims overread $r = 0.75$. A correlation of 0.75 means students who revise more tend to score higher, and the association is strong and positive — but it is observational: students who revise more differ in other ways (prior ability, attendance, sleep, motivation), any of which could be the real cause, so the data show association, not causation. To establish causation you would need a controlled experiment with random assignment. Second, "75% of the score" confuses r with percentages: $r = 0.75$ is not a proportion of anything. The related quantity is the coefficient of determination, $r^2 = 0.5625$, meaning about 56% of the variation in scores is explained by the linear relationship with revision time — variation in scores, not the scores themselves, and the remaining 44% is unexplained. Even that says nothing about cause.

Marking guidance: [8] — correct reading of r 2; association versus causation with a named confounder 2; correct use and interpretation of r^2 (including that 75% is wrong) 2; clear, coherent, correctly termed written English within the word limit 2.

Criterion D · Applying mathematics in real-life contexts

1 Daily consumption is 1800 litres (the gradient). The model makes sense while $0 \leq d \leq \dfrac{20000}{1800} = \dfrac{100}{9} \approx 11.1$ days, i.e. until the tank is empty.

Marking guidance: [2] — rate 1, domain 1.

2 $20000 - 1800d < 5000$ gives $d > \dfrac{15000}{1800} = 8.33$. Check: after 8 days $V = 5600$; after 9 days $V = 3800$. The tank first falls below 5000 L during the 9th day ($8\frac{1}{3}$ days after Monday).

Marking guidance: [3] — inequality set up 1, solved correctly 1, stated as a day with a boundary check 1.

3 New consumption $1800 \times 1.15 = 2070$ L/day, so $V(d) = 20000 - 2070d$. Below 5000 when $d > \dfrac{15000}{2070} = 7.25$: during the 8th day — about a day earlier. Check: $V(7) = 5510$, $V(8) = 3440$.

Marking guidance: [3] — updated rate 1, new model stated 1, new day found and stated 1.

4 Assumptions: consumption is exactly 1800 L every day (no weekday/weekend pattern, no weather effects) and the only outflow is community use (no leakage, evaporation or other input). The linear model is appropriate for short horizons within the recorded week, since the data show a steady fall; it fails at the end of the run — once the tank reaches zero the volume cannot keep decreasing, and usage would be cut off or the tank refilled, neither of which the model represents.

Marking guidance: [4] — two stated assumptions 2; evaluation naming where the linear m

Note: the tank is modelled as draining at a constant rate; answering treats 1800 L/day as the net consumption while the tank drains.

Revision set 12E · Transition and reflection

1 **Model response.** "Explain" asks for reasons in words that make a given fact or result clear — e.g. 12B Q12 asks you to explain why the angle in a semicircle is a right angle, answered with the two isosceles triangles and the angle sum. "Justify" asks for mathematical support for a claim you have made or a decision you have taken — e.g. in 12C Q8, computing $P(A)P(B) = 0.2 = P(A \cap B)$ justifies the conclusion "independent". Explanation answers "why is this true?"; justification answers "what evidence supports my claim?".

A strong answer distinguishes the two command terms and cites one real worked instance of each from the coursebook.

2 **Model response.** A DP mark scheme awards method marks because a correct procedure demonstrates the understanding the syllabus assesses; a slips in arithmetic shows the student could choose and carry out the right method, so those marks are earned even when the final answer is wrong. The final answer still matters: it carries its own accuracy mark, method marks usually require the method to be visible, and a wrong answer derived by a wrong method scores nothing. Writing the method protects the marks the answer alone cannot.

A strong answer explains why schemes reward visible method, and states that method marks are conditional and the answer keeps its own mark — not that the answer "does not matter".

3 **Model response.** (i) An MYP investigation is open — the question, resources and stopping point are partly yours, and it is assessed against criteria over days; a timed DP question is closed, with a fixed mark scheme awarding marks to specific steps within minutes. (ii) Feedback differs: MYP values the quality of the process and justification even without a final answer, while DP rewards the recorded steps under exam conditions, so planning and speed are part of performance.

Any two legitimate differences work if each names both sides (openness, assessment, time, feedback, independence).

4 **Model response.** (i) Practise retrieval without notes: close the book and re-solve worked examples from scratch, because watching a solution felt like understanding. (ii) Do timed, calculator-allowed past-style sets so the conditions of the test are rehearsed, not just the mathematics. (iii) Keep a three-column error log (question, idea tested, action) and retest the same idea a week later, so "understanding in class" is converted into independently reproducible method.

A strong answer gives specific, checkable habits that target the performance gap between recognition and recall — vague advice like "study harder" scores zero.

5 **Model response.** Calculus introduces objects and ideas the course has not used — limits, differentiation as a new operation on functions, integration as accumulation — rather than making existing MYP topics harder. Solving $x^2 - 5x + 6 = 0$'s difficulty class is unchanged; being asked for the gradient of $y = x^3$ at a point is new content, not deeper content. The jump students feel is usually the new vocabulary and notation, which is why readiness is about habits, not about doing harder quadratics.

A strong answer identifies at least one genuinely new calculus idea and separates "new" from "harder".

6 **Model response.** The claim would be misleading because readiness depends on factors a book cannot supply or certify: which DP course is taken (AA/AI, SL/HL), the school's teaching, and the student's own habits of retrieval, checking and self-diagnosis; the book practises some of these but guarantees none. It also contains no DP-specific content (calculus, vectors, complex numbers, distributions), and a promise that every completer succeeds would be an unfalsifiable marketing claim, not a mathematical one.

A strong answer cites the chapter's own caveats — course, school, habits — and the missing DP content.

7 **Model response.** Chapter 3 changed my thinking most: seeing that $y = ab^x$ models anything with a constant percentage change meant the same graph described decay of a drug and growth of a population, so a formula became a statement about a rate rather than a puzzle to solve. (Any chapter works if a specific idea and a before/after change are named.)

A strong answer names one specific idea and shows how it changed the student's approach, with a concrete example.

8 **Model response.** Developed: checking answers by substitution — I now test $x = 2 \pm \sqrt{7}$ in the original equation instead of moving on. Still needed: starting an unfamiliar problem without a labelled routine; I wait for a method to be named before writing anything. Both should be stated as specific procedures with conditions, as the ATL note requires.

A strong answer names one habit genuinely acquired and one gap precisely enough to act on ("I cannot yet..." plus the condition), not a mood.

9 **Model response.** "Why does the second difference of a quadratic equal twice the leading coefficient?" was my most useful question, because chasing it forced me to compute $a(x+1)^2 + b(x+1) + c - a(x^2 + bx)$ symbolically — turning a pattern I had memorised into something I could prove and then invent with.

A strong answer states a real mathematical question and says what it changed — the "useful" is explained, not assumed.

10 **Model response.** Start every set without notes, because fluency you can only show with the book open is recognition, not skill. Record each error with the named idea that failed and retest it a week later — the log is worth more than re-reading. When a question has no familiar routine, compute a small case and look at the pattern; most "impossible" questions open there.

Three sentences of concrete, usable advice; full marks require specifics, not encouragement.

Final reflection

1 **Model response.** At the start I needed the method naming on the page; now, facing $f(x+1) = 2f(x) + 3$ I generate values and look at differences without a prompt. That is a real, checkable change from Chapter 1's guided expansions to Chapter 12's unsignposted questions.

A strong answer compares two dated pieces of the student's own work and names the specific skill gained.

2 **Model response.** The division error where I wrote $x^3 - 3x^2 + 4 = (x-2)(x^2 + x - 2)$: the correction — always expand the quotient times the divisor back — caught the sign error, and the corrected factorisation $(x-2)^2(x+1)$ then checked out.

A strong answer describes one real error, its correction method, and why the correction works.

3 **Model response.** Of choosing, justifying, connecting and recognising limits, my strongest is connecting — I move between graphs and equations easily. Next I will work on recognising limits: stating what a proof has not shown, as in the four-terms-versus-induction point in Example 12.5.

A strong answer picks one of the four, evidences it from own work, and sets one actionable target.

4 **Model response.** In two years I want to answer questions like "does this series converge, and why?" — DP series and limits — which my current toolkit can state but not settle; that gap tells me what to build.

A strong answer names a question type beyond the course and links it to a concrete next step.

5 **Model response.** Mathematics is more than methods because the methods only become knowledge when you can say why they work and when they stop working: two students can both get $x = 2 \pm \sqrt{7}$, but only the one who can check it, explain the rejected root, and spot the quadratic underneath is doing mathematics. A collection of recipes fails the first unfamiliar question; understanding why a recipe works is what lets you invent a new one.

A strong paragraph for a younger student gives one example, one reason, and a honest statement of why justification matters — no slogans.

REFERENCE

R Mathematical notation

Notation is compressed language: it is only useful if reader and writer attach the same meaning to it. This table gives each symbol used in the course, with the reading that removes ambiguity. Where a symbol has more than one common meaning, the meaning used in this book is stated.

Sets, number and proof

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ natural numbers $1, 2, 3, \ldots$; integers; rationals; reals. This book does not include 0 in $\mathbb{N}$.

$\in$, $\notin$, $\subset$ is an element of; is not an element of; is a subset of. Thus $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

$\cup$, $\cap$, $\varnothing$ union; intersection; the empty set.

$\{x : \ldots\}$ the set of all x satisfying the stated condition, for example $\{x : x > 2\}$.

$[a, b]$, (a, b) closed interval $a \le x \le b$; open interval $a < x < b$. Mixed brackets combine the two.

$<$, $\le$, $>$, $\ge$ less than; at most; greater than; at least. The wide end of the sign faces the larger quantity.

$|x|$ modulus, the distance of x from 0, so $|-4| = 4$ and $|x| \ge 0$.

$\approx$, $\neq$, $\equiv$ approximately equal; not equal; identically equal, true for every permitted value.

$\Rightarrow$, $\Leftarrow$, $\Leftrightarrow$ implies; is implied by; is equivalent to. Use $\Leftrightarrow$ only when both directions are true.

$\therefore$, $\because$, $\blacksquare$ therefore; because; end of proof.

Δ a change in a quantity, or the discriminant $b^2 - 4ac$ of a quadratic. Context decides; this book writes "discriminant" where confusion is possible.

$a \times 10^n$ standard form, with $1 \le a < 10$ and n an integer.

Functions and algebra

$f(x)$ the value of function f at input x. $f(3)$ means substitute $x = 3$.

$f : A \to B$ a function with domain A and codomain B; its range is written $f(A)$ or $\operatorname{ran} f$.

$x \mapsto f(x)$ the rule sending x to $f(x)$, used when the function has no convenient name.

$f \circ g$ the composite function $(f \circ g)(x) = f(g(x))$: apply g first, then f.

f^{-1} the inverse function, defined by $f^{-1}(y) = x \Leftrightarrow f(x) = y$. It is not $\frac{1}{f}$.

D_f, R_f domain and range of f. A restriction is written $f(x) = x^2,\ x \ge 0$.

a^n, a^{-n}, $a^{m/n}$ n factors of a; the reciprocal of a^n; the nth root of a raised to the power m.

e the natural exponential base, $e = 2.718281828\ldots$, the limit of $\left(1 + \frac{1}{n}\right)^n$.

$\ln x$, $\log_a x$ natural logarithm, base e; logarithm base a. $\log x$ without a base means base 10 in this book.

$n!$ factorial, $n(n-1)\cdots 2 \cdot 1$, with $0! = 1$.

${}^nP_r$, ${}^nC_r$ permutations and combinations of r objects chosen from n. Also written $\binom{n}{r}$ for ${}^nC_r$.

Sequences and series

u_n the term in position n; u_1 is the first term, u_{n+1} the next.

S_n the sum of the first n terms; S_∞ is the sum to infinity when it exists.

d, r common difference of an arithmetic sequence; common ratio of a geometric sequence.

$\sum_{k=1}^{n} u_k$ $u_1 + u_2 + \cdots + u_n$. The letter k is a dummy variable.

Vectors and matrices

$\mathbf{a}$, $\overrightarrow{AB}$ a vector, as a column or in components; the displacement from A to B.

$|\mathbf{a}|$ the magnitude, or length, of $\mathbf{a}$. A unit vector has magnitude 1.

$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ unit vectors along the x-, y- and z-axes.

$\mathbf{a} \cdot \mathbf{b}$ scalar product $|\mathbf{a}||\mathbf{b}| \cos\theta$, a scalar, not a vector.

$\mathbf{A}$, $\det \mathbf{A}$, $\mathbf{A}^{-1}$, $\mathbf{A}^{\mathsf{T}}$ a matrix; its determinant; its inverse, existing only when $\det \mathbf{A} \neq 0$; its transpose.

$\mathbf{I}$ the identity matrix, with 1s on the leading diagonal and 0s elsewhere.

REFERENCE

R Notation: geometry, data and chance

Geometry and trigonometry

AB the length of segment AB; the same symbols may name the segment itself.

$\angle ABC$ the angle at vertex B, between BA and BC.

$\triangle ABC$ the triangle with vertices A, B, C; A, B, C also label its angles.

$\cong$, $\sim$ is congruent to, same shape and size; is similar to, same shape with lengths in a fixed ratio.

$\parallel$, $\perp$ is parallel to; is perpendicular to.

$^\circ$, rad degrees; radians. π rad $= 180^\circ$.

θ, α, β Greek letters used for unknown angles; θ is the default.

$\sin\theta$, $\cos\theta$, $\tan\theta$ trigonometric ratios. $\sin^{-1} x$ is the inverse sine, not $\dfrac{1}{\sin x}$.

$\overset{\frown}{AB}$ the arc from A to B; the minor arc unless stated.

$a : b$ the ratio of a to b; in a triangle, a is the side opposite angle A.

V, S, A volume; surface area; area. A_{base} is the area of the base.

r, h, l radius; perpendicular height; slant height, as used in cone and pyramid formulae.

Coordinate geometry in two and three dimensions

(x, y), (x, y, z) coordinates of a point in the plane; in space, in the order x, y, z.

m gradient, $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, the tangent of the angle of inclination.

$y = mx + c$, $y - y_1 = m(x - x_1)$ gradient–intercept and point–gradient forms of a line.

$(x - h)^2 + (y - k)^2 = r^2$ a circle with centre (h, k) and radius r.

d distance between points, $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ in the plane.

Statistics

$\bar{x}$ the sample mean; μ is the population mean.

f, cf frequency; cumulative frequency. $\sum f = n$ is the number of observations.

Q_1, Q_2, Q_3 lower quartile, median, upper quartile. $\text{IQR} = Q_3 - Q_1$.

σ, σ^2 standard deviation and variance of a population or data set. s is used for a sample estimate.

r Pearson's product-moment correlation coefficient, $-1 \le r \le 1$. In sequences, r is the common ratio.

$y = a + bx$ least-squares regression line of y on x; b is the gradient, a the intercept.

e_i, $\hat{y}_i$ residual $e_i = y_i - \hat{y}_i$, the observed value minus the value predicted by the model.

z standardised score, $z = \dfrac{x - \bar{x}}{\sigma}$, stating how many standard deviations a value lies from the mean.

Probability

$P(A)$ the probability of event A, a number from 0 to 1. $P(A')$ is the probability that A does not occur.

$A \cap B$, $A \cup B$ A and B both occur; A or B or both occur.

$P(A \mid B)$ the conditional probability of A given that B has occurred.

$E(X)$ the expected value of a discrete random variable X, $\sum xP(X = x)$.

$n(S)$, $n(A)$ the number of outcomes in the sample space; in event A.

$X \sim B(n, p)$ X has a binomial distribution with n trials and success probability p.

Modelling

x, y, z decision variables in a model; the objective is written $Z = ax + by$.

subject to the constraints of an optimisation problem, written as inequalities.

REFERENCE

R Formulae and relationships

A formula is a relationship, not a procedure. State what each letter represents, the conditions under which the relationship holds, and the units in which the quantities are measured.

Algebra and functions

Division algorithm $P(x) = D(x)Q(x) + R(x)$, where $\deg R < \deg D$. For a linear divisor $x - a$, the remainder is $P(a)$.

Factor theorem $x - a$ is a factor of $P(x)$ if and only if $P(a) = 0$.

Remainder theorem the remainder on dividing $P(x)$ by $ax - b$ is $P\left(\frac{b}{a}\right)$.

Quadratic formula for $ax^2 + bx + c = 0$, $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant $\Delta = b^2 - 4ac$: two distinct real roots if $\Delta > 0$; one repeated root if $\Delta = 0$; no real roots if $\Delta < 0$.

Vertex form $y = a(x - h)^2 + k$ has vertex (h, k). Completing the square gives $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$.

Binomial expansion $(a + b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$. The general term is $\binom{n}{r} a^{n-r} b^r$.

Difference of squares $a^2 - b^2 = (a - b)(a + b)$. **Sum and difference of cubes** $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$.

Index laws $a^m a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$, $(ab)^n = a^n b^n$, $a^0 = 1$ for $a \neq 0$.

Logarithm laws $\log_a(xy) = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$, $\log_a x^n = n \log_a x$.

Change of base $\log_a x = \frac{\log_b x}{\log_b a}$, so $\log_a x = \frac{\ln x}{\ln a}$.

Logarithm and exponential $y = a^x \Leftrightarrow x = \log_a y$; $\ln e^x = x$ and $e^{\ln x} = x$ for $x > 0$.

Transformations, composites and inverses

Transformation $y = af\big(b(x - h)\big) + k$: multiply y-values by a; divide x-values by b; translate right by h and up by k.

Composite $(f \circ g)(x) = f(g(x))$: the domain of $f \circ g$ contains those x in the domain of g for which $g(x)$ lies in the domain of f.

Inverse $f^{-1}(y) = x \Leftrightarrow f(x) = y$. The graph of f^{-1} is the reflection of the graph of f in $y = x$.

One-to-one an inverse function exists precisely when f is one-to-one on its domain; otherwise restrict the domain.

Sequences, series and finance

Arithmetic $u_n = u_1 + (n - 1)d$; $S_n = \frac{n}{2}\big(2u_1 + (n - 1)d\big) = \frac{n}{2}(u_1 + u_n)$.

Geometric $u_n = u_1 r^{n-1}$; $S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}$ for $r \neq 1$.

Sum to infinity $S_\infty = \frac{u_1}{1 - r}$, valid when $|r| < 1$.

Compound interest $A = P(1 + r)^n$, with r the decimal rate per period and n the number of periods. Depreciation uses $1 - r$.

Effective annual rate $(1 + i)^k - 1$ when a nominal annual rate i is compounded k times a year.

Sigma $\sum_{k=1}^{n} 1 = n$, $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

REF

REFERENCE

R Geometry, vectors, data and chance

Triangle trigonometry

Sine rule $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$, where R is the circumradius.

Cosine rule $a^2 = b^2 + c^2 - 2bc\cos A$, and $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$.

Area $A = \frac{1}{2}ab\sin C$. Heron's formula: $A = \sqrt{s(s-a)(s-b)(s-c)}$, $s = \dfrac{a+b+c}{2}$.

Ambiguous case when given two sides and a non-included angle, zero, one or two triangles may exist; check the angle sum.

Angles of elevation and depression measured from the horizontal, upward and downward respectively. A bearing is measured clockwise from north and written with three digits.

Radians arc $s = r\theta$; sector area $A = \frac{1}{2}r^2\theta$; θ in radians. π rad $= 180^\circ$.

Circle geometry

Angle at the centre is twice the angle at the circumference standing on the same arc.

Angle in a semicircle is a right angle; conversely, if the angle subtended by a chord is a right angle, the chord is a diameter.

Cyclic quadrilateral opposite angles sum to 180°; the exterior angle equals the interior opposite angle.

Tangent–radius a tangent is perpendicular to the radius at the point of contact. Tangents from an external point are equal in length.

Alternate segment the angle between a tangent and a chord equals the angle in the alternate segment.

Intersecting chords $AP \cdot PB = CP \cdot PD$; for a secant and tangent from an external point, $PT^2 = PA \cdot PB$.

Vectors and matrices

Magnitude $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$ in two dimensions, $\sqrt{a_1^2 + a_2^2 + a_3^2}$ in three.

Unit vector $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{|\mathbf{a}|}$, pointing in the same direction.

Scalar product $\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$; perpendicular vectors have $\mathbf{a}\cdot\mathbf{b} = 0$.

Angle between vectors $\cos\theta = \dfrac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.

Determinant for $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det\mathbf{A} = ad - bc$.

Inverse $\mathbf{A}^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, existing when $ad - bc \neq 0$.

Solving systems $\mathbf{Ax} = \mathbf{b}$ has the unique solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ when $\det\mathbf{A} \neq 0$.

Transformations $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ rotates 90° anticlockwise; $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ reflects in the x-axis; $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$ enlarges by k.

Statistics

Mean $\bar{x} = \dfrac{\sum x}{n}$ or $\bar{x} = \dfrac{\sum fx}{\sum f}$ for grouped or frequency data, using midpoints.

Variance $\sigma^2 = \dfrac{\sum (x-\bar{x})^2}{n} = \dfrac{\sum x^2}{n} - \bar{x}^2$; standard deviation $\sigma = \sqrt{\sigma^2}$.

Quartiles Q_1 and Q_3 at the medians of the lower and upper halves; IQR $= Q_3 - Q_1$.

Outliers values below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$.

Correlation $r = \dfrac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \sum (y-\bar{y})^2}}$, with $-1 \le r \le 1$.

Regression line $\hat{y} = a + bx$, $b = \dfrac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$, $a = \bar{y} - b\bar{x}$. The line passes through $(\bar{x}, \bar{y})$.

Residual $y - \hat{y}$. A residual plot should show no pattern; a curve indicates the linear model is unsuitable.

Probability

Complement $P(A') = 1 - P(A)$. **Addition** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Independent events $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$.

Conditional $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, $P(B) > 0$.

Multiplication $P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)$.

Expected frequency $nP(A)$ over n trials.

Counting ${}^nP_r = \dfrac{n!}{(n-r)!}$, ${}^nC_r = \dfrac{n!}{r!(n-r)!}$.

Binomial probability $P(X = r) = {}^nC_r\, p^r (1-p)^{n-r}$ for $r = 0, 1, \ldots, n$, when trials are independent with constant p.

Modelling and optimisation

Linear programming evaluate the objective at the vertices of the feasible region; the optimum of a linear objective occurs at a vertex or along an entire edge.

Sensitivity the amount an input may change before the optimal decision changes. Report the range, not only the point.

REFERENCE

G Glossary

Ambiguous case. The situation in which two sides and a non-included angle determine zero, one or two triangles.

Argument of a logarithm. The number whose logarithm is taken; it must be positive.

Asymptote. A line that a curve approaches without meeting, such as $x = 0$ for $y = \ln x$ or $y = 0$ for $y = 2^{-x}$.

Bearing. A direction measured clockwise from north, written with three figures, for example 047°.

Binomial coefficient. The number $\binom{n}{r} = {}^nC_r$ of ways to choose r objects from n; the coefficient of $a^{n-r}b^r$ in $(a + b)^n$.

Bivariate data. Data in which two variables are recorded for each individual, allowing association to be studied.

Causation. A genuine cause-and-effect link. Correlation alone does not establish it.

Centre of a circle. The fixed point equidistant from every point on the circle.

Chord. A straight line joining two points on a circle.

Common difference. The constant amount added to each term of an arithmetic sequence.

Common ratio. The constant multiplier from one term to the next in a geometric sequence.

Composite function. A function formed by applying one function to the output of another, written $f \circ g$.

Conditional probability. The probability of an event given that another event has occurred, $P(A \mid B)$.

Congruent. Identical in shape and size.

Constraint. A restriction on the variables of a model, usually written as an inequality.

Correlation coefficient. A number between -1 and 1 measuring the strength and direction of a linear association.

Cosine rule. The relationship $a^2 = b^2 + c^2 - 2bc\cos A$ in any triangle.

Cyclic quadrilateral. A quadrilateral whose vertices lie on a circle; its opposite angles sum to 180°.

Decay. A decrease by a constant ratio per period, modelled by $y = ka^x$ with $0 < a < 1$.

Degree of a polynomial. The highest power of the variable with a non-zero coefficient.

Determinant. A number associated with a square matrix; for a 2×2 matrix $ad - bc$. A zero determinant means no inverse.

Dispersion. The spread of data about a centre; measured by range, interquartile range or standard deviation.

Domain. The set of inputs for which a function is defined.

Dot product. See scalar product.

e. The natural exponential base, approximately 2.71828; the value of $\left(1 + \frac{1}{n}\right)^n$ for large n.

Exponential function. A function of the form $f(x) = ka^x$ with $a > 0$, $a \neq 1$.

Extrapolation. Using a model outside the range of the data; the further outside, the less reliable the prediction.

Factor theorem. $x - a$ is a factor of $P(x)$ exactly when $P(a) = 0$.

Feasible region. The set of points satisfying all the constraints of a linear programming problem.

Geometric series. The sum of the terms of a geometric sequence.

Gradient. The rate of change of y with respect to x; the tangent of the angle a line makes with the horizontal.

Identity. An equation true for every permitted value, written with $\equiv$.

Increasing function. A function for which $x_1 < x_2$ implies $f(x_1) < f(x_2)$.

Independence. Two events are independent when the occurrence of one does not change the probability of the other.

Index (exponent). The power to which a base is raised in a^n.

Interpolation. Estimating a value inside the range of the data, generally more reliable than extrapolation.

Inverse function. The function that reverses another, defined only when the original is one-to-one on its domain.

Inverse matrix. The matrix $\mathbf{A}^{-1}$ satisfying $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, existing when $\det \mathbf{A} \neq 0$.

IQR. Interquartile range, $Q_3 - Q_1$, the spread of the middle half of the data.

Leading term. The term of highest degree in a polynomial; it determines the end behaviour.

Linear programming. Optimising a linear objective subject to linear constraints.

Logarithm. The inverse of an exponential: $\log_a y$ is the power to which a must be raised to give y.

Magnitude. The length of a vector, written $|\mathbf{a}|$.

Matrix. A rectangular array of numbers; a 2×2 matrix represents a linear transformation of the plane.

Mean. The arithmetic average, the sum of the values divided by their number.

Median. The middle value of an ordered data set, or the mean of the two middle values.

Misconception. A plausible but incorrect belief about a mathematical idea, often revealed by a consistent error.

Model. A simplified mathematical description of a real situation, built to answer a question rather than to reproduce every detail.

Modelling cycle. Formulate, solve, interpret, evaluate, communicate, then revise.

Modulus. The absolute value of a number, its distance from zero, written $|x|$.

Mutually exclusive. Events that cannot occur together, so $P(A \cap B) = 0$.

Natural logarithm. A logarithm to base e, written $\ln x$.

Objective function. The quantity a linear programming model seeks to maximise or minimise.

One-to-one. A function in which distinct inputs always produce distinct outputs; such a function has an inverse.

Optimisation. Finding the best value of an objective subject to constraints.

Outlier. A value far from the rest of the data, often defined as below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$.

Period. The length of one complete cycle of a repeating pattern; in finance, the interval between interest payments.

Permutation. An arrangement in which order matters; ${}^{n}P_{r}$ counts them.

Perpendicular vectors. Vectors whose scalar product is zero.

Point of inflection. A point where a curve changes the direction of its bending.

Polynomial. An expression of the form $a_n x^n + \cdots + a_1 x + a_0$ with non-negative integer powers.

Population. The whole group about which data are wanted; a sample is drawn from it.

Position vector. The vector from the origin to a point, giving its coordinates as components.

Prerequisite. A skill or idea assumed before a new topic begins.

Principal value. The conventionally chosen value of an inverse trigonometric function, for example $-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2}$.

Probability distribution. The set of possible values of a random variable together with their probabilities.

Proof by contradiction. Assuming the opposite of what is to be proved and deriving an impossibility.

Quartile. One of the values dividing ordered data into four groups: Q_1, Q_2 (median), Q_3.

Radians. The angle subtended at the centre of a circle by an arc equal in length to the radius; π radians $= 180^\circ$.

Radian measure. Measurement of angles in radians rather than degrees.

Range (of a function). The set of outputs the function actually takes.

Range (of data). The largest value minus the smallest.

Rational function. A quotient of two polynomials, defined where the denominator is non-zero.

Reciprocal. The multiplicative inverse, $\dfrac{1}{x}$ for $x \neq 0$.

Regression line. The least-squares line used to predict one variable from another.

Remainder theorem. The remainder when $P(x)$ is divided by $x - a$ is $P(a)$.

Residual. The difference between an observed value and the value predicted by a model.

Root. A solution of an equation; on a graph, an x-intercept.

Sample. A subset of a population, selected to represent it.

Sample space. The set of all possible outcomes of a probability experiment.

Scalar product. $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$, a number, not a vector.

Secant. A line that cuts a circle at two points; also the reciprocal of cosine.

Sequence. An ordered list of numbers, often generated by a rule.

Series. The sum of the terms of a sequence.

Sigma notation. The use of $\sum$ to write a sum compactly.

Sign diagram. A number-line diagram showing where an expression is positive, zero or negative.

Sine rule. The relationship $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ in any triangle.

Skewed distribution. A distribution that is not symmetric; the mean is pulled towards the long tail.

Slant height. The distance from the apex of a cone or pyramid to the edge of the base, measured along a face.

Standard deviation. The square root of the variance; a measure of spread in the units of the data.

Standard form. The notation $a \times 10^n$ with $1 \leq a < 10$ and n an integer.

Stationary point. A point where the gradient is zero; a maximum, minimum or point of inflection.

Statement of inquiry. A sentence that frames a unit by linking a key concept, related concepts and a global context.

Substitution. Replacing a variable by a value or expression, as in evaluating $P(2)$.

Sum to infinity. The limit of the partial sums of an infinite geometric series, existing when $|r| < 1$.

Symmetry. A transformation that maps a figure onto itself.

Tangent (to a circle). A line meeting the circle at exactly one point, perpendicular to the radius there.

Term. A single number or product of numbers and variables within an expression; also an element of a sequence.

Transformation. A change of position, size or orientation of a graph or figure.

Translation. A shift of every point by the same vector.

Transpose. A matrix formed by interchanging rows and columns.

Tree diagram. A branching diagram showing successive events and their probabilities.

Trend. The general direction of a relationship in bivariate data.

Trigonometric ratio. A ratio of two sides of a right-angled triangle, depending only on the angle.

Unit vector. A vector of length 1, obtained by dividing a vector by its magnitude.

Variable. A quantity that can change or take different values; in a model, a quantity the designer can choose or that varies with conditions.

Variance. The mean of the squared deviations from the mean; its square root is the standard deviation.

Vector. A quantity with both magnitude and direction, represented by a directed line segment or a column of components.

Venn diagram. A diagram using overlapping regions to show sets and their intersections.

Vertex (of a parabola). The turning point, (h, k) in $y = a(x - h)^2 + k$.

Vertex (of a region). A corner point of a feasible region, where a linear objective may be optimised.

Weighted mean. A mean in which values are multiplied by weights reflecting their importance before summing.

z-score. The number of standard deviations a value lies from the mean.

REFERENCE

I Index

A

algebraic fraction 11
alternate segment 120
annuity 78
arc 111
arithmetic sequence 65
arithmetic series 68
assumption 219

B

back-bearing 138
boundary line 94
box plot 175

C

change of base 51
chord 111
clinometer 143
collinear 156
combination 208
common difference 65
common ratio 71
complement 197
composite function 30
compound interest 78
concyclic 111
conditional probability 200
confounding variable 188
constant term 3
constraint 226
converge 75
converse 122
correlation coefficient 183
cosine rule 134
critical values 91
cyclic quadrilateral 114

D

decreasing function 25
degree 3
determinant 163
dihedral angle 140
discriminant 91
dispersion 175
displacement vector 156
diverge 75
domain 25
doubling time 45

E

e 48
event 197
excluded value 12
experimental probability 198
explanatory variable 183
exponential equation 53
exponential function 45
extraneous solution 14
extrapolation 186

F

factor 3
factor theorem 9
factorial 208
feasible region 96
five-number summary 175
function 25

G

geometric sequence 71
geometric series 75
grouped data 178
growth factor 45

H

half-life 45
half-plane 94
horizontal line test 32
hypothesis 122

I

identity matrix 161
included angle 136
increasing function 25
independent events 203
interquartile range 175
intersection 197
invariant 124
inverse function 32

inverse matrix 163

L

leading coefficient 3
line of best fit 185
locus 126
logarithm 50
logarithmic scale 55

M

magnitude 153
mathematical model 219
matrix 160
modelling cycle 217
multiplicity 4

N

natural domain 25
natural logarithm 51

O

objective 226
objective function 98
one-to-one 25
orientation 166
outlier 175

P

parallax 143
parameter 219
parent graph 27
permutation 208
point of contact 116
polynomial 3
position vector 156
projection 140
proof 122

Q

quadratic inequality 91

R

range 25, 175
rational equation 14
recurrence relation 78
remainder theorem 9
residual 186
root 3

S

sample space 197
scalar 153
scatter graph 183
secant 116
section formula 156
sensitivity 100
sequence 65
series 68
shear 161
sign diagram 91
sine rule 131
singular matrix 163
standard deviation 177
subtend 111
sum to infinity 75
synthetic division 7

T

tangent 116
term 65
three-figure bearing 138
total probability 205
transformation 161
tree diagram 205
triangulation 143

U

unbounded 96
union 197
unit vector 153

V

validation 219
variance 177
vector 153
vertex 96

SUPPLEMENTARY SECTION

MYP Mathematics 5 — Student Coursebook

Spec-gap additions and verified corrections · 2026 verification pass

- How to use this section
- Chapters 1–10 — Try-it checks and Stretch & Apply boxes, with answers
- Criterion strand lines for the assessment tasks
- Answers-section entries to add
- Corrections applied in this edition

Appended to the coursebook — the original pagination is unchanged.

How to use this section

Same gap as MYP1–4: the front matter ("How the book works", p vii) describes the chapter rhythm without **Try-it checks** or **Stretch & Apply (SaA)** boxes, and the book has none. One of each per content chapter (Chapters 1–10; Chapter 11 is the capstone modelling project and Chapter 12 is cumulative revision — both excluded, as MYP3's Part E was).

Placement convention

- **Try-it check**: after the first worked example of the chapter's central section

(section noted per chapter).

- **Stretch & Apply**: at the end of the concept sequence, before the chapter's

investigation / modelling task.

Label-page text to add (front matter, "How the book works"): same two definitions as ../MYP1/chapter_additions.md.

> Try-it checks ask one or two rapid questions the moment a method is first shown. They test whether the new move is safe before you carry on; answers are given, so you can correct yourself in seconds. Stretch & Apply boxes move the method one step into a new context. They are deliberately short, sit at the end of a concept sequence, and reward connecting the idea to something you have already seen.

Chapter 1 — Polynomials, Factor Theorem and Algebraic Fractions

Trial placement: after the first worked example in 1.3 (remainder and factor theorems); SaA before the investigation.

Try-it check

Let $f(x) = x^3 - 6x^2 + 11x - 6$.

1. Find $f(1)$.
2. What does $f(1) = 0$ tell you about $x - 1$?

Answers: (1) 0. (2) $x - 1$ is a factor of f (factor theorem), so $x = 1$ is a root.

Stretch & Apply

An open box is made from a 12 cm square sheet by cutting squares of side x from the corners; its volume is $V = x(12 - 2x)^2$. Show that V is a cubic polynomial in x, then find V when $x = 2$.

Answer: $V = x(144 - 48x + 4x^2) = 4x^3 - 48x^2 + 144x$; $V(2) = 32 - 192 + 288 = 128$ cm^3.

Criterion strand lines — insert directly under each task heading

A · "Polynomial structure and division"

Strands: A i — select between division, remainder and factor-theorem methods; A ii — apply them accurately (exact and 3 d.p. answers as required); A iii — solve the set correctly.

B · "Remainders that follow a pattern"

Strands: B i — tabulate R_n, the remainder of $x^n - 1$ on division by $x - 2$, for $n = 1 \ldots 5$; B ii — describe the

pattern of the remainders as a general rule; B iii — verify the rule on further n without a calculator.

C · "Explaining the factor theorem"

Strands: C i — state the division identity with correct notation; C ii — show why substituting $x = a$ gives the remainder; C iii — explain the special case when the remainder is zero; C iv — a structured explanation with one worked example and one non-example.

D · "Designing the strongest box"

Strands: D i — identify the relevant elements (30 cm × 20 cm sheet, squares of side x, minimum volume 800 cm^3); D ii — select and apply the polynomial volume model with its domain; D iii — justify the accuracy of the recommended cut size; D iv — judge the design in the manufacturer's context.

Chapter 2 — Functions: Transformations, Composites and Inverses

Trial placement: after the first worked example in 2.3 (composite functions); SaA before the investigation.

Try-it check

Let $f(x) = 2x + 3$ and $g(x) = x^2$.

1. Find $f(g(x))$.
2. Find $g(f(x))$.

Answers: (1) $2x^2 + 3$. (2) $(2x + 3)^2$ — the order changes the result.

Stretch & Apply

$f(x) = 3x - 2$. Find $f^{-1}(x)$, then use the inverse (not inspection) to solve $3x - 2 = 17$.

Answer: $f^{-1}(x) = (x + 2)/3$; $x = f^{-1}(17) = 19/3$.

Criterion strand lines — insert directly under each task heading

A · "Functions and their properties"

Strands: A i — select methods for domain, range, composites and inverses; A ii — apply them accurately; A iii — solve the set correctly.

B · "Iterating a function"

Strands: B i — tabulate $f^{(1)}$, $f^{(2)}$, … for $f(x) = 2x + 1$; B ii — describe the pattern in the coefficients and constant terms as a general rule; B iii — verify the rule on further iterates.

C · "Explaining the order of transformations"

Strands: C i — correct notation for $y = f(2x - 6)$; C ii — move between the algebraic form and the sequence of transformations; C iii — a complete correction of the "compress then shift 6 right" reading; C iv — a structured explanation with a sketch.

D · "Modelling a theme-park ride"

Strands: D i — identify the relevant elements (60 m drop, $h(t) = 60 - 5t^2$, the domain $0 \le t \le 2\sqrt{3}$); D ii — select and apply the quadratic height model; D iii — justify the accuracy of the times and heights; D iv — judge the model's validity for the ride.

Chapter 3 — Exponential and Logarithmic Functions

Trial placement: after the first worked example in 3.1 (exponential equations); SaA before "how many digits?".

Try-it check

1. Solve $2^x = 32$.
2. Solve 5^(2x) = 125.

Answers: (1) $x = 5$. (2) $2x = 3$, so $x = 1.5$.

Stretch & Apply

A population doubles every 40 years. Write a model for P (in initial-value units) after t years, and find the annual growth factor to 2 decimal places.

Answer: P = 2^(t/40); annual factor 2^(1/40) = e^(ln 2/40) ≈ 1.02.

Criterion strand lines — insert directly under each task heading

A · "Exponentials and logarithms"

Strands: A i — select the exponential and logarithm rules; A ii — apply them accurately (exact and 3 s.f. answers); A iii — solve the set correctly.

B · "Powers, digits and leading digits"

Strands: B i — tabulate $\log_{10}(2^n)$, the number of digits and the leading digit for $n = 1 \ldots 12$; B ii — describe how the digit count changes as n grows, as a general rule; B iii — verify with further n.

C · "Why logarithms solve exponential equations"

Strands: C i — define a logarithm as an inverse operation; C ii — show the steps that isolate x in a base-30 equation; C iii — explain why the method works for any positive base; C iv — a structured explanation with a worked example.

D · "A cooling drink and a warming room"

Strands: D i — identify the relevant elements ($T(t) = 21 + 74(0.91)^t$, served at 95 °C, drinking window 55–35 °C); D ii — select and apply the exponential decay model; D iii — justify the accuracy of the times to the minute; D iv — judge the model in the café context.

Chapter 4 — Sequences and Series

Trial placement: after the first worked example in 4.3 (geometric sequences); SaA before "infinite sums".

Try-it check

The geometric sequence 3, 6, 12, …

1. Find u_8.
2. Find the sum of the first 8 terms.

Answers: (1) $3 \times 2^7 = 384$. (2) $S_8 = 3(2^8 - 1) = 765$.

Stretch & Apply

A ball that bounces to ¾ of its previous height is dropped from 1 m. Find the total distance it travels (down and up) before coming to rest.

Answer: $1 + 2(0.75 + 0.75^2 + \cdots) = 1 + 2 \times (0.75/0.25) = 7$ m.

Criterion strand lines — insert directly under each task heading

A · "Sequences and series"

Strands: A i — select rules for arithmetic and geometric sequences, annuities and infinite sums; A ii — apply them accurately; A iii — solve the set correctly.

B · "A tower of cans"

Strands: B i — tabulate the can counts for a tower whose rows grow 2, 5, 8, …; B ii — describe each pattern in words before writing the general formula; B iii — verify the formula on further rows.

C · "Explaining an infinite sum"

Strands: C i — define a partial sum and convergence with correct notation; C ii — use a table of partial sums of $1 + ½ + ¼ + \cdots$ to show convergence; C iii — address the $0.999\ldots = 1$ objection; C iv — a structured explanation for a classmate.

D · "Save now or borrow?"

Strands: D i — identify the relevant elements (£9,500 equipment, 36 months, 3.6% and 7.2% compounded monthly); D ii — select and apply the annuity and amortisation formulas; D iii — justify the accuracy of the monthly amounts to the penny; D iv — judge which option the group should take.

Chapter 5 — Quadratic Inequalities and Linear Programming

Trial placement: after the first worked example in 5.1 (quadratic inequalities in one variable); SaA before "integer solutions and the vertex".

Try-it check

1. Solve $x^2 - 5x + 6 < 0$.
2. Solve $x^2 + 2x - 8 \geq 0$.

Answers: (1) $2 < x < 3$. (2) $x \leq -4$ or $x \geq 2$.

Stretch & Apply

A farmer grows wheat (profit £100 per tonne) and maize (profit £60 per tonne). She can grow at most 30 tonnes in total, must grow at least 5 tonnes of maize, and can grow at most 20 tonnes of wheat. What mix maximises profit?

Answer: profit $100w + 60m$ over vertices $(0, 5) \to 300$; $(0, 30) \to 1800$; $(20, 10) \to 2600$; $(20, 5) \to 2300$. Maximum £2600 at 20 t wheat, 10 t maize.

Criterion strand lines — insert directly under each task heading

A · "Inequalities and feasible regions"

Strands: A i — select methods for quadratic inequalities and feasible regions; A ii — apply them accurately; A iii — solve the set correctly.

B · "How the objective chooses a corner"

Strands: B i — tabulate $P = x + ky$ at the four vertices (A(0, 0), B(10, 0), C(4, 6), D(0, 8)) for $k = 0.5, 1, 1.5, 2, 3$; B ii — describe when the optimal vertex changes, as a general rule; B iii — verify with further values of k.

C · "Why the best point is a corner"

Strands: C i — correct terminology for objective function, constraints and vertices; C ii — move between the graph of the level lines and the table of vertex values; C iii — a complete correction of the "largest coordinates" idea; C iv — a structured explanation with a sketch.

D · "The school canteen"

Strands: D i — identify the relevant elements ($x + y \leq 28$, $x + 2y \leq 44$, $y \leq 18$, contributions £1.50 and £2.00); D ii — select and apply the linear-programming method (vertices, binding constraints); D iii — justify the accuracy of the profit figure; D iv — judge the recommended daily menu.

Chapter 6 — Circle Geometry and Geometric Proof

Trial placement: after the first worked example in 6.1 (angles in circles); SaA before "circle theorems under transformation".

Try-it check

1. An angle at the circumference subtends an arc of 100°. Find the angle.
2. A cyclic quadrilateral has one angle of 72°. Find the opposite angle.

Answers: (1) 50° (angle at centre is twice the angle at circumference). (2) 108° (opposite angles of a cyclic quadrilateral sum to 180°).

Stretch & Apply

AB is a diameter of a circle and C is on the circle. If $\angle CAB = 34°$, find $\angle ABC$.

Answer: $\angle ACB = 90°$ (angle in a semicircle), so $\angle ABC = 180° - 90° - 34° = 56°$.

Criterion strand lines — insert directly under each task heading

A · "Angles, tangents and chords"

Strands: A i — select the circle theorems that apply (angles, tangents, chords); A ii — apply them with labelled diagrams; A iii — solve the set correctly.

B · "The arc of constant angle"

Strands: B i — record angle measurements for points on a fixed arc in a table; B ii — describe the constant-angle behaviour as a general rule; B iii — verify with further points, including the reflex case.

C · "Explaining why the centre angle doubles"

Strands: C i — state the centre-angle theorem and identify the arc; C ii — use two isosceles triangles and the exterior-angle step; C iii — explain the reflex case; C iv — a structured proof with a labelled diagram.

D · "A circular stained-glass panel"

Strands: D i — identify the relevant elements (1.5 m circular window, cyclic quadrilateral pane, proposed angles 80°, 100°, 95°, 85°); D ii — select and apply the opposite-angles supplementary condition; D iii — justify the corrected angle order; D iv — judge the design for the window.

Chapter 7 — Sine and Cosine Rules; Trigonometry in Three

Dimensions

Trial placement: after the first worked example in 7.1 (the sine rule); SaA before "measuring the unmeasurable".

Try-it check

In a triangle, $a = 8$, $A = 30^\circ$ and $B = 45^\circ$.

1. Find b.
2. Find C.

Answers: (1) $b = 8 \sin 45^\circ / \sin 30^\circ = 8\sqrt{2} \approx 11.3$. (2) $C = 105^\circ$.

Stretch & Apply

A vertical tower is observed from two points 100 m apart on a straight horizontal line; the angles of elevation are 30° and 50° (from the nearer point). Find the tower's height to 1 decimal place.

Answer: $h = 100 \tan 30^\circ \tan 50^\circ / (\tan 50^\circ - \tan 30^\circ) \approx 112.0$ m.

Criterion strand lines — insert directly under each task heading

A · "Solving triangles"

Strands: A i — select between the sine and cosine rules (and 3D setups); A ii — apply them accurately; A iii — solve the set correctly.

B · "The ambiguous triangle"

Strands: B i — tabulate the cases where SSA gives two, one or no triangle; B ii — describe the boundary between the cases as a general rule; B iii — verify with further data sets, keeping angles to 1 d.p.

C · "Why two rules?"

Strands: C i — derive the cosine rule from coordinates with correct notation; C ii — show it reduces to Pythagoras when $A = 90^\circ$; C iii — explain which information each rule requires; C iv — a structured comparison with an ambiguous example resolved by the cosine rule.

D · "A coastal survey"

Strands: D i — identify the relevant elements (A, B 4.2 km apart, B due east; L on bearings 062° and 318°; C 2.8 km from A on 145°); D ii — select and apply the bearing diagram and the sine and cosine rules; D iii — justify the accuracy of the distances in km; D iv — judge the patrol boat's route in context.

Chapter 8 — Vectors and Matrices

Trial placement: after the first worked example in 8.1 (vector notation and operations); SaA before section 8.6.

Try-it check

$\mathbf{a} = (3, -2)$ and $\mathbf{b} = (-1, 4)$.

1. Find $\mathbf{a} + \mathbf{b}$.
2. Find $|\mathbf{a}|$.

Answers: (1) (2, 2). (2) $\sqrt{13} \approx 3.6$.

Stretch & Apply

A = (1 2; 3 4). Find A^{-1} and verify that AA^{-1} is the identity.

Answer: det A = −2, so A^{-1} = (−2 1; 1.5 −0.5); AA^{-1} = (−2+3, 1−1; −6+6, 3−2) = (1 0; 0 1).

Criterion strand lines — insert directly under each task heading

A · "Vectors, matrix products and inverses"

Strands: A i — select methods for vector operations, matrix multiplication and inverse; A ii — apply them accurately; A iii — solve the set correctly.

B · "Repeating a shear"

Strands: B i — tabulate the effect of the shear S, S^2, S^3 on fixed test points; B ii — describe the pattern in the matrix entries as a general rule; B iii — verify the rule on S^4 and further powers.

C · "Explaining why order matters"

Strands: C i — state the correct meaning of ABv (B first, then A); C ii — give a numerical counterexample using a rotation and a reflection; C iii — explain why the order changes the result; C iv — a structured explanation with a diagram.

D · "Mapping a survey by drone"

Strands: D i — identify the relevant elements (the camera-to-map matrix and the survey points); D ii — select and apply the matrix mapping (with its translation); D iii — justify the accuracy of the map coordinates in metres; D iv — judge the mapped positions for the survey.

Chapter 9 — Statistics: Dispersion, Correlation and Regression

Trial placement: after the first worked example in 9.2 (standard deviation and grouped data); SaA before "correlation, causation and reliability".

Try-it check

1. Find the mean of 2, 4, 6, 8, 10.
2. Find the standard deviation of 2, 4, 6, 8, 10 (using the ÷ n convention), to 2 decimal places.

Answers: (1) 6. (2) deviations −4, −2, 0, 2, 4; squared sum 40; 40/5 = 8; √8 ≈ 2.83.

Stretch & Apply

A least-squares line $\hat{y} = 4 + 0.8x$ has $r = 0.6$. Interpret r^2 in context, and state one thing it cannot tell you.

Answer: $r^2 = 0.36$ — the linear model explains 36% of the variation in y; it cannot establish that x causes y (or that the relationship is linear beyond the data range).

Criterion strand lines — insert directly under each task heading

A · "Measuring and interpreting spread"

Strands: A i — select the dispersion, correlation and regression methods; A ii — apply them accurately (no rounding of intermediate totals); A iii — solve the set correctly.

B · "One point, four statistics"

Strands: B i — keep a full table of totals for the ten (revision, score) pairs; B ii — compute the four summary statistics and describe what each one shows (and misses) about the outlier score 18; B iii — verify each statistic by a second route.

C · "What the newsletter may claim"

Strands: C i — explain every statistical term before using it, for a non-specialist reader; C ii — separate association from causation in the breakfast-club claim; C iii — state what the data may and may not support; C iv — continuous prose with a clear bottom line for the council.

D · "What is a used car worth?"

Strands: D i — identify the relevant elements (eight (age, price) pairs in thousand pounds); D ii — select and apply the regression model (line, r, its meaning); D iii — justify the accuracy of price predictions; D iv — judge the model at the extremes of the age range.

Chapter 10 — Probability: Conditional Probability and Counting

Trial placement: after the first worked example in 10.2 (conditional probability); SaA before "the birthday problem".

Try-it check

A bag holds 4 red and 6 blue balls; two are drawn without replacement.

1. Find P(first ball is red).
2. Find P(both balls are red).

Answers: (1) 4/10. (2) (4/10)(3/9) = 2/15 ≈ 0.133.

Stretch & Apply

How many 4-digit codes can be made from the digits 1–9 with no digit repeated? How many of them end in an even digit?

Answer: $9 \times 8 \times 7 \times 6 = 3024$; codes ending in an even digit: $4 \times 8 \times 7 \times 6 = 1344$.

Criterion strand lines — insert directly under each task heading

A · "Sets, conditionals and counting"

Strands: A i — select the set, conditional and counting methods; A ii — apply them accurately (exact fractions and 3 d.p. answers); A iii — solve the set correctly.

B · "The sum of two dice"

Strands: B i — complete the 6×6 addition table and count outcomes for each total; B ii — describe the pattern of the probabilities as a general rule; B iii — verify the total of the probabilities is 1.

C · "Explaining a positive test result"

Strands: C i — distinguish $P(+ \mid C)$ from $P(C \mid +)$ in plain language; C ii — use a natural-frequency table (10,000 people, 2% prevalence, 90% sensitivity); C iii — compute the true probability of the condition after a positive test; C iv — a structured explanation for a non-specialist.

D · "A screening decision"

Strands: D i — identify the relevant elements (10,000 adults, 1.5% prevalence, 90% detection, 2% false-positive

rate, confirmatory test); D ii — select and apply the two-way natural-frequency table and conditional probabilities; D iii — justify the accuracy of the percentages reported; D iv — judge the screening policy in context.

Section 12.5 — Cumulative Examination

(one timed 90-minute examination, criterion-mapped in four blocks)

Criterion A · "Core methods" — add:

> Strands: A i — select across the Year 5 methods (polynomials, functions, exponential and logarithm rules, sequences, circles, trigonometry, statistics, probability); A ii — apply them accurately under timed conditions; A iii — show reasoning for every mark above 1.

Criterion B · "A pattern in the differences" — add:

> Strands: B i — compute the first and second differences of the tabulated quadratic function; B ii — describe the constant second difference as a general rule for quadratics; B iii — verify the recovered formula against the table.

Criterion C · "Explaining a decision" — add:

> Strands: C i — correct terminology for correlation ($r = 0.75$, not a percentage); C ii — separate association from causation; C iii — state what the correlation does and does not show; C iv — a 200–250 word response that corrects both claims.

Criterion D · "A community water tank" — add:

> Strands: D i — identify the relevant elements (20,000 L tank, $V(d) = 20{,}000 - 1{,}800d$, one week of consumption); D ii — select and apply the linear drain model; D iii — justify the accuracy of the day and time estimates; D iv — judge the tank's adequacy for the week.

Answers-section entries to add

Append under a new heading in the answers part (book p245 onwards):

> TRY-IT & STRETCH & APPLY — CH 1: 1 (0; $x - 1$ is a factor) 2 ($4x^3 - 48x^2 + 144x$; 128 cm^3). CH 2: 1 ($2x^2 + 3$; $(2x+3)^2$) 2 ($(x+2)/3$; $x = 19/3$). CH 3: 1 (5; 1.5) 2 ($P = 2\hat{}(t/40)$; ≈ 1.02). CH 4: 1 (384; 765) 2 (7 m). CH 5: 1 ($2 < x < 3$; $x \leq -4$ or $x \geq 2$) 2 (20 t wheat, 10 t maize; £2600). CH 6: 1 (50°; 108°) 2 (56°). CH 7: 1 ($8\sqrt{2} \approx 11.3$; 105°) 2 (≈ 112.0 m). CH 8: 1 ((2, 2); $\sqrt{13}$) 2 ($A^{-1} = (-2\ 1;\ 1.5\ -0.5)$; verified). CH 9: 1 (6; ≈ 2.83) 2 (36% explained; no causation). CH 10: 1 (4/10; $2/15 \approx 0.133$) 2 (3024; 1344).

Corrections applied in this edition

Answer-key corrections applied in this edition (5)

- Chapter 3 · Exercise 3B, Q7 (p261): $2.7^{0.4} = 1.48661$ → **1.48780**; the substitution 2973.22 → **2975.61**; "an error of about 10" → "about 8" ($e^{0.4} = 1.49182$ and $2000e^{0.4} = 2983.65$ were already correct; the true difference is 8.04).
- Chapter 7 · Mixed Practice 7, Q3 (p287): $a^2 = 64.49$ → **79.98** ($9.5^2 + 7.2^2 - 2{\cdot}9.5{\cdot}7.2{\cdot}\cos$

$63^\circ = 142.09 - 62.11$); a = 8.94 cm was already correct.

- Chapter 7 · Mixed Practice 7, Q4 (p287): numerator 16 → **4** ($36 + 49 - 81$); 0.1905 → **0.0476**; A = 87.3° was already correct.
- Chapter 7 · Mixed Practice 7, Q15 (p287): the key used the half-diagonal of an 8 m base ($4\sqrt{2}$); the base is 16 m, so the corrected working reads Half-diagonal = $8\sqrt{2} = 11.31$ m, $l = \sqrt{228} = 2\sqrt{57} \approx$ **15.1 m**, $\tan\alpha = 10/(8\sqrt{2}) = 0.8839$, $\boldsymbol{\alpha} = \mathbf{41.5}^\circ$; the dihedral angle $\beta = 51.3^\circ$ was already correct.
- Criterion D task, Q5 (p289): the reliability sentence is corrected — (2.85, 1.42) is a shift of 0.09 km ($\approx$ 90 m); opposite-signed errors shift the position up to about 0.11 km, so it is reliable to $\pm$0.12 km; then L_C = 3.92 km and the journey is 7.10 km in 17.0 minutes — neither distance nor plan changes.

Body corrections applied in this edition (3)

- Prerequisites for Chapter 11 (p218): a "3 × 3" label over a 2 × 2 array → "2 × 2".
- Section 11.4, GP Q2 hint (p226): "900 − 4a" → "900 + 4a".
- Criterion D, Q5 marking note (p315): the italic note no longer quotes a preamble sentence that is not printed in the book; it now states that the tank is modelled as draining at a constant rate and that answering treats 1800 L/day as the net consumption while the tank drains.

Flagged for a future edition — text left unchanged (1)

- Chapter 11 · Milestone Q4: the key's "10-year payback $\approx$ £3.67/m³" divides by 10 terms; the true 10-year payback is $\approx$ £0.92–1.22/m³.

Cleared on audit — no change

- Mixed Practice 7, Q18 and Chapter 9 Criterion A, Q3: both keys verified correct on re-derivation.